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UPSC CSE Prelims

Topic-wise Solved Paper II

CSAT (2011-2026)



*Source: The Indian Edtech Story: Q1 2021 Round-Up by Markelytics

**Answer keys are based on information available on the official UPSC website at the time of publication.

PREFACE

Unacademy Every year, the Union Public Service Commission (UPSC) conducts the Civil Services Examination (CSE) to recruit for various Civil Services of the Government of India, including the Indian Administrative Service (IAS), Indian Foreign Service (IFS), and Indian Police Service (IPS) and other Central Services.

One of the most difficult exams in the world, it aims to select the best of the best through a rigorous testing process. Consequently, it becomes challenging for the candidates to prepare for the exam because the previous year's papers and guidance are not readily available.

Therefore, we at Unacademy have designed this topic-wise UPSC Prelims Paper II (CSAT) containing more than 1000 questions featured in the last 16 years, along with official answer key and explanations.

While due care has been taken to ensure accuracy in the preparation of this book.

Nonetheless, if there are any corrections, feedback or suggestions, email us at ***learnersupport.upsc@unacademy.com.***

We appreciate the work done by the content team, headed by Ajit Satapathy, for releasing one of the most accurate answer keys by any coaching institute.

Let's Crack It!



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PART III: TOPICWISE SOLVED PAPER PRELIMS CSAT PAPER 2 (2011-2026)

CSAT

1 Comprehension

Directions for the next 3 (three) items:

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

When SARS-CoV-2 was first detected in 2019, it was a truly novel virus for the world. At that time, no one in the world had been exposed to SARS-CoV-2 or had specific immunity against it. In contrast, people across the world have been exposed to HMPV for decades and the virus is well-studied. HMPV and SARS-CoV-2 belong to two very different virus families with fundamentally different characteristics and epidemiology, with strong seasonality seen in HMPV, unlike SARS-CoV-2. Both viruses cause different severity of symptoms, particularly over the long term, and the affected population segments do not fully overlap. In general, HMPV causes milder illness with deaths being very rare and with no long-term post-viral symptoms.

1. Which of the following conclusions is/are valid? (CSAT-2026)

1. Though SARS-CoV-2 and HMPV are similar viruses with somewhat different epidemiology, the former became a pandemic because it was novel and people had not been exposed to it in the past.
2. The two viruses have fundamentally different impacts on human populations and should not therefore be dealt with in a similar manner.

Select the answer using the code given below.

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

2. Which of the following reflect the intent of the writer in the above passage? (CSAT-2026)

1. To evolve methodologies for objective analysis of the two viruses
2. To establish the epidemiological similarities and differences between the two viruses
3. To offer a better understanding of the remedies of HMPV when analysed in conjunction with SARS-CoV-2

Select the answer using the code given below.

(a) 1 and 2 (b) 2 and 3

(c) 1 and 3 (d) None of the above

3. Which of the following statements reflect the logical and rational inferences that can be drawn from the passage? (CSAT-2026)

1. HMPV has, historically, had longer documentation of studies when compared with SARS-CoV2
2. The two viruses are different from each other and this results in markedly different outcomes amongst those affected.
3. Long-term impacts of the two viruses are dissimilar and this is an important differentiator between them.
4. One of the common factors between the two viruses is seasonal specificity.

Select the answer using the code given below.

(a) 1 and 2 only (b) 1, 2 and 3

(c) 3 and 4 (d) 1 and 3 only

Directions for the next 2 (two) items:

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

India is starting to deploy AI for critical use cases such as weather forecasting, pest detection and control, and crop yield optimisation. However, penetration is limited to a small subset of tech-savvy farmers. In the US and in Europe, generative Al tools have started offering precision farming at scale, integrating large datasets to provide real-time agronomic insights. For at-scale integration and accessibility of AI tòols in India, it would be helpful to develop Indian languages-based AI tools for smallholder farmers, partner with AgTechs to create affordable Al solutions, and disseminate AI-based advisory services through government programmes.

4. Which of the following assumptions is/are valid? (CSAT-2026)

1. Agricultural productivity has marched ahead in the West because of the economies of scale facilitated by the adoption of AI tools.
2. Affordable AI tools rendered available in local languages can help AI-based solutions reach more and more small farmers.
3. Though penetration is as yet low, critical application areas deploying AI tools are already in use in India.

Select the answer using the code given below.

(a) 1, 2 and 3 (b) 2 only

(c) 1 and 3 only (d) 2 and 3 only

5. Which of the following statements is/are not correct? (CSAT-2026)

1. Tech-savvy farmers will drive the AgTech companies of the future.
2. The development of advisory services by the government programmes for the use of AI tools in agriculture would be helpful.
3. In the US and Europe, AI tools have replaced traditional agricultural practices.
4. The integration of large datasets for use in real-time agronomic analysis is already a reality.

Select the answer using the code given below.

(a) 1 and 3 only (b) 1, 3 and 4

(c) 2 and 4 (d) 3 only

Directions for the next 2 (two) items:

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

The Juvenile Justice (Care and Protection of Children) Act, or the JJ Act, 2015 allows for the possibility for trying adolescents above 16 as adults if they are accused of committing a heinous offence. A heinous offence is one with a minimum punishment of seven years. Offences such as culpable homicide and causing death by negligence, which are common in drunken driving cases, are not heinous offences because they do not have a prescribed minimum punishment. The JJ Act, amended in 2021, now categorises an offence that has no minimum sentence, but has a maximum sentence of seven years or more as a serious offence which nonetheless, in the opinion of activists, does not merit the transfer of a case to the adult criminal justice system.

6. Which of the following conclusions is/are valid? (CSAT-2026)

1. Only a serious offence as categorised by the revised JJ Act, justifies the transfer of a case to the adult judicial system.
2. The JJ Act, 2021, categorises an offence as a serious offence based on the maximum sentence it carries, rather than on the minimum sentence.

Select the answer using the code given below.

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

7. Which of the following statements is/are correct? (CSAT-2026)

1. If an offence has no minimum prescribed punishment, it cannot be considered heinous as per the JJ Act, 2015.
2. As per the JJ Act, 2021, an offence for which there is a provision for a maximum sentence of seven years or more, but no minimum sentence, is to be considered a serious offence.

Select the answer using the code given below.

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Directions for the next 2 (two) items :

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

The key source of the battle for clean skies and clear lungs is the fuel we burn, from household chulhas to the thermal power plants. In most cases, it is biomass or coal. The Supreme Court banned the use of pet coke, the dirtiest of such fuels. The Delhi Government banned the use of coal, which was later extended to the entire National Capital Region. It was also agreed that the thermal power plants would clean up or shut down. Action on this has been patchy to say the least. The lesson from the transition to CNG is that people need alternatives for a ban to be effective. When diesel buses were stopped, CNG supply had to be assured. It also had to be feasible in terms of cost. The Supreme Court agreed that fiscal measures were needed to keep clean fuel cheaper than dirty fuel. Now even as coal is banned, the price of natural gas makes industry uncompetitive.

8. Which of the following inferences is/are correct? (CSAT-2026)

1. The source of the energy we consume is the key to the battle for cleaner air.
2. Bans are effective where the will is strong and the people are convinced that such bans are for the greater good of society.
3. There is judicial approval for a policy that intervenes fiscally to facilitate benevolent pricing for cleaner fuel.

Select the answer using the code given below.

(a) 1 and 2 (b) 2 and 3

(c) 1 and 3 (d) 1 only

9. Which of the following statements is/are correct? (CSAT-2026)

1. Thermal power stations in Delhi were required to summarily shut down.
2. CNG supplies had to be assured once diesel vehicles were prohibited from plying.
3. The Supreme Court banned the use of coal across the National Capital Region.

Select the answer using the code given below.

(a) 1 and 2 (b) 3 only

(c) 2 only (d) 2 and 3

Directions for the next 2 (two) items:

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

Previous waves of customer-service technology, including email and those pesky voice menus, stoked concerns of job losses, only for them to fail to materialise. AI could yet prove different. And if it does, its effects may be salutary. Human agents could be freed up to spend more time on creative and rewarding tasks, like using feedback to make products and services better, and thereby spend less time listening to irate customers!

10. Which one among the following statements most appropriately reflects the point of view of the given passage? (CSAT-2026)

(a) If AI were to take over customer service, there would be no work left for human subjects to do.

(b) Irritating voice menus and email could not achieve human redundancy to the extent that AI might.

(c) The value of human intervention in the workplace affected by AI might be enhanced through redirection towards more fulfilling tasks.

(d) Unlike previous waves in customer-service technology, AI has raised the alarm of worker replacement.

11. Which of the following conclusions, made on the basis of the given passage, is/are correct? (CSAT-2026)

1. The advent of new customer-service technology had invariably sparked fears about job losses.
2. Often it is found that instead of job losses, alternative channels

for employee engagement are discovered while certain tasks are replaced by technology.

3. The advent of technology inevitably leads to stressful outcomes.

Select the answer using the code given below.

(a) 1 and 3 (b) 2 only

(c) 3 only (d) 1 and 2

Directions for the next 2 (two) items:

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

Cattle from the nearby villages came to the common ground to graze, and there was still a cool freshness in the air. Hori took several deep breaths and thought of sitting down for a while, since he'd be dying of heat in the scorching 'loo' wind the rest of the day. A number of farmers were eager to lease this bit of land and had offered a good price, but Rai Sahib, God bless him, had plainly told them it was reserved for grazing and would not be relinquished for any price. If he'd been one of those selfish Zamindars, he'd have said the cattle could go to hell, that there was no reason for him to miss the chance to make a little money. But the Rai Sahib still held to the old values, feeling that any landlord who didn't look after his tenants was less than human.

12. Which of the following conclusions is/are correct? (CSAT-2026)

1. All landlords essentially have some goodness trapped within them.
2. The common grazing grounds of a village are intended for use by the cattle of that village.
3. Landlords who believe in tradition tend to be more concerned about their tenants.
4. Winds later in the day tend to be cooler post the hot winds of the morning.

Select the answer using the code given below.

(a) 1 and 3 (b) 2 and 4

(c) 3 only (d) 2 only

13. Which of the following statements are not correct? (CSAT-2026)

1. The landholdings of Rai Sahib were currently not being used for farming.
2. Temperamentally, Rai Sahib was as greedy as other landlords.
3. It cannot be ascertained that Rai Sahib could have made some money by leasing out the grazing land.
4. It may be asserted that Rai Sahib valued his tenants and wanted to protect their livelihood.

Select the answer using the code given below.

(a) 1 and 2 only (b) 1 and 3 only

(c) 3 and 4 (d) 1, 2 and 3

Directions for the next 2 (two) items:

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

Was it the sun-dappled ambience, the strawberries and cream, the frustration of Flavio Cobolli's unforced errors against Serbian Novak Djokovic on Centre Court or simply the crushing weight of being a 64-year-old man in the third act of a very public life? Whatever the reason, Hugh Grant, the actor, deserves empathy. There he was, in the Royal Box at Wimbledon, flanked by Britain's well-dressed and well-rested spectators, watching the men's singles quarterfinals, when the actor did something quietly radical: head at a tilt, eyes closed, utterly unbothered, he took a nap. So praise be to Grant for serving up an unexpected ace. In that small, delicious moment, he didn't merely catch forty winks, he made an elegant case for surrender. Not to laziness, but to limits. To the body's quiet wisdom over society's relentless performance metrics. Wimbledon had its tennis. The perpetually sleep-deprived discovered a leading man, not of action, but of rest.

14. Which of the following statements is/are correct? (CSAT-2026)

1. Radical action can also be attributed to mild surrender where one acts against societal expectations.
2. Submitting to one's limitations, given the effect of age and other factors, ought not to be conflated with laziness.
3. 'Leading man' usually refers to one who plays the lead role in a movie; in this instance the implication is that Hugh Grant is performing the role of not an action hero, but that of a resting one!

Select the answer using the code given below.

(a) 1 and 2 only (b) 2 and 3 only

(c) 1, 2 and 3 (d) 3 only

15. Which of the following statements is/are correct? (CSAT-2026)

1. Hugh Grant was watching, from the Royal Box, the men's semifinal match on Centre Court between Flavio Cobolli and Novak Djokovic.
2. The phrase 'unexpected ace' in the context uses a term from the game of tennis to highlight Hugh Grant's somewhat uncharacteristic act of catching 'forty winks'; an act that is viewed with opprobrium.
3. Grant subjects the demands of society to the wisdom of his body.

Select the answer using the code given below.

(a) 1 and 2 (b) 3 only

(c) 2 and 3 (d) 2 only

Directions for the next 2 (two) items:

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

The process by which countries close their labour-productivity gap with the technology leader is based on convergence theory. The convergence model divides economic eras into three phases: the breakaway, the catch-up, and the fine-tuning phase. It also divides economic entities into two categories: the technology leaders and the technology followers.

The process begins with the development of a new technology, such as scavenging three million years ago (MYA), hunting one MYA, farming 12 thousand years ago, and industrial technology a little more than 200 years ago. During the breakaway phase, the per capita income of the technology leaders (e.g., Western Europe and North America in the industrial era) rises, but is unchanged for the technology followers. In the catch-up phase, the followers adopt the new technology and close their per capita income gap with the technology leaders. In the fine-tuning phase, where participants try to extract the remaining benefits from an increasingly exhausted technology, leaders and followers have similar per capita incomes.

16. Which of the following conclusions are correct? (CSAT-2026)

1. In the breakaway phase, economic progress is slow for the technology followers.
2. In the catch-up phase, leaders stagnate and followers, therefore, close the gap between them and the leaders.
3. In the fine-tuning phase, technology is exhausted, as it were, and both leaders and followers attempt to extract leftover benefits, leading to more or less similar per capita income levels.
4. Industrial technology followed scavenging, which preceded hunting, which itself was followed by farming.

Select the answer using the code given below.

(a) 3 and 4 only (b) 1, 2, 3 and 4

(c) 1 and 2 only (d) 2, 3 and 4 only

17. Which of the following statements is/are correct? (CSAT-2026)

1. The convergence model divides nations into three phases of economic progress.
2. At the heart of the convergence theory is the closing of the gap between labour and productivity.
3. Technology leaders typically have arrived earlier at different economic eras.
4. The time period covered by the convergence theory presented herein encompasses, as mentioned, 4012200 years.

Select the answer using the code given below.

(a) 2 and 4 (b) 2 and 3

(c) 3 only (d) 1, 3 and 4

Directions for the next 2 (two) items :

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

'Kalagram', the cultural village set up at the Maha Kumbh Mela, unfolded as a mosaic of India's diverse regions, each represented by seven meticulously crafted 'Sanskriti Angans'; stepping through the grand portal was like entering another world. These thematic zones, inspired by iconic temples like the Dakshineshwar Kali Temple and the Brahma Mandir, were treasure troves of regional artistry. Bengal's Pattachitra paintings, Assam's bamboo crafts, Tamil Nadu's Thanjavur paintings, and Madhya Pradesh's tribal sculpture, all were showcased in these living galleries where 230 master artisans breathed life into them using age-old techniques, their hands shaping India's ancient history into creations to behold.

18. Which of the following conclusions are valid? (CSAT-2026)

1. Seven Sanskriti Angans, representing different regions of India, had been showcased in Kalagram.
2. Regional artistry was recognised via the inspiration drawn from iconic temples.
3. India's ancient history had been crafted by the contemporary craftsmanship of 230 artisans into creations to behold.
4. Art forms from all regions of India had been showcased in these living galleries.

Select the answer using the code given below.

(a) 1 and 2 only (b) 2, 3 and 4

(c) 1, 2 and 4 (d) 1 and 3

19. Which one of the following statements is not correct? (CSAT-2026)

(a) Paintings from four States of India have been mentioned.

(b) Stepping into Kalagram is likened to stepping into another world.

(c) The Angans have been described as living galleries.

(d) Kalagram is divided into thematic zones, inspired by well-known temples of India.

Directions for the next 2 (two) items:

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

Sport is not just about winning medals or getting jobs through a sports quota. Today's generation is struggling with issues like depression and anxiety. Parents often say that their children are inactive and rarely leave the house. Sport can help tackle these problems. From sport you can learn time management, fitness, teamwork, coordination, and so much more. We need to develop a culture in which sport is seen as a way of life, a path to building a healthier, happier society.

20. Which of the following conclusions are valid? (CSAT-2026)

1. Sport is more than just games; it is a way of life.
2. Sport can help mitigate the problems of seclusion among the young.
3. Earning laurels in sport can help one secure a job on the basis of an assigned quota.
4. Standard corporate sector skills cannot be learnt through sport.

Select the answer using the code given below.

(a) 1 and 3 only (b) 2 and 4 only

(c) 1, 2 and 4 (d) 1, 2 and 3

21. Which of the following statements is/are correct? (CSAT-2026)

1. Parents are not encouraging enough when it comes to children playing sport.

2. Participation in sporting activities can help develop life skills.

3. Sport as a way of life can help evolve the very nature of society itself.

Select the answer using the code given below.

(a) 1 and 3 (b) 2 and 3

(c) 2 only (d) 3 only

Directions for the next 2 (two) items :

Read the following passage and answer the items that follow. Your answers should be based solely on the passage.

Passage

How is deflation done? Most countries use a method called 'double deflation', where input and output prices are deflated separately. Consider a manufacturer importing oil for use in production. If oil prices fall, output prices do not and quantities remain the same, real value added should not change. But if the same deflator is used for inputs and outputs, as in India, it would look as if the manufacturer had become more productive.

22. Which of the following statements is/are correct? (CSAT-2026)

1. Real value should not change in the instance of static output cost and unchanged quantities against falling oil prices.

2. Deflators are to be used separately for inputs and outputs, and this is a practice universally adopted by all economies.

Select the answer using the code given below.

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

23. Which of the following assumptions is/are valid? (CSAT-2026)

1. Deflation strategies can be used to make manufacturers appear to be doing better than they actually are.

2. When input and output prices are both deflated against a single input price, it is referred to as 'double deflation'.

Select the answer using the code given below.

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Directions for the following 4 (four) items:

Read the following **two passages** and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage - 1

Maintaining an ecosystem just to conserve biodiversity will affect its commercial potential as well as the livelihoods dependent on the ecosystem. There is also a conflict between using an ecosystem only for livelihoods, for commercial exploitation, or strictly for conservation. Deforestation caused due to commercial exploitation will lead to indirect harm like floods, siltation problems and microclimatic instability, apart from adversely affecting livelihoods dependent on forests. These conflicts are particularly acute in developing countries where the dependence of people on the ecosystem is significant, and commercial exploitation has the potential to boost national income.

24. Which one of the following statements best reflects the critical message conveyed by the author of the passage? (CSAT-2025)

(a) Conservation of biodiversity is not an issue to be worried about when some people depend on ecosystems for their livelihoods.

(b) Commercial exploitation of forests goes against the fundamental rights of the people dependent on forests for food and shelter.

(c) Sustenance of livelihood and degradation of ecosystem while being together exacerbate one another, leading to conflicts and imbalance.

(d) Commercial exploitation of ecosystems should be completely stopped.

25. With reference to above passage, the following assumptions have been made: (CSAT-2025)

I. No country needs to depend on ecosystems to boost national income.

II. Resource-rich countries need to share their resources with those of scant resources so as to prevent the degradation of ecosystems,

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Passage -2

The history of renewable energy suggests there is a steep learning curve, meaning that, as more is produced, costs fall rapidly because of economies of scale and learning by doing. The firms' green innovation is path-dependent: the more a firm does, the more it is likely to do in the future. The strongest evidence for this is the collapse in the price of solar energy, which became about 90% cheaper during the 2010s, repeatedly beating forecasts. Moving early and gradually gives economies more time to adjust, allowing them to reap the benefits of path-dependent green investment without much disruption. A late, more chaotic transition is costlier.

26. Which one of the following statements best reflects the central idea of the passage? (CSAT-2025)

(a) Economies of scale is essential for transition to green growth.

(b) Modern technological progress is intensely linked to path-dependent innovations.

(c) Countries with large economies are in a better position to adopt green technologies.

(d) Timing plays a crucial role in the case of green technology development.

27. With reference to the above passage, the following assumptions have been made: (CSAT-2025)

I. Path-dependent green investments will eventually most likely benefit growth as well as public finances in a country like

India.

II. If other green technologies follow the same pattern as that of solar energy, there will most likely be an easy green transition.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Directions for the following 4 (four) items:

Read the following **two passages** and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage - 1

Each State in India faces a distinctive set of challenges regarding the impact of warming, but also offers its own set of opportunities for reducing emissions depending on its natural resources. For example, coastal States need to take action to protect their shores from sea level rise, districts that are drier need to prepare for variable monsoon precipitation. Himalayan regions have their own unique challenges, and selected parts of peninsular India and offshore areas offer great opportunities for harnessing wind power. These various aspects need to be considered for developing clear and sustainable goals for the future.

28. Which one of the following statements best reflects the most logical, rational and pragmatic message conveyed by the author of the passage? (CSAT-2025)

(a) The mitigation and adaptation strategies to address/tackle the climate change is essentially the responsibility of each State.

(b) India is too diverse to implement any effective strategy or programme to address/tackle the climate change.

(c) It is basically the responsibility of the Union Government to implement the climate action plans and ensure net zero emissions.

(d) India needs to formulate effective climate change mitigation and adaptation strategies at the State/region

29. With reference to the passage, the following assumptions have been made: (CSAT-2025)

I. Green energy production can be linked to/integrated with the climate change mitigation and adaptation strategies.

II. Effects of climate change are much more severe in coastal and mountainous regions.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Passage-2

If the social inequality is the most acutely felt social problem in India, insecurity, more than poverty, is the most acutely felt economic problem. Besides those below the official poverty line, even those just over the poverty line are subject to multiple economic insecurities of various kinds (due to wealth and/or health risks, market fluctuations, job-related uncertainties).

Many Government policies are actually intended towards mitigating these insecurities.

30. Which one of the following statements best reflects the critical message conveyed by the passage? (CSAT-2025)

(a) India's political executive should be aware that poverty and social inequality and the consequent sense of insecurity is the main social problem.

(b) In India, poverty is the primary reason for social inequality and insecurity.

(c) Poverty and social inequality are so intricately linked that they pose an unmanageable crisis for India.

(d) Insecurity, more than poverty, is the main economic issue that Government policies must address.

31. With reference to the above passage, the following assumptions have been made: (CSAT-2025)

I. People above the poverty line also are prone to suffer from anxiety about economic insecurity.

II. Eradication of poverty can result in peace and social equality in the country.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Directions for the following 4 (four) items:

Read the following two passages and answer the items that follow the passages. Your answers to these Items should be based on the passages only.

Passage-1

One of the dismal realities of the agricultural sector in independent India has been that it never experienced a high-growth phase, unlike the non-agricultural economy. The highest decadal growth (compound annual growth rate or CAGR) for agriculture has been just 3-5% in the 1980s. Also, after experiencing a spurt in decadal growth during the 1980s, agricultural growth offered relative stagnation thereafter. This is in harp contrast to non-agricultural growth, which consistently increased from the 1980s to 2000s.

32. Which one of the following statements best reflects the corollary to the above passage? (CSAT-2025)

(a) The benefit of economic reforms percolates down more slowly to the agriculture sector than in other sectors of the economy.

(b) For India, the green revolution was not as useful as it was expected to be.

(c) India lagged behind other countries in adapting mechanized and modern farming.

(d) Rural-to-urban migration resulted in the stagnant agriculture sector.

33. With reference to the passage, the following assumptions have been made: (CSAT-2025)

The growing divergence between the fortunes of the agricultural and non-agricultural economy in India could have been reduced/ contained by:

I. adapting large-scale cultivation of commercial crops and viable corporate farming.

II. providing free insurance for all crops and heavily subsidizing seeds, fertilizers, electricity and farm machinery at par with developed countries.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Passage-2

In our country, handlooms are equated with a culture that ensures a continuity of tradition. This idea has become part of the public policy-framing and provides a legitimate basis for the State to support the sector. But the notion of tradition as a single, linear entity is being strongly contested today. The narratives dominant in defining culture/ tradition in a particular way are seen to have emerged as the identities and histories of large sections. The discounted and, at times, forcibly stifled identities are fighting for their rightful place in history. Against this backdrop, when we promote handloom as a traditional industry, it is not surprising that large sections of our population choose to ignore it.

34. Which one of the following statements best reflects the most logical and rational message conveyed by the author of the passage? (CSAT-2025)

(a) We need to free the handloom industry from the limited narrative linked to preserving cultural heritage.

(b) Continued State support to the handloom industry ensures the preservation of some of our glorious art forms and old traditions.

(c) Household units of the handloom sector should be modernized and made an economically viable organized industry.

(d) Handloom products need to be converted to machine-made designer products so as to make them more popular.

35. With reference to the above passage, the following assumptions have been made: (CSAT-2025)

I. There is no need for the State to be involved in any manner in the handloom sector.

II. Handloom products are no longer appealing and attractive in the rapidly changing modern world.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Directions for the following 4 (four) items:

Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

In our country, regrettably, teaching and learning for the examination have been our forte but the new demands of society and the future of work require critical and independent thinking, learning through doing, asking questions from multiple disciplinary perspectives on the same issue, using evidence for building arguments, and reflecting and articulation. Higher education should not "either be a mere servant of the government policy or a passive respondent to public mood." Higher learning is all about how to think rather than what to think. Teaching has to be re-invented.

36. Which one of the following statements best reflects the central idea conveyed by the passage? (CSAT-2025)

(a) India does not have enough resources for promoting quality education in its universities.

(b) The institutions of higher learning in the country should not be under the control of the Government.

(c) Classroom approach to higher education should be done away with.

(d) Classroom needs to be reimagined and teaching needs to be re-invented.

37. With reference to the above passage, the following assumptions have been made: (CSAT-2025)

I. Higher education is a constantly evolving subject that needs to align towards new developments in all spheres of society.

II. In our country, sufficient funds are not allocated for promoting higher education.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Passage-2

If there is inequality in the pattern of population growth, there is greater inequality in food production and utilization. As societies become wealthier, their consumption of animal products increases. This means that a greater proportion of such basic foodstuff as grains and legumes that could feed humans directly is instead being converted into feed for poultry and large farm animals. Yet this conversion of plant-based food into animal food for humans is far from efficient. Only 16% of the calories fed to chickens are recovered by us when we eat them. This conversion rate goes down to five to seven per cent in large animals that are fed grain to add fat and some protein before slaughter.

38. Which one of the following statements best reflects the crux of the passage? (CSAT-2025)

(a) There is an urgent need for a public policy to promote the consumption of cereal-based foods in wealthier societies.

(b) Animal-based food is far less efficient than grain/plant-based food in terms of production and utilization.

(c) Plant-based protein should replace animal-based protein in our daily diets.

(d) Inequality in food production and consumption is inevitable in any fast changing society.

39. With reference to the above passage, the following assumptions have been made: (CSAT-2025)

I. The food manufacturing and processing industries in every country should align their objectives and processes in accordance with the changing needs of the societies.

II. Wealthier societies tend to incur great loss of calories of food materials due to indirect utilization of their agricultural produce.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Passage-1

Over the next 30 years, many countries are promising to move to net-zero carbon, implying that household emissions will have to be cut to close to nothing. A leading climate scientist reckons that, at best, half the reduction might be achieved through demand-side measures, such as behavioural changes by individuals and households. And even that would require companies and governments to provide more incentives to change through supply-side investments to make low-carbon options cheaper and more widely available.

40. Which one of the following statements best reflects the central idea conveyed by the passage? (CSAT-2025)

(a) Moving to net-zero carbon is possible only by the reduction in household emissions.

(b) Low-carbon behaviour in people can be brought about by incentivising them.

(c) Cheaper goods and services can be made available to people by using low-carbon technologies.

(d) Manufacturing industries that use low-carbon technologies should be provided with subsidies.

41. With reference to the above passage, the following assumptions have been made: (CSAT-2025)

I. Supply-side investments in companies can result in low-carbon behaviour in people.

II. People are not capable of adapting low-carbon behaviour without the involvement of Government and Companies.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Passage -2

In only 50 years, the world's consumption of raw materials has nearly quadrupled, to more than 100 billion tons. Less than 9% of this is reused. Batteries of old vehicles contain materials such as lithium, cobalt, manganese and nickel that are pricey and can be hard to obtain. Supply chains are long and complicated. Buyers' risks are being aggravated by their suppliers' poor environmental and labour standards. Reusing materials makes sense. Once batteries reach the ends of their lives, they should go back to a factory where their ingredients can be recovered and put into new batteries.

42. Which one of the following statements best reflects the most logical, rational and pragmatic message conveyed by the passage? (CSAT-2025)

(a) Green economy is not possible without reusing critical minerals.

(b) Every sector of economy should adapt the reuse of material resources immediately.

(c) Circular economy can be beneficial for sustainable growth.

(d) Circular use of material resources is the only option for some industries for their survival.

43. With reference to the above passage, the following assumptions have been made: (CSAT-2025)

I. Automobile factories are examples of the circular economy.

II. Economic growth is compatible with circular use of mineral resources.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Directions for the following 4 (four) items:

Read the following two passages and answer the items that follow the passages. Your answer to these items should be based on the passages only.

Passage - 1

It is hard to predict how changes in the climate and the atmosphere's chemistry will affect the prevalence and virulence of agricultural diseases. But there is a risk that such changes will make some plant infections more common in all climatic zones, perhaps catastrophically so. Part of the problem is that centuries of selective breeding have refined the genomes of most high-value crops. They are spectacular at growing in today's conditions but genetic variations that are not immediately useful to them have been bred out. This is good for yields but bad for coping with changes. A minor disease or even an unknown one could suddenly rampage through a genetically honed crop.

44. Which one of the following statements best reflects the central idea conveyed by the passage? (CSAT-2025)

(a) Global climate change adversely affects the productivity of crops.

(b) Our total dependence on genetically honed crops entails possible food insecurity.

(c) Our food security should not depend on agricultural productivity alone.

(d) Genetically honed crops should be replaced with their wild varieties in our present cultivation practices.

45. With reference to the above passage, the following assumptions have been made: (CSAT-2025)

I. Global climate change can result in the migration of several plant diseases to new areas.

II. Scientific understanding of the wild relatives of our present crops would enable us to strengthen food security.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Passage - 2

"A good statesman, like any other sensible human being, learns more from his opponents than from his fervent supporters. For his supporters will push him to disaster unless his opponents show him where the dangers are. So if he is wise he will often pray to be delivered from his friends, because they will ruin him. But, though it hurts, he ought also to pray never to be left without opponents; for they keep him on the path of reason and good sense. The national unity of free people depends upon a sufficiently even balance of political power to make it impracticable for the administration to be arbitrary and for opposition to be revolutionary and irreconcilable."

46. Which one of the following statements best reflects the critical message conveyed by the author of the passage? (CSAT-2025)

(a) Without opposition parties, the administration in a democracy gets to become more responsible.

(b) Democracy needs to have revolutionaries in opposition to keep the government alert.

(c) Rulers in a democracy need the support of opposition for their political survival.

(d) In a democracy, the opposition is indispensable for the balance of political power and good governance.

47. With reference to the above passage, the following assumptions have been made: (CSAT-2025)

I. In a democracy, a strong opposition is required only if the Head of Government is indifferent.

II. The more aggressive the opposition, the better is the governance in a democracy.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Directions for the following 3 (three) items:

Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

There has been no democracy that has grown economically without corporate capitalism. It helps in modernizing the economy and enabling the transition from rural to urban, and agriculture to industry and services, which are inevitable with growth. It generates jobs - and there is no other way to fix a country's unemployment challenge without a further impetus to private business. Big companies can operate on a large scale and become competitive both domestically and externally. A vibrant corporate capitalist base also leads to additional revenues for the State - which in turn, can be used for greater welfare for the marginalized and creating a more level playing field in terms of opportunities.

48. Which one of the following statements best reflects the critical message conveyed by the author of the passage? (CSAT-2025)

(a) Corporate capitalism is important for economic growth of a State and also for democracy.

(b) Corporate capitalism is imperative for a modern State to achieve its political objectives.

(c) No State can ensure its economic survival for long without the role of corporate capitalism.

(d) Corporate capitalism and democracy have mutual dependence for their continued existence.

49. With reference to the above passage, the following assumptions have been made (CSAT-2025)

I. Corporate capitalism promotes the growth of labour force and provides more employment opportunities.

II. Poor and marginalized sections of population are benefited by corporate capitalism due to trickle-down effect.

Which of the above assumptions is/are valid?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Passage-2

"A network of voluntary associations stands as a 'buffer' between the relatively powerless individual and the potentially powerful State."

50. Which one of the following statements reflects the best explanation of the above passage? (CSAT-2025)

(a) It emphasizes the inability of the State to enforce its will in practice against the opposition of certain groups within it.

(b) It is a cooperative organization for the promotion of the well-being and development of the personality of its members.

(c) It takes individuals out of a state of isolation and gives them a chance to participate in the common endeavour.

(d) It permits citizens to have a variety of loyalties and allegiance.

Directions for the following 2 (two) items:

Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage - 1

A single number for inflation is an aggregate across different commodities and services - the price rise differs for different items of consumption. So, the single number is arrived at by assigning weights to different commodities and services. For WPI, the weights in production are used; for CPI, the consumption basket is used. But people are not homogeneous. The consumption basket is vastly different for the poor, the middle classes, and the rich. Hence, the CPI is different for each of these classes and a composite index requires averaging the baskets.

51. Which one of the following statements best reflects the most logical, rational and crucial message conveyed by the passage? (CSAT-2025)

(a) We must use WPI exclusively in measuring price rise and CPI should be done away with.

(b) The present calculation of inflation rate does not correctly measure price rise of individual item/commodity.

(c) Inflation data under-presents services in the consumption basket.

(d) Knowledge of inflation rate is not really of any use to anybody in the country.

Passage-2

Trust stands commonly defined as being vulnerable to others. Entrepreneurship implies trust in others and willingness to expose oneself to betrayal. Trust in expert systems is the essence of globalizing behaviour; trust itself emerges as a super-commodity in the social market and defines the characteristics of goods and services in a global market. Trusting conduct also means holding others in good esteem, and an optimism that they are, or will be, competent in certain respects.

52. Which one of the following statements best reflects the crux of the passage? (CSAT-2025)

(a) Trustworthiness cannot be expected in entrepreneurship.

(b) Trustworthy people are the most vulnerable people.

(c) No economic activity is possible without being exposed to betrayal.

(d) Trust is important though it entails risk.

Directions for the following 4 (four) items:

Read the following **two passages** and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

"The history of science is the real history of mankind." In this striking epigram, a nineteenth-century writer links science with its background. Like most epigrams, its power lies in emphasizing by contrast an aspect of truth which may be easily overlooked. In this case, it is easy to overlook the relations between science and mankind, and to treat the former as some abstract third party, which can sometimes be praised for its beneficial influences, but frequently and conveniently blamed for the horrors of war. Science and mankind cannot be divorced from time to time at men's convenience. Yet we have seen that, in spite of countless opportunities of improvement, the opening years of the present period of civilization have been dominated by international conflict. Is this the inevitable result of the progress of science or does the fault lie elsewhere?

53. Which of the following is/are emphatically conveyed by the author of the passage? (CSAT-2024)

1. Without science, mankind could not have continued to exist till today.
2. It is the science that will ultimately determine the destiny of mankind.

Select the correct answer using the code given below.

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

54. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. The horrors of modern life are the inevitable result of the progress of science.
2. The aspect of truth likely to be overlooked is that science is what man has made it.

Which of the assumptions given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage-2

Only with long experience and opening of his wares on many a beach where his language is not spoken, will the merchant come to know the worth of what he carries, and what is parochial and what is universal in his choice. Such delicate goods as justice, love and honour, courtesy, and indeed all the things we care for, are valid everywhere but they are variously moulded and often differently handled, and sometimes nearly unrecognizable if you meet them in a foreign land, and the art of learning fundamental common values is perhaps the greatest gain of travel to those who wish to live at ease among their fellows.

55. When we meet other people while we travel, we learn to differentiate between (CSAT-2024)

(a) imagination and understanding

(b) communities and nationalities

(c) local values and universal values

(d) friends and foes

56. With reference to the above passage, the following assumptions have been made: (CSAT-2024)

1. Travel leads to an understanding of humans.
2. Travel helps those who wish to learn fundamental common values.
3. A person with long experience in travel can resolve differences amongst people.

Which of the assumptions given above are valid?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Directions for the following 3 (three) items:

Read the following **two passages** and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

By the time children reach class 8, the bulk of them tend to be in the age range of 13 years to 15 years. But in our country, about a quarter of all children in class 8 struggle with reading simple texts and more than half are still unable to do basic arithmetic operations like division. Every year about 25 million young boys and girls from elementary school move into the life that lies for them beyond compulsory schooling. They cannot enter the workforce at least in the organized sector till they are 18. For many families, these children are the first from their families ever to get this far in school. Parents and children expect that such 'graduates' from school will go on to high school and college. Hardly anyone wants to go back to agriculture. On the other hand, abilities in terms of academic competencies are far lower than they should be even based on curricular expectations of class 8.

57. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. For effective school education, parents have greater role than the governments.
2. School curriculum that conforms to today's requirements and is uniform for the entire country may address the issues brought out.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

58. Which one of the following statements best reflects the central idea conveyed by the passage? (CSAT-2024)

(a) Total eradication of poverty in the country will resolve the issue of under-performance of our school- children.

(b) Monetary incentives to parents and teachers is a strategy to improve the children's academic performance.

(c) Public policy should ensure that competencies and achievements of young people are aligned with their expectations.

(d) India is not going to take advantage of the demographic dividend unless some school pass-outs go back to agriculture.

Passage-2

We take it for granted now that science has a social responsibility. The idea would not have occurred to Newton or Galileo. They thought of science as an account of the world as it is, and the only responsibility that they acknowledged was to tell the truth. The idea that science is a social enterprise is modern, and it begins at the industrial revolution. We are surprised that we cannot trace a social sense further back, because we nurse the illusion that the industrial revolution ended a golden age.

59. Which one of the following statements best reflects the thinking of the author about the science? (CSAT-2024)

(a) Science must value the commitment of the scientists.

(b) Science is a product of civilized society and must be used for the promotion of scientific awareness in people.

(c) Industrial revolution was made possible by the advancements in science.

(d) Science must pursue truth but be responsible for social welfare.

Directions for the following 4 (four) items:

Read the following **two passages** and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

Today, if we consider cities such as New York, London and Paris as some of the most iconic cities in the world, it is because plans carrying heavy systems approach were imposed on their precincts. The backbone of the systems theory is the process of translating social, spatial and cultural desirables into mathematical models using computing, statistics, optimization and an algorithmic way of formulating and solving problems. The early universities of the West which began to train professionals in planning, spawned some of the most ingenious planners, who were experts in these domains. This was because these very subjects were absorbed into the planning curriculum that had its roots in the social sciences, geography and architecture. Planning in India, and its education differ from the West.

60. Which one of the following statements best reflects the most logical and rational inference that can be made from the above passage? (CSAT-2024)

(a) Curriculum for urban planning courses should have diverse and interdisciplinary approach.

(b) In India, city administration is under bureaucracy which lacks formal training in urban planning and management.

(c) In India, the management of urban areas is a local affair with a chronic problem of insufficient funds.

(d) With high density of population and widespread poverty in our urban areas, planned development in them is very difficult.

61. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. India needs a new generation of urban professionals with knowledge relevant to modern urban practice.
2. Indian universities at present have no capacity or potential to impart training in systems approach.

Which of the assumptions given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage-2

Not every voice on the internet commands the same kind of audience.

When anonymous private entities with high capital can pay for more space for their opinions, they are effectively buying a louder voice. If political discourse in the digital sphere is a matter of outshining one's opponent till the election is won, then the quality of politics suffers. The focus of social media is restricted to the promotion of content that generates more user engagement, regardless of how inflammatory the content may be.

62. Which one of the following statements best reflects the central idea of the above passage? (CSAT-2024)

(a) Constructed as a marketplace of views, social media ensures instant access to information.

(b) Social media are not ideal or moral institutions but the products built by companies to make profits.

(c) Social media have been created to strengthen democracies.

(d) In today's world, social media are + inevitable for well-informed social life.

63. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. Internet is not inclusive enough.
2. Internet can adversely affect the quality of politics in a country.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Directions for the following 4 (four) items:

Read the following **two passages** and answer items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

According to the Food and Agriculture Organization, one-third of food produced for human consumption is lost or wasted globally. Food is lost or wasted throughout the supply chain, from initial agricultural production to final household consumption. The increasing wastage also results in land degradation by about 45%, mainly due to deforestation, unsustainable agricultural practices, and excessive groundwater extraction. The energy spent over wasted food results in about 35 billion tonnes of carbon dioxide production every year. Decay also leads to harmful emissions of other gases in the atmosphere. Addressing the loss and wastage of food in all forms is critical to complete the cycle of food sufficiency and food sustainability.

64. Which of the following statements best reflect the most logical and rational inferences that can be made from the passage? (CSAT-2024)

1. The current methods of food distribution are solely responsible for the loss and wastage of food.
2. Land productivity is adversely affected by the prevailing trend of food loss and wastage.
3. Reduction in the loss and wastage of food results in lesser carbon footprint.
4. Post-harvest technologies to present or reduce the loss and wastage of food are not available.

Select the correct answer using the code given below.

(a) 1, 2 and 3 (b) 2 and 3 only

(c) 1, 3 and 4 (d) 1, 2 and 4

65. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. The food distribution mechanism needs to be reimagined and made effective to reduce the loss and wastage of food.
2. Ensuring the reduction of wastage and loss of food is a social and moral responsibility of all citizens.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only

(c) Both and 2 (d) Neither 1 nor 2

Passage-2

As inflation rises, even governments previously committed to budget discipline are spending freely to help households. Higher interest rates announced by central banks are supposed to help produce modest fiscal austerity, because to maintain stable debts while paying more to borrow, governments must cut spending or raise taxes. Without the fiscal backup, monetary policy eventually loses traction. Higher interest rates become inflationary, not disinflationary, because they simply lead governments to borrow more to pay rising debt service costs. The risk of monetary mooring is greater when public debt rises, because interest rates become more important budget deficits.

66. Which of the following statements best reflects/reflect the most logical and rational inference/inferences that can be made from the passage? (CSAT-2024)

1. Central banks cannot bring down inflation without budgetary backing.
2. The effects of monetary policy depend on the fiscal policies pursued by the government.

Select the correct answer using the code given below.

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

67. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. Fiscal policies of governments are solely responsible for higher prices.
2. Higher prices do not affect the long-term government bonds.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Directions for the following 2 (two) items:

Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages

only.

Passage-1

Unlike religion and science, poetry does not posit or expect any belief in absolute truths. Those forces or individuals who claim to have absolute truths in their grasp tend to turn dictatorial and tyrannical. Truth usually does not admit any contradictions or imperfections. It is unitarian. It is, therefore, not of much use for poetry. Poetry abides by the plurality of life and existence. Perhaps poetry follows reality which is plural, anachronistic, full of contradictions. Against the tyranny of truth, Poetry remains a partisan of democratic reality. Against the arrogance of power, wealth and hierarchy, poetry proposes both humility and defiance.

68. Which one of the following statements best reflects the most logical and rational message conveyed by the above passage? (CSAT-2024)

(a) It is the poetry, not science or religion, which recognizes and accepts imperfections in humans.

(b) Truth is revealed through science or religion and poetry is anathema to truth.

(c) Poetry is romantic, imaginary and is about feeling whereas science and religion are about truth.

(d) In a world of violence, tyranny and bigotry, poetry is a form of dynamic resistance.

Passage-2

The flower was not invented to please us. It flaunted its petals and spread its perfume to attract an insect. The insect carries the pollen from flower to flower so that pollen is not carried away by wind and thus not wasted. What we call a flower's beauty is merely a by-product and a human invention. The perfume is not there to please us, it pleases us because it is there and we have been conditioned to it.

69. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. The author of the passage believes that flowers are creations of Nature's luxury.
2. The author of the passage does not believe in the usefulness of flowers except as things of beauty.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Directions for the following 3 (three) items:

Read the following **three passages** and answer De items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

In a robust democracy, reality, howsoever inconvenient it may be, finds its expression both in the speech of political leaders and the other social forms of assertion. The existence of the real has to be transparent, both through its circulation in and by the media as well as its argumentative articulation in deliberative democracy. A normatively responsible media through its communication effect has the responsibility to circulate the content of reality without distortion.

70. Which one of the following statements best reflects the crux of the above passage? (CSAT-2024)

(a) Responsible media should not distort the real in an ideal democracy.

(b) Fake news seems inherent in the life of an ideal democracy.

(c) There should not be any kind of restrictions on the freedom of expression in an ideal democracy.

(d) Irresponsible media and political leaders cannot be effectively controlled in an ideal democracy.

Passage-2

Now-a-days there is a growing trend to use interconnected home devices. As consumers increasingly network their homes, the connected home device manufacturers and service providers will seek to overcome "thin profit margins by gathering more of our personal data-with or without our agreement- turning the home into a corporate storefront". Corporate marketers will have powerful incentives to observe consumer behaviour to understand the buying needs and preferences of the device owners.

71. Which one of the following statements best reflects the most logical, rational and practical message implied by the passage? (CSAT-2024)

(a) Knowledge of consumer behaviour leads to more capital expenditure in manufacturing sector.

(b) Knowledge of consumer behaviour stimulates the growth of commerce and trade and thus helps in the overall economic development of the country.

(c) Interconnected devices give a lot of comfort to home users and improve the overall quality of life.

(d) Interconnected devices can be at security risk and home users may have privacy risk.

Passage-3

Green growth involves rethinking growth strategies with regard to the impacts on environmental sustainability and the environmental resources available to poor and vulnerable groups. In rethinking growth, we need to focus on the current reality of a resource-constrained world. Resource intensive and, in particular energy intensive processes will need to make way for more efficient and resource frugal development strategies if we are to avoid an economic dead end or a world in which only a small elite is able to enjoy affluence in the midst of a sea of poverty.

72. Which one of the following statements best reflects the crux of the above passage? (CSAT-2024)

(a) Environmental sustainability is inimical to our objective of achieving a high rate of GDP growth.

(b) Poverty eradication is not possible without a rapid economic growth and the consequent environmental degradation.

(c) Maintaining high environmental standards is now a prerequisite for achieving a steady, sufficient and inclusive growth.

(d) With large populations, rampant poverty and limited resources of today's world, environmental degradation cannot be prevented and inequalities are inevitable.

Directions for the following 3 (three) items:

Read the following **three passages** and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

When an international team of scientists pumped a carbon dioxide and water mix into underground basalt rocks, basic chemistry took over. The acidic mixture dissolved rocks' calcium and magnesium and formed limestone. Basically carbon dioxide is converted into stone, exclaimed the scientists.

73. Which one of the following statements best reflects the most logical, rational and practical suggestion implied by the passage? (CSAT-2024)

(a) It is a cheap and practical method to produce limestone at commercial level for building purposes.

(b) This can be used as one of the methods of carbon sequestration.

(c) Basalt rock can be made a good source of calcium and magnesium minerals by this method.

(d) Good rock-dissolving acid can be produced by mixing carbon dioxide and water.

Passage-2

Geographers analyzed 175 satellite images of ocean colour, which is an indicator of phytoplankton productivity at the ocean's surface, and found that giant icebergs are responsible for storing up to 20 percent of carbon in the Southern Ocean. The researchers discovered that melting water from giant icebergs which contains iron and other nutrients, supports hitherto unexpectedly high levels of phytoplankton growth.

74. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. Giant icebergs have a bearing on primary productivity and food chains of the Southern Ocean.
2. Melting of giant icebergs can produce climate change effects and impact world fisheries.

Which of the assumptions given above is/are valid?

(a) 1 only. (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage-3

Evolution has endowed caterpillars that feed on corn with a unique ability, they can induce the plant to turn off its defence against insect predators. This helps caterpillars to eat more and grow faster. The agent that causes this effect is the caterpillar's faeces or frass. The find could throw new light on compounds associated with plant response to pathogens like fungi or bacteria.

75. Which one of the following statements best reflects the most logical, rational and practical message conveyed by the passage? (CSAT-2024)

(a) Farmers can use caterpillars to feed on weeds in their crop fields/ plantations.

(b) This finding can help in the development of clinically useful antimicrobial compounds.

(c) This finding can help in the development of organic, ecologically sustainable pesticides.

(d) Caterpillars can be genetically modified to be predators of the other plant pests.

Directions for the following 4 (four) items:

Read the following **two passages** and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage-1

Conventional classrooms, by emphasizing fixed duration over learning effectiveness, resign themselves to variable outcomes. The tyranny of the classroom is that every learner is subjected to the same set of lectures in the same way for the same duration. In the end, a few learners shine, some survive, and the rest are left behind. After the fixed duration, the classroom model moves on, with not a thought spared for those left behind. This is how we end up with 10 percent employability in our graduates after a decade and half of formal education. Repeating the same ineffectual script in the realm of skill education will not produce different results.

76. Which of the following statements best reflects/reflect the most logical and rational inference/inferences that can be made from the passage? (CSAT-2024)

1. In conventional classroom learning, the central goal is duration of learning rather than attainment of competency.
2. Conventional classrooms encourage one-size-fits-all approach and stamp out all differentiation.

Select the correct answer using the code given below.

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

77. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. As a large number of workers in our country are employed in unorganised sector, India does not need to change its present conventional classroom system of education.
2. Even with its present conventional classroom system of education, India products sufficient number of skilled workers to fully realize the benefits of demographic dividend.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage-2

When a child reaches adolescence, there is apt to be a conflict between

the parents and the child, since the latter considers himself to be by now quite capable of managing his own affairs, while the former are filled with parental solicitude, which is often a disguise for love of power. Parents consider, usually, that the various moral problems which arise in adolescence are peculiarly their province. The options they express, however, are so dogmatic that the young seldom confide in them. and usually go their own way in secret.

78. Based on the above passage, the following assumptions have been made: (CSAT-2024)

1. The adolescent does not feel comfortable with his parents because they tend to be dominating and assertive.
2. The adolescent of modern times does not have much respect for parents.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

79. Which one of the following statements best reflects the central idea of the above passage? (CSAT-2024)

(a) Parents in general may not be of much help when children are on their way to becoming adults.

(b) When children reach adolescence, involvement of parents in their lives is unnecessary.

(c) Modern-day nuclear families are not capable of bringing up children properly.

(d) In modern societies, adolescents tend to be stubborn, disobedient and careless.

Passage

We often hear about conflicts among different States in India over river waters. Of the 20 major river systems, 14 are already water-stressed; 75% of the population lives in water-stressed; regions, a third of whom live in water-scare areas. Climate change, the demands of rising population and the need for agriculture to keep pace, and increased rate of urbanization and industralization will exacerbate water stress. According to the Constitution of India, water is a State subject and not that of the Union, except for regulation of inter-State rivers. Key to ensuring balance between competing demands of various stakeholders is a basin-based approach to allocate water amongst constituent regions and States. Allocating fair share of water to them requires assessments based on objective criteria, such as specificities of the river basin, size of dependent population, existing water use and demand, efficiency of use, projected future use, etc. while ensuring the environmental needs of the river and aquifers.

80. Which one of the following statements best reflects the most rational, practical and immediate action required to ensure fair and equitable allocation of water to different stakeholders? (CSAT-2023)

(a) A national, pragmatic, legal and policy framework for water allocation should be made.

(b) All river systems of the country should be linked and huge aquifers created.

(c) Water channels between regions of water surplus and regions of water deficit should be created.

(d) To mitigate water crisis, water demand of sectors such as agriculture and industry should be reduced.

Passage

More than half of Indian women and almost a quarter of Indian men of working age suffer from anaemia. According to studies, they are anywhere from 5-15% less productive than they could be, as a result thereof. India also has the largest tuberculosis burden in the world, costing 170 million workdays to the country annually. But what is just as important as lost productivity now is lost potential in the future. It is becoming increasingly clear that on many measures of cognitive ability, malnourished Indian children perform, two or three times worse than their adequately nourished peers. For an economy that will be more dependent on highly skilled workers, this Poses a significant challenge. And it is one that really should be addressed given India's demographic outlook.

81. Which one of the following statements best reflects what is implied by the passage? (CSAT-2023)

(a) Education system must be strengthened in rural areas.

(b) Large scale and effective implementation of skill development programme is the need of the hour.

(c) For economic development, health and nutrition of only skilled workers needs special attention.

(d) For rapid economic growth as envisaged by us, attention should be paid to health and nutrition of the people.

Passage

In India, a majority of farmers are marginal and small, less educated and possess low adaptive capabilities to climate change, perhaps because of credit and other constraints. So, one cannot expect autonomous adaptation to climate change. Even if it was possible, it would not be sufficient to offset losses from climate change. To deal with this, adaptation to climate change is paramount, alongside a fast mitigation response. Another solution is to have a planned or policy driven adaptation, which would require the government to come up with policy recommendations. Perception is a necessary prerequisite for adaptation. Whether farmers are adapting agricultural practices to climate change depends on whether they perceive it or not. However, this is not always enough for adaptation. It is important how a farmer perceives the risks associated with climate change.

82. Which one of the following statements best reflects the most logical and rational message conveyed by the author of the passage? (CSAT-2023)

(a) Adaptation to climate change and mitigation response are basically the responsibilities of the government.

(b) Climate change causes a change in government policies regarding land use patterns in the country.

(c) Risk perceptions of farmers are important for motivating them for taking adaptation decisions.

(d) Since mitigation is not possible, governments should come up with policies for quick response to climate change.

Passage

The emissions humans put into the atmosphere now will affect the climate in the middle of the century and onwards. Technological change, meanwhile, could make a future transition away from fossil fuels cheap or it might not, leaving the world with a terrible choice between sharply reducing emissions at huge cost or suffering through the effects of unabated warming. Businesses that do not hedge against the threat of uncertain outcomes fail. The world cannot afford such recklessness on climate change.

83. Which one of the following statements best reflects the crucial message conveyed by the author of the passage? (CSAT-2023)

(a) Businesses that cause emissions may need to close down or pay for pollution in future.

(b) The only solution is technological development related to the issues of climate change.

(c) Waiting to deal with carbon emissions until technology improves is not a wise strategy.

(d) Since future technological change is uncertain, new industries should be based on renewable energy sources.

Passage

Environmental problems cause health problems. Substantial changes in lifestyle can reduce environmental or health problems, but this idea appears almost impossible to adopt. With environmental problems, individual efforts can be perceived as having a negligible effect and therefore lead to inertia. With health, on the other hand, individual choices can make the difference between life and death, literally. And yet, barring a few, there seems to be the same collective lethargy towards making their choices.

84. Which one of the following statements best implies the most rational assumption that can be made from the passage? (CSAT-2023)

(a) We are likely to spend more money on cure than prevention.

(b) It is the job of the government to solve our environmental and public health problems.

(c) Health can be protected even if environmental problems go on unattended.

(d) Loss of traditional lifestyle and the influence of western values led to some unhealthy way of living.

Passage

Many people are not eating the right food. For some, itis simply a decision to stick with food they enjoy but which is not too healthy. This is leading to an increase in non-communicable diseases. This in turn leads to major burden on our healthcare systems that have the potential to derail the economic progress which is essential for the poor to improve their lives. For others, it is about limited access to nutritious food or a lack of affordability, leading to monotonous diets that do not provide the daily nutrients for them to develop fully. Part of the reason nutrition is under threat worldwide is that our food systems are not properly responding to nutritional needs. Somewhere along that long road from farm to fork, there are serious detours taking place.

85. Which one of the following statements best reflects the crux of the passage? (CSAT-2023)

(a) The scheme of Universal Basic Income should be implemented worldwide as a way of poverty alleviation.

(b) We must place food-based nutrition at the centre of our policy debate.

(c) Nutritional status of food should be improved by creating appropriate genetically modified crops.

(d) Using modern food processing technologies, we must fortify food items with required nutrient elements.

Passage

To tackle the problem of pollution in cities, policy makers think that drastic actions like temporary use of odd-even number scheme for vehicles, closing schools, factories, construction activities, and banning the use of certain type of vehicles are a way forward. Even then the air is not clean. Vehicles more than 15 years old comprise one percent of the total; and taking them off the road will not make any difference. Banning certain fuels and car types arbitrarily is not proper. Diesel engines produce more PM 2.5 and less CO2 than petrol or CNG engines. On the other hand, both diesel and CNG engines produce more NOx than petrol engines. No one has measured the amount of NOx that CNG engines are emitting. Arbitrary bans on vehicles that have passed mandated fitness tests and periodic pollution tests are unfair. What is needed is the scientific and reliable information about the source of pollutants on a continuing basis and the technologies that will work to reduce pollution from them.

86. Which one of the following statements best reflects the most logical and rational implication conveyed by the passage? (CSAT-2023)

(a) Arbitrary curbs on vehicles to reduce pollution are difficult to implement.

(b) Knee-jerk reactions cannot but an evidence-based approach will be more effective.

(c) A heavy penalty should be enforced on those driving without periodic pollution tests.

(d) In the absence of laws to deal with the problem of pollution, the administration tends to make arbitrary decisions.

Passage

Good corporate governance structures encourage companies to provide accountability and control. A fundamental reason why corporate governance has moved onto the economic and political agenda worldwide has been the rapid growth in international capital markets. Effective corporate governance enhance access to external financing by firms, leading to greater investment, higher growth and employment. Investors look to place their funds where the standards of disclosure, of timely and accurate financial reporting, and of equal treatment to all stakeholders are met.

87. Which of the following statements best reflects the logical inference from the passage given above? (CSAT-2023)

(a) It is an important agenda of the countries around the world to ensure access to good external financing.

(b) Good corporate governance improves the credibility of the firms.

(c) International capital markets ensure that the firms maintain good corporate governance.

(d) Good corporate governance paves the way for robust supply chains.

Passage

Elephants are landscape architects, creating clearings in the forest, preventing overgrowth of certain plant species and allowing space for the regeneration of others, which in turn provide sustenance to others herbivorous animals. Elephants eat plants, fruits and seeds, propagating the seeds when they defecate in other places as they travel. Elephant dung provides nourishment to plants and animals and acts as a breeding ground for insects. In times of drought, they access water by digging holes which benefits other wildlife.

88. Which one of the following statements best reflects the most logical and rational inference that can be drawn from the passage? (CSAT-2023)

(a) The home range of elephants needs to be a vast area of rich biodiversity.

(b) Elephants are the keystone species and they benefit the biodiversity.

(c) Rich biodiversity cannot be maintained in the forests without the presence of elephants.

(d) Elephants are capable of regenerating forests with species as per their requirement.

Passage

In India, the segregation of municipal waste at source is rare. Recycling is mostly with the informal sector. More than three-fourths of the municipal budget goes into collection and transportation, which leaves very little for processing/ resource recovery and disposal. Where does waste-to-energy fit into all this? Ideally it fits in the chain after segregation (between wet waste and rest), collection, recycling, and before getting to the landfill. Which technology is most appropriate in converting waste to energy depends on what is in the waste (that is biodegradable versus non-biodegradable component) and its calorific value. The biodegradable component of India's municipal solid waste is a little over 50 per cent, and biomethanation offers a major solution for processing this.

89. Based on the above passage, the following assumptions have been made:

1. Collection, processing and segregation of municipal waste should be with government agencies.
2. Resource recovery and recycling require technological inputs that can be best handled by private sector enterprises.

Which of the assumptions given above is/are correct? (CSAT-2023)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

90. Which one of the following statements best reflects the crux of the passage? (CSAT-2023)

(a) Generation of energy from municipal solid waste is inexpensive.

(b) Biomethanation is the most ideal way of generating energy from municipal solid waste.

(c) Segregation of municipal solid waste is the first step in ensuring the success of waste-to energy plants.

(d) The biodegradable component of India's municipal solid waste is not adequate to provide energy from waste efficiently/ effectively.

Passage

There is a claim that organic farming is inherently safer and healthier. The reality is that because the organic farming industry is still young and not well-regulated in India, farmers and consumers, alike, are not only confused about what products are best for them, but sometimes use products in ways that could harm them as well. For example, since organic fertilizers are difficult to obtain on a large scale in India, farmers often use farmyard manure, which may contain toxic chemicals and heavy metals. Certain plant sprays, such as Datura flower and leaf spray, have an element called atropine. If it is not applied in the right dose, it can act on the nervous system of the consumer. Unfortunately, how much and when to use it are not well-researched or regulated issues.

91. Based on the above passage, the following assumptions have been made:

1. Organic farming is inherently unsafe for both farmers and consumers.
2. Farmers and consumers need to be educated about eco-friendly food. Which of the assumptions given above is/are correct? (CSAT-2023)

(a) 1 only (c) Both 1and 2

(b) 2 only (d) Neither 1 nor 2

92. Which one of the following statements best reflects the most logical, rational and practical message conveyed by the author of the passage? (CSAT-2023)

(a) In India, organic farming should not be promoted as a substitute for conventional farming.

(b) There are no safe organic alternatives to chemical fertilizers.

(c) In India, farmers need to be guided and helped to make their organic farming sustainable.

(d) The aim of organic farming should not be to generate huge profits as there is still no global market for its products.

Passage

Food consumption patterns have changed substantially in India over

the past few decades. This has resulted in the disappearance of many nutritious foods such as millets. While food grain production has increased over five times since independence, it has not sufficiently addressed the issue of malnutrition. For long, the agriculture sector focused on increasing food production particularly staples, which led to lower production and consumption of indigenous traditional corps/ grains, fruits and other vegetables, impacting food and nutrition security in the process. Further, intensive, monoculture agriculture practices can perpetuate the food and nutrition security problem by degrading the quality of land, water and food derived through them.

93. Based on the above passage, the following assumptions have been made:

1. To implement the Sustainable Development Goals and to achieve zero-hunger goal, monoculture agriculture practices are inevitable even if they do not address malnutrition.
2. Dependence on a few crops has negative consequences for human health and the ecosystem
3. Government policies regarding food planning need to incorporate nutritional security.
4. For the present monoculture agriculture practices, farmers receive subsidies in various ways and government offers remunerative prices for grains and therefore they do not tend to consider crop diversity. Which of the above assumptions are valid? (CSAT-2023)

(a) 1, 2 and 4 only (b) 2 and 3 only

(c) 3 and 4 only (d) 1, 2, 3 and 4

Passage

Scientists studied the vernal window transition period from winter to the growing season. They found that warmer winters with less snow resulted in a longer lag time between spring events and a more protracted vernal window. This change in the spring timetable has ecological, social and economic consequences for agriculture, fisheries and tourism. As the ice melts earlier, the birds don't return, causing a delay, or lengthening in springtime ecological events.

94. With reference to the above passage, the following assumptions have been made:

1. Global warming is causing spring to come early and for longer durations.
2. Early spring and longer period of spring is not good for bird populations.

Which of the above assumptions is/are correct? (CSAT-2023)

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

Passage

A global analysis of nitrogen use efficiency—measure of the amount of nitrogen a plant takes in to grow versus what is left behind as pollution- says that using too much fertilizers will lead to increased pollution of waterways and the air. Currently, the global average for nitrogen use efficiency is approximately 0.4, meaning 40 per cent of the total nitrogen added to cropland goes into the harvested crop while 60 per cent is lost to the environment, says a study. More than half of the world's population is nourished by food grown with fertilizers containing synthetic nitrogen, which is needed to produce high crop yields. Plants take the nitrogen they need to grow, and the excess is left in the ground, water and air. This results in significant emissions of nitrous oxide, a potent greenhouse and ozone depleting gas, and other forms of nitrogen pollution, including eutrophication of lakes and rivers and contamination of river water.

95. Which one of the following statements best reflects the most logical, rational and crucial message implied by the passage? (CSAT-2023)

(a) An enhanced efficiency of use of nitrogen is imperative for both food production and environment.

(b) Production of synthetic nitrogen fertilizers cannot be stopped as it will adversely affect global food security.

(c) Alternatives to crops that require excess of nitrogen should be identified and cultivated

(d) Conventional agriculture using synthetic fertilizers should be replaced with agroforestry, agroecosystems and organic farming.

Passage

Along with sustainable lifestyles, climate justice is regarded as a significant principle in environmental parlance. Both the principles have bearings on political and economic choices of the nation. So far, in our climate change summits or compacts, both the principles have eluded consensus among nations. Justice, in the judicial sense, is well defined. However, in the context of climate change, it has scientific as well as sociopolitical connotations. The crucial question in the next few years will be how resources, technologies and regulations are used to support the victims of climate change. Justice in climate is not confined to actions relating to mitigation, but includes the wider notion of support for adaptation to climate change and compensation for loss and damage.

96. Which one of the following statements best reflects the most logical, rational and crucial message conveyed by the passage? (CSAT-2023)

(a) Climate justice should be ingrained in detail in the rules of all the new climate compacts/agreements.

(b) Environmental resources are unevenly distributed and exploited across the globe.

(c) There is an impending issue of dealing with a huge number of climate change victims/climate refugees.

(d) Climate changes in all its connotations is mostly due to developed countries and therefore their share of burden should be more.

Passage

Sourcing food from non-agricultural lands (uncultivated systems such as forests, wetlands, pastures, etc) in addition to agricultural lands enables a systemic approach to food consumption. It allows rural and tribal communities to sustain themselves for the whole year and steer clear of natural disasters and season-inducted shortfalls of agricultural

food. Since the productivity of trees is often more resilient to adverse weather conditions than annual crops, forest foods often provide a safety net during periods of food shortages caused by crop failure; forest foods also make important contributions during seasonal crop production gaps.

97. Which one of the following statements best reflects the most logical and rational message conveyed by the author of the passage? (CSAT-2023)

(a) Food yielding trees should replace other trees in rural and tribal areas and community owned lands.

(b) Food security cannot be ensured in India with the present practice of conventional agriculture.

(c) Wastelands and degraded areas in India should be converted into agroforestry systems to help the poor.

(d) Agroecosystems should be developed in addition to or along with conventional agriculture.

Passage

While awareness on use/misuse and abuse of antibiotics is common knowledge, as is the impact of dosing poultry with antibiotics, the environmental impact of antibiotics manufacturing companies not treating their waste has scarcely been discussed at any length or seriousness thus far. Pollution from antibiotics factories is fuelling the rise of drug-resistant infections. The occurrence of drug-resistant bacteria surrounding the pharma manufacturing plants is well known.

98. Which one of the following statements best reflects the most logical and practical message conveyed by the passage? (CSAT-2023)

(a) It is necessary to put proper effluent treatment protocols in place.

(b) It is necessary to promote environmental awareness among people.

(c) Spread of drug-resistant bacteria cannot be done away with, as it is inherent in modern medical care.

(d) Pharma-manufacturing companies should be set up in remote rural areas, away from crowded towns and cities.

Passage

Benefits of good quality school education accrue only when students complete and leave school after having acquired the gateway skills. Like one learns to walk before running, similarly one picks up advanced skills only after picking the basic foundational skills. The advent of the knowledge economy poses new challenges, and one of the severe consequences of having an uneducated workforce will be our inability to keep pace with the global economy. Without a strong learning foundation at the primary level, there can be no improvement in higher education or skill development.

99. Which one of the following statements best reflects the crux of the passage? (CSAT-2023)

(a) To become a global power, India needs to invest in universal quality education.

(b) India is unable to become a global power because it is not focusing or promoting knowledge economy.

(c) Our education system should focus more on imparting skills during higher education.

(d) Parents of many school children are illiterate and are unaware of the benefits of quality education.

Passage

The paradox of choice is illustrated by the story of Buridan's ass. Jean Buridan, the 14" century philosopher, wrote about free will and the inability to choose due to numerous choices and uncertainties. In the story, a donkey stands between two equally appealing stacks of hay. Unable to decide which to eat, it starves to death. Changes in technology and innovations such as smart phones and tablets only exacerbate our glut of choices. Constant connectivity and overconsumption of real-time data and social media can leave little room for self-reflection and rest, making decisions more difficult. Life is about choices. Many people are overwhelmed with attractive life choices, yet find themselves unhappy and anxious.

100. Which one of the following statements best reflects the most logical message implied by the above passage? (CSAT-2023)

(a) Modern technology enfeebles societal structure and makes life difficult.

(b) Modern life is full of uncertainties and endless difficult choices.

(c) We are influenced by the opinion of others and have no courage to follow our own convictions.

(d) In our lives, having too few choices may not be a good thing, but having too many can be equally as difficult.

Passage

Household finance in India is unique. We have a tendency to invest heavily in physical assets such as gold and property. Steps to encourage the financialization of savings are critical. A populace accustomed to traditional processes will not simply jump into financialization. Hurdles to change include onerous bureaucracy, a scepticism of organized financial institutions, a lack of basic information about which of the myriad services and providers is best for each family, and how (and even if) one can make the transition between them if necessary.

101. Regarding the financialization of household savings, which of the following statements best reflect the solutions that are implied by the passage?

1. A flexible environment is needed to develop solutions.
2. Households need customized solutions.
3. Innovations in financial technology are required.

Select the correct answer using the code given below:

(CSAT-2023)

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Passage

Pharmaceutical patents grant protection to the patentee for the duration of the patent term. The patentees enjoy the liberty to determine the

price of medicines, which is time-limited to the period of monopoly, but could be unaffordable to the public. Such patent protection offered to the patentees is believed to benefit the public over the longer term through innovations and research and development (R&D), although it comes at a cost, in the nature of higher prices for patented medicine, The patent— provide the patentee with a legitimate mechanism to get returns on the costs incurred in innovation and research.

102. Based on the above passage, the following assumptions have been made:

1. Patent protection given to patentees puts a huge burden on public's purchasing power in accessing patented medicines.
2. Dependence on other countries for pharmaceutical products is a huge burden for developing and poor countries.
3. Providing medicines to the public at affordable prices is a key goal during the public health policy design in many countries.
4. Governments need to find and appropriate balance between the rights of patentees and the requirements of the patients. Which of the above assumptions are valid? (CSAT-2023)

(a) 1 and 2 (b) 1 and 4

(c) 3 and 4 (d) 2 and 3

Passage

India should ensure the growth of the digital economy while keeping personal data of citizens secure and protected. No one will innovate in a surveillance-oriented environment or in a place where an individual's personal information is compromised. The ultimate control of data must reside with the individuals who generate it; they should be enabled to use, restrict or monetize it as they wish. Therefore, data protection laws should enable the right kind of innovation— one that is user-centric and privacy protecting.

103. Based on the above passage, the following assumptions have been made:

1. Protection of privacy is not just a right, but it has value to the economy.
2. There is a fundamental link between privacy and innovation.

Which of the above assumptions is/are valid? (CSAT-2023)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

In India, while the unemployment rate is a frequently used measure of poor performance of the economy, under conditions of rising school and college enrolment, it paints an inaccurate picture. The reported unemployment rate is dominated by the experience of younger Indians who face higher employment challenges and exhibit greater willingness to wait for the right job than their older peers. The unemployment challenge is greater for people with secondary or higher education, and rising education levels inflate unemployment challenges.

104. Which one of the following statements most likely reflects as to what the author of the passage intends to say? (CSAT-2023)

(a) Enrolment in schools and colleges is high but there is no quality education.

(b) Unemployment must be seen as a function of rising education and aspirations of young Indians.

(c) There are no labour-intensive industries to accommodate the huge number of unemployed people.

(d) The education system should be properly designed so as to enable the educated people to be self-employed.

Passage

"Science by itself is not enough, there must be a force and discipline outside the sciences to coordinate them and point to a goal. It is not possible to run a course right when the goal itself has not been rightly placed. What science needs is philosophy — the analysis of scientific method and the coordination of scientific purposes and results; without this, any science must be superficial. Government suffers, precisely like science, for lack of philosophy. Philosophy bears to science the same relationship which statesmanship bears to politics: movement guided by total knowledge and perspective, as against aimless and individual seeking. Just as the pursuit of knowledge becomes scholasticism when divorced from the actual needs of men and life, so the pursuit of politics becomes a destructive bedlam when divorced from science and philosophy."

105. Which one of the following statements best reflects the most rational, logical and practical message conveyed by the passage? (CSAT-2023)

(a) Modern statesmen need to be well trained in scientific methods and philosophical thinking to enable them to have a better perspective of their roles, responsibilities and goals.

(b) It is not desirable to have Governments managed by empirical statesmen unless well mixed with others who are grounded in learning and reflect wisdom.

(c) As the statesmen/bureaucrats are the products of a society, it is desirable to have a system of education in a society that focuses on training its citizens in scientific method and philosophical thinking from a very early age.

(d) It is desirable that all scientists need to be philosophers as well to make their work goal oriented and thus purposeful and useful to the society.

Passage

"The last end of the state is not to dominate men, nor to restrain them by fear; rather it is so to free each man from fear that he may live and act with full security and without injury to himself or his neighbour. The end of the state, | repeat, is not to make rational beings into brute beasts and machines. It is to enable their bodies and their minds to function safely. Itis to lead men to live by, and to exercise, a free reason; that they may not waste their strength in hatred, anger and guile, nor act unfairly toward one another."

106. Based on the above passage, which one of the following terms expresses the ultimate goal of the state? (CSAT-2023)

(a) Personal safety (b) Health of body and mind

(c) Communal harmony (d) Liberty

Passage

The main threat to maintaining progress in human development comes from the increasingly evident unsustainability of production and consumption patterns. Current production models rely heavily on fossil fuels. We now know that this is unsustainable because the resources are finite. The close link between economic growth and greenhouse gas emissions needs to be served for human development to become truly sustainable. Some developed countries have begun to alleviate the worst effects by expanding recycling and investing in public transport and infrastructure. But most developing countries are hampered by the high costs and low availability of clean energy sources. Developed countries need to support developing countries' transition to sustainable human development.

107. Unsustainability in production pattern is due to which of the following? (CSAT-2022)

1. Heavy dependence on fossil fuels.
2. Limited availability of resources.
3. Expansion of recycling.

Select the correct answer using the code given below.

(a) 1 and 2 only (a) 2 only

(c) 1 and 3 only (d) 1, 2 and 3

108. Consider the following statements: Developed countries can support developing countries' transition to sustainable human development by (CSAT-2022)

1. Making clean energy sources available at low cost.
2. Providing loans for improving their public transport at nominal interest rates.
3. Encouraging them to change their production and consumption patterns.

Which of the statements given above is/are correct?

(a) 1 only (c) 2 and 3 only

(b) 1 and 2 only (d) 1, 2 and 3

Passage

Unless the forces and tendencies which are responsible for destroying the country's environment are checked in the near future and afforestation of denuded areas is taken up on a massive scale, the harshness of the climatic conditions and soil erosion by wind and water will increase to such an extent that agriculture, which is the mainstay of our people, will gradually become impossible. The desert countries of the world and our own desert areas in Rajasthan are a grim reminder of the consequences of large-scale deforestation. Pockets of desert like landscape are now appearing in other parts of the country including the Sutlej-Ganga Plains and Deccan Plateau. Where only a few decades back there used to be lush green forests with perennial streams and springs, there is only brown earth, pare of vegetation, without any water in the streams and springs except in the rainy season.

109. According to the passage given above, deforestation and denudation will ultimately lead to which of the following? (CSAT-2022)

1. Depletion of soil resource.
2. Shortage of land for the common man.
3. Lack of water for cultivation.

Select the correct answer using the code given below.

(a) 1 and 2 only (c) 1 and 3 only

(b) 2 and 3 only (d) 1, 2 and 3

Passage

"In simple matters like shoemaking, we think only a specially trained person will serve our purpose, but in politics, we presume that everyone who knows how to get votes knows how to administer a State. When we are ill, is a guarantee of specific preparation and technical competence-we do not ask for the handsomest physician, or the most eloquent one: well then, when the whole State is ill should we not look for the service and guidance of the wisest and the best?"

110. Which one of the following statements best reflects the message of the author of the passage? (CSAT-2022)

(a) We assume that in a democracy, any politician is qualified to administer a State.

(b) Politicians should be selected from those trained in administration.

(c) We need to devise a method of barring incompetence from public office.

(d) As voters select their administrators, the eligibility of politicians to administer a State cannot be questioned.

Passage

The poverty line is quite unsatisfactory when it comes to grasping the extent of poverty in India. It is not only because of its extremely narrow definition of 'who is poor' and the debatable methodology used to count the poor, but also because of a more fundamental assumption underlying it. It exclusively relies on the notion of poverty as in sufficient income or insufficient purchasing power. One can better categorize it by calling it income poverty. If poverty is ultimately about deprivations affecting human well-being, then income poverty is only one aspect of it. Poverty of a life, in our view, lies not merely in the impoverished state in which the person actually lives, but also in the lack of real opportunity given by social constraints as well as personal circumstances-to choose other types of living. Even the relevance of low incomes, meagre possessions, and other aspects of what are standardly seen as economic poverty relate ultimately to their role in curtailing capabilities, i.e., their role in severely restricting the choices people have t8 lead variable and valued lives.

111. Why is the methodology adopted in India to count the 'poor' debatable? (CSAT-2022)

(a) There is some confusion regarding what should constitute the 'poverty line'.

(b) There are wide diversities in the condition of the rural and urban poor.

(c) There is no uniform global standard for measuring income poverty.

(d) It is based on the proposition of poverty as meagre income or buying capacity.

112. Why is income poverty only one measure of counting the 'poor'? (CSAT-2022)

(a) It talks of only one kind of deprivation ignoring all others.

(b) Other deprivations in a human life have nothing to do with lack of purchasing power.

(c) Income poverty is not a permanent condition; it changes from time to time.

(d) Income poverty restricts human choices only at a point of time.

113. What does the author mean by 'poverty of a life'? (CSAT-2022)

(a) All deprivations in a human life which stem not only from lack of income but lack of real opportunities.

(b) Impoverished state of poor people in rural and urban areas

(c) Missed opportunities in circumstances.

(d) Material as well as non-material deprivations in a human life which restrict human choices permanently.

Passage

In some places in the world, the productivity of staples such as rice and wheat has reached a plateau. Neither new strains nor fancy agrochemicals are raising the yields. Nor is there much unfarmed land left that is suitable to be brought under the plough. If global temperature continues to rise, some places will become unsuitable for farming. Application of technology can help overcome these problems. Agricultural technology is changing fast. Much of this change is brought about by affluent farmers in the West/Americas. Techniques developed in the West are being adapted in some places to make tropical crops more productive. Technology is of little use if it is not adapted. In the developing world, that applies as much to existing farming techniques as it does to the latest advances in genetic modification. Extending to the smallholders and subsistence farmers of Africa and Asia the best of today's agricultural practices, in such simple matters as how much fertilizers to apply and when, would lead to a greatly increased availability of food for humanity. So would things like better roads and storage facilities, to allow for the carriage of surpluses to markets and reduce wastage.

114. Based on the above passage; the following assumptions have been made: (CSAT-2022)

1. Development of agricultural technology is confined to developed countries.
2. Agricultural technology is not adapted in developing countries.

Which of the above assumptions is/are valid?

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

115. Based on the above passage; the following assumptions have been made: (CSAT-2022)

1. Poor countries need to bring about change in their existing farming techniques.
2. Developed countries have better infrastructure, and they waste less food.

Which of the above assumptions is/are valid?

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

116. Based on the above passage; the following assumptions have been made: (CSAT-2022)

1. Growing enough food for future generations will be a challenge.
2. Corporate farming is a viable option for food security in poor countries.

Which of the above assumptions is/are valid?

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

Passage

Natural selection cannot anticipate future environments on the earth. Therefore, the set of existing organisms can never be fully prepared for environmental catastrophes that await life. An outcome of this is the extinction of those species which cannot overcome environmental adversity. This failuré to survive, in modern terms, can be attributed to the genomes which are unable to withstand geological vagaries or biological mishaps infections, diseases and so on). In biological evolution on the earth, extinction of species has been a major feature. The earth may presently have up to ten million species, yet more than 90% of species that have ever lived on the earth are now extinct. Once again, the creationist doctrines fail to satisfactorily address why a divine creator will firstly bother to create millions of species and then allow them to perish. The Darwinian Explanation: for extinct life is once again simple, elegant and at once convincing organisms go extinct as a function of environmental or biological assaults for which their inheritance deems them ill-equipped. Therefore, the so-called Darwinian theory of evolution is not a theory at all. Evolution happens-this is a fact. The mechanism of evolution (Darwin proposed natural selection) is amply supported by scientific data. Indeed, to date no single zoological, botanical, geological, paleontological, genetic or physical evidence has refuted either of the central two main Darwinian ideas. If religion is not taken into consideration, Darwinian laws are acceptable just like the laws proposed by Copernicus, Galileo, Newton and Einstein-sets of natural laws that explain natural phenomena in the universe.

117. According to the passage, natural selection cannot anticipate future environments on the earth as (CSAT-2022)

1. species not fully prepared to face the environmental changes that await them will face extinction.
2. all the existing species would get extinct as their genomes will not withstand biological mishaps.
3. inability of the genome to withstand environmental changes would result in extinction.
4. extinction of species is a common feature.

Select the correct answer using the code given below.

(a) 1, 2 and 3 (c) 1,3 and 4

(b) 2,3 and 4 (d) 1,2 and 4

118. The passage suggests that Darwinian theory of evolution is not a theory at all because (CSAT-2022)

(a) it does not satisfy the creationist doctrine.

(b) extinction is a function of environment and biological assaults.

(c) there is no evidence to refute it.

(d) existence of organisms is attributed to a creator.

119. With reference to the passage, the following assumptions have been made: (CSAT-2022)

1. Only species that have the ability to overcome environmental catastrophes will survive and perpetuate.
2. More than 90% of the species on the earth are in the danger of getting extinct due to drastic changes in the environment.
3. Darwin's theory explains all the natural phenomena.

Which of the above assumptions is/are valid?

(a) 1 only (c) 3 only

(b) 1 and 2 only (d) 1, 2 and 3

Passage

The demographic dividend, which has begun in India and is expected to last another few decades, is a great window of opportunity. The demographic dividend is basically a swelling in the working age population, which conversely means that the relative ratio of very young and very old will, for a while, be on the decline. From the experience of Ireland and China, we know that this can be a source of energy and an engine of economic growth, The demographic dividend tends to raise a nation's savings rate since in any nation, it is the working age population that is the main saver. And since the savings rate is an important driver of growth, this should help elevate our growth rate. However, the benefits of demographic dividend depend on the quality of the working age population. And this implies bringing back the importance of education, acquisition of skills and human capital.

120. Which of the following would invariably happen in a country when the demographic dividend has begun to operate? (CSAT-2022)

1. The number of illiterate people will decrease.
2. The ratio of very old and very young will decrease for a while.
3. Population growth rate will quickly stabilize.

Select the correct answer using the code given below.

(a) 1 and 2 only (c) 1 and 3 only

(b) 2 only (d) 1,2 and 3

121. With reference to the passage, which of the following inferences can be drawn? (CSAT-2022)

1. Demographic dividend is an essential condition for a country to rapidly increase its economic growth rate.
2. Promotion of higher education is an essential condition for a country for its rapid economic growth.

Select the correct answer using the code given below.

(a) 1only (c) Both 1and 2

(b) 2 only (d) Neither 1 nor 2

Passage

The majority of people who fail to accumulate money sufficient for their needs, are generally, easily influenced by the opinions of others. They permit the newspapers and the gossiping neighbours to do their thinking for them. Opinions are the cheapest commodities on the earth. Everyone has a flock of opinions ready to be wished upon by anyone who will accept them. If you are influenced by opinions when you reach decisions, you will not succeed in any undertaking.

122. Which one of the following is implied by the passage? (CSAT-2022)

(a) Most of the people do not accumulate money for their needs.

(b) Most of the people never fail to accumulate money for their needs.

(c) There are people who fail to accumulate money for their needs.

(d) There is no need to accumulate money.

123. What is the main idea of the passage? (CSAT-2022)

(a) People should not be influenced by the opinions of others.

(b) People should accumulate as much money as they can.

(c) People should neither give nor accept the opinions.

(d) People will succeed in any undertaking if they do not accept any opinion at all.

Passage

To encourage research is one of the functions of a university. Contemporary universities have encouraged research, not only in those cases where research is necessary, but on all sorts of entirely unprofitable subjects as well. Scientific research is probably never completely valueless. However silly and insignificant it may seem, however mechanical and unintelligent the labours of the researchers, there is always a chance that the results may be of value to the investigator of talent, who can use the facts collected for him by uninspired but industrious researchers as the basis of some fruitful generalization. But where research is not original, but consists in the mere rearrangement of existing materials, where its objects is not scientific but literary or historical, then there is a risk of the whole business becoming merely futile.

124. The author's assumption about scientific research is that (CSAT-2022)

(a) it is never very valuable.

(b) it is sometimes very valuable.

(c) it is never without some value.

(d) it is always very valuable.

125. According to the author (CSAT-2022)

(a) not many research results can be of value to an intelligent investigator.

(b) a research result is always valuable to an intelligent investigator.

(c) any research result can be of value to an intelligent investigator.

(d) a research result must always be of some value to an intelligent investigator.

Passage

"The social order is a sacred right which is the basis of all other rights. Nevertheless, this right does not come from nature, and must therefore be founded on conventions."

126. With reference to the above passage which of the following statements is/are correct (CSAT-2022)

1. Conventions are the sources of rights of man.
2. Rights of man can be exercised only when there is a social order.

Select the correct answer using the code given below.

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

Passage

With steady economic growth, higher literacy and increasing skill levels, the number of Indian middleclass families has gone up exponentially. Direct results of the affluence have been changes in dietary patterns and energy consumption levels. People have moved to a higher protein-based diet like milk products, fish, and meat, all of which need significantly more water to produce than cereal-based diets. Increasing use of electronic and electric machines/gadgets and motor vehicles needs more and more energy and generation of energy needs water.

127. Which one of the following best reflects the crux of the passage? (CSAT-2022)

(a) People should be persuaded to continue with the mainly Indian traditional cereal-based diets.

(b) India needs to focus on developing agricultural productivity and capacity for more energy generation in the coming years.

(c) Modern technological developments result in the change of cultural and social behaviour of the people.

(d) Water management practices in India need to change dramatically in the coming years.

Passage

For two or three generations past, ever-increasing number of Individuals have been living as workers merely, not as human beings. An excessive amount of labour is rule today in every circle of society, with the result that man's spiritual element cannot thrive. He finds it very difficult to spend his little leisure in serious activities. He does not want to think; or he cannot even if he wants to. He seeks not Self-improvement, but entertainment which would enable him to be mentally idle and to forget his usual activities. Therefore, the so-called culture of our age is dependent more on cinema than on theatre, more on newspapers, magazines and crime stories than on serious literature.

128. The passage is based on the idea that (CSAT-2022)

(a) man should not work hard.

(b) the great evil of our age is overstrain.

(c) man cannot think well.

(d) man cannot care for his spiritual welfare.

129. Man does not seek self-improvement because he (CSAT-2022)

(a) is not intellectually capable.

(b) has no time to do so.

(c) is distracted by materialism.

(d) loves amusement and is mentally idle.

Passage

In an economic organization, allowing mankind to benefit by the productivity of machines should lead to a very good life of leisure, and much leisure is apt to be tedious except to those who have intelligent activities and interests. If a leisured population is to be happy, it must be an educated population, and must be educated with a view to enjoyment as well as to the direct usefulness of technical knowledge.

130. Which of the following best reflects the underlying tone of the passage? (CSAT-2022)

(a) Only an educated population can best make use of the benefits of economic progress.

(b) All economic development should be aimed at the creation of leisure.

(c) An increase in the educated population of a country leads to an increase in the happiness of its people.

(d) Use of machines should be encouraged in order to create a large leisured population.

Passage

How best can the problems of floods and droughts be addressed so that the losses are minimal, and the system becomes resilient? In this context, one important point that needs to be noted is that India gets "too much' water (about 75% of annual precipitation) during 120 days (June to September) and "too little' for the remaining 245 days. This skewed water availability has to be managed and regulated for its consumption throughout the year.

131. Which one of the following best reflects the practical, rational, and lasting solutions? (CSAT-2022)

(a) Constructing huge concrete storage tanks and canals across the country.

(b) Changing the cropping patterns and practices.

(c) Interlinking of rivers across the country.

(d) Buffer stocking of water through dams and recharging aquifers.

Passage

If presents bring less thrill now that we are grown up, perhaps it is because we have too much already; or perhaps it is because we have lost the fullness of the joy of giving, and with it the fullness of the joy of

receiving. Children's fears are poignant, their miseries are acute, but they do not look too forward nor too far backward. Their joys are clear and complete, because they have not yet learnt always to add 'but' to every proposition. Perhaps we are too cautious, too anxious, too sceptical. Perhaps some of our cares would shrink if we thought less about them and entered with more single-minded enjoyment into the happiness that come our way.

132. With reference to the passage, which one of the following statements is correct? (CSAT-2022)

(a) It is not possible for adults to feel thrilled by presents.

(b) There can be more than one reason why adults feel less thrilled by presents.

(c) The author does not know why adults feel less thrilled by presents.

(d) Adults have less capacity to feel the joy of loving or being loved.

133. The author of the passage is against (CSAT-2022)

(a) worrying too much about the past and future

(b) being in the habit of thinking about presents

(c) not being thrilled by new things

(d) giving and receiving joy only partially

Passage

With respect to what are called denominations of religion, if everyone is left to be a judge of his own religion, there is no such things as religion that is wrong; but if they are to be a judge of each other's religion, there is no such thing as a religion that is right, and therefore all the world is right or all the world is wrong in the matter of religion.

134. What is the most logical assumption that can be made from the passage given above? (CSAT-2021)

(a) No man can live without adhering to some religious denomination.

(b) Itis the duty of everyone to propagate one's religious denomination.

(c) Religious denominations tend to ignore the unity of man.

(d) Men do not understand their own religious denomination.

Passage

It is certain, that seditions, wars, and contempt or breach of the laws are not so much to be imputed to the wickedness of the subjects, as to the bad state of a dominion. For men are not born fit for citizenship but must be made so. Besides, men's natural passions are everywhere the same; and if wickedness more prevails, and more offences are committed in one commonwealth than in another, it is certain that the former has neither enough pursued the end of unity, nor framed its Jaws with sufficient forethought; and that, therefore, it has failed in making quite good its right as a commonwealth.

135. Which among the following is the most logical and rational inference that can be made from the passage given above? (CSAT-2021)

(a) Seditions, wars, and breach of the laws are inevitable in every dominion.

(b) It is not the people, but the sovereign who is responsible for all the problems of any dominion.

(c) That dominion is the best which pursues the aim of unity and has laws for good citizenship.

(d) Itis impossible for men to establish a good dominion.

Passage

Inequality violates a basic democratic norm: the equal standing of citizens. Equality is a relation that obtains between persons in respect of some fundamental characteristic that they share in common. Equality is, morally speaking, a default principle. Therefore, persons should not be discriminated on grounds such as race, caste, gender, ethnicity, disability, or class. These features of human condition are morally irrelevant. The idea that one should treat persons with respect not only because some of these persons some special features or talent, for example skilled cricketers, gifted musicians, or literary giants, but because persons are human being, is by now part of common-sense morality.

136. With reference to the above passage, the following assumptions have been made: (CSAT-2021)

1. Equality is a prerequisite for people to participate in the multiple transactions of society from a position of confidence.
2. Occurrence of inequality is detrimental to the survival of democracy.
3. Equal standing of all citizens is an idea that cannot actually be realised even in a democracy.
4. Right to equality should be incorporated into our values and day-to-day political vocabulary. Which of the above assumptions are valid?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 4 only (d) 3 and 4 only

Passage

Aristocratic government ruins itself by limiting too narrowly the circle within which power is confined; oligarchic government ruins itself by the incautious scramble for immediate wealth. But even democracy ruins itself by excess of democracy. Its basic principle Is the equal right of all to hold office and determine public policy. This is, at first glance, a delightful arrangement; it becomes disastrous because the people are not properly equipped by education to select the best rulers and the wisest courses. The people have no understanding and only repeat what their rulers are pleased to tell them. Such a democracy is tyranny or autocracy. — Plato

137. Which one of the following statements best reflects the crux of the passage given above? (CSAT-2021)

(a) Human societies experiment with different forms of governments.

(b) Any form of government tends to deteriorate by excess of its basic principle.

(c) Education of all citizens ensures a perfect, functional and

sustainable democracy.

(d) Having a government is a necessary evil because tyranny is inherent in any form of government.

Passage

India faces a challenging immediate future in energy and climate policymaking. The problems are multiple: sputtering fossil fuel production capability; limited access to electricity and modern cooking fuel for the poorest; rising fuel imports in an unstable global energy context; continued electricity pricing and governance challenges leading to its costly deficits or surplus supply; and not least, growing environmental contestation around land, water and air. But all is not bleak: growing energy efficiency programmes; integrated urbanisation and transport policy discussions; inroads to enhancing energy access and security; and bold renewable energy initiatives, even if not fully conceptualized, suggest the promise of transformation.

138. Which one of the following statements best reflects the critical message conveyed by the passage given above? (CSAT-2021)

(a) India's energy decision-making process is ever more complex and interconnected.

(b) India's energy and climate policy is heavily tuned to sustainable development goals.

(c) India's energy and climate actions are not compatible with its broader social, economic and environmental goals.

(d) India's energy decision-making process is straight forward supply-oriented and ignores the demand side.

Passage

There are reports that some of the antibiotics sold in the market are fed to poultry and other livestock as growth promoters. Overusing these substances can create superbugs, pathogens that are resistant to multiple drugs and could be passed along humans. Mindful of that, some farming companies have stopped using the drugs to make chickens gain weight faster. Since Denmark banned antibiotic growth promoters in the 1990s, the major pork exporters says it is producing more pigs—and the animals get fewer diseases.

139. Which one of the following statements best reflects the critical message conveyed by the passage given above? (CSAT-2021)

(a) People should avoid consuming the products of animal farming.

(b) Foods of animal origin should be replaced with foods of plant origin.

(c) Using antibiotics on animals should be banned.

(d) Antibiotics should only be used to treat diseases.

Passage

Policy makers and media have placed the blame for skyrocketing food prices on a variety of factors, including high fuel prices, bad weather in key food producing countries, and the diversion of land to non-food production. Increased emphasis, however, has been placed on a surge in demand for food from the most populous emerging economies. It seems highly probable that mass consumption in these countries could be well poised to create a food crisis.

140. With reference to the above passage, the following assumptions have been made: (CSAT-2021)

1. Oil producing countries are one of the reasons for high food prices.
2. If there is a food crisis in the world in the near future, it will be in the emerging economies.

Which of the above statements is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

A central message of modern development economics is the importance of income growth, by which is meant growth in Gross Domestic Product (GDP). In theory, rising GDP creates employment and investment opportunities. As incomes grow in a country where the level of GDP was once low, households, communities, and governments are increasingly able to set aside some funds for the production of things that make for a good life. Today GDP has assumed such a significant place in the development lexicon, that if someone mentions "economic growth", we know they mean growth in GDP.

141. With reference to the above passage, the following assumptions have been made: (CSAT-2021)

1. Rising GDP is essential for a country to be a developed country.
2. Rising GDP guarantees a reasonable distribution of income to all households.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

Fig trees (genus Ficus) are considered sacred in India, East Asia and Africa and are common in agricultural and urban landscapes where other large trees are absent. In natural forests, fig trees provide food for wildlife when other resources are scarce and support a high density and diversity of frugivores (fruit-eating animals). If frugivorous birds and bats continue to visit to fig trees located in sites with high human disturbances, sacred fig trees may promote frugivore abundance. Under favourable microclimate, plenty of seedlings of other tree species would grow around fig trees.

142. On the basis of the passage given above, the following assumption have been made: (CSAT-2021)

1. Fig trees can often be keystone species in natural forests.
2. Fig trees can grow where other large woody species cannot grow.
3. Sacred trees can have a role in biodiversity conservation.
4. Fig trees have a role in the seed dispersal of other tree species.

Which of the above assumptions is/are valid?

(a) 1 and 2 only (b) 3 only

(c) 2 and 4 only (d) 1, 3 and 4 only

Passage

At the heart of agroecology is the idea that agroecosystems should mimic the biodiversity levels and functioning of natural ecosystems. Such agricultural mimics, like their natural models, can be productive, pest-resistant, nutrient conserving, and resilient to shocks and stresses. In ecosystems there is no 'waste', nutrients are recycled indefinitely. Agroecology aims at closing nutrient loops, i.e., returning all nutrients that come out of the soil back to the soil such as through application of farmyard manure. It also harnesses natural processes to control pests and build soil fertility i.e., through intercropping. Agroecological practices include integrating trees with livestock and crops.

143. Consider the following: (CSAT-2021)

1. Cover crops
2. Fertigation
3. Hydroponics
4. Mixed farming
5. Polyculture
6. Vertical farming.

Which of the above farming practices can be compatible with agroecology, as implied by the passage?

(a) 1, 4 and 5 only (b) 2, 3, 4 and 5 only

(c) 1, 2, 3 and 6 only (d) 4 and 6 only

Passage

Computers increasingly deal not just with abstract data like credit card details and databases, but also with the real world of physical objects and vulnerable human bodies. A modern car is a computer on wheels; an aeroplane is a computer on wings. The arrival of the "Internet of Things" will see computers baked into everything from road signs and MRI scanners to prosthetics and insulin pumps. There is little evidence that these gadgets will be any more trustworthy than their desktop counterparts. Hackers have already proved that they can take remote control of internet connected cars and pacemakers.

144. Which one of the following statements best reflects the most critical inference that can be made from the passage given above? (CSAT-2021)

(a) Computers are not completely safe.

(b) Companies producing the software do not take cyber security seriously.

(c) Stringent data security laws are needed.

(d) The present trend of communication technologies will affect our lives in future.

Passage

A social and physical environment riddled with poverty, inequities, unhygienic and insanitary conditions generate the risk of infectious diseases. Hygiene has different levels: personal, domestic and community hygiene. There is no doubt that personal cleanliness brings down the rate of infectious diseases. But the entry of the market into this domain has created a false sense of security that gets conditioned and reinforced by the onslaught of advertisements. Experience in Western Europe shows that along with personal hygiene, general improvements in environmental conditions and components like clean water, sanitation and food security have brought down infant/child death/infections rates considerably. The obsession with hand hygiene also brings in the persisting influence of the market on personal health, overriding or marginalizing the negative impact on ecology and the emergence of resistant germs.

145. On the basis of the passage given above, the following assumptions have been made: (CSAT-2021)

1. People who are obsessed with personal hygiene tend to ignore the community hygiene.
2. Emergence of multi-drug resistant germs can be prevented by personal cleanliness.
3. Entry of the market in the domain of hygiene increases the risk of infectious diseases.
4. Scientific and micro-level interventions are not sufficient to bring down the burden of infectious diseases.
5. It is community hygiene implemented through public health measures that is really effective in the battle against infectious diseases.

Which of the above assumptions are valid?

(a) 1 and 2 only (b) 3 and 4 only

(c) 4 and 5 only (d) 1, 2 and 4 only

Passage

Researchers simulated street lighting on artificial grassland plots containing pea-aphids, sap-sucking insects, at night. These were exposed to two different types of light-a white light similar to newer commercial LED lights and an amber light similar to sodium streetlamps. The low intensity amber light was shown to inhibit, rather than induce, flowering in a wild plant of the pea family which is a source of food for the pea-aphids in grasslands. The number of aphids was also significantly suppressed under the light treatment due to the limited amount of food available.

146. Which one of the following statements best reflects the most critical inference that can be made from the passage given above? (CSAT-2021)

(a) Low intensity light has more adverse effect on the plants as compared to high intensity light.

(b) Light pollution can have a permanent adverse impact on an ecosystem.

(c) White light is better for the flowering of plants as compared to the light of other colours.

(d) Proper intensity of light in an ecosystem is important not only for plants but for animals too.

Passage

Approximately 80 percent of all flowering plant species are pollinated by animals, including birds and mammals, but the main pollinators are insects. Pollination is responsible for providing us with a wide variety of food, as well as many plant-derived medicines. At least one-third of

the world's agricultural crops depend upon pollination. Bees are the most dominant taxa when it comes to pollination, and they are crucial to more than four hundred crops. Pollination is an essential service that is the result of intricate relationship between plants and animals, and the reduction or less of either affects the survival of both. Effective pollination requires resources, such as refuges of pristine natural vegetation.

147. On the basis of the passage given above, the following assumptions have been made: (CSAT-2021)

1. Sustainable production of India's cereal food grains is impossible without the diversity of pollinating animals.
2. Monoculture of horticultural crops hampers the survival of insects.
3. Pollinators become scarce in cultivated areas devoid of natural vegetation.
4. Diversity in insects induces diversity of plants. Which of the above assumptions is/are valid?

(a) 1 only (b) 2, 3 and 4 only

(c) 1 and 2 only (d) 3 and 4 only

Passage

A study conducted on the impacts of climate change over the Cauvery basin of Tamil Nadu using regional climate models showed an increasing trend for maximum and minimum temperatures, and a decrease in the number of rainy days. These climatic shifts will have an impact on the hydrological cycles in the region, lead to more run-off and less recharge, and affect the groundwater tables. Further, there has been an increase in the frequency of droughts in the State. This has driven farmers to increase dependency on groundwater resources to secure their crops.

148. Which one of the following statements best reflects the crux of the passage given above? (CSAT-2021)

(a) Development of regional climate models helps in choosing climate-smart agricultural practices.

(b) Heavy dependence on groundwater resources can be reduced by adopting dry-land cropping systems.

(c) Climate changes increase the criticality of water resources while simultaneously threatening it.

(d) Climate changes cause the farmers to adopt unsustainable livelihoods and risky coping strategies.

Passage

Researchers were able to use stem cells to gauge the neurotoxic effects of the environmental pollutant Bisphenol A (BPA). They used a combination of biochemical and cell based assays to examines the gene expression profile during the differentiation of mouse embryonic stem cells upon treatment with BPA, a compound known to cause heart diseases, diabetes, and developmental abnormalities in humans. They were able to detect and measure BPA toxicity towards the proper specification of primary germ layers, such as endoderm and ectoderm, and the establishment of neural progenitor cells.

149. On the basis of the passage given above, the following assumptions have been made: (CSAT-2021)

1. BPA may alter embryonic development in vivo.
2. Biochemical and cell-based assays are useful in finding out treatments for pollution induced diseases.
3. Embryonic stem cells could serve as a model to evaluate the physiological effects of environmental pollutants.

Which of the above assumptions are valid?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Passage

Medieval merchants risked the hazards of the Silk Road to reach the markets of China; Portuguese caravels in the 15th century sailed beyond the bounds of the known world, searching less for knowledge than for gold and spices. Historically, the driver for opening frontiers has always been the search for resources. Science and curiosity are weaker drivers. The only way to open up space, whether the space of solar system or interstellar space is to create an economic engine and that engine is resource extraction.

150. Which one of the following statements best sums up the passage given above? (CSAT-2021)

(a) Wealth generation is the primary motive for any human endeavour.

(b) Space, whether space in solar system or interstellar space, will govern our future economy.

(c) Human beings are motivated to explore new frontiers principally by economic considerations.

(d) Wealth generation is based on the risk-taking behaviour of some men.

Passage

"…….. most people would agree that telling deliberate lies is wrong, except perhaps in certain special situations where more harm will be done by telling the truth. Even the most truthful people probably tell a good many more lies that might be regarded as semantic lies; their use of words contains some measure of falsehood, more or less deliberate."

151. The idea which the first part of the passage mention is (CSAT-2021)

(a) agreement about telling lies.

(b) disagreement about telling lies.

(c) disagreement about telling the truth.

(d) disagreement about the harm in telling the truth.

152. Which one of the following habits is found more often in good people? (CSAT-2021)

(a) Mixing up the true and false

(b) Intentional mixing up of truth with the false.

(c) Falsification of facts

(d) Complete concealment of truth

Passage

Can a democracy avoid being a welfare state for long? Why cannot mass welfare be left entirely to the markets? There is a built-in tension between markets and democracy. Markets do not work on a one-person-one-vote principle as democracies do. What one gets out of the marketplace depends on one's endowments, skills, purchasing power and the forces of demand and supply. Markets reward individual initiative and skill and may also lift many from the bottom rungs of society, but some people never get the opportunity to develop skills that markets demand; they are simply too poor and too handicapped; or skill formation takes too long. By creating jobs, markets may be able to help even unskilled people, but capitalism has always witnessed bursts of unemployment.

153. With reference to the above passage, the following assumptions have been made: (CSAT-2021)

1. Modern democracies rely on the market forces to enable them to be welfare states.
2. Markets ensure sufficient economic necessary for democracies to be effective.
3. Government programmes are needed for those left behind in economic growth.

Which of the above assumptions is/are valid?

(a) 1 and 3 only (b) 3 only

(c) 2 and 3 only (d) 1,2 and 3

154. Which one of the following best reflects the central idea of the passage given above? (CSAT-2021)

(a) School curriculum is not compatible with the expectations of children and parents

(b) Emphasis on academic achievements does give time for development of personality and skills.

(c) Preparing the children to be better citizens should be the responsibility of the education system.

(d) To be a better citizen, the present world order demands societal and life-coping skills in addition to academic content.

Passage

The best universities like Harvard and MIT, despite having the luxury of having some truly excellent teachers on their payroll, are increasingly embracing the "flipped classroom" format, where students listen to video lectures at home, and spend class time applying their knowledge, solving problems, discussing examples, etc. Professors guide that discussion and fill in wherever necessary, explaining those bits that seem to be eluding the students and throwing in advanced ideas that happen to be topical. These universities have made their video lectures available free for anyone in the world. They are also encouraging colleges and universities all over the world to integrate these online courses into their own pedagogy, picking the pieces that are appropriate for their needs and building a package around them.

155. Which one of the following statements best reflects the central idea of the passage given above? (CSAT-2021)

(a) Efficacy of universities would be better in online mode of conducting classroom tuition as compared conventional method.

(b) Availability of higher education can be made easier and cheaper without diluting the content.

(c) We need not invest much in infrastructure related to higher education and yet develop better human and social capital.

(d) Private sector institutions in higher education as well as coaching institutes can take advantage of this opportunity and thrive well.

Passage

Our cities are extremely vulnerable to climate change because of large concentrations of populations and poor infrastructure. Moreover, population densities are increasing in them, but we have not yet developed the systems to address climate change impacts. Our cities contribute to 65 percent of the GDP, but there*are not enough facilities to cater to the needs of the people. It is important to address the issues of air quality, transport, etc., that are vital to identifying sustainable solutions. We need to involve citizens in city planning and create an ecosystem that meets the needs of people.

156. Which among the following is the most logical and rational inference that can be made from the passage given above? (CSAT-2021)

(a) Our cities need to have well-defined administrative set-up with sufficient autonomy.

(b) Ever increasing population densities is a hindrance in our efforts to achieve sustainable development.

(c) To maintain and develop our cities we need to adopt sustainability related interventions.

(d) Public-Private Partnership mode of development is the viable long-term solution for the infrastructure and sustainability problems of India.

Passage

Nothing can exist in a natural state which can be called good or bad by common assent, since every man who is in natural state consults only his own advantage, and determines what is good or bad according to his own fancy and insofar as he has regard for his own advantage alone, and holds himself responsible to no one save himself by any law; and therefore sin cannot be conceived in a natural state, but only a civil state, which is decreed by common consent what is good or bad, and each one holds himself responsible to the state.

157. Which one of the following statements best reflects the central idea of the passage given above? (CSAT-2021)

(a) The conceptions of what is right or wrong exist due to the formation of a state

(b) Unless a ruling authority decides as to what is right or wrong, no man would be morally right

(c) Man is inherently immoral and selfish in a natural state.

(d) The idea of what is right or wrong is necessary for the survival of human species.

Passage

In the immediate future, we will see the increasing commodification of many new technologies-artificial intelligence and robotics, 3D manufacturing, custom made biological and pharmaceutical products, lethal autonomous weapons and driverless cars. This will pose conundrums. The moral question of how a driverless car will decide between hitting jaywalker and swerving and damaging the car has often been debated. The answer is both simple- save the human life- and complex. At which angle should the car swerve-just enough to save the jaywalker or more than enough? If the driverless car is in Dublin, who would take the decision? The Irish Government, or the car's original code writer in California, or a software programmer in Hyderabad to whom maintenance is outsourced? If different national jurisdictions have different fine print on prioritizing a human life, how will it affect insurance and investment decisions, including transnational ones?

158. Which of the following statements best reflect the rational, plausible, and practical implications that can be derived from the passage given above? (CSAT-2021)

1. Too much globalization is not in the best interests of any country.
2. Modern technologies are increasingly blurring the economic borders.
3. Innovation and capital have impinged on the domain of the state.
4. Public policy of every country should focus on developing its own supply chains.
5. Geopolitics will have to reconcile to many ambiguities and uncertainties.

Select the correct answer using the code given below:

(a) 1, 4 and 5 only (b) 1, 2, 3 and 4 only

(c) 2, 3 and 5 only (d) 1, 2, 3, 4 and 5

Passage

The resolution of bankruptcy cases of Indian banks under the Insolvency and Bankruptcy Code should help bring nonperforming assets (NPA) situation under some control. Despite the slow pace of resolution by the National Company Law Tribunal, the Code can be helpful in cleaning up bank books in future credit cycles. The recapitalization of public sector banks too can help increase the capital cushion of banks and induce them to lend more and boost economic activity. But bad debt resolution and recapitalization are only a part of the solution as they, by themselves, can do very little to rein in reckless lending that has pushed the Indian banking system to its current sorry state. Unless there are systemic reforms that address the problem of unsustainable lending, future credit cycles will continue to stress the banking system.

159. Which one of the following statements best reflects the most logical, rational and practical suggestion implied by the passage given above? (CSAT-2021)

(a) Lending by the banks should be closely monitored and regulated by the central Government.

(b) Interest rates should be kept low so as to induce banks to lend more, promote credit growth and thereby boost economic activity.

(c) Merger of many banks into a few large banks alone in the long-term solution to make them viable and prevent their bad performance

(d) Indian banking system requires structural reforms as a long-term solution for bad loans problem.

Passage

In India, the objective of macroeconomics policy is to enhance the economic welfare of the people, and any one wing of such macro policy, monetary or fiscal, cannot independently work without active support of another.

160. Which one of the following statements best reflects the corollary to the passage given above? (CSAT-2021)

(a) The central bank cannot work independently of the Government.

(b) Government should regulate financial markets and institutions closely.

(c) Market economy is not compatible with the socialist policies of the Government.

(d) Financial sector reforms are required for enhancing the economic welfare of the people.

Passage

Private investment in general is volatile. Foreign private investment is more volatile because the available investment avenues are significantly greater (ie., the entire world). Therefore, the responsibility of providing employment cannot be left to Foreign District Investment (FDI). The current FDI inflows are volatile over time and across sectors and regions, which is a necessary consequence of their search for the highest returns. The adverse consequences are unstable employment and an accentuation of income and regional inequalities. A probable positive consequence of foreign investment is the inflow of new technology and its subsequent diffusion. However, the technology diffusion is not at all certain because the existing state of physical and human capital in India may prove inadequate for the diffusion.

161. With reference to the above passage, the following assumptions have been made: (CSAT-2020)

1. Relying on foreign investment in the long run is not an economically sound policy.
2. Policies must be undertaken to reduce volatility in foreign private investment.
3. Policies must be undertaken to strengthen domestic private investment.
4. Public investment should be given priority over private investment.
5. Substantial public investment in education and health should be undertaken.

Which of the above assumptions is/are valid?

(a) 1, 2 and 4 (b) 1, 3 and 5

(c) 2, 4 and 5 (d) 3 only

Passage

Many opportunities to harness the highly skewed, seasonal and spatial distribution of monsoon floes, which occur in a four-month period from June to September annually, have been lost. Since these few months account for most of the rainfall and consequent freshwater availability, the need for holding rainwater in reservoirs, for subsequently releasing it for use over the year, is a necessity nobody can afford to overlook. Climate change will continue to affect weather conditions and create water shortages and excesses. While millions suffer from droughts and floods, waters in the country's many rivers flow unutilized, and are discharged into the sea every year.

162. With reference to the above passage, which of the following could be the most rational and. practical implications for India? (CSAT-2020)

1. Inter-linking of rivers should be undertaken.
2. A network of dams and canals should be built across the country for proper distribution of water.
3. Farmers should be provided easy loans for digging borewells.
4. Usage of water for agriculture should be regulated by law.
5. Distribution of river water among regions should be regulated by the Union Government.

Select the correct answer using the code given below.

(a) 1 and 2 (b) 2, 4 and 5

(c) 1, 3 and 4 (d) 2, 3 and 5

Passage

People will invest in education whenever they are granted the economic freedom to fully enjoy its benefits. Again, this is for the obvious reason that the return on education increases as the level of economic freedom rises. When people, thanks to lower tax rates, are allowed to retain most of the higher income that they gain from each incremental level of education, it makes eminent sense to invest in education. On the other hand, when the government decides to tax the higher income of educated individuals at even higher rates, it makes very little sense to invest in educating oneself further. The same incentives apply to parents who decide on whether to invest in their children's education.

163. Which reference to the above passage, the following assumptions have been made: (CSAT-2020)

1. Lower tax rates in a country invariably translate into greater investments in higher education
2. Investment in the education of children ensures their economic freedom.
3. Economic freedom has a positive impact on building up human capital.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) 3 only (d) 1, 2 and 3

Passage

Our urban bodies cannot possibly ensure sustainable delivery of water in our cities unless financing mechanisms are put in place. Water delivery requires heavy investment in collecting it from a natural source, treating it to make it potable, and laying a distribution network of pipes for delivery to the users. It also requires investments in sewerage infrastructure and sewage treatment plants so that the sewers can carry the wastewater to these plants to ensure that no untreated sewage is discharged back into natural water bodies. If our cities were rich enough to meet the entire cost, water could be delivered free. They are not.

164. What is the most logical and crucial message conveyed by the passage? (CSAT-2020)

(a) Urban local bodies must recover costs through user charges.

(b) Urban local bodies are not efficient enough to meet the water requirements of our cities.

(c) Water shortage in our cities is a perennial problem that cannot be solved.

(d) In view of the water crisis in our cities, there is an urgent need to limit the population of cities by adopting an upper limit of population size.

165. Which reference to the above passage, the following assumptions have been made: (CSAT-2020)

1. Rich cities only can ensure sustainable delivery of water.
2. Sustainable delivery of water in cities means much more than supplying water to households.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

In India, agriculture still engages about half of its workforce, and about 85 per cent of its farms are small and marginal Compared to China and Vietnam, which have experienced fast structural and rural transformation, India's story is of slow transformation. As a result, poverty reduction in India was at much slower pace during 1988-2014, compared to China and Vietnam. India's poverty reduction was slow during 1988- 2005, but during 2005-2012, it accelerated dramatically—almost three times faster than during the earlier period. What did India do during this period? Research reveals that the relative price scenario changed significantly (by more than 50%) in favour of agriculture in the wake of rising global prices. This boosted private investments in agriculture by more than 50%. As a result, Agri-GDP growth touched 4.1% during 2007- 2012 as against 2.4% during 2002-2007. The net surplus of Agri-trade touched $25 billion in 2013-2014; real farm wages rose by 7% per annum. All this led to unprecedented fall in poverty.

166. With reference to the above passage, the following assumptions have been made: (CSAT-2020)

1. Structural and rural transformation is impossible when farms are mainly small and marginal.

2. A good price incentive can trigger investments in agriculture.
3. India needs to build value chains for high-value agriproducts like livestock and horticulture.
4. Higher global prices of agricultural commodities are essential for India's poverty reduction.

Which of the above assumptions are valid?

(a) 1 and 3 (b) 2 and 4
(c) 2 and 3 (d) 3 and 4

167. Which one of the following statements best reflects the critical message of the passage? (CSAT-2020)

(a) India should create large-scale off-farm rural employment to reduce poverty in the near future.
(b) India should create a large number of farmer producer companies.
(c) Private investment in agriculture should be given priority over public investment.
(d) Inclusive agricultural growth is key to reduce poverty in the near future.

Passage

In India, over the last decade or so, labour has been departing agriculture, but is only going to construction and unregistered manufacturing which are not markedly better jobs. Services, where labour tends to be most productive, are not generating the additional jobs the country needs. India will need 24 million or so jobs over the next decade. The new sector, e-commerce, can at best close only half the jobs gap. Only those sectors that drive domestic demand such as health and education can comfortably fill the other half.

168. Which one of the following is best implied in the passage? (CSAT-2020)

(a) Strong measures need to be taken to reduce the rural to urban migration of labour.
(b) The working condition in construction and unregistered manufacturing needs to be improved.
(c) Service sector has been reducing the problem of unemployment.
(d) Increased social sector spending is imperative for large-scale job creation.

Passage

In India, the current focus on the right to privacy is based on some new realities of the digital age. A right is a substantive right only if it works in all situations, and for everyone. A right to free expression for an induvial about her exploitation, for instance, is meaningless without actual availability of security that guarantees that private force cannot be used to thwart this right. The role of the State, therefore, is not just to abstain from preventing rightful free expression, but also to actively ensure that private parties are not able to block it.

169. On the basis of the above passage, the following assumptions have been made: (CSAT-2020)

1. State should have some institutions to ensure its appropriate role in a digital society.
2. State should ensure that private parties do not violate the citizens' right to privacy.
3. Digital economy is not compatible with the idea of not violating the citizens' privacy.

Which of the above assumptions is/are valid?

(a) 1 and 2 (b) 3 only
(c) 1 and 3 (d) 2 only

Passage

One of the biggest ironies around water is that a comes from rivers and other wetlands. Yet it is seen as divorced from them. While water is used as a resource, public policy does not always grasp that it is a part of the natural ecosystem. Efforts at engineering water systems are thus effort at augmenting water supply rather than strengthening the capacities of ecological systems.

170. Which one of the following is the most logical and rational inference that can be made from the above passage? (CSAT-2020)

(a) Rivers and other wetlands should be protected under Ramsar Convention.
(b) Engineering water systems should be modernized and further augmented.
(c) Wetlands need to be reinforced as more than just open sources of water.
(d) Water supply should not be free of cost so as to prevent its misuse or overuse.

Passage

Asset allocation is the most important investment decision we will ever make, and sadly, most of us do not give that decision the importance it deserves. We are adamant about seeking predictability with our future. We tend to think of investing in risky assets as extremely volatile and value eroding. We also dislike fluctuating returns and the loss of control of investment. We think our money is best left idle, unproductive but safe. There is no asset that is risk free. We could lose our jobs, our homes can lose value, our banks can go bankrupt, our bonds can default, the government can collapse and companies we chose fondly may cease to exist. But we cannot live life assuming that all these extreme events are waiting to happen, and all at the same time. All these extreme forms of risks we know will not manifest at the same time.

171. Which one of the following statements best implies the suggestion given by the author of the passage (CSAT-2020)

(a) Distribute your wealth across different kinds of assets 50 that your risks would be minimized.
(b) Risk-taking behaviour should be a necessary component of your personality if you want to generate wealth.
(c) While making investments, find a trustworthy asset management organization which would manage your wealth for you.

(d) You should know that investing your money is a risky business.

Passage

Although most of the Genetically Modified (GM) crops cultivated now are genetically engineered for a single trait, in future, crops genetically engineered for more than one trait will be the norm. Thus, biotechnology's role in agriculture and the regulation of the same cannot be generation of GM crops. Instead, there is a need to take a comprehensive look, taking into account various aspects, including socioeconomic impacts, so that the potential of the technology can be harnessed while minimizing negative impacts. Given the importance of biotechnology in developing varieties that can help in climate change mitigation and adaptation, not using biotechnology as a part of the climate change action plan cannot be an option. Domestic regulation of biotechnology cannot be viewed in isolation of trade policy and obligations under various international treaties and conventions.

172. With reference to the above passage, the following assumptions have been made: (CSAT-2020)

1. Biotechnology regulation is an evolving process.
2. Participation of people is needed in policy decisions regarding biotechnology regulation.
3. Biotechnology regulation should take into account socio-economic aspects in decision-making.
4. Wider involvement of political executive in biotechnology regulation improved its effectiveness in dealing with the country's trade policies and international obligations.

Which of the above assumptions are valid?

(a) 1, 2 and 4 only (b) 1 and 3 only

(c) 2, 3 and 4 only (d) 1, 2, 3 and 4

173. Which one of the following statements best implies the crux of the passage? (CSAT-2020)

(a) Precautionary principle is not given importance in current debate on developing GM crops.

(b) Biotechnology is not currently used in climate change mitigation and adaptation mechanisms.

(c) Biotechnology's role is not confined to the current priorities of developing GM crops.

(d) The negative impacts of biotechnology are not properly understood.

Passage

Bank credit to the industrial sector has started shrinking. Its decline has been a serious concern as credit growth is essential to revive investment. The problem's origins lie in the incomplete reforms of the last 25 years. An institutional change that should have followed the 1991 reforms should have been setting up of a resolution corporation for banks. In a market economy with booms and busts, banks should be allowed to be set up and to fail. Today, we cannot shut down banks because there is not proper system to shut them down. Weak loss-making banks continue to need more capital.

174. Which one of the following is the most logical and rational inference that can be made from the above passage? (CSAT-2020)

(a) Indian banking system is not able to help the country in its economic growth.

(b) Economic reforms that started in 1991 have not helped in improving the economy to expected levels.

(c) India lacks the institutional mechanisms to deal with the failure of banks.

(d) Encouraging the foreign investments in our industrial sector is a good alternative to this sector's dependence on banks for credit.

Passage

India has tremendous potential for solar energy. We all realize that we have to stop burning fossil fuels to meet our energy needs. But certain renewable resources are still going through their cost curves and learning curves to get the required amount of output. The Indian Government has strongly committed to its targets of reducing emissions by 33 per cent by 2030, and towards this, it has initiated a strong push towards a gas-based economy-and has also invested heavily in renewable energy. However, business houses are wary of investing too heavily in renewable energy at a time when the technology is not yet ready.

175. Which one of the following is the most logical and rational inference that can be made from the above passage? (CSAT-2020)

(a) India's commitment to reduce emissions by 33% is unlikely to be achieved.

(b) India should import gas rather than invest in renewable resources.

(c) Getting renewable resources to market too soon may be costly.

(d) India should put in more efforts in the exploration of natural gas.

176. With reference to the above passage, the following assumptions have been made: (CSAT-2020)

1. Governments often provide inefficient and costly subsidies for technologies that may not be ready in the near future.
2. India's commitment of reducing emissions by 33% by 2030 shall be on the basis of gas-based economy.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

Genome editing is different from genome modification. Genome editing typically involves finding the part of a plant genome that could be changed to render it less vulnerable to disease, or resistant to certain herbicides, or to increase yields. Researchers use 'molecular scissors' to dissects the genome and repair it, which is a process that occurs naturally when plants are under attack from diseases and can throw up new mutations that enable the plant to survive future attacks. This evolutionary process can effectively be speeded up now that it is possible to examine plant

genome in detail in laboratories and create mechanisms through which the relevant genes can be altered very precisely.

177. With reference to the above passage, the following assumptions have been made: (CSAT-2020)

1. Genome editing does not require the transfer of genes from one plant to another.
2. Through genome editing, the chosen genes can be altered precisely in a manner akin to the natural process that helps plants to adapt to the environmental factors.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

Many people understand the connection between solid waste management and health in terms of the consequences of unattended heaps of dry garbage which become home for flies and other vermin. However, there is another aspect that is not well-understood, that is, what happens when unscientific solid waste management combines with poor drainage and dumping of untreated sewage into drains which are meant to carry storm water during rains. The result is choked drains which are full of stagnant water breeding mosquitoes, resulting in the spread of water-borne diseases.

178. In the context of India, which one of the following statements best reflects the critical message of the passage? (CSAT-2020)

(a) In India, the drainage networks are not separate for sewerage and storm water.

(b) Urban local bodies do not have enough resources and legislative authority to deal with the problems of waste management.

(c) Solid waste management should be integrated with the maintenance of drainage and sewerage networks.

(d) Bad management of solid waste and sewerage systems by our municipalities is the reason for drinking water shortages in our cities.

Passage

In Part III of the Constitution, which assures people certain fundamental rights, Article 25 proclaims that "all persons are equally entitled to freedom of conscience and the right freely to profess, practise and propagate religion". What people fail to notice is that this proclamation Is prefixed with the words "subject to public order, morality, health and to the other provisions of this Part", which set conditions precedent for the legal protection of religious practices of any community. The closing words of this prefatory rider in Article 25 virtually constitute a subordination clause placing other fundamental rights mentioned in Part Ill over and above the right to religious freedom. Among those other fundamental rights is the right to equality before law and equal protection of laws—assured at the outset and elaborated in later articles to mean, inter alia, that the State shall not deny equal protection of laws to any person or group of persons on the basis of religion alone.

179. What is the most logical inference from the above passage? (CSAT-2020)

(a) State shall not interfere with the religious affairs of the citizens.

(b) Religious freedom under the Constitution is open to State intervention.

(c) Religious freedom of the citizens is not covered under fundamental rights.

(d) Religious practices of any community are immune to State laws.

Passage

Spanish ships in the late 16th century first brought the potato tuber from South America to Europe whereby in the early 19th century, it had become a reliable backup to cereal crops, particularly in the cold, rain-soaked soils of Ireland. The Irish were soon almost wholly dependent on the potato as their staple food. And they were planting primarily one prodigious variety, the 'Lumper' potato, whose genetic frailty would be cruelly exposed by the fungus 'Phytophthora infestans'. In 1845, spores of the deadly fungus began spreading across the country, destroying nearly all the Lumpers in its path. The resulting famine killed or displaced millions.

180. Which one of the following statements best reflects the critical message of the passage? (CSAT-2020)

(a) For introducing any foreign plant into a country, the soil and climate conditions of that country should be suitable.

(b) As a staple food of a country, tuber crops like potato cannot replace cereal crops.

(c) Some of the fungal infections of plants cannot be prevented or stopped from spreading across large areas.

(d) Relying on a homogeneous food source is not desirable.

Passage

India is at once among the fastest growing global economies and home to the largest number of malnourished children in world. There are regions where malnutrition is not the exception but the norm. And across the country, malnutrition is the cause of death for roughly half the 1.3 million children who die before their fifth birthday each year. Even those children who survive suffer permanently from the damage that has already been done to their bodies and minds from not getting enough of the right foods and nutrients. Around 24 million children under 5 are stunted. That makes it harder for them to learn in school and subsequently earn a living as adults. Their lifetime earnings potential is almost a quarter less than of their healthy peers.

181. With refence to the above passage, which of the following is/are the most rational and practical implication/ implications? (CSAT-2020)

1. India's Public Distribution System should be monitored by the Union Government.
2. Girls should be encouraged to delay marriage and first pregnancy.
3. Mothers should be encouraged to breastfeed their children

immediately after birth.

4. The supply of safe drinking water and proper sanitation facilities to all should be ensured

5. Authorities should ensure the vaccination should ensure the vaccination as prescribed.

Select the correct answer using the code given below.

(a) 1, 2, 3 and 4 (b) 2, 3, 4 and 5

(c) 1 only (d) 3 and 5 only

Passage

The pulse variety 'Pusa Arhar 16' has the potential to be grown in the paddy-growing regions of Punjab, Haryana and Uttar Pradesh and eventually in all of India. It's yield (about 2000 kg/hectare) will be significantly greater than those of the existing varieties and because its size will be uniform, it will be amenable to mechanical harvesting, an attractive feature for farmers in northern India who currently use this technology for paddy. Most important, Arhar straw, unlike paddy straw, is green and can be ploughed back into the soil. In paddy straw, the problem is the high silica content, which does not allow for easy decomposition. In the case or Arhar, the farmer, even after combine harvesting, just needs to run a rotovator to cut the leftover straw into pieces, which can be ploughed back and will decompose very fast. All this is difficult with leftover paddy stalks that cannot be easily salvaged or ploughed back. Farmers, therefore, choose the easiest option of simply burning it.

182. Which of the following are the most rational inferences that can be made from the passage? (CSAT-2020)

1. Farmers' income will be higher with pulse cultivation than with paddy cultivation.

2. Pulse cultivation causes less pollution as compared to paddy cultivation.

3. Pulse straw can be used to improved soil quality.

4. In the context of northern Indian agriculture, paddy straw has no usefulness.

5. Mechanized agriculture is the main cause for stubble burning.

Select the correct answer using the code given below.

(a) 2, 3 and 5 (b) 1, 4 and 5

(c) 2 and 3 only (d) 1 and 4 only

Passage

In India, authorities always look to store the maximum amount of water in reservoirs during the monsoon season, which is then used for irrigation and generation of electricity during the summer months. It is an internationally accepted practice that the water level of a reservoir should be kept below a certain level before the onset of monsoon season. This is so that when monsoon rains come, there is space to store the excess rainwater and also so that water can be released in a regulated manner. But the authorities store the maximum amount of water in reservoirs even before the close of the monsoon, only to ensure greater electricity generation and irrigation.

183. With reference to the above passage, the following assumptions have been made (CSAT-2020)

1. High risks involved in holding maximum water in reservoirs are due to our over-dependence on hydropower projects.

2. Storage capacity of dams should not be fully used before or during monsoon season.

3. Role of dams in flood control is underestimated in India.

Which of the above assumptions is/are valid?

(a) 1 and 2 only (b) 2 only

(c) 3 only (d) 1, 2 and 3

Passage

Economic liberalization in India was shaped largely by the economic problems of the government than by the economic priorities of the people or by the long-term development objectives. Thus, there were limitations in conception and design which have been subsequently validated by experience. Jobless growth, persistent poverty and rising inequality have mounted as problems since economic liberalization began. And all these years later, four quite crises confront the economy, agriculture, infrastructure, industrialization and education as constraints on the country's future prospects. These problems must be resolved in economic growth has to be sustained and transformed into meaningful development.

184. Which of the following is/are the most rational and logical inference/inferences that can be made from the passage? (CSAT-2020)

1. It is essential to rethink and redefine the economic role of the State in the quest for development.

2. India has not made effective implementation of its policies in social sectors nor made sufficient investments in them.

Select the correct answer using the code given below.

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

185. With reference to the above passage, the following assumptions have been made: (CSAT-2020)

1. India's economy needs to be greatly integrated with global economy so as to create large number of jobs and to sustain its growth momentum.

2. Economic liberalization would cause large economic growth which would reduce poverty and create sufficient employment in the long run.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

India's economic footprint, given its population, still remains small compared to the US, the European Union or China. It has much to learn from other economies yet must implement solutions that fit its

unique circumstances. India especially needs an effective long term regulatory system based on collaboration rather than the current top-down approach. Regulations seek desirable outcomes yet are repeatedly used as political tools to push one agenda or another. Often, regulations fail to consider impacts on jobs and economic growth or less restrictive alternatives. Regulations may be used to protect local markets at the expense of more widely shared prosperity in the future. Additionally, regulations inevitably result in numerous unintended consequences. In today's hyper competitive global economy, regulations need to be viewed as

186. Which one of the following is the most logical, rational and crucial inference that can be derived from the above passage? (CSAT-2019)

(a) A better regulatory system will help India achieve the size of economy appropriate to its population

(b) In a competitive global economy, India must use regulations strategically.

(c) Regulations in India do not favour its integration with today's hyper competitive global economy.

(d) Job creation and economic growth should be dominant considerations in developing India's regulatory system.

187. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

In today's global economy,

1. regulations are not effectively used to protect local markets.
2. social and environmental concerns are generally ignored by the governments across the world while implementing the regulations.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both1and 2 (d) Neither 1 nor 2

Passage

In a study, scientists compared the microbiomes of poorly nourished and well-nourished infants and young children. Gut microbes were isolated from faecal samples of malnourished and healthy children. The microbiome was "immature" and less diverse in malnourished children compared to the better developed "mature" microbiome found in healthy children of the same age. According to some studies, the chemical composition of mother's milk has shown the presence of a modified sugar (sialylated oligosaccharides). This is not utilized by the baby for its own nutrition. However, the bacteria constituting the infant's microbiome thrive on this sugar which serves as their food. Malnourished mothers have low levels of this sugar in their milk. Consequently, the microbiomes of their infants fail to mature. That in turn, leads to malnourished babies.

188. Which one of the following is the most logical, rational, and crucial inference that can be derived from the above passage? (CSAT-2019)

(a) If malnourished condition in children is caused by gut bacteria, it cannot be treated

(b) The guts of malnourished babies should be inoculated with mature microbiomes

(c) Babies of malnourished mothers should be fed with dairy milk fortified with sialylated oligosaccharides instead of mother's milk

(d) Research on benign effects of gut bacteria on nutrition has policy implications.

189. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. Processed probiotic foods are a solution to treat the children suffering from malnutrition due to immature gut bacteria composition
2. The babies of malnourished mothers generally tend to be' malnourished.

Which of the above assumptions is/are valid?

(a) 1only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

Temperatures have risen nearly five times as rapidly on the Western Antarctic Peninsula than the global average over the past five decades. Researchers have now found that melting glaciers are causing a loss of species diversity among benthos in the coastal waters off the Antarctic Peninsula, impacting an entire seafloor ecosystem. They believe increased levels of suspended sediment in water to be the cause of the dwindling biodiversity in the coastal region.

190. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. Regions of glaciers warm faster than other regions due to global warming.
2. Global warming can lead to seafloor sedimentation in some areas.
3. Melting glaciers can reduce marine biodiversity in some areas.

Which of the above assumptions is/are valid?

(a) 1 and 2 only (b) 3 only

(c) 2 and 3 only (d) 1,2 and3

Passage

A research team examined a long-term owl roost. Owls prey on small mammals and the excreted remains of those meals that accumulated over the time, provide us an insight into the composition and structure of small mammals over the past 'millennia. The research suggested that when the Earth went through a period of rapid warming about 13,000 years ago, the small mammal community was stable and resilient. But, from the last quarter of the nineteenth century, human-made changes to the environment had caused an enormous drop in biomass and energy flow. This dramatic decline in energy flow means modern ecosystems are not adapting as easily as they did in the past.

191. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. Global warming is a frequently occurring natural phenomenon.
2. The impending global warming will not adversely affect small mammals.
3. Humans are responsible for the loss of the Earth's natural resilience.

Which of the above assumptions is/are valid?

(a) 1 and 2 only (b) 3 only

(c) 2 and 3 only (d) 1, 2 and 3

Passage

Food varieties extinction is happening all over the world - and it is happening fast. For example, of the 7,000 apple varieties that were grown during the nineteenth century, fewer than o hundred remain. In the Philippines, thousands of varieties of rice once thrived; now only up to a hundred are grown there. In China, 90 percent of the wheat varieties cultivated just a century ago have disappeared. Farmers in the past painstakingly bred and developed crops well suited to the peculiarities of their local climate and environment. In the recent past, our heavy dependence on a few high yielding varieties and technology-driven production and distribution of food is causing the dwindling of diversity in food crops. If some mutating crop disease or future climate change decimates the few crop plants, we have come to depend on to feed our growing population, we might desperately need some of those varieties we have let go extinct.

192. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. Humans have been the main reason for the largescale extinction of plant species.
2. Consumption of food mainly from locally cultivated crops ensures crop diversity.
3. The present style of production and distribution of food will finally lead to the problem of food scarcity in the near future.
4. Our food security may depend on our ability to preserve the locally cultivated varieties of crops. Which of the above assumptions are valid?

(a) 1 and 3 (b) 2 and 4

(c) 2 and 3 (d) 1 and 4

Passage

What stands in the way of the widespread and careful adoption of 'Genetic Modification (GM)' technology is an Intellectual Property Rights regime that seeks to create private monopolies for such technologies. If GM technology is largely corporate driven, it seeks to maximize profits and that too in the short run. That is why corporations make major investments for herbicide-tolerant and pest-resistant crops. Such properties have only a short window, as soon enough, pests and weeds will evolve to overcome such resistance. This suits the corporations. The National Farmers Commission pointed out that priority must be given in genetic modification to the incorporation of genes that can help impart resistance to drought, salinity and other stresses.

193. Which one of the following is the most logical, rational and crucial message conveyed by the above passage? (CSAT-2019)

(a) Public research institutions should take the lead in GM technology and prioritise the technology agenda.

(b) Developing countries should raise this issue in WTO and ensure the abolition of Intellectual Property Rights.

(c) Private corporations should not be allowed to do agribusiness in India, particularly the seed business.

(d) Present Indian circumstances do not favour the cultivation of genetically modified crops.

194. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. The issue of effects of natural calamities on agriculture is not given due consideration by GM technology companies.
2. In the long run, GM technology will not be able to solve agricultural problems arising due to global warming.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

Most invasive species are neither terribly successful nor very harmful. Britain's invasive plants are not widespread, not spreading especially quickly, and 'often less of a nuisance than vigorous natives such as bracken. The arrival of new species almost always increases biological diversity in region; in many cases, a flood of newcomers drives no native species to extinction. One reason is that invaders tend to colonise disturbed habitats like polluted lakes and post-industrial wasteland, where little else lives they are nature's opportunists.

195. Which one of the following is the most logical and rational inference that can be made from the above passage? (CSAT-2019)

(a) Invasive species should be used to rehabilitate desert areas and wastelands of a country.

(b) Laws against the introduction of foreign plants are unnecessary.

(c) Sometimes, the campaigns against foreign plants are pointless.

(d) Foreign plants should be used to increase the biodiversity of a country.

Passage

Diarrhoeal deaths among Indian children are mostly due to food and water contamination. Use of contaminated groundwater and unsafe chemicals in agriculture, poor hygiene in storage and handling of food items to food cooked and distributed in unhygienic surroundings; they are myriad factors that need regulation and monitoring. People need to have awareness of adulteration and ways of complaining to the relevant authorities. Surveillance of food-borne diseases involves a number of government agencies and entails good training of inspection staff. Considering the proportion of the urban population that depends on street food for its daily meals, investing in training and education of street vendors is of great significance.

196. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. Food safety is a complex issue that calls for a multipronged solution.
2. Great investments need to be made in developing the manpower for surveillance and training.
3. India needs to make sufficient legislation for governing food processing industry.

Which of the above assumptions is/are valid?

(a) 1 and 2 only (b) 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Passage

The interests of working and poor people have historically been neglected in the planning of our cities. Our cities are increasingly intolerant, unsafe and unliveable places for large numbers of citizens and yet we continue to plan via the old ways the static Development Plan—that draws exclusively from technical expertise, distanced from people's live experiences and needs, and actively excluding large number of people, places, activities and practices that are an integral part of the city.

197. The passage seems to argue (CSAT-2019)

(a) against the monopoly of builders and the interests of elite groups.

(b) against the need for global and smart cities.

(c) in favour of planning cities mainly for working class and poor people.

(d) in favour of participation of peoples' groups in city planning.

Passage

A vast majority of Indians are poor, with barely 10 percent employed in the organised sector. We are being convinced that vigorous economic growth is generating substantial employment. But this is not so. When our economy was growing at 3 percent per year, employment in the organised sector was growing at 2 percent per year. As the economy began to grow at 7-8 percent per year, the rate of growth of employment in the organised sector actually declined to 1 percent per year.

198. The above passage seems to imply that (CSAT-2019)

1. most of modern economic growth is based on technological progress.
2. much of modern Indian economy does not nurture sufficient symbiotic relationship with labourintensive, natural resource-based livelihoods.
3. service sector in India is not very labour-intensive.
4. literate rural population is not willing to enter organised sector.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 3 and 4 only

(c) 1, 2 and 3 only (d) 1, 2, 3 and 4

Passage

India has banking correspondents, who help bring people in the hinterland into the banking fold. For them to succeed, banks cannot crimp on costs. They also cannot afford to ignore investing in financial education and literacy. Banking correspondents are way too small to be viewed as a systemic risk. Yet India's banking regulator has restricted them to serving only one bank, perhaps to prevent arbitrage. Efforts at banking outreach may succeed only if there are better incentives at work for such last-mile workers and also those providers who ensure not just basic bank accounts but also products such as accident and life insurance and micro pension schemes.

199. Which one of the following is the most logical, rational and crucial inference that can be derived from the above passage? (CSAT-2019)

(a) Efforts to bring people in India's hinterland into the banking system are not successful,

(b) for meaningful financial inclusion, India's banking system needs more number of banking correspondents and other such last-mile workers.

(c) Meaningful financial inclusion in India requires that banking correspondents have diverse skills.

(d) Better banking outreach would be impossible unless each banking correspondent is allowed to serve a number of banks.

Passage

Political theorists no doubt have to take history of injustice, for example, untouchability, seriously. The concept of historical injustice takes note of a variety of historical wrongs that continue into the present in some form or the other and tend to resist repair. Two reasons might account for resistance to repair. One, not only are the roots of injustice buried deep in history, injustice itself constitutes economic structures of exploitation, ideologies of discrimination and modes of representation. Two, the category of historical injustice generally extends across a number of wrongs such as economic deprivation, social discrimination, and lack of recognition. This category is complex, not only because of the overlap between a number of wrongs, but because one or the other wrong, generally discrimination, tends to acquire partial autonomy from others. This is borne out by the history of repair in India.

200. What is the main idea that we can infer from the passage? A (CSAT-2019)

(a) Untouchability in India has not been taken seriously by political theorists.

(b) Historical injustice is inevitable in any society and is always beyond repair.

(c) Social discrimination and deprivation have their 100ts in bad economies.

(d) It is difficult, if not impossible, to repair every manifestation of historical injustice.

201. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. Removal of economic discrimination leads to removal of social

discrimination.

2. Democratic polity is the best way to repair historical Wrongs,

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2.

Passage

Education plays a great transformatory role in life, particularly s0 in this rapidly changing and globalizing world. Universities are the custodians of the intellectual capital and promoters of culture and specialized knowledge. Culture is an activity of thought, and receptiveness to beauty and human feelings. A merely well-informed man is only a bore on God's earth. What we should aim at is producing men who possess both culture and expert knowledge. Their expert knowledge will give them a firm ground to start from and their culture will lead them as deep as philosophy and as high as art. Together it will impart meaning to human existence.

202. 0n the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. A society without well educated people cannot be transformed into a modern society.
2. Without acquiring culture, a person's education is not complete.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1and 2 (d) Neither 1 nor 2.

Passage

Soil, in which nearly all our food grows, is a living resource that takes years to form. Yet it can vanish in minutes. Each year 75 billion tonnes of fertile soil is lost to erosion. That is alarming and not just for food producers. Soil can trap huge quantities of carbon dioxide in the form of organic carbon and prevent it from escaping into the atmosphere.

203. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. large scale soil erosion is a major reason for widespread food insecurity in the world.
2. Soil erosion is mainly anthropogenic.
3. Sustainable management of soils helps in combating climate change.

Which of the above assumptions is/are valid?

(a) 1 and 2 only (b) 3 only

(c) 2 and 3 only (d) 1, 2 and 3

Passage

Inequality is visible, even statistically measurable in many instances, but the economic power that drives it is invisible and not measurable... Like the force of gravity, power is the organising principle of inequality, be it of income, or wealth, gender, race, religion and region. Its effects are seen in a pervasive manner in all spheres, but the ways in which economic power pulls and tilts visible economic variables remain invisibly obscure.

204. On the basis of the above passage, the following assumptions have been made (CSAT-2019)

1. Economic power is the only reason for the existence of inequality in a society.
2. Inequality of different kinds, income, wealth, etc, reinforces power.
3. Economic power can be analysed more through its effects than by direct empirical methods.

Which of thy above assumptions is/are valid?

(a) 1 and 2 only (b) 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Passage

Climate change may actually benefit some plants by lengthening growing seasons and increasing carbon dioxide. Yet other effects of a warmer world, such as more pests, droughts, and flooding, will be less benign. How will the world adapt? Researchers project that by 2050, suitable croplands for four commodities - maize, potatoes, rice and wheat-will shift, in some cases pushing farmers to plant new crops. Some farmlands may benefit from warming, but others won't. Climate alone does not dictate yields; political shifts, global "demand, and agricultural practices will influence how farms fare in the future.

205. Which one of the following is the most logical and rational inference that can be made from the above passage? (CSAT-2019)

(a) Farmers who modernize their methods and diversify their fields will be in an advantageous position in future.

(b) Climate change will adversely affect crop diversity.

(c) Shifting crops to new croplands will lead to a great increase in the total area under cultivation and thus an increase in overall agricultural production

(d) Climate change is the most important factor affecting the agricultural economy in the future.

Passage

A bat's wings may look like sheets of skin. But underneath, a bat has the same five fingers as an orangutan or a human, as well as a wrist connected to the same cluster of wrist bones connected to the same long bones of the arm. What can be more curious than that the hand of a man, formed for grasping, that of a mole for digging, the leg of the horse, the paddle of the porpoise, and the wing of the bat, should all be constructed on the same pattern?

206. Which one of the following is the most logical, scientific and rational inference that can be made from the above passage? (CSAT-2019)

(a) Different species having similar structure of hands is an example of biodiversity.

(b) Limbs being used by different species for different kinds of work is an example of biodiversity.

(c) Man, and the aforementioned animals having similar structure of limbs is an example of coincidence in evolution.

(d) Man, and the aforementioned animals have a shared evolutionary history.

Passage

Around 56 million years ago, the Atlantic Ocean had not fully opened and animals, perhaps including our primate ancestors, could walk from Asia to North America through Europe and across Greenland. Earth was warmer than it is today, but as the Palaeocene epoch gave way to Eocene, it was about to get much warmer still -rapidly and radically. The cause was a massive geologically sudden release of carbon, During this period called Palaeocene - Eocene Thermal Maximum or PETM, the carbon injected into the atmosphere was roughly the amount that would be injected today if humans burned all the Earth's reserves of coal, oil and natural gas. The PETM lasted for about 1,50,000 years, until the excess carbon was reabsorbed. It brought on drought, floods, insect plagues and a few extinctions. Life on Earth survived-indeed, it prospered— but it was drastically different.

207. Based on the above passage the following assumptions have been made: (CSAT-2019)

1. Global warming has a bearing on the planet's biological evolution.
2. Separation of land masses causes the release of huge quantities of carbon into the atmosphere.
3. Increased warming of Earth's atmosphere can change the composition of its flora and fauna.
4. The present man-made global warming will finally lead to conditions similar to those which happened 56 million years ago.

Which of the assumptions given above are valid?

(a) 1 and 2 (b) 3 and 4

(c) 1 and 3 (d) 2 and 4

Passage

Low-end IoT (Internet of Things) devices are cheap commodity items: addressing security would add to the cost. This class of items is proliferating with new applications; many home appliances, thermostats, security and monitoring devices and personal convenience devices are part of the IoT. So are fitness trackers, certain medical implants and computer-like devices in automobiles. The IoT is expected to expand exponentially - but new security challenges are daunting.

208. Which one of the following statements is the most logical and rational inference that can be made from the above passage? (CSAT-2019)

(a) Development of enabling technologies in India can be a big boost to its manufacturing sector.

(b) India is not yet fully ready to adopt IoT in view of the imminent security challenges.

(c) life becomes more comfortable with the development of cheap low-end IoT devices.

(d) As we go digital, we must recognise the huge threat to Internet security from some IoT devices.

Passage

with the digital phenomenon restructuring most social sectors, itis little surprise that global trade negotiations are now eyeing the digital area | an attempt to pre-emptively colonise it. Big Data is freely collected or mined from developing countries, and converted into digital intelligence in developed countries. This intelligence begins to control different sectors and extract monopoly rents. A large foreign company providing cab service, for instance, is not a work of cars and drivers, it is digital intelligence about commuting, public transport, roads, traffic, city events, personal behavioural characteristics of commuters and driver and so on

209. Which one of the following is the most logical and rational corollary to the above passage? (CSAT-2019)

(a) Globalization is not in the interests of India as it undermines its socio-economic structures.

(b) India should be careful to protect its digital sovereignty in global trade talks.

(c) India should charge monopoly rents from multinational companies in exchange for Big Data.

(d) The loss of Big Data from India is proportional to the degree/ value of its foreign trade

210. Which of the following is most definitively implied by the above passage? (CSAT-2019)

(a) 7 Big Data is the key resource in the digital space.

(b) Big economies create Big Data

(c) Access to Big Data is the prerogative of developed countries.

(d) Access to and possession of Big Data is a characteristic of developed countries.

Passage

The rural poor across the world, including India, have contributed little to human-induced climate change, yet they are on the frontline in coping with its effects. Farmers can no longer rely on historical averages for rainfall and temperature, and the more frequent and extreme weather events, such as droughts and floods, can spell disaster. And there are new threats, such as sea level rise and the impact of melting glaciers on water supply. How significant are small farms? As many a$ two billion people worldwide depend on them for their food and livelihood. Small- holder farmers in India produce 41 percent of the country's food grains, and other food items that contribute to local and national food security.

211. What is the most logical and rational corollary to the above passage? (CSAT-2019)

(a) Supporting small farmers is an important part of any agenda regarding environmentally sustainable development

(b) Poor countries have little role to play in the mitigation of global warming.

(c) Due to a large number of farmer households, India will not have food security problem in the foreseeable future.

(d) Only small holder farmers in India can ensure food security.

212. The above passage implies that (CSAT-2019)

1. There is a potential problem of food insecurity in India.
2. India will have to strengthen its disaster management capabilities.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

A changing climate, and the eventual efforts of governments (however reluctant) to deal with it, could have a big impact on investors' returns. Companies that produce or use large amounts of fossil fuels will face higher taxes and regulatory burdens. Some energy producers may find it impossible to exploit their known reserves and be left with "stranded assets" deposits of oil and coal that have to be left in the ground. Other industries could be affected by the economic damage caused by more extreme weather-storms, floods, heat waves and droughts.

213. On the basis of the above passage, the following assumptions have been made: (CSAT-2019)

1. Governments and companies need to be adequately prepared to face the climate change.
2. Extreme weather events will reduce the economic growth of governments and company's future.
3. Ignoring climate change is a huge risk for investors

Which of the above assumptions is/are valid?

(a) 1 and 2 only (b) 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Passage

Access to schooling for those coming of school age is close to universal, but access to quality exhibits a sharp gradient with socio-economic status. Quotas for the weaker sections in private schools is a provision introduced by the Right of Children to Free and Compulsory Education Act, 2009. The quotas have imposed a debate on issues of social integration and equity in education that private actors had escaped by and large, The idea of egalitarian education system with equality of opportunity as its primary goal appears to be outside the space that private school principals inhabit. Therefore, the imposition of the quotas has led to resistance, sometimes justified.

214. With reference to the above passage, the following assumptions have been made: (CSAT-2019)

1. Making equality of opportunity a reality is the fundamental goal of the Indian education system.
2. The present Indian school system is unable to provide egalitarian education.
3. Abolition of private schools and establishment of more government schools is the only way to ensure egalitarian education.

Which of the above assumptions is/are valid?

(a) 1 and 2 only (b) 2 only

(c) 2 and 3 only (d) 3 only

Passage

A majority of the TB infected in India are poor and lack sufficient nutrition, suitable housing and have little understanding of prevention. TB then devastates families, makes the poor poorer, particularly affects women and children, and leads to ostracization and loss of employment. The truth is that even if TB does not kill them, hunger and poverty will. Another truth is that deep-seated stigma, lack of counselling, expensive treatment and lack of adequate support from providers and family, coupled with torturous side effects demotivate patients to continue treatment with disastrous health consequences.

215. Which one of the following is the most logical, rational, and crucial message conveyed by the above passage? (CSAT-2019)

(a) TB s not a curable disease in Indian circumstances.

(b) Curing TB requires more than diagnosis and medical treatment.

(c) Government's surveillance mechanism is deficient, and poor people have no access to treatment.

(d) India will be free from diseases like T8 only when its poverty alleviation programmes are effectively and successfully implemented.

Passage

Global population was around 1.6 billion in 1990—today it is around 7.2 billion and growing. Recent estimates on population growth predict a global population of 9.6 billion in 2050 and 10.9 billion in 2100. Unlike Europe and North America, where only three to four percent of population is engaged in agriculture, around 47 per cent of India's population is dependent upon agriculture. Even if India continues to do well in the service sector and the manufacturing sector picks up, it is expected that around 2030 when India overtakes China as the world's most populous country, nearly 42 percent of India's population will still be predominantly dependent on agriculture.

216. Which of the following is the most logical and rational inference that can be made from the above passage? (CSAT-2018)

(a) Prosperity of agriculture sector is of critical importance to India.

(b) Indian economy greatly depends on its agriculture.

(c) India should take strict measures to control its rapid population growth.

(d) India's farming communities should switch over to other occupations to improve their economic conditions

Passage

Many pathogens that cause foodborne illnesses are unknown. Food contamination can occur at any stage from farm to plate. Since most cases of food poisoning go unreported, the true extent of global foodborne illnesses is unknown.Improvements in international monitoring have led to greater public awareness, yet the rapid globalization of food

production increases consumers' vulnerability by making food harder to regulate and trace. "We have the world on our plates", says an official of WHO.

217. Which of the following is the most logical corollary to the above passage? (CSAT-2018)

(a) With more options for food come more risks.

(b) Food processing is the source of all foodborne illnesses.

(c) We should depend on locally produced food only.

(d) Globalization of food production should be curtailed.

Passage

I am a scientist, privileged to be somebody who tries to understand nature using the tools of science. But it is also clear that there are some really important questions that science cannot really answer, such as: Why is there something instead of nothing? Why are we here? In those domains, | have found that faith provides a better path to answers. I find it oddly anachronistic that in today's culture there seems to be a widespread presumption that scientific and spiritual views are incompatible.

218. Which of the following is the most logical and rational inference that can be made from the above passage? (CSAT-2018)

(a) It is the faith and not science that can finally solve all the problems of mankind.

(b) Science and faith can be mutually complementary if their proper domains are understood.

(c) There are some very fundamental questions which cannot be answered by either science or faith.

(d) In today's culture, scientific views are given more importance than spiritual views.

Passage

Though I have discarded much of past tradition and custom and am anxious that India should rid herself of all shackles that bind and contain her and divide her people, and suppress vast numbers of them, and prevent the free development of the body and the spirit; though I seek all this, yet I do not wish to cut myself off from that past completely. I am proud of that great inheritance that has been and is, ours and I am conscious that I too, like all of us, am a link in that unbroken chain which goes back to the dawn of history in the immemorial past of India.

219. The author wants India to rid herself of certain past bonds because (CSAT-2018)

(a) He is not able to see the relevance of the past

(b) There is not much to be proud of.

(c) He's not interested in the history of India.

(d) They obstruct her physical and spiritual growth

Passage

It is no longer, enough for us to talk about providing for universal access to education. Making available schooling facilities is an essential prerequisite but is insufficient to ensure that all children attend school and participate in the learning process. The school may be there, but children may not attend, or they may drop out after a few months. Through school and social mapping, we must address the entire gamut of social, economic, cultural and indeed linguistic and pedagogic issues, factors that prevent children from weaker sections and disadvantaged groups, as also girls, from regularly attending and complementing elementary education. The focus must be on the poorest and most vulnerable since these groups are the most disempowered and at the greatest risk of violation or denial of their right to education. The right to education goes beyond free and compulsory education to include quality education for al. Quality is an integral part of the right to education. If the education process lacks quality, children are being denied their right. The Right of Children to Free and Compulsory Education Act lays down that the curriculum should provide for learning through activities, exploration and discovery. This places an obligation on us to change our perception of children as passive receivers of knowledge, and to move beyond the convention of using textbooks as the basis of examinations. The teaching-learning process must become stress-free; and a massive programme for curricular reform should be initiated to provide for a child-friendly learning system that is more relevant and empowering. Teacher accountability systems and processes must ensure that children are learning. And that their right to learn in a child- friendly environment is not violated. Testing and redesigned to ensure that these do not force children to struggle between school and tuition centres, and bypass childhood.

220. According to the passage, which of the following is/are of paramount importance under the Right to Education? (CSAT-2018)

1. Sending of children to school by al parents
2. Provision of adequate physical infrastructure in schools
3. Curricular reforms for developing child-friendly learning system

Select the correct answer using the code given below.

(a) 1 only (b) 1 and 2 only

(c) 3 only (d) None of the above

221. With reference to the above passage, the following assumptions have been made: (CSAT-2018)

1. The Right to Education guarantees teachers accountability for the learning process of children
2. The Right to Education guarantees 100% enrolment of children in the schools.
3. The Right to Education intends to take full advantage of demographic dividend

Which of the above assumptions is/are valid?

(a) 1only (b) 2 and 3 only

(c) 3 only (d) 1,2 and 3

222. According to the passage, which one of the following is critical in bringing quality in education? (CSAT-2018)

(a) Ensuring regular attendance of children as well as teachers in school

(b) Giving pecuniary benefits to teachers to motivate them

(c) Understanding the socio-cultural background of children

(d) Inculcating learning through activities and discovery

223. What is the essential message in this passage? (CSAT-2018)

(a) The Right to Education now is a Fundamental Right

(b) The Right to Education enables the children of poor and weaker sections of the society to attend schools.

(c) The Right to Free and Compulsory Education should include quality education for all

(d) The Government as well as parents should ensure that all children attend schools.

Passage

'Desertification' is a term used to explain a process of decline in the biological productivity of an ecosystem, leading to total los of productivity. While this phenomenon is often linked to the arid, semi-arid and sub-humid ecosystems, even in the humid tropics, the impact could be most dramatic Impoverishment of human-impacted terrestrial ecosystems may exhibit itself in a variety of ways:accelerated erosion as in the mountain regions of the country, salinization of land as in the semi-arid and arid 'green revolution' areas of the country, e.g., Haryana and western Uttar Pradesh, and site quality decline—a common phenomenon due to general decline in tree cover and monotonous monoculture of rice/ wheat across the Indian plains. A major consequence of deforestation is that it relates to adverse alterations in the hydrology and related soil and nutrient losses. The consequences of deforestation invariably arise out of site degradation through erosive loses. Tropical Asia, Africa and South America have the highest levels of erosion. The already high rates for the tropics are increasing at an alarming rate (e.g., through the major river systems—Ganga and Brahmaputra, in the Indian context), due to deforestation and ill-suited land management practices subsequent to forest clearing. In the mountain context, the declining moisture retention of the mountain soils, drying up of the underground springs and smaller rivers in the Himalayan region could be attributed to drastic changes in the forest cover. An indirect consequence is drastic alteration in the upland-lowland interaction, mediated through water. The current concern the tea planter of Assam has is about the damage to tea plantations due to frequent inundation along the flood-plains of Brahmaputra, and the damage to tea plantation and the consequent loss in tea productivity is due to rising level of the river bottom because of siltation and the changing course of the river system The ultimate consequences of site desertification are soil degradation, alteration in available water and its quality, and the consequent decline in food, fodder and fuel-wood yields essential for the economic well-being of rural communities.

224. According to the passage, which of the following are the consequences of decline in forest cover? (CSAT-2018)

1. Loss of topsoil
2. Loss of smaller rivers
3. Adverse effect on agricultural production
4. Declining of groundwater

Select the correct answer using the code given below.

(a) 1, 2 and 3 only (b) 2, 3 and 4 only

(c) 1 and 4 only (d) 1, 2, 3 and 4

225. Which of the following is/are the correct inference/ inferences that can be made from the passage? (CSAT-2018)

1. Deforestation can cause changes in the course of rivers.
2. Salinization of land takes place due to human activities only.
3. Intense monoculture practice in plains is a major reason for desertification in Tropical Asia, Africa and South America. Select the correct answer using the code given below.

(a) 1 only

(b) 1 and 2 only

(c) 2 and 3 only

(d) None of the above is a correct inference.

226. With reference to 'desertification', as described in the passage, the following assumptions have been made: (CSE-2018)

1. Desertification is a phenomenon in tropical areas only.
2. Deforestation invariably leads to floods and desertification.

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

A diversity of natural assets will be needed to cope with climate change and ensure productive agriculture, forestry, and fisheries. For example, crop varieties are needed that. perform well under drought, heat, and enhanced CO2. But the private-sector and farmerled process of choosing crops favours homogeneity adapted to past or current conditions, not varieties capable of producing consistently high yields in warmer, wetter, or drier conditions. Accelerated breeding programmes are needed to conserve a wider pool of genetic resources of existing crops, breeds, and their wild relatives, Relatively intact ecosystems, such as forested catchments, mangroves, wetlands, can buffer the impacts of climate change. Under a changing climate, these ecosystems are themselves at risk, and management approaches will need to be more proactive and adaptive. Connections between natural areas, such as migration corridors, may be needed to facilitate species movements to keep up with the change in climate.

227. With reference to the above passage, which of the following would assist us in coping with the climate change? (CSAT-2018)

1. Conservation of natural water sources
2. Conservation of wider gene pool
3. Existing crop management practices
4. Migration corridors

Select the correct answer using the code given below.

(a) 1, 2 and 3 only (b) 1,2 and 4 only

(c) 3 and 4 only (d) 1, 2, 3 and 4

228. With reference to the above passage, the following assumptions have been made: (CSAT-2018)

1. Diversification of livelihoods acts as a coping strategy for climate change.
2. Adoption of mono cropping practice leads to the extinction of plant varieties and their wild relatives. Which of the above assumptions is/are valid?

Which of the above assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

Today, the top environmental challenge is a combination of people and their aspirations. If the aspirations are more like the frugal ones we had after the Second World War, a lot more is possible than if we view the planet as a giant shopping mall. We need to get beyond the fascination with glitter and understand that the planet works as a biological system.

229. Which of the following is the most crucial and logical inference that can be made from the above passage? (CSAT-2018)

(a) The Earth can meet only the basic needs of humans for food, clothing, and shelter.

(b) The only way to meet environmental challenge is to limit human population.

(c) Reducing our consumerism is very much in our own interest.

(d) Knowledge of biological systems can only help us save this planet.

Passage

Some people believe that leadership is quality which you have at birth or not at all. This theory is false, for the art of leadership can be acquired and can indeed be taught. This Discovery is made in time of war and the results achieved can surprise even the instructors. Faced with the alternatives of going left or right, every soldier soon grasps that a prompt decision either way is better than an endless discussion. A firm choice of direction has an even chance of being right while to do nothing will be almost certainly wrong.

230. The author of the passage holds the view that: (CSAT-2018)

(a) leadership can be taught through war experience only

(b) leadership can be acquired as well as taught

(c) the results of training show that more people acquire leadership than are expected

(d) despite rigorous instruction, very few leaders are produced.

Passage

All actions to address climate change ultimately involve costs. Funding is vital in order for countries like India to design and implement adaptation and mitigation plans and projects. The problem is more severe for developing countries like India, Which would be one of the hardest hit by climate change, given its need to finance development. Most countries do indeed treat climate change as real threat and are striving to address it in a more comprehensive and integrated manner with the limited resources at their disposal

231. With reference to the above passage, the following assumptions have been made (CSAT-2018)

1. Climate change is not a challenge for developed countries
2. Climate change is complex policy issue and also a development issue for many countries.
3. Ways and means of finance must be fount to enable developing countries to enhance their adaptive capacity.

Which of the above assumptions is/are valid?

(a) 1 and 2 only (b) 3 only

(c) 2 and 3 only (d) 1,2 and 3

Passage

Cooking with biomass and coal in India is now recognized to cause major health problems, with women and children in poor populations facing the greatest risk. There are more than 10 lakh premature deaths each year from household air pollution due to polluting cooking fuels with another 1.5 lakh due to their contribution to general air pollution in the country. Although the fraction of the Indian population using clean cooking fuels, such as LPG, natural gas and electricity, is slowly rising, the number using polluting solid fuels as their primary cooking fuel has remained static for nearly 30 years at about 70 crore.

232. Which of the following is the most crucial and logical inference that can be made from the above passage? (CSAT-2018)

(a) Rural people are giving up the use of polluting solid fuels due to their increasing awareness of health hazards.

(b) Subsidizing the use of clean cooking fuels will solve the problem of India's indoor air pollution.

(c) India should increase its import of natural gas and produce more electricity.

(d) Access to cooking gas can reduce premature deaths in poor households.

Passage

Scientific knowledge has its dangers, but so has every great thing. Over and beyond the dangers with which it threatens the present, it opens up as nothing else can, the vision of a possible happy world; a world without poverty, without war, with little illness. Science, whatever unpleasant consequences it may have by the way, is in its very nature a liberator.

233. Which one of the following is the most important implication of the passage? (CSAT-2018)

(a) A happy world is a dream of science

(b) Science only can build a happy world, but it is also the only major threat.

(c) A happy world is not possible without science

(d) A happy world is not at all possible with or without science.

Passage

The Arctic's vast reserves of fossil fuel, fish and minerals are now accessible for a longer period in a year. But unlike Antarctica, which is protected from exploitation by the Antarctic Treaty framed during the Cold War and is not subject to territorial claims by any country,there is no legal regime protecting the Arctic from industrialization, especially at a time when the world craves for more and more resources. The distinct possibility of ice-free summer has prompted countries with Arctic coastline to scramble for great chunks of the melting ocean.

234. Which one of the following is the most important implication of the passage? (CSAT-2018)

(a) India can have territorial claims in the Arctic territory and free access to its resources.

(b) Melting of summer ice in the Arctic leads to changes in the geopolitics.

(c) The Arctic region will solve the world's future problem of resource crunch.

(d) The Arctic region has more resources than Antarctica.

Passage

Being a member of the WTO, India is bound by the agreements that have been signed and ratified by its members, including itself. According to Article 6 of the Agriculture Agreement, providing minimum support prices for agricultural products is considered distorting and is subject to limits. The subsidy arising from 'minimal supports' cannot exceed 10 percent of the value of agricultural production for developing countries. PDS in India entails minimum support prices and public stockholding of food grains. It is possible that, in some years, the subsidy to producers will exceed 10 percent of the value of agricultural production.

235. What is the crucial message conveyed by the above passage? (CSAT-2018)

(a) India should revise its PDS.

(b) India should not be a member of WTO.

(c) For India, food security collides with trade.

(d) India provides food security to its poor.

Passage

India's educational system is modelled on the mass education system that developed in the 19th century in Europe and later spread around the world. The goal of the system is to condition children as 'good' citizens and productive workers. This suited the industrial age that needed the constant supply of a compliant workforce with a narrow set of capabilities. Our educational institutes resemble factories with bells, uniforms and batch processing of learners, designed to get learners to conform. But, from an economic point of view, the environment today is very different. It is a complex, volatile and globally interconnected world.

236. With reference to the above passage, the following assumptions have been made: (CSAT-2018)

1. India continues to be a developing country essentially due to its faulty education system.
2. Today's learners need to acquire new-age skill-sets.
3. A good number of Indians go to some developed countries for education because the educational systems there are a perfect reflection of the societies in which they function.

Which of the above assumptions is/are valid?

(a) 1 and 3 only (b) 2 only

(c) 2 and 3 only (d) 1, 2 and 3

Passage

The practice of dieting has become an epidemic; everyone is looking out for a way to attain that perfect body. We are all different with respect to our ethnicity, genetics, family history, gender, age, physical and mental and spiritual health status, lifestyles, and preferences. Thereby we also differ in what foods we tolerate or are sensitive to. So, we really cannot reduce so many complexities into one diet or diet book. This explains the failure of diets across the world in curbing obesity. Unless the reasons for weight gain are well understood and addressed and unless habits are changed permanently, no diet is likely to succeed.

237. What is the most logical and rational inference that can be made from the above passage? (CSAT-2018)

(a) Obesity has become an epidemic all over the world.

(b) A lot of people are obsessed with attaining a perfect body.

(c) Obesity is essentially an incurable disease.

(d) There is no perfect diet or one solution for obesity

Passage

Monoculture carries great risks. A single disease or pest can wipe out swathes of the world's food production, an alarming prospect given that it's growing, and wealthier population will eat 70% more by 2050. The risks are magnified by the changing climate. As the planet warms and monsoon rains intensify, farmlands in Asia will flood. North America will suffer more intense droughts, and crop diseases will spread to new latitudes.

238. Which of the following is the most logical, rational and crucial message given by the passage? (CSAT-2018)

(a) Preserving crop genetic diversity is an insurance against the effects of climate change.

(b) Despite great risks, monoculture is the only way to ensure food security in the world.

(c) More and more genetically modified crops only can save the world from impending shortages of food.

(d) Asia and North America will be the worst sufferers from climate change and the consequent shortage of food.

Passage

The quest for cheap and plentiful meat has resulted in farms where more and more animals are squeezed into smaller lots in cruel and shocking conditions. Such practices have resulted in many of the world's health pandemics such as the avian flu. Worldwide, livestock increasingly raised in cruel, cramped conditions, where animals spend their short live under artificial light, pumped full of antibiotics and growth hormones, until the day they are slaughtered. Meat production is water intensive. 15000 litres

of water is needed for every kilogram of meat compared with 3400 liters for rice, 3300 litres for eggs and 256 for a kilogram of potatoes.

239. What is the most rational and crucial message given by the passage? (CSAT-2018)

(a) Mass production of meat through industrial farming is cheap and is suitable for providing protein nutrition to poor countries.

(b) Meat-producing industry violates the laws against cruelty to animals.

(c) Mass production of meat through industrial farming is undesirable and should be stopped immediately.

(d) Environmental cost of meat production is unsustainable when it is produced through industrial farming.

Passage

A male tiger vas removed from Pench Tiger Reserve and was relocated in Parma National Park. Later, this tiger trekked toward his home 250 miles away. The trek of this solitary tiger highlights a crisis. Many wildlife reserves exist as islands of fragile habitat in a vast sea of humanity, yet tigers can range over a hundred miles, seeking prey, mates and territory. Nearly a third of India's tigers live outside tiger reserves, a situation that is dangerous for both human and animal. Prey and tigers can only disperse if there are recognized corridors of land between protected areas to allow unmolested passage.

240. Which of the following is the most rational and crucial message given by the passage? (CSAT-2018)

(a) The conflict between man and wildlife cannot be resolved, no matter what efforts we make.

(b) Safe wildlife corridors between protected areas is an essential aspect of conservation efforts.

(c) India needs to declare more protected areas and setup more tiger reserves.

(d) India's National Parks and Tiger Reserves need to be professionally managed.

241. With reference to the above passage, the following assumptions have been made: (CSAT-2018)

1. The strategy of conversation of wildlife by relocating them from one protected area to another is not often successful
2. India does not have suitable legislation to save the tigers, and

   its conservation efforts have failed which forced the tigers to live outside protected areas.

Which of the above assumptions is/are valid?

(a) 1only (b) 2 only

(c) Both 1 And 2 (d) Neither 1 nor 2

Passage

Disruption of traditional institutions, identifications and loyalties is likely to lead to ambivalent situations. It is possible that some people may renew their identification with traditional groups whereas others align themselves with new groups and Symbols emergent from processes of political development. In addition, political development tends to foster group awareness of a variety of class, tribe, region, clan, language, religion, occupation, and others.

242. Which one of the following is the best Explanation of the above Passage? (CSAT-2017)

(a) Political development is not a unilinear process for it involves both growth and decay.

(b) Traditional societies succeed in resisting positive aspects of political development.

(c) It is impossible for traditional societies to break away from lingering loyalties.

(d) Sustenance of traditional loyalties is conducive to political development.

Passage

There has been a significant trend worldwide towards regionalism in government, resulting in a widespread transfer of powers downwards towards regions and communities since 1990s. This process, which involves the creation of new political entities and bodies at a sub-national level and an increase in their content and powers, is known as devolution. Devolution has been characterized as being made up of three factors—political legitimacy, decentralization of authority and decentralization of resources. Political legitimacy here means a mass demand from below for the decentralization process, which is able to create a political force for it to take place. In many cases, decentralization is initiated by the upper tier of government without sufficient political mobilization for it at the grassroots level, and in such cases the decentralization process often does not fulfil its objectives.

243. Which among the following is the most logical, rational, and critical inference that can be made from the above Passage? (CSAT-2017)

(a) Emergence of powerful mass leaders is essential to create sub-national political entities and thus ensure successful devolution and decentralization

(b) The upper tier of government should impose devolution and decentralization on the regional communities by law or otherwise.

(c) Devolution, to be successful, requires a democracy in which there is free expression of the will of the people at lower level and their active participation at the grassroots level.

(d) For devolution to take place, a strong feeling of regionalism in the masses is essential.

Passage

We live in digital times. The digital is not just something we use strategically and specifically to do a few tasks. Our very perception of who we are, how we connect to the world around us, and the ways in which we define our domains of life, labour and language are hugely structured by the digital technologies. The digital is everywhere and like air, invisible. We live within digital systems, we live with intimate gadgets, we interact through digital media, and the very presence and

imagination of the digital has dramatically restructured our lives. The digital, far from being a tool, is a condition and context that defines the shapes and boundaries of our understanding of the self, the society, and the structure of governance.

244. Which among the following is the most logical and essential message conveyed by the above passage? (CSAT-2017)

(a) All problems of governance can be solved by using digital technologies.

(b) Speaking of digital technologies is speaking of our life and living.

(c) Our creativity and imagination cannot be expressed without digital media.

(d) Use of digital systems is imperative for the existence of mankind in future.

Passage

The IMF has pointed out that the fast-growing economies of Asia face the risk of falling into 'middle-income trap'. It means that average incomes in these countries, which till now have been growing rapidly, will stop growing beyond a point—a point that is well short of incomes in the developed West. The IMF identifies a number of causes of middle-income trap—none of which is surprising—from infrastructure to weak institutions, to less than favourable macroeconomic conditions. But the broad, overall cause, says IMF, is a collapse in the growth of productivity.

245. Which among the following is the most logical, rational, and critical inference that can be made from the above passage? (CSAT-2017)

(a) Once a country reaches middle-income stage, it runs the risk of falling productivity which leads to stagnant incomes.

(b) Falling into middle-income trap is a general characteristic of fast-growing economies.

c) There is no hope at all for emerging Asian economies to sustain the growth momentum.

(d) As regards growth of productivity, the performance of Asian economies is not satisfactory.

Passage

An innovative India will be inclusive as well as technologically advanced, improving the lives of all Indians. Innovation and R&D can mitigate increases in social inequality and relieve the pressures created by rapid urbanization. The growing divergence in productivity between agriculture and knowledge-intensive manufacturing and services threatens to increase income inequality. By encouraging India's R&D labs and universities to focus on the needs of poor people and by improving the ability of informal firms to absorb knowledge, an innovation and research agenda can counter this effect. Inclusive innovation can lower the costs of goods and services and create income - earning opportunities for the poor people.

246. Which among the following is the most logical and rational assumption that can be made from the above passage? (CSAT-2017)

(a) Innovation and R&D is the only way to reduce rural to urban migration.

(b) Every rapidly growing country needs to minimize the divergence between productivity in agriculture and other sectors.

(c) Inclusive innovation and R&D can help create an egalitarian society.

(d) Rapid urbanization takes place only when a country's economic growth is rapid.

Passage

Climate change is likely to expose a large number of people to increasing environmental risks forcing them to migrate. The international community is yet to recognize this new category of migrants. There is no consensus on the definition and status of climate refugees owing to the distinct meaning the term refugees carry under international laws. There are still gaps in understanding how climate change will work as the root cause of migration. Even if there is recognition of climate refugees, what is going to provide protection? More emphasis has been given to international migration due to climate change. But there isa need to recognize the migration of such people within the countries also so that their problems can be addressed properly.

247. Which of the following is the most rational inference from the above passage? (CSAT-2017)

(a) The world will not be able to cope with large scale migration of climate refugees.

(b) We must find the ways and means to stop further climate change.

(c) Climate change will be the most important reason for the migration of people in the future.

(d) Relation between climate change and migration is not yet properly understood.

Passage

Many farmers use synthetic pesticides to kill infesting insects. The consumption of pesticides in some of the developed countries is touching 3000 grams/hectare. Unfortunately, there are reports that these compounds possess inherent toxicities that endanger the health of the farm operators, consumers and the environment. Synthetic pesticides are generally persistent in the environment. Entering in the food chain they destroy the microbial diversity and cause ecological imbalance. Their indiscriminate use has resulted in development of resistance among insects to insecticides, upsetting of balance in nature and resurgence of treated populations. Natural pest control using the botanical pesticides is safer to the user and the environment because they break down into harmless compounds within hours or days in the presence of sunlight. Plants with pesticidal properties have been in nature for millions of years without any ill or adverse effects on the ecosystem. They are easily decomposed by many microbes common in most soil. They help in the maintenance of biological diversity, of predators and the reduction of environmental contamination and human health hazards. Botanical pesticides formulated from plants are biodegradable and their use in crop protection is a practical sustainable alternative.

248. On the basis of the above passage, the following assumptions have been made: (CSAT-2017)

1. Synthetic pesticides should never be used in modem agriculture.
2. One of the aims of sustainable agriculture is to ensure minimal ecological imbalance.
3. Botanical pesticides are more effective as compared to synthetic pesticides.

Which of the assumptions given above is/are correct?

(a) 1 and 2 only (b) 2 only

(c) 1 and 3 only (d) 1,2 and 3

249. Which of the following statements is/are correct regarding biopesticides? (CSAT-2017)

1. They are not hazardous to human health.
2. They are persistent in the environment.
3. They are essential to maintain the biodiversity of any ecosystem.

Select the correct answer using the code given below.

(a) 1 only (b) 1 and 2 only

(c) 1 and 3 only (d) 1, 2 and 3

Passage

An air quality index (AQI) is a way to combine measurements of multiple air pollutants into a single number or rating. This index is ideally kept constantly updated and available in different places. The AQI is most useful when lots of pollution data are being gathered and when pollution levels are normally, but not always, low. In such cases, if pollution levels spike for a few days, the public can quickly take preventive action (like staying indoors) in response to an air quality warning. Unfortunately, that is not urban India. Pollution levels in many large Indian cities are so high that they remain well above any health or regulatory standard for large part of the year. If our index stays in the Red/Dangerous' region day after day, there is not much any one can do, other than getting used to ignoring it.

250. Which among the following is the most logical and rational inference that can be made from the above passage? (CSAT-2017)

(a) Our governments are not responsible enough to keep our cities pollution free

(b) There is absolutely no need for air quality indices in our country.

(c) Air quality index is not helpful to the residents of many of our large cities.

(d) In every city, public awareness about pollution problems should increase.

Passage

Productive jobs are vital for growth and a good lob is the best form of inclusion. More than half of our population depends on agriculture, but the experience of other countries suggests that the number of people dependent on agriculture will have to shrink if per capita incomes in agriculture are to go up substantially. While industry is creating jobs, too many such jobs are low-productivity non-contractual jobs in the unorganized sector, offering low incomes, little protection, and no benefits. Service jobs are relatively of high productivity, but employment growth in services has been slow in recent years

251. Which among the following is the most logical and rational inference that can be made from the above passage? (CSAT-2017)

(a) We must create conditions for the faster growth of highly productive service jobs to ensure employment growth and inclusion.

(b) We must shift the farm workers to the highly productive manufacturing and service sectors to ensure the economic growth and inclusion.

(c) We must create conditions for the faster growth of productive jobs outside of agriculture even while improving the productivity of agriculture.

(d) We must emphasize the cultivation of high-yielding hybrid varieties and genetically modified crops to increase the per capita income in agriculture

Passage

A landscape-scale approach to land use can encourage greater biodiversity outside protected areas. During hurricane 'Mitch' in 1998, farms using eco agricultural practices suffered 58 percent, 70 percent and 99 percent less damage in Honduras, Nicaragua and Guatemala, respectively, than farms using conventional techniques. In Costa, vegetative windbreaks and fencerows boosted farmers' income from pasture and coffee while also increasing bird diversity. Bee pollination is more effective when agricultural fields are closer to natural or seminatural habitat, a finding that matters because 87 percent of the world's 107 leading crops depend on animal pollinators In Costa Rica, Nicaragua and Colombia silvopastoral systems that integrate trees with pastureland are improving the sustainability of cattle production, and diversifying and increasing farmers' income.

252. Which among the following is the most logical and rational inference that can be made from the above passage? (CSAT-2017)

(a) Agricultural practices that enhance biodiversity can often increase farm output and reduce the vulnerability to disasters.

(b) All the countries of the world should be encouraged to replace eco agriculture with conventional agriculture.

(c) Eco agriculture should be permitted in protected areas without destroying the biodiversity there.

(d) The yield of food crops will be very high if eco agricultural practices are adopted to cultivate them.

Passage

The medium-term challenge for Indian manufacturing is to move from lower to higher tech sectors, from lower to higher value-added sectors, and from lower to higher productivity sectors. Medium tech industries are primarily capital intensive and resource processing; and high tech industries are mainly capital and technology intensive. In order to

push the share of manufacturing in overall GDP to the projected 25 per cent, Indian manufacturing needs to capture the global market in sectors showing a rising trend in demand. These sectors are largely high technology and capital intensive.

253. Which among the following is the most logical and rational inference that can be made from the above passage? (CSAT-2017)

(a) India's GDP displays high value-added and high productivity levels in medium tech and resource processing industries.

(b) Promotion of capital and technology intensive manufacturing is not possible in India

(c) India should push up the public investments and encourage the private investments in research and development, technology upgradation and skill development.

(d) India has already gained a great share in global markets in sectors showing a rising trend in demand.

Passage

Over the last decade, Indian agriculture has become more robust with record production of food grains and oilseeds. Increased procurement, consequently, has added huge of food grains in the granaries. India is one of the world's top producers of rice, wheat, milk, fruits and vegetables. India is still home the quarter of all undernourished people in the world. On an average, almost half of the total expenditure of nearly half of the households is on food.

254. Which among the following is the most logical corollary to the above passage? (CSAT-2017)

(a) Increasing the efficiency of farm to-fork value chain is necessary to reduce the poverty and malnutrition.

(b) Increasing the agricultural productivity will automatically eliminate the poverty and malnutrition in India.

(c) India's agricultural productivity is already great and it is not necessary to increase it further.

(d) Allocation of more funds for social welfare and poverty alleviation programmes will ultimately eliminate the poverty and malnutrition in India.

Passage

The States are like pearls and the Centre is the thread, which turns them into a necklace; if the read snaps, the pearls are scattered.

255. Which one of the following views corroborates the above statement? (CSAT-2017)

(a) A strong Centre and strong States make the federation strong

(b) A strong Centre is a binding force for national integrity.

(c) A strong Centre is a hindrance to State autonomy.

(d) State autonomy is a prerequisite for a federation.

Passage

Really, I think that the poorest he that is in England has a life to live, as the greatest he, and therefore truly, I think it is clear that every man that is to live under a government ought first by his own consent to put himself under the government, and I do think that the poorest man in England is not at all bound in a strict sense to that government that he has not had a voice to put himself under.

256. The above statement argues for (CSAT-2017)

(a) distribution of wealth equally to all.

(b) rule according to the consent of the governed.

(c) rule of the poor.

(d) expropriation of the rich.

Passage

We have hard work ahead. There is no resting for any of us till we redeem our pledge in full fill we make all the people of India what destiny intends them to be. We are citizens of a great country, on the verge of bold advance, and we have to live up to that high standard. All of us, to whatever religion we may belong, are equally the children of India with, equal rights, privileges and obligations. We cannot encourage communalism or narrow mindedness, for no nation can be great whose people are narrow in thought or action.

257. The challenge the author of the above passage throws to the public is to achieve (CSAT-2017)

(a) a high standard of living, progress and privileges

(b) equal privileges, fulfilment of destiny and political tolerance.

(c) spirit of adventure and economic parity.

(d) hard work, brotherhood out national unity.

Passage

"The individual, according to Rousseau, puts his person and all his power in common under the supreme direction of the General Will and in our corporate capacity we receive each member as an indivisible part of the whole"

258. In the light of the above passage, the nature of General Will is best described as (CSAT-2017)

(a) the sum total of the private wills of the individuals.

(b) what is articulated by the elected representatives of the individuals.

(c) the collective good as distinct from private wills of the individuals.

(d) the material interests of the community.

Passage

In a democratic State, where a high degree of Political maturity of the people obtains, the conflict between the will of the sovereign law-making body and the organized will of the people seldom occurs.

259. What does the above passage imply? (CSAT-2017)

(a) In a democracy, force is the main phenomenon in the actual exercise of sovereignty.

(b) In a mature democracy, force to a great extent is the main phenomenon in the actual exercise of sovereignty.

(c) In a mature democracy, use of force is irrelevant in the actual

exercise of sovereignty.

(d) In a mature democracy, force is narrowed down to a marginal phenomenon in the actual exercise of sovereignty.

Passage

A successful democracy depends upon widespread interest and participation in politics, in which voting is an essential part. To deliberately refrain from taking such an interest, and from voting, is a kind of implied anarchy, it is to refuse one's political responsibility while enjoying the benefits of a free political society.

260. This passage relates to (CSAT-2017)

(a) duty to vote.

(b) right to vote.

(c) freedom to vote.

(d) right to participate in politics.

Passage

In a free country, the man who reaches the position of leader is usually one of outstanding, character and ability. Moreover, it is usually possible to foresee that he will reach such a position, since early in life one can see his qualities of character. But this is not always true in the case of a dictator; often he reaches his position of power through chance, very often through the unhappy state of his country.

261. The passage seems to suggest that (CSAT-2017)

(a) a leader foresees his future position.

(b) a leader is chosen only by a free country.

(c) a leader must see that his country is free from despair.

(d) despair in a country sometimes leads to dictatorship.

Passage

The greatest blessing that technological progress has in store for mankind is not, of course, an accumulation of material possessions. The amount of these that can be effectively enjoyed by one individual in one lifetime is not great. But there is not the same narrow limit to the possibilities of the enjoyment of leisure. The gift of leisure may be abused by people who have had no experience of making use of it. Yet the creative use of leisure by a minority in societies has been the mainspring of all human progress beyond the primitive level.

262. With reference to the above passage, the following assumptions have been made: (CSAT-2017)

1. People always see the leisure time as a gift and use it for acquiring more material possessions.
2. Use of leisure by some people to produce new and original things has been the chief source of human progress.

Which of these assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1and2 (d) Neither 1 nor 2

Passage

There is more than a modicum of truth in the assertion that "a working knowledge of ancient history is necessary to the intelligent interpretation of current events". But the sage who uttered these words of wisdom might well have added something on the benefits of studying particularly the famous battles of history for the lessons they contain for those of us who lead or aspire to leadership. Such a study will reveal certain qualities and attributes which enabled the winners to win—and certain deficiencies which caused the losers to lose, and the student will see that the same pattern recurs consistently, again and again, throughout the centuries.

263. With reference to the above passage, the following assumptions have been made: (CSAT-2017)

1. A study of the famous battles its history would help us understand the modern warfare.
2. Studying the history is essential for anyone who aspires to be a leader.

Which of these assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1and 2 (d) Neither 1 nor 2

Passage

What climate change will undeniably do is cause or amplify events that hasten the reduction of resources. Competition over these diminishing resources would ensue in the form of political or even violent conflict. Resource based conflicts have rarely been overt and are thus difficult to isolate. Instead, they take on veneers that appear more politically palatable. Conflicts over resources like water are often cloaked in the guise of identity or ideology.

264. What does the above passage imply? (CSAT-2017)

(a) Resource-based conflicts are always politically motivated.

(b) There are no political solutions to resolve environmental and resource-based conflicts.

(c) Environmental issues contribute to resource stresses and political conflict.

(d) Political conflict based on identity or ideology cannot be resolved.

Passage

The man who is perpetually hesitating which of the two things he will do first, will do neither. The man who resolves, but suffers his resolution to be changed by the first counter Suggestion of a friend—who fluctuates from opinion to opinion and veers from plan to plan-can never accomplish anything. He will at best be stationary and probably retrograde in all. It is only the man who first consults wisely, then resolves firmly and then executes his purpose with inflexible perseverance, undismayed by those petty difficulties which daunt a weaker spirit—that can advance to eminence in any line.

265. The keynote that seems to be emerging from the passage is that (CSAT-2017)

(a) we should first consult wisely and then resolve firmly.

(b) we should reject suggestions of friends and remain unchanged.

(c) we should always remain broad-minded

(d) we should be resolute and achievement-oriented

Passage

During the summer in the Arctic Ocean, sea ice has been melting earlier and faster, and the winter freeze has been coming later. In the last three decades, the extent of summer ice has declined by about 30 per cent. The lengthening period of summer melt threatens to undermine the whole Arctic food web, atop which stand polar bears.

266. Which among the following is the most crucial message conveyed by the above passage? (CSAT-2017)

(a) Climate change has caused Arctic summer to be short but temperature to be high

(b) Polar bears can be shifted to South Pole to ensure their survival.

(c) Without the presence of polar bears, the food chains in Arctic region will disappear.

(d) Climate change poses a threat to the survival of polar bears.

Passage

Why do people prefer open defecation and not want toilets or, if they have them, only use them sometimes? Recent research has shown two critical elements: ideas of purity and pollutions, and not wanting pits or septic tanks to fill they have to be emptied. These are the issue that nobody wants to talk about, but if we want to eradicate the practice of open defection, they have to be confronted and dealt properly.

267. Which among the following is the most crucial message conveyed by the above passage? (CSAT-2017)

(a) The ideas of purity and pollutions are so deep-rooted that they cannot be removed from the minds of the people.

(b) People have to perceive toilet use and pit-emptying as clean and not polluting.

(c) People cannot change their old habits

(d) People have neither civic series nor a sense of privacy.

Passage

In the last two decades, the world's gross domestic product (GDP) has increased 50 percent, whereas inclusive wealth has increased by a mere 6 percent. In recent decades, GDP-driven economic performance, has only harmed inclusive wealth like human capital, and natural capital like forests, land and water. While the world's human capital which stands at 57 percent of total inclusive wealth grew by only 8 percent, the natural which is 23 percent of total inclusive wealth declined by 30 per cent worldwide in the last two decades.

268. Which of the following is the most crucial inference from the above passage? (CSAT-2017)

(a) More emphasis should be laid on the development of natural capital

(b) The growth driven by GDP only is neither desirable nor sustainable

(c) The economic performance of the countries of the world is not satisfactory.

(d) The world needs more human capital under the present circumstances.

Passage

By 2020, when the global economy is expected to run short of 56 million young people, India, with its youth surplus of 47 million, could fill the gap. It is in this context that labour reforms are often cited as the way to unlock double-digit growth in India. In 2014, India's labour force was estimated to be about 40 per cent of the population, but 93 per cent of this force was in unorganized sector. Over the last decade, the compound annual growth rate (CAGR) of employment has slowed to 0.5 per cent, with about 14 million jobs created during last year when the labour force increased by about 15 million.

269. Which of the following is most rational inference from the above passage? (CSAT-2017)

(a) India must control its population growth so as to reduce its unemployment rate

(b) Labour reforms are required in India to make optimum use of its vast labour force productively.

(c) India is poised to achieve the double-digit growth very soon.

(d) India is capable of supplying skilled young people to oil, countries.

Passage

The very first lesson that should be taught to us when are old enough to understand it, is that complete freedom from the obligation to work is unnatural, and ought to be illegal, as we can escape our share of the burden of work only by throwing it on someone else's shoulders. Nature ordains that the human race shall perish of famine if it stops working. We cannot escape from this tyranny. The question we have to settle is how much leisure we can afford to allow ourselves

270. The main idea of the passage is that (CSAT-2017)

(a) it is essential for human beings to work.

(b) there should be a balance between work and leisure.

(c) working is a tyranny which we to face.

(d) human's understanding of the nature of work is essential

Passage

There is no harm in cultivating habits so long as they are not injurious. Indeed, most of us are little more than bundle of habits. Take away our habits and the residuum would hardly be worth bothering about. We could not get on without them. They simplify the mechanism of life. They enable us to do a multitude of things automatically, which, if we had to give fresh and original thought to them each time, would make existence an impossible confusion.

271. The author suggests that habits (CSAT-2017)

(a) tend to make our lives difficult.

(b) add precision to our lives

(c) make it easier for us to live.

(d) tend to mechanize our lives.

272. Rights are certain advantageous conditions of social less well-being indispensable to the true development of the citizen." In the light of this statement, which one of the following is the correct understanding of rights? (CSAT-2017)

(a) Rights aim at individual good only.

(b) Rights aim at social good only.

(c) Rights aim at both individual and social good.

(d) Rights aim at individual good devoid of social well- being.

Passage

By killing transparency and competition, crony capitalism is harmful to free enterprise, opportunity, and economic growth. Crony capitalism, where rich and the influential are alleged to have received land and natural resources and various licenses in return for payoffs to venal politicians, is now a major issue to be tackled. One of the greatest dangers to growth of developing economies like India is the middle income where crony capitalism creates oligarchies that slow down the growth.

273. Which among the following is the most logical corollary to the above passage? (CSAT-2016)

(a) Launching more welfare schemes and allocating more finances for the current schemes are urgently needed.

(b) Efforts should be made to push up economic growth by other means and provide licenses to the poor.

(c) Greater transparency in the functioning of the government and promoting the financial inclusion are needed at present.

(d) We should concentrate more on developing manufacturing sector than service sector.

Passage

Climate adaptation may be rendered ineffective if policies are not designed in the context of other development concerns. For instance, a comprehensive strategy that seeks to improve food security in the context of climate change may include a set of coordinated measures related to agricultural extension, crop diversification, integrated water and pest management and agricultural information series. Some of these measures may have to do with climate changes and others with economic development.

274. What is the most logical and rational inference that can be made from the above passage? (CSAT-2016)

(a) It is difficult to pursue climate adaptation in the developing countries.

(b) Improving food security is a far more complex issue than climate adaptation.

(c) Every developmental activity is directly or indirectly linked to climate adaptation.

(d) Climate adaptation should be examined in tandem with other economic development options.

Passage

Understanding of the role of biodiversity in the hydrological cycle enables better policymaking. The term biodiversity refers to the variety of plants, animals, microorganisms, and the ecosystems in which they occur. Water and biodiversity are interdependent. In reality, the hydrological cycle decides how biodiversity functions. In turn, vegetation and soil drive the movement of water. Every glass of water we drink has, at least in part, passed through fish, trees, bacteria, soil and other organisms. Passing through these ecosystems, it is cleansed and made fit for consumption. The supply of water is a critical service that the environment provides.

275. Which among the following is the most critical inference that can be made from the above passage? (CSAT-2016)

(a) Biodiversity sustains the ability of nature to recycle water.

(b) We cannot get potable water without the existence of living organisms.

(c) Plants, animals and microorganisms continuously interact among themselves.

(d) Living organisms could not have come into existence without hydrological cycle.

Passage

In the last decade, the banking sector has been restructured with a high degree of automation and products that mainly serve middle-class and upper middle-class society. Today there is need for a new agenda for the banking and non-banking financial services that does not exclude the common man.

276. Which one of the following is the message that is essentially implied in the above passage? (CSAT-2016)

(a) Need for more automation and more products of bank.

(b) Need for a radical restructuring of our entire public finance system.

(c) Need to integrate banking and non-banking institutions.

(d) Need to promote financial inclusion.

Passage

Safe and sustainable sanitation in slums has immeasurable benefits to women and girls in terms of their health, safety, privacy and dignity. However, women do not feature in most of the schemes and policies on urban sanitation. The fact that even now the manual scavenging exists, ones to show that not enough has been done to promote pour-flush toilets and discontinue the use of dry latrines. A more sustained and rigorous campaign needs to be launched towards the right to sanitation on a very large scale. This should primarily focus on the abolition of manual scavenging.

277. With reference to the above passage, consider the following statements: (CSAT-2016)

1. Urban sanitation problems can be fully solved by the abolition of manual scavenging only.
2. There is a need to promote greater awareness on safe sanitation practices in urban areas.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

To understand the nature and quantity of Government proper for man, it is necessary to attend to his character. As nature created him for social life, she fitted him for the station she intended. In all cases she made his natural wants greater than his individual powers. No one man is capable, without the aid of society, of supplying his own wants; and those wants, acting upon every individual, impel the whole of them into society.

278. Which among the following is the most logical and rational inference that can be made from the above passage? (CSAT-2016)

(a) Nature has created a great diversity in human society.

(b) Any given human society is always short of its wants.

(c) Social life is a specific characteristic of man.

(d) Diverse natural wants forced man towards social system.

Passage

The nature of the legal imperatives in any given state corresponds to the effective demands that state encounters, and that these, in their turn, depend, in a general way, upon the manner in which economic power is distributed in the society which the state controls.

279. The statement refers to: (CSAT-2016)

(a) the antithesis of Politics and Economics.

(b) the interrelationship of Politics and Economics.

(c) the predominance of Economics over Politics.

(d) the predominance of Politics over Economics.

Passage

About 15 percent of global greenhouse gas emissions come from agricultural practices. This includes nitrous oxide fertilizers; methane from livestock, rice production, and manure storage; and carbon dioxide (CO_2) from burning biomass, but this excludes CO_2 emissions from soil management practices, savannah burning and deforestation. Forestry and land-use change account for another percent of greenhouse gas emissions each ear, three quarters of which come from tropical deforestation. The remainder is largely from draining and burning tropical peatland. About the same amount of carbon is stored in the world's peatlands as is stored in the Amazon rainforest.

280. Which among the following is the most logical and rational inference that can be made from the above passage? (CSAT-2016)

(a) Organic farming should immediately replace mechanised and chemical dependant agricultural practices all over the world.

(b) It is imperative for us to modify our land use practices in order to mitigate climate change.

(c) There are no technological solutions to the problem of greenhouse gas emissions.

(d) Tropical areas are the chief sites of carbon sequestration.

Passage

As we look to 2050, when we will need to feed two billion more people, the question of which diet is best has taken on new urgency. The foods we choose to eat in the coming decades will have dramatic ramifications for the planet. Simply put, a diet that revolves around meat and dairy a way of eating that is on the rise throughout the developing. world, will take a greater toll on the world's resources than one that revolves around unrefined grains, nuts, fruits and vegetables.

281. What is the critical message conveyed by the above passage? (CSAT-2016)

(a) Our increasing demand for foods sourced from animals puts a greater burden on our natural resources.

(b) Diets based on grains, nuts, fruits and vegetables are best suited for health in developing countries

(c) Human beings change their food habits from time to time irrespective of the health concerns

(d) From a global perspective, we still do not know which type of diet is best for us.

Passage

All humans digest mother's milk as infants, but until cattle began being domesticated 10,000 years ago, children once weaned no longer needed to digest milk. As a result, they stopped making the enzyme lactase, which breaks down the sugar lactose into simple sugars. After humans began herding cattle, it became tremendously advantageous to digest milk, and lactose tolerance evolved independently among cattle herders in Europe, the middle East and Africa. Groups not dependant on cattle, such as the Chinese and Thai, remain lactose intolerant.

282. Which among the following is the most logical assumption that can be made from the above passage? (CSAT-2016)

(a) About 10,000 years ago, the domestication of animals took place in some parts of the world

(b) A permanent change in the food habits of a community can bring about a genetic change in its members

(c) Lactose tolerant people only are capable of getting simple sugars in their bodies

(d) People who are not lactose tolerant cannot digest any dairy product.

Passage

"The conceptual difficulties in National Income comparisons between underdeveloped and industrialized countries are particularly serious because a part of the national output in various underdeveloped countries is produced without passing through the commercial channels."

283. In the above statement, the author implies that: (CSAT-2016)

(a) The entire national output produced and consumed in industrialized countries passes through commercial channels.

(b) The existence of a non-commercialized sector in different underdeveloped countries renders the national income comparisons over countries difficult.

(c) No part of national output should be produced and consumed without passing through commercial channels.

(d) A part of the national output being produced and consumed without passing through commercial channels is a sign of

underdevelopment.

Passage

An increase in human-made carbon dioxide in the atmosphere could initiate a chain reaction between plant and microorganisms that would unsettle one of the largest carbon reservoirs on the planet soil In a study, it was found that the soil, which contains twice the amount of carbon present in a plants and Earth's atmosphere combined, could become increasingly volatile people add more carbon dioxide to the atmosphere. This is largely because of increased plant growth. Although a greenhouse gas and a pollutant, carbon dioxide also supports plant growth. As trees and other vegetation flourish in a carbon dioxide-rich future, their roots could stimulate microbial activity in soil that may in turn accelerate the decomposition of soil carbon and its release into the atmosphere as carbon dioxide.

284. Which among the following is the most logical corollary to the above passage? (CSAT-2016)

(a) Carbon dioxide is essential for the survival of microorganisms and plants

(b) Humans are solely responsible for the release of carbon dioxide into the atmosphere

(c) Microorganisms and soil carbon are responsible for the increased plant growth

(d) Increasing green cover could trigger the release of carbon trapped in soil.

Passage

Historically, the biggest Challenge to world agriculture has been to achieve a balance between demand for and supply of food. At the level of individual countries, the demand supply balance can be a critical issue for a closed economy, especially if it is a populous economy and its domestic agriculture is not growing sufficiently enough to ensure food supplies, on an enduring basis; it is not so much and not always, of a constraint for an open, and growing economy, which has adequate exchange surplus to buy food abroad. For the world as a whole, Supply-demand balance is always an inescapable prerequisite for warding off hunger and starvation. However, global availability of adequate supply does not necessarily mean that food would automatically move from countries of surplus to of deficit if the latter lack in purchasing power. The uneven distribution of hunger, starvation, under or malnourishment, etc., at the world-level, thus owes itself to the presence of empty-pocket hungry mouths, overwhelmingly confined to the underdeveloped economies. Inasmuch as 'a two-square meal' is of elemental significance to basic human existence, the issue of worldwide supply' of food has been gaining significance, in recent times, both because the quantum and the composition of demand has been undergoing big changes, and because, in recent years, the capabilities of individual countries to generate uninterrupted chain of food supplies have come under strain. Food production, marketing and prices, especially price affordability by the poor in the developing world, have become global issues that need global thinking and global solutions.

285. According to the above passage, which of the following are the fundamental solutions for the world food security problem? (CSAT-2016)

1. Setting up more agro-based industries.
2. Improving the price affordability by the poor.
3. Regulating the conditions of marketing.
4. Providing food subsidy to one and all.

Select the correct answer using the code given below:

(a) 1 and 2 (c) 1, 3 and 4 only

(b) 2 and 3 only (d) 1, 2, and 4

286. According to the above passage, the biggest challenge to world agriculture is: (CSAT-2016)

(a) to find sufficient land for agriculture and to expand food processing industries.

(b) to eradicate hunger in underdeveloped countries.

(c) to achieve a balance between the production of food and non-food items.

(d) to achieve a balance between demand for and supply of food.

287. According to the above passage, which of the following helps/help in reducing hunger and starvation in the developing economies? (CSAT-2016)

1. Balancing demand and supply of food.
2. Increasing imports of food.
3. increasing purchasing power of the poor.
4. Changing the food consumption patterns and practices.

Select the correct answer using the code given below:

(a) 1 only (c) 1 and 3 only

(b) 2, 3 and 4 only (d) 1, 2, 3 and 4

288. The issue of worldwide supply of food has gained importance mainly because of:

1. overgrowth of the population worldwide.
2. sharp decline in the area of food production
3. limitation in the capabilities for sustained supply of food.

Select the correct answer using the code given below: (CSAT-2016)

(a) 1 and 2 only (c) 2 and 3 only

(b) 3 only (d) 1, 2 and 3

Passage

Accountability, or the lack of it, in governance generally, and civil services, in particular, is a major factor underlying the deficiencies in governance and public administration. Designing an effective framework for accountability has been a key element of the reform agenda. A fundamental issue is whether civil services should be accountable to the political executive of the day or to society at large. In other words, how should internal and external accountability be reconciled? Internal accountability is sought to be achieved by internal performance monitoring, official supervision by bodies like the * -Central-Vigilance Commission-and-Comptroller and Auditor—

General, and judicial review of executive decisions. Articles 311 and 312 of the Indian Constitution provide job security and safeguards to the civil services, especially the All-India Services. The framers of the Constitution had envisaged that provision of these safeguards would result in a civil service that is not totally subservient to the political executive but will have the strength to function in larger public interest. The need to balance internal and external accountability is thus built into the Constitution. The issue is where to draw the line. Over the years, the emphasis seems to have tilted in favour of greater internal accountability of the civil services to the political leaders of the day who in turn are expected to be externally accountable to the society at large through the election process. This system for seeking accountability to Society has not worked out and has led to several adverse consequences for governance. Some special measures can be considered for improving accountability in civil services. Provisions of articles 311 and 312 should be reviewed and laws and regulations framed to ensure external accountability of civil services. The proposed Civil Services Bill seeks to address some of these requirements. The respective roles of professional civil services and the political executive should he defined so that professional managerial functions and management of civil services are depoliticized. For this purpose, effective statutory civil service boards should be created at the centre and in the states, Decentralization and devolution of authority to bring government and decision making closer to the people also helps to enhance accountability.

289. According to the passage, which of the following factor/ factors led to the adverse consequences for governance/public administration? (CSAT-2016)

1. Inability of civil services to strike a balance between internal and external accountabilities.
2. Lack of sufficient professional training to the officers of All India Services.
3. Lack of proper service benefits in civil services. lack of Constitutional provisions to define the respective roles of professional civil services vis-a-vis political executive in this context.

Select the correct answer using the code given below:

(a) 1 only (c) 1 and 4 only

(b) 2 and 3 only (d) 2, 3 and 4

290. With reference to the passage, the following assumptions have been made:

1. Political executive is an obstacle to the accountability of the civil services to the society.
2. In the present framework of Indian polity, the political executive is no longer accountable to the society.

Which of these assumptions is/are valid? (CSAT-2016)

(a) 1 only (c) Both 1 land 2

(b) 2 only (d) Neither 1 nor 2

291. Which one of the following is the essential message implied by this passage? (CSAT-2016)

(a) Civil services are not accountable to the society they are serving.

(b) Educated and enlightened persons are not taking up political leadership.

(c) The framers of the Constitution did not envisage the problems being encountered. by the civil services.

(d) There is a need and scope for reforms to improve the accountability of civil services.

292. According to the passage, which one of the following is not a means of enhancing internal accountability of civil services? (CSAT-2016)

(a) Better job security and safeguards.

(b) Supervision by Central Vigilance Commission.

(c) Judicial review of executive decisions.

(d) Seeking accountability ~ through enhanced participation by people in decision. making process.

Passage

In general, religious traditions stress our duty to God, or to some universal ethical principle. Our duties to one another derive from these. The religious concept of rights is primarily derived from our relationship to this divinity or principle and the implication it has on our other relationships. This correspondence between rights and duties is critical to any further understanding of justice. But, for justice to be practiced; rights and duties cannot remain formal abstraction. They must be grounded in a community (common unity) bound together by a sense of common union (communion). Even as a personal virtue, this solidarity is essential to the practice and understanding of justice.

293. With reference to the passage, the following assumptions have been made:

1. Human relationships are derived from their religious traditions.
2. Human beings can be duty bound only if they believe in God.
3. Religious traditions are essential to practice and understand justice.

Which of these assumption(s) is/are valid? (CSAT-2016)

(a) 1 only (c) 1 and 3 only

(b) 2and 3 only (d) 1, 2 and 3

294. Which one of the following is the crux of this passage? (CSAT-2016)

(a) Our duties to one another derive from our religious traditions.

(b) Having relationship to the divine principle is a great virtue.

(c) Balance between and duties is crucial to the delivery of justice in a society.

(d) Religious concept of rights is primarily derived from our relationship to God.

Passage

Biomass as fuel for power, heat, and transport has the highest mitigation potential of all renewable sources. It comes from agriculture and forest

residues as well as from energy crops. The biggest challenge in using biomass residues is a long-term reliable supply delivered to the power plant at reasonable costs; the key problems are logistical constraints and the costs of fuel collection. Energy crops, if not managed properly, compete with food production, and may have undesirable impacts on food prices. Biomass production is also sensitive to the physical impacts of a changing climate. Projections of the future role of biomass are probably overestimated, given the limits to the sustainable biomass supply, unless breakthrough technologies substantially increase productivity. Climate energy models project that biomass use could increase nearly four-fold to around 150 — 200 exajoules, almost a quarter of world primary energy in 2050. However, the maximum sustainable technical potential of biomass resources (both residues and energy crops) without disruption of food and forest resources ranges from 80 — 170 exajoules a year by 2050, and only part of this is realistically and economically feasible. In addition, some climate models rely on biomass-based carbon capture and storage, an unproven technology, to achieve negative emissions and to buy some time during the first half of the century. Some liquid biofuels such as corn-based ethanol, mainly for transport, may aggravate rather than ameliorate carbon emissions on a life-cycle basis. Second generation biofuels, based on lingo-cellulosic feedstocks — such as straw, bagasse, grass, and wood — hold the promise of sustainable production that is high-yielding and emit low levels of greenhouse gases, but these are still in the R & D stage.

295. What is/are the present constraint/constraints in using biomass as fuel for power generation?

1. Lack of sustainable supply of biomass. (CSAT-2016)
2. Biomass production competes with food production.
3. Bioenergy may not always be low carbon on a lifecycle basis.

Select the correct answer using the code given below:

(a) 1 and 2 only (c) 2 and 3 only

(b) 3 only (d) 1, 2 and 3

296. Which of the following can lead to food security problem? (CSAT-2016)

1. Using agricultural and forest residues as feedstock for power generation.
2. Using biomass for carbon capture and storage.
3. Promoting the cultivation of energy crops.

Select the correct answer using the code given below:

(a) 1 and 2 only (c) 2 and 3 only

(b) 3 only (d) 1, 2 and 3

297. In the context of using biomass, which of the following is/are the characteristic/characteristics of the sustainable production of biofuel? (CSAT-2016)

1. Biomass as a fuel for power generation could meet all the primary energy requirements of the world by 2050.
2. Biomass as a fuel for power generation does not necessarily disrupt food and forest resources.
3. Biomass as a fuel for power generation could help in achieving negative emissions, given certain nascent technologies.

Select the correct answer using the code given below:

(a) 1 and 2 only (c) 2 and 3 only

(b) 3 only (d) 1,2 and 3

298. With reference to the passage, following assumptions have been made:

1. Some climate-energy models suggest that the use of biomass as a fuel for power generation helps in mitigating greenhouse gas emissions. (CSAT-2016)
2. 1t is not possible to use biomass as a fuel for power generation without disrupting food and forest resources.

Which of these assumptions is/are valid?

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

Passage

We are witnessing a dangerous dwindling of biodiversity in our food supply. The green revolution is a mixed blessing. Over time farmers have come to rely heavily on broadly adapted, high yield crops to the exclusion of varieties adapted to the local conditions. Mono cropping vast fields with the same genetically uniform seeds helps boost yield and meet immediate hunger needs. Yet high-yield varieties are also genetically weaker crops that require expensive chemical fertilizers and toxic pesticides. In our focus on increasing the amount of food we produce today; we have accidentally put ourselves at risk for food shortages in future.

299. Which among the following is the most logical and critical inference that can be made from the above passage? (CSAT-2016)

(a) In our agricultural practices, we have become heavily dependent on expensive chemical fertilizers and toxic pesticides only due to green revolution.

(b) Mono cropping vast fields with high-yield varieties is possible due to green revolution.

(c) Mono cropping with high-yield varieties is the only way to ensure food security to millions.

(d) Green revolution can pose a threat to biodiversity in food supply and food security in the long run.

Passage

Human history abounds in claims and theories confining the right of governing to a few select citizens. Exclusion of the many is justified on the ground that human beings may be. rightfully segregated for the good of society and viability of the political process.

300. Which one of the following statements is least essential as a part of the argument in the above passage? (CSAT-2015)

(a) Man seeks control over external things affecting him.

(b) In society, there are 'super' and 'sub'" human beings

(c) Exceptions to universal citizen participation are conducive to systemic efficacy.

(d) Governing implies recognition of disparities in individual capacities.

Passage

By 2050, the Earth's population will likely have swelled from seven to nine billion people. To fill all those stomachs-while accounting for shifting consumption patterns, climate change, and a finite amount of arable land and potable water-some experts say food production will have to double. How can we make the numbers add up? Experts say higher yielding crop varieties and more efficient farming methods will be crucial. So will waste reduction. Experts urge cities to reclaim nutrients, water from waste streams, and preserve farmland. Poor countries, they say, can improve crop storage and packaging and rich nations could cut back on resource-intensive foods like meat.

301. Which one of the following statements best sums up the above passage? (CSAT-2015)

(a) The population of the world is growing very fast.

(b) Food security is a perennial problem only in developing countries.

(c) The world does not have enough resources to meet the impending food scarcity.

(d) Food security is increasingly a collective challenge

Passage

Many people in India feel if we cut our defence expenditure on weapon-building, we can create a climate of peace with our neighbours, subsequently reducing the conflict or creating a no-war situation. People who proclaim such ideas are either the victims of war or the propagators of false argument.

302. With reference to the above passage, which of the following is the most valid assumption? (CSAT-2015)

(a) Building of weapons systems by us has instigated our neighbours to wage wars against us.

(b) The greater spending on weapon-building by us would lessen the possibility of armed conflict with our neighbours.

(c) It is necessary to have state of the art weapons systems for national security.

(d) Many people in India believe that we are wasting our resources on weapon building

Passage

India accounts for nearly a fifth of the world's child deaths. In terms of numbers, it is the highest in the world-nearly 16 |akhs every year. Of these, more than half die in the first month of life. Officials believe that the reason for this is the absence of steps to propagate basic health practices relating to breast feeding and immunization. Also, the large reproductive population of 2.6 crore remains bereft of care during the critical phases of pregnancy and post-delivery. Added to this s the prevalence of child marriages, anaemia among young women and lack of focus on adolescent sanitation, all of which impact child death rates.

303. Which is the critical inference that can be made from the above passage? (CSAT-2015)

(a) A lot of Indians are illiterate and hence do not recognize the value of basic health practices.

(b) India has a very huge population, and the government alone cannot manage public health services.

(c) Universalization and integration of maternal health and child health services can effectively address the problem.

(d) The nutrition of women in childbearing age does not affect child mortality rate.

Passage

Food travel more than the people who eat them. Grocery stores and supermarkets are loaded with preserved and processed foods. This, however, often leads to environmental threats, such as pollution generated by long distance food transportation and wastage of food during processing and transportation, destruction of rain forests, reduced nutritional content, increased demand for preservation and packaging. Food insecurity also increases as the produce comes from regions that are not feeding their own population properly.

304. With reference to the above passage, which of the following statements is / are true? (CSAT-2015)

1. Consuming regionally grown food and not depending on long travelled food is a part of eco-friendly behaviour.
2. Food processing industry puts a burden on our natural resources.

Select the correct answer using the code given below:

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

I must say that, beyond occasionally exposing me to laughter, my constitutional shyness has been of no disadvantage Whatever. |, fact, | can see that, on the contrary, it has been all to my advantage. My hesitancy in speech, which was once an annoyance, is now a pleasure. Its greatest benefit has been that it has taught me the economy of words. I have naturally formed the habit restraining my thoughts. And | can now give myself the certificate that a thoughtless. word hardly ever escapes my tongue or pen. | do not recollect ever having had to regret anything in my speech or writing. | have thus been spared many amishap and waste of time. Experience has taught me that silence is part of the spiritual discipline of a votary of truth. Proneness to exaggerate, to suppress or modify the truth, wittingly or unwittingly, is a natural weakness of man, and silence is necessary in order to surmount it. A man of few words will rarely be thoughtless in his speech; he will measure every word. We find so many people impatient to talk. There is no chairman of a meeting who is not pestered with notes for permission to speak. And whenever the permission is given the speaker generally exceeds the time-limit, asks for more time, and keeps on talking without permission. Al this talking can hardly be said to be of any benefit to the world. It is so much waste of time. My shyness has been in reality my shield and buckler. It has allowed me to grow. It has helped me in my discernment of truth.

305. The author says that a thoughtless word hardly ever escapes his tongue or pen. Which one of the following is not a valid reason for

this? **(CSAT-2015)**

(a) He has no intention to waste his time.

(b) He believes in the economy of words.

(c) He believes in restraining his thoughts.

(d) He has hesitancy in his speech.

306. The most appropriate reason for the author to be spared many a mishap is that **(CSAT-2015)**

(a) He hardly utters or writes a thoughtless word.

(b) He is a man of immense patience.

(c) He believes that he is a spiritual person.

(d) He is a votary of truth.

307. For the author, silence is necessary in order to surmount **(CSAT-2015)**

(a) Constitutional shyness. (b) Hesitancy in speech.

(c) Suppression of thoughts. (d) Tendency to overstate.

Passage

The richer States have a responsibility to cut down carbon emissions and promote clean energy investments. These are the States that got electricity, grew faster and now have high per capita income, making them capable of sharing India's burden of becoming eco-friendly. Delhi, for example, can help by generating its own clean electricity using solar rooftop panels or even help poor States finance their clean energy projects. It is no secret that State Electricity Boards, which control 95% of the distribution network, are neck-deep in losses. These losses further discourage State utilities from adopting renewable energy as it is more expensive than fossil fuels.

308. Which among the following is the most logical and rational assumption that can be made from the above passage? **(CSAT-2015)**

(a) The richer States must lead in the production and adoption of renewable energy.

(b) The poor States always have to depend on rich States for electricity.

(c) The State Electricity Boards can improve their finances by undertaking clean energy projects.

(d) The high economic disparity between the rich and poor States is the major cause of high carbon emissions in India.

Passage

Set against a rural backdrop, 'Stench of kerosene' is the story of a couple, Guleri and Manak, who have been happily married for several years but do not have a child. Manak's mother is desperate to have a grandchild to carry on the family name. Hence, she gets Manak remarried in Guleri's absence. Manak, who acts as a reluctant but passive spectator, is meanwhile, informed by a friend that Guleri, on hearing about her husband's second marriage, poured kerosene on her clothes and set fire to them. Manak is heartbroken and begins to live as if he were a dead man. When his second wife delivers a son, Manak stares at the child for a long time and blurts out, "Take him away! He stinks of kerosene".

309. This is a sensitive issue-based story which tries to sensitize the readers about **(CSAT-2015)**

(a) Male chauvinism and infidelity.

(b) Love and betrayal.

(c) Lack of legal safeguards for women.

(d) Influence of patriarchal mindset.

Passage

The ultimate aim of government is not to rule or control by fear, nor to demand obedience, but conversely, to free every man from fear, that he may live in all possible security. In other words, to strengthen his natural right to exist and work without injury to himself or others. The object of government is not to change men from rational beings into beasts or puppets. It should enable them to develop their minds and bodies in security, and to employ their reason unshackled.

310. Which among the following is the most logical and rational inference that can be made from the above passage? (CSAT-2015)

(a) The true aim of government is to secure the citizens their social and political freedom.

(b) The primary concern of government is to provide absolute social security to all its citizens.

(c) The best government is the one that allows the citizens to enjoy absolute liberty in all matters of life.

(d) The best government is the one that provides absolute physical security to the people of the country.

Passage

Our municipal corporations are understaffed. The issue of skills and competencies of the staff poses an even greater challenge. Urban services delivery and infrastructure are complex to plan and execute. They require a high degree of specialization and professionalism. The current framework within which municipal employees, including senior management, are recruited does not adequately factor in the technical and managerial competencies required. Cadre and recruitment rules only specify the bare minimum in academic qualifications. There is no mention of managerial or technical competencies, or of relevant work experience. This is the case with most municipal corporations. They also suffer from weak organization design and structure.

311. Which among the following is the most logical and rational assumption that can be made from the above passage?(CSAT-2015)

(a) The task of providing urban services is a complex issue, which requires the organizational expansion of municipal bodies all over the country.

(b) Our cities can provide better quality of life if our local government bodies have adequate staff with required skills and competencies.

(c) Lack of skilled staff is due to the absence of institutions, which offer the requisite skills in city. management.

(d) Our country is not taking advantage of the demographic

dividend to manage the problems associated with rapid urbanization.

Passage

Flamingos in large flocks in the wild are social and extremely loyal. They perform group mating dances. Parents are very fond of their chicks, gathering them into creches for protection while both males and females fly off to search for food.

312. Which among the following is the most logical corollary. to the above passage? (CSAT-2015)

(a) Mass nesting in all species of birds is essential to ensure complete survival of their offspring.

(b) Only birds have the capacity to develop social behaviour and thus can do mass nesting to raise their chicks in safety.

(c) Social behaviour in some species of birds increases the odds of survival in an unsafe world.

(d) All species of birds set up creches for their chicks to teach them social behaviour and loyalty.

Passage

Vast numbers of Indian citizens without bank accounts live in rural areas, are financially and functionally illiterate, and have little experience with technology. A research study was conducted in a particular area in which electronic wage payments in Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS) are meant to go directly to the poor. It was observed that recipients often assume that the village leader needs to mediate the process, as was the case under the previous paper-based system. Among households under this research study area who claimed to have at |east one bank account, over a third reported still receiving MGNREGS wages in cash directly from a village leader.

313. What is the most logical, rational, and crucial message that is implied in the above passage? (CSAT-2015)

(a) `MGNREGS should be extended only to those who have a bank account.

(b) The paper-based system of payments is more efficient than electronic payment in the present scenario.

(c) The goal of electronic wage payments was not to eliminate mediation by village leaders.

(d) It is essential to provide financial literacy to the rural poor.

Passage

Individuals, groups, and leaders who promote human development operate under strong institutional, structural, and political constraints that affect policy options. But experience suggests broad principles for shaping an appropriate agenda for human development. One important finding from several decades of human development experience is that focusing exclusively on economic growth is problematic. While we have good knowledge about how to advance health and education, the causes of growth are much less certain, and growth is often elusive. Further, an unbalanced emphasis on growth is often associated with negative environmental consequences and adverse distributional effects. The experience of China, with its impressive growth record, reflects these broader concerns and underlines the importance of balanced approaches that emphasize investments in the non-income aspects of human development.

314. With reference to the above passage, consider the following statements (CSAT-2015)

1. In developing countries, a strong institutional framework is the only requirement for human development and policy options.
2. Human development and economic growth are not always positively interrelated.
3. Focusing only on human development should be the goal of economic growth.

Which of the above statements is/are correct?

(a) 1 only (b) 2 and 3 only

(c) 2 only (d) 1,2 and 3

315. With reference to the above passage, the following assumptions have been made: (CSAT-2015)

1. Higher economic growth is essential to ensure reduction in economic disparity.
2. Environmental degradation s consequence of economic growth.

Which of the above is/are valid assumption/assumptions?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

Climate change is already making many people hungry all over the world, by disrupting crop yields and pushing up prices. And it is not just food but nutrients that are becoming scarcer as the climate changes. It is the poorest communities that will suffer the worst effects of climate change, including increased hunger and malnutrition as crop production and livelihoods are threatened. On the other hand, poverty is a driver of climate change, as desperate communities' resort to unsustainable use of resources to meet current needs.

316. Which among the following is the most logical corollary to the above passage? (CSAT-2015)

(a) Government should allocate more funds to poverty alleviation programs and increase food subsidies to the poor communities.

(b) Poverty and climate impacts reinforce each other and therefore we have to reimagine our food systems.

(c) All the countries of the world must unite in fighting poverty and malnutrition and treat poverty as a global problem.

(d) We must stop unsustainable agricultural practices immediately and control food prices.

Passage

The Global Financial Stability Report finds that the share of portfolio investments from advanced economies in the total debt and equity investments in emerging economies has doubled in the past decade to 12 percent. The phenomenon has implications for Indian policy makers

as foreign portfolio investments in the debt and equity markets have been on the rise. The phenomenon is also flagged as a threat that could compromise global financial stability in a chain reaction, in the event of United States Federal Reserve's imminent reversal of its "Quantitative Easing" policy.

317. Which among the following is the most rational and critical inference that can be made from the above passage? (CSAT-2015)

(a) Foreign portfolio investments are not good for emerging economies.

(b) Advanced economies undermine the global financial stability.

(c) India should desist from accepting foreign portfolio investments in the future.

(d) Emerging economies are at a risk of shock from advanced economies.

Passage

Open defecation is disastrous when practiced in very densely populated areas, where it is impossible to keep away human faces from crops, wells, food and children's hands. Groundwater is also contaminated by open defecation. Many ingested germs and worms spread diseases. They prevent the body from absorbing calories and nutrients. Nearly one-half of India's children remain malnourished. Lakhs of them die from preventable conditions. Diarrhoea leaves Indians' bodies smaller on average than those of people in some poorer countries where people eat fewer 'calories. Underweight mothers produce stunted babies prone to sickness who may fail to develop their full cognitive potential. The germs released into environment harm rich and poor alike, even those who use latrines.

318. Which among the following is the most critical inference that can be made from the above passage? (CSAT-2015)

(a) The Central and State governments in India do not have enough resources to afford a latrine for each household.

(b) Open defecation is the most important public health problem of India.

(c) Open defecation reduces the human capital of India's workforce.

(d) Open defecation is a public health problem in all developing countries.

Passage

We generally talk about democracy but when it comes to any particular thing, we prefer a belonging to our caste or community or religion. So long as we have this kind of temptation, our democracy will remain a phoney kind of democracy. We must be in a position to respect a man as a man and to extend opportunities for development to those who deserve them and not to those who happen to belong to our community or race. This fact of favouritism has been responsible for much discontent and ill-will in our country.

319. Which one of the following statements best sums up the above passage? (CSAT-2015)

(a) Our country has a lot of diversity with its many castes, communities and religions.

(b) True democracy could be established by providing equal opportunities to all.

(c) So far none of us have actually understood the meaning of democracy.

(d) 1t will never be possible for us to establish truly democratic governance in our country.

Passage

The existence/establishment of formal financial institutions that offer safe, reliable, and alternative financial instruments is fundamental in mobilizing savings. To save, individuals need access to safe and reliable financial institutions, such as banks, and to appropriate financial instruments and reasonable financial incentives. Such access is not always available to all people in developing countries like India and more s0, in rural areas. Savings help poor households manage volatility in cash flow, smoothen consumption, and build working capital. Poor households without access formal savings mechanism encourage immediate spending temptations.

320. With reference to the above passage, consider the following statements: (CSATV-2015)

1. Indian financial institutions do not offer any financial instruments to rural households to mobilize their savings.
2. Poor households tend to spend their earnings/ savings due to lack of access to appropriate financial instruments.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

321. What is the crucial message conveyed in the passage (CSAT-2015)

(a) Establish more banks

(b) ncrease the Gross Domestic Product (GDP) growth rate

(c) ncrease the interest rate of bank Deposits

(d) Promote financial inclusion

Passage

Governments may have to take, steps which would otherwise be an infringement on the Fundamental Rights of individuals, such as acquiring a person's land against his will, or refusing permission for putting up a building, but the larger public interest for which these are done must be authorized by the people (Parliament). Discretionary powers to the administration can be done away with. It is becoming more and more difficult to keep this power within limits as the government has many numbers of tasks to perform. Where discretion has to be used, there must be rules and safeguards to prevent misuse of that power. Systems have to be devised which minimize, if not prevent, the abuse of discretionary power. Government work must be conducted within a framework of recognized rules and principles, and decisions should be similar and predictable.

322. Which among the following is the most logical assumption that can be made from the above passage? (CSAT-2015)

(a) Government should always be given wide discretionary power in all matters of administration

(b) The supremacy of rules and safeguards should prevail as opposed to the influence of exclusive discretion of authority.

(c) Parliamentary democracy is possible only if the Government has wider discretionary power.

(d) None of the above statements is a logical assumption that can be made from this passage.

Passage

India has suffered from persistent high inflation. Increase in administered prices, demand, and supply imbalances, imported inflation aggravated by rupee depreciation, and have combined to keep high inflation going. If India i import speculation there is an element common to all of them, it is that many of them are the outcomes. of economic reforms. India's vulnerability to the effects of changes in international prices has increased with trade liberalization. The effort to reduce subsidies has resulted in a continuous increase in the prices of commodities that are administered.

323. What is the most logical, rational, and crucial message that is implied in the above passage? (CSAT-2015)

(a) Under the present circumstances, India should completely avoid all trade liberalization policies and all subsidies.

(b) Due to its peculiar socio-economic situation, India is not yet ready for trade liberalization process.

(c) There is no solution in sight for the problems of continuing poverty and inflation in India in the near future.

(d) Economic reforms can often create a high inflation economy.

Passage

No Right is absolute, exclusive or inviolable. The Right of personal property, similarly, has to be perceived in the larger context of its assumed legitimacy. The Right of personal property should unite the principle of liberty with that of equality, and both with the principle of cooperation.

324. In the light of the argument in the above passage, which one of the following statements is the most convincing Explanation? (CSAT-2015)

(a) The Right of personal property is a Natural Right duly supported by statutes and scriptures.

(b) Personal property is a theft and an instrument of exploitation. The Right of personal property is therefore violative of economic justice.

(c) The Right of personal property is violative of distributive justice and negates the principle of cooperation.

(d) The comprehensive idea of economic justice demands that the Right of each person to acquisition of property has to be reconciled with that of others.

Passage

The conflict between man and State is as old as State history. Although attempts have been made for centuries to bring about a proper adjustment between the competing claims of State and the individual, the solution seems to be still far off. This is primarily because of the dynamic nature of human society where old values and ideas constantly yield place to new ones. It i4obvious that if individuals are allowed to have absolute freedom of speech and action, the result would be chaos, ruin and anarchy.

325. The author's viewpoint can be best summed up in which of the following statements? (CSAT-2015)

(a) The conflict between the claims of State and individual remains unresolved.

(b) Anarchy and chaos are the obvious results of democratic traditions.

(c) Old values, ideas and traditions persist despite the dynamic nature of human society.

(d) Constitutional guarantee of freedom of speech is not in the interest of society.

Passage

Climate change is a complex policy issue with major implications in terms of finance. All actions to address climate change ultimately involve costs. Funding is vital for countries like India to design and implement adaptation and mitigation plans and projects. Lack of funding is a large impediment to implementing adaptation plans. The scale and magnitude of the financial support required by developing countries to enhance their domestic mitigation and adaptation actions are a matter of intense debate in the multilateral negotiations under the United Nations Framework Convention on Climate Change (UNFCCC). The Convention squarely puts the responsibility for provision of financial support on the developed countries, taking into account their contribution to the stock of greenhouse gases (GHGS) in the atmosphere Given the magnitude of the task and the funds required, domestic finances are likely to fall short of the current and projected needs of the developing countries. Global funding through the multilateral mechanism of the Convention will enhance their domestic capacity to finance the mitigation efforts.

326. According to the passage, which of the following is/are a matter of intense debate in the multilateral negotiations under UNFCCC regarding the role of developing countries in climate change? (CSAT-2015)

1. The scale and size of required financial support.
2. Thecrop loss due to climate change in the developing countries.
3. To enhance the mitigation and adaptation actions in the developing countries.

Select the correct answer using the code given below:

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

327. In this passage, the Convention puts the responsibility for the provision of financial support on the developed countries because of (CSAT-2015)

1. Their higher level of per capita incomes.
2. Their large quantum of GDP.
3. Their large contribution to the stock of GHGS in the

atmosphere.

Select the correct answer using the code given below:

(a) 1 only (b) 1 and 2 only

(c) 3 only (d) 1,2 and 3

328. With regards to developing countries, it can be inferred from the passage that climate change is likely to have implications on their (CSAT-2015)

1. Domestic finances.
2. Capacity for multilateral trade.

Select the correct answer using the code given below:

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

329. Which one of the following is essentially discussed in the passage? (CSAT-2015)

(a) Conflict between developed and developing countries regarding support for mitigation.

(b) Occurrence of climate change due to excessive exploitation of natural resources by the developed countries.

(c) Lack of political will on the part of all the countries to implement adaptation plans.

(d) Governance problems of developing countries as a result of climate change.

Passage

The Himalayan ecosystem is highly vulnerable to damage, both due to geological reasons and on account of the stress caused by increased pressure of population, exploitation of natural resources and other related challenges. These aspects may be exacerbated due to the impact of climate change. It is possible that climate change may adversely impact the Himalayan ecosystem through increased temperature altered precipitation patterns, episodes of drought and biotic influences. This would not only impact the very sustenance of the indigenous communities in uplands but also the life of downstream dwellers across the country and beyond. Therefore, there is an urgent need for giving special attention to sustain the Himalayan ecosystem. This would require conscious efforts for conserving all the representative systems.

Further, it needs to be emphasized that the endemics with restricted distribution, and most often with specialized habitat requirements, are among the most vulnerable elements. In this respect the Himalayan biodiversity hotspot, with rich endemic diversity, is vulnerable to climate change. The Threats include possible loss of genetic resources and species, habitats and concomitantly a decrease in ecosystem services. Therefore, conservation of endemic elements in representative ecosystems/habitats assumes a great significance while drawing conservation plans for the region.

Towards achieving the above, we will have to shift towards contemporary conservation approaches, which include a paradigm of landscape level interconnectivity between protected area systems. The concept advocates a shift from the species-habitat focus to an inclusive focus on expanding the biogeographic range so that natural adjustments to climate change can proceed without being restrictive.

330. Consider the following statements:

According to the passage, **the adverse impact of climate change on an ecosystem can be a (CSAT-2014)**

1. Permanent disappearance of some of its flora and fauna.
2. Permanent disappearance of ecosystem itself.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

331. Which one of the following statements best implies the need to shift toward contemporary conservation approach? (CSAT-2014)

(a) Exploitation of natural resources causes a stress on the Himalayan ecosystem.

(b) Climate change alters precipitation patterns, causes episodes of drought and biotic interference.

(c) The rich biodiversity, including endemic diversity, makes the Himalayan region a biodiversity hotspot.

(d) The Himalayan biogeographic region should be enabled to adapt to climate change smoothly.

332. What is the most important message conveyed by the passage? (CSAT-2014)

(a) Endemism is a characteristic feature of Himalayan region.

(b) Conservation efforts should emphasize on biogeographic ranges rather than on some species or habitats.

(c) Climate change has adverse impact on the Himalayan ecosystem.

(d) Without Himalayan ecosystem, the life of the communities of uplands and downstream will have no sustenance.

333. With reference to the passage, the following assumptions have been made: (CSAT-2014)

1. To maintain natural ecosystems, exploitation of natural resources should be completely avoided.
2. Not only anthropogenic but also natural reasons can adversely affect ecosystems.
3. Loss of endemic diversity leads to the extinction of ecosystems.

Which of the above assumptions is/are correct?

(a) 1 and 2 (b) 2 only

(c) 2 and 3 (d) 3 only

Passage

It is often forgotten that globalization, is not only about policies on international economic relationships and transactions but has equally to do with domestic policies of a nation. Policy changes necessitated by meeting the internationally set conditions (by WTO etc.) of free trade and investment flows obviously affect domestic producers and investors. But the basic philosophy underlying globalization emphasizes

absolute freedom to markets to determine prices and production and distribution patterns, and view government interventions as processes that create distortions and bring in inefficiency. Thus, public enterprises have to be privatized through disinvestments and sales; sectors and - activities hitherto reserved for the public sector have to be opened to the private sector. This logic extends to the social services like education and health. Any restrictions on the' adjustments in workforce by way of retrenchment of workers should also pe removed and exit should be made easier by removing any restrictions on closures. Employment and wages should be governed by free play of market forces, as any measure to regulate them can discourage investment and also create inefficiency in production. Above all, in line with the overall philosophy of reduction in the role of the state, fiscal reforms should be undertaken to have generally low levels of taxation and government expenditure should be kept to the minimum to abide by the principle of fiscal prudence. All these are policy actions on the domestic front and are not directly related to the core items of the globalization agenda, namely free international flow of goods and finance.

334. According to the passage, under the globalization, government interventions are viewed as processes leading to (CSAT-2014)

(a) distortions and inefficiency in the economy.

(b) optimum use of resources.

(c) more profitability to industries.

(d) free play of market forces with regard to the industries.

335. According to the passage, the basic philosophy of globalization is to (CSAT-2014)

(a) give absolute freedom to producers to determine prices and production.

(b) give freedom to producers to evolve distribution patterns,

(c) give absolute freedom to markets to determine prices, production, and employment.

(d) give freedom to producers to import and export.

336. According to the passage, which of the following is/are necessary for ensuring globalization? (CSAT-2014)

1. Privatization of public enterprises
2. Expansionary policy of public expenditure
3. Free play of market forces to determine wages and employment
4. Privatization of social services like education and health.

Select the correct answer using the code given below:

(a) 1 only (b) 2 and 3 only

(c) 1, 3 and 4 (d) 2, 3 and 4

337. According to the passage, in the process globalization the State should have (CSAT-2014)

(a) expanding role (b) reducing role

(c) statutory role (d) none of the above roles.

Passage

In recent times, India has grown fast not only compared to its own past but also in comparison with other nations. But there cannot be any room for complacency because it is possible for the Indian economy to develop even faster and also to spread the benefits of this growth more widely than has been done thus far. Before going into details of the kinds of microstructural changes that we need to conceptualize and then proceed to implement, it is worthwhile elaborating on the idea of inclusive growth that constitutes the defining concept behind this Government's various economic policies and decisions. A nation interested in inclusive growth views the same growth differently depending on whether the gains of the growth are heaped primarily on a small segment or shared widely by the population. The latter is cause for celebration but not the former. In other words, growth must not be treated as an end in itself but as an instrument for spreading prosperity to all. India's own past experience and the experience of other nations suggests that growth is necessary for eradicating poverty but it is not a sufficient condition. In other words, policies for promoting growth need to be complemented with policies to ensure that more and more people join' in the growth process and, further, that there are mechanisms in place to redistribute some of the gains to those who are unable to partake in the market process and, hence, get left behind.

A simple way of giving this idea of inclusive growth a sharper form is to measure a nation's progress in terms of the progress of its poorest segment, for instance the bottom 20 per cent of the' population. One could measure the per capita income of the bottom quintile of the population and also calculate the growth rate of income; and evaluate our economic success in terms of these measures that pertain to the poorest segment. This approach is attractive because it does not ignore growth like some of the older heterodox criteria did. It simply looks at the growth of income of the poorest sections of the population. It also ensures that those who are outside of the bottom quintile do not get ignored. If that were done, then those people would in all likelihood drop down into the bottom quintile and so would automatically become a direct target of our policies. Hence the criterion being suggested here is a statistical summing up of the idea of inclusive growth, which, in turn, leads to two corollaries: to wish that India must strive to achieve high growth and that we must work to ensure that the weakest segments benefit from the growth.

338. The author's central focus is on (CSAT-2014)

(a) applauding India's economic growth not only against its own past performance, but against other nations.

(b) emphasizing the-need for economic growth which is the sole determinant of a country's prosperity.

(c) emphasizing inclusive growth where gains of growth are shared widely by the population.

(d) emphasizing high growth.

339. The author supports policies which will help (CSAT-2014)

(a) develop economic growth

(b) better distribution of incomes irrespective of rate of growth.

(c) develop economic growth and redistribute economic gains to

those getting left behind.

(d) put an emphasis on the development of the poorest segments of society.

340. Consider the following statements (CSAT-2014)

According to the author, India's economy has grown but there is no room for complacency as

1. growth eradicates poverty.
2. growth has resulted in prosperity for all.

Which of the statements given above is/a correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage

It is easy for the government to control State-owned companies through nods and winks. So what really needs to be done as the first step is to put petrol pricing on a transparent formula - if the price of crude is x and the exchange rate y, then every month or fortnight, the government announces a maximum price of petrol, which anybody can work out from the x and Y. The rule has to be worked out to make sure that the oil marketing can, in general, cover their costs. This will mean that if one company can innovate and cut costs, it will make greater profits. Hence, firms will be more prone to innovate and be efficient under this system. Once the rule is announced, there should be no interference by the government. If this is done for a while, private companies will re-enter this market. And once a sufficient number of them are in the fray, we can remove the rule-based pricing and leave it truly to the market (subject to, of course, the usual relations of antitrust and other competition laws).

341. Consider the following statements:

According to the passage, **an oil company can make greater profits, if a transparent formula for petrol pricing is announced every fortnight or month, by (CSAT-2014)**

1. promoting its sales
2. undertaking innovation
3. cutting costs.
4. selling its equity shares at higher prices.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 and 3

(c) 3 and 4 (d) 1, 2 and 4

342. Consider the following statements: (CSAT-2014)

According to the passage, private oil companies re-enter the oil producing market if

1. A transparent rule-based petrol pricing exists.
2. There is no government interference in the oil producing market.
3. subsidies are given by the government.
4. regulations of anti-trust are removed.

Which of the statements given above are correct?

(a) 1 and 2 (b) 2 and 3

(c) 3 and 4 (d) 2 and 4

Passage

Many nations now place their faith in capitalism and governments choose it as the strategy to create wealth for their people. The spectacular economic growth seen in Brazil, China and India after the liberalisation of their economies is proof of its enormous potential and success. However, the global banking crisis and the economic recession have left many bewildered. The debates tend to focus on free market operations and forces, their efficiency, and their ability for self-correction. Issues of justice, Integrity and honesty are rarely elaborated to highlight the failure of the global banking system. The apologists of the system continue to justify the success of capitalism and argue that the recent crisis was a blip. Their arguments betray an Ideological bias with the assumptions that an unregulated market is fair and competent, and that the exercise of private greed will be in the larger public interest. Few recognize the bidirectional relationship between capitalism and greed; that each reinforces the other. Surely, a more honest conceptualisation of the conflicts of interest among the rich and powerful players who have benefited from the system, their biases and ideology is needed; the focus on the wealth. creation should also highlight the resultant gross inequity.

343. The apologists of the "Free Market System", according to the passage, believe in (CSAT-2014)

(a) market without control by government authorities.

(b) market without protection by the government.

(c) ability of market to self-correct.

(d) market for free goods and services

344. With reference to "ideological bias", the passage implies that (CSAT-2014)

(a) free market is fair but not competent.

(b) free market is not fair but competent.

(c) free market is fair and competent

(d) free market is neither fair nor biased

345. "The exercise of private greed will be in the larger public interest" from the passage (CSAT-2014)

1. refers to the false ideology of capitalism.
2. underlies the righteous claims of the free market.
3. shows the benevolent face of capitalism.
4. ignores resultant gross inequity.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 and 3

(c) 1 and 4 (d) 4 only

Passage

Net profits are only 2.2% of their total assets for central public sector undertakings, lower than for the private corporate sector. While the

public sector or the State-led entrepreneurship played an important role in triggering India's industrialization, our evolving development needs, comparatively less-than-satisfactory performance of the public sector enterprises, the maturing of our private sector, a much larger social base now available for expanding entrepreneurship and the growing institutional capabilities to enforce competition policies would suggest that the time has come to review the role of public sector.

What should the portfolio composition of the government be? It should not remain static all times. The airline industry works well as a purely private affair. At the opposite end, rural roads, whose sparse traffic makes tolling unviable, have to be on the balance-sheet of the State. If the government did not own rural roads, they would not exist. Similarly, public health capital in our towns and cities will need to come from the public sector. Equally, preservation and improvement of forest cover will have to be a new priority for the public sector assets.

Take the example of steel. With near-zero tariffs, India is a globally competitive market for the metal. Indian firms export steel into the global market which demonstrates there is no gap in technology. Indian companies are buying up global steel companies, which shows there is no gap in capital availability. Under these conditions, private ownership works best.

Private ownership is clearly desirable in regulated industries, ranging from, finance to infrastructure, where a government agency performs the function of regulation and multiple competing firms are located in the private sector. Here, the simple and clean solution-government as the umpire and the private sector as the players is what works best. In many of these industries, we have a legacy of government ownership, where productivity tends to be lower, fear of bankruptcy is absent, and the risk of asking for money from the taxpayer is ever present. There is also the conflict of interest between government as an owner and as the regulator. The formulation and implementation of competition policy will be more vigorous and fair if government companies are out of action.

346. According to the passage, what is/are the reason/ reasons for saying that the time has come to review the role of public sector? (CSAT-2014)

1. Now public sector has lost its relevance in the industrialization process.
2. Public sector does not perform satisfactorily.
3. Entrepreneurship in private sector is expanding.
4. Effective competition policies are available now.

Which of the statements given above is/are correct in the given context?

(a) 1 and 3 only (b) 2 only

(c) 2, 3 and 4 only (d) 1, 2, 3 and 4

347. According to the passage, rural roads should be in the domain of public sector only. Why? (CSAT-2014)

(a) Rural development work is the domain of government only.

(b) Private sector cannot have monetary gains in this.

(c) Government takes money from tax payers and hence it is the responsibility of government only.

(d) Private sector need not have any social responsibility.

348. The portfolio composition of the government refers to (CSAT-2014)

(a) Public sector assets quality.

(b) Investment in liquid assets.

(c) Mix of government investment in different industrial sectors.

(d) Buying Return on Investment yielding capital assets.

349. The author prefers government as the umpire and private sector as players because (CSAT-2014)

(a) Government prescribes norms for a fair play by the private sector.

(b) Government is the ultimate in policy formulation.

(c) Government has no control over private sector players.

(d) None of the above statements is correct in this context.

Passage

Climate change poses potentially devastating effects on India's agriculture. While the overall parameters of climate change are increasingly accepted a 1°C average temperature increase over the next 30 years, sea level rise of less than 10 cm in the same period, and regional monsoon variations and corresponding droughts - the impacts in India are likely to be quite site and crop specific. Some crops may respond favourably to the changing conditions, others may not. This emphasizes the need to promote agricultural research and create maximum flexibility in the system to permit adaptations.

The key ingredient for "drought proofing" is the managed recharge of aquifers. To ensure continued yields of important staple crops (e.g. wheat), it may also be necessary to shift the locations where these crops are grown, in response to temperature changes as well as to water availability. The latter will be a key factor in making long term investment decisions.

For example, water runoff from the Himalayas is predicted to increase over the next 30 years as glaciers melt, but then decline substantially thereafter. It will be critical to provide incentives to plan for these large-scale shifts in agro-ecological conditions. India needs to make long term investment in research and development in agriculture. India is likely to experience changed weather patterns in future.

350. Consider the following statements: (CSAT-2014)

Climate change may force the shifting of locations of the existing crops due to

1. melting of glaciers.
2. water availability and temperature suitability at other locations.
3. poor productivity of crops.
4. wider adaptability of crop plants.

Which of the statements given above are correct?

(a) 1, 2 and 3 (b) 2 and 3 only

(c) 1 and 4 only (d) 1, 2, 3 and 4

351. According to the passage, why is it important to promote agricultural research in India? (CSAT-2014)

(a) To predict variations in monsoon patterns and to manage water resources.

(b) To make long-term investment decisions for economic growth

(c) To facilitate wider adaptability of crops.

(d) To predict drought conditions and to recharge aquifers.

Passage

It is essential that we mitigate the emissions of greenhouse gases and thus avoid some of the worst impacts of climate change that would take place in coming years and decades. Mitigation would require a major shift in the way we produce and consume energy. A shift away from overwhelming dependence on fossil fuels is now long overdue, but unfortunately, technological development has been slow arid inadequate largely because government policies have not promoted investments in research and development, myopically as a result of relatively low prices of oil. It is now, therefore, imperative for a country like India treating the opportunity of harnessing renewable energy on a large scale as a national imperative. This country is extremely well endowed with solar, wind and biomass sources of energy. Where we have lagged, unfortunately, is in our ability to develop and to create technological solutions for harnessing these resources.

One particular trajectory for carrying out stringent mitigation of greenhouse gas emissions assessed by the Intergovernmental Panel on Climate Change (IPCC) clearly shows the need for ensuring that global emissions of greenhouse gases peak no later than 2015 and reduce rapidly thereafter. The cost associated with such a trajectory is truly modest" and would amount, in the estimation of IPCC, to not more than 3 percent of the global GDP in 2030. In other words, the level of prosperity that the world would have reached without mitigation would at worst be postponed by a few months or a year at the most. This is clearly not a very high price to pay for protecting hundreds of millions of people from the worst risks associated with climate change. Any such effort, however, would require lifestyles to change appropriately also. Mitigation of greenhouse gas emissions is not a mere technological fix, and clearly requires changes in lifestyles and transformation of a country's economic structure, whereby effective reduction in emissions is brought about, such as through the consumption of much lower quantities of animal protein. The Food and Agriculture Organization (FAO) has determined that the emissions from the livestock sector amount to 18 percent of the total. The reduction of emissions from this source is entirely in the hands of human beings, who have never questioned the impacts that their dietary habits of consuming more and more animal protein are bringing about. Mitigation overall has huge co-benefits, such as lower air pollution and health benefits, higher energy security and greater employment.

352. According to the passage, which of the following would help in the mitigation of greenhouse gases? (CSAT-2014)

1. Reducing the consumption of meat.
2. Rapid economic liberalization.
3. Reducing the consumerism
4. Modern management practices of livestock

Select the correct answer using the code given below:

(a) 1, 2 and 3 (b) 2, 3 and 4

(c) 1 and 3 only (d) 2 and 4 only

353. Why do we continue to depend on the fossil fuels heavily? (CSAT-2014)

1. Inadequate technological development.
2. Inadequate funds for research and development.
3. Inadequate availability of alternative sources of energy.

Select the correct answer using the code given below:

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

354. According to the passage, how does the mitigation of greenhouse gases help us? (CSAT-2014)

1. Reduces expenditure on public health.
2. Reduces dependence on livestock.
3. Reduces energy requirements.
4. Reduces rate of global climate change.

Select the correct answer using the code given below:

(a) 1 ,2 and 3 (b) 1, 3 and 4

(c) 2, 3 and 4 (d) 1 and 4 only

355. What is the essential message of the passage? (CSAT-2014)

(a) We continue to depend on fossil fuel heavily.

(b) Mitigation of the greenhouse gases is imperative.

(c) We must invest in research and development.

(d) People must change their lifestyle.

Directions for the following 6 (six items):

The following six items are based on two

Passages in English to test the comprehension of English language and therefore these items do not have Hindi version. Read each

Passage and answer the items that follow.

Passage

In front of us was walking a bare-headed old man in tattered clothes. He was driving his beasts. They were all laden with heavy loads of clay from the hills and looked tired. The man carried a long whip which perhaps he himself had made. As he walked down the road he stopped now and then to eat the wild berries that grew on bushes along the uneven road. When he threw away the seeds, the bold birds would fly to peck at them. Sometimes a stray dog watched the procession philosophically and then began to bark. When this happened, my two little sons would stand still holding my hands firmly. A dog can sometimes be dangerous indeed.

356. The author's children held his hands firmly because (CSAT-2014)

(a) they were scared o f the barking dogs.

(b) they wanted him to pluck berries.

(c) they saw the whip in the old man's hand.

(d) the road was uneven.

357. The expression "a stray dog watched the procession philosophically" means that (CSAT-2014)

(a) the dog was restless and ferocious.

(b) the dog stood aloof, looking at the procession with seriousness.

(c) the dog looked at the procession with big, wondering eyes.

(d) the dog stood there with his eyes closed.

Passage

Cynthia was a shy girl. She believed that she was plain and untalented. One day her teacher ordered the entire class to show up for audition for the school play. Cynthia nearly died of fright when she was told that she would have to stand on stage in front of the entire class and deliver dialogues. The mere thought of it made her feel sick. But a remarkable transformation occurred during the audition. A thin, shy girl, her knees quaking, her stomach churning in terror, began to stun everyone with her excellent performance. Her bored classmates suddenly stopped their noisy chat to stare at her slender figure on the stage. At the end of her audition, the entire room erupted in thunderous applause.

358. Cynthia was afraid to stand on stage because (CSAT-2014)

(a) she felt her classmates may laugh at her.

(b) her stomach was churning.

(c) she lacked self-confidence.

(d) she did not like school plays.

359. Cynthia's classmates were chatting because (CSAT-2014)

(a) it was their turn to act next.

(b) they were bored of the performances.

(c) Cynthia did not act well.

(d) the teacher had no control over them.

360. Cynthia's knees were quaking because (CSAT-2014)

(a) she felt nervous and shy.

(b) the teacher scolded her.

(c) she was very thin and weak.

(d) she was afraid of her classmates.

361. The transformation that occurred during the audition refers to (CSAT-2014)

(a) the nervousness of Cynthia.

(b) the eruption of the entire room in thunderous applause.

(c) the surprise on the faces of her classmates.

(d) the stunning performance of Cynthia.

Directions for the following 8 (eight) items:

Read the following four passages and answer the items that follow each passage. Your answers to these items should be based on the passages only.

Passage -1

The subject of democracy has become severely muddled because of the way the rhetoric surrounding it has been used in recent years, There is, increasingly, an oddly confused dichotomy between those who want to 'impose' democracy on countries In the non-Western world (in these countries' 'own interest', of course) and those who are opposed to such 'imposition' (because of the respect for the countries' 'own ways'). But the entire language of 'imposition', used by both sides, is extraordinarily inappropriate since it makes the implicit assumption that democracy belongs exclusively to the West, taking it to be a quintessentially 'Western' idea which has originated and flourished only in the West.

But the thesis and the pessimism it generates about the possibility of democratic practice in the world would be extremely hard to justify. There were several experiments in local democracy in ancient India. Indeed, in understanding the roots of democracy in the world, we have to take an interest in the history of people participation and public reasoning in different parts of the world. We have to look beyond thinking of democracy only in terms of European and American evolution. We would fail to understand the pervasive demands for participatory living, on which Aristotle spoke with far-reaching insight, if we take democracy to be a kind of a specialized cultural product of the West.

It cannot, of course, be doubted that the institutional structure of the contemporary practice of democracy is largely the product of European and American experience over the last few centuries. This is extremely important to recognize since these developments in institutional formats were immensely innovative and ultimately effective. There can be little doubt that there is a major 'Western' achievement here.

362. Which of the following is closest to the view of democracy as mentioned in the above passage? (CSAT-2013)

(a) The subject of democracy is a muddle due to a desire to portray it as a Western concept, 'alien' to non-Western countries.

(b) The language or imposition of democracy is inappropriate. There is, however, a need to consider this concept in the backdrop of culture of 'own ways' of non-Western society.

(c) While democracy is not essentially a Western idea belonging exclusively to the West, the institutional structure of current democratic practices has been their contribution.

(d) None of the statements (a), (b) and (c) given above is correct.

363. With reference to the passage, the following assumptions have been made: (CSAT-2013)

1. Many of the non-Western countries are unable to have democracy because they take democracy to be a specialized cultural product of the West.
2. Western countries are always trying to impose democracy on non-Western countries. Which of the above is/are valid assumption/assumptions?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

364. According to the passage, which of the following should be the practice/practices in good corporate governance? (CSAT-2013)

1. Companies should always comply with labour and tax laws of the land.
2. Every company in the country should have a government 'representative as one of the independent directors on the board to ensure transparency.
3. The manager of a company should never invest his personal, funds in the company. Select the correct answer using the codes given below:

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

365. According to the passage, which of the following is/are the major benefit/benefits of good corporate governance? (CSAT-2013)

1. Good corporate governance leads to increase in share price of the company.
2. A company with good corporate governance always increases its business turnover rapidly.
3. Good corporate governance is the main criterion for foreign institutional investors when they decide to buy a company.

Select the correct answer using the codes given below:

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Passage - 3

Malnutrition most commonly occurs between the ages of six months and two years. This happens despite the child's food requirements being less than that of an older child. Malnutrition is often attributed to poverty, but it has been found that even in households where adults eat adequate quantities of food, more than 50 per cent of children-under-five do not consume enough food. The child's dependence on someone else to feed him/her is primarily responsible for the malnutrition. Very often the mother is working and the responsibility of feeding the young child is left to an older sibling. It is therefore crucial to increase awareness regarding the child's food needs and how to satisfy them.

366. According to the passage, malnutrition in children can be reduced (CSAT-2013)

(a) if the children have regular intake of food.

(b) after they cross the age of five.

(c) if the food needs of younger children are known.

(d) if the responsibility of feeding younger children is given to adults. Solution: c)

367. According to the author, poverty is not the main cause of malnutrition, but the fact that (CSAT-2013)

1. taking care of younger ones is not a priority for working mothers.
2. awareness of nutritional needs is not propagated by the Public Health authorities. Select the correct answer using the code given below:

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage - 4

A number of empirical studies find that farmers are risk-averse, though only moderately in many cases. There is also evidence to show that farmers' risk aversion results in cropping patterns and input use designed to reduce risk rather than to maximize income. Farmers adopt a number of strategies to manage and cope with agricultural risks. These include practices like crop and field diversification, non-farm employment, storage of stocks and strategic migration of family members. There are also institutions ranging from share tenancy to kinship, extended family and informal credit agencies. One major obstacle to risk sharing by farmers is that the same type of risks can affect a large number of farmers in the region. Empirical studies show that the traditional methods are not adequate. Hence there is a need for policy interventions, especially measures that cut across geographical regions.

Policies may aim at tackling agricultural risks directly or indirectly. Examples of risk-specific policies arc crop insurance, price stabilization and the development of varieties resistant to pests and diseases. Policies which affect risk indirectly are irrigation, subsidized credit and access to information. No single risk-specific policy is sufficient to reduce risk and is without side-effects, whereas policies not specific to risk influence the general situation and affect risks only indirectly. Crop insurance, as a policy measure to tackle agricultural risk directly, deserves careful consideration in the Indian context and in many other developing countries because the majority of farmers depend on rain-fed agriculture and in many areas yield variability is the predominant cause of their income instability.

368. The need for policy intervention to mitigate risks in agriculture is because (CSAT-2013)

(a) farmers are extremely risk-averse.

(b) farmers do not know how to mitigate risks.

(c) the methods adopted by farmers and existing risk sharing institutions are not adequate.

(d) majority of farmers depend on rain-fed agriculture.

369. Which of the following observations emerges from the above passage? (CSAT-2013)

(a) One can identify a single policy that can reduce risk without any side-effect.

(b) No single risk-specific policy is sufficient to reduce agricultural risk.

(c) Policies which affect risk indirectly can eliminate it.

(d) Government's policy intervention can mitigate agricultural risk completely.

370. Consider the following statements: (CSAT-2013)

I. A primary group is relatively smaller in size.

II. Intimacy is an essential characteristic of a primary group.

III. A family may be an example of a primary group.

In the light of the above statements, which one of the following is true?

(a) All families are primary groups.

(b) All primary groups are families.

(c) A group of smaller size is always a primary group.

(d) Members of a primary group know each other intimately.

371. Mineral oil deposits under the sea do not get completely decomposed because they (CSAT-2013)

(a) are constantly washed by the ocean currents.

(b) become rock and prevent oxygen from entering them.

(c) contain a mixture 'of hydrogen and carbon.

(d) are carcasses of organisms lying in saline conditions.

372. Sedimentary rock leads to the formation of oil deposits because (CSAT-2013)

(a) there are no saline conditions below it.

(b) it allows some dissolved oxygen to enter the dead organic matter below it.

(c) weight of overlying sediment layers causes the production of heat.

(d) it contains the substances that catalyze the chemical reactions required to change dead organisms into oil.

373. According to the passage, why should the discharge of agricultural slurry into watercourses be restricted? (CSAT-2013)

1. Losing nutrients in this way is not a good practice economically.
2. Watercourses do not contain the microorganisms that can decompose organic components of agricultural slurry.
3. The discharge may lead to the eutrophication of water bodies.

Select the correct answer using the codes given below

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

374. The passage refers to the conversion of "pollutant to fertilizer". What is pollutant and what is fertilizer in this context? (CSAT-2013)

(a) Decomposed organic component of slurry is pollutant and microorganisms in soil constitute fertilizer.

(b) Discharged agricultural slurry is pollutant and decomposed slurry in soil is fertilizer.

(c) Sprayed slurry is pollutant and watercourse is fertilizer.

(d) None of the above expressions is correct in this context.

375. According to the passage, what are the effects of indiscriminate use of fertilizers? (CSAT-2013)

1. Addition of pollutants to the soil and water.
2. Destruction decomposer of microorganisms in soil.
3. Nutrient enrichment of water bodies.
4. Creation of algal blooms.

Select the correct answer from the codes given below:

(a) 1, 2 and 3 only (b) 1, 3 and 4 only

(c) 2 arid 4 only (d) 1, 2, 3 and 4

376. What is/are the characteristics of a water body with cultural eutrophication? (CSAT-2013)

1. Loss of ecosystem services
2. Loss of flora and fauna
3. Loss of mineral nutrients

Select the correct answer using the code given below

(a) 1 only (b) 1 and 2 only

(c) 2 and 3 only (d) 1, 2 and 3

377. What is the central theme of this passage? (CSAT-2013)

(a) Appropriate legislation is essential to protect the environment.

(b) Modern agriculture is responsible for the destruction of environment.

(c) Improper waste disposal from agriculture can destroy the aquatic ecosystems.

(d) Use of chemical fertilizers is undesirable in agriculture.

Passage - 2

The miseries of the world cannot be cured by physical help only. Until man's nature changes, his physical needs will always arise, and miseries will always be felt, and no amount of physical help will remove them completely. The only solution of the problem is to make mankind pure. Ignorance is the mother of evil and of all the misery we see. Let men have light, let them be pure and spiritually strong and educated; then alone will misery cease in the world. We may convert every house in the country into a charitable asylum, we may fill the land with hospitals, but human misery will continue until man's character changes.

378. According to the passage, which of the following statements is most likely to be true as the reason for man's miseries? (CSAT-2013)

(a) The poor economic and social conditions prevailing in society.

(b) The refusal on the part of man to change his character.

(c) The absence of physical and material help from his society.

(d) Ever increasing physical' needs due to changing social structure.

379. With reference to the passage, the following assumptions have been made: (CSAT-2013)

1. The author gives primary importance to physical and material help in eradicating human misery.
2. Charitable homes, hospitals, etc. can remove human misery to a great extent. Which of the assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor

Directions for the following 2 (two) items:

Read the following passage and answer the two items that follow. Your answers to these items should be based on the passage only.

Passage

Ecological -research over the last quarter of the century has established the deleterious effects of habitat fragmentation due to mining, highways and such other intrusions on forests. When a large block of forests gets fragmented into smaller bits, the edges of all these bits come into contact with human activities resulting in the degradation of the entire forests. Continuity of forested landscapes and corridors gets disrupted 'affecting several extinction-prone species of wildlife. Habitat fragmentation, is therefore considered as the most serious threat to biodiversity conservation. Ad hoc grants of forest lands to mining companies coupled with rampant, illegal mining is aggravating this threat.

380. What is the central focus of this passage? (CSAT-2013)

(a) Illegal mining in forests

(b) Extinction of wildlife

(c) Conservation of nature

(d) Disruption of habitat

381. What is the purpose of maintaining the continuity of forested landscapes and corridors? (CSAT-2013)

1. Preservation of biodiversity.
2. Management of mineral resources.
3. Grant of forest lands for human activities.

Select the correct answer using the codes given below:

(a) 1 only (b) 1 and 2

(c) 2 and 3 (d) 1, 2 and 3

Directions for the following 8 (eight), items:

The following eight items (items 67 to 74) are based on three passages in English to test the comprehension of English language and therefore these items do not have Hindi version. Read each passage and answer the items that follow.

Passage - 1

Seven-year-old Jim came home from the park without his new bicycle. "An old man and a little boy borrowed it," he explained. "They are going to bring it back at four o'clock." His parents were upset that he had given his expensive new bicycle, but were secretly proud of his kindness and faith. Came four o'clock, no bicycle. The parents were anxious. But at 4:30, the door bell rang, and there stood a happy man and a boy, with the bicycle and a box of chocolates. Jim suddenly disappeared into his bedroom, and then came running out. "All right," he said, after examining the bicycle. ''You can have your watch back!"

382. When Jim came home without his bicycle, his parents (CSAT-2013)

(a) were angry with him.

(b) were worried.

(c) did not feel concerned.

(d) were eager to meet the old man and the little boy. Solution: b)

383. Jim returned the watch to the old man and the little boy because (CSAT-2013)

(a) they gave him chocolates.

(b) his father was proud of him.

(c) he was satisfied with the condition of his bicycle.

(d) they were late only by 30 minutes. Solution: c)

Passage - 2

It was already late when we set out for the next town, which according to the map was about fifteen kilometres away on the other side of the hills. There we felt that we would find a bed for the night. Darkness fell soon after we left the village, but luckily we met no one as we drove swiftly along the narrow winding road that led to the hills. As we climbed higher, it became colder and rain began to fall, making it difficult at times to see the road. I asked John, my companion, to drive more slowly. After we had travelled for about twenty kilometres, there was still no sign of the town which was marked on the map. We were beginning to get worried. Then without warning, the car stopped and we found we had run out of petrol.

384. The author asked John to drive more slowly because (CSAT-2013)

(a) the road led to the hills.

(b) John was an inexperienced driver.

(c) the road was not clearly visible.

(d) they were in wilderness.

385. The travellers set out for the town although it was getting dark because (CSAT-2013)

(a) they were in a hurry.

(b) the next town was a short distance away and was a hill-resort.

(c) they were in wilderness.

(d) the next town was a short distance away and promised a good rest for the night.

386. The travellers were worried after twenty kilo metres because (CSAT-2013)

(a) it was a lonely countryside.

(b) they probably feared of having lost their way.

(c) the rain began to fall

(d) it was getting colder as they drove.

Passage - 3

A stout old lady was walking with her basket down the middle of a street in Petrograd to the great confusion of the traffic and no small peril to herself. It was pointed out to her that the pavement was the place for foot-passengers, but she replied, "I m going to walk where I like. We've got liberty now." It did not occur to the dear lady that if liberty entitled

the foot-passenger to walk down the middle of the road it also entitled the taxi-driver to drive on the pavement, and that the end of such liberty would be universal chaos. Everything would be getting in everybody else's way and nobody would get anywhere. Individual liberty would have become social anarchy.

387. It was pointed out to the lady that she should walk on the pavement because she was (CSAT-2013)

(a) a pedestrian.

(b) carrying a basket.

(c) stout.

(d) an old lady.

388. The lady refused to move from the middle of the street because (CSAT-2013)

(a) she was not afraid of being killed.

(b) she felt that she is entitled to do whatever she liked.

(c) she did not like walking on the pavement.

(d) she was confused.

389. The old lady failed to realise that (CSAT-2013)

(a) she was not really free.

(b) her liberty was not unlimited.

(c) she was an old person.

(d) roads are made for motor vehicles only

Directions for the following 6 (six) items:

Given below are six items. Each item describes a situation and is followed by four possible responses. Indicate the response you find most appropriate. Choose only one response for each item. The responses will be evaluated based on the level of appropriateness for the given situation.

Please attempt all the items. There is no penalty for, wrong answers for these six items.

Directions for the following 15 (fifteen) items:

Read the following three passages and solve the items that follow each passage. Your Solutions to these items should be based on the passages only.

Passage I

Education, without a doubt, has an important functional, instrumental and utilitarian dimension. This is revealed when one asks questions such as 'what is the purpose of education?'. The Solutions, too often, are 'to acquire qualifications for employment/upward mobility', 'wider/higher (in terms of income) opportunities', and 'to meet the needs for trained human power in diverse field for national development'. But in its deepest sense education is not instrumentalist. That is to say, it is not to be justified outside of itself because it leads to the acquisition of formal skills or of certain desired psychological – social attributes. It must be respected in itself. Education is thus not a commodity to be acquired or possessed and then used, but a process of inestimable importance to individuals and society, although it can and does have enormous use value. Education then, is a process of expansion and conversion, not in the sense of conversion turning students into doctors or engineers, but the widening and turning out of the mind – the creation, sustenance and development of self-critical awareness and independence of thought. It is an inner process of moral-intellectual development.

390. What do you understand by the 'instrumentalist' view of education? (CSAT-2012)

(a) Education is functional and utilitarian in its purposes.

(b) Education is meant to fulfill human needs

(c) The purpose of education is to train the human intellect

(d) Education is meant to achieve moral development

391. According to the passage, education must be respected in itself because (CSAT-2012)

(a) It helps to acquire qualifications for employment.

(b) It helps in upward mobility and acquiring social status.

(c) It is an inner process of moral and intellectual development

(d) All the a), b) and c) given above are correct in this context.

392. Education is a process in which (CSAT-2012)

(a) Students are converted into trained professionals.

(b) Opportunities for higher income are generated

(c) individuals develop self-critical awareness and independence of thought

(d) qualifications for upward mobility are acquired

Passage II

Chemical pesticides lose their role in sustainable agriculture if the pests evolve resistance. The evolution of the pesticide resistance is simply natural selection in action. It is almost certain to occur when vast numbers of a genetically variable population are killed. One or a few individuals may be unusually resistant (perhaps because they possess an enzyme that can detoxify the pesticide). If the pesticide is applied repeatedly, each successive generation of the pest will contain a larger proportion of resistant individuals. Pests typically have a high intrinsic rate of reproduction, and so a few individuals in one generation may give rise to hundreds or thousands in the next, and resistance spreads very rapidly in a population.

This problem was often ignored in the past, even though the first case of DDT (dichlorodiphenyltrichloroethane) resistance was reported as early as 1946. There is an exponential increase in the numbers of invertebrates that have evolved resistance and in the number pesticides against which resistance has evolved. Resistance has been recorded in every family of arthropod pests (including dipterans such as mosquitoes and house flies, as well as beetles, moths, wasps, fleas, lice and mites) as well as in weeds and plant pathogens. Take the Alabama leaf worm, a moth pest of cotton, as an example. It has developed resistance in one or more regions of the world to aldrin, DDT, dieldrin, endrin, lindane and toxaphene.

If chemical pesticides brought nothing but problems, - if their use was intrinsically and acutely unsustainable – then they would already have fallen out of widespread use. This has not happened. Instead, their rate of production has increased rapidly. The ratio of cost to benefit for the

individual agricultural producer has remained in favour of pesticide use. In the USA, insecticides have been estimated to benefit the agricultural products to the tune of around $5 for every $1 spent.

Moreover, in many poorer countries, the prospect of imminent mass starvation, or of an epidemic disease, are so frightening that the social and health costs of using pesticides have to be ignored. In general the use of pesticides is justified by objective measures such as 'lives saved', 'economic efficiency of food production' and 'total food produced'. In these very fundamental senses, their use may be described as sustainable. In practice, sustainability depends on continually developing new pesticides that keep at least one step ahead of the pests – pesticides that are less persistent, biodegradable and more accurately targeted at the pests.

393. "The evolution of pesticide resistance is natural selection in action." What does it actually imply? (CSAT-2012)

(a) It is very natural for many organisms to have pesticide resistance.

(b) Pesticide resistance among organisms is a universal phenomenon.

(c) Some individuals in any given population show resistance after the application of pesticides

(d) None of the statements a), b) and c) given above is correct.

394. With reference to the passage, consider the following statements: (CSAT-2012)

1. Use of chemical pesticides has become imperative in all the poor countries of the world.
2. Chemical pesticides should not have any role in sustainable agriculture
3. One pest can develop resistance to many pesticides.

Which of the statements given above is/are correct?

(a) 1 and 2 only (b) 3 only

(c) 1 and 3 only (d) 1, 2 and 3

395. Though the problems associated with the use of chemical pesticides is known for a long time, their widespread use has not waned. Why? (CSAT-2012)

(a) Alternatives to chemical pesticides do not exist at all.

(b) New pesticides are not invented at all.

(c) Pesticides are biodegradable.

(d) None of the statements a), b) and c) given above is correct.

396. How do pesticides act as agents for the selection of resistant individuals in any pest population? (CSAT-2012)

1. It is possible that in a pest population the individuals will behave differently due to their genetic makeup.
2. Pests do possess the ability to detoxify the pesticides.
3. Evolution of pesticide resistance is equally distributed in pest population.

Which of the statements given above is/are correct?

(a) 1 only (b) 1 and 2 only

(c) 3 only (d) 1, 2 and 3

397. Why is the use of chemical pesticides generally justified by giving the examples of poor and developing countries? (CSAT-2012)

1. Developed countries can afford to do away with use of pesticides by adapting to organic farming, but it is imperative for poor and developing countries to use chemical pesticides.
2. In poor and developing countries, the pesticide addresses the problem of epidemic diseases of crops and eases the food problem.
3. The social and health costs of pesticide use are generally ignored in poor and developing countries.

Which of the statements given above is/are correct?

(a) 1 only (b) 1 and 2 only

(c) 2 only (d) 1, 2 and 3

398. What does the passage imply? (CSAT-2012)

(a) Alternative options to chemical pesticides should be promoted.

(b) Too much use of chemicals is not good for the ecosystem.

(c) There is no scope for the improvement of pesticides and making their use sustainable

(d) Both the statements a) and b) above are correct.

Passage III

Today's developing economies use much less energy per capita than developed countries such as the United State did at similar incomes, showing the potential for lower-carbon growth. Adaptation and mitigation need to be integrated into a climate-smart development strategy that increases resilience, reduces the threat of further global warming, and improves development outcomes. Adaptation and mitigation measures can advance development, and prosperity can raise incomes and foster better institutions. A healthier population living in better – built houses and with access to bank loans and social security is better equipped to deal with a changing climate and its consequences. Advancing robust, resilient development policies that promote adaptation is needed today because changes is the climate, already begun, will increase even in the short term.

The spread of economic prosperity has always been intertwined with adaptation to changing ecological conditions. But as growth has altered the environment and as environmental change has accelerated, sustaining growth and adaptability demands greater capacity to understand our environment, generate new adaptive technologies and practices, and diffuse them widely. As economic historians have explained, much of humankind's creative potential has directed at adapting to the changing world. But adaptation cannot cope with all the impacts related to climate change, especially as larger changes unfold in the long term.

Countries cannot grow out of harm's way fast enough to match the changing climate. And some growth strategies, whether driven by the government or the market, can also add to vulnerability – particularly if they over exploit natural resources. Under the Soviet development plan,

irrigated cotton cultivation expanded in water-stressed Central Asia and led to the near disappearance of the Aral Sea, threatening the livelihoods of fishermen, herders and farmers. And clearing mangroves- the natural coastal buffers against storm surges – to make way for intensive farming or housing development, increases the physical vulnerability of coastal settlements, whether in Guinea or in Louisiana.

399. Which of the following conditions of growth can add to vulnerability? (CSAT-2012)

1. When the growth occurs due to excessive exploitation of mineral resources and forests
2. When the growth brings about a change in humankind's creative potential.
3. When the growth is envisaged only for providing houses and social security to the people.
4. When the growth occurs due to emphasis on farming only.

Select the correct Solution using the codes given below:

(a) 1 only (b) 2, 3 and 4 only

(c) 1 and 4 only (d) 1, 2, 3 and 4

400. What does low-carbon growth imply in the present context? (CSAT-2012)

1. More emphasis on the use of renewable sources of energy.
2. Less emphasis on manufacturing sector and more emphasis on agricultural sector.
3. Switching over from monoculture practices to mixed farming
4. Less demand for goods and services.

Select the correct Solution: using the codes given below:

(a) 1 only

(b) 2, 3 and 4 only

(c) 1 and 4 only

(d) None of the above implies low-carbon growth

401. Which of the following conditions is/are necessary for the sustainable economic growth? (CSAT-2012)

1. Spreading of economic prosperity more.
2. Popularizing/spreading of adaptive technologies widely
3. Investing on research in adaptation and mitigation technologies.

Select the correct Solution using the codes given below:

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2, and 3

402. Which of the following inferences can be made from the passage? (CSAT-2012)

1. Rainfed crops should not be cultivated in irrigated areas
2. Farming under water-deficient areas should not be a part of development strategy.

Select the correct Solution using the codes given below:

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

403. Consider the following assumptions: (CSAT-2012)

1. Sustainable economic growth demands the use of creative potential of man.
2. Intensive agriculture can lead to ecological backlash.
3. Spread of the economic prosperity can adversely affect the ecology and environment.

With reference to the passage, which of the above assumptions is/ are valid?

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

404. Which one of the following statements constitutes the central theme of this passage? (CSAT-2012)

(a) Countries with greater economic prosperity are better equipped to deal with the consequences of climate change.

(b) Adaptation and mitigation should be integrated with development strategies

(c) Rapid economic growth should not be pursued by both developed and developing economies.

(d) Some countries resort to over exploitation of natural resources for the sake of rapid development.

405. With reference to the passage, which of the following statements is correct? (CSAT-2012)

(a) Introduction of exotic species into new geographical areas always leads to reduced biodiversity.

(b) Exotic species introduced by man into new areas have always greatly altered the native ecosystems.

(c) Man is the only reason to convert a hugely diverse range of local community compositions into more homogenous ones.

(d) None of the statements (a), (b), and (c) is correct in this context.

406. Why does man introduce exotic species into new geographical areas? (CSAT-2012)

1. To bread exotic species with local varieties.
2. To increase agricultural productivity.
3. for beautification and landscaping

Which of the above statements is/are correct?

(a) 1 only (b) 2and3only

(c) 1 and 3 only (d) 1, 2 and 3

407. How is homogenization prevented under natural conditions? (CSAT-2012)

(a) Evolution of groups of species specific to local habitats.

(b) Presence of oceans and mountain ranges

(c) Strong adaptation of groups of specific to local physical and climatic conditions

(d) All the statements (a), (b) and (c) given above are correct in this context.

408. How have the human beings influenced the biodiversity? (CSAT-2012)

1. By smuggling live organism
2. By building highways
3. By making ecosystems sensitive so that new species are not allowed
4. By ensuring that new species do not have major impact on local species.

Which of the statements given above are correct?

(a) 1 and 2 (b) 2 and 3

(c) 1 and 3 (d) 2 and 4

409. What can be the impact of invasion of exotic species on an ecosystem? (CSAT-2012)

1. Erosion of endemic species.
2. Change in the species composition of the community of the ecosystem

Select the correct Solution: using the codes given below:

(a) 1 only (b) 2 only

(c) both 1 and 2 (d) neither 1 nor 2

Passage II

Most champions of democracy have been rather reticent in suggesting that democracy would itself promote development and enhancement of social welfare – they have tended to see them as good but distinctly separate and largely independent goals. The detractors of democracy, on the other hand, seemed to have been quite willing to express their diagnosis of what they see as serious tensions between democracy and development. The theorists of the practical spirit - "Make up your mind: do you want democracy, or instead, do you want development?"- often came ,at least to start with , from East Asian countries, and their voice grew in influence as several of these countries were immensely successful – through the 1970s and 1980s and even later – in promoting economic growth without pursuing democracy.

To deal with these issues we have to pay particular attention to both the content of what can be called development and to the interpretation of democracy (in particular to the respective roles of voting and of public reasoning). The assessment of development cannot be divorced from the lives that people can lead and the real freedom that they enjoy. Development can scarcely be seen merely in terms of enhancement of inanimate objects of convenience, such as a rise in the GNP (or in personal incomes), or industrialisation – important as they may be as means to the real ends. Their value must depend on what they do to the lives and freedom of the people involved, which must be central to the idea of development.

If development is understood in a broader way, with a focus on human lives, then it becomes immediately clear that the relation between development and democracy has to be seen partly in terms of their constitutive connection, rather than only through their external links. Even though the question has often been asked whether political freedom is "conductive to development", we must not miss the crucial that political liberties and democratic rights are among the "constitutive components" of development does not have to be established indirectly through their contribution to the growth of GNP.

410. According to the passage, why is a serious tension perceived between democracy and development by the detractors of democracy? (CSAT-2012)

(a) Democracy and development are distinct and separate goals

(b) Economic growth can be promoted successfully without pursuing a democratic system of governance

(c) Non-democratic regimes deliver economic growth faster and far more successfully than democratic ones.

(d) All the statements (a), (b) and (c) given above are correct in this context.

411. According to the passage, what should be the ultimate assessment/ aim/view of development? (CSAT-2012)

(a) Rise in the per capita income and industrial growth rates.

(b) Improvement in the Human Development Index and GNP.

(c) Rise in the savings and consumption trends.

(d) Extent of real freedom that citizens enjoy.

412. What does a "constitutive" connection between democracy and development imply? (CSAT-2012)

(a) The relation between them has to be seen through external links.

(b) Political and civil rights only can lead to economic development

(c) Political liberties and democratic rights are essential elements of development.

(d) None of the statements (a), (b) and (c) given above is correct in this context.

Passage III

The need for Competition Law becomes more evident when foreign direct investment (FDI) is liberalised. The impact of FDI is not always pro-competitive. Very often FDI takes the form of a foreign corporation acquiring a domestic enterprise or establishing a joint venture with one. By making such an acquisition the foreign investor may substantially lessen competition and gain a dominant position in the relevant market, thus charging higher prices. Another scenario is where the affiliates of two separate multinational companies (MNCs) have been established in competition with one another in a particular developing economy, following the liberisation of FDI. Subsequently, the parent companies overseas merge. With the affiliates no longer remaining independent, competition in the host country may be artificially inflated. Most of these adverse consequences of mergers and acquisitions by MNCs can be avoided if an effective competition law is in place. Also, an economy that

has implemented an effective competition law is in a better position to attract FDI than one that has not. This is not just because most MNCs are expected to be accustomed to the operation of such a law in their home countries and know how to deal with such concerns but also that MNCs expect competition authorities to ensure a level playing field between domestic and foreign firms.

413. With reference to the passage, consider the following statements: (CSAT-2012)

1. It is desirable that the impact of Foreign Direct investment should be pro-competitive.
2. The entry of foreign investors invariably leads to the inflated prices in domestic markets.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

414. According to the passage, how does a foreign investor dominate the relevant domestic market? (CSAT-2012)

1. Multinational companies get accustomed to domestic laws.
2. Foreign companies establish joint ventures with domestic companies.
3. Affiliates in a particular market/sector lose their independence as their parent companies overseas merge.
4. Foreign companies lower the cost of their products as compared to that of products of domestic companies.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1, 2 and 3 only (d) 1, 2, 3 and 4

415. What is the inference from this passage? (CSAT-2012)

(a) Foreign investors and multinational companies always dominate domestic market.

(b) It is not in the best interest of domestic economy to allow mergers company.

(c) With competition law, it is easy to ensure a level playing field between domestic and foreign firms.

(d) For countries with open economy Foreign Direct investment is essential for growth

Directions for the following 6 (six) items:

Read the following two passages and Solution: the items that follow each passage. Your Solutions to these items should be based on the passages only.

Passage I

The poor especially in market economics need the strength that collectivities offer for creating more economic, social and political space for themselves, for enhancing their socio- economic well-being and voice, and as a protection against free market individualism. It has been argued that a group approach to farming, especially in the form of bottom up agricultural production collectivities, offers substantial scope for poverty alleviation and empowering the poor as well as enhancing agricultural productivity. To realize this potential, however, the groups would need to be voluntary in nature, small in size, participative in decision making and equitable in work sharing and benefit distribution. There are many notable examples of such collectivities to be found in varied contexts, such as in transitions economies. All of them bear witness to the possibility of successful cooperation under given conditions. And although the gender impact of the family cooperatives in the transition economies are uncertain, the Indian examples of women-only groups farming offer considerable potential for benefiting women.

416. Agricultural collectivities such as group based farming can provide the rural poor (CSAT-2012)

1. Empowerment
2. Increased agricultural productivity.
3. Safeguard against exploitative markets.
4. Surplus production of agricultural commodities.

Select the correct Solution: using the codes given below:

(a) 1, 2, 3 and 4 (b) 1, 2 and 3 only

(c) 2 and 4 only (d) 1, 3 and 4 only

417. What does the author imply by "gender impact"? (CSAT-2012)

(a) Women are doubtful participants in cooperatives.

(b) Family cooperatives may not include women

(c) Women benefitting from group farming.

(d) Women's role in transition economies is highly restrictive.

418. Consider the following assumptions: (CSAT-2012)

1. It is imperative for transition economies to have agricultural collectivities.
2. Agricultural productivity can be increased by group approach to farming.

With reference to the above passage which of these assumptions is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage II

In a typical Western liberal context, deepening of democracy invariably leads to consolidation of 'liberal values'. In the Indian context, democratization is translated into greater involvement of people not as 'individuals' which is a staple to liberal discourse, but as communities or groups. Individuals are getting involved in the public sphere not as 'atomized' individuals but as members of primordial communities drawn on religious or caste identity. Community-identity seems to be the governing force. It is not therefore surprising that the so-called peripheral groups continue to maintain their identities with reference to the social groups {caste, religion or sect} to which they belong while getting involved in the political processes despite the fact that their political goals remain more or less identical. By helping to articulate

the political voice of the marginalized, democracy in India has led to 'a loosening of social strictures' and empowered the peripherals to be confident of their ability to improve the socio economic conditions in which they are placed. This is a significant political process that had led to a silent revolution through a meaningful transfer of power from the upper caste cities to various subaltern groups within the democratic framework of public governance.

419. According to the passage, what does "deepening of democracy" mean in the Western context? (CSAT-2012)

(a) Consolidation of group and class identities.

(b) Democratization translated as greater involvement of people.

(c) Democratization as greater involvement of 'atomized' individuals in the public sphere.

(d) None of the statements a, b and c given above is correct in this context.

420. Greater democratization in India has not necessarily led to (CSAT-2012)

(a) The dilution of caste and communal identities in the public sphere.

(b) Irrelevance of community identity as governing force in Indian politics.

(c) Marginalization of elite groups in society.

(d) Relative unimportance of hereditary identities over class identities.

421. What is the "silent revolution" that has occurred in the Indian democratic process? (CSAT-2012)

(a) Irrelevance of caste and class hierarchies in political processes

(b) Loosening of social strictures in voting behavior and patterns.

(c) Social change through transfer of power from upper caste elites to subaltern groups.

(d) All the statements a), b) and c) given above are correct in this context.

Passage-1

For fourteen and half months I lived in my little cell or room in Dehradun jail, and I began to feel as if I was almost a part of it. I was familiar with every bit of it, I knew every mark and dent on the whitewashed walls and on the uneven floors and on the ceiling with the moth eaten rafters. In the little yard outside I greeted little tufts of grass and odd bits of stone as old friends. I was not alone in my cell, for several colonies of wasp and hornets lived there, and many lizards found the home behind the rafters, emerging in the evening in the search of prey.

422. Which of the following explains best the sentence in the passage "I was almost a part of it"? (CSAT-2012)

(a) I was not alone in the cell

(b) I was familiar with every bit of the cell

(c) I greeted little tufts of grass like old friends.

(d) I felt quite at home in the cell

423. The passage attempts to describe: (CSAT-2012)

(a) The general conditions of the country's jail

(b) The prisoner's capacity to notice the minute details of his surroundings

(c) The prisoner's conscious efforts to overcome the loneliness

(d) The prisoner's ability to live happily with other creatures

424. The author of the passage seems to suggest that (CSAT-2012)

(a) It is possible to adjust one-self to uncongenial surroundings.

(b) The conditions in Indian prisons are not bad

(c) It is not difficult to spend one's time in prison

(d) There is a need to improve the conditions in our jails.

Passage-2

We started pitching the highest camp that has been ever made. Everything took five times as long as it would have taken in the place where there was enough air to breathe; but at last we got tent up, and when we crawled in, it was not too bad. There was only a light wind and inside it was not too cold for us to take off our gloves. At night most climbers take off their boots; but I prefer to keep them on. Hilary, on the other hand took his off and lain them next to his sleeping bag.

425. What does the expression "pitching the highest camp" imply? (CSAT-2012)

(a) They reached the summit of the highest mountain in the world.

(b) Those who climbed that far earlier did not pitch any camp.

(c) So far nobody climbed that high.

(d) They were too many climbers and needed to pitch a big camp

426. They took a long time to finish the work because: (CSAT-2012)

(a) They were very tired.

(b) There was not enough air to breathe

(c) It was very cold

(d) It was very dark

427. When they crawled into the tent (CSAT-2012)

(a) They took off their gloves because it was not very cold

(b) They could not take off their gloves because it was very cold.

(c) They took of their gloves though it was very cold.

(d) They did not take off their gloves though it was not very cold.

Passage-3

A local man, staying on the top of the floor of an old wooden house, was awakened at midnight by fire. Losing his way in the smoke-filled passage, he missed the stairway and went into another room. He picked a bundle to protect his face from fire and immediately fell through the floor below where he managed to escape through a clear doorway. The "bundle" proved to be the baby of the Mayor's wife. The "hero" was congratulated.

428. The man went to another room because (CSAT-2012)

(a) He did not know where the stairway was

(b) The passage was full of smoke

(c) He was extremely nervous

(d) He stumbled on bundle

429. The man was called hero because (CSAT-2012)

(a) Expressed his willingness to risk his life to save others

(b) Managed to escape from fire

(c) Showed great courage in fighting the fire.

(d) Saved a life

Directions for the following 8 (eight) items:

Read each of the following two passages and answer the items that follow. Your answers to these items should be based on the passages only.

Passage-I

For achieving inclusive growth there is-a critical need to rethink the-role of the State. The early debate among economists about the size of the Goverment can be misleading. The need of the hour is to have an enabling Government. India is too large and complex a nation for the State to be able to deliver all that is needed. Asking the Government to produse all the essential goods, create all the necessary jobs, and keep a curb on the prices of all goods is to lead to a large cumbersome bureaucracy and widespread corruption. The aim must be to stay with the objective of inclusive growth that was laid down by the founding fathers of the nation and also to take a more modern view of what the State can realistically deliver. This is what leads to the idea of an enabling State, that is, a Government that does not try to directly deliver to the citizens everything that they need. Instead, it (1) creates an enabling ethos for the market so that individual enterprise can flourish and citizens can, for the most part, provide for the needs of one another, and (2) steps in to help those who do not manage to do well for themselves, for there will always be individuals, no matter what the system, who need support and help. Hence we need a Government that, when it comes to the market, sets effective, incentive-compatible rules and remains on the sidelines with minimal interference, and, at the same time, plays an important role in directly helping the poor by ensuring that they get basic education and health services and receive adequate nutrition and food .

430. According to passage: (CSAT-2011)

1. The objective of inclusive growth was laid down by the founding fathers of the nation.
2. Need of the hour is to have an enabling Government.
3. The Government should engage in maximum interference in market processes.
4. There is a need to change the size of the Government.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 4 only (d) 1, 2, 3 and 4

431. According to the passage, the strategy of inclusive growth can be effected by focussing on (CSAT-2011)

(a) Meeting all the needs of every citizen in the country.

(b) Increasing the regulations over the manufacturing sector.

(c) Controlling the distribution of manufacturing goods.

(d) Delivery of the basic services to the deprived sections of the society.

432. What constitutes an enabling Government? (CSAT-2011)

1. A large bureaucracy.
2. Implementation of welfare programmes through representatives.
3. Creating an ethos that helps individual enterprise
4. Providing resources to those who are underprivileged.
5. Offering direct help to the poor regarding basic services.

Select the correct answer from the codes given below :

(a) 1, 2 and 3 only (b) 4 and 5 only

(c) 3, 4 and 5 only (d) 1, 2, 3, 4 and 5

433. Why is the State unable to deliver "all that is needed"? (CSAT-2011)

1. It does not have sufficient bureaucracy.
2. It does not promote inclusive growth.

Select the correct answer from the codes given below:

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

434. What is essential message being conveyed by the author of the passage? (CSAT-2011)

(a) The objectives of inclusive growth laid down by the founding fathers of the nation should be remembered.

(b) The Government needs to make available more schools and health services.

(c) The Government needs to establish markets and industries to meet the needs of the poor strata of the society.

(d) There is a need to rethink the role of the State in achieving inclusive growth.

Passage-2

The concept of 'creative society' refers to a phase of development of a society in which a large number of potential contradictions become articulate and active. This is most evident when oppressed social groups get politically mobilised and demand their rights. The upsurge of the peasants and tribals, the movements for regional autonomy and self-determination, the environmental movements, and the women's movements in the developing countries are signs of emergence of creative society in contemporary times. The forms of social movements and their intensity may vary from country to country and place to place within a country. But the very presence of movements for social transformation in various spheres of a society indicates the emergence of a creative society in a country.

435. What does the author imply by "creative society"? (CSAT-2011)

1. A society where diverse art forms and literary writings seek incentive.
2. A society where social inequalities are accepted as the norm.
3. A society where a large number of contradictions are recognised.
4. A society where' the exploited and the oppressed groups grow conscious of their human rights and upliftment.

Select the correct answer using the codes given below:

(a) 1, 2 and 3 (b) 4 only

(c) 3 and 4 (d) 2 and 4

436. What according to the passage are the manifestations of social movements? (CSAT-2011)

1. Aggressiveness and being incendiary.
2. Instigation by external forces.
3. Quest for social equality and individual freedom.
4. Urge for granting privileges and self-respect to disparaged sections of the society.

Select the correct answer using the codes given below:

(a) 1 and 3 only (b) 2 and 4 only

(c) 3 and 4 only (d) 1, 2, 3 and 4

437. With reference to the passage consider the following statements: (CSAT-2011)

1. To be a creative society, it is essential to have a variety of social movements.
2. To be a creative society, it is imperative to have potential contradictions and conflicts.

Which of the statements given above is/are correct ?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage-1

Ecosystems provide people with a variety of goods and services; food, clean water, clean air, flood control, soil stabilization, pollination, climate regulation, spiritual fulfilment and aesthetic enjoyment, to name just a few. Most of these benefits either are irreplaceable or the technology necessary to replace them is prohibitively expensive. For example, potable fresh water can be provided by desalinating sea-water, but only at great cost. The rapidly expanding human population has greatly modified the Earth's ecosystems to meet their increased requirements of some of the goods and services, particularly food, fresh water, timber, fibre and fuel. These modifications have contributed substantially to human well being and economic development. The benefits have not been equally distributed. Some people have actually been harmed by these changes. Moreover, short-term increases in some ecosystem goods and services have come at the cost of the long-term degradation of others. For example, efforts to increase the production of food and fibre have decreased the ability of some ecosystems to provide clean water, regulate flooding and support biodiversity.

438. With reference to the passage, consider the following statements. Expanding human' population has an adverse effect on: (CSAT-2011)

1. Spiritual fulfilment
2. Aesthetic enjoyment
3. Potable fresh water
4. Production of food and fibre
5. Biodiversity

Which of the statements given above are correct ?

(a) 1, 2 and 3 only (b) 2, 4 and 5 only

(c) 3 and 5 only (d) 1, 2, 3, 4 and 5

439. The passage mentions that "some people have actually been harmed by these changes." What does it imply? (CSAT-2011)

1. The rapid expansion of population has adversely affected some people.
2. Sufficient efforts have not been made to increase the production of food and fibre.
3. In the short term some people may be harmed, but in the long term everyone will benefit from modifications In the Earth's ecosystems.

Which of the statements given above is/are correct?

(a) 1 only

(b) 2

(c) 1 and 3

(d) None of the statements given above

440. With reference to the passage, consider the following statements: (CSAT-2011)

1. It is imperative to modify the Earth's ecosystems for the well being of mankind.
2. Technology can never replace all the goods and services provided by ecosystems.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Paasage-B

A moral act must be our own act; must spring from our own will. If we act mechanically, these is no moral content in our act. Such action would be moral, If we think it proper to act like a machine and do so. For, in doing so, we use our discrimination. We should bear in mind the distinction between acting mechanically and acting intentionally. It may be moral of a king to pardon a culprit. But the messenger carrying out the order of pardon plays only a mechanical part in the king's moral act. But if the messenger were to carry out the king's order considering it to be his duty,

his action would be a moral one. How can a man understand morality who does not use his own intelligence and power of thought, but lets himself be swept along like a log of wood by a current? Sometimes a man defies convention and acts on his own with a view to absolute good.

441. Which of the following statements best describe/describes the thought of the writer? (CSAT-2011)

1. A moral act calls for using our discretion.
2. Man should react to a situation immediately.
3. Man must do his duty.
4. Man should be able to defy convention in order to be moral.

Select the correct answer from the codes given below :

(a) 1 only (b) 1 and 3

(c) 2 and 3 (d) 1 and 4

442. Which of the following statements is the nearest definition of moral action, according to the writer? (CSAT-2011)

(a) it is a mechanical action based on official orders from superiors.

(b) It is an action based on our sence of discretion.

(c) It is a clever action based on the clarity of purpose.

(d) It is a religious action based on understanding.

443. The passage contains a statement "lets himself be swept along like a log of wood by a current." Among the following statements, which is/are nearest in meaning to this? (CSAT-2011)

1. A person does not use his own reason.
2. He is susceptible to influence/pressure.
3. He cannot withstand difficulties/challenges.
4. He is like a log of wood.

Select the correct answer using the codes given below:

(a) 1 only (b) 1 and 2

(c) 2 and 3 (d) 1 and 4

Directions for the following 4 (four) items:

Read the following passage and answer the items that follow. Your answers to these items should be based on the passage only.

Passage

A country under foreign domination seeks escape from the present in dreams of a vanished age, and finds consolation in visions of past greatness. That is a foolish and dangerous pastime in which many of us indulge. An equally questionable practice for us in India is to imagine that we are still spiritually great though we have come down, in the world in other respects. Spiritual or any other greatness cannot be founded on lack of freedom and opportunity, or on starvation and misery. Many western writers have encouraged that notion that Indians are other-worldly. I suppose the poor and unfortunate in every country become to some extent other-worldly, unless they become revolutionaries, for this world is evidently not meant for them. So also subject peoples.

As a man grows to maturity he is not entirely engrossed in, or satisfied with, the external objective world. He seeks also some inner meaning, some psychological and physical satisfactions. So also with peoples and civilizations as they mature and grow adult. Every civilization and every people exhibit these parallel streams of an external life and an internal life. Where they meet or keep close to each other, there is an equilibrium and stability. When they diverge conflict arises and the crises that torture the mind and spirit.

444. The passage mentions that "this world is evidently not meant for them". It refers to people who (CSAT-2011)

1. seek freedom from foreign domination.
2. live in starvation and misery.
3. become revolutionaries .

Which of the statements given above is/are correct ?

(a) 1 and 2 (b) 2 only

(c) 2 and 3 (d) 3 only

445. Consider the following assumptions : (CSAT-2011)

1. A country under foreign domination cannot indulge in spiritual pursuit.
2. Poverty is an impediment in the spiritual pursuit.
3. Subject peoples may become other-wordly.

With reference to the passage, which of the above assumptions is/ are valid ?

(a) 1 and 2 (b) 2 only

(c) 2 and 3 (d) 3 only

446. The passage thematically centres on (CSAT-2011)

(a) the state of mind of oppressed people

(b) starvation and misery

(c) the growth of civilization

(d) body, mind and spirit of people in general

447. According to the passage, the torture of the mind and spirit is caused (CSAT-2011)

(a) by the impact of foreign domination.

(b) by the desire to escape from foreign domination and find consolation in visions of past greatness.

(c) due to lack of equilibrium between an external life and an internal life.

(d) due to one's inability to be either revolutionary or other-worldly.

Directions for the following 4 (four) items:

Read the followuing passage and answer the items that follow. Your answers to these items should be based on the passage only.

Passage

A species that exerts. an influence out of proportion to its abundance in an ecosystem is called a keystone species. The keystone species

may influence both the species richness of communities and the flow of energy and materials through ecosystems. The sea star Pisaster the flow of energy and materials through ecosystems. The sea star Pisaster ochraceus, which lives in rocky intertidal ecosystems on the Pacific coast of North America, is also an example of a keystone species. Its preferred prey is the mussel Mytilus californianus . In the absence of sea-stars, these mussels crowd out other competitors in a broad belt of the intertidal zone. By consuming mussels, sea star creates bare spaces that are taken over by a variety of other species. A study at the University of washington demonstrated the influence of Pisaster on species richness by removing sea stars from selected parts of the intertidal zone repeatedly over a period of five years. Two major changes occured in the areas from which sea stars were removed. First, the lower edge of the 46. mussel bed extended farther down into the intertidal zone, showing that sea stars are able to eliminate mussels completely where they are covered with water most of the time. Second, and more dramatically, 28 species of animals and algae disappeared from the sea star removal zone. Eventually only Mytilus, the dominant competitor, occupied the entire substratum. Through its effect on competitive relationships, predation by Pisaster largely determines which species live in these rocky intertidal ecosystems.

448. What is the crux of the passage? (CSAT-2011)

(a) Sea star has a preferred prey.

(b) A preferred prey determines the survival of a keystone species.

(c) Keystone species ensures species diversity.

(d) Sea star is the only keystone species on the Pacific coast of North America.

449. With reference to the passage, consider the following statements : (CSAT-2011)

1. Mussels-are generally the dominant species in intertidal ecosystems.
2. The survival of sea stars is generally determined by the abundance of mussels.

Which of the statements given above is /are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

450. Which of the following is/are implied by the passage? (CSAT-2011)

1. Mussels are always hard competitors for sea stars.
2. Sea stars of the Pacific coast have reached the climax of their evolution.
3. Sea stars constitute an important component in the energy flow in intertidal ecosystems.

Which of the statements given above is/are correct?

(a) 1 and 2 (b) 2 only

(c) 1 and 3 (d) 3 only

451. Consider the following assumptions: (CSAT-2011)

1. The food chains/food web in an ecosystem are influenced by keystone species.
2. The presence of keystone species is a specific characteristic of aquatic ecosystems.
3. If the keystone species is completely removed from an ecosystem, it will lead to the collapse of the ecosystem.

With reference to the passage, which of the above assumptions is/ are valid?

(a) 1 only (b) 2 and 3 only

(c) 1 nad 3 only (d) 1, 2 and 3

452. Consider the following argument: (CSAT-2011)

"In, order to be a teacher one must graduate from college. All poets are poor. Some Mathematicians are poets. No college graduate is poor."

Which one of the following is not a valid conclusion regarding the above argument?

(a) Some Mathematicians are not teachers.

(b) Some teachers are not Mathematicians.

(c) Teachers are not poor.

(d) Poets are not teachers

Directions for the following 5 (five) items:

Read the following passage and answer the items that follow. Your answers to these items should-be based on the passage only.

Passage

Now India's children have a right to receive at least eight years of education, the gnawing question is whether' it will remain 'on paper' or 'become a reality. One hardly needs a reminder that this right is different from the others enshrined in the Constitution, that the beneficiary - a six year old child cannot demand it, nor can she or he fight a legal battle when the right is denied or violated. In all cases, it is the adult society which must act on behalf of the child. In another peculiarity, where a child's right to education is denied, no compensation offered later can be adequate or relevant. This is so because childhood does not last if a legal battle fought on behalf of a child is eventually won, it may be of little use to the boy or girl because the opportunity missed at school during childhood cannot serve the same purpose later in life. This may be painfully true for girls because our society permits them only a short childhood, if at all. The Right to Education (RTE) has become law at a point in India's history when the ghastly practice of female infanticide has resurfaced in the form of foeticide. This is "symptomatic of a deeper turmoil" in society which compounding the traditional obstacles to the education of girls. "Tenacious prejudice against the intellectual potential of girls runs across our cultural diversity and the system of education has not been able to address it.

453. With reference to the passage, consider the following statements: (CSAT-2011)

1. When children are denied education, adult society does not act on behalf of them.
2. Right to Education as a law cannot be enforced in the country.

Which of the statements given above is/are correct ?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

454. According to the passage, what could be the traditional obstacles to the education of girls? (CSAT-2011)

1. Inability of parents to fight a legal battle when the Right to Education is denied to their children.
2. The traditional way of thinking about girl's role in society.
3. The prejudice against the intellectual potential of girls.
4. Improper system of education.

Select the correct answer from the codes given below:

(a) 1 and 2 only (b) 2, 3 and 4 only

(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

455. On the basis of the passage, consider the following statements: (CSAT-2011)

1. Right to Education is a legal right and not a fundamental right.
2. For realising the goal of universal education, the education system in the country must be made identical to that of developed countries.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

456. Which one of the following statements conveys the key message of the passage ? (CSAT-2011)

(a) India has declared that education is compulsory for its children.

(b) Adult society is not keen on implementing the Right to Education.

(c) The Right to Education, particularly of a girl child, needs to be safeguarded.

(d) The system of education should be address the issue of right to education.

457. Which one of the following statements conveys the inference of the passage? (CSAT-2011)

(a) The society has a tenacious prejudice against the intellectual potential of girls.

(b) Adults cannot be relied upon to fight on behalf of children for their Right to Education.

(c) The legal fight to get education for children is often protracted and prohibitive.

(d) There is no sufficient substitute for education received in childhood.

Directions for the following 9 (nine) items:

The following nine items (Questions 64 to 72) are based on three passages in English to test the comprehension of English language and therefore these items do not have Hindi version. Read each passage and answer the items that follow.

Passage-I

He walked several miles that day but could not get anything to eat or drink except some dry bread and some water, which he got from cottagers and farmers. As night fell, he slept under a haystack lying in a meadow. He felt frightened at first, for the wind blew awfully over the empty fields. He felt cold and hungry, and was feeling more lonely than he had ever felt before. He however, soon fell asleep, being much tired with his long walk. When he got up next day, he was feeling terribly hungry so he purchased a loaf of bread with a few coins that he had.

458. When the night fell, he slept (CSAT-2011)

(a) in the open field

(b) under a pile of dry grass

(c) in a farmer's cottage

(d) under a tree

459. He soon fell asleep because (CSAT-2011)

(a) he was exhausted

(b) he was all alone

(c) he had not slept for days

(d) he was very frightened

460. With reference to the passage, consider the following statements : (CSAT-2011)

1. He was walking through the countryside,
2. The cottagers and farmers gave his enough food so that he could sleep at night without feeling hungry.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Passage - II

I opened the bag and packed the boots in; and then, just as I was going to close it, a horrible idea occurred to me - Had I packed my toothbrush ? I don't know how it is, but I never do know whether I've packed my toothbrush. My toothbrush is a thing that haunts me when I'm travelling, and makes my life a misery, I dream that haven't packed it, and wake up in a cold perspiration, and get out of bed and hunt for it. And, in the morning, I pack it before I have used it, and it is always the last thing I turn out of the bag; and then repack and forget it, and have to rug upstairs for it at the last moment and carry it to the railway station, wrapped up in my pocket-handkerchief.

461. When he was going to close the bag, the idea that occurred to him was (CSAT-2011)

(a) unpleasant (b) sad

(c) fantastic (d) amusing

462. What makes his life miserable whenever he undertakes travelling? (CSAT-2011)

(a) Going to railway station

(b) Forgetting the toothbrush

(c) Packing his bag

(d) Bad dreams

463. His toothbrush is finally **(CSAT-2011)**

(a) in his bag (b) in his bed

(c) in his handkerchief (d) lost

Passage-III

In spring, polar bear mothers emerge from dens with three month old cubs. The mother bear has fasted for as long as eight months but that does not stop the young from demanding full access to her remaining reserves. If there are triplets, the most persistent stands to gain an extra meal and it may have the meal at the expense of others. The smallest of the litter forfeits many meals to stronger siblings. Females are protective of their cubs but tend to ignore family rivalry over

food. In 21 years of photographing polar bears, I've only once seen the smallest of triplets survive till autumn.

464. Female polar bears give birth during **(CSAT-2011)**

(a) spring (b) summer

(c) autumn (d) winter

465. Mother bear **(CSAT-2011)**

(a) takes sides over cubs

(b) lets the cubs fend for themselves

(c) feeds only their favourites

(d) see that all cubs get an equal share

466. With reference to the passage, the following assumptions have been made: **(CSAT-2011)**

1. Polar bears fast as long as eight months due to non-availability of prey.
2. Polar bears always give birth to triplets.

Which of the assumptions given above is/are valid?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

1 Comprehension_Explanations

1. Answer: (b)

Explanation:

Statement 1 is invalid: The passage explicitly states that HMPV and SARS-CoV-2 "belong to two very different virus families with fundamentally different characteristics and epidemiology." Calling them "similar viruses" directly contradicts the passage. While the passage does support that SARS-CoV-2 was novel and no one had specific immunity to it (which fits the second half of statement 1), the premise that the two viruses are "similar" undermines the conclusion as stated.

Statement 2 is valid: The passage establishes that the two viruses have fundamentally different characteristics, different epidemiology (with strong seasonality in HMPV but not SARS-CoV-2), different severity of symptoms, particularly over the long term, and "affected population segments [that] do not fully overlap." This collectively supports the inference that they have different impacts on human populations and therefore should not be dealt with in a similar manner.

2. Answer: (d)

Explanation:

Statement 1 is not correct. The passage describes existing knowledge about the two viruses (exposure history, epidemiology, symptom severity). It does not propose or develop methodologies for analysis. So "evolving methodologies for objective analysis" is not the intent.

Statement 2 is plausible but not paired correctly: The passage does contrast the two viruses' epidemiology (HMPV has strong seasonality, SARS-CoV-2 was novel, families differ, exposure history differs), so this somewhat reflects the author's intent. However, no option offers "2 only".

Statement 3 is not correct: The passage discusses neither remedies nor a comparative treatment framework. It is purely descriptive about how the two viruses differ.

Since 1 and 3 are clearly not the writer's intent, the only consistent choice is None of the above.

3. Answer: (b)

Explanation:

Statement 1 is correct: The passage states people have been exposed to HMPV "for decades" and the virus "is well-studied," whereas SARS-CoV-2 was "a truly novel virus" in 2019. So HMPV has a longer history of documentation and study than SARS-CoV-2.

Statement 2 is correct: The passage notes that the two viruses cause "different severity of symptoms" and that "affected population segments do not fully overlap", i.e., the differences between the viruses result in markedly different outcomes among those affected.

Statement 3 is correct: The passage explicitly says symptoms differ "particularly over the long term," and that HMPV has "no long-term post-viral symptoms" (contrasted with SARS-CoV-2's known long-term sequelae). This makes long-term impact a key differentiator between the two.

Statement 4 is not correct: The passage notes "strong seasonality seen in HMPV, unlike SARS-CoV-2." So seasonality is a difference, not a common factor between the two viruses.

4. Answer: (d)

Explanation:

Based on the passage, let us evaluate each statement:

Statement 1 is invalid: The passage mentions that in the US and Europe, generative AI tools have started offering precision farming at scale. It does not state or imply that the overall agricultural productivity in the West has "marched ahead" historically because of AI tools. The causal link is an overreach and cannot be assumed from the text.

Statement 2 is valid: The passage explicitly suggests that for "at-scale integration and accessibility" of AI tools in India, it is helpful to develop them in Indian languages and partner with AgTechs to create affordable solutions. This logically assumes that affordability and local language availability are key drivers to help these solutions reach a wider base of smallholder farmers.

Statement 3 is valid: The passage directly states that "India is starting to deploy AI for critical use cases such as weather forecasting, pest detection and control..." and follows it up by noting that "penetration is limited to a small subset of tech-savvy farmers." This confirms that while penetration is low, critical applications of AI are indeed already in use in India.

5. Answer: (a)

Explanation:

Based on the logical analysis of the statements in the context of artificial intelligence and agriculture:

Statement 1 is not correct: In the context of agricultural technology, it is typically the innovations and advancements by AgTech companies that drive the transformation of agriculture and empower farmers (making them tech-savvy), rather than the farmers driving the AgTech companies. The statement reverses the general cause-and-effect relationship highlighted in such contexts.

Statement 2 is correct: The development of government advisory services to train and assist farmers in using AI tools is a highly practical and positive step toward agricultural modernization.

Statement 3 is not correct: This is an extreme statement. Even in developed regions like the US and Europe, AI tools and precision agriculture augment and assist traditional agricultural practices to improve efficiency and yield; they have not completely replaced traditional farming practices.

Statement 4 is correct: The integration of large datasets (Big Data) for real-time agronomic analysis (such as weather forecasting, soil health monitoring, and crop predictive analytics) is already a reality in modern precision agriculture.

Since the question asks for the statements that are not correct, the correct combination is 1 and 3.

6. Answer: (b)

Explanation:

Statement 1 is invalid: The passage explicitly states that the JJ Act

allows for the possibility of trying adolescents above 16 as adults if they are accused of committing a heinous offence, not a serious offence. Furthermore, the passage notes that in the opinion of activists, a serious offence does not merit the transfer of a case to the adult criminal justice system. Therefore, the claim that only a serious offence justifies this transfer is completely contradictory to the passage.

Statement 2 is valid: The passage mentions, "The JJ Act, amended in 2021, now categorises an offence that has no minimum sentence, but has a maximum sentence of seven years or more as a serious offence." This clearly indicates that for an offence to be classified as a "serious offence," the determining factor is the maximum sentence it carries (seven years or more), given that there is no prescribed minimum sentence.

Since only the second conclusion logically follows from the text, the correct option is B.

7. Answer: (c)

Explanation:

Statement 1 is correct: The passage defines a heinous offence as "one with a minimum punishment of seven years," and explicitly cites culpable homicide and death by negligence as offences that "are not heinous offences because they do not have a prescribed minimum punishment." So if an offence has no minimum prescribed punishment, it cannot be classified as heinous under the JJ Act, 2015.

Statement 2 is correct: The passage states, "The JJ Act, amended in 2021, now categorises an offence that has no minimum sentence, but has a maximum sentence of seven years or more as a serious offence." This is exactly what Statement 2 paraphrases.

Since both statements are directly supported by the passage, Option C is correct.

8. Answer: (c)

Explanation:

Statement 1 is correct: The passage opens: "The key source of the battle for clean skies and clear lungs is the fuel we burn, from household chulhas to the thermal power plants." This directly supports the inference that the source of energy we consume is key to the battle for cleaner air.

Statement 2 is not correct: The passage states that for a ban to be effective, "people need alternatives" (citing CNG availability after diesel buses were stopped, and fiscal measures to keep clean fuel cheaper than dirty fuel). The argument is about practical alternatives, not about public will or conviction. The statement misattributes the cause of effectiveness.

Statement 3 is correct: The passage says: "The Supreme Court agreed that fiscal measures were needed to keep clean fuel cheaper than dirty fuel." This is exactly judicial approval for fiscal intervention to facilitate benevolent pricing for cleaner fuel.

9. Answer: (c)

Explanation:

Statement 1 is not correct: The passage says thermal power plants "would clean up or shut down", shutting down was one of two options. It then notes that "action on this has been patchy". So they were not "required to summarily shut down"; the requirement was either clean up or shut down, and even that has not been fully enforced.

Statement 2 is correct: The passage explicitly states: "When diesel buses were stopped, CNG supply had to be assured." This matches the statement directly.

Statement 3 is not correct: Per the passage, the Supreme Court banned pet coke, while the Delhi Government banned coal (later extended to the rest of NCR). The coal ban is therefore attributable to the Delhi Government, not the Supreme Court.

10. Answer: (c)

Explanation:

To determine the most appropriate reflection of the passage's point of view, let us evaluate each option based on the text:

Option A: The passage explicitly states that human agents could be "freed up to spend more time on creative and rewarding tasks." Therefore, it is incorrect to say there would be no work left for humans to do.

Option B: While the passage does suggest that AI might succeed in replacing certain human tasks where previous technologies failed, this option misses the author's core message. The author's main point of view is optimistic, focusing on the salutary (beneficial) effects of this shift, rather than just the redundancy itself.

Option C: This statement perfectly captures the central theme of the passage. The author argues that if AI takes over certain customer-service roles, its effects "may be salutary" because human workers can be redirected toward "creative and rewarding tasks" (more fulfilling tasks), thereby enhancing the value of their intervention (e.g., "using feedback to make products and services better").

Option D: The passage explicitly states that previous waves of technology also "stoked concerns of job losses." Therefore, it is incorrect to claim that AI has raised the alarm of worker replacement unlike previous waves.

Conclusion: Option C is the only statement that accurately and comprehensively reflects the author's optimistic point of view regarding AI in the workplace.

11. Answer: (d)

Explanation:

Statement 1 is correct: The passage opens with: "Previous waves of customer-service technology, including email and those pesky voice menus, stoked concerns of job losses…" and then says "AI could yet prove different." The pattern of fears being sparked applies across previous waves and the current one, supporting the claim that this has happened invariably with new customer-service technology.

Statement 2 is correct: The passage notes that earlier fears of job loss "fail[ed] to materialise," and that AI could free human agents up to "spend more time on creative and rewarding tasks." So technology has historically replaced specific tasks while opening up alternative channels of employee engagement, exactly as the statement says.

Statement 3 is not correct: The passage's tone is the opposite, the author thinks AI's effects "may be salutary" because workers can shift to "rewarding tasks." Nothing in the passage supports the sweeping claim that technology inevitably leads to stressful outcomes.

12. Answer: (c)

Explanation:

Let us examine each conclusion in the context of the passage:

1. **All landlords essentially have some goodness trapped within them.** This is not supported by the passage. The passage praises Rai Sahib specifically and contrasts him with selfish Zamindars. It does not say that all landlords have goodness within them.
2. **The common grazing grounds of a village are intended for use by the cattle of that village.** This does not follow exactly from the passage. The passage says that cattle from nearby villages came to the common ground to graze. It does not say that the grazing ground was meant only for the cattle of that particular village.
3. **Landlords who believe in tradition tend to be more concerned about their tenants.** This follows from the passage. Rai Sahib is described as holding to the old values, and these old values include the belief that a landlord should look after his tenants.
4. **Winds later in the day tend to be cooler post the hot winds of the morning.** This is incorrect. The passage says there was cool freshness in the air at that time, but Hori expected the rest of the day to be hot because of the scorching loo wind.

Therefore, only conclusion 3 is correct.

13. Answer: (d)

Explanation:

The question asks which statements are not correct based on the passage.

Statement 1 is not correct: The passage refers specifically to a single common grazing ground reserved by Rai Sahib. It says nothing about whether the rest of his landholdings are or aren't used for farming. The sweeping claim about "the landholdings" (all of them) cannot be drawn from the passage.

Statement 2 is not correct: The passage explicitly contrasts Rai Sahib with "selfish Zamindars" and notes that he held to "the old values" of looking after tenants. Far from being "as greedy as other landlords," he is presented as an exception to that greed.

Statement 3 is not correct: The passage states that "a number of farmers were eager to lease this bit of land and had offered a good price." So we can ascertain that Rai Sahib could have made money by leasing it out, he simply chose not to.

Statement 4 is correct: The line "any landlord who didn't look after his tenants was less than human" supports this assertion directly.

Since statements 1, 2 and 3 are the ones that are not correct, Option D is correct.

14. Answer: (c)

Explanation:

Let us evaluate each statement based on the passage:

Statement 1 is correct: The passage describes Hugh Grant's nap as "quietly radical" and an "elegant case for surrender… to the body's quiet wisdom over society's relentless performance metrics." This directly supports the idea that a radical action can be a form of mild surrender against societal expectations.

Statement 2 is correct: The author explicitly mentions the "crushing weight of being a 64-year-old man" (effect of age) and states that his nap was a surrender "Not to laziness, but to limits." This means submitting to one's physical limitations should not be mistaken for laziness.

Statement 3 is correct: The passage concludes with a clever play on words: "The perpetually sleep-deprived discovered a leading man, not of action, but of rest." Since Hugh Grant is a famous actor, the term "leading man" usually refers to a movie's main star (often an action hero). Here, the author implies that by napping publicly, Grant metaphorically performed the role of a "resting hero" for sleep-deprived people to look up to.

Since all three statements accurately reflect the passage's contents and tone, Option C is the correct answer.

15. Answer: (b)

Explanation:

Let us evaluate each statement based on the passage:

- Hugh Grant was watching the men's semifinal match between Flavio Cobolli and Novak Djokovic. This is incorrect. The passage clearly says that Hugh Grant was watching the men's singles quarterfinals, not the semifinals.
- The phrase 'unexpected ace' highlights Grant's act of catching 'forty winks'; an act viewed with opprobrium. This is incorrect. The phrase does use a tennis term to describe Grant's unexpected nap, but the act is not viewed with opprobrium in the passage. The author praises Grant and says he deserves empathy.
- Grant subjects the demands of society to the wisdom of his body. This is correct. The passage says Grant's nap was a surrender not to laziness, but to limits, and to the body's quiet wisdom over society's relentless performance metrics.

Therefore, only statement 3 is correct.

16. Answer: (a)

Explanation:

Let us evaluate the given statements based on the passage:

Statement 1 is not correct: The passage explicitly states that during the breakaway phase, the per capita income is "unchanged for the technology followers." "Unchanged" implies zero economic progress, not "slow" progress.

Statement 2 is not correct: The passage mentions that in the catch-up phase, "the followers adopt the new technology and close their per capita income gap with the technology leaders." It does not state that the leaders stagnate; leaders might still be experiencing economic growth, but the followers grow at a faster rate to close the gap.

Statement 3 is correct: The passage clearly states that in the fine-tuning phase, "participants try to extract the remaining benefits from an increasingly exhausted technology, leaders and followers have similar per capita incomes." This perfectly matches the phrasing of the statement.

Statement 4 is correct: The passage provides a clear chronological timeline of technological development: Scavenging (3 million years ago) → Hunting (1 million years ago) → Farming (12 thousand years ago) → Industrial technology (200 years ago). Industrial technology followed scavenging (true), scavenging preceded hunting (true), and hunting was followed by farming (true).

Since only statements 3 and 4 are correct, Option A is the correct answer.

17. Answer: (b)

Explanation:

Statement 1 is not correct: The passage says that the convergence model divides economic eras into three phases: the breakaway phase, the catch-up phase, and the fine-tuning phase. It does not say that it divides nations into three phases of economic progress.

Statement 2 is correct: The passage begins by saying that convergence theory explains the process by which countries close their labour-productivity gap with the technology leader. Hence, the idea of closing this productivity gap is central to the theory.

Statement 3 is correct: Technology leaders are the ones that move ahead first when a new technology develops. The passage says that during the breakaway phase, the per capita income of technology leaders rises while that of followers remains unchanged. This supports the idea that technology leaders typically arrive earlier at different economic eras.

Statement 4 is not correct: The passage mentions technologies such as scavenging three million years ago, hunting one million years ago, farming twelve thousand years ago, and industrial technology a little more than two hundred years ago. These are separate reference points in history and should not be added together to get 4012200 years.

Therefore, statements 2 and 3 are correct.

18. Answer: (a)

Explanation:

Statement 1 is valid: The passage says that Kalagram represented India's diverse regions through seven meticulously crafted Sanskriti Angans. Hence, it is valid to conclude that seven Sanskriti Angans representing different regions of India were showcased in Kalagram.

Statement 2 is valid: The passage states that the thematic zones were inspired by iconic temples such as the Dakshineshwar Kali Temple and the Brahma Mandir, and that these zones displayed regional artistry. Thus, regional artistry was presented through zones whose design drew inspiration from iconic temples.

Statement 3 is not valid: The passage says that 230 master artisans shaped India's ancient history into creations using age-old techniques. However, it does not say that India's ancient history itself was crafted by contemporary craftsmanship. This statement overstates the idea.

Statement 4 is not valid: The passage gives examples from Bengal, Assam, Tamil Nadu, and Madhya Pradesh, but it does not say that art forms from all regions of India were showcased. Therefore, this is an overgeneralisation.

Therefore, only statements 1 and 2 are valid.

19. Answer: (a)

Explanation:

Option A is not correct: The passage names art forms from four states, but only two of them are paintings: Bengal's Pattachitra paintings and Tamil Nadu's Thanjavur paintings. The other two are Assam's bamboo crafts and Madhya Pradesh's tribal sculpture, neither is a painting. So "paintings from four States" is factually inconsistent with the passage.

Option B is correct: The passage says: "stepping through the grand portal was like entering another world." So Kalagram is indeed likened to another world.

Option C is correct: The passage explicitly calls the Angans "living galleries", "…showcased in these living galleries…"

Option D is correct: The passage says: "These thematic zones, inspired by iconic temples like the Dakshineshwar Kali Temple and the Brahma Mandir…", directly supporting that Kalagram is divided into thematic zones inspired by well-known temples.

Since Option A is the only statement not correct, Option A is the answer.

20. Answer: (d)

Explanation:

Based on the passage, let us evaluate each conclusion:

Statement 1 is valid: The passage explicitly states that "Sport is not just about winning medals" and emphasizes the need to develop a culture where sport is "seen as a way of life."

Statement 2 is valid: The passage highlights that today's youth are "inactive and rarely leave the house" (which implies seclusion) and directly states that "Sport can help tackle these problems."

Statement 3 is valid: The opening line, "Sport is not just about winning medals or getting jobs through a sports quota," acknowledges that securing a job through a sports quota is indeed a reality and a benefit of sports, even though it is not the only benefit.

Statement 4 is invalid: The passage mentions that sport teaches "time management, fitness, teamwork, coordination," which are quintessential skills required in the standard corporate sector. Therefore, the claim that these skills cannot be learnt through sport contradicts the passage.

Since statements 1, 2, and 3 are valid conclusions, the correct option is D.

21. Answer: (b)

Explanation:

Statement 1 is not correct: The passage notes that "parents often say that their children are inactive and rarely leave the house", this is parents reporting their observations, not the passage criticising parents for being unsupportive. The claim that parents are not encouraging is not made in the passage.

Statement 2 is correct: The passage explicitly says that "from sport you can learn time management, fitness, teamwork, coordination, and so much more", these are life skills, so participation in sport helping develop life skills is directly supported.

Statement 3 is correct: The passage closes with: "We need to develop a culture in which sport is seen as a way of life, a path to building a healthier, happier society." Treating sport as a way of life is presented as a path that reshapes society, which matches the statement.

22. Answer: (d)

Explanation:

Statement 1 is not correct: The passage says that if oil prices fall, output prices do not change, and quantities remain the same, real value added should not change. However, the statement says real value should not change. The passage is specifically about real value added after deflating inputs and outputs, not about real value in general. Hence, the statement is not accurately framed.

Statement 2 is not correct: The passage says that most countries use double deflation, where input and output prices are deflated separately.

It does not say that this practice is universally adopted by all economies. In fact, the passage states that India uses the same deflator for inputs and outputs, which shows that the practice is not universal.

Therefore, neither statement 1 nor statement 2 is correct.

23. Answer: (a)

Explanation:

Statement 1 is valid: In economic accounting, the choice of deflation strategy (such as single deflation versus double deflation) can significantly impact the calculated real value added. For instance, if input prices fall sharply and a single deflator is used instead of separate ones, the real value added might be overestimated, making manufacturers appear to be performing better than they actually are.

Statement 2 is invalid: Double deflation is a method where gross output and intermediate inputs are deflated separately using their own distinct price indices (an output price index for output and an input price index for inputs). Deflating both against a single price index is contrary to the concept of double deflation.

Therefore, only assumption 1 is valid.

24. Answer: C

Explanation:

This captures the central idea:

- There's a conflict between livelihood sustenance and ecosystem degradation.
- Each affects the other, causing imbalance.
- This is exactly what the passage highlights.

25. Answer D

Explanation:

- Assumption I is invalid because the passage clearly mentions that in developing countries, dependence on ecosystems for income is significant.
- Assumption II is also invalid as the passage does not mention sharing resources between countries as a solution.

26. Answer: B

Explanation: The passage emphasizes *path-dependence*—firms innovate more the more they already do—and shows how continued green innovation (like solar energy becoming cheaper) follows this pattern. While timing and economies of scale are mentioned, the core message is that innovation builds on itself over time, making path-dependence central.

27. Answer D

Explanation: Neither assumption is directly supported by the passage. It avoids predictions about specific countries or ease of transition.

28. Answer: (d)

Explanation:

The passage discusses state-specific challenges and opportunities in tackling climate change. It emphasizes that different regions require tailored solutions (e.g., coastal vs Himalayan vs peninsular wind zones).

29. Answer: D

Explanation : The passage does not explicitly link green energy to mitigation strategies (I), nor claim effects are *more severe* in specific regions (II).

30. Answer: (d)

Explanation: The author clearly states that **insecurity is a more acutely felt economic issue** than poverty. This is further supported by the mention that **even those above the poverty line** face economic uncertainties.

31. Answer: (a)

Explanation:

Statement I is valid: The passage explicitly says that **even those just over the poverty line** face **multiple economic insecurities**.

Statement II is not valid: The passage does **not mention peace or social equality**, nor does it imply that **poverty eradication alone would achieve that**.

32. Correct Answer: A

(a) The benefit of economic reforms percolates down more slowly to the agriculture sector than in other sectors of the economy.

Explanation:

This option best captures the essence of the passage, as it reflects that agricultural growth lagged behind the non-agricultural sector—implying that reforms and growth-inducing changes impacted agriculture at a much slower pace.

33. Answer : (d)

Explanation : Neither I nor II is a valid assumption as per the passage. Both are possible policies, but the passage never indicates that they would reduce the divergence.

(d) Neither I nor II

34. Correct answer: (a)

Explanation:

The passage critiques the narrow focus on tradition, calling for a broader, more inclusive view that reflects diverse identities and histories. This aligns with the passage's main idea of challenging the restrictive heritage-based narrative and promoting inclusivity.

35. Answer: Option (d) is correct.

Explanation:

Assumption I: The passage critiques the nature of State involvement, not its existence. It calls for a broader approach, not a complete withdrawal.

Assumption II: The passage blames the narrow traditional narrative for the lack of public interest, not the appeal of the products themselves.

36. Answer: D

Explanation: The passage emphasizes that traditional teaching focused on exam preparation must change to foster critical thinking, creativity, and active learning, meaning teaching methods need to be reinvented.

37. Answer: D

Explanation: The passage does not mention evolving higher education

aligning with societal changes (I) nor discuss funding issues (II). Hence, neither assumption is supported.

38. Answer: B

Explanation: The passage highlights the inefficiency in converting plant-based food into animal products, showing low calorie recovery from animals compared to direct plant consumption.

39. Answer: B

Explanation: The passage supports assumption II, as it explains calorie loss in wealthier societies due to converting plant food into animal feed. Assumption I is not mentioned or implied.

40. Answer: B

Explanation: The passage highlights that reducing emissions needs both individual behaviour change and government or company incentives to promote low-carbon choices.

41. Answer: A

Explanation: The passage supports assumption I, stating supply-side investments help encourage low-carbon behaviour. It does not claim that people cannot change without the government or companies, so assumption II is not valid.

42. Answer: C

Explanation: The passage emphasizes reusing materials, especially in batteries, to recover valuable resources and reduce environmental risks, highlighting the benefits of a circular economy.

43. Answer: D

Explanation: The passage does not state that automobile factories currently follow circular economy (I) nor explicitly claim that economic growth is always compatible with circular use of minerals (II).

44. Answer: C

Explanation: The passage warns that relying solely on genetically refined crops risks food security due to vulnerability to diseases, highlighting the need for broader strategies beyond just productivity.

45. Answer: C

Explanation: The passage implies that climate change may spread diseases to new zones (I) and suggests that genetic diversity, including wild relatives, is important for crop resilience and food security (II).

46. Answer: D

Explanation: The passage highlights that opponents keep leaders cautious and balanced, preventing arbitrary rule and ensuring healthy democracy.

47. Answer: D

Explanation:

The passage does not state that opposition is needed only if the leader is indifferent (I), nor does it suggest that more aggressive opposition always leads to better governance (II).

48. Answer: A

Explanation:

The passage highlights how corporate capitalism drives economic growth by modernizing the economy and creating jobs, which supports democracy's development.

49. Answer: C

Explanation: The passage states corporate capitalism generates jobs (supporting I) and increases state revenue for welfare and leveling opportunities (supporting II).

50. Answer: A

Explanation: The passage highlights how voluntary associations act as a check on State power, limiting its ability to impose will when opposed by organized groups.

51. Answer: B

Explanation: The passage explains that inflation as a single number averages diverse price changes across goods and services, which vary for different income groups, so the current measure doesn't accurately reflect price rises for individual items or groups.

52. Answer: D

Explanation: The passage highlights that trust is fundamental to entrepreneurship and global markets, even though it involves vulnerability and risk of betrayal.

53. Answer: (d)

Explanation:

Statement 1 is not correct: The passage highlights science's influence on human history but doesn't claim mankind's existence depends solely on it. So, the statement overstates the argument.

Statement 2 is not correct: While acknowledging science's impact, the passage doesn't assert it will shape mankind's destiny. It presents a nuanced view, raising questions rather than making definitive claims.

54. Answer: (b)

Explanation:

Assumption 1 is invalid: The author questions, rather than affirms, that modern horrors are due to science. The passage suggests the cause may lie elsewhere, not in science itself.

Assumption 2 is valid: The passage stresses that science isn't separate from human agency. It supports the idea that science is shaped by human choices and actions.

55. Answer: (c)

Explanation:

Option (c) is correct: Option (c) is correct: The passage highlights that values like justice and love are universal but expressed differently across cultures. Travel helps one distinguish between local and universal values, aligning perfectly with this option.

56. Answer: (a)

Explanation:

Assumption 1 is valid: The passage suggests that travel helps one understand values like justice and honor across cultures, supporting the idea that it enhances awareness of human differences.

Assumption 2 is valid: It clearly states that learning universal values is a key benefit of travel, directly affirming the assumption.

Assumption 3 is invalid: While travel builds understanding of values, the passage doesn't claim it enables one to resolve conflicts—only to better comprehend them.

57. Answer: (d)

Explanation

Assumption 1 is invalid: The passage discusses educational challenges faced by families but doesn't compare the roles of parents and government. So, assuming parents play a greater role is unfounded.

Assumption 2 is invalid: While the passage mentions academic lags, it doesn't suggest a cause or propose a uniform curriculum as a solution. This assumption goes beyond the passage.

58. Answer: (c)

Explanation:

Option (c) is correct: The passage highlights a gap between students' academic abilities and educational expectations. Option (c) rightly suggests the need for policies to bridge this mismatch.

59. Answer: (d)

Explanation:

Option (d) is correct: : It best reflects the author's view of science's dual role—seeking truth like Newton and Galileo, and promoting social welfare, a modern idea from the Industrial Revolution era.

60. Answer: (a)

Explanation:

Option (a) is correct: In this passage, it is mentioned that "Planning in India, and its education differ from the West". So, the statement 1 which says - "Curriculum for urban planning courses should have a diverse and interdisciplinary approach" best reflects the most logical and rational inference of the passage.

61. Answer: (a)

Explanation:

Assumption 1 is correct: From the above given passage, we can assume that India needs a new generation of urban professionals with knowledge relevant to modem urban practice.

Assumption 2 is not correct: This is an extreme statement.

62. Answer: (b)

Explanation:

Option (b) is correct: From the given para "When anonymous private entities with high capital can pay for more space for their opinions, they are effectively buying a louder voice", it can be said that the Social media are not ideal or moral institutions but the products built by companies to make profits.

63. Answer: (b)

Explanation:

Statement 1 is incorrect: The author focuses on equity, not inclusivity. While social media allows everyone to voice opinions (inclusivity), those with more money get greater visibility, highlighting inequality in influence, not access.

64. Answer: (b)

Explanation:

Statement 1 is incorrect: The passage doesn't blame current food distribution methods for wastage.

Statement 2 is correct: Food wastage leads to land degradation, harming productivity.

Statement 3 is correct: Reducing food loss lowers the carbon footprint.

Statement 4 is incorrect: Post-harvest technologies are not discussed in the passage.

65. Answer: (c)

Both assumptions are correct:

1. The passage implies food distribution must be improved to reduce wastage.
2. Food wastage involves human actions, making its reduction a social and moral duty.

66. Answer: (c)

Statement 1 is correct: The passage explicitly mentions: "Without the fiscal backup, monetary policy eventually loses traction." So, monetary policy alone is insufficient to tackle inflation

Statement 2 is correct: Monetary policy and fiscal policy is interdependent. So it is obvious that the effects of monetary policy depend on the fiscal policies pursued by the government.

67. Answer: (d)

Statement 1 is not correct: Fiscal policies of governments not solely responsible for higher prices. Monetary policies of the central government are also responsible for higher and lower prices.

Statement 2 is not correct: In this passage, the author did not mention anything about effects of higher prices on government bonds. So, the assumption can not be made.

68. Answer: (a)

Explanation:

Option (a) is correct: It captures the passage's message that poetry embraces life's plurality and imperfections, unlike science or religion, which seek absolute truths.

69. Answer: (d)

Explanation:

Assumption 1 is incorrect: The passage shows flowers serve a practical role in pollination, not as Nature's luxury.

Assumption 2 is incorrect: The author views flowers primarily for their function, not just their beauty.

70. Answer: (a)

Explanation:

Option (a) is correct: From the sentence "A normatively responsible media through its communication effect has the responsibility to circulate the content of reality without distortion.", we can clearly say that the crux of the passage is that responsible media should not distort the real in an ideal democracy.

71. Answer: (d)

Explanation:

Option (d) is correct: Amongst the given options, the statement " Interconnected devices can be at security risk and home users may have privacy risk" best addresses the most logical, rational and practical message implied by the passage, i.e. the privacy and security risks associated with interconnected home devices.

72. Answer: (c)

Explanation:

Option (c) is correct: The passage emphasizes that sustainable, resource-efficient development is essential for inclusive growth and avoiding extreme inequality.

73. Answer: (b)

Explanation:

Option (b) is correct: The passage explains a method where injecting CO_2 into basalt forms limestone, effectively capturing and storing carbon dioxide as stone.

74. Answer: (a)

Explanation:

Assumption 1 is valid: The passage links iceberg meltwater to phytoplankton growth, which supports the marine food chain, making this a reasonable inference.

Assumption 2 is not valid: The passage doesn't discuss wider climate effects or impacts on world fisheries, so this assumption isn't supported.

75. Answer: (c)

Explanation:

Option (c) is correct: This option aligns well with the passage. The ability of caterpillars to turn off plant defences using their frass suggests the potential for developing new types of pesticides that could be organic and ecologically sustainable. This is a logical and practical application of the finding.

76. Answer: (c)

Explanation:

Inference 1 is correct: The passage criticizes fixed class durations and uniform lectures that leave many students behind, leading to low graduate employability.

Inference 2 is correct: The passage highlights how conventional classrooms eliminate differentiation by treating all learners the same way for the same time.

77. Answer: (d)

Explanation:

Inference 1 is incorrect: The passage criticizes fixed-duration classes for poor employability and calls for education reform to improve outcomes.

Inference 2 is incorrect: The passage highlights problems in the current system and warns that repeating the same approach won't improve skill education or employability.

78. Answer: (a)

Explanation:

Assumption 1 is valid: The passage shows parents' controlling nature makes adolescents uncomforta-ble and less likely to confide in them.

Assumption 2 is not valid: The passage doesn't indicate that adolescents lack respect for their parents.

79. Answer: (a)

Explanation:

Option (a) is correct: It captures the passage's central idea that parental control often causes conflict and secrecy during adolescence, weakening parental influence.

80. Answer: (a)

Option (a) is correct: A national legal and policy framework for water allocation ensures fair and equitable distribution based on objective criteria.

Options (b) and (c) are incorrect: Linking rivers and creating channels are large, complex projects with environmental and social risks, not immediate practical solutions.

Option (d) is incorrect: The passage does not recommend this as a solution.

81. Answer: (d)

The statement "For rapid economic growth as envisaged by us, attention should be paid to health and nutrition of the people" best reflects what is implied by the passage.

82. Answer: (c)

Option (a) is incorrect: The passage recognizes the government's role but also highlights the importance of individual actions by farmers.

Option (b) is incorrect: This statement isn't supported; the passage focuses on farmers' challenges and the need for adaptation.

Option (c) is correct: Farmers' risk perceptions are key to motivating adaptation decisions, especially given their constraints.

Option (d) is incorrect: The passage supports both adaptation and mitigation; it doesn't say mitigation is impossible.

83. Answer: (c)

Option (c) is correct: Delaying action on carbon emissions until technology improves is unwise, as emissions must be addressed now. The passage doesn't claim technology alone will solve climate change or that new industries must only use renewable energy.

84. Answer: (a)

Option (a) is correct: The passage implies society is more likely to spend on curing problems than preventing them, due to reluctance in adopting

lifestyle changes despite their benefits.

85. Answer: (b)

Explanation:

Option (a) is incorrect: This is not supported by the passage.

Option (b) is correct: The passage highlights poor nutrition and its health impacts, emphasizing food-based nutrition in policy.

Option (c) is incorrect: The passage doesn't advocate genetically modified crops as the solution.

Option (d) is incorrect: Food fortification is not explicitly proposed as a solution in the passage.

86. Answer: (b)

Option (b) is correct: The passage highlights that knee-jerk pollution measures are ineffective, advocating for evidence-based policies instead.

Other options are incorrect: The passage doesn't focus on penalties, lack of laws, or administrative issues.

87. Answer: (b)

Option (a) is incorrect: The passage doesn't explicitly say access to external financing is a global agenda.

Option (b) is correct: Good corporate governance boosts firm credibility, improving access to financing and growth.

Option (c) is incorrect: The passage doesn't claim international markets ensure good governance.

Option (d) is incorrect: Supply chains are not mentioned in the passage.

88. Answer: (b)

Option (b) is correct: Elephants are keystone species that support biodiversity by shaping landscapes, preventing overgrowth, and dispersing seeds to aid plant regeneration.

89. Answer: (d)

Option (d) is correct: The passage does not make any explicit statements regarding the involvement of government agencies or private sector enterprises in the collection, processing, segregation, resource recovery, or recycling of municipal waste. Therefore, both assumptions 1 and 2 cannot be confirmed based on the information provided in the passage.

90. Answer: (c)

Option (a) is incorrect: The passage doesn't support that energy from municipal waste is inexpensive.

Option (b) is incorrect: Biomethanation is mentioned but not as the "most ideal" energy source.

Option (c) is correct: Segregation of municipal waste is crucial for effective waste-to-energy processes.

Option (d) is incorrect: The passage states over 50% of waste is biodegradable, supporting energy generation potential.

91. Answer: (b)

Statement 1 is incorrect: It does not make a definitive statement about organic farming being inherently unsafe.

Statement 2 is correct: It highlights the confusion among them and mentions the lack of research and regulation regarding the use of certain products in organic farming. This implies that there is a need for more education to ensure the safe and eco-friendly production and consumption of organic food.

92. Answer: (c)

Option (c) is correct: The passage highlights challenges in organic farming and stresses the need to guide farmers toward sustainable practices.

93. Answer: (b)

Statement 1 is incorrect: Monoculture is not inevitable for achieving SDGs and zero hunger.

Statement 2 is correct: Monoculture reduces nutritious indigenous food and harms land, water, and food quality.

Statement 3 is correct: Increased food production hasn't solved malnutrition; policies must focus on nutrition security.

Statement 4 is incorrect: The passage doesn't mention subsidies or prices tied to monoculture practices.

94. Answer: (a)

Option (a) is correct: The correct option that fits the question's framing (in terms of assumptions) is (a) 1 only, as it is primarily focused on the warmer and longer springs due to global warming.

95. Answer: (a)

Option (a) is correct: Improving nitrogen use efficiency is vital for sustainable food production and reducing environmental harm.

Options (b), (c), and (d) are incorrect: These statements go beyond or are not supported by the passage.

96. Answer: (a)

Option (a) is correct: Climate justice—including adaptation, compensation, and support—must be clearly included in future climate agreements.

Options (b), (c), and (d) are incorrect: These points are not discussed or supported by the passage.

97. Answer: (d)

Option (a) is incorrect: The passage supports adding non-agricultural food sources, not replacing other trees.

Options (b) and (c) are incorrect: These points are not stated or implied in the passage.

Option (d) is correct: Agroecosystems should complement conventional agriculture to improve food security.

98. Answer: (a)

Option (a) is correct: Proper effluent treatment protocols are needed to prevent pollution and drug-resistant infections from pharmaceutical waste.

Options (b), (c), and (d) are incorrect: These do not address the specific issue or go beyond the passage's scope.

99. Answer: (a)

Option (a) is correct: The passage stresses the need for universal quality education with foundational skills for India's global growth.

Options (b), (c), and (d) are incorrect: These introduce claims or ideas not supported by the passage.

100. Answer: (d)

Option (a), (b), and (c) are incorrect: They either misinterpret or oversimplify the passage's message.

Option (d) is correct: The passage highlights that having too many choices can be overwhelming and stressful.

101. Answer: (d)

Statement 1 is correct: People used to traditional methods need a flexible environment for financialization.

Statement 2 is correct: Customized solutions can help overcome hurdles like lack of information and skepticism.

Statement 3 is correct: The passage implies modern solutions are needed, even if financial technology isn't explicitly mentioned.

102. Answer: (b)

Statement 1 is correct: Patent protection burdens public purchasing power for accessing medicines.

Statement 2 is incorrect: Dependence on other countries isn't directly discussed as a burden.

Statement 3 is incorrect: Public health policy design isn't mentioned.

Statement 4 is correct: There must be a balance between patentees' rights and patients' access to affordable medicines.

103. Answer: (c)

Statement 1 is correct: Data control by individuals shows privacy is both a fundamental right and economically important.

Statement 2 is correct: The passage links strong data privacy with encouraging innovation in a user-centric environment.

104. Answer: (b)

Option (a) is incorrect: It not supported by the passage as it does not explicitly discuss the quality of education.

Option (b) is correct: The most likely statement that reflects what the author intends to say is: (b) Unemployment must be seen as a function of rising education and aspirations of young Indians.

Option (c) is incorrect: The passage focuses on rising education levels and employment related challenges faced by young Indians, rather than the availability of labor-intensive industries.

Option (d) is incorrect: It does not specifically suggest that the education system should be designed to enable self-employment.

105. Answer: (a)

Option (a) is correct: Training statesmen in scientific methods and philosophy is ideal, though the passage doesn't specify how to achieve it.

Options (b), (c), and (d) are incorrect: They either misinterpret or overstate points not explicitly supported by the passage.

106. Answer: (d)

Option (d) is correct: Based on the passage, the ultimate goal of the state can be expressed as: (d) Liberty The passage emphasizes that the end of the state is not to dominate or restrain individuals, but rather to free each person from fear and provide them with full security. It aims to enable individuals to live and act with freedom, without causing harm to themselves or others. All this suggests liberty as a core goal.

107. Answer: (a) is correct.

Statement 1 is correct: Heavy reliance on fossil fuels makes current production unsustainable.

Statement 2 is correct: Finite resources make the current production and consumption patterns unsustainable.

Statement 3 is incorrect: Recycling reduces, not worsens, the unsustainability of production.

108. Answer: (b)

Statement 1 is correct: Developed countries must support developing countries' shift to sustainable development.

Statement 2 is correct: Providing loans for public transport at low interest is one way developed countries can help.

Statement 3 is incorrect: The passage doesn't specify how developed countries should encourage changes in production and consumption.

109. Answer: (c) is correct.

Statement 1 is correct: Deforestation leads to soil depletion, turning green areas into barren land.

Statement 2 is incorrect: The passage does not mention impacts on the common man.

Statement 3 is correct: The passage highlights harsher climatic conditions and the loss of perennial water sources.

110. Answer: (c)

Option (c) is correct: The statement that best reflects the message of the author of the passage is that "We need to devise a method of barring incompetence from public office."

111. Answer: (d) is correct.

Option (a) is incorrect: The passage does not mention confusion about methodology or classification, only that current poverty measures are too narrow.

Option (b) is incorrect: The passage does not discuss rural or urban poor.

Option (c) is incorrect: The passage does not talk about a lack of uniform global standards for income poverty measurement.

Option (d) is correct: The key message is that poverty cannot be measured solely by income or purchasing power; it also involves limited choices and opportunities beyond income.

112. Answer: (a) is correct.

Option (a) is correct: The passage explains that income poverty is just one part of poverty, which fundamentally involves deprivation and lack of genuine opportunities due to social and personal constraints.

Option (b) is incorrect: The passage does not discuss factors affecting purchasing power.

Option (c) is incorrect: The passage does not address whether poverty is temporary or permanent.

Option (d) is incorrect: The passage does not consider the temporal dimension of poverty (whether it occurs at certain times or always).

113. Answer: (a) is correct.

Option (a) is correct: The passage emphasizes deprivation as a lack of real choice or opportunity, not just income.

Option (b) is outside the passage's scope: No mention of rural or urban poor.

Option (c) is incorrect: The passage discusses lack of opportunity, not missed opportunities, or multiple personal circumstances.

Option (d) is incorrect: The passage does not imply that limitations on choice are permanent — the goal is to overcome these deprivations.

114. Answer: (d) is correct.

Assumption 1 is invalid: The passage states that many advances come from the West, but this doesn't imply they are exclusive to developed countries. Technological progress in agriculture also occurs elsewhere.

Assumption 2 is invalid: The author cautions against blindly copying Western methods due to differing climates, not against adaptation altogether. He supports context-specific improvements in developing nations.

115. Answer: (a) is correct.

Assumption 1 is valid: The passage highlights the need to adapt both traditional and modern techniques, including basics like fertilizer use, showing a clear call for changing agricultural practices in developing countries.

Assumption 2 is invalid: The passage does not discuss infrastructure in developed countries or their food wastage, making this assumption beyond its scope.

116. Answer: (a)

Statement 1 is correct: The passage highlights that traditional methods of increasing yields for staples are reaching their limits, and that climate change may further reduce farming areas. It suggests that applying technology and adapting practices—especially among smallholders and subsistence farmers—can help increase food availability. **This implies that feeding future generations will be challenging.**

Statement 2 is not correct: The passage does not suggest that corporate farming is a viable solution for food security in poor countries.

117. Answer: (c) is correct.

Statement 1 is correct: As per the passage, natural selection means that species that have not adapted to their environment become extinct.

Statement 2 is incorrect: The phrase "All the existing species" is not correct in this option. Itis clearly conveyed in the passage that only those species that cannot adapt to environmental changes will go extinct.

Statement 3 is correct: Because the passage says, "This failure to survive, in modern terms, can be attributed to the genomes which are unable to withstand geological vagaries or biological mishaps (infections, diseases and so on)"

Statement 4 is correct: Because the passage says, "In the biological evolution on the earth, extinction of the species has been a major feature". Therefore, we can consider this as a common feature.

118. Answer: (c)

Option (a) is incorrect: The passage rejects the doctrine of creation as an explanation for evolution, so its validity isn't being "undermined"—it's irrelevant to the scientific explanation.

Option (b) is incorrect: While extinction relates to environmental and biological factors, this doesn't explain why Darwin's theory is not considered just a theory.

Option (c) is correct: The passage affirms evolution as a fact, with Darwin's ideas strongly backed by diverse scientific evidence, establishing its credibility beyond mere theory.

Option (d) is incorrect: The passage refutes creationism, attributing the survival of organisms to adaptation, not to divine design.

119. Answer: (a) is correct.

Statement 1 is correct: The passage states that species unable to adapt to environmental or biological challenges face extinction, linking it to genetic limitations.

Statement 2 is incorrect: The passage does not make any prediction about future extinctions, so such a conclusion is unwarranted.

Statement 3 is incorrect: Darwin's theory addresses the process of evolution, not all natural phenomena, making this statement overly broad.

120. Answer: (b) is correct.

Statement 1 is incorrect: A decline in illiteracy depends on policy efforts, not directly on the operation of the demographic dividend.

Statement 2 is correct: A demographic dividend implies a larger working-age population, which reduces the proportion of dependents like the very young and old.

Statement 3 is incorrect: The passage does not link the demographic dividend to controlling or stabilizing population growth, making such a claim unfounded.

121. Answer: (d) is correct.

Statement 1 is incorrect: While the passage highlights the demographic dividend as a valuable opportunity for growth, it does not claim it is essential, meaning growth can occur without it.

Statement 2 is incorrect: The passage emphasizes skill development and human capital but does not specifically mention higher education as a requirement for rapid economic growth.

122. Answer: (c) is correct.

Option (a) is incorrect: The passage does not quantify how many people fail to accumulate money, so claiming that most do is not supported.

Option (b) is incorrect: The passage clearly discusses people who do fail to save money, so saying most never fail is inaccurate.

Option (c) is correct: The passage implies that some people are unable to save enough to meet their needs, which supports this option.

Option (d) is incorrect: While the passage emphasizes the importance of saving, it does not claim that anyone who fails to do so will never succeed—just that they are more likely to be misled or unsuccessful.

123. Answer: (a) is correct.

Option (a) is correct: The passage advises that to succeed, one should avoid being influenced by others' opinions when making decisions.

Option (b) is incorrect: The focus is on saving enough money to meet needs, not accumulating the maximum amount possible.

Option (c) is incorrect: Though opinions are called cheap, the passage does not reject all advice—only warns against being easily influenced by unhelpful opinions.

Option (d) is incorrect: While accepting others' opinions may hinder success, the passage does not claim that rejecting them guarantees success.

124. Answer: (c) is correct.

Option (a) is incorrect: The author states that even seemingly irrelevant research can be useful when applied by a skilled investigator.

Option (b) is incorrect: The passage clearly says research is always valuable, not just occasionally.

Option (c) is correct: The author emphasizes that no research is completely without value.

Option (d) is incorrect: The passage acknowledges some research, like mere reorganization of existing material, may be less worthwhile.

125. Answer: (c) is correct.

Option (a) is incorrect: The author says there is always a chance research may be valuable but doesn't specify how often results are useful.

Option (b) is incorrect: Not every result will be useful to an intelligent investigator; some research may be futile if unoriginal or purely literary/historical.

Option (c) is correct: The passage suggests research is rarely valueless and any result can be valuable to a talented investigator.

Option (d) is incorrect: The passage doesn't say research must be useful to the intelligent investigator, only that even seemingly pointless research might prove valuable to someone.

126. Answer: (c) is correct.

Statement 1 is correct: The passage states that social order is a foundational right based on conventions, not nature, enabling other rights.

Statement 2 is correct: Since social order underpins all rights, human rights can only be exercised when public order exists.

127. Answer: (d) is correct.

Option (a) is incorrect: The passage does not suggest dietary changes or advocate returning to a cereal-based diet.

Option (b) is incorrect: No recommendations for future actions are provided; it only highlights the water usage issue.

Option (c) is incorrect: Social behavior is not discussed; the focus is on economic development and its effect on food and water use.

Option (d) is correct: The passage emphasizes that water management in India must change dramatically to meet rising demands from both diet and energy production.

128. Answer: (b) is correct.

Option (a) is incorrect: The author doesn't oppose hard work but stresses the need for balance to allow spiritual growth.

Option (b) is correct: The passage highlights that excessive work prevents intellectual and spiritual development, calling overstraining the major problem.

Option (c) is incorrect: The focus isn't on poor thinking ability but on the lack of time after work for reflection or spiritual pursuits.

129. Answer: (d)

Option (d) is correct: The passage states that due to the overwhelming labor and constant busyness in modern life, individuals do not pursue self-improvement but rather seek entertainment that allows them to be mentally idle and forget their routine activities. This clearly suggests that the reason for not seeking self-improvement is that man prefers amusement, which keeps him mentally idle.

130. Answer: (a) is correct.

Option (a) is correct: The passage highlights that educated people appreciate technical knowledge, are happier, and engage in meaningful leisure activities.

Option (b) is incorrect: The passage does not claim all economic growth leads to more leisure, nor does it promote excessive leisure without knowledge.

Option (c) is incorrect: The passage specifically links education to happiness among leisured populations, not the entire population.

Option (d) is incorrect: The passage discusses how to best use leisure time among an existing leisured group, not that mechanization creates leisure time.

131. Answer: (d) is correct.

Option (a) is incorrect: Large dams face high costs, seismic risks, and social-environmental issues, making them impractical nationwide.

Option (b) is incorrect: Changing crops or methods alone won't effectively manage excess water, which would still be lost.

Option (c) is incorrect: River interlinking poses serious sustainability problems like displacement, ecosystem damage, and loss of farmland.

Option (d) is correct: Storing excess water by refilling aquifers and using dams is practical, economical, and sustainable, ensuring longer groundwater availability.

132. Answer: (b)

Explanation:

The passage **clearly suggests multiple possible reasons** why adults may feel less thrilled by presents:

- "Perhaps it is because we have too much already."

- "Perhaps it is because we have lost the fullness of the joy of giving, and with it the fullness of the joy of receiving."
- "Perhaps we are too cautious, too anxious, too sceptical."

This shows that the **author is offering multiple possible explanations, making option (b) correct**

133. Answer: (a)

Explanation:

The author writes:

- "Children's fears are poignant, their miseries are acute, but they do not look too forward nor too far backward."
- "Perhaps some of our cares would shrink if we thought less about them..."

This suggests that adults worry too much about what's behind and what's ahead, which prevents them from fully enjoying the present — unlike children who live more in the moment. The author is critical of this tendency.

134. Answer: (c)

Option (a) is incorrect: As the passage does not presuppose that, a person cannot survive without adhering to any particular religious denomination.

Option (b) is incorrect: The paragraph primarily addresses how religion is interpreted from various angles and not how religion is propagated.

Option (c) is correct: If we assume that our religion is the only true one and that all others are false, we are supposing that there are no shared values or principles that unite all men together.

Option (d) is incorrect: People have a tendency to reject and disapprove of other faiths because they think they fully understand their own religion and they base their opinions only on this understanding.

135. Answer: (c)

Option (a) is incorrect: The passage doesn't suggest that issues like sedition or wars are unavoidable or unrelated to subjects' behavior.

Option (b) is incorrect: The passage does not place the entire responsibility for resolving problems solely on the sovereign.

Option (c) is correct: The passage states that more crime arises from poor rules and lack of unity, implying a good state results from wise governance and unity.

Option (d) is incorrect: The passage doesn't discuss the sovereign's inherent ability to establish good dominion, so this can't be concluded.

136. Answer: (c)

Statement 1 is correct: The passage implies equality is essential for addressing social inequities.

Statement 2 is incorrect: While inequality violates democratic norms, the passage doesn't claim it threatens democracy's survival.

Statement 3 is incorrect: The passage aims to eradicate inequalities, so suggesting they're inevitable contradicts its purpose.

Statement 4 is correct: The passage describes equality as a moral default based on shared fundamental characteristics.

137. Answer: (b)

Option (a) is incorrect: While mentioned, it's not the passage's main argument or core message.

Option (b) is correct: The passage explains how democracy and oligarchy can both become tyrannies if their principles are taken to excess.

Option (c) is incorrect: The author doesn't argue that democracy is preferable or provide ways to achieve it; education is only one factor mentioned.

Option (d) is incorrect: The author doesn't oppose all forms of government despite acknowledging tyranny can arise in any system.

138. Answer: (a)

Option (a) is correct: The passage highlights the complex challenges in energy and climate policy, including fossil fuel limits and economic instability.

Option (b) is incorrect: The passage only briefly mentions India's energy policy and SDGs, without discussing their roles in decision-making.

Option (c) is incorrect: Focusing only on climate change conflicts misses the broader challenge of energy policymaking outlined in the passage.

Option (d) is incorrect: The passage acknowledges fuel imports but also considers demand-side issues, showing policymaking complexity beyond just imports.

139. Answer: (d)

Option (a) is incorrect: The passage doesn't specify any restrictions on what people should consume.

Option (b) is incorrect: The passage doesn't mention substituting plant-based foods for animal products; it focuses on antibiotic use.

Option (c) is incorrect: The passage does not call for banning antibiotics on animals, only reducing their use as growth promoters.

Option (d) is correct: The passage highlights the harmful effects of antibiotics as growth promoters and notes the ban on such use by some farming industries, like Denmark.

140. Answer: (d)

Statement 1 is incorrect: The passage links rising food prices to high fuel costs but does not blame oil-producing nations for these prices.

Statement 2 is incorrect: The passage identifies emerging economies' consumption as a cause of potential food crises, not as victims of such crises.

141. Answer: (d)

Statement 1 is incorrect: The passage says GDP growth alone isn't enough; income redistribution is also key for development.

Statement 2 is incorrect: Rising GDP doesn't guarantee fair income distribution among all families.

142. Answer: (d)

Option (a) is incorrect: The passage does not mention large woody species clearing space for fig trees, so statement 2 can't be confirmed.

Option (b) is incorrect: While the passage supports statement 3 about biodiversity around fig trees, it's not the only valid assumption.

Option (c) is incorrect: Although fig trees aid seed dispersal, there's no mention of woody species clearing the way for their expansion (statement 2).

Option (d) is correct: Only this option aligns with the passage and the explanations above.

143. Answer: (a)

Option (a) is correct: The passage highlights integrating trees with animals and crops—practices like polyculture and mixed farming—as key agroecological methods.

Option (b) is incorrect: Fertigation and hydroponics are artificial techniques, which don't align with the natural, ecological principles of agroecology described.

Option (c) is incorrect: For the same reasons as (b), these techniques don't fit agroecological systems.

Option (d) is incorrect: Vertical farming, like hydroponics, is soilless and thus not compatible with agroecology's focus on natural soil nutrient cycles.

144. Answer: (a)

Statement 1 is correct: The passage shows gadgets aren't fully secure as hackers can control devices like cars and pacemakers.

Statement 2 is incorrect: The passage doesn't blame software companies for neglecting cybersecurity.

Statement 3 is incorrect: While data security laws may be needed, the passage mainly highlights vulnerabilities due to complex technology.

Statement 4 is incorrect: This inference is not supported by the passage.

145. Answer: (c)

Option (a) is Incorrect: The passage doesn't link personal hygiene obsession to neglecting community hygiene or mention multi-drug resistant germs.

Option (b) is Incorrect: Personal hygiene reduces infections; the passage doesn't claim market entry raises disease risk but warns about false security.

Option (c) is Correct: Western Europe's experience shows community hygiene policies reduce diseases, proving personal hygiene alone isn't enough.

Option (d) is Incorrect: Since statements 1 and 2 are wrong, this option is invalid despite statement 4 being correct.

146. Answer: (b)

Statement 1 is Incorrect: Low-intensity light stops wildflowers from flowering, so it's not worse than high-intensity light.

Statement 2 is Correct: Light pollution reduces aphids by limiting food, potentially harming the ecosystem permanently.

Statement 3 is Incorrect: The passage doesn't say white light is better than other colors for flowering.

Statement 4 is Incorrect: Low light inhibits flowering and affects aphids, showing optimal light levels matter for the ecosystem.

147. Answer: (d)

Statement 1 is Incorrect: Saying cereal production is "impossible" without pollinator diversity is an extreme claim.

Statement 2 is Incorrect: The passage doesn't say monoculture harms insect survival or agricultural diversity directly.

Statement 3 is Correct: Pollinators need natural vegetation, so their scarcity in farmed areas without it is implied.

Statement 4 is Correct: Loss of insect diversity affects plant diversity, showing their interdependence.

148. Answer: (c)

Statement 1 is incorrect: Climate-smart agribusiness practices are not discussed in the passage.

Statement 2 is incorrect: The passage mentions groundwater reliance but not sustainable agriculture or dry-land cropping.

Statement 3 is correct: The crux is that climate change impacts water resources and hydrological cycles.

Statement 4 is incorrect: The passage does not mention risky coping strategies or unsustainable livelihoods.

149. Answer: (c)

Statement 1 is correct: BPA may affect embryonic development in vivo, as suggested by the passage.

Statement 2 is incorrect: The passage does not confirm that assays help identify treatments for pollution-related illnesses.

Statement 3 is correct: Embryonic stem cells can be used as models to assess the effects of environmental pollutants.

150. Answer: (c)

Option (a) is incorrect: Wealth generation is a reason for exploration but not the main driver; other motives like scientific curiosity exist.

Option (b) is incorrect: The passage only discusses initial exploration based on economic factors, not future predictions.

Option (c) is correct: The passage highlights economic reasons for exploration, including the spice trade and gold search.

Option (d) is incorrect: Risk-taking is mentioned but not as the sole way to generate wealth.

151. Answer: (d)

Option (a) is incorrect: The agreement is in the second part. Hence, this option is incorrect.

Option (b) is incorrect: Since it is not always possible to tell the truth, the first part agrees on telling lies.

Option (c) is incorrect: The first option, like option (b), is primarily focused on the agreement.

Option (d) is correct: We can infer from the passage that the first half of the discussion is on the idea of disagreement about the harm in telling the truth.

152. Answer: (b)

Option (a) is incorrect: The passage says people mix truth and lies to lessen harm but doesn't link this to good behavior.

Option (b) is correct: Good people deliberately use some falsehood to reduce harm from the truth, as highlighted in the passage.

Option (c) is incorrect: The passage does not advocate falsifying facts, only mixing truth with some lies.

Option (d) is incorrect: The passage doesn't support lying outright; it suggests only partial concealment of truth.

153. Answer: (b)

Statement 1 is incorrect: The passage explains the "built-in tension" between markets and democracy as systemic, not just a vague idea.

Statement 2 is incorrect: The passage notes unemployment and skill gaps, showing that markets alone don't ensure democratic efficiency.

Statement 3 is correct: Government programs are needed to address market failures and ensure equity for those left out.

154. Answer: (d)

Option (a) is incorrect: The passage does not mention the expectations of parents or kids.

Option (b) is incorrect: There is no reference to the time spent on academics versus skill-building activities.

Option (c) is incorrect: The passage doesn't specify who should be responsible for holistic education, only that schools currently fall short.

Option (d) is correct: The passage emphasizes teaching beyond academics—covering societal issues and life skills needed in the current world order.

155. Answer: (b)

Option (a) is incorrect: The passage highlights the benefits of online learning but doesn't discuss flaws in traditional teaching.

Option (b) is correct: The passage clearly states universities share free video lectures and encourage global integration of these online courses.

Option (c) is incorrect: The passage does not claim investment in higher education is unnecessary; free content is a result of such investments.

Option (d) is incorrect: The passage doesn't specify types of institutions but focuses on improving education through online pedagogy.

156. Answer: (c)

Option (a) is incorrect: The passage does not mention administrative issues from lack of autonomy; it focuses on climate change due to population and infrastructure.

Option (b) is incorrect: Though population growth is noted, the main focus is climate change requiring a holistic approach, which (b) lacks.

Option (c) is correct: The passage emphasizes climate problems from population and infrastructure growth, advocating sustainable solutions.

Option (d) is incorrect: The passage mentions public participation but not the private sector's role, so no conclusion about public-private partnerships can be drawn.

157. Answer: (a)

Option (a) is correct: The passage links the concept of sin to the formation of a civil state, which defines right and wrong by common consent.

Option (b) is incorrect: The reference is to a civil state created by consensus, not to a distinct ruling authority.

Option (c) is incorrect: The passage discusses self-interest in a natural state but does not label humans as immoral.

Option (d) is incorrect: While self-interest is emphasized in the natural state, it is not directly connected to survival, but rather to the absence of external accountability.

158. Answer: (c)

Statement 1 is incorrect: The passage does not address globalization but focuses on ethical and jurisdictional challenges posed by modern technologies like driverless cars.

Statement 2 is correct: The passage explicitly questions who should make decisions in case of accidents involving autonomous vehicles across jurisdictions.

Statement 3 is correct: It highlights cross-border dilemmas in ethics, legal responsibility, and the impact on global insurance and investment decisions.

Statement 4 is incorrect: The author raises questions but does not propose any concrete solutions to the issues discussed.

Statement 5 is correct: The entire passage centers around the ambiguities and ethical uncertainties created by emerging technologies.

159. Answer: (d)

Option (a) is incorrect: The author emphasizes the need for systemic reforms, which go beyond mere regulation to address unsustainable lending.

Option (b) is incorrect: The passage supports recapitalization of public banks, not lowering interest rates, making this option irrelevant.

Option (c) is incorrect: Tax incentives for new borrowers are not discussed in the passage and fall outside its scope.

Option (d) is correct: The author clearly states that without systemic reforms, unsustainable lending will continue to burden the banking system.

160. Answer: (a)

Option (a) is correct: The passage highlights that monetary and fiscal policies must work together, implying coordination between the government and the Central Bank.

Option (b) is incorrect: While potentially true, this statement is beyond the scope of the passage.

Option (c) is incorrect: This option is both contextually irrelevant and makes an exaggerated claim.

Option (d) is incorrect: The passage does not discuss financial sector reforms, only the interplay of macroeconomic policies.

161. Answer: (b)

Statement 1 is correct: Because the passage makes it obvious how negative foreign investment results into.

Statement 2 is incorrect: Since it suggests that the State has the power to intervene and lessen volatility in foreign private investment. Also, the passage makes no mention of the state's capacity to lessen such volatility.

Statement 3 is correct: Domestic private investment can be strengthened using adequate policies.

Statement 4 is incorrect: Passage has no mention of this.

Statement 5 is correct: Since it emphasises the significance of public investment in specific areas.

162. Answer: (a)

Statement 1 is correct: The passage supports optimizing uneven river water distribution across the country.

Statement 2 is correct: The mention of droughts and floods implies the need for better water distribution.

Statement 3 is incorrect: The passage does not mention borewell credit issues; it's irrelevant to the main argument.

Statement 4 is incorrect: There is no mention of a lack of regulation on agricultural water use.

Statement 5 is incorrect: The passage does not attribute unequal distribution to a lack of cooperation between governments.

163. Answer: (c)

Statement 1 is incorrect: Because it suggests that lower tax rates will always result in increased investment in higher education. Although the sentence describes the relationship between the two directly, the phrase "invariably" makes it sound extreme.

Statement 2 is incorrect: Rather than the other way around, economic freedom brought about by reduced tax rates ensures investment in higher education.

Statement 3 is correct: Increased economic freedom allows for investments in education, which are essential to creating human capital.

164. Answer: (a)

Option (a) is correct: It is implied that user charges are one way to create financial mechanisms needed for ULBs to function.

Option (b) is incorrect: The issue is the lack of financing methods, not the inefficient use of existing resources.

Option (c) is incorrect: The passage suggests that sustainable water delivery is possible with proper financing, so water scarcity isn't inevitable.

Option (d) is incorrect: Population growth is not mentioned as a cause of water scarcity in the passage.

165. Answer: (b)

Statement 1 is incorrect: The passage implies financial challenges in less affluent cities but does not suggest they cannot ensure sustainable water service.

Statement 2 is correct: Sustainable water delivery includes investment in infrastructure like sewers and treatment plants, as mentioned in the passage.

166. Answer: (c)

Statement 1 is an invalid assumption: The passage only states that 85% of farms are small and marginal, without linking them to hindering structural or rural transformation.

Statement 2 is a valid assumption: The passage clearly states that a favorable price scenario led to a rise in private agricultural investments.

Statement 3 is a valid assumption: The passage implies the importance of developing value chains for high-value agricultural products to aid rural transformation.

Statement 4 is an invalid assumption: There is no mention of global prices being necessary to alleviate poverty in India.

167. Answer: (d)

Option (a) is incorrect: Focus is not on general productivity.

Option (b) is incorrect: Price shift is a factor, not the main idea.

Option (c) is correct: Highlights inclusive agri-growth as the core idea.

Option (d) is incorrect: Too broad; not limited to agriculture.

168. Answer: (d)

Option (a) is incorrect: The passage mentions labor moving to unregistered manufacturing, not rural-to-urban migration.

Option (b) is incorrect: The passage notes poor job quality in construction/manufacturing but doesn't specify working conditions as the cause.

Option (c) is incorrect: The passage states services aren't generating enough jobs; it doesn't support the claim made here.

Option (d) is correct: The author suggests boosting sectors like health and education to create needed jobs.

169. Answer: (a)

Statement 1 is a valid assumption: The government must prevent private parties from infringing on citizens' rights and audit its role in protecting digital privacy.

Statement 2 is a valid assumption: The state must actively stop private interference to ensure citizens can exercise their digital privacy rights.

Statement 3 is an invalid assumption: The passage does not claim the digital economy harms privacy; it highlights risks that can be managed by proper government action.

170. Answer: (c)

Option (a) is incorrect: The Ramsar Convention is neither mentioned nor implied in the passage.

Option (b) is incorrect: The passage advocates increasing ecological system capacity, not modernizing water systems as stated.

Option (c) is correct: The passage highlights that water is part of natural ecosystems, so wetlands must be reinforced beyond being mere water sources.

Option (d) is incorrect: The author does not suggest or imply this measure.

171. Answer: (a)

Option (a) is correct: The passage clearly states that choosing an asset allocation strategy is the most important investing decision, linking it to predictability and risks.

Option (b) is incorrect: The passage does not discuss how to create wealth; it only focuses on asset allocation.

Option (c) is incorrect: There is no mention or implication of asset management companies; the passage suggests individuals make their own decisions.

Option (d) is incorrect: This statement cannot be concluded from the passage.

172. Answer: (b)

Statement 1 is a valid assumption: The passage highlights biotechnology's evolving role beyond GM crops, indicating ongoing regulation due to its broad potential.

Statement 2 is an invalid assumption: The passage does not mention public participation in biotechnology policy decisions.

Statement 3 is a valid assumption: The passage calls for a comprehensive approach considering socio-economic impacts to harness biotechnology's benefits while minimizing harms.

Statement 4 is an invalid assumption: There is no reference to the political executive's role in regulating biotechnology.

173. Answer: (c)

Option (c) is correct: The passage highlights biotechnology's current role in producing genetically modified (GM) crops with specific traits and emphasizes its broader potential to develop varieties that help mitigate and adapt to climate change. This makes option (c) the best reflection of the passage's main idea.

174. Answer: (c)

Option (a) is incorrect: The passage discusses the decrease in loans to the industrial sector but does not infer the banking system's overall contribution to economic growth.

Option (b) is incorrect: The passage highlights shortcomings in reforms and mentions missing elements like a resolution corporation for banks, so the conclusion in option (b) cannot be logically drawn.

Option (c) is correct: The passage discusses the failure to create a resolution corporation for banks during the 1991 reforms, leading to unresolved bank failures.

Option (d) is incorrect: The passage addresses the lack of institutional safeguards against bank failures but does not mention promoting foreign investment in the industrial sector as a solution.

175. Answer: (c)

Option (a) is incorrect: The government is committed to reducing emissions with gas and renewable energy.

Option (b) is incorrect: Renewables are a better long-term choice than imported gas.

Option (c) is correct: Renewables face short-term cost and tech challenges, so investment hesitancy is valid.

Option (d) is incorrect: The passage doesn't support option (d) as a main conclusion.

176. Answer: (a)

Statement 1 is a valid assumption: The passage hints at government subsidies and overinvestment in renewables before the technology is fully ready.

Statement 2 is an invalid assumption: The government plans to reduce emissions not only through gas but also by increasing renewable energy use.

177. Answer: (c)

Statement 1 is correct: Genome editing involves pinpointing DNA regions in plants to improve traits like disease resistance and productivity.

Statement 2 is correct: The passage states that detailed genome analysis and precise gene alteration now speed up this evolutionary process.

178. Answer: (c)

Option (a) is incorrect: The passage explains the combination of unscientific solid waste management with poor drainage and sewage dumping, referring to different types of drains, not just drainage.

Option (b) is incorrect: The passage discusses problems from waste and sewage but does not mention lack of funding as a cause.

Option (c) is correct: It highlights that people mainly understand only that dry garbage attracts pests, missing other harmful effects of dumping trash into storm drains.

Option (d) is incorrect: The passage mentions clogged drains causing waterborne diseases but does not mention the lack of drinking water.

179. Answer: (b)

Option (a) is incorrect: The passage states that Article 25 is subordinated to other fundamental rights in Part III, so the state can intervene if those rights are affected.

Option (b) is correct: The passage explicitly says Article 25 is subject to other fundamental rights in Part III, allowing state intervention when those rights are at stake.

Option (c) is incorrect: Article 25 guarantees freedom of religion as a fundamental right in Part II.

Option (d) is incorrect: The passage notes that religious practices are subject to state regulations under public order, morality, and health conditions.

180. Answer: (d)

Statement (d) is correct: The passage highlights how the lack of diversity in staple foods made the country vulnerable to a fungus, causing famine without substitutes. Though other statements convey messages, only (d) captures the main point opposing food source uniformity.

181. Answer: (b)

Statement 1 is incorrect: Improving child nutrition cannot happen by just monitoring the PDS system.

Statement 2 is correct: Delaying marriage and childbirth improves the mother's and child's health, leading to better nutrition.

Statement 3 is correct: Mothers transfer immunity proteins (immunoglobulins) to unborn children, boosting their nutrition and survival.

Statement 4 is correct: Less than 50% of Indians have access to clean water and sanitation, contributing to high neonatal mortality; improving WASH is crucial.

Statement 5 is correct: Poor child health stems from malnutrition and related diseases, so vaccination drives are essential for child survival.

182. Answer: (c)

Option (c) is correct: Statements 1, 4, and 5 are incorrect based on the passage's points about pulses, straw burning, and causes of stubble burning.

183. Answer: (d)

Option (d) is correct: The validity of the assumptions is asked by the question. All three assumptions are true in this case. Statement 2 is true because the entire passage's main point is this. Statement 1 and 3 are consistent with international standards for managing dams and water.

184. Answer: (c)

Statement 1 is correct: The passage highlights the government's key role in economic development, showing liberalization was driven more by economic needs than public priorities.

Statement 2 is correct: The passage implies social sector policies have not been effectively implemented.

185. Answer: (d)

Statement 1 is invalid: The passage blames economic liberalization (globalization) for jobless growth, poverty, and inequality, so the assumption is wrong.

Statement 2 is invalid: The passage suggests meaningful development—not liberalization—can reduce poverty and create jobs; liberalization is seen as causing unemployment.

186. Answer: The correct option is option (b).

Option (c) isn't the main inference, and option (d) is only a secondary issue. Option (b) is the most logical main inference. Option (a) discusses regulatory problems but isn't the key point. So, (b) is correct.

187. Answer: (d).

Option (a) is incorrect because it mentions the protection of local markets but says nothing about how effective the regulations are.

Option (b) is incorrect since the passage states governments prioritize social and environmental concerns over commercial ones.

188. Answer: Answer: The correct option is (d).

Option (a) is wrong because treatment is a possibility if the reason is identified. Option (b) seems illogical, and nowhere is it addressed (inoculation etc.). This alternative doesn't appear anywhere and seems too improbable. (Inoculation etc.) Option (c) is incorrect because this we do not know from the passage). Option (d) - It is a logical, sensible, and important conclusion regarding the decisions that need to be made in terms of nutrition policy, etc.

189. Answer: The correct option is option (b)

Option (b) is correct: Assumption 1 doesn't make sense because the passage does not talk about the benefits of processed probiotic foods. Assumption 2 according to the text, assumption 2 is unquestionably true. (The last two sentences).

190. Answer: The correct option is option (c).

Statement 1 is incorrect: It is a general statement which does not find a mention in the above passage. Statements 2 and 3 are correct: As both, statements are explicitly mentioned in the passage.

191. Answer: The correct option is option (b).

Assumption 1 cannot be supported by this paragraph because only one case is given. Therefore, choices (a) and (d) are eliminated. Assumption 2 is obviously incorrect as the opposite is mentioned, So, choice (C) is not a possibility. Therefore, (b) is accurate.

192. Answer: The correct option is option (b).

Assumption 1 is incorrect: Talks about the "extinction of plant species" but the passage is about varieties". Assumptions 2 and 4 are correct: As they are the main points of the passage. Therefore, the answer is option (b). Assumption 3 is incorrect: If you check assumption 3, it is not necessarily valid always, but only when a crisis (mutating crop disease or future climate change) happens. Hence, eliminate options (a) and (c).

193. Answer: The correct option is option (a)

Option (a) is correct: The passage criticises the strategy of private corporations and identifies what India needs which private firms won't do, so only public enterprises can or must do). Option (d) is not incorrect; however, the best response is option (a).

194. Answer: The correct option is option (a)

Option (a) is correct: Assumption 1 is valid since that is expressly stated in the passage. Assumption 2 infer that GM technology will "never" be able to do it is an overly general assumption that may not be true. Thus, choice (a) is the most valid choice.

195. Answer: The correct option is option (c).

Option (a) is not the most logical conclusion. Option (b} is too extreme even if we do not use thig reasoning (trick), as there may be situations where we need such laws. Option (c) as per the passage option (c) is accurate, making it the best choice. Option (d) inappropriate because it refers to "foreign plants."

196. Answer: The correct option is option (a)

Option (a) is correct: Assumption 2 is correct as it talks about "Great investments need to be made". Options (b) and (c) are therefore incorrect. Assumption 1 can be said as true. (Start of the passage). So, the optimal response is option (a).

197. Answer: The correct option is option (d).

Option (d) is correct: Options (a) and (b) are incorrect, given that is one of the arguments made at the outset, option(c)is not incorrect. The optimal choice is option (d), as it also addresses option (c). The concerns of working- class and poor people will be taken care of naturally if we begin involving people's groups in city planning. In reality, there is a lot

more room for improvement.

198. Answer: The correct option is option (c).

Option (a) is correct: Nowhere does Statement 4 appear. Therefore, options (b) and (d) are eliminated. Now statement 3 - Given that the growth of the economy has been driven primarily by services, it is a decent assumption to make to equate the service sector with the organised sector. Therefore, 3 is correct. This eliminates makes option (c) the preferred choice.

199. Answer: The correct option is option (c).

Option (c) is correct because it places emphasis on the banking correspondents' abilities. Option (a) and (d) are incorrect as well because no conclusion has been reached anywhere in the text. Option (c) is preferable to option (b) because it addresses the main issue under discussion.

200. Answer: The correct option is option (d).

Option (d) is correct: Option (b) is categorically disqualified because it is "beyond repair" That is never mentioned in the text. This approach does not discuss Option (c). The first statement can also imply that political theorists have chosen option (a), despite the fact that it sounds correct. The optimal choice is option (d).

201. Answer: The correct option is option (d)

Option (d) is correct: The passage makes it abundantly clear that historical injustice encompasses a wide range of phenomena, including economic hardship, social discrimination, and lack of acknowledgement. Assumption 1 as a result, may not be accurate. Assumption 2 Nowhere is clearly stated. Therefore, neither is true.

202. Answer: The correct option is option (b)

Option (b) is correct: In assumption 1 we assume that being "Well Educated" refers to having both information and culture and it is an extreme statement. So, it is not accurate. Assumption 2 is explicitly stated, so assumption 2 is unquestionably true.

203. Answer: The correct option is option (b)

Option (b) is correct: Assumption 1 is incorrect because the text makes no mention of the connection between soil erosion and food insecurity. The use of the term "alarming" does not imply the presence of food insecurity. Assumption 2 is incorrect because the passage makes no note of how man-made soil erosion came to be. The finalize supports assumption 3 as being true.

204. Answer: The correct option is option (b)

Option (b) is correct: Assumption 1 is incorrect because the phrase "only reason" is used. Assumption 2 is incorrect because it inverts the connection; power is what fuels and reinforces inequality. Assumption 3 is 1 accurate is confirmed by the first statement.

205. Answer: The correct option is option (a)

Option (a) is correct: Option (a) should be generally w true, though. Even after taking all those steps, other variables might prevent farmers from benefiting. There is no reference of option (b). Since the passage is not addressed in this way, option (c) is incorrect. The ideal es option is (a). The main idea of the text is that

206. Answer: The correct option is option (d)

Option (d) is correct: Option (a) is not explicitly mentioned. Option (b) mentions "biodiversity" once more, but biodiversity is not what happens when appendages are used for various purposes. Option (c) appears to be right but is incorrect because "coincidence es in evolution" is not mentioned explicitly. Option (d) is best as it is a "very curious situation" due to "a shared evolutionary history."

207. Answer: The correct option is option (c)

Option (c) is correct: Assumption 1 is undoubtedly true. Assumption 2 is incorrect because it is implausible, at assumption 2 cannot be proven. Assumption 3 is correct as it is explicitly mentioned. Assumption 4 is incorrect n because it's unrealistic due to the excessively large sums at stake.

208. Answer: The correct option is option (d)

Option (d) is correct: Option (a) is unrelated to the passage. Option (b) is also unrelated because the text makes no mention of "India". Option (c) is incorrect because this is not the passage's subject. The best answer is option (d).

209. Answer: The correct option is option (b)

Option (b) is correct: Obviously, the optimal conclusion is option (b). (i.e., guidance for the future).

210. Answer: The correct option is option (a)

Option (a) is correct: It is the main concept. Option (b) is just a fact and not an implication of the passage. Option (c) is incorrect because developed nations do not have this privilege. Option (d) is not completely accurate.

211. Answer: The correct option is option (a)

Options (b), (c) and (d) are incorrect. Option (b) is incorrect: It misinterprets the passage; while the poor may not have caused climate change, they may now play a part in its mitigation. Option (c) is incorrect: It makes a significant presumption that might be incorrect. Option (d) is incorrect: There is no reference of Option (d) in any place. It's incorrect because "only" is present. The best alternative is. (a).

212. Answer: The correction option is option (c)

Explanation: Assumption 1 is correct: Since small farms generate a significant amount of food and are in danger, assumption 1 is correct. Assumption 2 is correct: Given the tone of the passage, assumption number two is a direct warning to India.

213. Answer: The correct option is option (c)

Option (c) is correct: Assumption 3 is correct, so choice (a) is not a possibility. Assumption 2 is incorrect because the phrase "economic growth of governments" in assumption 2 is strangely worded. We thus refuse it. This eliminates options (a) and (d). Only (c) is available as an option.

214. Answer: The correct option is option (b)

Option (b) is correct: Assumption 2 according to the text is unquestionably true. Option (d) is thus eliminated. The wording of Assumption 3 is overly strong, and it is not stated. The passage's conclusion includes a word of support for private schools. This eliminates option (c). The only

options remaining are (a) and (b). Assumption 1 is not an assumption of the passage. Therefore, the solution is (b).

215. Answer: The correct option is option (b)

Option (b) is correct: Option (a) is incorrect because it is never stated in such a fatalistic manner. Option (c) is incorrect despite appearing to be ineffective, treatment is available to the impoverished. It's incorrect to say, "have no access." Option (d) is incorrect as it deviates off course in a different path. The best response is (b).

216. Answer: (a)

Option (a) is correct: The passage states that 47% of India's population works in agriculture, with at least 42% expected in the future, making (a) the most logical conclusion.

Option (b) isn't mentioned anywhere, so it's not a valid inference.

Option (c) suggests a policy decision not implied by the passage.

Option (d) is also not supported by the text.

217. Answer:(a)

Option (a) is correct: The passage highlights that as food production becomes harder to regulate and trace, consumer vulnerability increases.

Options (b) and (c) are incorrect as they are not mentioned in the passage.

Option (d) is incorrect because the passage criticizes the globalization of food production but does not call for its stoppage.

218. Answer:(b)

Option (b) is correct: The author finds it odd that science and culture are seen as incompatible, implying disagreement with that view.

Options (a), (c), and (d) are incorrect as these points are not supported by the passage.

219. Answer: (d)

Option (d) is correct: The author uses words like "shackles," "bind," and "suppress" to show that past ties hinder India's physical and spiritual growth.

Options (a), (b), and (c) are incorrect as they don't reflect this view

220. Answer: (c)

Option (c) is correct: The passage highlights that the Right to Education focuses more on improving education quality—like curriculum reforms and teacher accountability than just infrastructure or parental attitudes. The main emphasis is on the teaching-learning process.

221. Answer:(a)

Option (a) is correct: The passage's mention of "teacher accountability systems must ensure…" shows it's an assumption, not a guaranteed fact. The author hopes for this in the future, so (b), (c), and (d) are incorrect.

222. Answer:(d)

Explanation:

Options (a) and (b) are incorrect as they are simple to rule out.

Option (c) is incorrect, as the main point is not mentioned.

Option (d) is the best response

223. Answer:(c)

The correct option is (c).

The "essential message" of the paragraph is what is demanded of us.

Option (a) is incorrect: It Does not convey the main idea. Option (b) is incorrect: Although the statement is true in itself, it does not convey the main idea. option (c) is correct : Because the text attempts to convey the idea of quality education.option (d) is incorrect: Because primary message is not mentioned.

224. Answer: (d)

Option (d) is correct: The passage links deforestation and human activities like agriculture to topsoil loss, drying rivers, and damage to groundwater and crops. All four points (1, 2, 3, and 4) are valid, making option (d) the right answer.

225. Answer:(a)

The correct option is (a).

Option (a)is correct: Statement 1 is stated in a passage about agriculture, the question is very well-founded and the conclusion is thus accurate.

Option (b) is incorrect: Statement 2 is untrue because it asserts that only human intervention can cause the salinization of land, which is not what the passage states Option (c) is incorrect: Statement 3 is untrue because the text only discusses India's extreme monoculture practices and makes no mention of the negative effects on Africa and South America.

226. Answer:(d)

The correct option is (d).

Option (a) is incorrect: Statement 1 is untrue Because desertification also happens in subtropical and temperate areas.

Option (b) is incorrect: Different negative effects of deforestation combine to ultimately cause flash floods and desertification, but this is not an assumption that is made in the passage.

Thus option (c) is also incorrect and hence by elimination option(d)is the best choice.

227. Answer:(b)

Statement 3 contradicts the idea of coping with climate change, so options A, C, and D (which include it) are eliminated.

Option (a) is also eliminated.

Option (b) is correct.

228. Answer: (c)

The correct option is (c).

Assumption 1 is correct: The passage's first sentence refers to the "diversity of natural assets."

Assumption 2 is correct: The text discusses the risks that mono-cropping practices pose to native plant varieties. "Choosing crops favours homogeneity," it states

Hence, both the assumptions are valid.

229. Answer: (c)

Option (a) is not selected Since it is not specifically mentioned

Option (b) is incorrect.

Option (c) is correct as it is mentioned.

Option (d) is incorrect because knowing that the Earth is "a biological system" is distinct from having "knowledge of biological systems".

230. Answer: (b)

The correct option is (b).

Option (a) is incorrect: Because it's not the only option

Option (b) is correct: As according to the passage,leadership skills can be taught or organically acquired, refuting the idea that people are born with them (or not at all).

Option (c) is incorrect: Because no numerical contrast is given.

Option (d) is incorrect: As there is no reference of it

231. Answer:(c)

Option (c) is correct: The passage highlights climate change as a complex issue mainly affecting developing countries that need funding to adapt, especially India. Claims 2 and 3 are true, so (c) fits best.

Options (a), (b), and (d) are incorrect.

232. Answer:(d)

Option (d) is correct: The passage highlights health risks from biomass and coal cooking fuels and the low use of clean fuels like LPG. So, providing cleaner cooking fuel through schemes like PMUY can reduce preventable deaths.

Options (a), (b), and (c) are incorrect or irrelevant.

233. Answer:(a)

The correct option is (a).

Option (a) is correct: Because science offers the possibility of a happy world, this can be an important implication.

Option (b) is incorrect: As it says only that science can. Option (c) is incorrect: As the passage mentions science for a happy world but is not adequate in and of itself. Option (d) is incorrect: As it reads "..with or without science,"

234. Answer: (b)

The correct option is (b).

Option (a) is incorrect: As nowhere addressed.

Option (b)is correct: As The passage makes it abundantly obvious that there is no legal system in place to protect the Arctic, and an ice-free summer could be the starting point for geopolitical changes as nations vie for control of the region.

Option (c) is incorrect: Because the text makes no mention of Arctic resources being able to satisfy the world's resource needs,

Option (d) is incorrect: Because an assessment of the resource quantities in the Arctic and Antarctica is omitted.

235. Answer:(c)

Option (c) is correct: It best captures the conflict India faces between WTO obligations and protecting food security, which is the key issue discussed.

Options (a) and (b) omit strong policy suggestions, and (d) only states a fact without conveying the main message.

236. Answer:(b)

Option (b) is correct: The passage highlights the outdated, factory-like education system that produces conforming students, pointing to a disconnect with today's market needs.

Statements 1 and 3 are not supported by the passage, so options (a), (c), and (d) are incorrect.

237. Answer:(d)

Option (d) is correct: The passage emphasizes the complexity of individual body types and why generic diets often fail, advocating instead for understanding weight gain causes and changing habits.

Options (a), (b), and (c) are incorrect because they are either vague, not the main inference, or too extreme.

238. Answer:(a)

The correct option is (a).

Option (a) is correct: As it makes more sense than other options.

Options (b) & (c) are incorrect: Only the dangers associated with monoculture and climate change are discussed in this passage.

Option (d) is incorrect: Because North America and Asia's potential unfavourable impacts of climate change are mentioned in the passage, this does not necessarily imply that these regions will experience the worst food shortages.

239. Answer:(d)

Option (d) is correct: It best captures the core message of the passage — that factory farming is unsustainable due to both cruelty and environmental costs.

Option (a) is incorrect because it contradicts the passage.

Option (b) is incorrect as it doesn't highlight the main point.

Option (c) is incorrect since it introduces a new idea not discussed in the passage.

240. Answer:(b)

Option (b) is correct: The passage emphasizes the need for **safe wildlife corridors** to allow animals like tigers to move freely and safely, reducing conflict with humans.

Option (a) is incorrect: It's overly extreme and not supported by the passage.

Option (c) is incorrect: There is no mention of creating **more** reserves.

Option (d) is incorrect: It unfairly generalizes a **negative outlook** on India that is not present in the text.

241. Answer:(a)

The correct option is (a).

Option (a) is correct: As moving wild animals frequently fails because the nation lacks wildlife corridors.

Option (b) is incorrect: The law in India is adequate and aids in preserving the tiger population. Furthermore, that isn't the subject of this passage's debate.

242. Answer: The correct option is option (a)

Option (a) is correct: The **passage** as a whole makes option (a) the best choice because it discusses both growth—new identities emerging—and decay—old identities disrupted. So, the appropriate response is (a).

243. Answer: The correct option is option (c)

Option (c) is correct: Option (a) is incorrect because the passage makes no mention of the notion that strong mass leaders could establish sub-national political entities. Option (b) is incorrect as the passage indicates the opposite. Option (d) is incorrect as it is not possible to reliably infer "strong regionalism" from the passage. Hence, the correct answer is (c). It can be deduced from the passage easily.

244. Answer: The correct option is option (b)

Options (b) is correct: Because the language and tone of the text make it obvious that choice (b) is the right choice. "Digital is everywhere", "condition and context that defines the shapes and boundaries of our understanding of the self, the society" etc. favour this choice Options (a), (c) and (d) are extreme statements and generalisation

245. Answer: The correct option is option (a)

Option (a) is correct: It can be inferred from the text, which states that a collapse in productivity growth is the main reason for stagnant incomes. So, the appropriate answer is (a). Options (b), (c) and (d) are extreme statements and generalisations.

246. Answer: The correct option is option (c)

Option (c) is correct: It is the best choice as it reflects the broad idea presented in the passage. Options (a) and (d) are incorrect as they can be ruled out due to their extreme wording, i.e., "only". The passage is not definitive about these. Option (b) is incorrect as it is too generalist as the context of the passage is limited to India and we have a better choice in the form of option (c).

247. Answer: The correct option is option (d)

Option (d) is correct: It can be inferred from the passage. In fact, this idea is the crux of the passage. Option (a) and (c) are incorrect as they are not implied by the passage in question. They are unrealistic. Option (b) is incorrect because it cannot be directly concluded from the passage

248. Answer: (b)

Statement 1 is not correct: It says 'Synthetic pesticides should never be used in modern agriculture' - this is a very extreme statement and not mentioned anywhere in the passage.

Statement 2 is correct: From the passage, we can draw the assumption 'One of the aims of sustainable agriculture is to ensure minimal ecological imbalance' from the concern the author is showing.

Statement 3 is not correct: in this passage, the author is saying the advantages of biopesticides but not comparing them with synthetic ones.

249. Answer: The correct option is option (a)

Statement 1 is correct: It is mentioned in the passage "... have been in nature for millions of years without any ill or adverse effects on the ecosystem."

Statement 2 is incorrect: Because it is mentioned in the passage "Synthetic pesticides are generally persistent in the environment." and not biopesticides.

Statement 3 is incorrect: Because it cannot be concluded from the passage.

250. Answer: The correct option is option (c)

Option (c) is correct: Because it is implied by the content of the passage. So, the appropriate response is (c). Options (a) and (d) are incorrect because they are not covered by the passage. Option (b) is incorrect because it conflicts with the content of the passage.

251. Answer: The correct option is option (c)

Option (c) is correct: The first line states that productive employment is essential for inclusion and growth, and the last line notes that while service jobs are reasonably productive, they are not expanding quickly. This leads to the conclusion that by fostering the rapid development of highly productive service jobs, employment growth and inclusion will be ensured Option (a) is a close to correct option but it does not talk about agriculture, hence it is not chosen Option (b) and (d) are incorrect because as they are too general in nature and even far-fetched

252. Answer: The correct option is option (a)

Option (a) is correct: Because it can be inferred from the passage. Option (b) is incorrect because it says the opposite of what the passage conveys. Option (c) and Option (d) are incorrect because they are far-fetched

253. Answer: The correct option is option (c)

Option (c) is correct: Between Option (a) and option (c), option (c) is better in keeping with the passage. More focus on public investments would increase the private investments in research and development, etc. Hence, the correct answer is (c). Option (b) and (d) are incorrect because it can be clearly concluded as per the passage.

254. Answer: The correct option is option (a)

Option (a) is correct: Because it is the best choice as it logically follows the point made in the passage. Therefore, the right answer is (a). Option (b) is incorrect because it states, "eliminate poverty," which the passage does not, because it is too specific. Option (c) is incorrect because it omits a discussion of productivity. Option (d) is incorrect because it is out of scope of passage.

255. Answer: The correct option is option (b)

Option (b) is correct: Consider the following phrase - "States will scatter in the absence of a centre", According to the passage's underlying meaning, option (b) is a correct option to choose. Other options are out of scope of the passage

256. Answer: The correct option is option (b)

Option (b) is correct: Because "Poorest are not bound to government as he didn't had a say in its formation," the passage claims. This, along with the passage's overall argument, supports "rule according to the consent

of the governed." Therefore, the right answer is (b). Other options are out of scope of the passage.

257. Answer: The correct option is option (d)

Option (d) is correct: It is evident clearly from the passage that the challenge the author of the above passage throws to the public is to achieve brotherhood, national unity and hard work.

258. Answer: The correct option is option (c)

Option (c) is correct: It is evident clearly from the **passage** that it is the sum total of the private wills of the individuals.

259. Answer: The correct option is option (d)

Option (d) is correct: It is evident clearly from the passage that in a mature democracy, force is narrowed down to a marginal phenomenon in the actual exercise of sovereignty.

260. Answer: The correct option is (a)

Option (a) is correct: Because the passage talks about voting as a duty and describes refraining from it as an act of anarchy and refusing one's responsibility. Hence, the correct answer is the best choice i.e., option (a). Other options are out of scope of the passage

261. Answer: The correct option is option (d)

Option (d) is correct: Because it is the only one that makes sense based on the passage, as can be seen from the words "...very often through the unhappy state of his country." Options (a), (b) and (c) are incorrect because it cannot be concluded from the passage. Additionally, the limiting terms "only," "must," etc. compel one to reconsider these choices.

262. Answer: The correct option is option (b)

Statement 1 is incorrect: Because it can be seen from the, last two sentences of the passage, it is in contradiction with it.

Statement 2 is correct: Because it is a reasonable assumption in light of the passage, particularly the final sentence. Consequently, option (b) is the right answer.

263. Answer: The correct option is option (d)

Statement 1 is incorrect: It is an incorrect assumption because studying historical battles is discussed in terms gf developing leadership skills rather than comprehending modern warfare.

Statement 2 is incorrect: Despite the passage's emphasis on the value of learning history, it never makes the assumption that it is necessary for all leaders, Consequently, (d) is the right option.

264. Answer: The correct option is option (c)

Option (c) is correct: It is the only choice that the: Passage supports. Additionally, this assertion sums up: the passage's main point. As a result, Option (c) is the- right answer. Other options are out of the scope of the passage.

265. Answer: The correct option is option (a)

Keynote could be interpreted as the passage's main idea, or essence. Option (a) is correct: it includes the main idea of "not. being perpetually hesitant" and "advance to eminence," which stand in for the passage's opening and closing ideas, respectively. Consequently, the right answer is Option (a).

Option (b) and (c) are out of the scope of the passage and are contradictory.

Option (d) accurately states the facts, but it does not convey the main idea.

266. Answer: The correct option is option (d)

Option (d) is correct: It can be deduced from the passage. and emerges as the **passage**'s most important takeaway:

Option (a), (c) and (b) are incorrect because it is not possible to infer them from the passage.

267. Answer: The correct option is option (b)

Option (b) is correct: The passage discusses facing and overcoming the concepts of purity and pollution.

The choice (b), "People have to perceive toilet use and pit-emptying as clean and not polluting," is the most important message in this context. Other options are out of the scope of the passage.

268. Answer: The correct option is (b)

Option (b) is correct: It seems to be the best alternative and the most important inference. Option (c) is incorrect because it does not make sense in light of the passage. Options (a) and (d) are incorrect because they refer to either natural or human capital

269. Answer: The correct option is option (b)

Option (b) is correct: The passage as a whole, particularly line two of the passage, supports option (b) as being correct. Option (d) is incorrect because it can be disregarded where the passage is silent for being overly specific. Options (a) and (c) are incorrect because they cannot be deduced directly from the text, making them incorrect.

270. Answer: The correct option is option (a)

Option (a) is correct: A passage's main concept is encapsulated in a concise but comprehensive summary. It doesn't address anything in specific, just everything the paragraph discusses. Although we desire leisure, working is necessary, as the text explains before challenging the reader to consider how much leisure time they actually have. Therefore, it is essential for human beings to work" is the main idea here. Option (a) is the right answer. options (b), (c), (d) are out of scope of the passage.

271. Answer: The correct option is option (c)

Option (c) is correct: According to the passage, habits make it less difficult for us to live, as evidenced by the last line "...make it easier for us to live" As a result, option (c) is accurate. Other options are out of the scope of the **passage**

272. Answer: The correct answer is option (c)

The given statement implies the social well-being is essential for individual development. So, rights aim at both individual and social good. Hence, option (c) is correct.

273. Answer: (c) is correct.

Theme: Problems with governance and crony capitalism:

Option (c) is correct: The first phrase of the passage, "By destroying transparency.....," makes it very evident that more transparency is required in the way that the government functions. India's economy is

expanding, yet crony capitalism might be detrimental to rapid economic expansion. A "corollary" is a claim that naturally follows from an established one. **Hence, option (c) is correct.**

274. Answer: (d) is correct.

Option (d) is correct: Concerns about development and climate adaption are the theme of the paragraph. Only option (d) comes across as reasonable, comprehensive, and neutral. There is no mention of other options. **Hence, option (d) is correct.**

275. Answer: (a) is correct.

Option (a) is correct: Theme: Hydrology and biodiversity, Water and biodiversity are interrelated, according to the passage. Although they are not incorrect, options (c) and (d) are unrelated to the query. Option (b) is an extreme choice. Thus, the best critical inference is **option (a), which is correct.**

276. Answer: (d) is correct.

Option (d) is correct: Theme: Automation in the banking industry. It is implied that "financial inclusion" must be promoted in the final sentence of the passage, "Now there is a need for a new that does not exclude the common man." The **passage** also mentions current items that serve the upper middle class and above. Thus, the **correct option is (d).**

277. Answer: (b)

Option (b) is correct: Theme: Manual scavenging and sanitation Statement 1 is obviously incorrect because nothing can be guaranteed to be "completely solved. "Clearly, the **passage** implies statement 2. **Hence, option (b) is correct.**

278. Answer: (d)

Option (d) is correct: Theme: Man's nature and proper government Option (a) is not even briefly, mentioned Option (b) is incorrect since people may not be able to fully satisfy their own needs individually, but society may be able to do so (that is why people join together to form societies). Although option (c) is correct in itself but this might also apply to other animals. Therefore, the most accurate inference is **option (d).**

279. Answer: (b)

Option (b) is correct: Theme: The role of economics and politics in society. A society's legal framework is determined by the demands that society's citizens make over time. Demands are influenced by how economic power is distributed within a society. The state runs society. Hence, the connection of (State) politics and economics is being researched. Best **option is (b)** The passage does not imply that one of them is more important than the other.

280. Answer: (b)

Option (b) is correct: Theme: Greenhouse gas emissions and agriculture Provided how extreme it is, **Option (a)** can be eliminated. **Option (b)** is expressly mentioned. **Option (c)** is incorrect. Therefore, the best response is option (b)

281. Answer: (a)

Option (a) is correct: Theme: Dietary habits in a world with a lot of people The paragraph does not only discuss developing nations. It is about how the world's resources will be impacted by the population expansion in developing nations and their use of meat and milk.

Options (c) and (d) are therefore incorrect. **Option (b)** is hot an option because it exclusively discusses developing countries. The best **Option is (a)**. It is the message that must be acknowledged the most.

282. Answer: (b)

Option (b) is correct: Theme: Different societies' lactose tolerance The **passage** discusses how various societies have evolved genetically over time as a result of their historical reliance on milk and cattle. Options (c) and (d) c are not viable due to their limitations. Option (a) is not an assumption and is too straight forward. Hence, **Option (b)** is the best option.

283. Answer: (b)

Option (b) is correct: Theme: Challenges with measuring national production. Here, every option seems to be in order! Let's examine.

Option (a) is indicated but not explicitly stated (suggested). Consider the option (b), The author is trying to communicate exactly this.

Option (b) is preferable to (a).

Option (c) can be eliminated because it involves a value judgement. Option (d) should be rejected because it is a strongly implied assertion. The text discusses the conceptual challenges of comparing country incomes. Hence, Option (b) is the response that is implied the most.

284. Answer: (d)

Option (d) is correct: Theme: The delicate balance of carbon in the atmosphere and soil. Because "solely responsible" is used, **Option (b)** is obviously incorrect. Volcanoes and other natural phenomena also do this. **Option (c)** ignores how these components are interdependent. Option (a) is invalid, not incorrect, but a fact and not a corollary (a proposition that follows from one already proved). The most suitable corollary is **Option (d).** More soil-trapped carbon will be released as the amount of green cover increases. **Hence, option (d) is correct.**

285. Answer: (b)

Option (b) is correct: Theme - Global food production, supply, demand & poverty The **passage** does not talk about setting up more agro based industries. So, statement 1 is wrong. 4 is incorrect as well because no mention of subsidies is made. Hence**, Options (a), (c), and (d) are incorrect.** The best answer is option (b). The **passage** undoubtedly makes reference to the importance of marketing regulation and pricing affordability for the underprivileged.

286. Answer: (d)

Option (d) is correct: World food production, supply, demand, and poverty are the theme. **Options (a) and (c)** can be immediately eliminated. The fact that is brought up in **option (b)** is one that affects management rather than global agriculture. The main idea is in option (d). As a result, **option (d)** is the right response.

287. Answer: (c)

Option (c) is correct: World food production, supply, demand, and poverty are the theme. As an explicit policy objective, Statement 4 is not specified. **Options (b) and (d)** are thus no longer correct options. Both

1 and 3 are made quite clear. As a result, **option (c)** is the correct option.

288. Answer: (b)

Option (b) is correct: World food production, supply, demand, and poverty are the theme. Interestingly, neither **statements 1 nor 2** are explicitly stated. Just the third statement applies to the passage. As a result, **option (b)** is the correct option.

289. Answer: (c)

Option (c) is correct: Theme: Concerns about civil. services and accountability. Statement 1 is definitely, a factor that leads to adverse consequences for governance. Statement 4 is also one of the factors that creates the issue of public administration. Hence, only. option (c) comes across as reasonable, comprehensive, and neutral.

290. Answer: (a)

Option (a) is correct: Theme: Concerns about civil services and accountability Statement 1 is correct not stated clearly but can be implied from the passage, The **passage** made the following **statement 2**: "By being answerable to the political executive, which was. accountable to the public/society, the civil services can be made accountable to the society. "That mechanism currently does not function. That does not imply that the political executive is not accountable to the general population. **Hence, option (a) is correct.**

291. Answer: (d)

Option (d) is correct: Theme: Concerns about civil services and accountability The **passage** makes indirect but minute references to the third choice (c). **Option (d)** is unambiguously expressed, while Option (a) states a fact (not implied). Not specified in **option (b). Option (d)** is clearly the correct option.

292. Answer: (d)

Option (d) is correct: Theme: Concerns about civil services and accountability You'll be confused by **Option (d) and (a).** The **passage** ends with a mention of option (d), which would increase accountability by bringing decision-making closer to the people. However, option (a)—better job security and safety measures—were implemented to ensure a feeling of security. This separated the civil service from political processes and, as a result, guaranteed external accountability. The best option is **Option (d).**

293. Answer: (a)

Option (a) is correct: Religion, justice, communities, rights, and responsibilities are the theme. The final sentence invalidates **statement 3**. The finest response, option (a), is left as the only choice.

294. Answer: (c)

Option (c) is correct: Religion, justice, communities, rights, and responsibilities are the theme. Crux (main message) of this passage is "balance between rights and responsibilities is important to the delivery of justice in society." **Hence, option (c) is correct.**

295. Answer: (d)

Option (d) is correct: Biomass fuels and the evolving debate on emissions is theme of the passage. All the three constraints given in the question are mentioned in the passage. **Hence, option (d) is correct.**

296. Answer: (b)

Option (b) is correct: Theme: The evolving debate over emissions and biomass fuels, "Energy crops compete with food production and have an impact on prices," the first paragraph is the fifth line reads. Thus, statement 3 is true. Now consider other options:

Option (a) is not an option (as statement 3 is true) The use of biomass for carbon capture and storage is mentioned in Statement 2 but this is nowhere linked directly to the issue of food security in the passage. So, the best option is (b).

297. Answer: (c)

Option (c) is correct: Theme: The ongoing debate over emissions and biomass fuels Clearly, statement 1 is incorrect. **Options (a) and (d)** are thus no longer choices. The second claim is true. Biomass as a fuel has a limited impact on food and forest supplies If those experimental emerging technologies succeed, statement 3 might be true. The best **options (c)**.

298. Answer: (a)

Option (a) is correct: The **passage** makes implicit **assumption 1**. As a result, option (a) is correct. **Assumption 2** is true only when improperly handled, not always.

299. Answer: (d)

Option (d) is correct: The theme is "The Green Revolution's High Yield Curse" "Our food supply is experiencing a dangerous decline in biodiversity. The green revolution is a mixed blessing". **Option (a), (b) and (c) is not incorrect**, but they are stated as fact. As a result, **option (d)** is the most logical and critical inference.

300. Answer: (a) is correct.

Option (a) is correct: Option (a) is the least essential as a part of the argument. The passage talks about the disparities between people's abilities and how the polity is affected by this. All other options are out of the scope of the passage.

301. Answer: (d) is correct

Option (d) is correct: Option (d) best captures the Explanation because the passage discusses the world's growing population and the efforts that must be made to ensure that the population increase does not outpace food production. Other options are nowhere mentioned in the passage.

302. Answer: (b) is correct

Option (b) is correct: Option (b) is correct because as we Consider the following lines "if we cut our defence expenditure on weapon-building, we can create a climate of peace with our neighbours". The author calls this a "false argument". This simply means that the author assumes that a greater amount of spending on weapons (opposite to what is argued) would reduce a war like possibility.

303. Answer: (c)

Option (c) is correct: Because one of the main causes of high mortality is mentioned in the passage as the absence of healthcare services. Therefore, it can be concluded that expanding access to maternal and infant health services will help to solve the issue. **Consequently, choice is the right response. (c).**

304. Answer: (c)

Option (c) is correct: Because the passage discusses how processed foods harm the ecosystem. The first claim is assumed. Additionally, the processing of food places a significant burden on our natural resources. It makes clear reference to Statement 2. **Consequently, choice is the right response. (c).**

305. Answer: (a)

Option (a) is correct: Because the answer can be deduced from the following lines of the passage—"And | can now give myself the certificate that a thoughtless word hardly ever escapes my tongue or pen. | do not recollect ever having had to regret anything in my speech or writing. I have thus been spared many a mishap and waste of time."

306. Answer: (a)

Option (a) is correct: Because the fact that the author had never written a careless word is the most suitable Explanation: for why he had avoided an accident**. Answer (a) is the finest choice.**

307. Answer: (d)

Option (d) is correct: Because according to the author, a human's propensity for exaggeration is normal. He can overcome and overcome this tendency with the aid of silence. **The best response is choice. (d).**

308. Answer: (a)

Option (a) is correct: Because according to the passage, "the richer states have a responsibility to reduce carbon emissions and promote investments in clean energy." **Consequently, choice is the right response (a).**

309. Answer: (d)

Option (d) is correct: Because there is no infidelity involved, option (a} is ruled out. Manak had a forced second marriage. Furthermore, the passage's main subject is not betrayal. The passage highlights the impact of a patriarchal mindset more so than the absence of legislative protections. **Consequently, option (d).**

310. Answer: (a)

Option (a) is correct: Because the real goal of government is to guarantee the social and political independence of the people. The optimal response is option (a).

311. Answer: (b)

Option (b) is correct: Because the passage's main topic is the understaffing of municipal corporations and the lack of specialization and professionalism among municipal employees. Consequently, option (b) is the right response.

312. Answer: (c)

Option (c) is correct: Because the text talks about how flamingos are social and how they look out for their chicks. Therefore, the optimal response is option (c).

313. Answer: (d)

Option (d) is correct: Because the passage primarily discusses the findings of a research on MGNREGS, but it also serves as an illustration of the negative effects of financial illiteracy. Consequently, choice is the right response is (d).

314. Answer: (c)

Option (c) is correct: Because the passage makes it clear that concentrating solely on economic progress has hampered human development. It does not, however, state that the sole goal should be to promote human growth. As a result, statement 2 is true and statements 1 and 3 are incorrect. The appropriate answer is choice (c).

315. Answer: (b)

Option (b) is correct: Because only statements 2 is explicitly mentioned in the passage. Sometimes, economic development has a negative impact on the environment. Therefore, the correct answer is option (b).

316. Answer: (b)

Option (b) is correct: Because the text discusses how poverty and climate change are related on a cause- and-effect basis. Only option (b) connects these two problems, making it the best response. **Option (a), (b) and (c) are not the logical proposition to the above passage.**

317. Answer: (d)

Option (d) is correct: Because according to the passage— "total debt and equity investments in emerging economies has doubled in the past decade to 12 percent". The author further states that this phenomenon is flagged as a threat that could compromise global financial stability. Therefore, **option (d) is the most appropriate response.**

318. Answer: (c)

Option (c) is correct: Because the passage discusses the negative effects of open defecation, including malnutrition, mortality, and stunted development in children. This has an effect on worker productivity. The Passage says nothing about the Centre's and the other developing nations' lack of means. Therefore, the **best response is (c).**

319. Answer: (b)

Option (b) is correct: Because the central idea of this passage is that democracy's fundamental duty should be to ensure that everyone has an equitable opportunity to succeed. As a result, **option (b) is the right response.**

320. Answer: (b)

Option (b) is correct: Because India is only mentioned in passing and not explicitly discussed in the passage. Therefore, statement 1 can be regarded as being false. The passage's final sentence notes that households spend money when there are no formal saving mechanisms in place. Therefore, statement 2 is accurate; and **option (b) is the right response.**

321. Answer: (d)

Option (d) is correct: Because the author states in the very beginning that "the existence/establishment of formal financial institutions that offer safe, reliable and alternative financial instruments is fundamental in mobilizing savings". This would help poor households access formal savings mechanism. Thus, **option (d) is the most crucial message** of the passage. Rest of the options are not relevant with respect to the passage.

322. Answer: (b)

Option (b) is correct: Because the most logical inference from the passage is what the query asks. There is no way. to consider **options (a)**

or (c) as logical premises. As 3 result, option (b) is the best alternative.

323. Answer: (d)

Option (d) is correct: Because the passage claims that economic reforms have led to price increases and high inflation. One of the many reasons for high inflation is the attempt to cut subsidies. The passage's most important lesson is thus revealed in Option (d), which highlights the link between price increases and economic changes.

324. Answer: (d)

Explanation

Option (d) is correct: Because the passage discusses how individual liberties, and the more general concept of equality can work together. **The most appropriate response is (d).**

325. Answer: (a)

Option (a) is correct: Because the passage only expresses one point of view and is narrative in character. It claims that the struggle between man and state is as old as society itself but makes no note of how the issue has been solved. As a result, option (a) is the right solution.

326. Answer: (c)

Option (c) is correct: Because according to the passage, the amount of money needed to improve mitigation and adaptation strategies in developing nations is a topic of heated speculation and discussion within the UNFCC. As a result, statements 1 and 3 are true, and option is the best solution (c).

327. Answer: (c)

Option (c) is correct: Because the passage makes a clear reference to Statement 3. The higher level of per capita income and substantial GDP of the developed nations are not mentioned. Consequently, the choice is the correct solution. (c).

328. Answer: (a)

Option (a) is correct: Because the "Domestic finances are expected to fall short of the current and projected needs of the developing countries," the passage states clearly. As a result, statement 1 is true.

329. Answer: (a)

Option (a) is correct: Because the passage highlights the differences between established and developing nations in terms of climate change.

330. Answer: The correct option is option (a)

Statement 1 is correct: It is referenced in the **passage**

Statement 2 is incorrect: The phrase "disappearance of the ecosystem itself" in statement 2 is wrong and extreme.

331. Answer: The correct option is option (d)

Option (d) is correct: Because if we carefully read the question being asked. It requests the "need" for a shift rather than the "cause" of changing to modern conservation practises. All except (d) would be correct if the question of cause was dealt with

332. Answer: The correct option is option (b)

Option (b) is correct: Because even though the passage covered all the topics covered in the question, the question explicitly asks for the "important message" and not things like "concern" or "fact," etc. Analyze this "keyword" carefully. The answers should have been (c) or {d) if the question asked about the "most important concern" (d). Answer would have been (a) if asked about the "most important fact". The most important message conveyed by the passage is mentioned in option (b).

333. Answer: The correct option is option (b)

Option (b) is correct: Because statements 1and 3 cannot be correct as it is clearly seen from the passage. So, the answer has to be (b). Simple elimination will save you a lot of time and makes your job easy.

334. Answer: The correct option is Option (a)

Option (a) is correct: Because it is directly mentioned in the passage. Option (b), (c) and (d) are incorrect as they are nowhere mentioned in the passage.

335. Answer: The correct option is option (c)

Option (c) is correct: Despite not receiving a straight mention, it is hinted at in the passage in passing. You ought to be able to understand the main idea of the text as you read it. confusion is lessened. Important point to remember is - Focus on capturing the essence (central idea/ central core/message/view) of the text as you read.

336. Answer: The correct option is option (c)

Option (c) is correct: Again, use removal to shorten the process. Of course, statement 2 is untrue. B and D are thus eliminated. Statement 3 is undoubtedly true. (You know from the previous answer). So, the solution must be (c).

337. Answer: The correct option is option (b)

Option (b) is correct: Because it is evident from the passage and is self-explanatory. Option (a), (c) and (d) are incorrect-they are out of the scope of the passage

338. Answer: The correct option is option (c)

Option (c) is correct: Because if we just concentrate on getting the main idea and overview of the text, it emphasizes on inclusive growth where gains of growth are shared widely by the population. Option (a), (b) and (d) are incorrect-they are out of the scope of the passage.

339. Answer: (c)

Option (c) is correct: The author stresses that growth should be a means to ensure inclusive prosperity, not an end in itself.

Options (a) and (b) are incorrect: Both ignore the importance of growth as a driver for inclusion; (b) wrongly assumes inclusion irrespective of growth.

Option (d) is incorrect: It misrepresents the author's balanced view by implying growth alone is sufficient.

340. Answer: The correct option is option (d)

Option (d) is correct: Because none of these are mentioned by the author. In contrast, he notes that distribution needs to be examined and that development has not been able to lift a sizable number of people out of poverty.

341. Answer: (b)

The passage suggests that an oil company can make greater profits if a

transparent formula for petrol pricing is announced every fortnight or month by undertaking innovation and cutting costs. The passage does not mention promoting sales or selling equity shares at higher prices as ways to make greater profits under the proposed system.

342. Answer: The correct option is option (a)

Option (a) is correct: Because it is clearly stated in the content of the passage. Option (b), (c) and (d) are incorrect as they are nowhere mentioned in the passage.

343. Answer: The correct option is option (c)

Options (d) and (b) in this case are readily discarded because they are not discernible from the text. Given that, to your understanding, both of the other two options exhibit traits of a free market system, aspirants may be confused between the two. And both are named in the **passage** somewhere. You now need to choose the choices with a little caution and objectivity. You must choose the answer "according to the **passage**" This sentence will makes it possible to choose option (c) — "The debates tend to focus on free market operations and forces, their efficiency and their ability for self correction". Option (a) s incorrect: Because it is not explicitly mention in the passage.

344. Answer: The correct option is option (c)

Option (c) is correct: Because the apologists' ideological presumption is that open markets are fair and competent. This is the "ideological bias", the **passage** implies Option (a), (b) and (c) are incorrect as they are out of the scope of the passage.

345. Answer: The correct option is option (c)

Option (c) is correct: Because statement 1 and 4 are correct as they talk about the private greed and it's ignorance.

346. Answer: The correct option is option (c)

Option (c) is correct: Because statements. 2, 3 and 4 can be easily deduced from the passage.

347. Answer: The correct option is option (b)

Option (b) is correct: Because the phrase "tolling unviable" clearly states that they cannot recover costs. Option (a), (c) and (d) are incorrect as they are nowhere mentioned in the passage.

348. Answer: The correct option is option (c)

Option (c) is correct: Because it can be easily deduced from the **passage** that the portfolio composition of the government refers to mix of government investment in different industrial sectors.

349. Answer: The correct option is option (a)

Option (a) is correct: Because it is obvious when the passage discusses the absence of government action and competition rules. (Last line).

350. Answer: The correct option is option (b)

Option (b) is correct: Because statements 1 and 4 would not be true as they are not stated in the passage.

351. Answer: The correct option is option (c)

Option (c) is correct: Because the central concept of the passage is to facilitate broader adaptability of crops for which it advocates promoting agricultural research. This has been stated in the passage quite explicitly.

352. Answer: The correct option is option (c)

Option (c) is correct: Because it is explicitly stated in the passage. Option (a), (b) and (d) are incorrect as they are out of the scope of the passage.

353. Answer: The correct option is option (d)

Option(d)is correct: All statements 1, 2and 3 are incorrect because without proper technological development, R&D and alternative use of energy, dependence on fossil fuels in inevitable.

354. Answer: The correct option is option (b)

Option (b) is correct: Because it is explicitly stated in the passage.

355. Answer: The correct option is option (b)

Option (b) is correct: Because according to the passage, "It is essential that we mitigate the emissions of greenhouse gases and thus avoid some of the worst impacts of climate change" and "Mitigation overall has huge co-benefits, such as lower air pollution and health benefits, higher energy security and greater employment."

356. Answer: X

Note: UPSC dropped the question after the exam.

357. Answer: X

Note: UPSC dropped the question after the exam.

358. Answer: X

Note: UPSC dropped the question after the exam.

359. Answer: X

Note: UPSC dropped the question after the exam. (CSAT-2014)

360. Answer: X

Note: UPSC dropped the question after the exam.

361. Answer: X

Note: UPSC dropped the question after the exam.

362. Solution: (c) This is quite clear from the passage.

Answer: (c)

- The **passage** critiques the idea that democracy is a **Western imposition**, asserting that this notion is flawed because it wrongly assumes democracy originated exclusively in the West.
- It emphasizes that democratic practices and **public reasoning** have existed in **non-Western societies** too, such as **ancient India**.
- It also acknowledges that although **modern institutional structures** of democracy are primarily a **Western contribution**, democracy as a concept has **broader historical and cultural roots**.

363. Answer: (d)

Statement 1 is incorrect: The passage refutes the idea that democracy is purely Western and highlights ancient Indian democratic practices, rejecting the notion that non-Western nations are incapable of democracy.

Statement 2 is incorrect: The passage critiques the idea of "imposing" democracy, not claiming Western nations always do so but challenging the view that democracy is exclusive to the West.

364. Answer: (c)

Statement 1 is correct: The passage affirms that good corporate governance involves following all laws, including labor and tax laws.

Statement 2 is incorrect: It supports the need for independent directors but does not say they should be government representatives.

Statement 3 is incorrect: It emphasizes separating personal and corporate funds, not banning personal investments altogether.

365. Answer: (a)

Statement 1 is correct: The passage states that good corporate governance positively affects a company's share price.

Statement 2 is incorrect: The passage does not claim it always leads to a rapid turnover increase; this is an exaggeration.

Statement 3 is incorrect: It says governance is one of the factors for foreign investors, not the main one.

366. Answer: (c)

The passage emphasizes that:

- Malnutrition often affects children **between 6 months and 2 years**, despite their lower food requirements.
- It is **not always due to poverty**, since even in well-fed households, children under five may not get enough food.
- The **key reason is the child's dependence** on others for feeding, often left to an **older sibling** when the mother is working.
- Therefore, **awareness about the child's food needs** and **how to meet them** is **crucial**.

367. Answer: (d)

Statement 1 is incorrect: The passage notes that working mothers are often away, affecting feeding, but doesn't imply they don't prioritize care.

Statement 2 is incorrect: It stresses the importance of awareness but does not blame Public Health authorities for failing to spread it.

368. Answer: (c)

- The passage highlights that while f**armers employ various strategies to manage agricultural risks**—such as crop diversification, non-farm employment, and storage of stocks—empirical studies indicate these traditional methods are insufficient.
- Additionally, existing institutions like share tenancy and informal credit agencies face challenges, especially when widespread risks affect large numbers of farmers in a region. This inadequacy underscores the need for policy interventions that address agricultural risks more effectively.

369. Answer: (b)

- The passage emphasizes that while individual risk-specific policies—such as crop insurance, price stabilization, and development of pest-resistant crop varieties—can address certain aspects of agricultural risk, **none are entirely sufficient on their own and may have side effects**. Therefore, a combination of both direct and indirect policy interventions is necessary to effectively mitigate agricultural risks.

370. Solution: (d)

371. Answer: (b)

- The passage explains that crude mineral oil forms from the remains of tiny marine organisms and plants that, over millions of years, accumulate under the seabed. These organic materials are covered by layers of sand and silt, which **harden into sedimentary rock.** This rock acts as a barrier, **preventing oxygen from reaching the organic matter beneath**, thereby inhibiting complete decomposition. The pressure from the overlying sediments generates heat, transforming the buried organic material into crude oil.

372. Answer: (c)

- The formation of oil deposits is significantly influenced by the accumulation of sedimentary rock layers. As these layers build up over time, **their increasing weight exerts pressure** on the organic-rich sediments beneath them. This pressure, combined with geothermal heat from the Earth's interior, **raises the temperature of these buried sediments**. The elevated heat and pressure conditions facilitate the transformation of organic matter into hydrocarbons, leading to the formation of oil deposits.

373. Answer: (d)

Statement 1 is correct: Returning slurry as manure is economically better than losing nutrients through water discharge.

Statement 2 is incorrect: The passage doesn't say water lacks microorganisms; decomposition happens but excess nutrients still cause harm.

Statement 3 is correct: Excess nutrients from runoff cause eutrophication, damaging lake ecosystems.

374. Answer: (b)

- The passage discusses how **agricultural slurry**, when **discharged into watercourses**, becomes a **pollutant** because it leads to problems like **eutrophication** and **dead zones** in aquatic ecosystems.
- However, when this same slurry is **returned to the land**, it is **decomposed by soil microorganisms**, and the **mineral nutrients** become **available to plants**. In this form, it acts as a **fertilizer**.

375. Answer: (b)

Statement 1 is correct: Agricultural runoff and slurry pollute water bodies with nutrients like nitrogen and phosphorus.

Statement 2 is incorrect: Fertilizers don't harm soil microorganisms; they help decompose slurry when returned to land.

Statement 3 is correct: Excess nitrogen and phosphorus cause eutrophication in lakes and oceans.

Statement 4 is correct: Nutrient overload leads to toxic algal blooms and dead zones in oceans.

376. Answer: (b)

Statement 1 is correct: Cultural eutrophication leads to the **degradation of ecosystem services** such as nutrient regulation, water clarity, and recreational value.

Statement 2 is correct: The process results in reduced biodiversity due to

oxygen depletion and the proliferation of harmful algal blooms, which can lead to the loss of aquatic plants and animals.

Statement 3 is not correct: This is not a characteristic of cultural eutrophication. Instead, **cultural eutrophication is characterized by an increase in mineral nutrients** (such as nitrogen and phosphorus) that lead to excessive plant and algal growth.

377. Answer: (c)

- The passage primarily **discusses the negative impacts of agricultural runoff,** particularly the discharge of agricultural slurry, on water bodies and aquatic ecosystems. It highlights how improper waste disposal leads to eutrophication, which in turn causes significant ecological problems such as loss of biodiversity, oxygen depletion, and the creation of "dead zones."
- While the passage touches on other aspects like the economic considerations of waste management and the role of microorganisms in **decomposing organic matter,** the primary focus remains on the environmental consequences of improper agricultural waste disposal on aquatic ecosystems.

378. Answer: (b)

The **central idea** of the passage is that **material aid or physical help alone** cannot eliminate human misery. The author emphasizes that:

- **Miseries will continue** as long as **man's inner nature remains unchanged.**
- **Ignorance**, impurity, and **lack of spiritual strength and education** are seen as the **root causes** of suffering.
- Only by **transforming human character**—making mankind **pure, spiritually strong, and enlightened**—can true and lasting relief from misery be achieved.

379. Answer: (d)

Statement 1 is not correct: The passage clearly states that **physical and material help alone cannot completely remove human miseries**. The author emphasizes the need for spiritual purity and education to address the root causes of human suffering.

Statement 2 is not correct: The passage suggests that while charitable homes and hospitals may provide temporary relief, they **cannot fundamentally solve the problem of human misery.** The author argues that true relief comes from changing human character and eliminating ignorance.

380. Answer: (d)

- The **central idea** of the passage revolves around the **impact of habitat fragmentation**—primarily caused by mining, highways, and human intrusion—on forests and biodiversity. It discusses how the **disruption of continuous forest landscapes** leads to **degradation** and poses a **major threat to biodiversity**, particularly extinction-prone species.
- While illegal mining and extinction are mentioned, they are **consequences** or **contributing factors**, not the core theme. The main concern is the **disruption of natural habitats** due to fragmentation.

381. Answer: (a)

The passage discusses **habitat fragmentation**—when large blocks of forests are broken into smaller patches due to **mining, highways, and other human intrusions**. This fragmentation causes the following effects:

- **Edges of forest patches come into contact with human activity**, leading to **degradation** of ecosystems.
- **Continuity of forest landscapes and corridors gets disrupted**, which is harmful to **wildlife species**, especially those that are **extinction-prone**.
- This disruption is described as **"the most serious threat to biodiversity conservation."**

382. Answer: (b)

- When Jim came home without his bicycle, his parents were **upset** that he had given away his expensive new bicycle but were secretly **proud** of his kindness and faith. This indicates that they were **concerned about his decision** yet admired his generosity.

383. Answer: (c)

- Jim returned the watch to the old man and the little boy because he was satisfied with the **condition of his bicycle upon its return**. This indicates that Jim had held onto the man's watch as a form of security to ensure his bicycle was returned in good condition. Upon verifying that his bicycle was unharmed, he returned the watch.

384. Answer: (c)

- In the passage, the narrator and John are driving on a narrow, winding road through hilly terrain at night when it begins to rain, **reducing visibility.** The narrator asks John to drive more slowly due to the difficulty in seeing the road clearly under these conditions.

385. Answer: (d)

- The travelers decided to set out for the next town despite the onset of darkness because they believed it was only about fifteen kilometers away and anticipated **finding accommodation there for the night**. This **expectation of a short journey** to secure rest motivated their decision to proceed.

386. Answer: (b)

- The travelers became worried after covering twenty kilometers because they had not yet encountered the town that was indicated on their map to be fifteen kilometers away. This discrepancy led them to suspect that **they might have lost their way.**

387. Answer: (a)

- The passage describes a stout old lady walking down the middle of a street in Petrograd, causing confusion and endangering herself. She was advised to use the pavement, which is designated for pedestrians. Therefore, the reason it was pointed out to her to **walk on the pavement** is because she was a **pedestrian**.

388. Answer: (b)

- The stout old lady **refused to move from the middle of the street** because she believed that **liberty entitled her to walk wherever she pleased**. She stated, "I'm going to walk where I like. We've got liberty now." This indicates that she felt entitled to do whatever she liked.

389. Answer: (b)

- The old lady failed to realize that **her liberty was not unlimited**. While she believed that newfound freedom allowed her to walk wherever she pleased, she overlooked the fact that **unrestricted individual liberty can lead to social chaos**. As the passage illustrates, if everyone exercised their freedom without considering others—such as pedestrians walking in the streets and drivers operating on sidewalks—universal disorder would ensue. Therefore, individual liberties must be balanced with social responsibilities to maintain order.

390. Answer: (a)

Explanation:

The passage discusses the **instrumentalist** view of education, which sees education as having a **functional, instrumental, and utilitarian dimension**. It mentions that this view answers the purpose of education in terms of **employment, upward mobility, and national development**, implying that education is **a means to an end** rather than an end in itself.

391. Answer: (c)

Explanation:

The passage argues that education should not be seen merely as a means to an end (such as employment or social mobility). Instead, it **must be respected in itself** because it is an **inner process of moral and intellectual development**. The passage explicitly states that education is about **the widening and turning out of the mind, creating self-critical awareness, and fostering independence of thought**.

392. Answer: (c)

Explanation:

The passage emphasizes that **education is not merely instrumentalist**—it should not be viewed as a means to employment, higher income, or social mobility. Instead, it describes education as **a process of expansion and conversion, where individuals develop self-critical awareness and independence of thought**.

393. Answer: (c)

Explanation:

The passage explains that **pesticide resistance evolves through natural selection**. When pesticides kill a large number of pests, a **few individuals that have a natural resistance survive and reproduce**, leading to a growing resistant population over successive generations.

394. Answer: (b)

Explanation:

Statement 1 is incorrect: The passage says poorer countries may use pesticides due to starvation fears but doesn't say it's always necessary.

Statement 2 is incorrect: Pesticides can play a role in sustainable agriculture if improved to be safer and targeted.

Statement 3 is correct: The Alabama leaf worm example shows one pest can develop resistance to many pesticides.

395. Answer: (d)

The actual answer is there in the last paragraph: "Moreover, in many poorer countries, the prospect of imminent mass starvation, or of an epidemic disease, are so frightening that the social and health costs of using pesticides have to be ignored. In general the use of pesticides is justified by objective measures such as 'lives saved', 'economic efficiency of food production' and 'total food produced'."

396. Answer: (b)

Explanation:

Statement 1 is correct: The passage states that pest populations are genetically variable, meaning individuals within the population may have different responses to pesticides due to their genetic makeup.

Statement 2 is correct: The passage explains that some pests may possess an enzyme that can detoxify the pesticide, making them resistant.

Statement 3 is incorrect: The passage describes that resistance spreads over generations, meaning it is not equally distributed in the population from the start. Instead, resistant individuals become more common over time due to natural selection.

397. Answer: (c)

Explanation:

Statement 1 is correct: The passage mentions that chemical pesticides are still widely used because the ratio of cost to benefit remains in favor of their use. It also suggests that while developed countries may explore organic farming, poor and developing countries heavily rely on pesticides to ensure food security.

Statement 2 is correct: The passage explicitly states that pesticides help address food shortages and epidemic diseases in poorer nations, making their use necessary despite potential drawbacks.

Statement 3 is correct: The passage highlights that social and health costs are often ignored in poorer countries due to the pressing need to prevent starvation and disease outbreaks.

398. Answer: (d)

Explanation:

Statement (a) is correct: The passage implies promoting alternatives to chemical pesticides is necessary for sustainability.

Statement (b) is correct: Excessive use of chemicals harms the ecosystem by causing pest resistance.

Statement (c) is incorrect: The passage clearly states there is scope to improve pesticides for sustainable use.

399. Answer: (a)

Explanation:

Statement 1 is correct: Overexploitation of resources increases vulnerability, as shown by the Aral Sea example.

Statement 2 is incorrect: The passage says human creativity helps adaptation; growth doesn't harm it.

Statement 3 is incorrect: Growth providing housing and social security reduces vulnerability.

Statement 4 is incorrect: Harmful practices are criticized, but farming itself isn't always a cause of vulnerability.

400. Answer: (a)

Explanation:

Statement 1 is correct: Low-carbon growth means reducing emissions by using renewable energy for sustainable development.

Statement 2 is incorrect: It doesn't require cutting manufacturing or shifting to agriculture; cleaner industrial processes are possible.

Statement 3 is incorrect: Mixed farming isn't highlighted as a key factor in low-carbon growth.

Statement 4 is incorrect: Low-carbon growth focuses on sustainable production, not reducing demand.

401. Answer: (b)

Explanation:

Statement 1 is not correct: The passage warns that growth alone doesn't ensure sustainability and can increase vulnerability.

Statement 2 is correct: Diffusing adaptive technologies widely is crucial to sustain growth amid environmental changes.

Statement 3 is correct: Investing in research for adaptation and mitigation technologies is necessary to build capacity.

402. Answer: (d)

Explanation:

Statement 1 is incorrect: The passage doesn't mention restricting rainfed crops in irrigated areas; sustainable agriculture often supports mixed cropping.

Statement 2 is incorrect: While over-exploitation is noted, the passage doesn't call for stopping all farming in water-scarce regions, only for avoiding unsustainable irrigation.

403. Answer: (d)

Explanation:

Statement 1 is correct: The passage highlights human creativity in adapting to environmental changes through new technologies for sustainable growth.

Statement 2 is correct: It cites cotton farming in Central Asia and mangrove clearing as examples of intensive agriculture causing serious environmental harm.

Statement 3 is correct: The passage notes that some growth strategies, like deforestation and resource overuse, increase environmental vulnerability.

404. Answer: (b)

Explanation:

The passage discusses how adaptation and mitigation must be incorporated into economic growth strategies to ensure sustainability. While options (a), (c), and (d) focus on specific observations or extreme cases, only option (b) captures the main idea of balancing development with environmental concerns. Hence, it best represents the central theme of the passage.

405. Answer: (d)

Explanation:

The passage states that **not all** exotic species introductions lead to biodiversity loss, so statement (a) is incorrect. Similarly, the passage mentions that **some** exotic species have altered ecosystems, but not all, making (b) incorrect. Lastly, humans have contributed to homogeneity, but natural processes also play a role, making (c) incorrect. Since all statements are extreme generalizations, the correct answer is (d).

406. Answer: (b)

Explanation:

Statement 1 is incorrect: The passage doesn't mention breeding exotic species with local varieties as a reason for introduction.

Statement 2 is correct: Exotic species are introduced to increase agricultural productivity and gain public benefits.

Statement 3 is correct: Some species are introduced for recreational purposes like landscaping and beautification.

407. Answer: (d)

Explanation:

Option (d) is correct: All statements (a), (b), and (c) are true.

(a) Species evolve locally in centers of endemism, maintaining unique biodiversity.

(b) Physical barriers like oceans and mountains restrict species spread.

(c) Locally evolved species are adapted to specific climates, limiting their natural spread.

408. Answer: (a)

Explanation:

Statement 1 is correct: Humans introduce exotic species accidentally or illegally, impacting biodiversity.

Statement 2 is correct: Infrastructure like highways removes barriers and aids species migration, affecting ecosystems.

Statement 3 is incorrect: Humans have not made ecosystems resistant; instead, they increase exotic species spread.

Statement 4 is incorrect: Introduced species often cause harm; humans haven't prevented their negative impact on local species.

409. Answer: (c)

Explanation:

Statement 1 is correct: Exotic species like the brown tree snake in Guam can cause the extinction of endemic species by predation or competition.

Statement 2 is correct: Introduced species can alternative communities, leading to biodiversity homogenization and loss of ecosystem uniqueness.

410. Answer: (b)

Explanation:

The passage explains that some theorists, especially from East Asian

countries, argued that **economic growth was achieved without democracy** in the 1970s and 1980s. While democracy and development are often seen as separate (statement a), and some authoritarian regimes have achieved rapid growth (statement c), the passage does **not** make a generalized claim that all non-democratic regimes deliver growth faster. Hence, the correct answer is **(b)**.

411. Answer: (d)

Explanation:

The passage argues that development is **not just about economic growth (GNP, per capita income, industrial growth, or consumption trends)** but about **the freedom and quality of life of individuals**. This is explicitly mentioned in the second paragraph:

"The assessment of development cannot be divorced from the lives that people can lead and the real freedom that they enjoy."

Thus, **(d)** is the correct answer.

412. Answer: (c)

Explanation:

The passage states that democracy is **not just an external factor affecting development** but an **intrinsic part of it**. The last paragraph mentions that **political freedoms and democratic rights are fundamental components of development** rather than just contributing factors. Statement **(a) is incorrect** because the passage refutes the idea that democracy and development are only externally linked. Statement **(b) is incorrect** because the passage does not claim that political and civil rights alone lead to economic development. Hence, the correct answer is **(c)**.

413. Answer: (a)

Explanation

statement 1 is correct: The passage explicitly mentions: The impact of FDI is not always pro-competitive." This implies that **it is desirable for FDI to be pro-competitive**, but it does not always happen.

Statement 2 is incorrect: because the passage states that higher prices occur **only when the parent companies of competing affiliates merge overseas**, not always.

414. Answer: (b)

Explanation:

Statement 1 is incorrect: Familiarity with home-country competition laws helps MNCs comply but does not lead to market dominance.

Statement 2 is correct: FDI through acquisitions or joint ventures enables foreign investors to reduce competition and dominate markets.

Statement 3 is correct: Mergers of parent companies abroad reduce affiliate independence, lessening competition and increasing market control locally.

Statement 4 is incorrect: The passage doesn't cite cost reduction as a dominance strategy; it focuses on mergers and acquisitions.

415. Answer: (c)

Explanation:

The passage highlights that effective competition laws prevent foreign investors from unfairly dominating the domestic market. It ensures that mergers and acquisitions do not harm competition and that both domestic and foreign firms compete on equal terms. The passage does not claim that foreign investors always dominate or that mergers are entirely bad. Thus, ensuring a level playing field through competition law is the correct inference.

416. Answer: (b)

Explanation:

Statement 1 is correct: Agricultural collectivities empower the poor by improving their economic, social, and political standing.

Statement 2 is correct: Group-based farming boosts productivity through collective effort and fair distribution.

Statement 3 is correct: Collectivities protect farmers from exploitative free-market pressures.

Statement 4 is incorrect: The passage doesn't claim collectivities guarantee surplus agricultural production.

417. Answer: (c)

Explanation:

The passage states that the **gender impact of family cooperatives in transition economies is uncertain**, but in India, **women-only farming groups have considerable potential to benefit women**. This suggests that the author is referring to how **women might benefit from group farming initiatives**.

418. Answer: (b)

Explanation:

Assumption 1 is not valid: The passage mentions that transition economies have examples of successful agricultural collectivities, but it does **not state that they are imperative** (i.e., absolutely necessary) for these economies. The passage only highlights the potential benefits of collectivities but does not claim that transition economies **must** have them.

Assumption 2 is valid: The passage explicitly states that a **group approach to farming offers substantial scope for enhancing agricultural productivity**. This directly supports the assumption that agricultural productivity can be increased through collective farming.

419. Answer: (c)

Explanation:

Democratization as greater involvement of 'atomized' individuals in the public sphere. This is directly supported by the line: "In the Indian context, democratization is translated into greater involvement of people not as 'individuals' which is a staple to liberal discourse…" which implies that in the Western context, involvement as individuals is central.

420. Answer: (b)

Explanation:

The passage highlights that **democratization in India has strengthened community identities** (such as caste and religion) rather than dissolving them. It states that individuals engage in the political process **not as atomized individuals but as members of primordial communities**. This

suggests that caste and communal identities continue to play a significant role in public life.

421. Answer: (c)

Explanation:

The passage describes the **silent revolution** as a process where democracy in India has led to a **loosening of social strictures** and **empowerment of marginalized groups**. It specifically highlights the **transfer of power from upper-caste elites to various subaltern groups** within the democratic framework.

Directions for the following eight items:

The following eight items (questions 66 to 73) are based on three passages in English to test the comprehension of the English language and therefore these items do not have Hindi version. Read each item and Solution: the items that follow.

422. Answer: (b)

Explanation:

The passage emphasizes how the prisoner became deeply familiar with his surroundings, noticing every mark, dent, and even the creatures around him. This suggests that he had **developed a close connection with the cell**. While **option (d) ("I felt quite at home in the cell") might seem relevant, it is more of an interpretation rather than an objective fact** stated in the passage. Therefore, **option (b) is the most accurate choice** as it directly aligns with the passage's description.

423. Answer: (c)

Explanation:

The passage describes **how the prisoner paid close attention to his surroundings, greeting tufts of grass, noticing marks on the walls, and observing small creatures**. This suggests that he was actively engaging with his environment to adapt and cope with isolation. **Option (b) ("The prisoner's capacity to notice the minute details of his surroundings") is a part of the passage but does not capture its overall theme**. Similarly, **option (d) ("The prisoner's ability to live happily with other creatures") is an exaggerated interpretation**. Thus, **option (c) best describes the passage's main idea**.

424. Answer: (a)

Explanation:

The passage describes how the prisoner gradually became familiar with his surroundings, including the walls, floor, ceiling, and even small creatures like wasps and lizards. This implies that **he adapted to the prison environment over time**. The passage does not comment on whether Indian prisons are good or bad (**option b and d**), nor does it suggest that spending time in prison is easy (**option c**). Thus, **option (a) is the best choice**.

425. Answer: (c)

Explanation:

The passage states that they were setting up the "highest camp that has ever been made," which means **no previous climbers had set up a camp at such an altitude before**. Option (a) is incorrect because they are not at the summit yet. Option (b) is also incorrect because it does not claim that no one had climbed that high, just that no camp had been set up. Option (d) is irrelevant to the passage. **Therefore, option (c) is the correct answer.** (c) is the right choice.

426. Answer: (b)

Explanation:

The passage clearly states, **"Everything took five times as long as it would have taken in a place where there was enough air to breathe."** This directly indicates that the lack of oxygen made their work much slower. While cold weather (option c) could be a factor, **the passage specifically attributes the delay to the thin air**. Therefore, **option (b) is the correct choice.**

427. Answer: (a)

Explanation:

The passage states, **"There was only a light wind and inside it was not too cold for us to take off our gloves."** This means they removed their gloves because **the temperature inside the tent was bearable**. Option (b) is incorrect because they were able to remove their gloves. Option (c) is misleading, as it was not "very cold." Option (d) contradicts the passage. **Thus, option (a) is the right answer.**

428. Answer: (b)

Explanation:

The passage states, "Losing his way in the smoke-filled passage, he missed the stairway and went into another room." This directly indicates that he couldn't find the correct way because of the smoke, making **option (b) the right choice**.

429. Answer: (d)

Explanation:

The man unknowingly picked up the Mayor's baby while trying to protect himself from the fire. As a result, he not only saved himself but also the baby. This act of saving a life made him a hero, making **option (d) the best answer**.

430. Answer: (a)

Explanation:

Statement 1 is correct: The passage confirms inclusive growth was a key goal set by India's founding leaders.

Statement 2 is correct: It emphasizes the need for an enabling government.

Statement 3 is incorrect: The passage advises against excessive government interference.

Statement 4 is incorrect: It focuses on redefining the government's role, not reducing its size.

431. Answer: (d)

Explanation:

The passage emphasizes the role of an **enabling Government** that does not attempt to provide everything to every citizen but rather **creates a favorable environment for individual enterprise** and **supports those in need.** It specifically states that the Government should "play an important

role in directly helping the poor by ensuring that they get basic education and health services and receive adequate nutrition and food."

432. Answer: (c)

Explanation:

An enabling Government is one that facilitates economic and social development without excessive direct control, focusing on creating opportunities and supporting the underprivileged rather than managing everything itself.

433. Answer: (d)

Explanation:

The passage explains that **India is too large and complex** for the State to deliver all that is needed. The issue is not the **lack of bureaucracy** but rather that **a large bureaucracy can become inefficient and lead to corruption**. Hence, **Statement 1 is incorrect.**

Additionally, the passage does not claim that the Government fails to **promote inclusive growth.** Instead, it suggests that the approach to governance should evolve to an **"enabling State"** model, where the Government focuses on **creating opportunities and directly helping the poor.** Therefore, **Statement 2 is also incorrect.**

434. Answer: (d)

Explanation:

The passage primarily emphasizes the need to **redefine the role of the Government** to ensure **inclusive growth**. It argues against excessive bureaucracy and direct control over the economy, instead advocating for a **Government that enables markets and supports the underprivileged.**

435. Answer: (c)

Explanation:

Statement 1 is not correct: The passage does not mention artistic or literary incentives as a defining feature of a "creative society." Instead, it focuses on social movements and contradictions.

Statement 2 is not correct: A creative society does not accept social inequalities as the norm; rather, it challenges them by mobilizing oppressed groups.

Statement 3 is correct: The passage states that a creative society is marked by the recognition and articulation of contradictions in society.

Statement 4 is correct: The passage highlights that oppressed and exploited groups become aware of their rights and mobilize for their upliftment in a creative society.

436. Answer: (c)

Explanation:

Statement 1 is not correct: The passage does not mention aggressiveness or being incendiary as characteristics of social movements. Instead, it discusses mobilization and demand for rights.

Statement 2 is not correct: There is no reference to social movements being instigated by external forces. The passage emphasizes internal contradictions and the struggle of oppressed groups.

Statement 3 is correct: The passage highlights the quest for social equality and individual freedom as key aspects of social movements.

Statement 4 is correct: The passage discusses the upliftment of the oppressed, which includes granting privileges and self-respect to disadvantaged groups.

437. Answer: (c)

Explanation:

Statement 1 is correct: The passage links diverse social movements to the emergence of a creative society.

Statement 2 is correct: It states that contradictions and conflicts are essential for a society to be creative.

438. Answer: (c)

Explanation:

Statement 1 is not correct: The passage does not mention any direct impact of human population expansion on spiritual fulfillment.

Statement 2 is not correct: While ecosystem degradation can affect aesthetic enjoyment, the passage does not explicitly state this as a consequence of human population growth.

Statement 3 is correct: The passage mentions that potable fresh water can be obtained through desalination but at a high cost, indicating that human activities have impacted fresh water availability.

Statement 4 is not correct: The passage states that human modifications have increased food and fibre production, not harmed it.

Statement 5 is correct: The passage explicitly states that efforts to increase food and fibre production have reduced biodiversity.

439. Answer: (a)

Explanation:

Statement 1 is correct: The passage says ecosystem changes for human needs have harmed some people.

Statement 2 is incorrect: The passage acknowledges increased production, not insufficient efforts.

Statement 3 is incorrect: It warns that long-term benefits are uncertain due to ecosystem degradation.

440. Answer: (d)

Explanation:

Statement 1 is incorrect: The passage describes ecosystem modification as a reality, not a necessity.

Statement 2 is incorrect: The passage acknowledges some technological replacements exist but notes they are costly, so technology can sometimes replace ecosystem services.

441. Answer: (d)

Explanation:

Statement 1 is correct: Moral acts must stem from one's own will and thoughtful decision-making.

Statement 2 is incorrect: The passage stresses deliberate action, not an immediate reaction.

Statement 3 is incorrect: Morality requires intelligent choice, not just duty-following.

Statement 4 is correct: Sometimes moral action means defying convention for the greater good.

442. Answer: (b)

Explanation:

The passage emphasizes that a **moral act must come from our own will and discretion**, rather than being performed mechanically or under orders. It distinguishes between **acting mechanically and acting intentionally with moral judgment,** which aligns with **option (b)**. The other options refer to **mechanical obedience (a), cleverness (c)**, and **religious action (d)**, which are not the main focus of the passage.

443. Answer: (b)

Explanation:

Statement 1 is correct: The phrase "lets himself be swept along like a log of wood by a current" implies that the person **does not use their own reason or intelligence** but simply follows along passively.

Statement 2 is correct: The phrase also suggests that the person is **easily influenced or pressured** by external forces rather than making independent decisions.

Statement 3 is not correct: The passage does not suggest that the person is unable to **withstand difficulties or challenges**; rather, it highlights a lack of independent thinking.

Statement 4 is not correct: While the phrase uses "log of wood" as a metaphor, simply stating "he is like a log of wood" does not fully capture the intended meaning of passivity and lack of discretion.

444. Answer: (b)

Explanation:

Statement 1 is not correct: The phrase **"this world is evidently not meant for them"** refers to those suffering from extreme poverty and misfortune, not necessarily to those seeking freedom from foreign domination.

Statement 2 is correct: The passage explicitly states that **"the poor and unfortunate in every country become to some extent other-worldly"**, implying that those living in **starvation and misery** often escape into spiritual or other-worldly thinking because they find little hope in their material conditions.

Statement 3 is not correct: The passage contrasts **other-worldliness** with **becoming a revolutionary**, implying that those who become revolutionaries do not resign themselves to suffering but actively fight against it.

445. Answer: (c)

Explanation:

Statement 1 is not correct: The passage does not say that a country under foreign domination **cannot** engage in spiritual pursuits. It only criticizes the tendency to **romanticize past greatness** or falsely claim spiritual superiority while suffering from oppression and poverty.

Statement 2 is correct: The passage states that **spiritual or any other greatness cannot be founded on lack of freedom and opportunity, or on starvation and misery,** indicating that poverty hinders true spiritual growth.

Statement 3 is correct: The passage mentions that **subject peoples** (those under foreign rule) often become **other-worldly** as a way to cope with their struggles.

446. Answer: (a)

Explanation:

The passage primarily discusses **how oppressed people, particularly those under foreign domination or living in poverty, respond mentally and emotionally to their circumstances.** It highlights their tendency to either **seek solace in past greatness, adopt an other-worldly mindset, or turn to revolutionary actions.** While civilization, starvation, and the body-mind-spirit connection are mentioned, the **central theme revolves around the mindset of the oppressed.**

447. Answer: (c)

Explanation:

The passage explains that **civilizations and people exhibit both an external (material) life and an internal (spiritual) life.** When these two aspects **diverge rather than maintaining balance, conflict arises, leading to mental and spiritual turmoil.** While foreign domination and past visions are discussed, the **core reason for the torture of the mind and spirit is the imbalance between external and internal life.**

448. Answer: (c)

Explanation:

The passage highlights the role of **keystone species in maintaining species richness and ecosystem balance.** It provides an example of the **sea star Pisaster**, which prevents mussels from dominating and allows other species to thrive. The experiment at the University of Washington showed that removing the sea star **led to a loss of 28 species**, proving that keystone species **ensure biodiversity by regulating competitive relationships in the ecosystem.**

449. Answer: (d)

Explanation:

Statement 1 is incorrect: Mussels dominate only when sea stars are absent, so their dominance depends on context.

Statement 2 is incorrect: Sea stars regulate mussel populations; mussel abundance doesn't determine sea star survival.

450. Answer: (d)

Explanation:

Statement 1 is incorrect: Mussels are prey for sea stars, not competitors with them.

Statement 2 is incorrect: The passage doesn't claim sea stars have reached an evolutionary climax.

Statement 3 is correct: Sea stars, as keystone species, affect species richness and energy flow in the ecosystem.

451. Answer: (c)

Explanation:

Statement 1 is correct: Keystone species impact species richness and energy flow, playing a vital role in ecosystems.

Statement 2 is incorrect: Keystone species exist in both aquatic and terrestrial ecosystems, not only aquatic.

Statement 3 is correct: Removing sea stars (a keystone species) caused species loss and mussel dominance, showing ecosystem collapse.

452. Solution:

Instead of using Venn diagrams and spending time, simply use logic to solve this.

Since teachers can only be college graduates and since no college graduates are poor, so C is correct. Moreover, since poets are poor, no poet can be a teacher. Also, since some mathematicians are also poets (and poor) they can't be teachers. So, A is also correct.

B can not be said with absolute certainty. Hence B is incorrect and the answer.

453. Answer: (d)

Explanation:

Statement 1 is not correct: The passage says adults must act to protect children's right to education because children can't fight legal battles themselves.

Statement 2 is not correct: The passage acknowledges challenges like societal prejudice in implementing the RTE law but does not claim the law is unenforceable.

454. Answer: : (b)

Explanation:

Statement 1 is not correct: The passage does not mention **parents' inability to fight legal battles** as a traditional obstacle to girls' education. Instead, it emphasizes that **adult society must act on behalf of children.**

Statement 2 is correct: The passage highlights **traditional thinking about the role of girls** in society as a major challenge, stating that **girls are given only a short childhood** and face **deep-rooted biases.**

Statement 3 is correct: The passage clearly mentions **"tenacious prejudice against the intellectual potential of girls",** which prevents their access to education.

Statement 4 is correct: The passage states that **the education system has failed to address gender-based prejudices,** indicating that an **improper system of education** is an obstacle.

455. Answer: (d)

Explanation:

Statement 1 is not correct: The **Right to Education (RTE) is a Fundamental Right** under **Article 21A** of the Indian Constitution. It guarantees **free and compulsory education for children aged 6-14 years.** Hence, it is **not just a legal right but a fundamental right.**

Statement 2 is not correct: The passage does not suggest that India's education system must be identical to that of developed countries. Instead, it highlights **gender biases, societal prejudices, and systemic flaws** as the major challenges to achieving universal education.

456. Answer: (c)

Explanation:

The passage emphasizes the **challenges faced by children, especially girls, in accessing education,** including **social prejudices, traditional barriers, and systemic failures.** It highlights the **need to actively safeguard and implement the Right to Education**, particularly for **girls,** making option (c) the most appropriate choice.

457. Answer: (d)

Explanation:

The passage stresses that **missing out on education during childhood cannot be compensated later,** as childhood is a crucial period for learning. Even if a legal battle is won, it may not serve the same purpose later in life, making option (d) the best inference.

458. Answer: (b)

Explanation:

The passage states that **"he slept under a haystack lying in a meadow."** A haystack is a pile of dry grass, making option (b) the correct answer.

459. Answer: (a)

Explanation:

The passage mentions that **"he however, soon fell asleep, being much tired with his long walk."** This clearly indicates that exhaustion, not fear or loneliness, caused him to sleep quickly.

460. Answer: (a)

Explanation:

Statement 1 is correct: The passage mentions that he got **dry bread and water from cottagers and farmers** and later **slept under a haystack in a meadow.** This suggests that he was **walking through the countryside.**

Statement 2 is incorrect: The passage states that he **felt cold and hungry at night,** indicating that the food he received was **not enough to satisfy his hunger.**

461. Answer: (a)

Explanation:

The idea of forgetting his **toothbrush** causes him **anxiety and distress,** making his life miserable when traveling. The phrase "a horrible idea occurred to me" suggests an unpleasant thought rather than a sad, fantastic, or amusing one.

462. Answer: (b)

Explanation:

The passage emphasizes how the **toothbrush haunts him** while traveling, causing him anxiety and sleeplessness. He **constantly worries** about whether he has packed it, making his life miserable.

463. Answer: (c)

Explanation:

The passage mentions that after multiple packing and unpacking attempts, he ultimately **carries the toothbrush to the railway station wrapped in his pocket-handkerchief.**

464. Answer: (d)

Explanation:

The passage states that **polar bear mothers emerge from dens with three-month-old cubs in spring.** This means that the cubs must have been born approximately **three months earlier, which would be in winter.**

465. Answer: (b)

Explanation:

The passage mentions that **female polar bears are protective of their cubs but tend to ignore family rivalry over food.** This implies that the mother does not interfere in the competition among cubs and lets them fend for themselves.

466. Answer: (d)

Explanation:

Statement 1 is not correct: The passage doesn't say fasting is due to lack of prey—it may be due to denning and nursing.

Statement 2 is not correct: Triplets are rare, not a regular occurrence.

2 Interpersonal Skills, Communication Skills

1. Match List-I with List-II and select the answer using the code given below the Lists: (CSAT-2026)

List-I (Relationship category)	List-II (Communication type)
A. Between cricket captain and team members	1. Informal and firm
B. Between judge and lawyers in court	2. Informal and open-ended
C. Between Vice Chancellor and Deputy Registrar	3. Formal and open-ended
D. Between peers and coworkers	4. Formal and firm

Code:

(a) A-1, B-3, C-4, D-2 (b) A-1, B-4, C-3, D-2

(c) A-2, B-4, C-3, D-1 (d) A-2, B-3, C-4, D-1

2. Match List-I with List-II and select the answer using the code given below the Lists: (CSAT-2026)

List-I (Tool of communication)	List-II (Purpose)
A. Memorandum	1. To record decisions
B. Flyer	2. To inform confidentially
C. Bcc	3. To intimate a directive
D. Minutes	4. To disseminate non-targeted information

Code:

(a) A-1, B-4, C-2, D-3 (b) A-1, B-2, C-4, D-3

(c) A-3, B-4, C-2, D-1 (d) A-3, B-2, C-4, D-1

3. Match List-I with List-II and select the answer using the code given below the Lists: (CSAT-2026)

List-I (Barrier to communication)	List-II (Example)
A. Semantic	1. Lack of feedback
B. Cognitive	2. Misunderstanding the meaning of a word
C. Organisational	3. Fear of social stigma
D. Affective	4. Information overload

Code:

(a) A-3, B-1, C-4, D-2 (b) A-3, B-4, C-1, D-2

(c) A-2, B-1, C-4, D-3 (d) A-2, B-4, C-1, D-3

4. With reference to 'circular letters', which of the following statements is/are correct? (CSAT-2026)

1. Circular letters are usually addressed to a group of people.
2. Non-standard and customized content is typical of circular letters.
3. Circular letters are used to intimate appraisals and increments of employees within organisations.
4. Circular letters are less cost-effective than personalised and specific recipient-directed letters.

Select the answer using the code given below.

(a) 1 and 2 (b) 2 and 4

(c) 1, 3 and 4 (d) 1 only

5. Consider the following statements: (CSAT-2012)

The Third World War, if it ever starts will end very quickly with possible end of civilization. It is only the misuse of nuclear power which will trigger it.

Based on the above statement which one of the following inferences is correct?

(a) Nuclear power will be used in Third World War.

(b) There will be no civilization left after the third world war.

(c) The growth of nuclear power will destroy civilization in the long run.

(d) The third world war will not take place.

Directions for the following 2 (two) items:

Each of the following two items consists of four statements. Of these four statements, two cannot both be true, but both can be false. Study the statements carefully and identify the two that satisfy the above condition. Select the correct answer using the codes given below each set of statements :

6. Examine the following statements: (CSAT-2011)

1. All animals are carnivorous.
2. Some animals are not carnivorous.
3. Animals are not carnivorous.
4. Some animals are carnivorous.

Codes:

(a) 1 and 3 (b) 1 and 2

(c) 2 and 3 (d) 3 and 4

7. Who are the females in the group? (CSAT-2011)

(a) C and E (b) C and D

(c) E and A (d) D and E

8. Whose wife is the teacher? (CSAT-2011)

(a) C (b) D

(c) A (d) B

2 Interpersonal Skills, Communication Skills_ Explanations

1. Answer: (b)

Explanation:

To determine the correct matching, let us analyze the nature of each relationship and its corresponding communication type:

1. **Cricket captain and team members (A):** The relationship among sports teammates is generally informal. However, the captain must lead, set fields, and make strategic decisions, which requires the communication to be firm. Thus, A matches with 1 (Informal and firm).
2. **Judge and lawyers in court (B):** A courtroom is a highly structured and formal environment. The judge acts as the ultimate authority, enforcing rules, sustaining or overruling objections, and giving verdicts. Therefore, the communication from the judge is firm. Thus, B matches with 4 (Formal and firm).
3. **Vice Chancellor and Deputy Registrar (C):** This is a formal hierarchical relationship within a university's administration. However, unlike a courtroom, their day-to-day communication involves discussing policies, administrative planning, and collaborative problem-solving, which makes it open-ended. Thus, C matches with 3 (Formal and open-ended).
4. **Peers and coworkers (D):** Coworkers operating at the same hierarchical level typically share an informal relationship. Their communication involves brainstorming, sharing ideas, and casual discussions, making it open-ended. Thus, D matches with 2 (Informal and open-ended).

Matching these up, we get A-1, B-4, C-3, D-2. Therefore, the correct option is B.

2. Answer: (c)

Explanation:

Let us evaluate the purpose of each communication tool given in List-I to find its correct match in List-II:

A. Memorandum (Memo): A memorandum is a written message used in a professional setting, typically to communicate policies, procedures, or to intimate a directive within an organization. Thus, A matches with 3.

B. Flyer: A flyer is a paper advertisement intended for wide distribution, typically handed out to individuals or posted in public places. Its primary purpose is to disseminate non-targeted information to a general audience. Thus, B matches with 4.

C. Bcc (Blind Carbon Copy): This is an email feature used to send a copy of a message to someone without the other recipients knowing. It is used to inform confidentially. Thus, C matches with 2.

D. Minutes: Minutes are the official written record of a meeting. Their primary purpose is to record decisions, discussions, and action items agreed upon during the meeting. Thus, D matches with 1.

Combining these deductions, the correct matching is A-3, B-4, C-2, D-1. This corresponds to Option C.

3. Answer: (d)

Explanation:

To find the correct match, let us analyze the meaning of each communication barrier:

Semantic barrier (A): 'Semantic' refers to the meaning of language, words, or symbols. Therefore, misunderstanding the meaning of a word (2) is a classic example of a semantic barrier.

Cognitive barrier (B): 'Cognitive' relates to mental processes such as thinking, understanding, and memory. Information overload (4) overwhelms an individual's mental processing capacity, making it a cognitive barrier.

Organisational barrier (C): These barriers arise from the structure, rules, policies, and culture of a workplace. A lack of feedback (1) is an organisational barrier because it typically stems from poor communication channels or rigid hierarchical structures within an organisation.

Affective barrier (D): 'Affective' refers to emotions, feelings, and attitudes. Fear of social stigma (3) is an emotional response that prevents a person from communicating openly, making it an affective barrier.

Matching these gives A-2, B-4, C-1, D-3, which corresponds to Option D.

4. Answer: (d)

Explanation:

Let us evaluate the given statements regarding 'circular letters':

Statement 1 is correct: A circular letter is a formal document intended for mass communication. It is usually addressed to a large group of people (such as employees, customers, or shareholders) to disseminate general information, announcements, or updates.

Statement 2 is not correct: Because circular letters are meant for a wide audience, their content is standard and uniform. Customized or non-standard content is a feature of personalized letters, not circulars.

Statement 3 is not correct: Appraisals, increments, and performance reviews are highly confidential and individualized matters. They are communicated through personalized, specific recipient-directed letters, never through a general circular.

Statement 4 is not correct: Circular letters are highly cost-effective and time-saving. Since the same message is duplicated and distributed to multiple recipients without the need for individual customization, it costs much less in terms of time and resources compared to personalized letters.

Since only Statement 1 is correct, the correct option is D.

5. Solution: (a)

Very clear when its mentioned that only the misuse of nuclear power will trigger world war. So nuclear power has to be used in the third world war.

6. Solution: (a)

The approach of solving such questions is that you check the validity of

each and every statement for each option. Consider any one statement to be true at one time, and then check whether that option combination can be correct. Check all 4 like this, you will know.

Consider S1: If all animals are carnivorous, then the statement 3 is wrong. And if 3 is right, 1 is wrong. So, both are not true together. But if some animals are carnivorous (as per S4), the both of these are false. Hence correct option is (a).

Consider S2: if 1 is correct, 2 is false. If 2 is correct, 1 has to be false. Now check other statements, 3 and 4. If 3 is correct, then 1 is incorrect, but 2 can be correct.

Consider S3: Both can be true at the same time. Hence incorrect option.

Consider S4: If 3 is correct, 4 is incorrect and vice versa. If 1 is correct, then 3 is incorrect, but 4 can be correct. If 2 is correct, then 3 can be correct, but 4 will be incorrect.

7. Solution: (c)

Clear from the arrangement.

8. Solution: (d)

Clear from the diagram.

3 Logical Reasoning and Analytical Ability

1. P, Q, R, S and T are ranked 1 to 5 (not necessarily in that order). The rank of P is 4, the rank of Q is not 5, the rank of R is 1, the rank of S is not 2, the rank of T is not 3. Then which of the following is/are correct? (CSAT-2026)

I. If the rank of S is 3, then that of T is 2.

II. If the rank of Q is 3, then that of T is 5.

Select the answer using the code given below.

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

2. Two identical straight rods are painted in five distinct colours so that each of them gets divided into five equal parts along the length. In one of them, the portions are marked P1, P2, P3, P4 and P5 (not necessarily in that order) whereas in the other, they are marked Q1, Q2, Q3, Q4 and Q5 (not necessarily in that order). When the rods are kept parallel to each other side by side, P1 and Q3 match, P4 matches Q1 or Q2, and Q4 matches P3 or P5. If Q3 and Q5 are adjacent, which of the following is/are possible? (CSAT-2026)

I. Q3 is marked at the middle portion of the straight rod.

II. P2 is marked at one of the extreme portions of the straight rod.

Select the answer using the code given below.

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

3. Seven persons A, B, C, D, E, F and G travel by three cars X, Y, Z. A and another two of them travel by X. Only E travels with G. C travels by Z, but B does not travel by Y. Besides, A and B do not travel by the same car. Then which of the following are correct? (CSAT-2026)

I. No one travels alone.

II. Only D travels with F.

III. Only C travels with B.

Select the answer using the code given below.

(a) I and II only (b) I and III only

(c) II and III only (d) All the three

4. There are four statements X, Y, Z and W. Their relations are as follows: If X is incorrect, then so is Z; if Y is incorrect, then W is correct; if W is correct, then X is incorrect. Which of the following is/are correct? (CSAT-2026)

I. If X is correct, then so is Y.

II. If Z is correct, then it is not necessary that Y is correct.

Select the answer using the code given below.

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

5. P has a son and a daughter. S is the mother of T. S is R's spouse. Q and R are children of P. Then how is Q related to S? (CSAT-2026)

(a) Qis a sister of S

(b) Q is a daughter of S

(c) Q is the mother of S

(d) Q is a sister of the husband of S

6. Consider the following statements: Every red is blue. Every blue is green. Every green is yellow. Which of the following statements denoted by P, Q and R are correct? (CSAT-2026)

P. Every blue is yellow.

Q. Every red is green.

R. Every red is yellow.

Select the answer using the code given below.

(a) P and Q only (b) Q and R only

(c) P and R only (d) P, Q and R

7. Seven cubes are identical in shape. Out of these, the weight of each of the six cubes is equal and the weight of the remaining cube is less than the weight of any other cube. A balance is used to identify the lightest cube. What is the minimum number of attempts required to distinguish the odd cube with certainty? (CSAT-2026)

(a) 2 (b) 3

(c) 4 (d) 1

8. If ZERO is encoded as ADSN, then how do you encode STOP? (CSAT-2026)

(a) SPOT (b) TSPO

(c) TSOP (d) POST

9. Consider the following three statements, namely S1, S2 and S3: (CSAT-2026)

S1. Protecting the environment is an existential exigency for humans, given the impact of environmental degradation on climate change.

S2. Scientific consensus has not been achieved with regard to the extent of the contribution of human intervention to climate change.

S3. Environmental activism includes climate alarmism and other extremist points of view that often become the focus of climate change deniers.

Which of the following relationships based on the statements given above is/are correct?

1. S3 is a counterpoint to S1
2. S3 is unconnected to S1 and S2
3. S2 could be the reason for S3

Select the answer using the code given below.

(a) 2 and 3 only (b) 1, 2 and 3

(c) 1 and 2 only (d) 3 only

10. Eight persons P, Q, R, S, T, U, V and W sit around a round table in eight different seats placed with equal distance between any two consecutive seats. Both P and R are adjacent to Q. Both T and R are adjacent to S. Both U and W are adjacent to V. S and W are on opposite chairs. If while going in the clockwise direction around the table from P, one meets R before T, then how many persons shall Q cross while moving in the clockwise direction around the table before meeting W? (CSAT-2026)

(a) 5 (b) 4

(c) 2 (d) 1

11. A pattern formed by two characters a and b is repeated more than once in the following string: (CSAT-2026)

x b x a x a x x a x a x bab

What is xx in the 7th and 8th positions from the left in the above string?

(a) aa (b) ab

(c) ba (d) bb

12. What comes at X and Y respectively in the following sequence?

January, January, December, October, X, March, October, Y, September **(CSAT-2025)**

(a) July, May (b) July, April

(c) June, May (d) June, April

13. A solid cube is painted yellow on all its faces. The cube is then cut into 60 smaller but equal pieces by making the minimum number of cuts. Which of the following statements is/are correct? (CSAT-2025)

I. The minimum number of cuts is 9.

II. The number of smaller pieces which are not painted on any face is 6.

Select the correct answer using the code given below:

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

14. Consider the sequence AB_CC_A_BCCC_BBC_C that follows a certain pattern. Which one of the following completes the sequence? (CSAT-2025)

(a) B, C, B, C, A (b) A, C, B, C, A

(c) B, C, B, A, C (d) C, B, B, A, C

15. What is X in the sequence (CSAT-2025)

24, X, 12, 18, 36, 90?

(a) 18 (b) 12

(c) 9 (d) 6

16. If P = +, Q = -, R = x, S = ÷, then insert the proper notations between the successive numbers in the equation 60_15_3_20_4 = 20: (CSAT-2025)

(a) SPRQ (b) QRPS

(c) QRSP (d) SPQR

17. Three teams P, Q, R participated in a tournament in which the teams play with one another exactly once. A win fetches a team 2 points and a draw 1 point. A team gets no point for a loss. Each team scored exactly one goal in the tournament. The team P got 3 points, Q got 2 points and R got 1 point. Which of the following statements is/are correct? (CSAT-2025)

I. The result of the match between P and Q is a draw with the score 0-0.

II. The number of goals scored by R against Q is 1.

Which of the statements given above is/are correct?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

18. P is the brother of Q and R. S is R's mother. T is P's father. How many of the following statements are definitely true? (CSAT-2025)

I. S and T are a couple. II. Q is T's son.

III. T is Q's father. IV. S is P's mother.

V. R is T's daughter. VI. P is S's son.

Select the correct answer using the code given below:

(a) Only two (b) Only three

(c) Only four (d) Only five

19. If NO is coded as 210, NOT is coded as 4200 and NOTE is coded as 21000, then how is NOTES coded? (CSAT-2025)

(a) 399000 (b) 420000

(c) 440000 (d) 630000

20. If FRANCE is coded as 654321 and GERMANY is coded as 9158437, then how is YEMEN coded? (CSAT-2025)

(a) 54321 (b) 81913

(c) 71913 (d) 71813

21. The 5-digit number PQRST (all distinct digits) is such that T≠0.P is thrice T. S is greater than Q by 4,while Q is greater than R by 3.

How many such 5-digit numbers are possible? **(CSAT-2025)**

(a) 3 (b) 4

(c) 5 (d) 6

22. In a certain code if 64 is written as 343 and 216 is written as 729, then how is 512 written in that code? (CSAT-2025)

(a) 1000 (b) 1331

(c) 1728 (d) 2197

23. A question is given followed by two Statements I and II. Consider the Question and the Statements and mark the correct option.
(CSAT-2025)

Question:

What is the smallest 1-digit number having exactly 4 distinct factors?

Statement I: 2 is one of the factors.

Statement II: 3 is one of the factors.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other statement alone.

(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.

(d) The Question can be answered even without using any of the Statements.

24. A question is given followed by two Statements I and II. Consider the Question and the Statements and mark the correct option.
(CSAT-2025)

Question:

Let P, Q, R, S be distinct non-zero digits. If PP x PQ = RRSS, where $P \leq 3$ and $Q \leq 4$, then what is Q equal to?

Statement I: R = 1.

Statement II: S=2.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other statement alone.

(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.

(d) The Question can be answered even without using any of the Statements.

25. A question is given followed by two Statements I and II. Consider the Question and the Statements and mark the correct option.
(CSAT-2025)

Question:

How is Q related to P?

Statement I: P has two sisters, R and S.

Statement II: R's father is the brother of Q.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other statement alone.

(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.

(d) The Question cannot be answered even using any of the Statements.

26. A question is given followed by two Statements I and II. Consider the Question and the Statements and mark the correct option.
(CSAT-2025)

Question:

In a football match, team P playing against Q was behind by 3 goals with 10 minutes remaining. Does team P win the match?

Statement I: Team P scored 4 goals in the last 10 minutes.

Statement II: Team Q scored a total of 4 goals in the match.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other statement alone.

(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.

(d) The Question cannot be answered even using any of the Statements.

27. A question is given followed by two Statements I and II. Consider the Question and the Statements and mark the correct option.
(CSAT-2025)

Question:

Is $(p+q)^2 - 4pq$, where p, q are natural numbers, positive?

Statement I: $p < q$

Statement II: $p > q$.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other statement alone.

(b) The Question can be answered by using either Statement

alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.

(d) The Question can be answered even without using any of the Statements.

28. A Main Statement is followed by four Statements labelled P, Q, R and S. Choose the ordered pair of the Statements where the first Statement implies the second, and the two Statements are logically consistent with the Main Statement. (CSAT-2024)

Main Statement:

Pradeep becomes either a Director or a Producer.

Statement P: Pradeep is a Director.

Statement Q: Pradeep is a Producer.

Statement R: Pradeep is not a Director.

Statement S: Pradeep is not a Producer.

Select the correct answer.

(a) SP only (b) RQ only

(c) Both SP and RQ (d) Neither SP nor RQ

29. A Statement is given followed by two Conclusions numbered I and II. Consider the Statement and the Conclusions. (CSAT-2024)

Statement:

India is the world's largest producer of milk.

Conclusion-1:

India is the world's largest exporter of milk.

Conclusion-II:

India does not import milk.

Which one of the following is correct?

(a) Only Conclusion-I follows

(b) Only Conclusion-II follows

(c) Both Conclusion-I and Conclusion-II follow

(d) Neither Conclusion-I nor Conclusion-II follows

30. If ABC and DEF are both 3-digit numbers such that A, B, C, D, E, and F are distinct non-zero digits such that ABC+ DEF= 1111, then what is the value of A+B+C+D+E+F? (CSAT-2023)

(a) 28 (c) 30

(b) 29 (d) 31

31. D is a 3-digit number such that the ratio of the number to the sum of its digits is least. What is the difference between the digit at the hundred's place and the digit at the unit's place of D? (CSAT-2023)

(a) 0 (c) 8

(b) 7 (d) 9

32. If 'ZERO' is written as 'CHUR', then how is 'PLAYER' written? (CSAT-2023)

(a) SOCAGT (b) SODBGT

(c) SODBHT (d) SODBHU

33. Consider the following statements:

1. A is older than B
2. C and D are of the same age
3. E is the youngest
4. F is younger than D
5. F is older than A

How many statements given above are required to determine the oldest person/persons? (CSAT-2023)

(a) Only two (b) Only three

(c) Only four (d) All five

34. What is the middle terms of the sequence Z, Z, Y, Y, Y, X, X,X, X, W, W, W, W, W,.....,A? (CSAT-2023)

(a) H (b) I

(c) J (d) M

35. In a party, 75 persons took tea, 60 persons took coffee and 15 persons took both tea and coffee. No one taking milk takes tea. Each person takes at least one drink. Question: how many persons attended the party?

Statement-1: 50 persons took milk.

Statement-2: Number of persons who attended the party is five times the number of persons who took milk only. Which one of the following is correct in respect of the above Question and the Statements? (CSAT-2023)

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.

(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.

(d) The Question cannot be answered even by using both the Statements together.

36. Consider the following:

I. A+B means A is neither smaller nor equal to B.

II. A-B means A is not greater than B.

III. A x B means A is not smaller than B.

IV. A÷ B means A is neither greater nor equal to B.

V. A+ B means A is neither smaller nor greater than B.

Statement: $P \times Q, P - T, T \div R, R \pm S$
Conclusion-1: $Q \pm T$
Conclusion-2: S+Q

Which one of the following is correct in respect of the above

Statement and the Conclusions? (CSAT-2023)

(a) Only Conclusion-1 follows from the Statement.

(b) Only Conclusion-2 follows from the Statement.

(c) Both Conclusion-1 and Conclusion-2 follow from the Statement.

(d) Neither Conclusion-1 nor Conclusion-2 follows from the Statement.

37. Consider the following statements in respect of five candidates P,Q,R,S, and T. Two statements are true and one statement is false.

True Statement: One of P and Q was selected for the job.

False Statement: At least one of R and S was selected for the job.

True Statement: At most two of R, S and T were selected for the job.

Which of the following conclusions can be drawn?

1. At least four were selected for the job.
2. S was selected for the job.

Select the correct answer using the code given below:(CSAT-2023)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

38. Let P,Q, R, S and T be five statements such that:

I. If P is true, then both Q and S are true.

II. If R and S are true, then T s false.

Which of the following can be concluded?

1. If T is true, then at least one of P and R must be false.
2. If Q is true, then P is true.

Select the correct answer using the code given below:(CSAT-2023)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

39. The letters of the word "INCOMPREHENSIBILITIES" are arranged alphabetically in reverse order. How many positions of the letter/letters will remain unchanged? (CSAT-2023)

(a) None (b) One

(c) Two (d) Three

40. Two friends X and Y start running, and they run together for 50 m in the same direction and reach a point. X turns right and runs 60 m, while Y turns left and runs 40m Then X turns left and runs 50m and stops, while Y turns right and runs 50 m and then stops. How far are the two friends from each other now? (CSAT-2022)

(a) 100m (c) 60m

(b) 90m (d) 50m

41. Which date of June 2099 among the following is Sunday? (CSAT-2022)

(a) 4 (c) 6

(b) 5 (d) 7

42. If the order of the letters in the English alphabet is reversed and each letter represents the letter whose position it occupies, then which one of the following represents 'LUCKNOW'? (CSAT-2022)

(a) OGXPMLD (c) OFXPMLE

(b) OGXQMLE (d) OFXPMLD

43. Eight students A, B, C, D, E, F, G and H sit around a circular table, equidistant from each other, facing the centre of the table, not necessarily in the same order. B and D sit neither adjacent to C nor opposite to C. A sits in between E and D, and F sits in between B and H. Which one of the following is definitely correct? (CSAT-2022)

(a) B sits in between A and G (c) E sits opposite to F

(b) C sits opposite to G (d) None of the above

44. An Identity Card has the number ABCDEFG, not necessarily in that order, where each letter represents a distinct digit (1, 2, 4, 5, 7, 8, 9 only). The number is divisible by 9. After deleting the first digit from the right, the resulting number is divisible by 6. After deleting two digits from the right of original number, the resulting number is divisible by 5. After deleting three digits from the right of original number, the resulting number is divisible by 4. After deleting four digits from the right of original number, the resulting number is divisible by 3. After deleting five digits from the right of original number, the resulting number is divisible by 2. Which of the following is a possible value for the sum of the middle three digits of the number? (CSAT-2022)

(a) 8 (b) 9

(c) 11 (d) 12

45. Consider the Question and two Statements given below: (CSAT-2022)

Question: Is Z brother of X?

Statement-1:" X is a brother of Y and Y is a brother of Z

Statement-2: X, Y and Z are siblings.

Which one of the following is correct in respect of the Question and the Statements?

(a) Statement-1 alone is sufficient to Answer the Questions.

(b) Statement-2 alone is sufficient to Answer the Question.

(c) Both Statement-1 and Statement-2 are sufficient to Answer the Question.

(d) Both Statement-1 and Statement-2 are not sufficient to Answer the Question.

46. How many seconds in total are there in x weeks, x days, x hours x minutes and X seconds? (CSAT-2022)

(a) 11580x (c) 694860x

(b) 11581x (d) 694861x

47. P, Q, R, S, T and U are six members of a family. R is the spouse of Q, U is the mother of T and S is the daughter of U. P's daughter is

T and R's son is P. There are two couples in the family. Which one of the following is correct? (CSAT-2022)

(a) Q is the grandfather of T (b) Qs the grandmother of T

(c) R is the mother of P (d) T is the granddaughter of Q

48. Two Statements followed by four Conclusions are given below. You have to take the Statements to be true even if they seem to be at variance from the commonly known facts. Read all the Conclusions and then decide which of the given Conclusions logically follows follow from the Statements, disregarding the commonly known facts: (CSAT-2022)

Statement-1: All pens are books.

Statement-2: No chair is a pen.

Conclusion-I: All chairs are books.

Conclusion-II: Some chairs are pens.

Conclusion-III: All books are chairs.

Conclusion-IV: No chair is a book.

Which one of the following is correct?

(a) Only Conclusion-I

(b) Only Conclusion-II

(c) Both Conclusion-lll and Conclusion-IV

(d) None of the Conclusion follows.

49. Three Statements followed by three Conclusions are given below. You have to take the Statements to be true even if they seem to be at variance from the commonly known facts. Read all the Conclusions and then decide which of the given Conclusions logically follows/ follow from the Statements, disregarding the commonly known facts: (CSAT-2022)

Statement-1: Some doctors are teachers.

Statement-2: All teachers are engineers.

Statement-3: All engineers are scientists.

Conclusion-I: Some scientists are doctors.

Conclusion-II: All engineers are doctors.

Conclusion-II: Some engineers are doctors.

Which one of the following is correct?

(a) Only Conclusion-I

(b) Only Conclusion-II

(c) Both Conclusions-I and Conclusion-lll

(d) Both Conclusions-I and Conclusion-II

50. Six Persons A, B, C, D, E and F are sitting equidistant from each other around a circular table (facing the center of the table). Consider the Question and two statements given below: (CSAT-2022)

Question: Who is sitting on the immediate left of A?

Statement-1: B is sitting opposite to C and D is sitting opposite to E.

Statement -2: F is sitting on the immediate left of B.

Which one of the following is correct in respect of the Question and the Statements?

(a) Statement-1 alone is sufficient to Answer the question.

(b) Statement-2 alone is sufficient to Answer the question.

(c) Both Statement-1 and Statement-2 are sufficient to Answer the Question.

(d) Both Statement-1 and Statement-2 are not sufficient to Answer the Question.

51. Consider the Question and two Statements given below: (CSAT-2022)

Question: What is the age of Manisha?

Statement-1: Manisha is 24 years younger than her mother.

Statement-2: 5 years later, the ages of Manisha and her mother will be in the ratio 3: 5.

Which one of the following is correct in respect of the Question and the Statement?

52. Six lectures A, B, C, D, E and F, each of one hour duration, are scheduled between 8:00 am, and 2:00 p.m. (CSAT-2022)

Consider the Question and two Statements given below: Question: Which lecture is in the third period?

Statement-1: Lecture F is preceded by A and followed by C.

Statement-2: There is no lecture after lecture B.

Which one of the following is correct in respect of the Question and the Statements?

(a) Statement-1 alone is sufficient to Answer the question.

(b) Statement-2 alone is sufficient to Answer the question.

(c) Both Statement-1 and Statement-2 are sufficient to Answer the Question.

(d) Both Statement-1 and Statement-2 are not sufficient to Answer the Question.

53. Consider the following statements: (CSAT-2022)

1. Between 3:16 p.m. and 3:17 p.m., both hour hand and minute hand coincide.
2. Between 4:58 p.m. and 4:59 p.m., both minute hand and second hand coincide.

Which of the above statements is/are correct?

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

54. A Statement followed by Conclusion-I and Conclusion-II is given below. You have to take the Statement to be true even if it seems to be at variance from the commonly known facts. Read all Conclusions and then decide which of the given Conclusion(s) logically follows/follow from the Statement, disregarding the commonly known facts. Statement: Some radios are mobiles. All mobiles are

computers. Some computers are watches. Conclusion-I: Certainly, some radios are watches. Conclusion-II: Certainly, some mobiles are watches. Which one of the following is correct? (CSAT-2021)

(a) Only Conclusion-I

(b) Only Conclusion-II

(c) Both Conclusion-I and Conclusion-II

(d) Neither Conclusion-I not Conclusion-II

55. A Statement followed by Conclusion-I and Conclusion-II is given below. You have to take the Statement to be true even if it seems to be at variance from the commonly known facts. Read all Conclusions and then decide which of the given Conclusion(s) logically follows/follow from the Statement, disregarding the commonly known facts.

Statement: Some cats are almirahs. Some almirahs are chairs. All chairs are tables.

Conclusion-I: Certainly, some almirahs are tables.

Conclusion-II: Some cats may not be chairs.

Which one of the following is correct? (CSAT-2021)

(a) Only Conclusion-I

(b) Only Conclusion-II

(c) Both Conclusion-I and Conclusion-II

(d) Neither Conclusion-I nor Conclusion-II

56. Images of consonants of the English alphabet (Capitals) are observed in a mirror. What is the number of images of these which do not look like their original shapes? (CSAT-2021)

(a) 13 (c) 15

(b) 14 (d) 16

57. A bank employee drives 10 km towards South from her house and turns to her left and drives another 20 km. She again turns left and drives 40 km, and then she turns to her right and drives for another 5 km. She again turns to her right and drives another 30 km to reach her bank where she works. What is the shortest distance between her bank and her house? (CSAT-2021)

(a) 20 km (b) 25 km

(c) 30 km (d) 35 km

58. A woman runs 12 km towards her North, then 6 km towards her South and then 8 km towards her East. In which direction is she from her starting point? (CSAT-2021)

(a) An angle less than 45° South of East

(b) An angle less than 45° North of East

(c) An angle more than 45° South of East

(d) An angle more than 45° North of East

59. Replace the incorrect term by correct term in the given sequence.

3,2,7,4,13,10, 21, 18, 31, 28, 43, 40

where odd terms and even terms follow the same pattern.

(CSAT-2021)

(a) 0 (b) 1

(c) 3 (d) 6

60. 42. Following is a matrix of certain entries. The entries follow a certain trend row-wise. Choose the missing entry (?) accordingly.

7B	**10A**	**3C**
3C	**9B**	**6A**
10A	**13C**	**?**

(CSAT-2021)

(a) 9B (b) 3A

(c) 3B (d) 3C

61. You are given two identical sequences in two rows.

Sequence-I: 8 4 6 15 52.5 236.5

Sequence-II: 5 A B C D E

What is the entry in the place of C for the Sequence-II? (CSAT-2021)

(a) 2.5 (b) 5

(c) 9.375 (d) 32.8125

62. Consider two Statements and a Question:

Statement-1: Priya is 4 ranks below Seema and is 31st from the bottom.

Statement-2: Ena is 2 ranks above Seema and is 37th from the bottom Question: What is Seema's rank from the top in the class of 40 students?

Which one of the following is correct in respect of the Statements and the Question? (CSAT-2021)

(a) Statement-1 alone is not sufficient to Answer the Question

(b) Statement-2 alone is not sufficient to Answer the Question.

(c) Either Statement-1 alone or Statement-2 alone is sufficient to Answer the Question.

(d) Both Statement-1 and Statement-2 are required to Answer the Questions.

63. Consider two Statements and a Question:

Statement-1: Each of A and D is heavier than each of B, and F, but one of them is the heaviest.

Statement-2: A is heavier than D, but is lighter than C.

Question: Who is the heaviest among A, B, C, D and E?

Which one of the following is correct in respect of the Statements and the Question? (CSAT-2021)

(a) Statement-1 alone is sufficient to Answer the Question

(b) Statement-2 alone is sufficient to Answer the Question

(c) Both Statement-1 and Statement-2 are required to Answer the Question

(d) Neither Statement-1 alone nor Statement-1 alone is sufficient to Answer the Question

64. In the English alphabet, the first 4 letters are written in opposite order, and the next 4 letters are written in opposite order and so on; and at the end Y and Z are interchanged. Which will be the fourth letter to the right of the 13" letter? (CSAT-2021)

(a) N (b) T

(c) H (d) I

65. Consider two Statements and a Question:

Statement 1: The last day of the month is a Wednesday.

Statement 2: The third Saturday of the month was the seventeenth day.

Question: What day is the fourteenth of the given month?

Which one of the following is correct in respect of the Statements and the Question? (CSAT-2021)

(a) Statement 1 alone is sufficient to the Answer the Question.

(b) Statement 2 alone is sufficient to Answer the Question.

(c) Both Statement 1 and Statement 2 are required to Answer the Question.

(d) Neither Statement 1 alone nor Statement 2 alone is sufficient to Answer the Question.

66. Which day is 10" October 2027 (CSAT-2021)

(a) Sunday (b) Monday

(c) Tuesday (d) Saturday

67. Consider two Statements and four Conclusions given below. You have to take the Statements to be true even if they seem to be at variance from the commonly known facts. Read all Conclusions and then decide which of the given Conclusion(s) logically follows/follow from the Statements, disregarding the commonly known facts.

Statement- 1: Some greens are blues.

Statement-2: Some blues are blacks.

Conclusion-1: Some greens are blacks.

Conclusion-2: No green is black.

Conclusion-3: All greens are blacks.

Conclusion-4: All blacks are greens.

Which one of the following is correct? (CSAT-2021)

(a) Conclusion-1 and Conclusion-2 only

(b) Conclusion-2 and Conclusion-3 only

(c) Conclusion-3 and Conclusion-4 only

(d) Neither Conclusion 1 nor 2 nor 3 nor 4

68. What is the value of 'X' in the sequence 2, 7, 22, 67,202, X, 1822? (CSAT-2021)

(a) 603 (b) 605

(c) 607 (d) 608

69. Jay and Vijay spent an equal amount of money to buy some pens and special pencils of the same quality from the same store. If Jay bought 3 pens and 5 pencils, and Vijay bought 2 pens and 7 pencils, then which one of the following is correct? (CSAT-2021)

(a) A pencil costs more than a pen.

(b) The price of a pencil is equal to that of a pen.

(c) The price of a pen is two times the price of a pencil.

(d) The price of a pen is three times the price of a pencil.

70. In a code language 'MATHEMATICS' is written as 'LBSIDNZUHDR'. How is 'CHEMISTRY' written in that code language? (CSAT-2021)

(a) DIDLHRSSX (b) BIDNHTSSX

(c) BIDLHTSSX (d) DGFLIRUQZ

71. At which one of the following times, do the hour hand and the minute hand of the clock make an angle of 180° with each other? (CSAT-2021)

(a) At 7:00 hours

(b) Between 7:00 hours and 7:05 hours

(c) At 7:05 hours

(d) Between 7:05 hours and 7:10 hours

72. In the series: _b_a_ba_b_abab_aab; fill in the six blanks (_) using one of the following given four choices such that the series follows a specific order. (CSAT-2021)

(a) bababa (b) baabba

(c) bbaabb (d) ababab

73. Joseph visits the club on every 5th day; Harsh visits on every 24th day, while Sumit visits on every 9th day. If all three of them met at the club on a Sunday, then on which day will all three of them meet again? (CSAT-2021)

(a) Monday (b) Wednesday

(c) Thursday (d) Sunday

74. Two Statements S1 and S2 are given below with regard to four members P, Q, R and S followed by a Question:

S1: R is greater than P as well as Q.

S2: S is not the largest one.

Question: Among four numbers P, Q, R and S, which one is the largest?

Which one of the following is correct in respect of the above Statements and the Question? (CSAT-2020)

(a) S1 alone is sufficient to Answer the Question.

(b) S2 alone is sufficient to Answer the Question.

(c) S1 and S2 together are sufficient to Answer the Question, but neither S1 nor S2 alone is sufficient to Answer the question.

(d) S1 and S2 together are not sufficient to Answer the Question.

75. Two Statements S1 and S2 are given below followed by a Question:

S1: n is a prime number.

S2: n leaves a remainder of 1 when divided by 4.

Question: If n is a unique natural number between 10 and 20, then what is n?

Which one of the following is correct in respect of above Statements and the Question? (CSAT-2020)

(a) S1 alone is sufficient to Answer the Question.

(b) 52 alone is sufficient to Answer the Question.

(c) S1 and S2 together are sufficient to Answer the Question, but neither 51 nor S2 alone is sufficient to Answer the Question.

(d) S1 and S2 together are not sufficient to Answer the Question.

76. Two Statements S1 and S2 are given below with regard to two numbers followed by a Question:

S1: Their product is 21.

S2: Their sum is 10.

Question: What are the two numbers?

Which one of the following is correct in respect of the above Statements and the Question? (CSAT-2020)

(a) S1alone is sufficient to Answer the Question.

(b) S2 alone is sufficient to Answer the Question.

(c) S1 and S2 together are sufficient to Answer the Question, but neither S1 nor S2 alone is sufficient to Answer the Question.

(d) S1 and S2 together are not sufficient to Answer the Question.

77. Two statements are given followed by two Conclusions:

Statements:

All numbers are divisible by 2.

All numbers are divisible by 3.

Conclusion-I:

All numbers are divisible by 6.

Conclusion-II:

All numbers are divisible by 4.

Which of the above Conclusions logically follows/follow from the two given Statements? (CSAT-2020)

(a) Only Conclusion-I

(b) Only Conclusion-II

(c) Neither Conclusion-I nor Conclusion-II

(d) Both Conclusions-I and Conclusion-II

78. Two Statements are given followed by two Conclusions:

Statements:

All cats are dogs.

All cats are black.

Conclusion-I:

All dogs are black.

Conclusion-II:

Some dogs are not black.

Which of the above Conclusions logically follows/follow, from the two given Statements, disregarding commonly known facts? (C5AT-2020)

(a) Only Conclusion-I

(b) Only Conclusion-II

(c) Neither Conclusion-I nor Conclusion-II

(d) Both Conclusions-I and Conclusion-II

79. Question 36. Consider the following sequence of numbers: 5 1 4 7 3 9 8 5 7 2 6 3 1 5 8 6 3 8 5 2 2 4 3 4 9 6

How many odd numbers are followed by the odd number in the above sequence? (CSAT-2020)

(a) 5 (b) 6

(c) 7 (d) 8

80. Three Statements S1, S2 and S3 are given below followed by a Question:

S1: C is younger than D, but older than A and B.

S2: D is the oldest.

S3: A is the older than B.

Questions: Who among A, B, C and D is the youngest?

Which one of the following is correct in respect of the above Statements and the Question? (CSAT-2020)

(a) S1alone is sufficient to Answer the Question.

(b) S1 and S2 together are sufficient to Answer the Question.

(c) S2 and S3 together are sufficient to Answer the Question.

(d) S1 and S3 together are sufficient to Answer the Question.

81. A family of two generations consisting of six members P, Q, R, S, T and U has three males and three females. There are two married couples and two unmarried siblings. U is P's daughter and Q is R's mother-in-law. T is an unmarried male and S is a male. Which one of the following is correct? (CSAT-2020)

(a) R is U's husband (b) R is S's wife

(c) S is unmarried (d) None of the above

82. Consider the following Statements and Conclusions: Statements: (CSAT-2019)

1. Some rats are cats.
2. Some cats are dogs.
3. No dog is a cow.

Conclusions:

I. No cow is a rat.

II. No dog is a rat.

III. Some cats are rats.

Which of the above conclusions is/are drawn from the statements?

(a) I, II and III (c) Only III

(b) Only I and II (d) Only II and III

83. In a conference, out of a total 100 participants, 70 are Indians. If 60 of the total participants are vegetarians, then which of the following statements is/are correct? 1. At least 30 Indian participants are vegetarians. 2. At least 10 Indian participants are non-vegetarians. Select the correct Answer using the codes given below: (CSAT-2019)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

84. All members of a club went to Mumbai and stayed in a hotel. On the first day, 80% went for shopping and 50% went for sightseeing, whereas 10% took rest in the hotel. Which of the following conclusion(s) can be drawn from the above data?

1. 40% members went for shopping as well as sightseeing.
2. 20% members went for only shopping.

Select the correct Answer using the code given below: (CSAT-2019)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

85. In a school, 60% students play cricket. A student, who does not play cricket, plays football. Every football player has got a two-wheeler. Which of the following conclusions cannot be drawn from the above data?

1. 60% of the students do not have two-wheelers.
2. No cricketer has a two-wheeler.
3. Cricket players do not play football.

Select the correct Answer using the given code below: (CSAT-2019)

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3 only

86. In a group of 15 people; 7 can read French, 8 can read English while 3 of them can read neither of these two languages. The number of people who can read exactly read one language is (CSAT-2019)

(a) 10 (b) 9

(c) 5 (d) 4

87. Consider the following sequence that follows some arrangement: c_accaa_aa_bc_b

The letters that appear in the gaps are (CSAT-2019)

(a) abba (b) cbbb

(c) bbbb (d) cecc

88. A joint family consists of seven members A, B, C, D, E, F and G with three females. G is a widow and sister-in-law of D's father F. B and D are siblings and A is daughter of B. C is cousin of B. Who is E? (CSAT-2019)

1. Wife of F
2. Grandmother of A
3. Aunt of C

Select the correct Answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

89. Which one of the following statements is correct? (CSAT-2019)

(a) Weight of Mohan is least

(b) Weight of Sohan is least

(c) Weight of Rohan is least

(d) Whose weight is least cannot be determined.

90. Consider the following sum:

$\bullet + 1 \bullet + 2 \bullet + 3 \bullet + 1 = 21 \bullet$

In the above sum, • stands for (CSAT-2018)

(a) 4 (b) 5

(c) 6 (d) 8

91. Consider following pattern of numbers:

8	10	15	13
6	5	7	4
4	6	8	8
6	11	16	?

What is the number at? in the above pattern? (CSAT-2018)

(a) 17 (b) 19

(c) 21 (d) 23

92. If LSJXVC is the code for MUMBAI, the code for DELHI is (CSAT-2018)

(a) CCIDD (b) CDKGH

(c) CCJFG (d) CCIFE

93. If RAMON is written as 12345 and DINESH as 675849, then HAMAM will be written as (CSAT-2018)

(a) 92233 (b) 92323

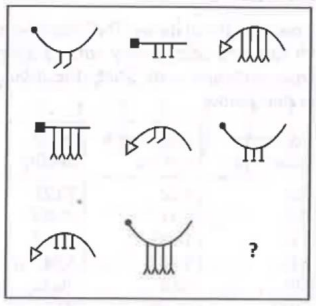
(c) 93322 (d) 93232

94. Consider the sequence given below: 4/12/95, 1/1/96, 29/1/96, 26/2/96 ... What is the next term of the series? (CSAT-2018)

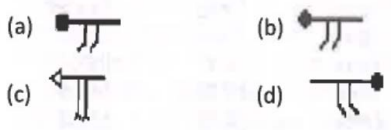
(a) 24/3/96 (b) 25/3/96

(c) 26/3/96 (d) 27/3/96

95. Consider the figures given below:



To fit the question mark, the correct Answer is (CSAT-2018)



96. Examine the following statements:

1. All colours are pleasant.
2. Some colours are pleasant.
3. No colour is pleasant.
4. Some colours are not pleasant.

Given that statement 4 is true, what can be definitely concluded? (CSAT-2017)

(a) 1 and 2 are true (b) 3 is true

(c) 2 is false (d) 1 is false

97. Consider the following relationships among members of a family of six persons A, B, C, D, E and F: (CSAT-2017)

1. The number of males equals that of females.
2. A and E are sons of F.
3. D is the mother of two, one boy and one girl.
4. B is the son of A.
5. There is only one married couple in the family at present.

Which one of the following inferences can be drawn from the above?

(a) A, B and C are all females.

(b) A is the husband of D.

(c) E and F are children of D.

(d) D is the daughter of F.

98. If 2 boys and 2 girls are to be arranged in a row so that the girls are not next to each other, how many possible arrangements are there? (CSAT-2017)

(a) 3 (b) 6

(c) 12 (d) 24

99. Consider the following:

A, B, C, D, E, F, G and H are standing in a row facing north.

B is not neighbour of G.

F is to the immediate right of G and neighbour of E.

G is not at the extreme end.

A is sixth to the left of E.

H is sixth to the right of C.

Which one of the following is correct in respect of the above? (CSAT-2017)

(a) C is to the immediate left of A.

(b) D is immediate neighbour of B and F.

(c) G is to the immediate right of D.

(d) A and E are at the extreme ends.

100. In a certain code, '256' means 'red colour chalk', '589' means 'green colour flower' and '254' means 'white colour chalk'. The digit in the code that indicates 'white' is (CSAT-2017)

(a) 2 (b) 4

(c) 5 (d) 8

Directions for the following 3 (three) items: Consider the given information and answer the three items that follow. A, B, C, D, E, F and G are Lecturers from different cities - Hyderabad, Delhi, Shillong, Kanpur, Chennai, Mumbai and Srinagar (not necessarily in the same order) who participated in a conference. Each one of them is specialized in a different subject, viz., Economics, Commerce, History, Sociology, Geography, Mathematics and Statistics (not necessarily in the same order). Further,

1. Lecturer from Kanpur is specialized in Geography.
2. Lecturer D is from Shillong.
3. Lecturer C is from Delhi is specialized in Sociology.
4. Lecturer 8 is specialized in neither History nor Mathematics.
5. Lecturer A who is specialized in Economics does not belong to Hyderabad.
6. Lecturer F who is specialized in Commerce belongs to Srinagar.
7. Lecturer G who is specialized in Statistics belongs to Chennai.

101. Who is specialized in Geography? (CSAT-2017)

(a) B

(b) D

(c) E

(d) Cannot be determined as data are inadequate

102. To which city does the Lecturer specialized in Economics belong? (CSAT-2017)

(a) Hyderabad

(b) Mumbai

(c) Neither Hyderabad nor Mumbai

(d) Cannot be determined as data are inadequate

103. Who of the following belongs to Hyderabad? (CSAT-2017)

(a) B

(b) E

(c) Neither B nor E

(d) Cannot be determined as data are inadequate

104. In a school, there are five teachers A, B, C, D and E. A and B teach Hindi and English. C and B teach English and Geography. D and A teach Mathematics and Hindi. E and B teach History and French. Who teaches maximum number of subjects? (CSAT-2017)

(a) A (b) B

(c) D (d) E

Directions for the following 3 (three) items: Consider the given information and answer the three items that follow. Eight railway stations A, B, C, D, E, F, G and H are connected either by two-way passages or one-way passages. One-way passages are from C to A, E to G, B to F, D to H, G to C, E to C and H to G. Two-way passages are between A and E, G and B, F and D, and E and D. (CSAT-2017)

105. While travelling from C to H, which one of the following stations must be passed through? (CSAT-2017)

(a) G (b) E

(c) B (d) F

106. In how many different ways can a train travel from F to A without passing through any station more than once? (CSAT-2017)

(a) 1 (b) 2

(c) 3 (d) 4

107. If the route between G and C is closed, which one of the following stations need not be passed through while travelling from H to C? (CSAT-2017)

(a) E (b) D

(c) A (d) B

108. In a group of six women, there are four tennis players, four postgraduates in Sociology, one postgraduate in Commerce and three bank employees. Vimala and Kamla are the bank employees while Amala and Komala are unemployed. Komala and Nirmala are among the tennis players. Amala, Kamla, Komala and Nirmala are postgraduates in Sociology of whom two are bank employees. If Shyamala is a postgraduate in Commerce, who among the following is both a tennis player and a bank employee? (CSAT-2017)

(a) Amala (b) Komala

(c) Nirmala (d) Shyamala

109. A clock strikes once at 1 o'clock, twice at 2 o'clock and thrice at 3 o'clock, and so on. If it takes 12 seconds to strike at 5 o'clock, what is the time taken by it to strike at 10 o'clock? (CSAT-2017)

(a) 20 seconds (b) 24 seconds

(c) 28 seconds (d) 30 seconds

110. Consider the given statement and the two conclusions that follow:

Statement:

Morning walk is good for health.

Conclusions:

1. All healthy people go for morning walk.
2. Morning walk is essential for maintaining good health.

What is/are the valid conclusion/ conclusions? (CSAT-2017)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

111. Six boys A, B, C, D, E and F play a game of cards. Each has a pack of 10 cards. F borrows 2 cards from A and gives away 5 to C who in turn gives 3 to B while B gives 6 to D who passes 1 to E. Then the number of cards possessed by D and E is equal to the number of cards possessed by (CSAT-2017)

(a) A, B and C (b) B, C and F

(c) A, B and F (d) A, C and F

112. There are 4 horizontal and 4 vertical lines, parallel and equidistant to one another on a board. What is the maximum number of rectangles and squares that can be formed? (CSAT-2017)

(a) 16 (b) 24

(c) 36 (d) 42

113. In a test, Randhir obtained more marks than the total marks obtained by Kunal and Debu. The total marks obtained by Kunal and Shankar are more than those of Randhir. Sonal obtained more marks than Shankar. Neha obtained more marks than Randhir. Who amongst them obtained highest marks? (CSAT-2017)

(a) Randhir (b) Neha

(c) Sonal (d) Data are inadequate

Directions for the following 2 (two) items:

Consider the given information and answer the two items that follow.

No supporters of 'party X', who knew Z and supported his campaign strategy, agreed for the alliance with 'party Y'; but some of them had friends in 'party Y'.

114. With reference to the above information, which one among the following statements must be true? (CSAT-2017)

(a) Some supporters of 'party Y' did not agree for the alliance with the 'party X'.

(b) There is at least one supporter of 'party Y' who knew some supporters of 'party X' as a friend.

(c) No supporters of 'party X' supported Z's campaign strategy.

(d) No supporters of 'party X' knew Z.

115. With reference to the above information, consider the following statement:

1. Some supporters of 'party X' knew Z.

2. Some supporters of 'party X', who opposed 2's campaign strategy, knew Z.

3. No supporters of 'party X' supported Z's campaign strategy.

Which of the statements given above is/are not correct? (CSAT-2017)

(a) 1 only (b) 2 and 3 only

(c) 3 only (d) 1, 2 and 3

116. If second and fourth Saturdays and all the Sundays are taken as only holidays for an office, what would be the minimum number of possible working days of any month of any year? (CSAT-2017)

(a) 23 (b) 22

(c) 21 (d) 20

117. Four tests-Physics, Chemistry, Mathematics and Biology are to be conducted on four consecutive days, not necessarily in the same order. The Physics test is held before the test which is conducted after Biology. Chemistry is conducted exactly after two tests are held. Which is the last test held? (CSAT-2017)

(a) Physics (b) Biology

(c) Mathematics (d) Chemistry

118. Consider the following:

Statement:

Good voice is a natural gift, but one has to keep practicing to improve and well in the field of music.

Conclusions:

1. Natural gifts need nurturing and care.

2. Even though one's voice is not good, one can keep practicing.

Which one of the following is correct, in respect of the above statement and conclusions? (CSAT-2017)

(a) Only conclusion I follows from the statement.

(b) Only conclusion II follows from the statement.

(c) Either conclusion I or conclusion II follows from the statement.

(d) Neither conclusion I nor conclusion II follows from the statement.

119. 15 students failed in a class of 52. After removing the names of failed students, a merit order list has been prepared in which the position of Ramesh is 22nd from the top. What is his position from the bottom? (CSAT-2017)

(a) 18th (b) 17th

(c) 16th (d) 15th

120. Consider the following:

A+ B means A is the son of B.

A-B means A is the wife of B.

What does the expression P+R-Q mean? (CSAT - 2017)

(a) Q is the son of P.

(b) Q is the wife of P.

(c) Q is the father of P.

(d) None of the above

121. A ate grapes and pineapple; B ate grapes and oranges; C ate oranges, pineapple and apple; D ate grapes, apple and pineapple. After taking fruits, B and C fell sick. In the light of the above facts, it can be said that the cause of sickness was: (CSAT-2016)

(a) Apple (c) Grapes

(b) Pineapple (d) Oranges

122. Consider the following statements:

1. The rate of population growth is increasing in the country.

2. The death rate is declining faster in the country compared to birth rate.

3. The birth rate is declining faster in the country compared to death rate.

4. Rural-urban migration is taking place regularly in the country.

Which one of the following conclusions may be true in the light of the above facts? (CSAT-2016)

(a) The rate of population growth is increasing due to rural-urban migration.

(b) The rate of population growth is increasing due to decline in death rate only.

(c) The rate of population growth is increasing due to increase in birth rate only.

(d) The rate of population growth is increasing due to faster decline in death rate than in birth rate.

123. Consider the following statement:

"We shall go either for a picnic or for trekking"

Which of the following, if true, would falsify this claim? (CSAT-2016)

(a) We go for a picnic but not for trekking.

(b) Activities such as picnic and trekking are encouraged by the health authorities.

(c) We go for trekking and not for picnic.

(d) We do not go either for picnic or for trekking.

124. There were 50 faculty members comprising 30 males and the rest females. No male faculty member knew music, but many of the female faculty members did. The Head of the institution invited six faculty members to a tea party by draw of lots. At the party it was discovered that no member knew music. The conclusion is that: (CSAT-2016)

(a) The party comprised male faculty members only.

(b) The party comprised only those female faculty members who could not give renderings in music.

(c) The party comprised both male and female faculty members.

(d) Nothing can be said about the gender composition of the party.

125. Five people A, B, C, D and E are seated about a round table. Every chair is spaced equidistant from adjacent chairs.

1. C is seated next to A
2. A is seated two seats from D
3. B is not seated next to A

On the basis of above information, which of the following must be true?

1. D is seated next to B
2. E is seated next to A
3. D and Care separated by two seats.

Select the correct Answer using the code given below: (CSAT-2016)

(a) 1 only (c) 3 only

(b) 1and 2 only (d) Neither 1 nor 2 nor 3

126. There are five hobby clubs in a college — photography, yachting, chess, electronics, and gardening. The gardening group meets every second day, the electronics group meets every third day, the chess group meets every fourth day, the yachting group meets every fifth day, and the photography group meets every sixth day. How many times do all the five groups meet on the same day within 180 days? (CSAT-2016)

(a) 5 (c) 10

(b) 18 (d) 3

Passage

Directions for the following 5 (five) items: Consider the following information and answer the five items that follow:

There are five persons in a group — P, Q, R, Sand T. The group has one doctor, one lawyer and one artist. P and S are unmarried students. T is a man married to one of the group members. Q is the brother of P and is neither doctor nor artist. R is not doctor.

127. Who is the doctor? (CSAT-2016)

(a) T (c) Q

(b) P (d) R

128. Who is the artist? (CSAT-2016)

(a) P (c) R

(b) Q (d) T

129. Who is the spouse of R? (CSAT-2016)

(a) P (c) Q

(b) T (d) S

130. Who is the lawyer? (CSAT-2016)

(a) P (c) R

(b) Q (d) S

131. Who of the following is definitely a man? (CSAT-2016)

(a) P (c) Q

(b) S (d) None of the above

132. Consider the following statements:

1. Either A and B are of the same age or A is older than B.
2. Either C and D are of the same age or D is older than C.
3. B is older than C.

Which of the following conclusions can be drawn from the above statements? (CSAT-2016)

(a) A is older than B.

(b) B and D are of the same age.

(c) D is older than C.

(d) A is older than C.

Passage

Direction for the following 3 (three) items: Consider the given information and answer the three items that follow.

Six boxes A, B, C, D, E and F have been painted with six different colours viz., violet, indigo, blue, green, yellow, and orange and arranged from left to right (not necessarily either kept or painted with the colours in the same order). Each box contains a ball of any one of the following six games: cricket, hockey, tennis, golf, football, and volleyball (not necessarily in the same order). The golf ball is in violet box and is not in the box D. The box A which contains tennis ball is orange in colour and is at the extreme right. The hockey ball is neither in box D nor in box E. The box C having cricket ball is painted green. The hockey ball is neither in the box painted blue nor in the box painted yellow. The box C is fifth from right and next to box B. The box B contains volleyball. The box containing the hockey ball is between the boxes containing golf ball and volleyball.

133. Which one of the following boxes contains the golf ball? (CSAT-2016)

(a) F (c) E

(b) D (d) None of the above

134. Which of the following statements is/are correct? (CSAT-2016)

(a) D is painted yellow (c) B is painted blue

(b) F is painted indigo (d) All of the above

135. The football is in the box of which colour? (CSAT-2016)

(a) Yellow

(b) Indigo

(c) Cannot be determined as data are inadequate

(d) Blue

136. A person climbs a hill in a straight path from point '0' on the ground in the direction of north-east and reaches a point 'A' after travelling a distance of 5 km. Then, from the point 'A' he moves to point 'B' in the direction of northwest. Let the distance AB be 12 km. Now, how far is the person away from the starting point '0'? (CSAT-2016)

(a) 7 km (c) 17 km

(b) 13 km (d) 11 km

Passage

Direction for the following 3 (three) items: Consider the given information and answer the three items that follow.

When three friends A, B and C met, it was found that each of them wore an outer garment of a different colour. In random order, the garments are jacket, sweater and tie; and the colours are: blue, white and black. Their surnames in random order are Ribeiro, Kumar and Singh. Further, we know that:

1. Neither B nor Ribeiro wore a white sweater
2. C Wore a tie
3. Singh's garment was not white
4. Kumar does not wear a jacket
5. Ribeiro does not like to wear the black colour
6. Each of the friends wore only one outer garment of only one colour

137. What is C's surname? (CSAT-2016)

(a) Ribeiro (c) Kumar

(b) Singh (d) Cannot be determined

138. What is the colour of the tie? (CSAT-2016)

(a) Black (c) White

(b) Blue (d) Cannot be determined

139. Who wore the sweater? (CSAT2016)

(a) A (c) B

(b) C (d) Cannot be determined

140. A military code writes SYSTEM as SYSMET and NEARER as AENRER. Using the same code, FRACTION can be written as: (CSAT-2016)

(a) CARFTION (c) NOITCARE

(b) FRACNOIT (d) CARFNOIT

141. Between 6 PM and 7 PM the minute hand of a clock will be ahead of the hour hand by 3 minutes at (CSAT-2015)

(a) 6:15 PM (b) 6:18 PM

(c) 6:36 PM (d) 6:48 PM

142. There are 5 tasks and 5 persons. Tasks-1 cannot be assigned to either person-1 or person-2. Task -2 must be assigned to either person-3 or person-4. Every person is to be assigned one task. In how many ways can the assignment be done? (CSAT-2015)

(a) 6 (b) 12

(c) 24 (d) 144

143. In a box of marbles, there are three less white marbles than the red ones and five more white marbles than the green ones. If there are a total of 10 white marbles, how many marbles are there in the box? (CSAT-2015)

(a) 26 (b) 28

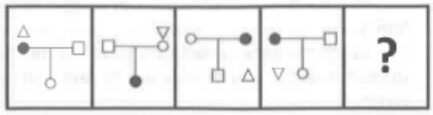
(c) 32 (d) 36

144. Twelve people form a club. By picking lots, one of them will host a dinner for all once in a month. The number of dinners a particular member has to host in one year is (CSAT-2015)

(a) One (b) Zero

(c) Three (d) Cannot be predicted

145. Consider the following given below: (CSAT-2015)



To fit the question mark, the correct Answer is: (CSAT-2015)

(a) 

(b) 

(c) 

(d) 

146. Consider the following matrix:

3	8	10	2	?	1
6	56	90	2	20	0

What is the missing number at '?' in the matrix? (CSAT-2015)

(a) 5 (b) 0

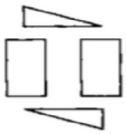
(c) 7 (d) 3

147. What is the missing number 'X' of the series 7, X, 21, 31, 4 (CSAT-2015)

(a) 11 (b) 12

(c) 13 (d) 14

148. Four cardboard pieces of specific shapes are shown in the following figure:

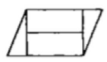


Which one of the following figures given can be formed by joining these pieces together? (CSAT-2015)

(a) 

(b) 

(c) 

(d) 

149. All good athletes want to win and all athletes who want to win eat a well - balanced diet therefore all athletes who do not eat a well-balanced diet are bad athletes.

The best conclusion from these statements is that (CSAT-2015)

(a) No bad athlete wants to win

(b) No athlete who does not eat a well- balanced diet is a good athlete

(c) Every athlete who eats a well — balanced diet is a good athlete

(d) All athletes who want to win are good athletes

150. If A runs less fast than B, and B runs as fast but not faster than C; then, as compared to A, C runs (CSAT-2015)

(a) Slower than A

(b) Faster than A

(c) With same speed as A

(d) Given data is not sufficient to determine

151. Examine the following statements:

1. Lady finger is tastier than cabbage.
2. Cauliflower is tastier than lady's finger.
3. Cabbage is not tastier than peas.

The conclusion that can be drawn from these statements is that (CSAT-2015)

(a) Peas are as tasty as lady's finger.

(b) Peas are as tasty as cauliflower and lady's finger.

(c) Cabbage is the least tasty of the four vegetables.

(d) Cauliflower is tastier than cabbage.

152. Shahid and Rohit start from the same point in Opposite directions. After each 1 km, Shahid always turns left and Rohit always turns right. Which of the following, statements is correct? (CSAT-2015)

(a) After both have travelled 2 km, the distance between them is 4 km.

(b) They meet after each has travelled 3 km.

(c) They meet for the first time after each has travelled 4 km.

(d) They go on without ever meeting again.

153. 62. Each of the six different faces of a cube has been coated with a different colour i.e., V, I, B, G, Y and O. Following information is given: (CSAT-2015)

1. Colors Y, O and B are on adjacent faces.
2. Colors I, G and Y are on adjacent faces.
3. Colors B, Gand Y are on adjacent faces.
4. Colors O, V and B are on adjacent faces.

Which is the color of the face opposite to the face colored with 0?

(a) B (b) V

(c) G (d) I

154. Consider the following statements followed by two conclusions:

Statements: Some men are great. Some men are wise.

Conclusion I: Men are either great or wise.

Conclusion II: Some men are neither great nor wise.

Which one of the following is correct? (CSAT-2015)

(a) Only conclusion I is valid.

(b) Only conclusion II is valid.

(c) Both the conclusions are valid.

(d) Neither of the conclusions is valid.

155. Usha runs faster than Kamala, Priti runs slower than Swati, Swati runs slower than Kamala. Who is the slowest runner? (CSAT-2015)

(a) Kamala (b) Priti

(c) Swati (d) Usha

Directions for the following 2 (two) items: Read the following passage and answer the 2 (two) items that follow:

A, B, C, D, E and F are cousins. No two cousins are of the same age, but all have birthdays on the same day of the same month. The youngest is 17 years old and the oldest E is 22 years old. F is somewhere between B and D in age. A is older than B. Cis older than D. A is one year older than C.

156. Which one of the following is possible? (CSAT-2015)

(a) D is 20 years old (b) F is 18 years old

(c) F is 19 years old (d) F is 20 years old

157. What is the number of logically possible orders of all six cousins in terms of increasing age? (CSAT-2015)

(a) 1 (b) 2

(c) 3 (d) 4

158. Two men, Anil, and David, and two women, Shabnam and Rekha are in a sales group. Only two speak Tamil. The other two speak Marathi. Only one man and one woman can drive a car. Shabnam speaks Marathi. Anil speaks Tamil. Both Rekha and David can drive. Which of the following statements is true? (CSAT-2015)

(a) Both the Tamil speakers can drive a car.

(b) Both the Marathi speakers can drive a car.

(c) Both of those who can drive a car speak Marathi.

(d) One of those who can drive a car speaks Tamil.

159. A society consists of only two types of people - fighters and cowards. Two cowards are always friends. A fighter and a coward are always enemies. Fighters are indifferent to one another. If A and B are enemies, C and D are friends, E and F are indifferent to each other, A and E are not enemies, while B and F are enemies. Which of the following statements is correct? (CSAT-2015)

(a) B, Cand F are cowards.

(b) A, E and F are fighters.

(c) B and E are in the same category.

(d) A and F are in different categories.

160. A father is nine times as old as his son and the mother is eight times as old as the son. The sum of the father's and mother's age is 51 years. What is the age of the son? (CSAT-2015)

(a) 7 years (b) 5years

(c) 4 years (d) 3 years

161. In a town, 45% population read magazine A, 55% read magazine B, 40% read magazine C, 30% read magazines A and B, 15% read magazines B and C, 25% read magazines A and C; and 10% read all the three magazines. What percentage does not read any magazine? (CSAT-2015)

(a) 10% (b) 15%

(c) 20% (d) 25%

162. Out of 130 students appearing in an examination, 62 failed in English, 52 failed in Mathematics, whereas 24 failed in both English and Mathematics. The number of students who passed finally is (CSAT-2015)

(a) 40 (b) 50

(c) 55 (d) 60

163. In a group of persons travelling in a bus, 6 persons can speak Tamil, 15 can speak Hindi and 6 can speak Gujarati. In that group, none can speak any other language. If 2 persons in the group can speak two languages and one person can speak all the three languages, then how many persons are there in the group? (CSAT-2015)

(a) 21 (b) 22

(c) 23 (d) 24

164. The number of persons who read magazine X only is thrice the number of persons who read magazine Y. The number of persons who read magazine Y only is thrice the number of persons who read magazine X. Then, which of the following conclusions can be drawn? (CSAT-2015)

1. The number of persons who read both the magazines is twice the number of persons who read only magazine X
2. The total number of persons who read either one magazine both the magazines is twice the number of persons who read both the magazines. Select the correct Answer using the code given below:

(a) 1 only (b) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

165. Consider the following statements:

1. Some claim to have seen UFOs (Unidentified Flying Objects).
2. Life on other heavenly bodies is considered to be a possibility.
3. Voyage to space is now an established fact.

From the above statements, it may be concluded that (CSAT-2015)

(a) UFOs are heavenly bodies

(b) UFOs are sent from other heavenly bodies

(c) Some living species in other heavenly bodies are more intelligent than man

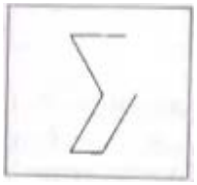
(d) Nothing definite can be said about the UFOs

166. In a society it is customary for friends of the same sex to hug and for friends of opposite sex to shake hands when they meet. A group of friends met in a party and there were 24 handshakes. Which one among the following numbers indicates the possible number of hugs? (CSAT-2015)

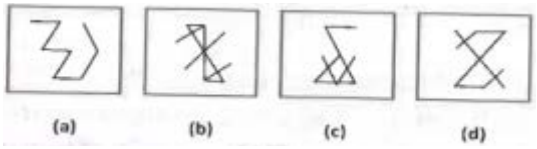
(a) 39 (b) 30

(c) 21 (d) 20

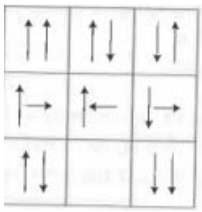
167. Examine the following figure:



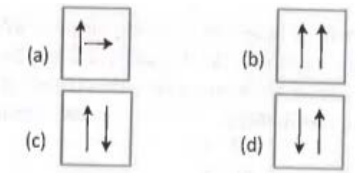
Which one of the following figures has the above figure embedded in it? (CSAT-2014)



168. Consider the following matrix:



Which one of the following figures fits into blank part of the above matrix? (CSAT-2014)



169. Consider the table given below in which the numbers bear certain relationship among themselves along the rows:

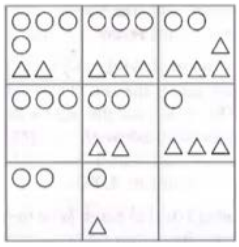
29	13	18
33	X	19
30	27	3

Which one of the following numbers is the missing number indicated above by X? (CSAT-2014)

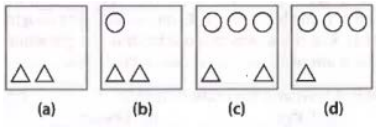
(a) 19 (b) 15

(c) 14 (d) 8

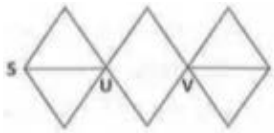
170. 36. Consider the following matrix with one empty block in the lower extreme corner:



Which of the following figures could fit in the empty block and thus complete the matrix? (CSAT-2014)



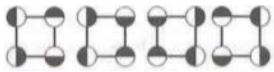
171. With reference to the figure given below number of different routes from S to T without retracing from U and/or V (T is the next node after V), is (CSAT-2014)



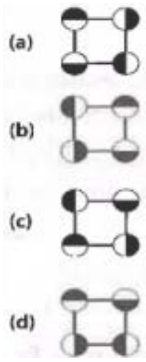
(a) 3 (b) 6

(c) 9 (d) 18

172. Consider the following figures:



Change in positions of beads in the figures above follows a sequence. Follows the same sequence, which of the figures look should appear as the fifth figure above? (CSAT-2014)



173. A and B decided to travel from place X to place Y by bus. A has Rs. 10 with him and he finds that it is 80% of the bus fare for two persons. B finds that he has Rs. 3 with him and hands it over to A. In this context, which one of the following statements is correct? (CSAT-2014)

(a) Now the money A has just enough to buy two tickets.

(b) A still needs Rs. 2 for buying the tickets.

(c) After buying the two tickets A will be left with 50 paisa.

(d) The money A now has is still not sufficient to buy two tickets.

174. As per agreement with a bank, a businessman had to refund a loan in some equal installments without interest. After paying 18 installments he found that 60 percent of his loan was refunded. How many installments were there in the agreement? (CSAT-2014)

(a) 22 (b) 24

(c) 30 (d) 33

175. A group of 630 children is seated in rows for a group photo session. Each row contains three less children than the row in front of it. Which one of the following number of rows is not possible? (CSAT-2014)

(a) 3 (b) 4

(c) 5 (d) 6

176. There are seven persons up on a ladder, A, B, C, D, E, F, and G (not in that order). A is further up than E but is lower than C. B is in the middle. G is between A and B. E is between B and F. If F is between E and D, the person on the bottom step of the ladder will be (CSAT-2014)

(a) B (b) F

(c) D (d) E

177. Consider that:

1. A is taller than B.
2. C is taller than A.
3. Distaller than C.
4. E is the tallest of all.

If they are made to sit in the above order of their height, who will occupy the mid position? (CSAT-2014)

(a) A (b) B

(c) C (d) D

178. Consider the following statements:

There are six villages A, B, C, D, E, and F.

F is 1 km to the west of D.

B is 1 km to the east of E.

A is 2 km to the north of E.

C is 1 km to the east of A.

D is 1 km to the south of A.

Which three villages are in a line? (CSAT-2014)

(a) A, C, B (b) A, D, E

(c) C, B, F (d) E, B, D

179. Four children are sitting in a row. A is occupying the seat next to B but not next to C. If C is not sitting next to B, who is/are occupying seat/seats adjacent to D? (CSAT-2014)

(a) B (b) A

(c) B and A (d) Impossible to tell

180. Assume that:

1. The hour and minute hands of a clock move without jerking.
2. The clock shows a time between 8 o'clock and 9 o'clock.
3. The two hands of the clock are one above the other.

After how many minutes (nearest integer) will the two hands be again lying one above the other? (CSAT-2014)

(a) 60 (b) 62

(c) 65 (d) 67

181. If the 3rd day of a month is Monday, which one of the following will be the fifth day from 21st of this month? (CSAT-2014)

(a) Monday (b) Tuesday

(c) Wednesday (d) Friday

182. For a charity show, the total tickets sold were 420. Half of these tickets were sold at the rate of Rs. 5 each, one-third at the rate of Rs. 3 each and the rest for Rs. 2 each. What was the total amount received? (CSAT-2014)

(a) Rs. 900 (b) Rs. 1,540

(c) Rs. 1,610 (d) Rs. 2,000

Directions for the following 3 (three) items: Read the passage given below and answer the items that follow.

A, B, C, D, E and F are members of a family. They are engineer, stenographer, doctor, draughtsman, lawyer and judge (not in order). A, the engineer is married to the lady stenographer. The judge is married to the lawyer. F, the draughtsman is the son of B and brother of E. C, the lawyer is the daughter-in-law of D. E is the unmarried doctor. D is the grandmother of f There are two married couples in the family.

183. What is the profession of B? (CSAT-2014)

(a) Judge (b) Lawyer

(c) Draughtsman (d) Cannot be determined

184. Which of the following is/are couple/couples? (CSAT-2014)

(a) AD only (b) BC only

(c) Both AD and BC (d) Both AC and BD

185. What is the profession of D? (CSAT-2014)

(a) Judge (b) Stenographer

(c) Doctor (d) Cannot be determined

186. Four persons, Alok, Bhupesh, Chander, and Dinesh have a total of Rs. 100 among themselves. Alok and Bhupesh between them have as much money as Chander and Dinesh between them, but Alok has more money than Bhupesh; and Chander has only half the money that Dinesh has. Alok has in fact Rs. 5 more than Dinesh has. Who has the maximum amount of money? (CSAT-2014)

(a) Alok (b) Bhupesh

(c) Chander (d) Dinesh

187. Examine the following statements: (CSAT-2014)

1. George attends Music classes on Monday.
2. He attends Mathematics classes on Wednesday.
3. His Literature classes are not on Friday.
4. He attends History classes on the day following the day of his Mathematics classes.
5. On Tuesday, he attends his Sports classes.

If he attends just one subject in a day and his Sunday is free, then he is also free on (CSAT-2014)

(a) Monday (b) Thursday

(c) Saturday (d) Friday

188. In a row 'A' is in the 11th position from the left and 'B' is in the 10th position from the right. If 'A' and 'B' interchange, then 'A' becomes 18th from the left. How many persons are there in the row other than 'A' and 'B'? (CSAT-2014)

(a) 27 (b) 26

(c) 25 (d) 24

189. Location of B is north of A and location of C is east of A. The distances AB and AC are 5 km and 12 km respectively. The shortest distance (in km) between the locations B and C is (CSAT-2014)

(a) 60 (b) 13

(c) 17 (d) 7

190. There are 50 students admitted to a nursery class. Some students can speak only English and some can speak only Hindi. 10 students can speak both English and Hindi. If the number of students who can speak English is 21, then how many students can speak Hindi, how many can speak only Hindi, and how many can speak only English? (CSAT-2014)

(a) 21, 11 and 29 respectively (b) 28, 18 and 22 respectively

(c) 37, 27 and 13 respectively (d) 39, 29 and 19 respectively

191. Six books are labeled A, B, C, D, E, and F and are placed side by side. Books B, C, E, and F have green covers while others have yellow covers. Books A, B, and D are new while the rest are old volumes. Books A, B, and C are law reports while the rest are medical extracts. Which two books are old medical extracts and have green covers? (CSAT-2014)

(a) C and E (b) D and E

(c) E and F (d) F and D

192. A straight-line segment is 36 cm long. Points are to be marked on the line from both the end points. From each end, the first point is at a distance of 1 cm from the end, the second point is at a distance of 2 cm from the first point and the third point is at a distance of 3 cm from the second point and so on. If the points on the segment are not counted and the common points are counted as one, what is the number of points? (CSAT-2014)

(a) 10 (b) 12

(c) 14 (d) 16

193. Six identical cards are placed on a table. Each card has number '1' marked on one side and number '2' marked on its other side. All the six cards are placed in such a manner that the number '1' is on the upper side. In one try, exactly four (neither more nor less) cards are turned upside down. In how many least number of tries, can the cards be turned upside down such that all the six cards show number '2' on the upper side? (CSAT-2014)

(a) 3 (b) 5

(c) 7 (d) This cannot be achieved

194. Five persons fire bullets at a target at an interval of 6, 7, 8, 9 and 12 seconds respectively. The number of times they would fire the bullets together at the target in an hour is (CSAT-2014)

(a) 6 (b) 7

(c) 8 (d) 9

195. Four friends, A, B, C and D distribute some money among themselves in such a manner that A gets one less than B, C gets 5 more than D, D gets 3 more than B. Who gets the smallest amount? (CSAT-2013)

(a) A (b) B

(c) C (d) D

Directions for the following 4 (four) items:

Read the following statements and answer the four items that follow:

Five cities P, Q, R, S and T are connected by different modes of transport as follows: P and Q are connected by boat as well as rail.

Sand R are connected by bus and boat. Q and T are connected by air only.

P and R are connected by boat only.

T and R are connected by rail and bus.

196. Which mode of transport would help one to reach R starting from Q, but without changing the mode of transport? (CSAT-2013)

(a) Boat (b) Rail

(c) Bus (d) Air

197. If a person visits each of the places starting from P and gets back to P, which of the following places must he visit twice? (CSAT-2013)

(a) Q (b) R

(c) S (d) T

198. Which one of the following pairs of cities is connected by any of the routes directly without going to any other city? (CSAT-2013)

(a) P and T (b) T and S

(c) Q and R (d) None of these

199. Between which two cities among the pairs of cities given below are there maximum travel options available? (CSAT-2013)

(a) Q and S (b) P and R

(c) P and T (d) Q and R

Directions for the following 3 (three) items:

Read the following passage and answer the three items that follow:

A tennis coach is trying to put together a team of four players for the forthcoming tournament. For this 7 players are available: males A, Band C; and females W, X, Y and Z. All players have equal capability and at least 2 males will be there in the team. For a team of four, all players must be able to play with each other. But, B cannot play with W, C cannot play with Z and W cannot play with Y.

200. If Y is selected and B is rejected, the team will consist of which one of the following groups? (CSAT-2013)

(a) A, C, Wand Y (b) A, C, X and Y

(c) A, C, Y and Z (d) A, W, Y and Z

201. If B is selected and Y is rejected, the team will consist of which one of the following groups? (CSAT-2013)

(a) A, B, C and W (b) A, B, C and Z

(c) A, B, C and X (d) A, W, Y and Z

202. The music director of a film wants to select four persons to work on "different aspects of the composition of a piece of music. Seven persons are available for this work; they are Rohit, Tanya, Shobha, Kaushal, Kunal, Mukesh and J aswant. Rohit and Tanya will not work together. Kunal and Shobha will not work together. Mukesh and Kunal want to work together. (CSAT-2013)

Which of the following is the most acceptable group .of people that can be selected by the music director?

(a) Rohit, Shobha, Kunal and Kaushal

(b) Tanya, Kaushal, Shobha and Rohit

(c) Tanya, Mukesh, Kunal and Jaswant

(d) Shobha, Tanya, Rohit and Mukesh

203. Five people A, B, C, D and E are, seated about a round table, Every chair is spaced equidistant from adjacent chairs, (CSAT-2013)

I. C is seated next to A.

II. A is seated two seats from D.

III. B is not seated next to A.

Which of the following must be true?

I. D is seated next to B.

II. E is seated next to A.

Select the correct answer from the codes given below:

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

Directions for the following 3 (three) items:

Examine carefully the following statements and answer the three items that follow:

Out of four friends A, B, C and D, A and B play football and cricket, Band C play cricket and hockey,

A and D play basketball and football,

C and D play hockey and basketball.

204. Who does not play hockey? (CSAT-2013)

(a) D (b) C

(c) B (d) A

205. Who plays football, basketball and hockey? (CSAT-2013)

(a) D (b) C

(c) B (d) A

206. Which game do B, C and D play? (CSAT-2013)

(a) Basketball (b) Hockey

(c) Cricket (d) Football

207. Geeta is older than her cousin Meena, Meena's brother Bipin is older than Geeta. When Meena and Bipin visit Geeta, they like to play chess. Meena wins the game more often than Geeta. Based on the above information, four conclusions, as given below, have been made. Which one of these logically follows from the information given above? (CSAT-2013)

(a) While playing chess with Geeta and Meena, Bipin often loses.

(b) Geeta is the oldest among the three.

(c) Geeta hates to 10 the game.

(d) Meena is the youngest of the three.

Directions for the following 4 (four) items:

Read the following passage and answer the four items that follow. YOUR' answers to these items should be based on the passage only.

Passage

Financial markets in India have acquired greater depth and liquidity over the years. Steady reforms since 1991 have led to growing linkages and integration of the Indian economy and its financial system with the global economy. Weak global economic prospects and continuing uncertainties the in international financial markets therefore, have had their impact on the emerging market economies. Sovereign risk concerns, particularly in the Euro area, affected financial markets for the greater part of the year, with the contagion of Greece's sovereign debt problem spreading to India and other economies by way of higher-than-normal levels of volatility.

The funding constraints in international financial markets could impact both the availability and cost of foreign funding for banks and corporates. Since the Indian financial system is bank dominated, banks' ability to withstand stress is critical to overall financial stability. Indian banks, however, remain robust, notwithstanding a decline in capital to risk-weighted assets ratio and a rise in non- performing asset levels in the recent past. Capital adequacy levels remain above the regulatory requirements. The financial market infrastructure continues to function without any major disruption. With further globalization, consolidation, deregulation, and diversification of the financial system, the banking business may become more complex and riskier. Issues like risk and liquidity management and enhancing skill therefore assume greater significance.

208. According to the passage, the financial markets in the emerging market economies including India had the adverse impact in recent years due to (CSAT-2013)

1. weak global economic prospects.
2. uncertainties in the international financial markets.
3. sovereign risk concerns in the Euro area.
4. bad monsoons and the resultant crop loss.

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 1, 2 and 3

(c) 2 and 3 only (d) 2, 3 and 4

209. The Indian financial markets are affected by global changes mainly due to the (CSAT-2013)

(a) increased inflow of remittances from abroad.

(b) enormous increase in the foreign exchange reserves.

(c) growing global linkages and integration of the Indian financial markets.

(d) contagion of Greece's sovereign debt problem.

210. According to the passage, in the Indian financial system, banks' ability to with stand stress is critical to ensure overall financial stability because Indian financial system is (CSAT-2013)

(a) controlled by the Government of India.

(b) less integrated with banks.

(c) controlled by Reserve Bank of India.

(d) dominated by banks.

211. Risk and liquidity management assumes more importance in the Indian banking system in future due to (CSAT-2013)

1. further globalization.
2. more consolidation and deregulation of the financial system.
3. further diversification of the financial system.
4. more financial inclusion in the economy.

Select the correct answer using the code given below:

(a) 1, 2 and 3 (b) 2, 3 and 4

(c) 1 and 2 only (d) 3 and 4 only

212. There are five hobby clubs in a college viz, photography, yachting, chess, electronics and gardening. The gardening group meets every second day, the electronics group meets every third day, the chess group meets every fourth day, the yachting group meets every fifth day and the photography group meets every sixth day. How many times do all the five groups meet on the same day within 180 days? (CSAT-2013)

(a) 3 (b) 5

(c) 10 (d) 18

213. A, B, C, D and E belong to five different cities P, Q, R, Sand T (not necessarily in that order). Each one of them comes from a different city. Further it is given that: (CSAT-2013)

1. B and C do not belong to Q.
2. B and E do not belong to P and R.
3. A and C do not belong to R, Sand T.
4. D and E do not belong to Q and T.

Which one of the following statements is not correct?

(a) C belongs to P (b) D belongs to R

(c) A belongs to Q (d) B belongs to S

214. Seven men, A, B, C, D, E, F and G are standing in a queue in that order. Each one is wearing a cap of a different colour like violet, indigo, blue, green, yellow, orange and red. D is able to see in front of him green and blue, but not violet. E can see violet and yellow, but not red. G can see caps of all colours other than orange. If E is wearing an indigo coloured cap, then the colour of the cap worn by F is (CSAT-2013)

(a) Blue (b) Violet

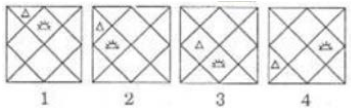
(c) Red (d) Orange

215. In a class of 45 students, a boy is ranked 20th. When two boys joined, his rank was dropped by one. What is his new rank from the end? (CSAT-2013)

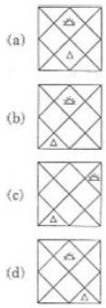
(a) 25th (b) 26th

(c) 27th (d) 28th

216. Consider the following figures 1, 2, 3 and 4: (CSAT-2013)



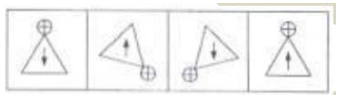
In the figure from 1 to 4 above, two symbols are shown to change their position in a regular direction. Following the same sequence, which one of the following will appear at the fifth stage?



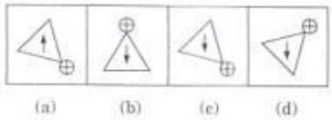
Directions for the following 2 (two) items:

In each item, there are two sets of figures; first four figures named Problem figures and next four figures named Answer figures indicated as (a), (b), (c) and (d). The problem figures follow a particular sequence. In accordance with the same, which one of the four answer figures should appear as the fifth figure?

217. Problem figures: (CSAT-2013)



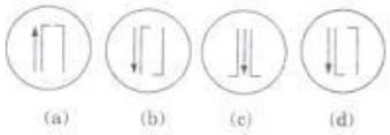
Answer figures:



218. Problem figures: (CSAT-2013)

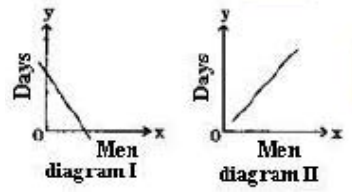


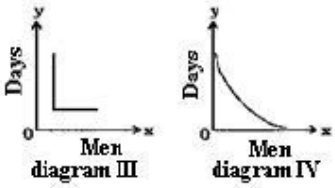
Answer figures:



219. Consider the following diagrams: (CSAT-2013)

x men, working at constant speed, do a certain job in y days. Which one of these diagrams shows the relation between x and y?





(a) diagram I (b) diagram II

(c) diagram III (d) diagram IV

220. Consider the following matrix: (CSAT-2013)

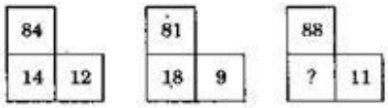
3	370	7
2	224	6
1	730	X

What is the number at 'X' in the above matrix?

(a) 5 (b) 8

(c) 9 (d) 11

221. Examine the following three figures in which the numbers follow a specific pattern: (CSAT-2013)

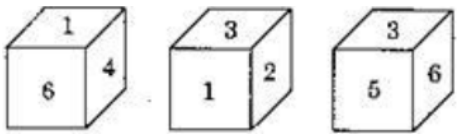


The missing number (?) in the third figure above is

(a) 7 (b) 16

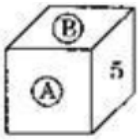
(c) 21 (d) 28

222. A cube has six numbers marked 1, 2, 3, 4, 5 and 6 on its faces. Three views of the cube are shown below: (CSAT-2013)



What possible numbers can exist on the two faces marked Ⓐ and Ⓑ, respectively on the cube ?

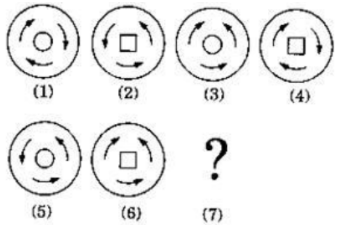
What possible numbers can exist on the two faces marked **A** and **B**, respectively on the cube?



(a) 2 and 3 (b) 6 and 1

(c) 1 and 4 (d) 3 and 1

223. Consider the following figures:



Which one of the following figures would logically come in the 7th position indicated above by a question mark? **(CSAT-2013)**

(a) 

(b) 

(c) 

(d) 

224. Consider the following statements: (CSAT-2012)

1. All artists are whimsical.
2. Some artists are drug addicts.
3. Frustrated people are prone to become drug addicts.

From the above three statements it may be concluded that:

(a) Artists are frustrated

(b) Some drug addicts are whimsical

(c) All frustrated people are drug addicts.

(d) Whimsical people are generally frustrated

225. Examine the following statements: (CSAT-2012)

1. Either A & B are of same age or A is older than B
2. Either C & D are of same age or D is older than C
3. B is older than C

Which of the following conclusions can be drawn from the above statements?

(a) A is older than B

(b) B and D are of the same age

(c) D is older than C

(d) A is older than C

226. Examine the following statements: (CSAT-2012)

1. Only those who have a pair of binoculars can become the member of the birdwatcher's club.
2. Some members of the birdwatcher's club have cameras.
3. Those members who have cameras can take part in photo-

contests.

Which of the following conclusions can be drawn from the above statements?

(a) All those who have a pair of binoculars are members of the birdwatcher's club.

(b) All members of the birdwatcher's club have a pair of binoculars.

(c) All those who take part in photo-contests are members of the birdwatcher's club.

(d) No conclusion can be drawn.

227. During the summer vacation Ankit went to a summer camp where he took part in hiking, swimming and boating. This summer, he is looking forward to a music camp where he hopes to sing, dance and learn to play the guitar. (CSAT-2012)

Based on the above information four conclusions as given below, have been made. Which one of these logically follows from the information given above?

(a) Ankit's parents want him to play guitar.

(b) Ankit prefers music to outdoor activities.

(c) Ankit goes to some type of camp every summer.

(d) Ankit likes to sing and dance

228. Three persons A, B & C wear shirts of Black, Blue and Orange colours (not necessarily in the order) and pants of green, yellow and orange (not necessarily in that order). No person wore shirt and pant of the same colour. Further, it is given that, (CSAT-2012)

1. A did not wear shirt of black colour.
2. B did not wear shirt of blue colour.
3. C did not wear shirt of orange colour.
4. A did not wear the pants of green colour
5. B wore pants of orange colour.

What were the colours of the pants and shirts worn by C respectively?

(a) Orange and black (b) Green and blue

(c) Yellow and blue (d) Yellow and black

229. Ten new TV shows started in January- 5 sitcoms, 3 drama and 2 news magazines. By April, only seven of the new shows were still on, five of them being sitcoms. (CSAT-2012)

Based on the above information, for conclusions, as given below, have been made. Which of these logically follows from the information given above?

(a) Only one news magazine show is still going on.

(b) Only one of the drama show is still going on.

(c) At least one discontinued show was a drama.

(d) Viewers prefer sitcoms over drama.

230. Read the passage given below and the two statements that follow (given on the basis of the passage): (CSAT-2012)

Four men are waiting at Delhi airport for Mumbai flight. Two are doctors and other two are businessman. Two speak Gujarati and two speak Tamil. No two of the same profession speak the same language. Two are Muslims and two are Christians. No two of the same religion are of the same profession, nor do they speak same language. The Tamil speaking doctor is Christian.

1. The Christian-Businessman speaks Gujarati.
2. The Gujarati-speaking doctor is a Muslim.

Which of the above statements is/are correct conclusion/ conclusions?

(a) 1 only (b) 2 only

(c) Both 1 & 2 (d) Neither 1 nor 2

231. Examine the following statements: (CSAT-2012)

1. I watch TV only if I am bored
2. I am never bored when I have my brother's company.
3. Whenever I go to the theatre I take my brother along.

Which one of the following conclusions is valid in the context of the above statements?

(a) If I am bored I watch TV

(b) If I am bored, I seek my brother's company.

(c) If I am not with my brother, than i'll watch TV.

(d) If I am not bored I do not watch TV.

232. Only six roads A, B, C, P, Q and R connect a military camp to the rest of the country. Only one out of A, P, and R are open at any one time. If B is closed so is Q. Only one of A & B is open during storms. P is closed during floods. In the context, which one of the following statements is correct? (CSAT-2012)

(a) Under normal conditions only three roads are open.

(b) During storms at least one road is open.

(c) During floods only three roads are open

(d) During calamities all roads are closed

233. Examine the following statements: (CSAT-2012)

1. None but students are the members of the club.
2. Some members of the club are married.
3. All married persons are invited for dance.

Which one of the conclusions can be drawn from the above statements?

(a) All students are invited for dance

(b) All married students are invited for dance

(c) All members of the club are married person

(d) None of the above conclusions can be drawn

234. Four political parties W, X, Y and Z decided to set up a joint candidate for the coming parliamentary elections. The formula agreed by them was the acceptance of a candidate of the most of the parties. For aspiring candidates A, B, C and D approached the parties for their ticket. (CSAT-2012)

A was acceptable to W but not Z

B was acceptable to Y but not X

C was acceptable to W and Y

D was acceptable to W and X

When candidate B was preferred by W and Z, candidate C was preferred by X and Z and candidate A was acceptable to X but not Y:

Who got the ticket?

(a) A (b) B

(c) C (d) D

235. Consider the following statements: (CSAT-2012)

1. All X-brand cars parked here are white.
2. Some of them have radial tyres
3. All X-brand cars manufactured after 1986 have radial tyres are parked here.
4. All cars are not X-brand.

Which one of the following conclusions can be drawn from the above statements?

(a) Only white cars are parked here.

(b) Some white X-brand cars with radial tyres are parked here.

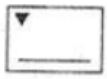
(c) Cars other than X-brand cannot have radial tyres.

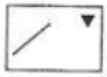
(d) Most of the X-brand cars are manufactured before 1986.

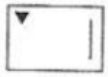
236. Figures given below are changing with certain rules as we observe them from left to right: (CSAT-2012)

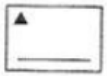


According to this rule which of the following would be the next figure if the changes were made in the same rule?

(a) 

(b) 

(c) 

(d) 

237. Consider the following information regarding the performance of class of 1000 students in four different tests: (CSAT-2012)

Tests	I	II	III	IV
Average marks	60	60	70	80
Range of marks	30 to 90	45 to 75	20 to 100	0 to 100

If a student scores 74 marks in each of the four tests, in which one of the following tests is her performance the best comparatively?

(a) Test I (b) Test II

(c) Test III (d) Test IV

Directions for the following 5 (five) items:

Examine the information given in the following paragraph and Solution: the items that follow:

Guest lectures on five subjects viz., Economics, History, Statistics, English and Mathematics have to be arranged in a week from Monday to Friday. Only one lecture can be arranged on each day. Economics cannot be scheduled on Tuesday. Guest faculty for History is available only on Tuesday. Mathematics lecture has to be schedules immediately after the day of Economics lecture. English lecture has to be scheduled immediately before the day of Economics lecture.

238. Which lecture is scheduled on Monday? (CSAT-2012)

(a) History (b) Economics

(c) Mathematics (d) Statistics

239. Which lecture is scheduled between Statistics and English? (CSAT-2012)

(a) Economics (b) History

(c) Mathematics (d) No lecture

240. Which lecture is the last one in the week? (CSAT-2012)

(a) History (b) English

(c) Mathematics (d) Economics

241. Which lecture is located scheduled on Wednesday? (CSAT-2012)

(a) Statistics (b) Economics

(c) English (d) History

242. Which lecture is scheduled before the Mathematics lecture? (CSAT-2012)

(a) Economics (b) History

(c) Statistics (d) English

243. In five flats, one above the other, live five professionals. The professor has to go up to meet his IAS officer friend. The doctor is equally friendly to all, and has to go up as frequently as go down. The engineer has to go up to meet his MLA friend above whose flat lives the professor's friend. (CSAT-2012)

From the ground floor to the top floor, in what order do the five

professionals live?

(a) Engineer, Professor, Doctor, IAS officer, MLA

(b) Professor, Engineer, Doctor, IAS officer, MLA

(c) IAS officer, Engineer, Doctor, Professor, MLA

(d) Professor, Engineer, Doctor, MLA, IAS officer

244. Consider the following three statements: (CSAT-2011)

1. Only students can participate in the race.
2. Some participants in the race are girls.
3. All girl participants in the race are invited for coaching.

Which one of the following conclusions can be drawn from the above statements?

(a) All participants in the race are invited for coaching.

(b) All students are invited for coaching.

(c) All participants in the race are students.

(d) None of the statements (a), (b) and (c) given above is correct.

245. Examine the following statements: (CSAT-2011)

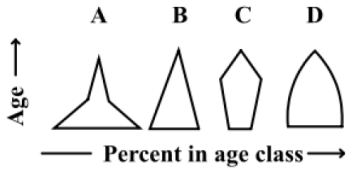
1. All trains are run by diesel engine.
2. Some trains are run by diesel engine.
3. No train is run by diesel engine.
4. Some trains are not run by diesel engine.

Codes:

(a) 1 and 2 (b) 2 and 3

(c) 1 and 3 (d) 1 and 4

246. Consider the four age pyramids given below namely A, B, C and D representing four different countries. (CSAT-2011)

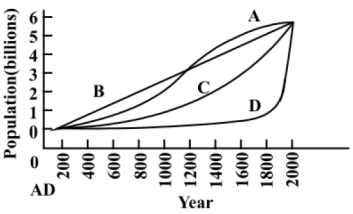


Which one of them indicates the declining population?

(a) A (b) B

(c) C (d) D

247. The followings figures has four curves namely A, B, C and D, Study the figure and answer the item that follows. (CSAT-2011)



Which curve indicates the exponential growth?

(a) A (b) B

(c) C (d) D

248. What can we say about persons with more than one disease from these graphs ? (CSAT-2011)

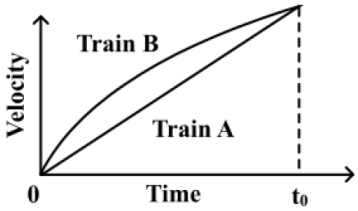
(a) There are likely to be persons with more than one disease in Town A.

(b) There are likely to be persons with more than one disease in Town B.

(c) There are likely to be persons with more than one disease in both Towns A and B.

(d) No inference can be drawn.

249. Consider the following Velocity-Time graph. It shows two trains starting simultaneously on parallel tracks. (CSAT-2011)



With reference to the above graph, which one of the following statements is not correct?

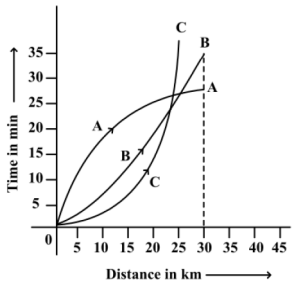
(a) Train B has an initial acceleration greater than that of Train A.

(b) Train B is faster than Train A at all times.

(c) Both trains have the same velocity at time to'

(d) Both trains travel the same distance in time to units.

250. Consider the following distance - time graph. The graph shows three athletes A, Band C running side by side for a 30 km race. (CSAT-2011)



With reference to the above graph consider the following statements :

1. the race was won by A.
2. B was ahead of A up to 25 km 26 mark.
3. C ran very slowly from the beginning.

Which of the statements given above is/are correct ?

(a) 1 only (b) 1 and 2 only

(c) 2 and 3 only (d) 1, 2 and 3

251. Consider the following figures: (CSAT-2011)

2	6
80	24

3	?
120	36

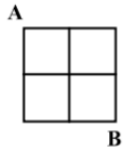
What is the missing number?

(a) 7 (b) 8

(c) 9 (d) 10

252. Study the following figure: (CSAT-2011)

A person goes from A to B always moving to the right or downwards along the lines. How many different routes can he adopt?

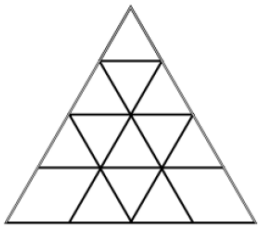


Select the correct answer from the codes given below:

(a) 4 (b) 5

(c) 6 (d) 7

253. Consider the following figure and answer the item that follow: (CSAT-2011)



What is total number of triangles in the above grid?

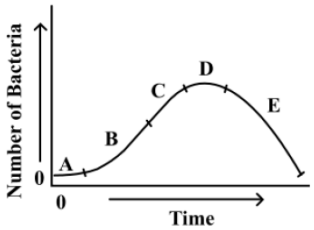
(a) 27 (b) 26

(c) 23 (d) 22

Directions for the following 3 (Three) items:

Read the passage given below, study the graph that follows and answer the three items given belwo the figure.

During a party, a person was exposed to contaminated water. A few days later, he developed fever and loose motions. He suffered for some days before going to a doctor for treatment. On starting the treatment, he soon became better and recovered completely a few days later. The following graph shows different phases of the person's disease condition as regions A, B, C, D and E of the curve.



254. Which region/regions of the curve correspond/corresponds to incubation phase of the infection? (CSAT-2011)

(a) A only

(b) B only

(c) Band C

(d) No part of the curve indicates the incubation phase

255. Which region of the curve indicates that the person began showing the symptoms of infection? (CSAT-2011)

(a) A (b) B

(c) C (d) D

256. Which region of the curve indicates that the treatment yielded effective relief? (CSAT-2011)

(a) C

(b) D

(c) E

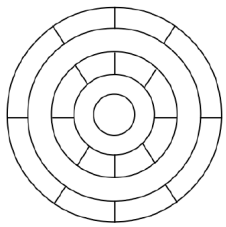
(d) The curve does not indicate the treatment

257. There are four routes to travel from city A to city B and six routes from city B to city C. How many routes are possible to travel from the city A to city C? (CSAT-2011)

(a) 24 (b) 12

(c) 10 (d) 8

258. Consider the following figure and answer the item that follows: (CSAT-2011)



What is the minimum number of different colours required to paint the figure given above such that no two adjacent regions have the same colour?

(a) 3 (b) 4

(c) 5 (d) 6

259. Consider the following figure and answer the items that follows:

15	
	48

A square is divided into four rectangles as shown above. The lengths of the sides of rectangles are natural n umbers. The areas of two rectangles are indicated in the figure. What is the length of each side of the square? **(CSAT-2011)**

(a) 10

(b) 11

(c) 15

(d) Cannot be determined as the given data are Insufficient

260. The houses of A and B face each other on a road going north-south, A's being on the western side. A comes out of his house, turns left, travels 5 km, turns right, travels 5 km to the front of D's house. B does exactly the same and reaches the front of C's house. In this context, which one of the following statements is correct? (CSAT-2011)

(a) C and D live on the same street.

(b) C's house faces south.

(c) The houses of C and D are less than 20 km apart.

(d) None of the above

261. How is D related to E? (CSAT-2011)

(a) Husband (b) Son

(c) Father (d) Wife

262. In a queue, Mr. X is fourteenth from the front and Mr. Y is seventeenth from the end, while Mr. Z is exactly in between Mr. X and Mr. Y. If Mr. X is ahead Mr. Y and there are 48 persons in the queue, how many persons are then between Mr. X and Mr. Z? (CSAT-2011)

(a) 6 (b) 7

(c) 8 (d) 9

3 Logical Reasoning and Analytical Ability_ Explanations

1. Answer: (d)

Explanation:

Given: P = 4, R = 1, Q ≠ 5, S ≠ 2, T ≠ 3. The remaining ranks {2, 3, 5} are distributed among Q, S, T.

Statement I: If S = 3, then T = 2.

If S = 3, then Q and T take {2, 5}. Since Q ≠ 5, we must have Q = 2, which forces T = 5. So T = 5, not 2. Statement I is incorrect.

Statement II: If Q = 3, then T = 5.

If Q = 3, then S and T take {2, 5}. Since S ≠ 2, we must have S = 5, which forces T = 2. So T = 2, not 5. Statement II is incorrect.

2. Answer: (c)

Explanation:

Let's break down the given information step by step:

1. **Understanding the Alignment:**

   Two identical straight rods are painted in five distinct colours. Because they are identical, the sequence of colours on both rods is exactly the same. When they are kept parallel side by side and their portions "match", it means they align perfectly and have the same colour. If they were shifted, none of the distinct colours would match. If they were placed in opposite directions, at most only the middle portion would match in colour. However, the problem gives us three matching pairs, which means the rods must be aligned perfectly in the same direction (Position 1 to 1, Position 2 to 2, etc.).

2. **Analyzing the Conditions:**

   Let the 5 positions on the rods be 1, 2, 3, 4, and 5. At each position, there is one P label (from Rod 1) and one Q label (from Rod 2). The given matching conditions tell us which labels share the same position:

   - P1 and Q3 are at the same position.
   - P4 and (Q1 or Q2) are at the same position.
   - Q4 and (P3 or P5) are at the same position.
   - Q3 and Q5 are adjacent (their position numbers differ by 1).

3. **Evaluating the Statements (Checking for Possibility):**

   The question asks which of the statements is/are possible. We just need to find if there is at least one valid arrangement that satisfies all conditions for each statement. Let's try to construct an arrangement where Q3 is in the middle (Position 3) and P2 is at an extreme (Position 1 or 5):

   - Position 1: P2 and Q2
   - Position 2: P3 and Q4
   - Position 3: P1 and Q3
   - Position 4: P5 and Q5
   - Position 5: P4 and Q1

   Let's verify if this arrangement violates any rules:

   - Does P1 match Q3? Yes, both are at Position 3.
   - Does P4 match Q1 or Q2? Yes, P4 matches Q1 at Position 5.
   - Does Q4 match P3 or P5? Yes, Q4 matches P3 at Position 2.
   - Are Q3 and Q5 adjacent? Yes, Q3 is at Position 3 and Q5 is at Position 4.
   - Are all labels P1–P5 and Q1–Q5 used exactly once? Yes.

4. **Conclusion:**

   In our perfectly valid arrangement, Q3 is at the middle portion (Position 3), so Statement I is possible; and P2 is at an extreme portion (Position 1), so Statement II is possible. Since both scenarios can occur without violating any given conditions, both I and II are possible.

3. Answer: (b)

Explanation:

Deduce the seating.

- A is in X (given).
- "Only E travels with G" → E and G share a car alone (just the two of them).
- C is in Z.
- B ∉ Y and B ≠ A's car, so B is in Z.
- E, G cannot be in Z (else C would be with them, contradicting "only E with G"). So E, G are in Y.
- X must hold A + two others. The only people left are D and F, so X = {A, D, F}.

Final assignment:

- X = {A, D, F}
- Y = {E, G}
- Z = {B, C}

Check the statements.

- **I. No one travels alone.** Smallest car has 2 people. ✓ True
- **II. Only D travels with F.** F is in X with both A and D, so F has two companions. ✗ False
- **III. Only C travels with B.** Z = {B, C}, so B's sole companion is C. ✓ True

4. Answer: (a)

Explanation:

Let us represent the statements being correct as X, Y, Z, and W, and the statements being incorrect as ¬X, ¬Y, ¬Z, and ¬W.

According to the question, the given relations can be written as logical implications:

- 1. If X is incorrect, then Z is incorrect: $\neg X \implies \neg Z$
- 2. If Y is incorrect, then W is correct: $\neg Y \implies W$
- . If W is correct, then X is incorrect: $W \implies \neg X$

By combining these three relations, we can form a continuous chain of implications:

$\neg Y \implies W \implies \neg X \implies \neg Z$

In logic, the contrapositive of a statement $A \implies B$ is $\neg B \implies \neg A$, and both always hold the exact same truth value. Taking the contrapositive of our entire chain of implications (reversing the order and negating each term), we get:

$Z \implies X \implies \neg W \implies Y$

Now, let us evaluate the given conclusions based on this contrapositive chain:

Statement I: If X is correct, then so is Y. From our contrapositive chain, we can clearly see that $X \implies Y$. This means that if X is correct, Y must definitely be correct. Therefore, Statement I is correct.

Statement II: If Z is correct, then it is not necessary that Y is correct. From our contrapositive chain, we can see that $Z \implies Y$. This means that if Z is correct, it is absolutely necessary that Y is correct. Therefore, Statement II is incorrect.

Since only Statement I is correct, the correct option is A.

5. Answer: (d)

Explanation:

Let us break down the given information step-by-step to determine the relationship between Q and S:

Step 1: Determine the genders of S and R

- We are given that S is the mother of T. This means S is a female.
- We are also given that S is R's spouse. Since S is a female, her spouse R must be a male (husband of S).

Step 2: Determine the gender of Q

- We are given that Q and R are the children of P.
- The first statement says P has exactly one son and one daughter.
- Since we already established that R is a male, R is the son of P.
- Therefore, the other child, Q, must be the daughter of P. This makes Q a female and the sister of R.

Step 3: Find the relationship between Q and S

- Q is the sister of R.
- R is the husband of S.
- Hence, Q is the sister of the husband of S (which also means Q is the sister-in-law of S).

Comparing this conclusion with the given options, Option D perfectly matches our deduction.

6. Answer: (d)

Explanation:

Based on the given statements, we can establish a logical chain of relationships using set theory or syllogism:

- 1. Every red is blue: This means the set of Red (R) is a subset of Blue (B). So, $R \subset B$.
- 2. Every blue is green: This means the set of Blue (B) is a subset of Green (G). So, $B \subset G$.
- 3. Every green is yellow: This means the set of Green (G) is a subset of Yellow (Y). So, $G \subset Y$.

Combining these premises, we get a continuous chain of subsets:

$R \subset B \subset G \subset Y$

Now, let us evaluate the given conclusions:

- **Statement P (Every blue is yellow):** Since $B \subset G$ and $G \subset Y$, it logically follows that $B \subset Y$. Thus, statement P is correct.
- **Statement Q (Every red is green):** Since $R \subset B$ and $B \subset G$, it logically follows that $R \subset G$. Thus, statement Q is correct.
- **Statement R (Every red is yellow):** Since $R \subset B$, $B \subset G$, and $G \subset Y$, it logically follows that $R \subset Y$. Thus, statement R is correct.

Since all three statements P, Q, and R are logically deduced from the premises, the correct option is D.

7. Answer: (a)

Explanation:

To find the minimum number of attempts required to identify the lightest cube with certainty, we must consider the worst-case scenario using an optimal weighing strategy.

Step 1: First Weighing

Divide the 7 cubes into three groups: two groups of 3 cubes each, and one group of 1 cube. Place the two groups of 3 cubes on either pan of the balance.

- **Case 1: The balance is equal.** This means all 6 cubes on the balance are of equal weight. The lightest cube is the 1 cube that was left out. Here, we find it in 1 attempt.
- **Case 2: The balance is unequal.** One pan will be lighter, meaning the lightest cube is among the 3 cubes on that lighter pan.

Step 2: Second Weighing

Take the 3 cubes from the lighter pan (from Case 2). Select any 2 cubes and place one on each pan of the balance, leaving the 3rd cube aside.

- **Case 2a: The balance is equal.** The lightest cube is the 1 cube left aside.
- **Case 2b: The balance is unequal.** The cube on the lighter pan is the lightest cube.

In every possible scenario, a maximum of 2 weighings is sufficient to guarantee finding the lightest cube. Therefore, the minimum number of attempts required to distinguish the odd cube with certainty is 2.

8. Answer: (b)

Explanation:

In the given code, ZERO is encoded as ADSN. Observe the pattern:

- Z is changed to A.

- E is changed to D.
- R is changed to S.
- O is changed to N.

So, the pattern is: 1st letter → next letter; 2nd letter → previous letter; 3rd letter → next letter; 4th letter → previous letter. Applying the same pattern to STOP:

- S becomes T.
- T becomes S.
- O becomes P.
- P becomes O.

Therefore, STOP is encoded as TSPO.

9. Answer: (d)

Explanation

To determine the correct relationships between the statements, let us analyze them individually and logically:

Understanding the Statements:

- **S1** asserts that protecting the environment is an absolute necessity (existential exigency) because of climate change.
- **S2** points out a lack of scientific consensus regarding exactly how much humans contribute to climate change.
- **S3** observes that environmental activism sometimes includes extreme views or alarmism, which climate change deniers then use as a target to discredit the movement.

Evaluating the Given Relationships:

1. **S3 is a counterpoint to S1:** A counterpoint to S1 would need to argue that protecting the environment is not an existential necessity. S3 merely comments on the nature of some environmental activism and the tactics of climate deniers; it does not negate the need to protect the environment. Therefore, Relationship 1 is incorrect.
2. **S3 is unconnected to S1 and S2:** If S3 were unconnected, it would share no logical or causal link with the other statements. However, S3 is directly related to S2. The lack of definitive scientific consensus (S2) creates an environment of uncertainty. This uncertainty is exactly why some activists might resort to alarmism to force urgent action, and why deniers exploit those extreme views to justify their skepticism (S3). Because a clear link exists, Relationship 2 is incorrect.
3. **S2 could be the reason for S3:** As established above, the absence of a precise scientific consensus on the extent of human intervention (S2) provides the very breeding ground for both climate alarmism (to compensate for the lack of consensus) and climate denial (exploiting the lack of consensus). Thus, S2 acts as a logical catalyst or reason for the phenomena described in S3. Therefore, Relationship 3 is correct.

Logical Shortcut (Elimination Method):

Notice the inherent logical contradictions in the options: Relationship 2 claims S3 is unconnected to S1 and S2; Relationship 1 claims S3 is a counterpoint to S1 (a connection); Relationship 3 claims S2 is the reason for S3 (a connection). Relationship 2 is mutually exclusive with both 1 and 3. So Options A (2 and 3), B (1, 2, and 3) and C (1 and 2) are each logically contradictory. By pure logic, they eliminate themselves, leaving Option D (3 only) as the only valid choice.

10. Answer: (a)

Explanation:

Let's break down the given conditions step by step to determine the seating arrangement of the 8 persons around the circular table.

- Both P and R are adjacent to Q: This means Q is sitting exactly between P and R. The sequence is either P – Q – R or R – Q – P.
- Both T and R are adjacent to S: This means S is sitting exactly between T and R. The sequence is either T – S – R or R – S – T.

Combining the first two conditions, since R is common to both, the 5 persons must sit together in a contiguous block. The sequence of these 5 persons is either P – Q – R – S – T or T – S – R – Q – P.

- Both U and W are adjacent to V: This means V is sitting exactly between U and W. The sequence is U – V – W or W – V – U. This forms a block of 3 persons.

Since there are 8 seats in total, these two blocks (one of 5 persons and one of 3 persons) will complete the circle.

- S and W are on opposite chairs: In a circular arrangement of 8 seats, opposite chairs have exactly 3 seats between them on either side. Assign position numbers 1 to 8 clockwise. Place the block P – Q – R – S – T at positions 1 to 5, so S is at position 4. For W to be opposite S, W must be at position 8. This leaves positions 6 and 7 for U and V. Since V must be between U and W, V takes position 7 and U takes position 6.

Verify the clockwise condition: In our arrangement (1:P, 2:Q, 3:R, 4:S, 5:T, 6:U, 7:V, 8:W), moving clockwise from P (1), we meet R (3) before T (5). This satisfies the condition. (If placed counter-clockwise, we would meet T before R, which would violate the rule.)

Finding the final answer:

The question asks how many persons Q crosses while moving clockwise before meeting W. Q is at position 2 and W is at position 8. Moving clockwise from Q (2) to W (8), Q passes the persons at positions 3, 4, 5, 6, and 7, occupied by R, S, T, U, and V. Therefore, Q crosses exactly 5 persons before meeting W.

11. Answer: (d)

Explanation:

The given string is:

x b x a x a x x a x a x b a b

There are 15 positions, so a pattern repeated more than once can have length 5, repeated 3 times. The last five positions are:

a x b a b

Using the fixed letters in the string, the repeating pattern is:

a b b a b

Repeating this pattern three times gives:

a b b a b a b b a b a b b a b

Now, the 7th and 8th positions are b and b. Therefore, xx = bb.

12. Answer: (b)

Explanation:

January - 0 = January

January -1 = December

December -2 = October

October - 3 = X = July

July - 4 = March

March -5 = October

October - 6 = April

13. Answer: (c)

Explanation:

Step 1: Minimum number of cuts to get 60 pieces

Let's assume the cube is divided into:

- **a parts along x-direction**
- **b parts along y-direction**
- **c parts along z-direction**

Then the total number of small cubes = a × b × c = 60.

We want to **minimize the number of cuts**, which will be:

(a−1) + (b−1) + (c−1) = a + b + c − 3

We now try to find integer triplets (a, b, c) such that a × b × c = 60 and minimize (a + b + c).

(3, 4, 5) → 3×4×5 = 60; cuts = 2+3+4 = **9**

No other integer combination gives a lower total number of cuts. So, **Statement I is correct.**

Step 2: Cubes not painted on any face

For a cube of dimension **3 × 4 × 5**:

- Only the **interior cubes** are not painted.
- Cubes not painted = (3−2) × (4−2) × (5−2) = **1×2×3 = 6**

So, **Statement II is also correct.**

14. Answer: (c)

Explanation: The sequence shows a repeating pattern of letters with specific placements of A, B, and C. The option (c) fits this pattern logically, completing the sequence correctly.

15. Answer: (b)

Explanation: Multiplying by 0.5, 1,1.5,2,2.5

16. Answer: B

Explanation:

Substituting Q = −, R = ×, P = +, S = ÷ in the order QRPS gives:

60 − 15 × 3 + 20 ÷ 4 = 20, which is correct.

17. Answer: C

Explanation : PQ: 0-0; QR: 1-1: PR: 1-0

18. Answer: C

Explanation : Definitely True statements: I, III, IV, VI

19. Answer: A

Explanation : The code is multiplication of alphanumeric values of letters

20. Answer: D

Explanation : Have to eliminate and find codes for each letter for eg: M= 8

21. Answer: B

Explanation : Numbers are 35921, 63072, 64182, 94183

22. Answer: B

Explanation: 4^3 becomes 7^3; 6^3 becomes 9^3,so 8^3 becomes 11^3 =1331

23. Answer: D

Explanation: The answer is 6 no need to know any further information

24. Answer: D

Explanation: only way we get 4 digits answer is if PP is 33 and then only possible value for other number is 34 and final value of 1122 so no need for any information

25. Answer: D

Explanation: We don't know the gender of Q

26. Answer: C

Explanation:

Statement I or II alone is insufficient to determine if team P won.

Together, they provide enough information to answer the question.

27. Answer: B

Explanation : the equation is $(p-q)^2$

28. Answer: (c)

Explanation:

The given main statement is : Pradeep becomes either a Director or a Producer.

It means Pradeep cannot be both.

Also, he must be one of them.

SP:

Statement S: Pradeep is not a Producer. It means he must be a Director, as given in statement P. So, statement P follows from S and they are logically consistent with the main statement.

RQ:

Statement R: Pradeep is not a Director. It means he must be a Producer, as given in statement Q. So, statement Q follows from R and they are logically consistent with the main statement.

29. Answer: (d)

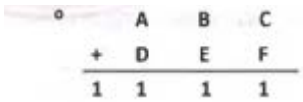
Explanation:

Here, the statement says that India is the world's largest producer of milk. However, we don't know anything about the consumption/demand of milk in India.

If domestic demand > production, it would mean that India will have to import milk. Even if domestic demand < production, and India gets to export the surplus milk, we cannot say that it would definitely be the largest exporter. That's because how much milk India gets to export will depend on the difference between the milk production and milk consumption in India. Thus, none of the conclusions is correct

30. Answer: (d)

Let,

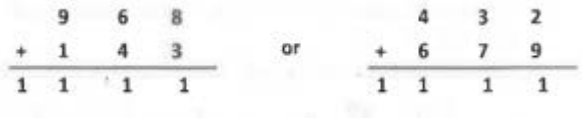


	A	B	C
+	D	E	F
1	1	1	1

Let C, F be 8, 3 respectively

So C+F =11, then take

B+ E = 10 i.e. 6, 4 respectively and A, D be 9 and 1 respectively to get 1111 as total addition. Be it,



	9	6	8
+	1	4	3
1	1	1	1

or

	4	3	2
+	6	7	9
1	1	1	1

Every time you get, A+ B + C+ D + E + F as 31. **Hence, option (d) is correct.**

31. Answer:(c)

To determine the smallest ratio, we need to consider different values for XYZ and calculate the corresponding ratios.

Starting with the range of 100-199, let's evaluate the ratios for the numbers in this range:

For XYZ =100, the ratiois 100 / (1+0+0) = 100/ 1 = 100.

For XYZ = 101, the ratio is 101 / (1 + 0 + 1) = 101 /2= 50.5.

For XYZ = 102, the ratio is 102 / (1 +0+2) = 102 / 3 = 34,

For XYZ = 199, the ratio is 199 / (1+9 + 9)=199/19= 10.47.

As we can see, the ratio gradually decreases as XYZ increases within the 100-199 range. The lowest ratio within this range is approximately 10.47, which occurs for XYZ =199.

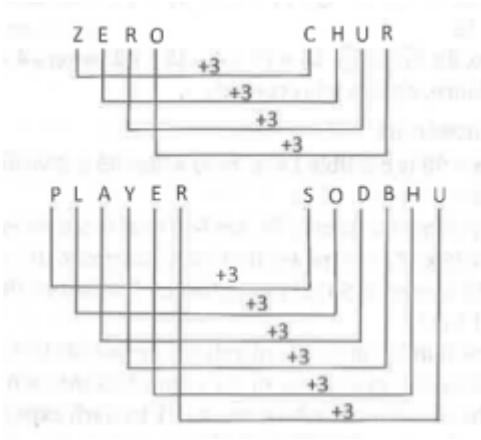
To verify that this is indeed the smallest ratio, we can continue exploring the ranges of 200-299, 300-399, and 50 on. However, we notice that the ratios in these ranges are gradually increasing.

For example, in the 200-299 range, the lowest ratio occurs for XYZ = 299, which is approximately 14.95. In the 300-399 range, the lowest ratio occurs for XYZ = 399, which is approximately 19. And so on.

Therefore, based on our analysis, we conclude that the smallest ratio occurs for XYZ = 199, with a value of approximately 10.47.

Finally, the difference between the digit at the hundred's place (1) and the digit at the unit's place (9) of the number 199 is 9-1 = 8. **Hence, option (c) is correct.**

32. Answer: (d)



Hence, option (d) is correct.

33. Answer: (d)

1 A > B

2 – C = D

3 - E < of all

4 – F < D

5 – F > A

To get who is the oldest here we must combine all these statements and get the correct ranking

E < B < A < F < D = C

Hence, option (d) is correct.

34. Answer: (b)

The given Sequence starts with two 'Z's and then continues with increasing repetitions of each letter, up to the letter 'A'.

To find the middle tern'w, we first need to determine the total nurnber of t.erms in the sequence.

The series here is, Z, Z,Y, Y, Y, X, X, X, X, W, W, W, W, W,……., A

2,3,4,5....,27 {where Zs are 2, Ys are 3, Xs are 4 and so on up to,A repeated 27 times)

The formula for the sum of an arithmetic series is given by:

S = (n/2) * (first term + last term),

where S is the sum of the series, n is the number of terms, and the first and last terms form the boundaries of the series.

In this case, the first term is 2 and the last term is 27. We need to find the sum of the numbers from the 2nd term to the 27th term, which means we have a total of 26 terms (27 -2+ 1= 26).

Plugging these values into the formula, we get: §=(26/2)x (2+27)=13x29=377.

From Z to A, there are 377 terms.

Middie Term of the series:

377/2 = 188.5 (Since the middle term can only be in whole numbers)

189th term (this is the middle term)

We know A is repeated 27, B 26 times, C 25 times and so on, thus we continue to subtract from 377, number of repetitions of these letters till we reach 189.

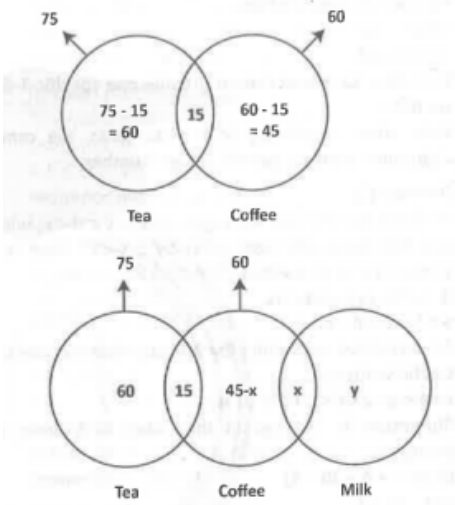
377-(27 +26 +25+24 +23 + 22 + 21 +20) = 189.

As A =27; B=26;C=25; D =24 and so on, 20 must mean H,

Next letter repeated 19 times must be 'I' which is 189th term. **Hence, option (b) is correct.**

35. Answer: (a)

Consider the following Venn Diagrams:



From the above expressed information, we can derive following equation:

60+60+y

120+y

Total number of people (N) in the party are 120 +y

Statement 1: 50 persons took milk

x+y =50

This statement alone is not sufficient in giving N

Statement 2: Number of persons who attended the party is five times the number of persons who took milk only.

- 120+y=5y
- 4y=120
- y=30

So, N =120 + 30 = 150

Statement 2 alone give the answer.

36. Answer: (b)

Given Notations:
A + B → A > B
A - B → A < B
A x B → A > B
A ÷ B → A < B
A ± B → A = B
Given Statement: P x Q, P - T, T ÷ R, R ± S
Translation of Given Statement:
P > Q, P < T, T < R, R = S
Comparative Analysis:
Based on these comparisons, we can infer the following order:
Q < P < T < R = S
Conclusion I: Q ± T or Q = T

This conclusion suggests that Q is equal to T. However, according to the given statement, Q is smaller than P, and P is smaller than T. Hence, Q cannot be equal to T.

Therefore, Conclusion I is false.

Conclusion II: S + Q or S>Q

This conclusion states that S is greater than Q. According to the given information, R is equal to S, and R is greater than T. Since T is greater than P, and P is greater than Q, we can conclude that S is indeed greater than Q. **Hence, Conclusion II is true.**

37. Answer: (d)

True statement: One of P and Q was selected for the job

P, Q, R, S, T are five candidates.

So it says either P and or Q only was selected, but not both.

False statement: At least one of R and S was selected for the job.

So we conclude here that no one among R and S was selected as it is a false statement.

True statement: At most two of R, S, and T were selected

for the job. But from the false statement above it is clear that neither R nor S was selected.

So we have T" here for his/her share of chances.

The selected candidates may be,

PQ, PT, QT

But, none of the conclusion can be down.

38. Answer: (a)

Conclusion 1 is correct: If '' is true, then at least one of 'P' and 'R must be false.

Let's check the validity of this conclusion:

If 'T is true, then at least one of 'R' and 'S' must be false. If 'S' is false, then 'P' must also be false.

Conclusion 2 is incorrect: If 'Q' is true, then 'P' is true.

Let's check the validity of this conclusion

If Q is true, then P might be true or false. So, conclusion is not correct.

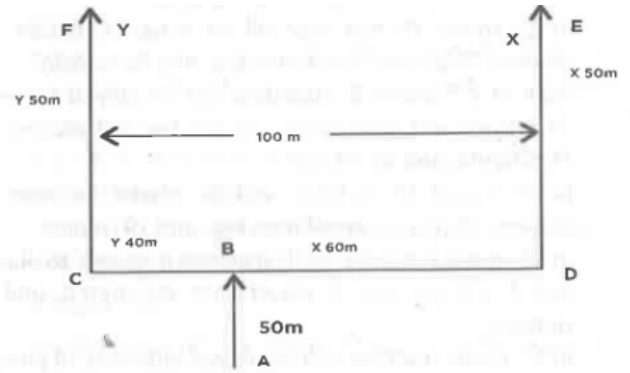
39. Answer: (c)

Only position of the two letters will remain unchanged.

I	N	C	O	M	P	R	E
H	E	N	S	I	B	I	L
I	T	I	E	S			
		↓	↓	↓	↓	↓	↓
↓	↓	↓	↓	↓	↓	↓	↓
↓	↓	↓	↓	↓	↓	↓	
		T	S	S	R	P	O
N	N	M	L	I	I	I	I
I	H	E	E	E	C	B	

Hence, option (c) is correct.

40. Answer: (a) is correct.



Now according to question X and Y start running from point A and run till point B.

AB = 50m

After reaching B, X turn right and reached at point D such that.

BD = 60m

After reaching point B, Y turn left and reached point C such that.

BC=40m

After reaching point C, Y turn left and reach point F such that.

CF=50m

After reaching point D, X take Right and reach point E such that.

DE=50m

Finally, X and Y are T E and F respectively to find distance between X and Y we need to find EF (distance between EandF)

EF=BC+BD EF = 60+40 = 100m.

Hence, option (a) is correct.

41. Answer: (d) is correct.

5th June 2022 was Sunday (UPSC prelims was on 51th June, and it was Sunday)

Difference between both the years = 2099 - 2022 = 77 years

We have a leap year after every 4 years in a century. So, Number of leap years in between 2022 and 2099 = 77/4 =19 (we will neglect the remainder because 2022 is not a leap year) Hence, till 2099 we will have 58 normal years and 19 leap years. Now in a normal year we have 1 odd day and in a leap year we have 2 odd days. (Odd Days: we know that a week contains 7 days counting from Monday to Sunday. So, any number of days, which are more than complete number of a week in a given period are called odd days. E.g., a period of 10 days contains 3 odd days) Total Number of odd days = (58 x 1) + (19 x 2) = 58 + 38 =96

96 give 5 as the remainder when divided by 7. (To check no. of odd days)

So, 5th June 2099 will be Sunday + 5 = Friday And, so 7th June 2099 will be a Sunday.

Hence, option (d) is correct.

42. Answer: (d) is correct.

According to the condition given in the question we need to find the opposite letter to the letters given in the word LUCKNOW. To understand the concept of opposite letter: opposite letter of A is Z, B is Y

We can do it in two ways 1st is by formula.

Opposite Letter Position = 27 - Letter Position

So, in case of LUCKNOW,

We get: letter opposite to L: 27-12=15=O

Letter opposite to U: 27 - 21 = F

Letter opposite to C: 27 - 3 = 24 = X

Letter opposite to K: 27 - 11 = 16 = P

Letter opposite to N: 27 – 14 = 13 = M

Letter opposite to 0: 27 -15 = 12 = L

Letter opposite to W: 27 -23 = 4 = D

The word associated to LUCKNOW is "OFXPMLD".

Hence, option (d) is correct. 2nd way to do it write all the alphabets in correct under and under them write alphabets in reverse order like Z under A Y under B etc.

Question Hack: Observe the options 1% letter is same in all options. 2" letter or last letter is different in every option. So, we just need to check the opposite letter of 2" and last letter which will be enough to find the Answer. Like opposite letter of 2" letter "U" is "F" and opposite letter of "W" is "D". Now check the options: there is only one option in which 2 letter is "F" and last letter is "D" that is option (d).

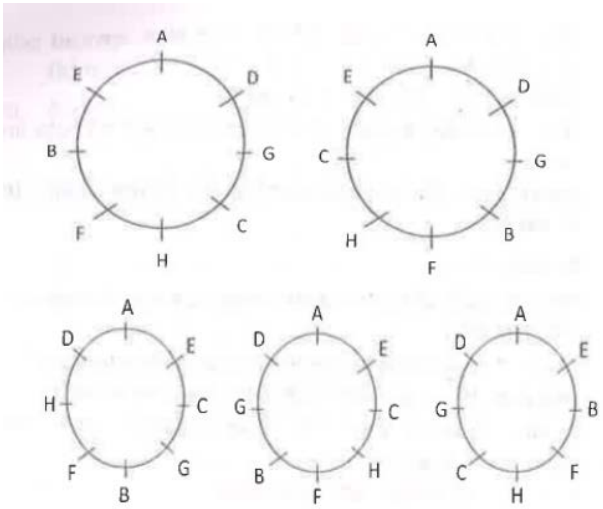
43. Answer: (d) is correct.

B and D sit neither adjacent to C nor opposite to C.

A sits in between E and D

F sits in between B and H.

Using the given statements, following arrangements can be possible.



We can clearly see that none of the option is correct. **Hence, option (d) is the correct.**

44. Answer: (a)

Given the Identity Card number ABCDEFG with distinct digits from {1, 2, 4, 5, 7, 8, 9} and divisibility constraints:

Divisible by 9: Sum of all digits = 36 (already satisfies, as 36 is divisible by 9).

After deleting G (last digit): ABCDEF must be divisible by 6.

F must be even (2, 4, 8) and sum(ABCDEF) = 36 - G must be divisible by 3. Since 36 is divisible by 3, G must be a multiple of 3. Only possible G = 9.

After deleting F and G: ABCDE must be divisible by 5. Thus, E = 5.

After deleting E, F, G: ABCD must be divisible by 4. The last two digits (CD) must form a number divisible by 4.

After deleting D, E, F, G: ABC must be divisible by 3. Sum(A+B+C) must be divisible by 3.

After deleting C, D, E, F, G: AB must be divisible by 2. Thus, B must be even.

F can only be 8 (since F ∈ {2, 4, 8}, and only F = 8 satisfies the divisibility-by-3 condition for ABC).

CD pairs for F = 8: Valid pairs are 12 and 72 (forming numbers divisible by 4).

Case 1: CD = 12 (C=1, D=2). Remaining digits: A=7, B=4. Number: 7412589.

Middle three digits (C, D, E): 1 + 2 + 5 = 8.

Case 2: CD = 72 (C=7, D=2). Remaining digits: A=1, B=4. Number: 1472589.

Middle three digits (C, D, E): 7 + 2 + 5 = 14 (not an option).

Option (a) is correct: The only valid sum of the middle three digits (positions 3, 4, 5) from the possible configurations is 8.

45. Answer: (d) is correct.

Let's consider the two statements one by one.

Statement 1: X is the brother of ₹, and Y is the brother of Z.

From statement 1 we can conclude that: X and Y are male. But we do not know whether Z is a male or a female. So, we cannot say anything about gender of Z Hence, we cannot conclude whether Z is the brother or sister of X.

Statement 2: X, Y, and Z are siblings. It does not tell us about the gender of X, Y and Z. So, we cannot say anything about gender of Z Hence, we cannot conclude whether Z is the brother or sister of X.

Even if we combine the two statements, then also we cannot conclude the gender of Z Hence, cannot Answer the question. **Hence, option (d) is correct.**

46. Answer: (d) is correct.

1 week = 7 days = (7x24) 168 hours = (168x60) 10080 minutes = (10080x60) 604800 seconds

Hence, 1 week = 604800 seconds

X weeks = 604800X seconds

X day = = 86400X seconds

X hours = 3600X seconds

X minutes = 60X seconds

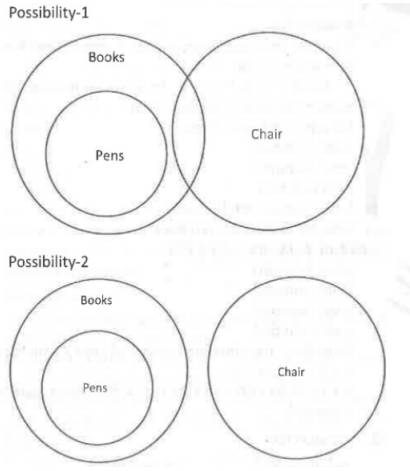
Hence, no. of seconds in in x weeks, x days, x hours x minutes and X seconds = 604800x + 86400x + 3600x + 60x + x = 694861x seconds. **Hence, option (d) is correct.**

47. Answer: (X)

UPSC has dropped this question.

48. Answer: (d) is correct.

According to the given two statements in the question, two possibilities arise which satisfies both the statements.



Now, come to the conclusion part using the above possibilities.

Conclusion-I: All chairs are books.

This conclusion is not true. Either no chair is book or some chairs are book.

Conclusion-Il: Some chairs are pens.

This conclusion is also not true because no chair is pen.

(Statement-2)

Conclusion-Ill: All books are chairs.

This conclusion is also not true because we can only conclude that some books are chair.

Conclusion-IV: No chair is a book.

This conclusion is also not true because some books are chairs.

Hence, none of the conclusion follows. **Hence, option (d) is correct.**

49. Answer: (c) is correct.

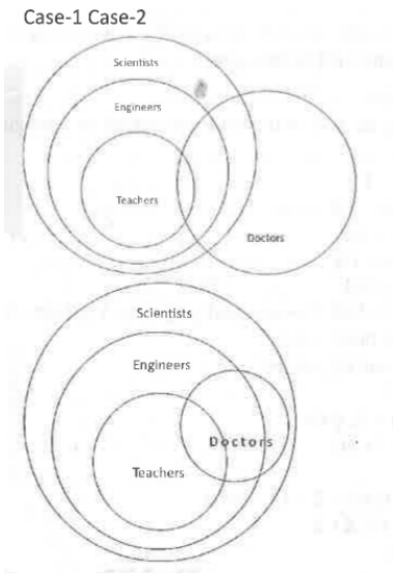
First consider all teachers are engineers and all engineers are scientists.

Then, from statement 1, some doctors are teachers.

Two possible cases arise following the statement 1.

In case 1, some teachers, some engineers and some doctors are engineers.

In case 2, all doctors are scientists.



Now, let's find out which conclusions are true.

Conclusion-I: Some scientists are doctors.

From both the cases, we can clearly see that some scientists are doctors. So, this conclusion is true.

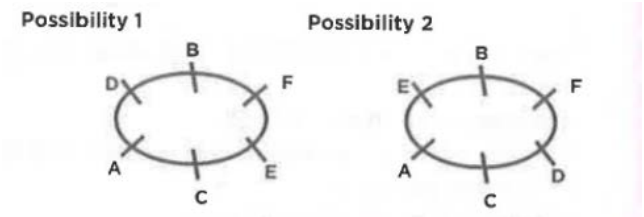
Conclusion-II: All engineers are doctors.

This conclusion is not true because some engineers are doctor not all engineers are doctors.

Conclusion-lIl: Some engineers are doctors.

This conclusion is true. **Hence, option (c) is the correct.**

50. Answer: (d) is correct.



It i§ evident that neither of the two statements is sufficient to Answer the question.

We can clearly see that two possibilities arise and either D or E can be on the immediate left of **A. Hence, option(d) is correct.**

51. Answer: (c) is correct.

It is clear that neither of the statements alone is sufficient to Answer the question. Now, let's check both the statements.

Let the age of Manisha's mother is x and age of Manisha be y.

Using statement 1, y =x—24 .... (a)

Using statement 2, 5 years later age of Manisha =y +5 and age of Manisha's mother =x + 5

$\frac{y+5}{x+5}=\frac{3}{5}$

5(y +5) = 3(x+5)

5y +25=3x+15 .... (b)

Using (a), (x = y + 24) equation (b) becomes

5y+25=3(y+24)+15

5y—3y=87-25

2y=62

y=31

So, age of Manisha = 31 years. Hence, both Statement-1 and Statement-2 are sufficient to Answer the question.

Hence, option (c) is correct.

52. Answer: (d) is correct.

It's evident that neither of the two statements is sufficient alone.

There is no lecture after lecture B means B is the last lecture.

While combining these two, we get multiple possible cases:

AFC_ _B

_AFC_B

_ _AFCB

A, F and C may not be consequent lectures. So, there can be even more cases. Hence, even combing both the statements we cannot determine the Answer. **Hence, option (d) is correct.**

53. Answer: (c) is correct.

Statement 1 is correct: At 3 o'clock, the minute hand is 15-minute spaces apart from hour hand.

To coincide, it must gain 15 minutes. We know that 55 minutes are gained

in 60 minutes. (when minute hand complete a round (60 minutes) hour hand completes just 30 degrees (5 minutes).

So, 15 minutes are gained in $\frac{60}{55}x15=\frac{180}{11}=16.36$ minute

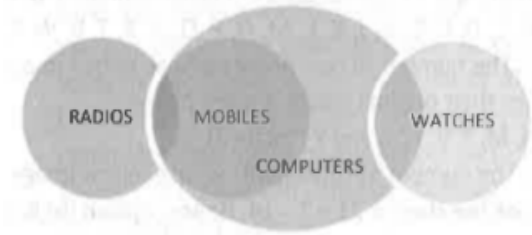
Hence, hour hand and minute hand will coincide at 3:16:36; This is between 3:16 pm and 3:17 pm. Hence, statement 1 is correct.

Statement 2 is correct: Every minute second hand cross the minute hand so it will do the same sometime between 4:58 p.m. and 4:59 p.m. Hence, statement 2 is also correct.

Hence, option (c) is correct.

54. Answer: (d)

We represent the given statement through Venn diagram.



In this Venn diagram Some radios are mobiles. All mobiles are computers. Some computers are watches. But clearly, we can see that nothing common in radio and watches.

Conclusion I is not followed.

For conclusion II:

by the above Venn diagram Some radios are mobiles. All mobiles are computers. Some computers are watches. But clearly, we can see that nothing common in mobile and watches.

Conclusion II is also not followed. Hence, option (d) is correct.

55. Answer: (c)

Given statement: Some cats are almirahs. Some almirahs are chairs. All chairs are tables.

We can represent the above statement though Venn diagram.



Conclusion-1: Certainly, some almirahs are tables. We can clearly infer from the Venn diagram that this conclusion is true.

Conclusion-2: Some cats may not be chairs. This conclusion is also true. Hence, option (c) is correct.

56. Answer: (b)

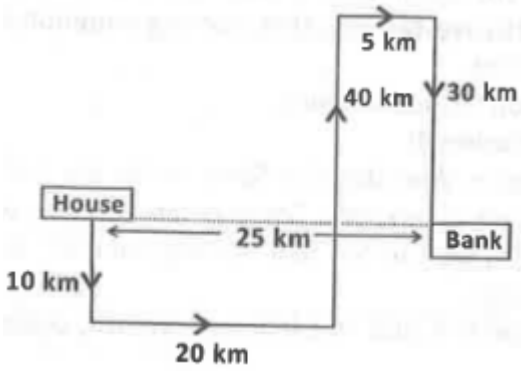
There are 21 consonants in English alphabet which are: B, C, D, F, G, H, J, K, L, M, N, P, Q, R, S, T, V, W, X, Y and Z. The number of consonants whose mirror image looks like their original image is given below:

H, M, T, V, W, X and Y (Total= 7)

$\therefore$ The number of consonants whose mirror image do not look like them =21-7=14. Hence, option (b) is correct.

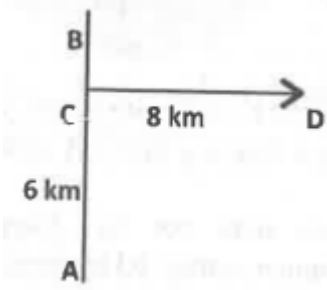
57. Answer: (b)

The route taken by a bank employee is given below:



It is clear from the figure, the shortest distance between her house and bank is 25 km. Hence, option (b) is correct.

58. Answer: (b)



A women start from point A and runs 12 km towards her north and reach at point B, then she run 6 km towards her south and reach at point C and then 8 km towards her East and reach at point D.

Now, we have to find her direction from the starting point i.e. AD

Angle made by AD with the axis is θ.

$\therefore \tan\theta=\frac{AC}{CD}=\frac{6}{8}=0.75<1$

$->\theta<45°$

AD is pointing in the direction North of East.

So, the woman is making an angle less than 45° North of East from her starting position. Hence, option (b) is correct.

59. Answer: (a)

Consider the odd series:

3+4=7

7+6=13

13+8=21

21+10=31

31+12=43

Similarly consider the even series:

2 +2=4 This term is incorrect.

4+6=10

10+8=18

18+10=28

28+12=40

∴ 2 should be replaced with 0. Hence, option (a) is correct.

60. Answer: (c)

First consider the alphabets in the given cells. We can clearly observe from each row that A, B and C are in cyclic manner.

So, in place of ?, alphabet B will come. Now consider the. numbers in each row. We see that 7 +3=10,3+6=9

So,10+?=13->?=3

∴ The required term (?) = 3B. Hence, option (c)is correct.

61. Answer: (c)

The pattern observed in Sequence-1 is:

8x0.5=4

4x1.5=6

6x2.5=15

15x3.5=52.2

52.2x4.5=236.25

Using the same pattern we will identify Sequence-2:

5x0.5=2.5(A)

2.5x1.5=3.75(B)

3.75x2.5=9.375(C)

9.375 x 3.5 = 32.8125(D)

32.8125x 4.5 = 147.65625(E)

The value of C = 9.375 Hence, option (c) is correct.

62. Answer: (c)

Total number of students in class = 40

According to Statement-1:

Priya is 31st from bottom. It means (40 — 31 = 9) students are above Priya i.e. Priya is 10th from top.

Priya is 4 ranks below Seema i.e. rank of Seema =10-4 = 6th from top

Rank of Seema = 4" from top

So, statement-1 alone is sufficient to Answer the question.

According to Statement-2

Ena is 37th from the bottom i.e. 3 students are above Ena Rank of Ena = 4th from top

Ena is 2 ranks above Seema.

∴ Rank of Seema = 4 + 2 = 6th from top.

So, statement-2 alone is sufficient to Answer the question. Hence, option (c) is correct.

63. Answer: (c)/(d)

We have to find out which among A, B, C, D and E is heaviest.

Statement-1: A, D > B, E, F

This statement alone is not sufficient to Answer the question.

Statement-2: C>A>D Using this statement alone we cannot determine the heaviest item.

If we combine both the statements, then C > A > D > B, E, F

∴ C is the heaviest.

So, both the statements are required to Answer the question and neither Statement-1 alone and Statement-1 alone is sufficient to Answer the Question.

Hence, both the options (c) and (d) are correct.

64. Answer: (b)

In the English alphabet, the first 4 letters are written in opposite order, and the next 4 letters are written in opposite order and so on; and at the end Y and Z are interchanged.

The English alphabet: ABCD EFGH IIKL MNOP QRST UVWX YZ

New arrangement: DCBA HGEF LKJI PONM TSRQ XWVU ZY

The 13th letter in the given arrangement is P.

Now, 4th letter to the right of 13th letter = T Hence, option (b) is correct.

65. Answer: (b)

We have to find the 14th day of month.

Consider Satetement-1: The last day of the month is Wednesday.

Here we do not know the number of days in the month. There can be 28, 29, 30 and 31 days.

So, statement-1 alone is not sufficient to Answer the question.

Consider Statement-2: The third Saturday of the month was the seventeenth day.

Here 17* day = Saturday. So, 14"" day = Saturday - 3 days = Wednesday

Statement-2 alone is sufficient to Answer the question. Hence, option (b) is correct.

66. Answer: (a)

We know that 10th October 2021 is Sunday.

1 year = 365 days = (52x7 + 1) days

It means, 10th October 2022 = Sunday + 1 = Monday

10th October 2023 = Monday + 1 = Tuesday

1 leap year = 366 days = (52x7 + 2) days

2024 s leap year. So, 10th October 2024 = Tuesday +2 = Thursday

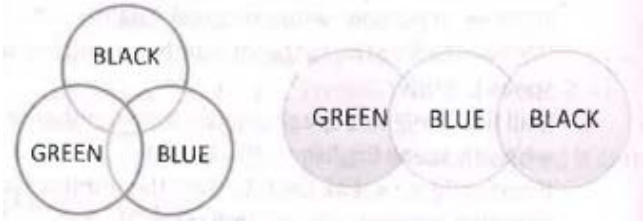
10th October 2025 = Thursday +1 = Friday

10th October 2026 = Friday + 1 = Saturday

10th October 2027 = Saturday + 1 = Sunday. Hence, option (a) is correct.

67. Answer: (d)

These are the two possibilities which satisfy both statement i.e. some greens are blues and some blues are blacks.



Conslusion-1 Some greens are black. This may or may not be true.

Conslusion-2: No green is black. This may or may not be true depending on which of the above case is taken.

Conslusion-3 All greens are blacks. This is definitely not true.

Conclusion-4 All blacks are greens. This is definitely not true.

So, none of the conclusion is correct. Hence, option (d) is correct.

68. Answer: (c)

The given sequence is 2, 7, 22, 67, 202, X, 1822. We observe the following pattern:

2x3+1=7

7x3+1=22

22x3+1=67

67x3+1=202

So, X=202x3+1=607

And 607 x 3 + 1 = 1822. Hence, option (c) is correct.

69. Answer: (c)

Let the price of pen be x and price of pencil be y .

According to question,

3x+5y =2x+7y

->x=2y

So, the price of pen is two times the price of pencil Hence, option (c) is correct.

70. Answer: (b)

In a certain code language, MATHEMATICS is written as LBSIDNZUHDR.

M	-1	L		C	-1	B
A	+1	B		H	+1	I
T	-1	S		E	-1	D
H	+1	I		M	+1	N
E	-1	D		I	-1	H
M	+1	N		S	+1	T

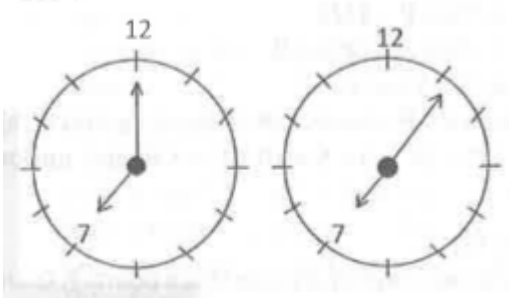
A	-1	Z		T	-1	S
T	+1	U		R	+1	S
I	-1	H		Y	-1	X
C	+1	D				
S	-1	R				

From the table we can clearly see the pattern.

So, CHEMISTRY is written as BIDNHTSSX. Hence, option (b) is correct.

71. Answer: (d)

In the given figure, two clocks are shown It is clearly visible that around 7:06 or 7:07, the angle between the hour hand and the minute hand is around 180°.



Hence, option (d) is correct.

72. Answer: (d)

The given series is _b_a_ba_b_abab_aab.

Now, we will break the series in 3 letters: _b_/a_b/a_b/_ab/ab_/aab

By inspection, we can find out the pattern being followed in this series i.e. abb/aab

Using this pattern, fill the blanks: - abb / aab / abb / aab / abb / aab. Hence, option (d) is correct.

73. Answer: (b)

Joseph visits the club on every 5th day.

Harsh visits on every 24th day.

Sumit visits on every 9th day.

Now, we have to take the L.C.M of 5,9 and 24 to find out the next day

when Joseph, Harsh and Sumit meet.

5=5x1

9=3x3

24=3x8

LCM (5,9,24) = 5x3x3x8 = 360

A week has 7 days.

357 is divisible by 7 (357 = 7x51)

We can write, 360th day = 357th day + 3 days

They all met on Sunday i.e. 357th day is also Sunday.

$\therefore$ 360th day = Sunday + 3 days = Wednesday

So, all three of them met again on Wednesday. Hence, option (b) is correct.

74. Answer: (c)

According to statement 1: R is greater than P as well as Q, i.e., R > P AND Q, here we do not know anything about S so cannot conclude about the largest.

According to statement 2: S is not the largest number, here we have no information about P, Q and R, Hence, cannot conclude about the largest.

Taking both the statements together: from 1% statement we can say that R is largest among P, Q and R and by 2nd statement we can say that S is not larger than R implies Ris larger than S. Hence, by taking both the statements together we can conclude R is largest. Hence, option (c) is correct.

75. Answer: (d)

According to statement 1: n is a prime number.

Prime numbers between 10 and 20: 11,13,17,19

Hence, we cannot have a unique number by statement 1

According to statement 2: n leaves a remainder 1 when divide by 4.

Numbers between 10 and 20 which leaves remainder 1 when divide by 4: 13, 17.

Hence, we cannot have a unique number by statement 2.

Taking both the statements together:

Prime numbers between 10 and 20 which leaves a remainder 1 when divide by 4: 13, 17

Both statements together are not sufficient to Answer the question. Hence, option (d) is correct.

76. Answer: (c)

According to statement 1: product of two numbers is 21. Numbers whose product is 21: (1, 21) and (3, 7)

Hence, statement 1 alone is not sufficient to Answer the question.

According to statement 2: the sum of two numbers is 10, Numbers whose sum is 10: (1, 9), (2, 8), (3, 7), (4, 6)

Hence, statement 2 alone is not sufficient to Answer the question.

Now taking both the statements together: Numbers whose sum is 10 and product is 21: (7,3) we got only 1 pair of numbers.

Hence, by combining both the statements we can find the two numbers. Hence, option (c) is correct.

77. Answer: (a)

Statements given in the question are:

1. All numbers are divisible by 2.
2. All numbers are divisible by 3.

Now by the rule of divisibility by 6: numbers are divisible by 6 if they are divisible by 2 and 3.

So, conclusion - I is true. Now, it is not necessary that if a number is divisible by 2 and 3 will be divisible by 4, for example 6 is divisible by 2 and 3 but 6 is not divisible by 4. Hence, conclusion - II is not followed. Hence, option (a) is correct.

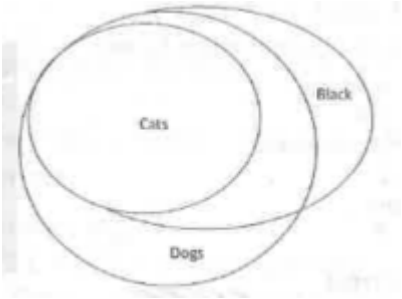
78. Answer: (c)

Given statements are:

All cats are dogs.

All cats are black.

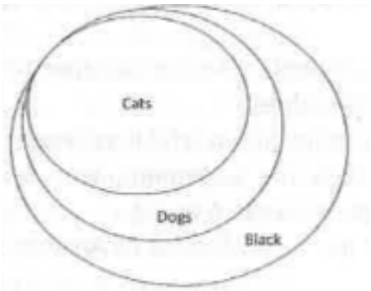
We can have two cases: For conclusion – I



By this Venn diagram: here we can see All cats are dogs and all cats are black. Here we can see clearly that some dogs are not black.

Conclusion - I is not followed.

For 2nd conclusion:



here, by this Venn diagram: here we can see All cats are dogs and all cats are black. Here we can see clearly that all dogs are black.

Conclusion — II is also not followed. Hence, option (c) is correct.

79. Answer: (b)

Given sequence 5 1 4 7 3 9 8 5 7 2 6 3 1 5 8 6 3 8 5 2 2 4 3 4 9 6

The odd numbers followed by another odd number are: 5, 7, 3, 5, 3, and 1 (as shown by bold and underlined numbers)

So, required number of odd numbers = 6. Hence, option (b) is correct.

80. Answer: (d)

From statement 1: C is younger than D, but older than A and B.

We can conclude that: D>C>A and B

From here we cannot say anything about who is younger between A and B.

So statement 1 is not sufficient to Answer the question.

From statement 2: D is the oldest.

We cannot conclude anything about who is younger.

Even if we combine both the statement, we cannot conclude who is younger between A and B.

So S1 and S2 together are not sufficient to Answer the Question.

From statement 3: A is the older than B.

We can conclude that A<B

Now by combining statement 1 and statement 3 we can conclude:

A<B<C<D

Hence, A is youngest.

Thus, S1 and S3 together are sufficient to Answer the Question. Hence, option (d) is correct.

81. Answer: (b)

Information given in question is: there is a two generation family.

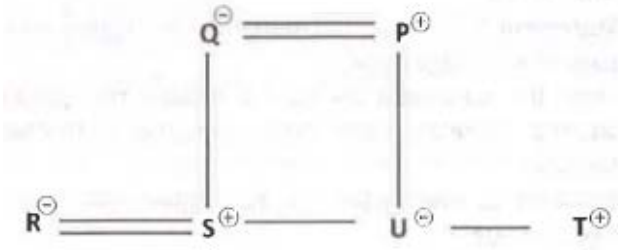
Total 6 members: 3 males (T, S and one more) and 3 females (U, Q and one more)

There are two couples and rest two is unmarried siblings.

P, Q, R are definitely married. T is definitely single. We do not know about S and U is married or not.

U is P's daughter and Q is R's mother-in-law, it means that P and Q belong to the senior generation and U and R belong to the junior generation.

Hence, the family tree is like:



U and T are the unmarried siblings. R is S's wife. Hence, option (b) is correct.

82. Answer: The correct answer is option (c)

Using the given statements, we will make a Venn diagram



Conclusions:

No cow is a rat — It may or may not be possible because we are given that no dog is cow, but some rats and cats can be cows.

So, conclusion | is not drawn from the statements.

No dog is rat — This may or may not be possible as some cats are dog, and some cats are rats also. So, there is a possibility that some dogs can be rats.

Some cats are rats — This conclusion is definitely valid because it is given that some rats are cats.

Only conclusion-Ill is valid. **Hence, option (c) is correct.**

83. Answer: The correct answer is option (c)

Total number of participants = 100

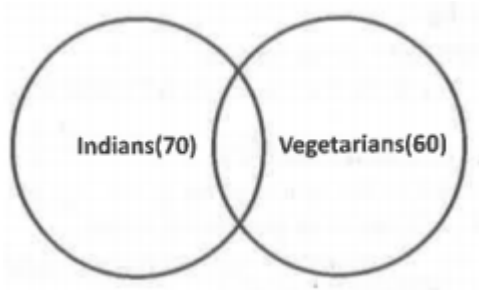
Number of Indians = 70

Number of vegetarians = 60

So, number of non-vegetarians = 100 - 60 = 40

So, total number of participants who are Indian as well as

vegetarians = (70 + 60) — 100 = 30



Another case, if all the 60 vegetarians are Indian then there will be at least 10 non-vegetarians Indian

participants. So, both the statements are correct. **Hence, option (c) is correct.**

84. Answer: The correct answer is option (a)

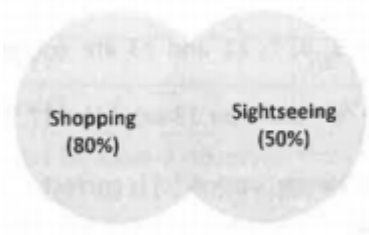
Let the total members of a club who went to Mumbai

are 100. Members who went for shopping = 80% = 80

Members who went for sightseeing = 50% = 50

Members who took rest in the hotel = 10% = 10

By using Venn diagram,



So, the number of members who went for both shopping and sightseeing = 80 + 50 - 100 = 40

40% members went for both shopping and sightseeing.

Also, the number of members who went for shopping alone = 80 - 40 = 40

40% members went for shopping alone. **Hence, option (a) is correct.**

85. Answer: The correct answer is option (d)

Let the total number of students be 100.

Number of students who play cricket = 60% = 60

Student who does not play cricket, plays football i.e. 100 - 60 = 40 students definitely plays Football.

Every football player has got a two-wheeler ie. 40 students definitely have two-wheeler.

Conclusion-I: 60% of the students do not have two - wheelers. This conclusion cannot be drawn because a cricketer may play football and can have a two-wheeler.

Conclusion-II: No cricketer has a two-wheeler. This conclusion cannot be drawn because a cricketer may have a two-wheeler. We have given that every football player has two-wheeler but the relationship between two-wheeler and cricket player is not given.

Conclusion-III: Cricket players do no play football. This conclusion cannot be drawn because it is not given that a cricketer cannot play Football. So, none of the conclusions can be drawn from the given data.

Hence, option (d) is correct.

86. Answer: The correct answer is option (b)

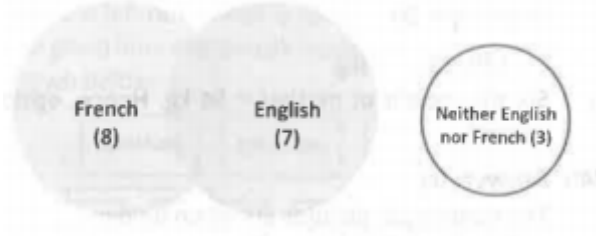
Total number of people in a group = 15

Those who can read French

Those who can read English = 3

Those who can read neither French nor English = 3 So, the number

of people who can read either French or English= 15 -3 =12



By using Venn diagram, Those who can read both French and English=8+7-12=3

Those who read only English =7 -3 = 4 And those who read only French=8-3=5 So, the number of people who can read exactly one language = 4 + 5 = 9,

Hence, option (b) is correct.

87. Answer: The correct option is option (b)

The given arrangement: c_accaa_aa_bc_b

There are 15 characters in this sequence. We will break

this sequence in 3 parts with 5 characters each in it.

The sequence is c_acc / aa_aa / _bc_b By analyzing with the given options, we find the correct

sequence: ccacc aabaa bbcbb.

So, cbbb is the correct sequence of letters which will fill

the gaps. **Hence, option (b) is correct.**

88. Answer: The correct option is option (d)

A joint family consists of 7 members with 3 females. This means, there are 4 males. Information:

G is a widow and sister-in-law of D's father F

B and D are siblings

Ais dauéhter of B

Cis cousin of B.

This implies G is a female who is a widow and sister-in-

law of F, where F is male.

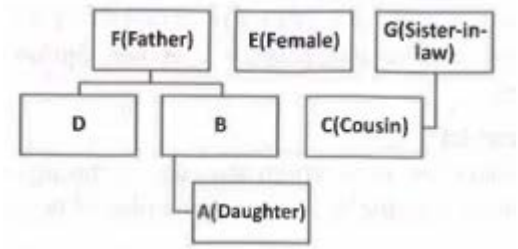
A is also a female, who is daughter of B.

We cannot determine the gender of B, D and C.

Only E is left out, which is female and wife of F.

The given information can be interpreted by the diagram

shown below:



So, we can conclude that E is wife of F, Grandmother of A and Aunt of C. **Hence, option (d) is correct.**

89. Answer: The correct option is option (b)

From S1, we can interpret that y is less than x and z. y is the weight of Sohan.

So, weight of Sohan is the least. **Hence, option (b) is correct.**

90. Answer:(d)

The correct option is (d).

The given expression • +1 • +2 • + 3 • + 1 =21• can be written as:

• +(10+ •) + (20+ •) + (10 •+3) + (10 •+1) = 210+ •

34+23(•)=210+•

22(•) =176

• = 8 So, • stands for 8.

91. Answer: (a)

The correct option is (a). We observe the following pattern: 1st and 3rd number is added and 2" number is subtracted from them to get the resultant.

In the 1st column: 8-6+4=6

In the 2nd column: 10 - 5+ 6= 11

In the 3 column: 15-7 +8 = 16

Similarly, in the 4™ column: 13 - 4 + 8 = 17.

92. Answer: (a)

The correct option is (a).

Let's interpret the code for MUMBAI that is LSIXVC.

M	L	M – 1 = L	D	C	D – 1 = C
U	S	U – 2 = S	E	C	E – 2 = C
M	J	M – 3 = J	L	I	L – 3 = I
B	X	B – 4 = X	H	D	H – 4 = D
A	V	A – 5 = V	I	D	I – 5 = D
I	C	I – 6 = C			

So, CCIDD is the code for DELHI.

93. Answer: (b)

The correct option is (b).

It is given that,

RAMON is written as 12345

DINESH is written as 675849

R	1	D	6	H	9
A	2	I	7	A	2
M	3	N	5	M	3
O	4	E	8	A	2
N	5	S	4	M	3
		H	9		

So, HAMAM will be written as 92323

94. Answer: (b)

The correct option is (b)

The given sequence is: 4/12/95, 1/1/96, 29/1/96,26/2/96 ...

2" term = 4/12/95 + 28 days = 1/1/96

3"term = 1/1/96 + 28 days = 29/1/96

4" term = 29/1/96 + 28 days = 26/2/96

The year 1996 is a leap year. So, there are 29 days in February 1996. So, 5" term will be 26/2/96 + 28 days = 25/3/96.

95. Answer: (a)

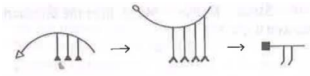
The correct option is (a)

In each row, there are three different shapes i.e., square,circle and triangle.

In the 3rd row: a rectangle and circle is present but the

square is missing. So, the missing figure must have a

Square.



96. Answer: The correct answer is option (d)

It is given that statement 4 is true. This implies some colours are not pleasant. It means there will be at least one colour which is not pleasant. So, all colours are not pleasant.

Statement-1 is definitely false. **Hence, option (d) is correct.**

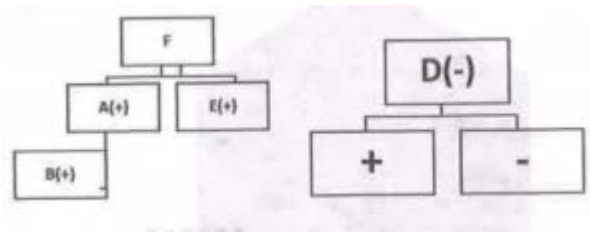
97. Answer: The correct answer is option (b)

In a family, there are 6 persons A, B, C, D, E and F.

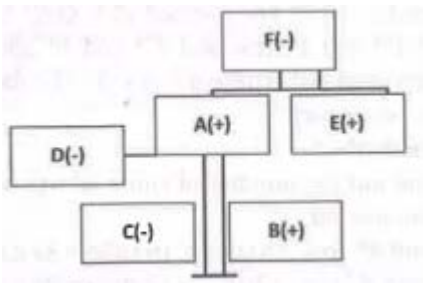
The number of males are equal to that of female i.e. 3 males and 3 females

A and E are sons of F. B is the son of A. D is the mother of two, one boy and one girl.

+ represent male and - represent female.



A, B and E are male. So, F, D and C are female. There is only one married couple i.e. A and D are married couple is the only possibility. B and C are their son and daughter respectively.



So, we can only infer that A is the husband of D. **Hence, option (b) is correct.**

98. Answer: The correct answer is option (c)

We have to arrange 2 boys and 2 girls in a row such that

the girls are not next to each other.

B1 B2 G1 G2

Total number of ways to arrange 2 boys and 2 girls in a row = 4! =4x3x2x1 =24

When two girls are sitting together, six cases arise:

G1G2B1B2/G1G2B2B1/B1G1G2B2/B2G1G2B1/B1B2G1G2/B2B1G1G2

Consider these two girls as one unit and now we have to arrange them in 3 places=31=6

These girls can also interchange their position. So, number of ways = 6 x 21 = 12

Number of ways when the girls are not sitting together = Total ways — When they are sitting together =24 - 12 = 12. So, total possible arrangements are 12.

Hence, option (c) is correct.

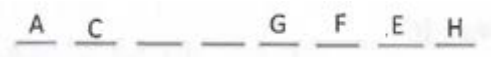
99. Answer: The correct answer is option (c)

The given information can be summarised as:

F is to the immediate right of G and neighbour of E and A is sixth to the left of E.



H s sixth to the right of C. This is possible only when H is on extreme right.



B and D are left and B is not neighbour of G, J the possible arrangement is:



We can clearly see that; C is immediate right of A and D is immediate neighbour of B and F. A and H are at extreme ends and G is to the immediate right of D. **Hence, option (c) is correct.**

100. Answer: The correct answer is option (b)

In a certain code:

256	Red colour chalk
589	Green colour flower
254	white colour chalk

Common between rows 1 and 2 is: 5 and colour.

->5 means colour.

Common between row 1 and 3 is: 2/5 and colour/chalk,

->2 means chalk,

4 means white because 2 means chalk and 5 means colour. **Hence, option (b) is correct.**

101. Answer: The correct answer is option (a)

Lecturer	Subject	Place
A	Economics	Hyderabad
B	History, Mathematics	
C	Sociology	Delhi
D		Shillong
E		
F	Commerce	Srinagar
G	Statistics	Chennai

From table, we can see that Lecturer D/E is specialized in History/ Mathematics.

So, Lecturer B is specialized in Geography.

Also, Lecturer from Kanpur is specialized in Geography. This means, Lecturer B is from Kanpur. So, Lecturer A is from Mumbai and Lecturer E is from

Hyderabad The final order is given below in the table

Lecturer	Subject	Place
A	Economics	Mumbai
B	Geography	Kanpur
C	Sociology	Delhi
D	History/Mathematics	Shillong
E	History/Mathematics	Hyderabad
F	Commerce	Srinagar
G	Statistics	Chennai

From table, Lecturer B is specialized in Geography.

Hence, option (a) is correct.

102. Answer: The correct answer is option (b)

From table given above, Lecturer A is specialized in Economics and he belongs to Mumbai. **Hence, option (b) is correct.**

103. Answer: The correct option is option (b)

From table given above, Lecturer E belongs to Hyderabad. **Hence, option (b) is correct.**

104. Answer: The correct answer is option (b)

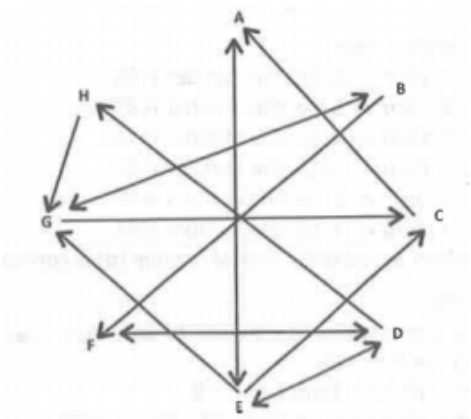
Let's combine the information given in a table format:

Clearly, Teacher B teaches the maximum number of subjects i.e. 5 subjects. **Hence, option (b) is correct.**

105. Answer: The correct answer is option (b)

One-way passages are shown by single arrow and two- way passages are

shown by double arrow.



From figure, while travelling from C to H, one must go through: C->A ->E->D->H

So, station E must be passed through. **Hence, option (b) is correct.**

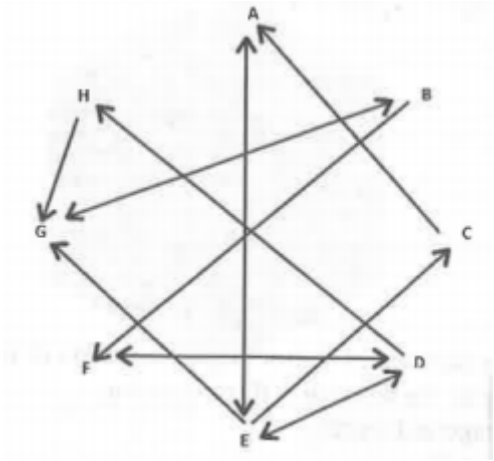
106. Answer: The correct answer is option (d)

Possible ways to move from F to A without passing through any station more than once FDEA FDECA FDHGCA FDEGCA

So, there are four possible ways. **Hence, option (d) is correct.**

107. Answer: The correct answer is option (c)

If the route between G and C is closed, then the one-way and two-way passages are:



Possible route while travelling from H to C is HGBFDEC.

So, one may not pass-through station A. **Hence, option (c) is correct.**

108. Answer: The correct answer is option (c)

Let's summarise the given information in table format:

Amala, Kamla, Komala and Nirmala are postgraduates in Sociology of whom two are bank employees. Kamala is a bank employee and Amala and Komala are unemployed So, Nirmala is also a bank employee.

Name	Tennis Player	Postgraduate	Bank Employee
Vimala			✓
Kamala		Sociology	✓
Amala		Sociology	× unemployed)
Komala	✓	Sociology	× nemployed)
Nirmala	✓	Sociology	✓
Shyamala		Commerce	

From table, we can clearly see that Nirmala is both a tennis player and a bank employee. **Hence, option (c) is correct.**

109. Answer: The correct answer is option (b)

A clock strikes once at 1 o'clock, twice at 2 o'clock and thrice at 3 o'clock, and so on.

It means the clock strikes 5 times at 5 o' clock.

So, at 10 o clock, clock strikes 10 times.

If the clock takes 12 seconds to strike at 5 o'clock 5 times, then the time taken by clock to strike 10 times at 10 o' clock = 12 x 2 = 24 seconds. **Hence, option (b) is correct.**

110. Answer: The correct answer is option (d)

Conclusion-1: All healthy people go for morning walk.

This conclusion is incorrect because morning walk is good for health. It does not mean all healthy people go for morning walk.

Conclusion-2: Morning walk is essential for maintaining good health. This conclusion is also incorrect because it

is not essential condition for good health. **Hence, option (d) is correct.**

111. Answer: The correct answer is option (b)

Initially, six boys A, B, C, D, E and F have 10 cards.

Now, F borrows 2 cards from A and gives away 5 to C who

in turn gives 3to B

A=10-2=8

F=10+2-5=7

C=10+5-3=12

B=10+3=13

Now, B gives 6 to D who passes 1 to E.

B=13-6=7

D=10+6-1=15

E=10+1=11

Finally, A = 8 cards, B = 7 cards, C = 12 cards, D = 15 cards,

E=11 cards and F =7 cards

Number of cards possessed by D and E = 15 + 11 = 26 cards

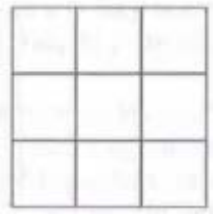
We can see that the number of cards possessed by B, C and F = 7+ 12 + 7 = 26 cards

The number of cards possessed by D and E is equal to the number of cards possessed by B, C and F. **Hence, option (b) is correct.**

112. Answer: The correct answer is option (c)

There are 4 horizontal and 4 vertical lines, parallel and equidistant to one another on a board.

The board will look like:



To make a square or rectangle, we need 2 horizontal and 2 vertical lines.

The number of ways to select 2 vertical lines

The number of ways to select 2 horizontal lines

So, the total number of squares and rectangles that can be formed

= 6 x 6 = 36

Hence, option (c) is correct.

113. Answer: The correct answer is option (d)

Using the given information, we have Randhir Kunal + Debu

-> Marks of Randhir is more than Kunal and Shankar alone.

Kunal + Shankar > Randhir

-> Marks of Shankar can be more than or less than Randhir.

Sonal > Shankar

Neha > Randhir

It is evident that we cannot determine who obtained the highest marks. **Hence, option (d) is correct.**

114. Answer: The correct answer is option (b)

From the given information, we can conclude that

Some supporters of party X knew Z and do not agree for alliance with party Y.

Some supporters of party X knew Z and have friends in party Y.

Option (a) is incorrect: Because no information is given related to supporters of party Y.

Option (b) is correct: That there is at least one supporter of 'party Y' who knew some supporters of 'party X' as a friend.

Option (c) is incorrect: Because there are some supporters of 'party X' who supported Z's campaign strategy. Option (d) is incorrect: It can be concluded from the given information. **Hence, option (b) is correct.**

115. Answer: The correct answer is option (c)

Statement-1: Some supporters of 'party X' knew Z. this is correct.

Statement-2: Some supporters of 'party X', who opposed Z's campaign strategy, knew Z. This statement cannot be concluded from the given data.

Statement-3: No supporters of 'party X' supported Z's campaign strategy. This statement is incorrect because there are some supporters of 'party X' who supported Z's campaign strategy.

So, statement 2 and 3 are incorrect. Hence, option (b) is correct answer.

116. Answer: The correct answer is option (b)

Let's take February (non-leap year) in order to minimise the number of working days.

In a non-leap year, February has 28 days.

Sun	Mon	Tues	Wed	Thur	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28

All the Sundays and 2nd and 4th Saturday are taken as holidays. It means, on 1, 8, 14, 15, 22 and 28 will be holiday.

Total number of working days = 28-6 = 22 days

So, the minimum number of possible working days of any month of any year = 22 days **Hence, option (b) is correct.**

117. Answer: The correct answer is option (c)

On four consecutive days Physics, Chemistry, Mathematics and Biology are to be conducted.

It is given that Chemistry is conducted exactly after two tests are held. It means Chemistry is conducted on 3rd day.

The Physics test is held before the test which is conducted after Biology. It means, neither Physics nor Biology test held on the last.

So, Mathematics is the last test. **Hence, option (c) is correct.**

118. Answer: The correct answer is option (a)

Conclusion-I: Natural gifts need nurturing and care. This is correct because it is given in the statement that one needs to keep practicing to improve and good voice is a natural gift.

Conclusion-Il: Even though one's voice is not good; one can keep practicing. This conclusion is incorrect with respect to the given statement. **Hence, option (a) is correct.**

119. Answer: The correct answer is option (c)

Total number of students = 52

Number of students failed = 15

The names of the failed students have been removed

So, students left = 52 - 15 = 37

Now, Ramesh is 22th from the top. So, his position from the bottom = $37 — 22 + 1 = 16^{th}$. **Hence, option (c) is correct.**

120. Answer: The correct answer is option (c)

Using the given statements,

P+ R means P is the son of R

R—Q means R is the wife of Q

It means P is the son of R (mother) and Q (Father). So, Q is the father of P. **Hence, option (c) is correct.**

121. Answer: (d)

Fruits eaten by A, B, C and D are:

A - grapes and pineapple

B - grapes and oranges

C - oranges, pineapple, and apple

D - grapes, apple, and pineapple.

After taking fruits, B and C fell sick. Common fruit taken by B and C is oranges.

So, it can be said that the cause of sickness was oranges. **Hence, option (d) is correct.**

122. Answer: (d)

From the given statements, it can be concluded that the growth of population is increasing in the country due to faster decline in death rate compared to birth rate. **Hence, option (d) is correct.**

123. Answer: (d)

From the given statement it can be concluded that they should either go for picnic or for trekking.

Option (a): We go for a picnic but not for trekking. If this is true, it will not falsify the claim.

Option (b): Activities such as picnic and trekking are encouraged by the health authorities. This statement, if true, would not falsify the claim.

Option (c): We go for trekking and not for picnic. If this is true, it will not falsify the claim.

Option (d): We do not go either for picnic or trekking. If this is true, then it would falsify the claim because it is claimed that they should either go for picnic or trekking. **Hence, option (d) is correct.**

124. Answer: (d)

Total faculty members = 50

Number of males = 30

So, number of females = 50 — 30 = 20

All the male faculty members do not know music.

Some female faculty members know music.

Now, the Head of the institution invited six faculty members to a tea party by draw of lots and party it was discovered that no member knew music.

Case-1: All 6 are male faculty members.

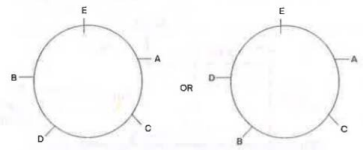
Case-2: All 6 are female faculty members as there are some females also who do not know music.

Case-3: There can be mix of male and female faculty members also. So, nothing can be said about the gender composition of the party.

Hence, option (d) is correct.

125. Answer: (b)

Five people A, B, C, D and E are seated about a round table. Using the given statements, following combination can be drawn:



There can be many other arrangements as the position of A and C can be interchanged.

From the above diagram, we can observe that E is seated next to A and D is seated next to B.

It means statement-1 and statement-2 are correct.

D and C can be seated next to each other or separated by one seat.

So, statetement-3 is not true.

Hence, option (b) is correct.

126. Answer: (d)

Number of hobby clubs = 5 i.e., photography, yachting, chess, electronics, and gardening. Gardening group meets — 2nd day.

Electronics group meets — 3rd day.

Chess group meets — 4th day.

Yachting group meets — 5th day.

Photography group meets — 6th day.

L.C.M of 2,3,4,5 and 6 = 60 All the five groups meet on 60th day. So, within 180 days, all the 5 group meet on the same day = 3 times i.e., on 60th day, 120th day and 180th day. **Hence, option (d) is correct.**

127. Answer: (a)

We will make a table using the given information:

Person	Profession	Gender	Married/ Unmarried
P	Student		Unmarried
Q	Lawyer	Male	
R	Artist	Female	Married
S	Student		Unmarried
T	Doctor	Male	Married

P and S are unmarried students.

Q is the lawyer.

T is the doctor.

R is the artist.

So, T - Doctor, Married, Male

T is married to R (since Q is male and P & S are unmarried students, the only possibility is R being female and married to T).

128. Answer: (c)

From table given above, R is the artist. **Hence, option (c) is correct.**

129. Answer: (b)

From table given above, the spouse of R is T. **Hence, option (b) is correct.**

130. Answer: (b)

From table given above, Q is the lawyer. **Hence, option (b) is correct.**

131. Answer: (c)

From table above, Q and T are definitely a man. **Hence, option (c) is correct.**

132. Answer: (d)

It is given that,

A>B

D>C

B>C

So, A>B>C

D's position cannot be determined.

So, definitely A is older than C.

Hence, option (d) is correct.

133. Answer: (b)

Let's summarise the given information in the table given below:

Box	Ball	Colour	Box Position from Left
A	Tennis	Orange	6
B	Volleyball	Yellow/Blue	3
C	Cricket	Green	2
D	Golf, Hockey Football	Yellow/Blue	1
E	Hockey Golf	Violet	5
F	Hockey	Blue, Yellow, Indigo	4

The hockey ball is not in box D and E. So, it will come in box F.

The golf ball is in violet box and not in the box D. 50, it will come in box E.

-> Box D has football ball.

The hockey ball is neither in the box painted blue nor in the box painted yellow. The left-out colour is Indigo. So, Box F is painted Indigo.

From table, Box E contains the golf ball.

Hence, option (b) is correct.

134. Answer: (b)

From table given above, it is clear that F is painted Indigo and the colour of box B/D is either Yellow/Blue.

Hence, option (b) is correct.

135. Answer: (c)

From table given above, football is in box D which is painted either yellow or blue.

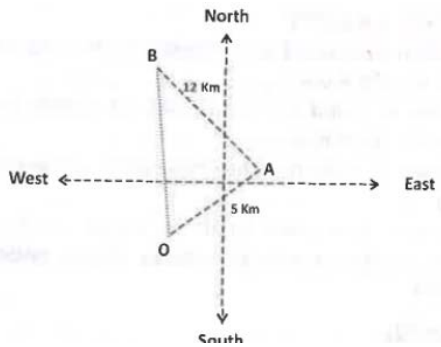
Hence, option (c) is correct.

136. Answer: (b)

A person starts from O and climbs the hill in northeast direction and reaches at point A after travelling a distance of 5 km and from there he moves in north-west direction and reaches at point B.

AB =12 km and OA = 5km

BAO is right-angled triangle at O.



Using Pythagoras Theorem,

$OB^2 = OA^2 + AB^2$

$OB^2 = 5^2 + 12^2 = 25 + 144 = 169$

OB = 13km

So, the person is 13 km away from the starting point O.

Hence, option (b) is correct.

137. Answer: (a)

	Jacket	Sweater	Tie	Blue	White	Black
A		✓			✓	
B	✓	×			×	✓
C			✓	✓		

Ribeiro		×	✓	✓	×	×
Singh	✓				×	✓
Kumar	×	✓			✓	

Ribeiro does not wear white and black colour. Singh does not wear white colour.

So, Ribeiro wore blue colour. Singh wore black colour and Kumar wore White colour.

Also, C wears a tie and B does not wear sweater. So, B wears jacket and A wear sweater.

Neither B nor Ribeiro wore a white sweater. This means sweater is white and Ribeiro is C

So, A is Kumar and B is Singh

From table, C's surname is Ribeiro.

Hence, option (a) is correct.

138. Answer: (b) From table given above, the colour of tie is blue.

Hence, option (b) is correct.

139. Answer: (a)

From table given above, A wore the sweater.

Hence, option (a) is correct.

140. Answer: (d)

In a military code,

SYSTEM — SYSMET

->SYS / TEM = SYS / MET

NEARER — AENRER

> NEA / RER — AEN / RER

We can clearly see that in the code first half is reversed and then the second half is reversed.

So, FRACTION is written is CARFNOIT.

FRAC / TION — CARF / NOIT.

Hence, option (d) is correct.

141. Answer: (c)

Go by option:

Option (a) is incorrect: Hour hand will be between 6 and 7, minute hand will be at 3 so hour hand is ahead.

Option (b) is incorrect: Hour hand will be between 6 and 7, minute hand will be between 3 and 4 so hour hand is ahead.

Option (d) is incorrect: Hour hand will be between 6 and 7, minute hand will be between 9 and 10 so minute hand is ahead of hour hand more than 3 minutes even more than 10 minutes.

Hence, option (c) is correct.

142. Answer: (c)

In this question, Tasks-1 cannot be assigned to either person-1 or person-2.

Hence, we can assign task 1 in two ways.

Task -2 must be assigned to either person-3 or person-4.

Hence, we can assign task 2 in 2 ways.

Now 3 tasks left which are to be assigned to 3 persons.

We can assign 3 tasks to 3 people in 6 ways.

So total no. of ways = 2x2x6 = 24 ways.

Hence, option (c) is correct.

143. Answer: (b)

No. of white marbles = W = 10

Given that red marbles are 3 more than white marbles.

Hence, no. of red marbles = 13.

White marbles are 5 more than green marbles.

Then no. of green marbles = 5

Total no. of marbles = 10+ 5 + 13 = 28.

Hence, option (b) is correct.

144. Answer: (d)

They are deciding host by picking lots so it cannot be predicted that lot of a person comes out how many times. Like it can be possible that name of a person comes out every time or doesn't come ever.

Hence, option (d) is correct.

145. Answer: (a)

Pattern of triangle: rotate by 90 degree and get inverted. Pattern of dot: rotate anti clockwise on the nodes. Same pattern is followed by square and circle.

By following the above stated patterns, the triangle will come out on the upper left side, and it will be straight. And position of others will be same as that in 2 figures.

Hence, option (a) is correct.

146. Answer: (a) We will establish a relation between 1% row and 2™ row:

6=3x2(3-1)

56=8x7(8-1)

90=10x9(10-1)

2=2x1(2-1)

Hence, 20=5x4(5-1)

Hence, the number which is missing = 5.

Hence, option (a) is correct.

147. Answer: (c)

Given series= 7,x,21,31,43

To find= the value of x

We will find the difference between the consecutive numbers:

Difference between 21 and 31=31-21= 10

Difference between 31 and 43=43-31= 12

We can see that the series follow a pattern of difference of 2. If the difference between 21 and 31 s 10,

Then the difference between x and 21 should be 8.

x=21-d

x= 21-8= 13.

Hence, option (c) is correct.

148. Answer: (d)

There are two rectangles and two triangles, and we need to join them.

As we see in the options (a), (b) and (c) we do not have any rectangle these are parallelograms.

Hence, these 3 options are eliminated. And as we see option (d) there are two triangles which are adjusted vertically, and two triangles are adjusted

left and right to rectangles.

Hence, option (d) is correct.

149. Answer: (b)

Option (a) is incorrect: Statement said that all good athletes want to win but it does not imply bad athlete want to win.

Option (b) is correct: Given that All good athletes want to win and all athletes who want to eat a well-balanced diet Hence, if an athlete who does not eat a good diet is not a good athlete.

Option (c) is incorrect: Not but the converse is true.

Option (d) is incorrect: No even bad athletes also want to

Hence, option (b) is correct.

150. Answer: (b)

Given that A runs less fast than B.

A<B

B does not rum faster than C. Hence, B < C

Hence, A< B < C, C is faster than A.

Hence, option (b) is correct.

151. Answer: (c)

Lady finger is tastier than cabbage. Hence, L>C

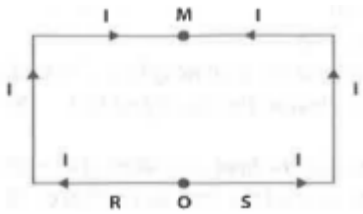
Cauliflower is tastier than lady's finger. Hence, Ca>L

So, Ca>L>C

Cauliflower is tastier than cabbage.

152. Answer: (b)

Let's understand the given problem by using a diagram:



They, started at 0.

Rohit takes right turn and shahid takes left after every 1 km.

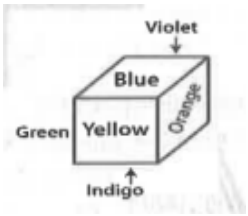
They meet at point M.

And each travelled 3kms before meeting.

Hence, option (b) is correct.

153. Answer: (c)

Given problem can be visualized through a picture as given below:



Color of face opposite to orange is green.

Hence, option (c) is correct.

154. Answer: (d)

Conclusion I is invalid: Given that some men are great some men are wise, But there could be men who are neither great nor wise.

Conclusion II is invalid: It could happen but not surely because let's suppose 50% men are wise and other 50% are great.

Hence, option (d) is correct.

155. Answer: (b)

By the information given in question:

Usha runs faster than Kamala Implies Usha>Kamala.

Priti runs slower than Swati implies Priti < Swati.

Swati runs slower than Kamala implies Swati < Kamala.

Overall, Priti < Swati < Kamala < Usha.

Priti is slowest. **Hence, option (b) is correct.**

156. Answer: (b)

There are 2 possibilities which are shown below:

(i) E - 22

A – 21

C – 20

B – 19

F – 18

D – 17

And 2nd possibility is:

(ii) E -22

B – 21

A – 20

C – 19

F – 18

D – 17

Hence, by both the possibilities we can conclude that age of Fis 18 years.

Hence, option (b) is correct.

157. Answer: (b)

Clearly by the explanation given above we can conclude the answer-2 logical possible orders of all six cousins in terms of increasing age.

Hence option (b) is correct.

158. Answer: (d)

Option (a) is not true: As Rekha and David only can drive. And Anil speaks Tamil who cannot drive.

Option (b) is not true: As Rekha and David only can drive. And Shabnam speaks Marathi who cannot drive.

Option (c) is not true: As Rekha and David only can drive. And Shabnam speaks Marathi who cannot drive.

Option (d) is true: As Rekha and David only can drive. And Anil speaks Tamil so one of Rekha and David speak Tamil.

Hence, option (d) is correct.

159. Answer: (b)

Given that two cowards are always friends, a fighter and a coward are always enemies, two fighters are indifferent to one another.

A, B are enemies implies one of them is a fighter and another is coward.

C, D are friends implies both C, D are cowards.

E, F are indifferent implies both are fighters.

A, E are not enemies implies A is also a fighter.

B, F are enemies implies B is a coward.

A, E, F are fighters and B, C, D are cowards.

Hence, option (b) is correct.

160. Answer: (d)

Let's take age of son = x years. Then age of father = 9x

Age of mother = 8. Given that sum of age of father and mother = 51

9x+8x=51

17x=51

X = 3.

Hence, option (d) is correct.

161. Answer: (c)

Let's suppose no. of people in town = 100

No. of people who read magazine A = n(A) = 45

No. of people who read magazine B = n(B) = 55

No. of people who read magazine C = n(C) = 40

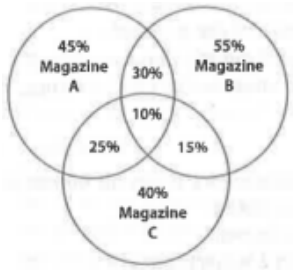
n (A and B)=30

n(B and C)=15

n(A and C)=25

n(A and Band C) = 10.

Let's understand this by using Venn's diagram.



We know the formula.

n(A or B or C) = n(A) + n(B)+ n(C) = n(A and B) - n(B and C)-n(A and C) + n(A and B and C)

n(A or B or C) = (45 + 55 + 40) — (30 + 25 +15) + 10 = 80

Hence, there are 20 people who don't read any magazine.

Hence, option (c) is correct.

162. Answer: (a)

Total no. of students = 130.

No. of students who failed in English = n(E) = 62

n(M) =52

n(Mand E) = 24

Now by formulae: n(E or M) = n(E) + n(M) - n(E and M)

n(E or M) = 62 +52-24 =90

Total 90 students are failed. No. of students who passes finally = 130 - 90 = 40.

Hence, option (a) is correct.

163. Answer: (c)

People can speak Tamil, Hindi and Gujrati

Tamil speakers are persons who can speak tamil is 6.

Only Tamil speakers are 6-2-1=3

Hindi speakers are persons who can speak is 15.

Only Hindi speakers are 15-2-1=12

Gujarati speakers are persons who can speak is 6.

Only Gujarati speakers are 6-1=5

Thus, the number of persons who can speak only one language is 3+12+5=20

Number of persons who can speak two languages =2

Number of persons who can speak all the languages = 1

Total number of persons = 23.

Hence, option (c) is correct.

164. Answer: X

Note: UPSC dropped the question after the exam.

165. Answer: (d)

Answer is option (d) because from these 3 statements we. cannot say anything about UFOs. **Hence, option (d) is correct.**

166. Answer: (c)

Let the no. of men = x

No. of women = y

Total no. of handshakes = xy= 24

Possible factors of 24 are 4 and 6.

X = 6 or 4 and y = 6 or 4. No. of hugs = X_{c2} +Y_{c2} or 6_{c2} + 4_{c2} = 21.

Hence, option (c) is correct.

167. Answer: (c)

The correct option is option C

No explanation is needed for such problems. It is based on pure imagination.

You can easily find that the given figure is embedded in option (c). **Hence, option (c) is correct.**

168. Answer: (b)

The correct option is option B

Observed pattern is: when you move to 2nd column 1st arrow is same and 2nd arrow rotate 180°. (With respect to 1st column). When you move to 3" column 1 arrow rotate 180° and 2 arrow is same (w.r.t 1%t column)

Now, we can find the figure of blank space check the 3¢ row and 1% column 1% arrow is heading upward and 2°¢

arrow is heading downward

Now according to pattern in next figure 1%t arrow will be same (upward) and 2" arrow will rotate 180 so will be heading upward.

Both the arrows will be heading upward. Hence, option b is correct.

169. Answer: (d)

The correct option is option D

Sum of numbers in each row = 60

As30+27+3=60and 29 + 13+ 18 = 60

So according to observation 33 + 19 + X = 60

Or X = 60— 52 = 8. **Hence, option (d) is correct.**

170. Answer: (a)

The correct option is option A

Observe the pattern: As we move left to right in each row no. of circles decreasing by 1

And triangles are increasing by 1.

By observing this pattern in 3" row

In 1% figure there are 2 circles and O triangles

In next figure 1 circle and 1 triangle

So, in next figure there must be O circles and 2 triangles.

Hence, option (a) is correct.

171. Answer: (d)

The correct option is option D

No. of routes fromS to T=3

No. of routes fromU to V =2

No. of routes fromV to T=3

So, no. of routes from S to T = 3x2x3 = 18. **Hence, option (d) is correct.**

172. Answer: (b)

The correct option is option B

Observe the pattern: Look at the upper left circle, in 1% figure left half of circle is highlighted, in 2™ figure right

half of circle is highlighted, in 3™ figure lower half of circle is highlighted, in 4 figure upper half of circle is

highlighted, Hence, by the same pattern in 5% image again left half of circle will be highlighted.

From this we can eliminate options (a) and (d).

Now we can also observe that upper left/right and lower left/right make complete circle highlighted.

By this we can eliminate option (c). Hence, option (b) is CORRECT

173. Answer: (c)

The correct option is option C

Money A has = 10 Rs.

Money B has = 3 Rs.

Now given that money A has is 80% of fare for 2.

Fair for two = x

Then 80% of x = 10

Then x = 12.50 Rs.

Total money A and B have = 10 + 3 = 13 Rs. After buying

ticket they left with 13 - 12.50 = 50 paisa. **Hence, option (c) is correct.**

174. Answer: (c)

The correct option is option C

60% of loan is paid in installments = 18

1% of loan is paid in installments = 18/60

100% Of loan is paid in installments = (18/60) x 100 = 30.

Hence, option (c) is correct.

175. Answer: (d)

The correct option is option D

Go by option

No. of children in 1% row = x

In 2" row = x-3 \4

In 3" row = x-6 E4—PB

In 4" row =x—9 and so on...

Now let's check 3 rows are possible or not

If no. of rows =3 1km.

Then x + x-3 +x— 6 = 630 OR 3x = 621 OR x = 207

Hence, 3 rows are possible.

By doing this we can conclude 6 rows are not possible

x + x - 3 + x - 6 + x - 9 + x - 12 + x - 15 = 630

6x= 675

Or x=112.5(not an integer). **Hence, option (d) is correct.**

176. Answer: (c)

The correct option is option C

Statement 1: "A is further up than E but is lower than C. B is in the middle."..... E (where we can't predict the person in between we use the dots.)

Statement 2: B is in the middle.

Statement 3: G is between A and B.

So, the first four on top of ladder have to be C>A>G>B.....E.

Statement 4: If F is between E and D.

So, sequence becomes like C>A>G>B>E>F>D.

D is on the bottom step of ladder. **Hence, option (c) is correct.**

177. Answer: (c)

The correct option is option C

Statement 1: A>B (A is taller than B)

Statement 2: C>A.

By combining both statements: C>A>B. Statement 3: D>C.

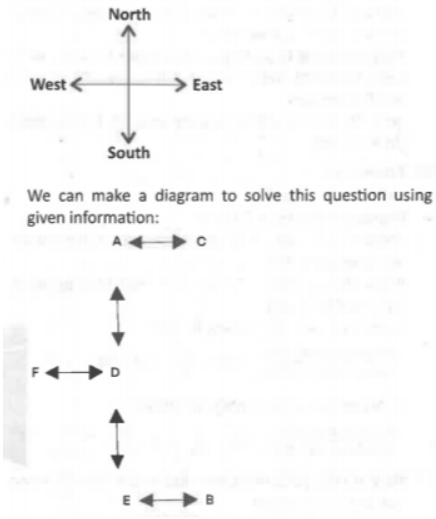
Statement 4: E is tallest.

Hence, the sequence become like: E>D>C>B>A.

C acquires middle position. **Hence, option (c) is correct.**

178. Answer: (B)

The correct option is option B



In the diagram arrow shows the direction and represent 1km.

We can clearly see that A, D and E are in a line. **Hence, option (b) is correct.**

179. Answer: (b)

The correct option is option B

Statement 1: A is occupying the seat next to B but not next to C

Possibilities can be: AB or BA, but not CAB or BAC. Statement 2: If C is not sitting next to D.

CD or DC is not possible. So, o C has to be next to B, and D next to A.

Hence, there are two possibilities: DABC or CBAD.

In both cases A is adjacent to D. **Hence, option (b) is correct.**

180. Answer: (c)

The correct option is option

Between 8 o'clock and 9 o'clock two hands of clock are one above each other when time is between 8:43 and 8:44.

After that both hands of clock are one above each other when time is between 9:48 and 9:49.

Hence, after 65 minutes two hands of clock are one above each other. **Hence, option (c) is correct.**

181. Answer: (c)

The correct option is option C

Given that 3rd day of month is Monday.

So, 3+7= 10 and 10+ 7 = 17 and 17 + 7 = 24th day is Monday.

So, 3, 10, 17 and 24 are Monday. Now 5th day from 21st is 26th

Now 24th is Monday and 26th will be Wednesday. **Hence, option (c) is correct.**

182. Answer: (c)

The correct option is option C

Total no. of tickets sold = 420.

Half of 420 = 210. Now 210 tickets are sold at rs.5/ticket.

Amount of money collected by selling 210 tickets = 210 x5 =1050.

One third of 420 = 140

Now 140 tickets are sold at rs.3/ticket.

Amount of money collected by selling these 140 tickets =140 x 3 = 420.

Now 420 — 140 — 210 = 70 tickets are sold at rs.2/ticket.

Amount of money collected by selling these 70 tickets = 70x 2 =140

Total amount of money collected = 1050 + 420 + 140 = 1610. **Hence, option (c) is correct.**

183. Answer: (a)

The correct option is option A

Statement 1: A, the engineer is married to the lady stenographer.

Implication: A is male and an engineer also stenographer is some female.

Statement 2: The judge is married to the lawyer.

Implication: judge and lawyer are married to each other.

Statement 3: F, the draughtsman is the son of B and brother of E

Implication: F is male and draughtsman.

Statement 4: C, the lawyer is the daughter-in-law of D,

Implication: C is lawyer and female also by 2" statement judge is male and also son of D.

Statement 5: D is the grandmother of F

implication: D is female and F is son/daughter of judge.

Hence, the final relationship chart of family is:

First Generation: A (Male - Engineer) — D (Female-Stenographer)

Second Generation: B (Male - judge) — C (Female - Lawyer)

Third Generation: F (Male — Draughtsman) & E (Male/ Female - Doctor)

Now by this explanation we will Answer this question and next 2 questions.

Now clearly profession of B is judge. **Hence, option (a)**

184. Answer: (c)

The correct option is option C

Explanation: By explanation given above we conclude the answer- Both AD and BC. Hence option (c) is correct.

185. Answer: (b)

The correct option is option B

Explanation: By explanation given above we conclude the answer- Stenographer. **Hence option (b) is correct.**

186. Answer: (a)

The correct option is option A

Let's take

Money Alok has = A,

Money Bhupesh has =B,

Money Chander has = C and

Money Dinesh has = D

Now according to question: A + B+ C + D = 100

Alok and Bhupesh between them have as much money

as Chander and Dinesh between them.

So, A+B =C+D

Alok has more money than Bhupesh.

So, A>B

Chander has only half the money that Dinesh has

So, D=2C

Now we have A>B and D>C, so chander and Bhupesh cannot have maximum money.

Alok has in fact Rs. 5 more than Dinesh has.

So, A>D

Alok has the maximum amount of money. **Hence, option (a) is correct.**

187. Answer: (d)

The correct option is option D

Day	Subject
Monday	Music (statement 1)
Tuesday	Sports (statement 5)
Wednesday	Mathematics (statement 2)
Thursday	History (statement 4, next to math)
Friday	Not literature (break)
Saturday	Literature (implication of 3rd statement)

By the table we can directly say George is free on Friday.

Hence, option (d) is correct.

188. Answer (c)

Initial:

Left: .....A (11th) ... .B(10th from right) ... Right

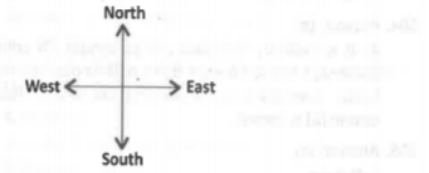
After Interchange:

Left: ... (17 people) ... A (18th) ... (9 persons) ... Right

This confirms that the total number of persons is 27, and the number of persons other than (A) and (B) is 25.

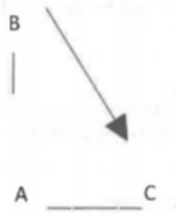
189. Answer B

The correct option is option B



South

Now the locations of A, B, and C are given in figure below:



AB = 5km.

AC = 12km.

We need to find BC.

By using Pythagoras theorem:

$Ab^2 + AC^2 = BC^2$

Or, $5^2 + 12^2 = B$

$169 = BC^2$

BC =13. **Hence, option (b) is correct.**

190. Answer: (d)

The correct option is option D

Total no. of students = 50 = n (T)

No. of students who speak English = 21 = n (E)

No. of students who speak Hindi and English = 10 = n(H ∩ E)

No. of people who speak English only = n (E only) = n (E) - n(H ∩ E); n (E only)=21-10=11

now go on options only option (d) tells us that no. of people who speak English only is 11. **Hence, option (d) is correct.**

To find other Answer if 11 students speak English only then 50 - 11 = 39 will speak Hindi.

To find only Hindi speaking student = 39— 10 = 29. **Hence, option (d) is correct.**

191. Answer: (b)

The correct option is option B

Lets make a table to understand the question:

Description	Books
Green cover	B, C, E, F
Yellow covers	A, D
New volumes	A, B, D
Old volumes	C, E, F
Law reports	A, B, C
Medical extract	D, E, F

See in the table and observe the rows of green cover and medical extracts. Books that are old medical extracts

and have green covers are: E and F. **Hence, option (b) is correct.**

192. Answer: (b)

The correct option is option B

First point is at a distance of 1 cm, 2" at 2 cm 3" point at 6cm and so on — therefore, the distance of each point from the left edge of the scale will be.

1%t point — 1 cm, 2" -3 cm, 3 - 6 cm, 4"~ 10 cm, 5t - 15 cm, 6t — 21 cm, 7" — 28 cm, 8" —36 cm.

Similarly, we can count 8 points starting from right edge.

Total no. of points = 8+ 8 = 16 There are 2 end points so we should reduce these so points remains = 14

5t point from the left; and 6™ point from the right are common and will repeat twice. So, they have to be counted only once. Reduce 2 more points from 14 which become 12 as the Answer. **Hence, option (b) is correct.**

193. Answer: (a)

The correct option is option A

To get all 2s we need to get 4 1s first by using this logic

let's solve this question.

Before flipping, the Cards show: 111111.

First flip (make any 4 cards show 2): 222211,

2nd flip (take one of those showing 1 and three of those showing two and flip): 211121.

3rd flip (Flip all the ones): 222222...required result.

Hence, option (a) is correct.

194. Answer: (b)

LCM of the intervals (6, 7, 8, 9, 12 seconds) = $2^3 \times 3^2 \times 7 = 504$ seconds

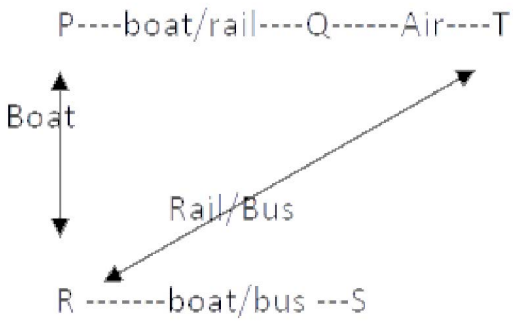
3600/504 = 7.14

so, they fire together 7 times in an hour.

195. Solution: (a)

196. Solution: (a)

Its a problem which can be solved best using a diagram.



Solution will be boat. Clear from the diagram.

197. Solution: (b)

Because the node at R leads to S from where there is no other way. So, one has to return to R in any possible way.

198. Solution: (d)

Clear from the diagram.

199. Solution:

Clear from the diagram. It will be bus, rail, boat, air – all.

200. Solution:

If Y is selected, W will not be selected. So, options (a) and (d) are wrong. As C cannot play with Z, option (c) is also wrong.

201. Solution:

If B is selected, W will not be selected. So options (a) and (d) are wrong. As C cannot play with Z, option (b) is also wrong.

202. Solution: c)

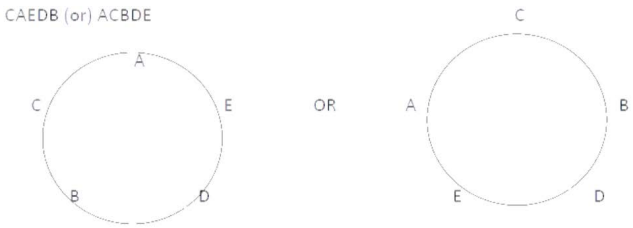
Option (a) is not possible as Shobha and Kunal will not work together
Option (b) is not possible as Rohit and Tanya will not work together
Option (d) is not possible as Rohit and Tanya will not work together
Hence, option (c)

203. Solution: c)

Easiest way is to draw a circle.

These are the two possibilities:

CAEDB (or) ACBDE C



In both the cases, D is placed next to B. And this is true for A and E.

204. Solution: (d)

Clear from the statements.

205. Solution: (c)

Simple enough.

206. Solution: (b)

207. Solution: (d)

Geeta is older than her cousin Meena => G > M
Meena's brother Bipin is older than Geeta => B > G
Therefore, B > G > M

208. Answer: (b)

- Based on the passage, the financial markets in emerging market economies, including India, have been adversely impacted in recent years due to:
  - **Weak global economic prospects**
  - **Uncertainties in the international financial markets**
  - **Sovereign risk concerns in the Euro area**
- These factors have contributed to higher-than-normal levels of volatility in financial markets. However, **bad monsoons and the resultant crop loss** are not mentioned in the passage as factors affecting financial markets in this context.

209. Answer: (c)

- The passage discusses how India's financial markets have deepened and become more integrated with the global economy since 1991. This growing integration means that international financial uncertainties, such as the Eurozone's sovereign debt issues, have impacted India's markets, leading to increased volatility. Therefore, the primary reason for the influence of global changes on Indian financial markets is their **growing global linkages and integration**.

210. Answer: (d)

- The passage emphasizes that the Indian financial system is predominantly **bank-dominated**, highlighting the critical role of banks in ensuring overall financial stability. This dominance underscores the importance of banks' resilience to withstand financial stress.

211. Answer: (a)

The passage highlights several factors that will make risk and liquidity management more important in the Indian banking system in the future:

1. **Further globalization**: The passage mentions that further globalization will make the banking business more complex and riskier. This implies that as the Indian financial system becomes more integrated with the global economy, managing risks and liquidity will become more critical.
2. **More consolidation and deregulation of the financial system**: The passage states that consolidation and deregulation will also contribute to making the banking business more complex and riskier. This suggests that these factors will increase the importance of risk and liquidity management.
3. **Further diversification of the financial system**: The passage indicates that diversification will similarly increase the complexity and riskiness of the banking business. Therefore, risk and liquidity management will become more important as a result of diversification.

4. **More financial inclusion in the economy**: While financial inclusion is generally a positive development, the passage does not link it to increased complexity or risk in the banking system.

212. Solution: (a)

Take the LCM of 2,3,4,5 and 6. You get 60 i.e. all the groups meet on the same day on every 60th day. Hence, only three times.

213. Solution: (d)

From the statements you can construct the following and then figure out what belongs to what.

214. Solution:

You can construct the following table from the statements.

Person	Cap
A/B/C	green/blue/yellow
D	violet
E	indigo
G	orange

So F is wearing red cap.

215. Solution: (c)

Initial Condition = 19 guys – The Boy (20th Rank) – 25 other guys

Now = 20 guys – The Boy – 26 other guys (Since his rank has shifted by one, one guy should be ahead and one behind)

So, his rank from end is 27th.

216. Solution: (b)

The 'sun' is following an anti-clockwise rotation in square figure. The triangle is following a zigzag pattern. Pretty simple to grasp.

217. Solution: (c)

The triangle is rotating clockwise by an angle of 120 degrees. And, the arrow is following a patten of up-down in subsequent figures.

218. Solution: (b)

Little tricky. But after observing for some time, you will be able to see the pattern.

The three segments of each figure (two Ls and one arrow); are

(i) Moving left

(ii) When they reach the extreme left (first position), they come back again from the other left (but with an inverted image)

So, when the arrow in the first figure moves to second figure, it becomes inverted. And until it goes again to the extreme left and moves, it would not invert.

For such problems, good observation power and practice is needed.

219. Solution:

You can see it diagram wise (Technique 1) or mathematically (Technique 2).

Diagram I: Even for zero x, there exists some y (i.e. it assumes some value). This means, even if there are no men, there exist some working days, which is wrong.

Diagram II: If x is increased y is also increased. This means if no. of men are increased then the no. of working days increased, which is wrong.

Diagram III: This has two parts in it, a vertical line and a horizontal line. Vertical line suggests infinite working days for some number of workers, which is wrong. Horizontal line suggests, same working days even if the no of men is increased/decreased which is also wrong.

Diagram IV: This shows the correct relation i.e. no. of working days reduces as no. Of men are increased.

You can see this in terms of a mathematical equation too (Technique 2):

Here x and y are inversely related. If x, the number of men increases, then, y, the number of days for completing the work decreases. So, mathematically they are inversely proportional.

=> x multiplied(1/y) => xy = constant

This is an equation of a rectangular hyperbola which is shown in diagram IV.

220. Solution: (c)

If you add all the digits in the central column, you get the addition of the numbers in the first and second column.

For e.g. 3+7+0=10; which is equal to 3+7 = 10

Same in Second row: 2+2+4 = 8 which is equal to 2+6 = 8

Same should be in third row: 7+3+0=10 which should be equal to 1+x = 10 => X = 9

221. Solution: (c)

Observe the first two figures carefully. You will see this pattern. 2*(84/12)=14

2*(81/9)=18

222. Solution: a)

Technique 1: (Simple drawing)

First you can follow a very simple method. Take a rubber (cuboid or cube shape) with you in the exam hall. Try writing up the numbers as they have been put in the question figure. You have the answer on that rubber in front of you.

Technique 2: (Logic)

To see logically, If we check the first two cubes, the common number 1 has numbers 4, 6, 2 and 3 on its four adjacent faces. So the remaining number 5 is on the opposite face of 1. .....Statement 1 (S1)

See the 4th cube. Here we need to find two adjacent numbers of 5.

Method I:

Looking at it in other way, (from statement 1) it can be said that, 2,3,4 and 6 are the adjacent numbers of 5. So out of all, only (a) is possible.

Method II:

If you see statement (1) itself, you can eliminate three answer options.

Number 1 should not be an adjacent number of 5 because it is the

opposite number of 5 (from statement (1)) .

So answer options (b),(c) and (d) can be eliminated as each of these options have 1 as one of its two numbers.

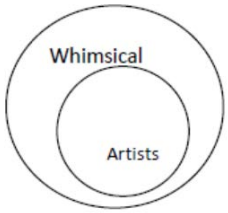
223. Solution: (d)

The pattern is: Outer arrows - first clockwise, then anti-clock wise, then two facing each other, with the below one in the anti-clockwise direction. Inner circle and square change alternatively. Follow this and you have the answer.

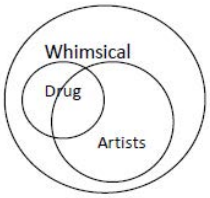
224. Solution: (b)

Again try drawing a Venn diagram.

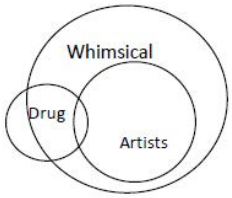
S1: If all artists are whimsical, it means that whimsical is a big category (or circle) and all artists will be one small circle in that.



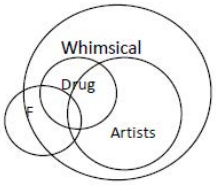
S2: If some artists are drug addicts, draw a circle that cuts the circle of Drug addicts. Now, that circle can either be completely inside whimsical OR outside it cutting it.



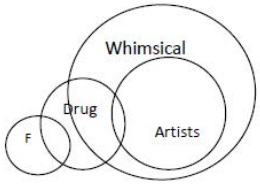
OR



S3: If frustrated people MAY become drug addicts (not all), then Frustrated (F) can be a smaller circle cutting the drug circle.



OR



Now, check the options.

Option (a) can be dismissed easily.

Option (b) has to be correct, because no matter what the drug circle has to cut the whimsical circle.

Option (c) too can be dismissed based on the diagram.

Option (d) is also wrong, because the diagram does not support it.

225. Solution: (d)

From S1: A>=B From S2: D>=C From S3: B>C

Now if B (which is either equal or older than A) is greater than C, then A has to be older than C.

226. Solution: (b)

You can use a Venn diagram but use logic, you can solve it quicker.

See, option 1 and 3 are too general and not necessarily true. But option 2 is a correct conclusion as all the club members definitely have a pair of binoculars.

227. Solution: (d)

Parents are not even in the picture. So, no to (a).

(b) conclusion cannot be made. We can't compare his likings with the information we have.

(c) would be too general to say, because we know only about two summer camps that he has been to.

(d) is the choice left which is evident from his joining the camps.

228. Solution: (b)

Make a small table from the given data.

	Shirt	Pant
A -	~~Black~~	~~Green~~
B -	~~Blue~~	Orange
C -	~~Orange~~	

Now, from the above information, this can be the deduced table.

	Shirt	Pant
A -	~~Black~~ ~~Blue~~/Orange	~~Green,~~Yellow
B -	~~Blue~~ Black/~~Orange~~	Orange
C -	~~Orange~~ Blue	Green

Answer is green and blue.

Answer is green and blue.

229. Solution: (c)

Three shows were discontinued as given. Since no sitcoms shows were discontinued, the discontinued ones must be either drama or news magazines. Now, there were only two news magazines. So, atleast one discontinued show was a drama.

230. Solution: C

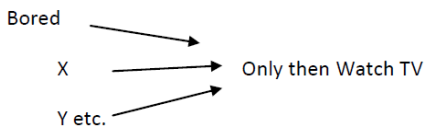
Assume these are the 4 people - Doc 1, Doc 2 , Business 1, Business 2

Now, if Doc 2(Tamil, Christian), then Doc 1 (has to be Muslim, Gujarati) and Business 1 has to be (Gujarati, Christian); so Business 2 will be (Muslim, Tamil)

231. Solution: (d)

This is a typical logic based reasoning.

To make such questions easier, always draw a logical flow diagram.



Here being bored or X or Y is leading to the action "Watch TV". But if the person is watching TV, it does not mean that he was necessarily bored. The cause could have been X or Y or anything. So, option (a) is incorrect. But, at the same time, he will not watch TV, if he is note bored.

Same diagram can be visualized for the relation between "not being bored" and "being in brother's company ". You will see that his brother's company is not the only way in which he can escape boredom. This tells that (c) is wrong because he may not be bored when he doesn't have his brother's company.

232. Solution: (b)

Verify logically for each option.

Take the first one - In this case, if R is open, then B, C and Q can also be open. 4 roads can be open. So its wrong.

Take the second one - In storm either A or B is open. Hence, it has to be correct.

Take the third one - P is closed during flooding. Therefore, if A is open then neither of B and Q will be open. So, at most two roads can be open. So, This may or may not be true.

Take the fourth one - From option 2, we know that at least one road is open. Hence, this is incorrect.

In all three given situations - Normal, Storms and Floods, there is no closed condition on road C. Hence, (b).

233. Answer: (b)

Explanation:

From the statements:

- None but students are the members of the club

– This implies that every member of the club must be a student.

- Some members of the club are married

– This indicates at least some members (who are also students) are married.

- All married persons are invited for dance

– Anyone who is married is invited, regardless of whether they are a student or not.

Because all married persons are invited, it follows that any married student (being a subset of married persons) is necessarily invited for dance. Therefore:

Conclusion (b): "All married students are invited for dance" can be drawn.

234. Solution: (c)

As it is given, A is accepted by W, Z and X.

B accepted by Y, W and Z.

C accepted by W, Y, X and Z. D accepted by W and X.

So, C can be accepted by all the four parties.

235. Solution: b)

You can draw a Venn diagram or solve logically. But in this case you can just approach logically verifying each statement option.

Take (a): Its wrong because only X-brand cars parked here are white.

Take (b): All X-brand cars are white and some have radial tyres which means its correct. Take (c): Its an out of the league statement. J

Take (d): An illogical and out of the context statement.

236. Solution: (d)

You need to pick up just the pattern of triangle and you have the answer. See the triangle is moving corners anti-clockwise and turning upside down successively. So in the answer figure it has to be at the left top and vertical up. Only one option (d) satisfies this condition.

Learn saving time. You need not solve the full question to arrive at the solution.

237. Solution: (b)

If she scores 74 marks, we can assume that this will be her best performance if he scored 74 marks in a subject which has least highest marks and the least higher range of marks. Test II satisfies this best.

Directions for the following 5 (five) items:

Examine the information given in the following paragraph and Solution: the items that follow:

Guest lectures on five subjects viz., Economics, History, Statistics, English and Mathematics have to be arranged in a week from Monday to Friday. Only one lecture can be arranged on each day. Economics cannot be scheduled on Tuesday. Guest faculty for History is available only on Tuesday. Mathematics lecture has to be schedules immediately after the day of Economics lecture. English lecture has to be scheduled immediately before the day of Economics lecture.

238. Solution: (d)

Simply write the days and cross the ones that are not applicable. You will get the answer. Since Tuesday's schedule is fixed – for English to come immediately before economics, and economics to come immediately before maths, this will be the arrangement.

Monday - Statistics Tuesday – E, H Wednesday – English Thursday – Economics Friday –Mathematics

239. Solution: (b)

As seen from the table we drew.

240. Solution: (c)

As from the table.

241. Solution: (c)

Straight from the table.

242. Solution: a)

This is clear from the question itself.

243. Solution: (d)

It is clear that the doctor is at the centre and the MLA stays above the Engineer.

It is also given that IAS officer is professor's friend. So, IAS officer must stay above Engineer and MLA. Only the last option is satisfying the above conditions.

You can also check it by drawing a diagram. S1: IAS>....>Prof

S2: Doc is in the middle S3: IAS>..MLA>..>Engineer

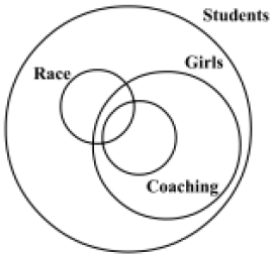
Now only way in which these can be satisfied is:

IAS>MLA>Doctor> Engineer> Professor

244. Solution: (c)

Option C can be easily seen from statement 1.

Others are not correct as if you can draw the Venn diagram. Let students be a bigger group out of which people participate in race. Out of students let Girls be a sub-group; and let these two groups intersect as given in the statement. Now draw the coaching group; it will look something like this.



245. Solution: (c)

This question is exactly similar to the one above. Only the wordings have been changed. Apply the exact same technique.

246. Solution: (c)

The base in the pyramid denotes the number of children and the top denotes number of senior citizens. Now clearly if the top is wider than the bottom, it means fewer births are taking place and more deaths are taking place. So population will reduce.

247. Solution: (c)

Exponential curve is the one where the slope of the curve increases exponentially. The equation is y= ex

Curve A is not exponential as it shows a slight decrease. Curve B is also not exponential (it is linear)

Curve D is not exponential as it remains constant for a period of almost upto 1000 years and then shows a sudden increase.

Curve C is exponential as it increases continuously and also slowly in the early stage and at a fast pace in later stages.

248. Solution: (b)

If you add all the percentages of the pie chart for both the towns, town A = 100%, but town B = 121%. Only if a person has more than one disease, can he be counted twice, which will increase the percentage.

249. Solution: (d)

Acceleration in a velocity time graph can be seen from the graph's slope (tilt with respect to horizontal). B's slope is more than A, which means for the same time period, it is changing velocity faster than that of A. Hence more acceleration.

Option (b) and (c) can be verified from y-axis.

To calculate distance in a velocity-time graph, simply take the area under the curve. It is more for B than A.

250. Solution: (b)

As per the graph, A has taken just below 30 min to complete the race whereas B has taken more than 35 min and C did not finish the race. So statement 1 is correct.

B has taken less time than A upto 25 km. Hence B was ahead of A upto 25 km. So statement 2 is also correct.

You can observe that C has taken less time than A and B upto 25 km. So, statement 3 is wrong.

251. Solution: (c)

6, second column = 2 multiplied by 3; 24 = 6 X 4 and 80 = 2 X 40

Use the same pattern. So, 36 should be multiplied by 4 to get 9 (which is also 3 times 3).

252. Solution: (c)

Technique 1:

You can check all the routes manually and count the total number.

Technique 2:

Avoid technique 1 if the diagram has more boxes. Use logic.

You can more from A to B in the given manner from point A node in two ways (right or downward). In each way there are 3 ways to go to B. Hence total 3+3 = 6.

Technique 3:

Use permutation & combination.

Consider a p x q rectangular grid with top left corner A and bottom right corner B. The number of distinct paths available to traverse from A to B moving downward and rightward is p + qCp or p + qCq.

We have a 2 X 2 rectangular grid in this case. So, the number of paths to traverse from A to B

= 4C2 = 6.

253. Solution: (c)

The triangles given are equilateral triangles of different lengths 1,2,3 and 4 units. The number of triangles of 1 unit length = 1 + 3 + 5 + 3 = 12

The number of triangles of length 2 units = 1 + 2 + 3 = 6. Plus there is one inverted triangle of length 2 units. So the number of triangles of length 2 units = 7

The number of triangles of length 3 units = 1 + 2 = 3 The number of triangles of length 4 units = 1

So, the total number of triangles = 12 + 7 + 3 + 1 = 23

254. Solution: (a)

Incubation means where the bacteria just started multiplying after insertion in the body. Thus A.

255. Solution: (b)

Clear from the diagram.

256. Solution: (c)

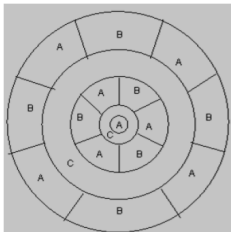
Relief is when the number of bacteria starts going down.

257. Solution: (a)

Simple arithmetic. For every one route, you will have six routes to take. So, total number of routes is 6 X 4 = 24

258. Solution: (a)

You can paint the figure using the colours A, B and C like this.



259. Solution: (b)

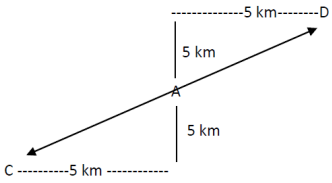
Area 15 units can be 15 X 1 or 5 X 3.

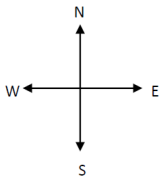
Area 48 units can be 48 X 1 or 24 X 2 or 16 X 3 or 12 X 4 or 8 X 6

Since the bigger figure is a square choose a combination with the length and breadth that forms a square: So if you take 5 units (from 15 units area) + 6 units (from 48 units area) or 8 units (from 48 units area) + 3 units (from 15 units area), it is equal to a side 11 square.

260. Solution: (c)

From the arrangement, it is clearly visible that





The distance between C and D, is twice the distance between C and A.

Between C and A, we use Pythagoras theorem i.e. square root of (52 + 52) = $\left(\sqrt{25}+25\right)=\sqrt{25}$

So, distance between C-D will $2\sqrt{25}$ = 14.12 km approx.

Read the following paesage and answer (three) items that follow:

A, B, C, D and E are members of the same family. There are two fathers, two sons, two wives, three males and two females. The teacher was the, wife of a lawyer who was the son of a doctor. E is not male, neither also a wife of a professional. C is the youngest person in the family and D is the eldest. B is a male.

261. Solution: a)

Clearly the family has three generations. We are putting the male first and then the female in the couple. Now, there can be 2 possibilities.

D – E (Doctor)	D (Doctor)
|	|
(Lawyer) B – A (teacher)	(Lawyer) B – A (teacher)
|	|
C	C – E

From the first arrangement, D is E's husband.

262. Solution: (c)

Since X is 14th from the front, Y is 17th from the end, and also X is ahead of Y, there must be 48 – (14 + 17) = 17 persons in between X and Y.

Now, as there are equal number of people between X and Z; and Z and Y. So, the number of persons between X and Z has to be 8.

It will be like – 13 ---X ----(8)----Y----(8)-----Z

4 Decision-Making and Problem-Solving

1. Which among the following actions would constitute the most appropriate directive(s) in resolving interpersonal conflict in an office with culturally diverse personnel? (CSAT-2026)

1. Direct personnel to practise activities that are the cultural markers of diverse groups
2. Allow conflicts to resolve naturally over time to set an appropriate precedent of leadership
3. Encourage personnel to seek each other's perspectives

Select the answer using the code given below.

(a) 1 and 3 (b) 2 only

(c) 3 only (d) 2 and 3

2. You are required to design a 'questionnaire' to be filled on-location by visitors, based on the following objective while writing a report: (CSAT-2026)

"To determine the feasibility of setting up a family-oriented vacation resort in the vicinity of a lake destination in the mountains"

Which of the following heads would you include in the questionnaire to make it most appropriate for your purpose?

1. Size of family
2. Budget
3. Number of earners in the family
4. Food allergies and dietary restrictions

Select the answer using the code given below.

(a) 1 and 3 (b) 1, 2 and 4

(c) 1 and 2 only (d) 2 and 3

3. Consider the following statements: (CSAT-2025)

I. If $A \leq B > C < D > E > F \geq G = H$; then B is always greater than E.

II. If $P > Q = R \geq S = T \leq U = V > W$; then S is always less than V.

Which of the statements given above is/are correct?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

4. With reference to the above table, the following assumptions have been made:

1. Nowadays, the prosperity of an already high-performing state cannot be sustained without making further large investments in its telecom infrastructure.
2. Nowadays, a very high Tele-density is the most essential condition for promoting business and economic growth in a state.

Which of the above assumptions is/are valid? (CSAT-2018)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Directions for the following 6 (six) items:

Given below are six items. Each item describes a situation and is followed by four possible responses. Indicate the response you find most appropriate. Choose only one response for each item. The responses will be evaluated based on the level of appropriateness for the given situation.

Please attempt all the items. There is no penalty for, wrong answers for these six items.

5. You are the head of your office. There are certain houses reserved for the allotment to the office staff and you have been given the discretion to do so. A set of rules for the allotment of the houses has been laid down by you and has been made public. Your personal secretary, who is very close to you, comes to you and pleads that as his father is seriously ill, he should be given priority in allotment of a house. The office secretariat that examined the request as per the rules turns down the request and recommends the procedure to be followed according to the rules. You do not want to annoy your personal secretary. In such circumstances, what would you do ? (CSAT-2013)

(a) Call him over to your room and personally explain why the allotment cannot be done.

(b) Allot the house to him to win his loyalty.

(c) Agree with the office note to show that you are not biased and that you do not indulge in favouritism.

(d) Keep the file with you and not pass any orders.

6. While travelling in a Delhi-registered commercial taxi from Delhi to an adjacent city (another State), your taxi driver informs you that as he has no permit for running the taxi in that city, he will stop at its Transport Office and pay the prescribed fee of Rs. forty for a day. While paying the fee at the counter you find that the transport clerk is taking an extra fifty rupees for which no receipt is being given. You are in a hurry for your meeting. In such circumstances, what would you do? (CSAT-2013)

(a) Go up to the counter and ask the clerk to give back the money which he has illegally taken.

(b) Do not interfere at all as this is a matter between the taxi driver and the tax authorities.

(c) Take note of the incident and subsequently report the matter to the concerned authorities.

(d) Treat it as a normal affair and simply forget about it.

7. A person lives in a far off village which is almost two hours by bus. The villager's neighbour is a very powerful landlord who is trying to occupy the poor villager's land by force. You are the District Magistrate and busy in a meeting called by a local Minister. The villager has come all the way, by bus and on foot, to see you and give an application seeking protection from the powerful landlord. The villager keeps on waiting outside the meeting hall for an hour.

You come out of the meeting and are rushing to another meeting. The villager follows you to submit his application. What would you do? (CSAT-2013)

(a) Tell him to wait for another two hours till you come back from your next meeting.

(b) Tell him that the matter is actually to be dealt by a junior officer and that he should give the application to him.

(c) Call one of your senior subordinate officers and ask him to solve the villager's problem.

(d) Quickly take the application from him, ask him a few relevant questions regarding his problem and then proceed to the meeting.

8. **There is a shortage of sugar in your District where you are the District Magistrate. The Government has ordered that only a maximum amount of 30 kg sugar is to be released for wedding celebrations. A son of your close friend is getting married and your friend requests you to release at least 50 kg sugar for his son's wedding. He expresses annoyance when you tell him about the Government's restrictions on this matter. He feels that since you are the District Magistrate you can release any amount. You do not want to spoil your friendship with him. In such circumstances, how would you deal with the situation? (CSAT-2013)**

(a) Release the extra amount of sugar which your friend has requested for

(b) Refuse your friend the extra amount and strictly follow the rules.

(c) Show your friend the copy of the Government instructions and then persuade him to accept the lower amount as prescribed in the rules.

(d) Advise him to directly apply to the allotting authority and inform him that you do not interfere in this matter.

9. **You are in-charge of implementing the Family Planning programme in an area where there is a strong opposition to the present policy. You want to convince the residents of the need for keeping small families. What would be the best way of communicating this message? (CSAT-2013)**

(a) By logically explaining to the residents the need for family planning to improve the health and living standards.

(b) By encouraging late marriages and proper spacing of children.

(c) By offering incentives for adopting family planning devices.

(d) By asking people who have been sterilized or are using contraceptives to directly talk to the residents.

10. **You are a teacher in a University and are setting a question paper on a particular subject. One of your colleagues, whose son is preparing for the examination on that subject, comes to you and informs you that it is his son's last chance to pass that examination and whether you could help him by indicating what questions are going to be in the examination. In the past, your colleague had helped you in another matter. Your colleague informs you that his son will suffer from depression if he fails in this examination. In such circumstances, what would you do? (CSAT-2013)**

(a) In view of the help he had given you, extend your help to him.

(b) Regret that you cannot be of any help to him.

(c) Explain to your colleague that this would be violating the trust of the University authorities and you are not in a position to help him.

(d) Report the conduct of your colleague to the higher authorities

11. **You have differences of the opinion regarding the final report prepared by your subordinate that is to be submitted urgently. The subordinate is justifying the information given in the report. You would (CSAT-2012)**

(a) Convince the subordinate that he is wrong

(b) Tell him to reconsider the results

(c) Revise the report on your own

(d) Tell him not to justify the mistake

12. **You are competing with your batch-mate for the prestigious award to be decided based on an oral presentation. You have been asked by the committee to finish on time. Your friend however, is allowed more than the stipulated time period. (CSAT-2012)**

(a) Lodge a complaint to chairperson against the discrimination.

(b) Not listen to any justification from the committee

(c) Ask for withdrawal of your name

(d) Protest and leave the place

13. **You are handling a time bound project. During the project review meeting, you find that project is likely to get delayed due to lack of cooperation of the team members. You would, (CSAT-2012)**

(a) Warn the team members for their non-cooperation

(b) Look into the reasons for non-cooperation

(c) Ask for the replacement of the team members

(d) Ask for the extension of time citing reasons

14. **You are a chairperson of a state sports committee. You have received a complaint and later it was found that an athlete in junior age category who has won a medal has crossed the age criteria by 5 days. You would... (CSAT-2012)**

(a) Ask the screening committee for a clarification

(b) Ask the athlete to return the medal

(c) Ask the athlete to get an affidavit from the court declaring his/her age.

(d) Ask the members of the committee for their views

15. **You are handling a priority project and have been meeting all the deadlines and planning your leave during the project. Your immediate boss does not grant leave citing the urgency of the project. You would... (CSAT-2012)**

(a) Proceed on leave without waiting for the sanction

(b) Pretend to be sick and take leave

(c) Approach higher authority to reconsider the leave application.

(d) Tell the boss that it is not justified.

16. You are involved in setting up a water supply project in remote area. Full recovery of cost is impossible in any case. The income levels in the area are low and 25% of the population is below poverty line (BPL). When a decision has to be taken on pricing you would.... (CSAT-2012)

(a) Recommended that the supply of water be free of charge in all respects

(b) Recommended that the users pay a onetime fixed sum for installation of taps and the usage of water be free

(c) Recommended that a fixed monthly charges be levied on the non-BPL families and for BPL families water should be free

(d) Recommended that the users pay a charge based on the consumption of water with differentiated charges for Non-BPL and BPL families.

17. As a citizen you have some work with a government department. The official calls you again and again; and without directly asking you, sends out feelers for a bribe. You want to get your work done. You would... (CSAT-2012)

(a) Give a bribe

(b) Behave as if you have not understood the feelers and persist with your application

(c) Go to the higher officer for help verbally complaining about feelers.

(d) Send in a formal complaint

18. You have been asked to give an explanation for not attending an important official meeting. Your immediate boss who has not informed you about the meeting is now putting pressure on you not to place an allegation against him / her. You would (CSAT-2011)

(a) send a written reply explaining the fact.

(b) seek an appointment with the top boss to explain the situation.

(c) admit your fault to save the situation.

(d) put the responsibility on the coordinator of the meeting for not informing.

19. A local thug (bad element) has started illegal construction on your vacant plot. He has refused your request to vacate and threatened you of dire consequences in case you do not sell the property at a cheap price to him. You would (CSAT-2011)

(a) sell the property at a cheap price to him.

(b) go to the police for necessary action.

(c) ask for help from your neighbours.

(d) negotiate with the goon to get a higher price.

20. You have to accomplish a very important task for your headquarters within the next two days. Suddenly you meet with an accident. Your office insists that you complete the task. You would (CSAT-2011)

(a) ask for an extension of deadline.

(b) inform Headquarters of your inability to finish on time.

(c) Suggest alternate person to headquarters who may do the needful.

(d) stay away till you recover.

21. You are an officer-in-charge for providing basic medical facilities to the survivors of an earthquake affected area. Despite your best possible effort, people put allegations against you for making money out of the funds given for relief. You would (CSAT-2011)

(a) let an enquiry be set up to look into the matter.

(b) ask your senior to appoint some other person in your place.

(c) not pay attention to allegations.

(d) stop undertaking any initiative till the matter is resolved.

22. You have been made responsible to hire boats at a short notice to be used for an area under flood. On seeing the price mentioned by the boat owners you found that the lowest price was approximately three times more than the approved rate of the Government. You would (CSAT-2011)

(a) reject the proposal and call for a fresh price.

(b) accept the lowest price.

(c) refer the matter to the Government and wait.

(d) threaten the boat owners about a possible cancellation of the licence.

23. You are the officer-in-charge of a village administering distribution of vaccine in an isolated epidemic hit village, and you are left with only one vaccine. There is a requirement of that vaccine from the Gram Pradhan and also a poor villager. You are being pressurised by the Gram Pradhan to issue the vaccine to him. You would (CSAT-2011)

(a) initiate the procedure to expedite the next supply without issuing the vaccine to either.

(b) arrange vaccine for the poor villager from the distributor of another area.

(c) ask both to approach a doctor and get an input about the urgency.

(d) arrange vaccine for the Gram Pradhan from the distributor of another area.

24. You have taken up a project to create night-shelters for homeless people during the winter season. Within a week of establishing the shelters, you have received complaints from the residents of the area about the increase in theft cases with a demand to remove the shelters. You would (CSAT-2011)

(a) ask them to lodge a written complaint in the police station.

(b) assure residents of an enquiry into the matter.

(c) ask residents to consider the humanitarian effort made.

(d) continue with the project and ignore their complaint.

25. You, as an administrative authority, have been approached, by the daughter-in-law of an influential person regarding harassment by her in-laws on account of insufficient dowry. Her parents are not able to approach you because of social pressures. You would (CSAT-2011)

(a) call the in-laws for an explanation.

(b) counsel the lady to adjust, given such a circumstance.

(c) take action after her parents approach you.

(d) ask her to lodge complaint with the police.

4 Decision-Making and Problem-Solving_ Explanations

1. Answer: (c)

Explanation:

To resolve interpersonal conflicts in a culturally diverse workplace, a leader must adopt proactive and empathetic strategies.

Statement 1 is not correct: Directing personnel to practice activities that are cultural markers of diverse groups is an artificial and forced approach. It does not address the root cause of any conflict and can even lead to resentment or be perceived as cultural appropriation. True cultural harmony comes from understanding, not forced participation in cultural practices.

Statement 2 is not correct: Allowing conflicts to "resolve naturally" is a laissez-faire approach that often backfires. In a diverse workplace, unaddressed conflicts can escalate, breed toxicity, and negatively impact team morale and productivity. Effective leadership requires timely and constructive intervention, not avoidance.

Statement 3 is correct: Encouraging personnel to seek each other's perspectives is a cornerstone of effective conflict resolution. It fosters empathy, active listening, and mutual respect. By understanding different cultural viewpoints and personal perspectives, employees can bridge communication gaps and resolve misunderstandings collaboratively.

Therefore, only Statement 3 constitutes an appropriate directive.

2. Answer: (c)

Explanation:

To determine the correct heads for the questionnaire, we must align them with the objective: determining the feasibility of setting up a family-oriented vacation resort.

1. **Size of family:** Highly relevant. Since the proposed resort is "family-oriented", knowing the average family size of the visitors is essential for planning room layouts, amenities, and recreational activities. This directly impacts the feasibility and design of the project.
2. **Budget:** Highly relevant. Understanding the visitors' vacation budget is crucial for determining the financial viability of the resort, deciding the pricing strategy, and identifying whether the target demographic can sustain the business.
3. **Number of earners in the family:** Irrelevant. The actual vacation budget (Head 2) is a direct and sufficient indicator of a family's spending capacity. Asking for the number of earners is unnecessarily intrusive and does not provide direct value for a feasibility study.
4. **Food allergies and dietary restrictions:** Irrelevant at this stage. This is micro-level operational data collected from guests when they actually book a stay or dine at the restaurant. It does not help in determining the macro-level feasibility of setting up the resort itself.

Therefore, only Size of family and Budget are appropriate for the feasibility questionnaire. Hence, the correct option is 1 and 2 only.

3. Answer: (d)

Explanation: The exact relation between B & E can not be determined and S & V can be equal also so S is not always less than V

4. Answer: (d)

The correct option is (d)

Both statements tries to connect GDP growth with growth in telecom sector which is not correct.

Directions for the following 6 (six) items:

Given below are six items. Each item describes a situation and is followed by four possible responses. Indicate the response you find most appropriate. Choose only one response for each item. The responses will be evaluated based on the level of appropriateness for the given situation.

Please attempt all the items. There is no penalty for, wrong answers for these six items.

5. Solution: More preferred (a), less preferred (c)

Option (b) would be abdicating duty, responsibility and decision-making. It gives a very bad signal about your administrative and decision-making acumen.

Option (a) is favouritism and going against the rules. Better avoid this. UPSC traps with such questions of loyalty, nepotism and favouritism.

Options (a) and (c) can be the answer. While (c) is correct, it would be a purely bureaucratic approach to handling situations. We recommend and prefer (a) because you will be able to balance out all things – decision-making; personal relations; organizational health; rules and procedures; integrity and impartiality etc.

6. Solution: More preferable (a); less preferable (c)

Even though you are in hurry, you can definitely ask back the money from the clerk and then later go and report the authorities. (c) is less preferable not because of the principle involved, but because (a) is more practical in that situation.

Choosing (a) and (c), you are acting against corruption. This shows uprightness; inclination towards justice; dutiful citizenship and zero tolerance for corruption.

Other options show carelessness; tolerance for corruption and apathy. You will have to balance ethics with pragmatism. Thats all.

7. Solution: More preferable (c); less preferable (d)

While all the options seem reasonable; you have to choose the one that reflects the qualities of a citizen-centric civil servant.

Option (a) is not appropriate; because waiting for 2 hours again for him will not be possible. He is from a far off village and has already waited enough. This would give a wrong signal about the administration and put the villager's morale down.

Option (b) will not be appropriate because a junior officer may not be placed well enough to handle the powerful landlord's issue. Moreover,

he has come so far to see you and speak to you about his problem. You should not do file and application pushing with him.

Option (c) is more appropriate because a senior subordinate officer will be better placed to deal with his problem. Moreover, you are personally calling the officer and giving him the application. This will show the villager that you are taking the matter seriously.

Option (d) is less preferable because the villager will not have time to discuss his problem. And it may give him an impression that you are not taking the matter seriously. Still it is better than (a) and (b).

8. Solution: More preferable (c); less preferable (b) Option (a) is nepotism and corruption straightaway.

Option (d) will be abdication of duty and shirking responsibility to save your image and friendship.

Option (c) is preferable because you will be following rules; avoiding favouritism; and yet be able to save the friendship.

Option (b) is less preferable as denying the quota rudely may cost your friendship. This question checks integrity, uprightness and emotional intelligence together.

9. Solution: More preferable (a); less preferable (c)

Option (b) might not go well with the traditional villagers; hence less pragmatic.

Option (d) might also not go well with the traditional villagers because they will not like to talk about rural taboos like contraception and sterilization.

Options (a) and (c) go with the cultural context.

This question checks your cultural sensitivity and emotional intelligence.

10. Solution: More preferable (c); less preferable (b)

Option (a) is out of question as it is a violation of conduct rules. Option (d) will be extreme and unnecessary that too with a colleague who had helped you. It will only spoil your personal relation with him.

Option (b) is straightforward, but (c) is better. Explaining a colleague is always better than outright denial.

The qualities that are being checked are uprightness; integrity; honesty and emotionally intelligence.

11. Solution: More preferred (c); less preferred (a)

The question tests your organizational, managerial and administrative skills alongwith emotional intelligence.

Since the report is to be submitted urgently, you don't have time for the results to be reconsidered or to keep arguing with the subordinate. So, best way out is (c), if not then (c).

B or D will not lead to any results.

12. Solution: More preferred (a); less preferred (d).

This question checks capacity and inclination for justice, and withstanding odd situations. Option (b) is a knee-jerk reaction and shows your emotional intelligence in poor light. Option (c) shows a lack of interest in fighting for justice.

Going through option (a), you are going by the official machinery of grievance redressal; fighting for justice as well withstanding the odds that will come in the way.

Option (d) shows the inclination towards fighting for justice, but it is not result-oriented. Leaving the place will not help in any way.

13. Solution: More preferred (b); less preferred (a)

The question too tests your organizational, managerial and administrative skills along with emotional intelligence.

Asking for replacement or extension without looking into reasons or talking to team members will be a wrong decision. Moreover, it is a time bound project. So, options (c) and (d)are out.

(a) is preferable because you will at least get the work done.

(b) is even more preferable because you will get to the core of the issue, and then ensure that the work is completed on time.

14. Solution: More preferable (a); less preferable (d).

This question checks your administrative acumen and sense of balance in decision-making.

Since the investigation in the case is over, asking for clarification from the committee that screened the athlete is the best way out. You will get to know if it was a procedural error or operational error or an unintended mistake. Then you can take action. Best is (a).

Or you can even ask the committee for views. This is a more democratic approach since you are heading the committee.

Other options are a little bit on the extreme side and not preferable in this case. The reputation of the athlete is also at stake.

Take a balanced, pragmatic approach.

15. Solution: More preferable (d); less preferable (c)

This question checks for balance in decision-making and handling organizational pulls and pressures on your personal life.

(a) and (b) are not right for obvious reasons like dishonesty; disobeying official orders and protocols; and irresponsible behaviour.

(c) can get you the leave, within the official channels, but your relationship with immediate boss will be spoiled. He will not like a higher authority overruling him on a matter of leave for you, especially when its an urgent project.

(d) is a more nuanced approach; because its an urgent project (and your leave is not all that urgent and important).

16. Solution: (d) more preferable; (c) less preferable.

This question checks your administrative acumen; sense of balance in decision-making and empathy for the marginalised.

Option (a) is implausible/impractical.

Option (b) - fixed installation charges will be difficult for BPL to pay. And later making it free for everyone is impractical.

C or D is a more humane and sympathetic approach because BPL are getting either water free or at a differentiated (lower cost) while others are paying to ensure economic viability.

D is preferable because otherwise there will be misuse of BPL water.

17. Solution: More preferable (c); less preferable (b)

This checks tolerance for corruption; uprightness; and integrity.

Sending in a formal complaint will be a wrong step as there is no evidence that the official has asked you for a bribe.

Giving a bribe will ofcourse be wrong.

Behaving as if you have not understood will be a less preferable option because the official has not done it just once. He has been doing it again and again.

So going to the higher officer for help will be the best approach. Because this way not only your case will be quickly resolved, that other official may stop asking future bribes from others in fear of action from higer-ups.

18. Solution: More preferred (a); less preferred (b).

Admitting your fault would be succumbing to unjust pressure. It is not expected of a civil servant. Putting responsibility on someone else would be unfair, wrong and irresponsible.

The other two are in line with upholding the truth and being sincere with your official responsibilities.

19. Solution: More preferred (b); less preferred (d)

Selling the property at cheap price would be succumbing to pressure. It shows lack of capacity to fight for justice.

Asking for help from neighbours would not be a good idea as he is a local thug. So the neighbours may not support you.

Negotiating for a higher price (though less preferable) will ensure that you don't get a bad deal. Willing to go to police shows that you can fight against injustice. This is a trait civil servants should have. UPSC asks such questions to test the same.

20. Solution: More preferred (c); less preferred (b).

Asking for an extension is impractical as it is a very important task that has to be submitted within two days. If you inform the headquarters of your inability, they will at least assign another person. The work will not be affected.

Staying away till you recover would be abdicating official responsibilities and duty; and reflects badly on your professional values and commitment to work.

Best is (c) as it shows your professionalism towards work keeping in mind your own personal limitations.

21. Solution: More preferred (a); less preferred (c)

Asking for someone else to be appointed would be succumbing to public pressure, which is like a daily affair in the bureaucracy.

Stopping any further initiative will affect the relief operations and is not advisable. Public interest should come first.

Letting an inquiry in the matter is a sign of highest integrity and is expected of civil servants. Integrity simple means subjecting oneself to the greatest scrutiny possible.

22. Solution: More preferred (b); less preferred (d).

Rejecting the proposal and calling for fresh prices is a long drawn process. You do not have time as its urgent for floods.

Same problem is there with referring to the government and waiting.

Since its very urgent best would be to accept the lowers price for now since its required very urgently for relief operations. You may be dragged into controversies later, but public interest should be put first. Service should be selfless.

Threatening the boat owners might take some time, and given the urgency may not be that practical.

23. Solution: More preferred (c); less preferred (b)

Best would be to get who needs it more urgently and give it to him. But, as its an epidemic hit village, you would need the pradhan to help you in the relief efforts. So instead of saving the poor villager (even out of sympathy and duty), you should keep the lives of others in mind too.

Rejecting vaccine to either would be foolish and inhuman.

This question checks your pragmatism and empathy towards the downtrodden.

24. Solution: More preferred (b); less preferred (c)

A complaint in the police station may lead to the shelters being removed by police. Its the winter season. It will result in undue harassment for the poor shelterless people.

Continuing with the project ignoring the complaint would be abdicating duty and is not a sign of integrity, fairness and impartiality.

Asking them to consider the humanitarian efforts made would at least save the homeless people's lives in the winter. Then you can monitor the situation and take appropriate action.

25. Solution: More preferred (a); less preferred (d)

(d) is less preferred because the girl might not go to police because of social pressures as mentioned in the question.

Since you are an administrative authority, you cannot handle dowry cases. Best would be to assist the lady by calling the in-laws for explanation.

This would improve the faith of people in the administration; it shows capacity for justice; and sensitivity towards important social issues.

5 General Mental Ability

1. For $1/3 < x < y < 2$, which of the following statements is/are always correct? (CSAT-2026)

I. $x + 1/x < y + 1/y$

II. $\sqrt{(1+y^2)} / y < \sqrt{(1+x^2)} / x$

Select the answer using the code given below.

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

2. What is the minimum number of times one needs to measure to get 298 litres of water from a tank, if the measuring cylinders have capacities 1 litre, 6 litres, 25 litres and 100 litres? (CSAT-2026)

(a) 4 (b) 5

(c) 9 (d) 13

3. Three variables x, y and z take values 2, 3, 4 or 5 such that their values are always distinct. If M and N denote the largest possible value and the smallest possible value, respectively, for the expression $\{(x \times y) + z\}$; then M − N is (CSAT-2026)

(a) 11 (b) 12

(c) 13 (d) 14

4. In a sequence of numbers, each number other than the first two is the sum of the two immediately preceding numbers from it. If the first two numbers in the sequence are 4 and 7, then the sixth number is (CSAT-2026)

(a) 29 (b) 37

(c) 43 (d) 47

5. A shopkeeper employs a delivery boy and gives him a motorcycle for home delivery. For every delivery, the boy is given ₹ 5. At the end of the day, he also gets ₹ 2 for every kilometre of the distance covered in the day. The boy wants to earn more than ₹ 500 a day, but does not want to travel more than 100 km. Which of the following numbers of deliveries would definitely meet his target? (CSAT-2026)

(a) 80

(b) 85

(c) 90

(d) The question cannot be answered due to insufficient data

6. The weight of X, in kg, is denoted by X. The weights of A, B, C, D, P, Q, R and S are measured. Given: (CSAT-2026)

$A + B + C + D = 17$

$A + C = 6$

$P + Q + S + D = 15$

$P + Q + R + B = 17$

$P = R$ and $Q = S$

Which one of the following statements is correct?

(a) B and D together weigh less than the total weight of P and Q.

(b) P and Q together weigh more than the total weight of A and C.

(c) P weighs more than Q.

(d) Q weighs more than P.

7. A 4-digit number N is such that when divided by 3, 5, 6, 9 leaves a remainder 1, 3, 4, 7 respectively. What is the smallest value of N? (CSAT-2025)

(a) 1068 (b) 1072

(c) 1078 (d) 1082

8. A mobile phone has been stolen. There are 3 suspects P, Q and R. They were questioned knowing that only one of them is guilty. Their responses are as follows: (CSAT-2025)

P: I did not steal. Q stole it.

Q: R did not steal. I did not steal.

R: I did not steal. I do not know who did it.

Who stole the mobile phone?

(a) P (b) Q

(c) R (d) Cannot be concluded

9. How many times the hour hand and the minute hand coincide in a clock between 10:00 a.m. and 2:00 p.m. (same day)? (CSAT-2024)

(a) 3 times (b) 4 times

(c) 5 times (d) 6 times

10. The calendar for the year 2025 is same for (CSAT-2024)

(a) 2029 (b) 2030

(c) 2031 (d) 2033

11. What is the angle between the minute hand and hour hand when the clock shows 4:25 hours? (CSAT-2024)

(a) 12.5° (b) 15°

(c) 17.5° (d) 20°

12. Consider the sequence (CSAT-2024)

A_BCD_BBCDABC_DABC_D

that follows a certain pattern. Which one of the following completes the sequence?

(a) B, A, D, C (b) B, A, C, D

(c) A, A, C, D (d) A, A, D, C

13. Consider the following: (CSAT-2024)

1. 1000 litres = 1 m^3
2. 1 metric ton = 1000 kg
3. 1 hectare = 10000 m^2

Which of the above are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

14. On January 1st, 2023, a person saved Rs. 1. On January 2nd, 2023, he saved Rs. 2 more than that on the previous day. On January 3rd, 2023, he saved Rs. 2 more than that on the previous day and so on. At the end of which date was his total savings a perfect square as well a perfect cube? (CSAT-2024)

(a) 7th January, 2023 (b) 8th January, 2023

(c) 9th January, 2023 (d) Not possible

15. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

Question:

What are the unique values of x and y, where x, y are distinct natural numbers?

Statement-I: x/y is odd.

Statement-II: xy = 12

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the Statements together

16. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

A certain amount was distributed among X, Y and Z.

Question:

Who received the least amount?

Statement-1:

X received 4/5 of what Y and Z together received.

Statement-II:

Y received 2/7 of what X and Z together received.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the Statements together

17. A person walks 100 m straight from his house, turns left and walks 100 m, again turns left and walks 300 m, then turns right and walks 100 m to reach his office. In which direction does he walk initially from his house if his office is exactly in the North-East direction? (CSAT-2024)

(a) North-West (b) West

(c) South (d) South-West

18. A person walks 100 m Westward, then turns left and walks 100 m. He then takes 225° turn clockwise. In which direction is he walking now? (CSAT-2024)

(a) South-West (b) South-East

(c) North-West (d) North-East

19. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

Question:

What is the time required to download the software?

Statement-I:

The size of the software is 12 megabytes.

Statement-ll:

The transfer rate is 2.4 kilobytes per second.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the Statements together

20. What is the number of fives used in numbering a 260-page book? (CSAT-2024)

(a) 55 (b) 56

(c) 57 (d) 60

21. If $7 \oplus 9 \oplus 10 = 8, 9 \oplus 11 \oplus 30 = 5, 11 \oplus 17 \oplus 21 = 13$, what is the value of $23 \oplus 4 \oplus 15$?

(CSAT-2023)

(a) 6 (b) 8

(c) 13 (d) 15

22. If today is Sunday, then which day is it exactly on 1010 th day? (CSAT-2023)

(a) Wednesday (b) Thursday

(c) Friday (d) Saturday

23. 40 children are standing in a circle and one of them (say child-1) has a ring. The ring is passed clockwise. Child-1 passes on to

child-2, child-2 passes on to child-4, child-4 passes on to child-7 and so on. After how many such changes (including child-1) will the ring be in the hands of child-1 again? (CSAT-2023)

(a) 14 (b) 15

(c) 16 (d) 17

24. Is p greater than q?

Statement-1: p x q is greater than zero.

Statement-2: p2 is greater than q2.

Which one of the following is correct in respect of the above Question and the Statements? (CSAT-2023)

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.

(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.

(d) The Question cannot be answered even by using both the Statements together.

25. Is (p + g- r) greater than (p - q + r), where p, q and r are integers?

Statement-1: (p - q) is positive.

Statement-2: (p - r) is negative.

Which one of the following is correct in respect of the above Question and the Statements? (CSAT-2023)

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.

(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.

(d) The Question cannot be answered even by using both the Statements together.

26. Consider a 3-digit number.

Question: What is the number?

Statement-1: The sum of the digits of the number is equal to the product of the digits.

Statement-2: The number is divisible by the sum of the digits of the number.

Which one of the following is correct in respect of the above Question and the Statements? (CSAT-2023)

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.

(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone.

(d) The Question cannot be answered even by using both the Statements together.

27. For five children with ages a < b < c < e; any two successive ages differ by 2 years.

What is the age of the youngest child?

Statement-1: The age of the eldest is 3 times the youngest.

Statement-2: The average age of the children is 8 years.

Which one of the following is correct in respect of the above Question and the Statements? (CSAT-2023)

(a) The Question can be answered by using one of the Statement alone.

(b) The Question can be answered by using either Statement alone.

(c) The Question can be answered by using both the Statement together, but cannot be answered using either Statement alone.

(d) The Question cannot be answered even by using both the Statements together.

28. There are large number of silver coins weighing 2gm, 5gm, 10gm, 25gm, 50gm each. Consider the following statements:

1. To buy 78 gm of coins one must buy at least 7 coins.
2. To weigh 78 gm using these coins one can use less than 7 coins.

Which of the statements given above is/are correct? (CSAT-2023)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

29. Consider the sequence ABC_ _ABC_DABBCD_ABCD that follows a certain pattern.

Which of the following completes the sequence? (CSAT-2023)

(a) DACB (b) CDAB

(c) DCCA (d) DDCA

30. AB and CD are 2-digit numbers. Muliplying AB with CD results in a 3-digit number DEF. Adding DEF to another 3-digit number GHI results in 975. Further A, B, C, D, E, F, G, H, I are distinct digits. If E= 0, F=8, then what is A+B+C equal to? (CSAT-2023)

(a) 6 (b) 7

(c) 8 (d) 9

31. A cuboid of dimensions 7cm x 5¢cm x 3cm is painted red, green and blue colour on each pair of opposite faces of dimensions 7cm x 5cm x Sem, S5cm x 3cm, 7em x 3cm respectively. Then the cuboid is cut and separated into various cubes each of side length 1cm. Which of the following statements is/are correct?

1. There are exactly 15 small cubes with no paint on any face.
2. There are exactly 6 small cubes with exactly two faces, one painted with blue and the other with green.

Select the correct answer using the code given below: (CSAT-2023)

(a) 1 Only (b) 2 Only

(c) Both 1and 2 (d) Neither 1 nor 2

32. **The letters A, B, C, D and E are arranged in such a way that there are exactly two letters between A and E. How many such arrangements are possible? (CSAT-2022)**

(a) 12 (b) 18

(c) 24 (d) 36

33. **A, B and C are three places such that there are three different roads from A to B, four different roads from B to C and three different roads from A to C. In how many different ways can one travel from A to C using these roads? (CSAT-2022)**

(a) 10 (c) 15

(b) 13 (d) 36

34. **In a tournament of chess having 150 entrants, a player is eliminated whenever he loses a match. It is given that no match results in a tie/ draw. How many matches are played in the entire tournament? (CSAT-2022)**

(a) 151 (b) 150

(c) 149 (d) 148

35. **Consider the Question and two Statements given below: (CSAT-2022)**

Question: Is x an integer?

Statement-1: x/3 is not an integer.

Statement-2: 3x is an integer.

Which one of the following is correct in respect of the Question and the Statements?

(a) Statement-1 alone is sufficient to Answer the Question.

(b) Statement-2 alone is sufficient to Answer the Question.

(c) Both Statement-1 and Statement-2 are sufficient to Answer the Question.

(d) Both Statement-1 and Statement-2 are not sufficient to Answer the Question.

36. **In the series AABABCABCDABCDE., which letter appears at the 100th place? (CSAT-2022)**

(a) G (b) H

(c) I (d) J

37. **A is 16th from the left end in a row of boys and V is 18th from the right end. G is 11th from A towards the right and 3rd from V towards the right end. How many boys are there in the row? (CSAT-2020)**

(a) 40

(b) 41

(c) 42

(d) Cannot be determined due to insufficient data

38. **Consider the following arrangement that has some missing letters: abab_b_bcb_dcdcded_d. The missing letters which complete the arrangement are (CSAT-2020)**

(a) a, b, c, d (b) a, b, d, e

(c) a, c, c, e (d) b, c, d, e

39. **In the particular year 12th January is a Sunday, then which one of the following is correct? (CSAT-2020)**

(a) 15th July is a Sunday if the year is a leap year.

(b) 15th July is a Sunday if the year is not a leap year.

(c) 12th July is Sunday if the year is a leap year.

(d) 12th July is a not Sunday if the year is a leap year.

40. **The letters from A to Z are numbered from 1 to 26 respectively. If GHI = 1578 and DEF = 912, then what is ABC equal to? (CSAT-2020)**

(a) 492 (b) 468

(c) 262 (d) 246

41. **What is the missing term @ in the following?**

ACPQ: BESU: MNGI: @ (CSAT-2020)

(a) NPJL (b) NOJM

(c) NPIL (d) NPJM

42. **Mr. 'X' has three children. The birthday of the first child falls on the 5 Monday of April that of the second one falls on the 5% Thursday of November. On which day is the birthday of his third child, which falls on 20th December? (CSAT-2019)**

(a) Monday (b) Thursday

(c) Saturday (d) Sunday

43. **If x is greater than or equal to 25 and y is less than or equal to 40, then which of the following is always correct? (CSAT-2019)**

(a) x is greater than y

(b) (y—x) is greater than 15

(c) (y=x) is less than or equal to 15

(d) (x +y) is greater than or equal to 65

44. **Ena was born 4 years after her parents' marriage. Her mother is three years younger than her father and 24 years older than Ena, who is 13 years old. At what age did Ena's father get married? (CSAT-2019)**

(a) 22 years (b) 23 years

(c) 24 years (d) 25 years

45. **A wall clock moves 10 minutes fast in every 24 hours, The clock was set right to show the correct time at 8:00 a.m on Monday. When the clock shows the time 6:00 p.m. on Wednesday, what is the correct time? (CSAT-2019)**

(a) 5:36 p.m (b) 5:30 p.m

(c) 5:24 p.m (d) 5:18 p.m

46. **'A' started from his house and walked 20 m towards East, where his friend B joined him. They together walked 10 m in the same direction. Then 'A' turned left while friend 'B' turned right and**

travelled 2 m and 8 m respectively. Again 'B' turned left to travel 4 m followed by 5 m to his right to reach his office. 'A' turned right and travelled 12 m to reach his office. What is the shortest distance between the two offices? (CSAT-2019)

(a) 15 m (b) 17 m

(c) 19 m (d) 20 m

47. Which year has the same calendar year as that of 2009? (CSAT-2019)

(a) 2018 (b) 2017

(c) 2016 (d) 2015

48. If every alternative letter of the English alphabet from B onwards (including B) is written in lower case (small letters) and the remaining letters are capitalized, then how is the first month of the second half of the year written? (CSAT-2019)

(a) JULY (b) jULy

(c) july (d) jUIY

49. Each face of a cube can be painted in black or white colour. In how many different ways can the cube be painted? (CSAT-2019)

(a) 9 (b) 10

(c) 11 (d) 12

50. If $ means 'divided by'; @ means 'multiplied by'; # means 'minus', then the value of 10#5@1$5 is (CSAT-2019)

(a) 0 (b) 1

(c) 2 (d) 9

51. If F stands second in the ranking, then the position of B is (CSAT-2019)

(a) Third (b) Fourth

(c) Fifth (d) Sixth

52. If B scores the least, the rank of C will be (CSAT-2019)

(a) Second (b) Third

(c) Fourth (d) Second or third

53. If E is ranked third, then which one of the following is correct? (CSAT-2019)

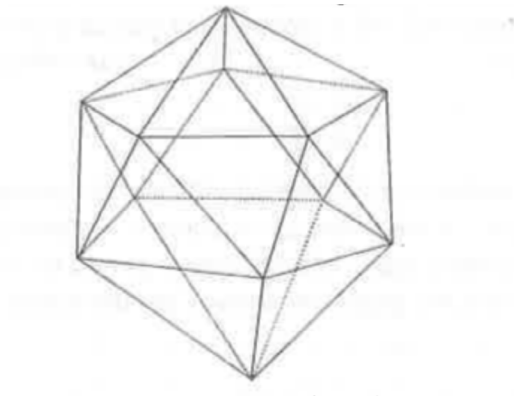
(a) E gets more marks than C

(b) C gets more marks than E

(c) A is ranked fourth

(d) D is ranked fifth

54. Consider the following three-dimensional figure:



How many triangles does the above figure have? (CSAT-2018)

(a) 18 (b) 20

(c) 22 (d) 24

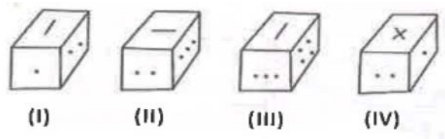
55. How many diagonals can be drawn by joining vertices of an octagon? (CSAT-2018)

(a) 20 (b) 24

(c) 28 (d) 64

Directions for the following 3 (three) items:

Rotated positions of a single solid are shown below. The various faces of the solid are marked with different symbols like dots, cross and line. Answer the three items that follow the given figures.



56. What is the symbol on the face opposite to that containing a single dot? (CSAT-2018)

(a) Four Dots (b) Three Dots

(c) Two Dots (d) Cross

57. What is the symbol on the face opposite to that containing two dots? (CSAT-2018)

(a) Single dot (b) Three dots

(c) Four dots (d) Line

58. What is the symbol on the face opposite to that containing the cross? (CSAT-2018)

(a) Single dot (b) Two dot

(c) Line (d) Four dots

59. A solid cube of 3 cm side, painted on all its faces, is cut up into small cubes of 1 cm side. How many of the small cubes of 1 cm side. How many of the small cubes will have exactly two painted faces? (CSAT-2018)

(a) 12 (b) 8

(c) 6 (d) 4

60. If Pen < Pencil, Pencil < Book and Book > Cap, then which of the following is always true? (CSAT-2018)

(a) Pen > Cap (b) Pen < Book

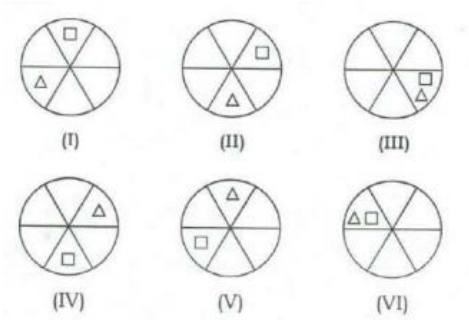
(c) Pencil = Cap (d) Pencil > Cap

61. There are 24 equally spaced points lying on the circumference of a circle. What is the maximum number of equilateral triangles that can be drawn by taking sets of three points as the vertices? (CSAT-2018)

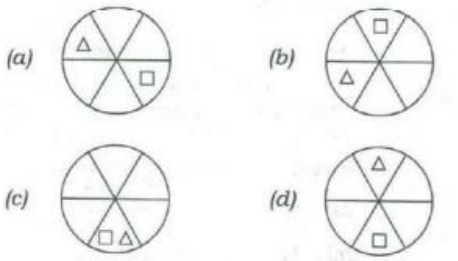
(a) 4 (b) 6

(c) 8 (d) 12

62. Consider the following figures:



In the figures (1) to (V1) above, some parts are shown to change their positions in regular directions. Following the same sequence, which of the figures given below will appear at (VIl) stage? (CSAT-2018)



63. A watch loses 2 minutes in every 24 while another watch gains 2 minutes, in 24 hours. At a particular instant, the two watches showed an identical time. Which of the following statements is correct if 24-hour clock is followed? (CSAT-2017)

(a) The two watches show the identical time again on completion of 30 days.

(b) The two watches show the identical time again on completion of 90 days.

(c) The two watches show the identical time again on completion of 120 days.

(d) None of the above statements are correct.

64. Anita's mathematics test had 70 problems carrying equal marks i.e., 10 arithmetic, 30 algebra and 30 geometry. Although she Answered 70% of the arithmetic, 40 % of algebra and 60% of the geometry problems correctly, she did not pass the test because she got less than 60% marks. The number of more questions she would have to Answer correctly to earn a 60% passing marks is: (CSAT-2016)

(a) 1 (c) 7

(b) 5 (d) 9

65. Two members X and Y are respectively 20% and 28% less than a third number Z. By what percentage is the number Y less than the number X? (CSAT-2016)

(a) 12% (c) 9%

(b) 10% (d) 8%

66. A piece of tin is in the form of a rectangle having length 12 cm and width 8 cm. This is used to construct a closed cube. The side of the cube is: (CSAT-2016)

(a) 2 cm (c) 4 cm

(b) 3 cm (d) 6 cm

67. In a question paper there are five questions to be attempted and Answer to each question has two choices —True (T) or False (F). It is given that no two candidates have given the Answer to the five questions in an identical sequence. For this to happen the maximum number of candidates is: (CSAT-2016)

(a) 10 (c) 26

(b) 18 (d) 32

68. The sum of the ages of 5 members comprising a family, 3 years ago was 80 years. The average age of the family today is the same as it was 3 years ago, because of an addition of a baby during the intervening period. How old is the baby? (CSAT-2016)

(a) 6 months (c) 2 years

(b) 1 year (d) 2 years and 6 months

69. A person is standing on the first step from the bottom of a ladder. If he has to climb 4 more steps to reach exactly the middle step, how many steps does the ladder have? (CSAT-2016)

(a) 8 (c) 10

(b) 9 (d) 11

70. A person walks 12 km due north, then 15 km due east, after that 19 km due west and then 15 km due south How far is he from the starting point? (CSAT-2016)

(a) 5 km (c) 37 km

(b) 9 km (d) 61km

71. How many numbers are there between 100 and 300 which either begin or end with 2? (CSAT-2016)

(a) 110 (c) 112

(b) 111 (d) None of these

72. Four-digit numbers are to be formed using the digits 1, 2, 3 and 4; and none of these four digits are repeated in any manner. Further,

1. 2 and 3 each not to immediately follow each other.
2. 1 is not to be immediately followed by 3.
3. 4 is not appear at the last place.
4. 1 is not appear at the first place.

How many different numbers can be formed? (CSAT-2016)

(a) 6 (c) 9

(b) 8 (d) None of the above

73. In a race, a competitor has to collect 6 apples which are kept in a straight line on a track and a bucket is placed at the beginning of the track which is starting point. The condition is that the competitor can pick only one apple at a time, run back with it and drop it in the bucket, how much total distance he has to run if the bucket is 5 meters from the first apple and all other apples are placed 3 meters apart? (CSAT-2016)

(a) 40 m (c) 75 m

(b) 50 m (d) 150 m

74. A round archery target of diameter 1m is marked with four scoring regions from the centre outwards as red, blue, yellow and white. The radius of the red band is 0.20m. The width of all the remaining bands is equal. If archers throw arrows towards the target, what is the probability that the arrows fall in the red region of the archery target? (CSAT-2016)

(a) 0.40 (c) 0.16

(b) 0.20 (d) 0.04

75. Four persons A, B, C and D consisting of two married couples are in a group. Both the women are shorter than their respective husbands. A is the tallest among the four. C is taller than B. D is B's brother. In this context, which one of the following statements is not correct? (CSAT-2015)

(a) All four have family ties.

(b) B is the shortest among the four.

(c) C is taller than D

(d) A is B's husband

76. Consider the following statements:

1. A man had a wife, two sons and two daughters in his family.
2. The daughters were invited to a feast and the male members of the family went out to take part in a picnic.
3. The man's father did not return from his work.

Which of the following statements is true? (CSAT-2015)

(a) Only the man's wife was left at home.

(b) Itis likely that the man's wife was left at home.

(c) None was left at home.

(d) More than one person was left at home.

77. Geeta: Naresh has become a better boxer since he started meditation.

Radha: Impossible. A boxer's most important asset is his aggressiveness.

Radha:'s statement reflects her belief that (CSAT-2015)

(a) meditation tends to make a person less aggressive.

(b) meditation has little or no effect on the person who practices it.

(c) Naresh was a poor boxer earlier because he was not aggressive enough.

(d) Naresh would not have taken to meditation as he was a boxer.

78. The mangroves can shed tons of leaves per acre every year; fungi and bacteria break down this leaf litter and consume it, they then are consumed by tiny worms and crustaceans, which in turn feed small fish, which feed larger fish and birds and crocodiles. (CSAT-2015)

Which among the following is the most logical inference of the above statement?

(a) Coastal areas cannot have food chains without mangroves.

(b) Mangroves are an essential component of all marine ecosystems.

(c) Mangroves have a crucial role in some of the coastal food chains.

(d) The composition of marine flora and fauna is largely determined by mangroves.

79. The number of deaths among the army personnel is 8 in 1000, but among the civilian population it is 20 per 1000. Which one of the following inferences can be drawn from this statement? (CSAT-2014)

(a) It is better to join the army.

(b) The relationship is fortuitous.

(c) Quality of Life Index is very high within the armed forces.

(d) The groups cannot be compared due to their heterogeneity.

80. Given the statement: "Buses are the cause of more accidents than cars, and trucks cause fewer accidents than buses," which of the following conclusions can we draw? (CSAT-2014)

(a) There are more buses on the road than trucks.

(b) Car drivers are more careful than bus drivers.

(c) Truck drivers are more skilled than either car or bus drivers.

(d) None of the above

81. "If political leadership fails to emerge, there is likelihood of military taking over power in developing countries. Radical student groups or labour may try to raise revolution, but they are not likely to compete with the military. Military intervention, rule, and withdrawal from politics is closely related to a society's level of political development." In the context of political development, the assumption in the above passage is that (CSAT-2014)

(a) Political leadership is not an effective instrument.

(b) Military fills in political vacuum.

(c) Military intervention is inevitable for development.

(d) None of the above

82. "Liberty, therefore, is never real unless the Government can be called to account when it invades rights." (CSAT-2014)

Which one of the following is the best justification of the above statement?

(a) In the realisation that the government can be brought to book in a court of law

(b) In identifying a man as a political unit in a way which distinguishes him from other citizens

(c) In a decentralized society wherein the basic needs of men can find satisfaction

(d) In the understanding that liberty and restraints are complementary

83. Consider the following statement: (CSAT-2012)

"Though quite expensive, television is not a luxury item, as one can learn many things through television".

Which of the following is the valid inference from the above given statement?

(a) All expensive things are regarded as luxury.

(b) All essential things for learning are not luxury.

(c) Television is essential for learning.

(d) Television is not a luxury item.

84. Gita is prettier than Sita but not as pretty as Rita. Then, (CSAT-2012)

(a) Sita is not as pretty as Gita

(b) Sita is prettier than Rita

(c) Rita is not as pretty as Gita

(d) Gita is prettier than Rita

85. Given that, (CSAT-2012)

1. A is the brother of B
2. C is the father of A.
3. D is brother of E.
4. E is the daughter of B

Then, the uncle of D is?

(a) A (b) B

(c) C (d) E

86. Examine the following statements: (CSAT-2012)

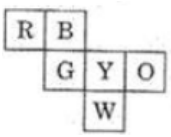
1. Rama scored more than Rani
2. Rani scored less than Ratna
3. Ratna scored more than Rama
4. Padma scored more than Rama but less than Ratna.

Who scored the highest?

(a) Rama (b) Padma

(c) Rani (d) Ratna

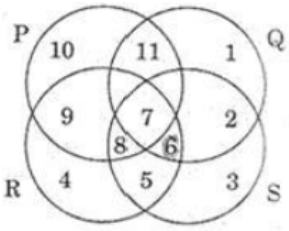
87. Six squares are coloured, front and back, red(R), blue(B), yellow (Y), green (G), white (W), orange (O) and are hinged together as shown in the figure given below. If they were folded to form a cube what would be the face opposite to white face? (CSAT-2012)



(a) R (b) G

(c) B (d) O

88.

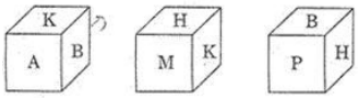


In the above figure, circle P represents hardworking people, circle Q represents intelligent people, Circle R represents truthful people and circle S represents honest people. Which region represents the people who are intelligent, honest and truthful but not hardworking? **(CSAT-2012)**

(a) 6 (b) 7

(c) 8 (d) 11

89. Three views of a cube following a particular motion are given below: (CSAT-2012)

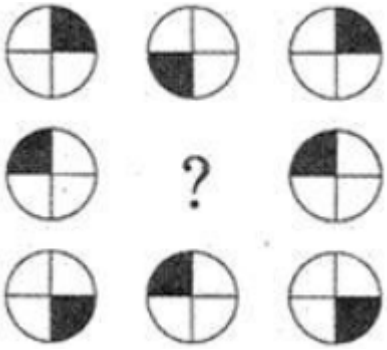


What is letter opposite to A?

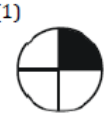
(a) H (b) P

(c) B (d) M

90.



Which one of the figures shown below occupies the blank space(?) in the matrix given below? **(CSAT-2012)**

(1) 

(2) 

(3) 

(4) 

91. Consider the following statements: **(CSAT-2012)**

1. All machines consume energy
2. Electricity provides energy
3. Electrically operated machines are cheap to maintain
4. Electrically operated machines do not cause pollution.

Which one of the following inferences can be drawn from the above statements?

(a) All machines are run by electric energy.

(b) There is no form of energy other than electricity

(c) Most machines are operated on electric energy.

(d) Electrically operated machines are preferable to use.

92. Examine the following statements: **(CSAT-2012)**

1. None but the rich ran afford air-travel.
2. Some of those who travel by air become sick
3. Some of those who become sick require treatment

Which one of the following conclusions can be drawn from the above statements?

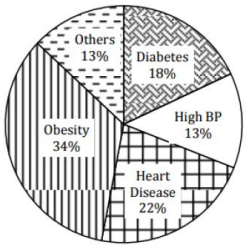
(a) All the rich persons travel by air.

(b) Those who travel by air become sick

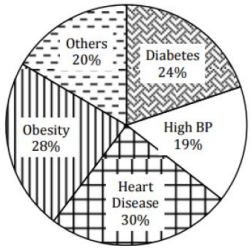
(c) All the rich persons become sick.

(d) All those who travel by air are rich

93. Which of the two towns has a higher number of persons with Diabetes? **(CSAT-2011)**



Distribution of diseases in Town-A



Distribution of diseases in Town-B

(a) Town A

(b) Town B

(c) Same in Town A and Town B

(d) No inference can be drawn

5 General Mental Ability_Explanations

1. Answer: (b)

Explanation:

Statement I is not always true:

Consider $f(t) = t + 1/t$. Its derivative is $f'(t) = 1 - 1/t^2$, which is negative on (0, 1) and positive on (1, ∞). So f decreases on (1/3, 1) and increases on (1, 2), not monotonic on the whole interval.

Counterexample: $x = 1/2$, $y = 3/4$. Then $x + 1/x = 2.5$ but $y + 1/y = 0.75 + 1.333\ldots \approx 2.083$. So $x + 1/x > y + 1/y$, contradicting I.

Statement II is always true:

Rewrite:

$$\sqrt{(1+t2)} / t = \sqrt{((1+t2)/t2)} = \sqrt{(1/t2 + 1)}$$

For $t > 0$, as t increases, $1/t^2$ decreases, so $\sqrt{(1/t^2 + 1)}$ decreases. Thus $g(t) = \sqrt{(1+t^2)}/t$ is strictly decreasing on (0, ∞).

Since $x < y$ (both positive), $g(x) > g(y)$, i.e.

$$\sqrt{(1+y2)} / y < \sqrt{(1+x2)} / x$$

2. Answer: (b)

Explanation:

To find the minimum number of measurements required to get exactly 298 litres, we can use a combination of drawing water from the tank (adding to our container) and pouring water back into the tank (subtracting from our container).

Approach 1: Only Adding Water

If we only draw water, we would use the greedy approach (using the largest possible cylinders first):

- 2 times 100 litres = 200 litres
- 3 times 25 litres = 75 litres
- 3 times 6 litres = 18 litres
- 5 times 1 litre = 5 litres

Total volume = 200 + 75 + 18 + 5 = 298 litres. Total measurements = 2 + 3 + 3 + 5 = 13 times.

Approach 2: Adding and Subtracting Water (Optimal)

To minimize the number of measurements, we can overshoot the target volume and then remove the excess water by pouring it back into the tank.

- We can draw 100 litres 3 times: 3 × 100 = 300 litres. (This takes 3 measurements)
- We have an excess of 2 litres (300 − 298 = 2).
- We can remove this excess by using the 1-litre cylinder to pour water back into the tank 2 times: 2 × 1 = 2 litres. (This takes 2 measurements)

Total volume = 300 − 2 = 298 litres. Total measurements = 3 (drawing) + 2 (pouring back) = 5 times.

Can we do it in 4 measurements?

To get close to 298 with 4 measurements, we must use the 100-litre cylinder 3 times (300 litres). We would then have only 1 measurement left to subtract the excess. The closest volumes we could achieve are:

- 3 × 100 − 1 × 1 = 299 litres
- 3 × 100 − 1 × 6 = 294 litres

Since neither gives exactly 298 litres, 4 measurements are not enough. Thus, the minimum number of times one needs to measure is 5.

3. Answer: (c)

Explanation:

To find the largest and smallest possible values for the expression (x × y) + z, we must choose distinct values for x, y, and z from the given set {2, 3, 4, 5}.

Step 1: Finding the largest possible value (M)

To maximize the value of (x × y) + z, we should maximize the product (x × y) because multiplication of numbers greater than 1 yields a larger result than addition.

- The two largest numbers available are 4 and 5. Therefore, we assign x = 4 and y = 5 (or vice versa), giving a product of 4 × 5 = 20.
- The remaining numbers are 2 and 3. To maximize the overall sum, we assign the largest remaining number to z, which is 3.
- Calculating the maximum value:

M = (4 × 5) + 3 = 20 + 3 = 23

Step 2: Finding the smallest possible value (N)

To minimize the value of (x × y) + z, we should minimize the product (x × y).

- The two smallest numbers available are 2 and 3. Therefore, we assign x = 2 and y = 3 (or vice versa), giving a product of 2 × 3 = 6.
- The remaining numbers are 4 and 5. To minimize the overall sum, we assign the smallest remaining number to z, which is 4.
- Calculating the minimum value:

N = (2 × 3) + 4 = 6 + 4 = 10

(Self-check: If we tried x = 2, y = 4, z = 3, the result would be (2 × 4) + 3 = 11, which is larger than 10. Thus, 10 is indeed the absolute minimum.)

Step 3: Finding the difference (M − N)

Now, we subtract the smallest value from the largest value:

M − N = 23 − 10 = 13

Therefore, the difference between the largest and smallest possible values is 13.

4. Answer: (d)

Explanation:

Each term is the sum of the two preceding terms.

- a1 = 4

- a2 = 7
- a3 = 4 + 7 = 11
- a4 = 7 + 11 = 18
- a5 = 11 + 18 = 29
- a6 = 18 + 29 = 47

5. Answer: (d)

Explanation:

Earnings E = 5d + 2k where d = deliveries and k = km. We need E > 500 to "definitely" hold for the given d.

Why d alone is not enough.

The number of kilometres k depends on the routes the boy actually takes; the only constraint stated is k ≤ 100. To guarantee E > 500 from the value of d alone, the inequality must hold even in the worst case k = 0, i.e. 5d > 500 ⇒ d > 100.

Check each option.

For d = 80, 85, 90, taking k = 0:

- d = 80: E = 400
- d = 85: E = 425
- d = 90: E = 450

None of these exceed 500 in the worst case, so none of A, B, or C "definitely" meets the target. The number of kilometres travelled is essential information that the question does not pin down.

6. Answer: (b)

Explanation:

Let's write down the given equations: (1) A + B + C + D = 17; (2) A + C = 6; (3) P + Q + S + D = 15; (4) P + Q + R + B = 17; (5) P = R and Q = S.

Step 1: Find the value of B + D

Substitute equation (2) into equation (1):

(A + C) + B + D = 17 ⇒ 6 + B + D = 17 ⇒ B + D = 11

Step 2: Simplify equations (3) and (4)

Using the given relations P = R and Q = S, we can rewrite equations (3) and (4). From equation (3):

P + Q + Q + D = 15 ⇒ P + 2Q + D = 15 — (Equation 6)

From equation (4):

P + Q + P + B = 17 ⇒ 2P + Q + B = 17 — (Equation 7)

Step 3: Find the value of P + Q

Add Equation (6) and Equation (7) together:

(P + 2Q + D) + (2P + Q + B) = 15 + 17 ⇒ 3P + 3Q + B + D = 32

We already know from Step 1 that B + D = 11. Substitute this into the equation:

3(P + Q) + 11 = 32 ⇒ 3(P + Q) = 21 ⇒ P + Q = 7

Step 4: Evaluate the given options

- **Option A:** States that B + D < P + Q. We found B + D = 11 and P + Q = 7. Since 11 is not less than 7, this statement is incorrect.
- **Option B:** States that P + Q > A + C. We found P + Q = 7 and we are given A + C = 6. Since 7 > 6, this statement is correct.
- **Options C and D:** We only know that P + Q = 7. The individual weights of P and Q cannot be uniquely determined from the given information (e.g., P could be 4 and Q could be 3, or vice versa). Thus, we cannot definitively say whether P > Q or Q > P.

Therefore, the only correct statement is Option B.

7. Answer: (c)

Explanation: LCM of 3,5,6,9 =90

The remainders is "-2" in each case so the number is of the form 90k-2

if k=12 we get the smallest 4 digit number of 1080-2= 1078

8. Answer: A

Explanation : Based on the analysis, only assuming P is guilty leads to a consistent scenario where the guilty person lies and the innocent people tell the truth.

9. Answer: (a)

Explanation:

Time frame: 10 a.m. to 2:00 p.m.

Hour and minute hands coincide:

1 time between 10 a.m. and 11:00 a.m.

1 time between 11 a.m. and 1:00 p.m.

1 time between 1 p.m. and 2:00 p.m.

So, hour and minute hands coincide 3 times between 10:00 a.m. and 2.00 p.m.

10. Answer: (c)

Explanation:

Option (c) is correct:

To determine if a year is a leap year, divide it by 4. If it's evenly divisible by 4, it's a leap year.

Now, 2025 is not a leap year because 2025 divided by 4 is 506.25.

2025: non-leap year, starts on let's say day X.

2026: starts on X + 1 day.

2027: starts on X + 2 days.

2028: starts on X + 3 days. But 2028 is a leap year, so 2029 will start on X + 5 days (because 2028 has 366 days, so 2 days shift).

Then 2029: non-leap year, so 2030 starts on X + 5 + 1 = X + 6 days.

2031: starts on X + 6 + 1 = X + 7 days, which is a full week later,

so 2031 would start on the same day as 2025.

11. Answer: (c)

Explanation:

Time = 4 : 25

Minute hand moves 6° per minute.

So, Angle traced by minutes hand in 25 minutes = 25 × 6° = 150°

The hour hand moves half a degree in one minute. So, angle traced by hour hand at 4 hours and 25 minutes, i.e. 265 minutes = 265 × 0.5° = 132.5°

Thus, the angle between two hands at 4 : 25 = 150° - 132.5° = 17.5°

12. Answer: (c)

Explanation:

Option (c) is correct:

The Sequence has the pattern of double alphabets:

=AABCC

=ABBCD

=ABCCD

=ABCDD

Thus, if we put A, A, C, D in the blank, then it will complete the sequence.

13. Answer: (d)

Explanation:

Option (d) is correct: All are correct.

1000 litres = 1 m^3

1 metric ton = 1000 kg

1 hectare = 10000 m^2

14. Answer: (b)

Explanation:

1st Jan	2nd	3rd	4th …	
↓	↓	↓		
1	1+2	3+2	…	64 → is perfect Cube and Square
		↓	↓	
		3	5	

So, series becomes1, 3, 5, …64

Common difference = 2

a = 1

$Sn=\frac{n}{2}[2a+(n-1)d]$

$64=\frac{n}{2}[2(1)+(n-1)2]$

$64=n(1+n-1)$

$n^2= 64$

n = 8

8th January 2023

15. Answer: (c)

Explanation:

It is given that x and y are the distinct natural numbers.

Also, given in statement 1, x/y is odd = 3/1 =3

Statement 1 is alone not sufficient to answer the question as there are many possibilities.

Using statement-II alone: xy = 12, the values of x and y may be 12 & 1, 4 & 3, or 6 & 2. So, this statement alone is also not sufficient.

On combining statement-I and statement-II: 12/1 = 12, which is not odd. 4/3 is a fraction. Now, 6/2 = 3, which is an odd number. Hence, unique values of x and y are 6 and 2 respectively.

The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

16. Answer: (c)

Explanation:

From Statement 1, X received 4/5 of what Y and Z together received

x= (⅘) (y+z)

5x= 4y +4z _____ (i)

From statement 2, Y received 2 / 7 of what X and Z together received.

y= (2/7) (x+z)

7y= 2x +2z _____ ii

Using (i) and (ii), we get

9x= 18y

x=2y

Putting the value of x in _____(ii)

7y= 4y + 2z

3y = 2z

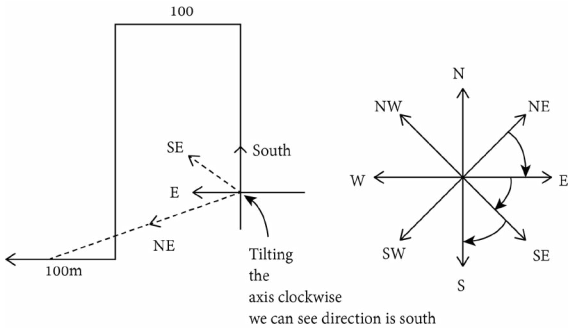
z>y

So, y received the least amount.

hence, we can say that the The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

17. Answer: (c)

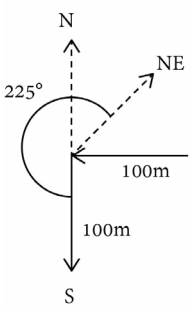
Explanation:

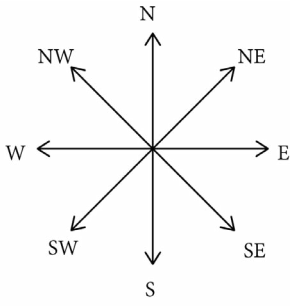


18. Answer: (d)

Explanation:

As per the information given in the question, the following can be drawn:





Form the diagram, it can be said that she is working in the North-East direction.

19. Answer: (c)

Explanation:

The given statements are:

Statement 1: The size of the software is 12 megabytes.

Statement 2: The transfer rate is 2.4 kilobytes per second.

The time required to download the software is = 12 Megabytes (12000)/2.4 = 5000 seconds.

Therefore, the Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

20. Answer: (b)

Explanation:

From 1 to 100, 5 comes 20 times.

From 101 to 200, 5 comes 20 times.

From 201 to 300, 5 comes 20 times.

So, we just need to check the numbers after 260. Such numbers having 5 are 265, 275, 285, 295 i.e. 4 numbers. So, the number of 5's used from 1 to 260 = (20 + 20 + 20) - 4 = 56

21. Answer: (a)

7 ⊕ 9 ⊕ 10 = 7 + 9 + 10 = **26.** 26 is expressed as, **2 + 6 = 8**
Similarly, 9 ⊕ 11 ⊕ 30 = 9 + 11 + 30 = 50, where 5 + 0 = 5
And 11 ⊕ 17 ⊕ 21 = 11 + 17 + 21 = 49, where 4 + 9 = 13
So, 23 ⊕ 4 ⊕ 15 = 23 + 4 + 15 = 42, where 4 + 2 = 6

Hence, option (a) is correct.

22. Answer: (a)

10^{10} = 10000000000

10,000,000,000 ÷ 7 = 1,428,571,428 remainder 4

This means there are 1,428,571,428 complete weeks in 10 billion days, and 4 days are remaining.

Now, if today is Sunday, then after 7 days (1 week), it will again be Sunday. After 14 days (2 weeks), it will again be Sunday, and so on. Therefore, after 1,428,571,428 weeks, it will again be Sunday.

But we still have 4 days remaining.

If today is Sunday, then the 4th day from today (inclusive) is Wednesday. (Sunday – Day 1, Monday – Day 2, Wednesday – Day 3, Thursday – Day 4)

Therefore, the correct answer is (a) Wednesday.

The modulo operation, denoted as "mod;" calculates the remainder when one number is divided by another. For example, 13 mod 5 = 3, as dividing 13 by 5 results in a quotient of 2 with a remainder of 3.

Let's assign numbers to the days of the week: Sunday = 0, Monday = 1, Tuesday = 2, and so on up to Saturday = 6. Since today is Sunday (day 0), we can calculate the day using modular arithmetic.

1010 mod 7:

Let's start by calculating 101 mod 7:

101 mod 7 = 3

Calculate (102 mod 7) using the previous result:

102 mod 7 = (101 mod 7 x 101 mod 7) mod 7 (3 x 3) mod 7 = 9 mod 7 = 2

Calculate (104 mod 7) using the previous result:

104 mod 7 = (102 mod 7)2 mod 7

22 mod 7 = 4 mod 7 = 4

Calculate 108 mod 7 using the previous result:

108 mod 7 = (104 mod 7)2 mod 7

42 mod 7 = 16 mod 7 = 2

Calculate 1010 mod 7 using the previous result:

1010 mod 7 = (108 mod 7 x 102 mod 7) mod 7 (2 x 2) mod 7 = 4

4 mod 7 = 4

Therefore, if today is Sunday, the day corresponding, to the 1010th day would be four days after Sunday, which is Thursday. **Hence, option (b) is correct.**

23. Answer: (b)

Ring is passed clockwise from child 1 to 2 to child 4 to child 7 and so on and comes back to child 1 again.

We observe the series is formed by adding +1, +2, +3, +4, +5, +6, +7,…… and so on.

Series:

1 2 4 7 11 16 22 29 37 6 16 27 39 12 26 1

So, after 15 changes the ring comes back to child 1. **Hence option (b) is correct.**

24. Answer: (d)

Statement-1: p x q is greater than zero.

If the product of p and q is greater than zero, it means either both p and q are positive or both are negative. In either case, we cannot determine

which one is greater.

Such as, For eg 2x3>0

Or-2x-3>0

Either p<q or p>q here.

Therefore, this statement alone is not sufficient to answer the question.

Statement-2: p2 is greater than q2.

This statement tells us that the squares of p and q are involved. If both p and q are positive or both are negative, then the comparison holds true. However, if p is positive and q is negative or vice versa, we cannot determine which one is greater.

Such as, For eg 32 > 22

or (-3)2>(-2)2

Either P<q or P>q here

Therefore, this statement alone is also not sufficient to answer the question.

Statement 1 and 2 combined:

If we combine both statements, we still don't have enough information to determine if p is greater than q. Therefore, the Question cannot be answered even by using both the Statements together.

25. Answer: (c)

As given in the question, we find that,

Statement 1:

p - q is positive

which means p>q

Statement 2:

p – r is negative

which means p<r

This implies that, q< p<r

Now let's see,

(p+q+r)=(p-q+r)=2q-2r=2(q-r)

Since r is greater than p, the term (p - q + r) will have a

larger positive value compared to (p +q - r).

Hence, (p + q - r) is not greater than (p - q + r) and question cannot be answered using either statement alone.

26. Answer: (d)

To answer the question, let's evaluate each statement individually.

Statement 1: The sum of the digits of the number is equal to the product of the digits.

The any combination of digits 1, 2 and 3 can fulfill this condition.

123

1x2x3=6

1+2+3=6

132

1x3x2=6

1+3+2=6

312

3x1x2=6

3+1+2=6

Hence statement 1 alone cannot answer the question.

Statement 2: The number is divisible by the sum of the digits of the number.

At least 2 combinations of the above given 3-digit numbers fit this condition.

132/6=22

312/6=52

Even this statement cannot provide one specific 3-digit number.

Even when combining both statements, we cannot determine a single, specific 3-digit number.

27. Answer: (b)

We have five children with ages a < b < c < d < e, where

any two successive ages differ by 2 years. Here 'a' is

youngest and 'e' is eldest.

It can be expressed as,

a<b<c<d<e

As successive ages differ by 2 years, above expression

can be written as:

Cc-4<c-2<c<c+2<c+4

Statement 1: The age of the eldest is 3 times the youngest.

Given c+4=3(c-4)

c+4 =3c-12

2c=16

c=8

From above conclusion we can calculate the ages of 5

Children as, a=4; b=6; c=8; d=10; e=12

This statement alone can give answer.

Statement 2: The average age of the children is 8 years

To find the average of all the ages we can write the

expression as:

(c-4+c-2+4c+c+2+4c+4)/5=8

5c=8x5

c=8

From the above conclusion we can calculate the ages of

5 children as, a=4;b=6;c=8;d=10;e=12

This statement alone too can give answer.

Both statements alone can answer the question.

28. Answer: (c)

Available silver coins' weight options are 2gm, 5gm, 25gm, 50gm each

Statement 1 is correct: To get 'minimum' number of coins we need one coin of 50gm, two coins of 10gm, four coins of 2gm, i.e. 1x50 + 2x10 + 4x2 = 78gm

So number of coins here = 142+4 =7

Statement 2 is correct: 78gm = (80 - 2)gm

It can be expressed as 50gm (one coin) + 25gm (one coin) + 5gm (one coin) - 2gm (one coin)

Here we need just 4 coins to weight 78gm.

29. Answer: (d)

This question is solved by Hit & Trial method.

Using option (a)

A B C D A A B C C D A B B C D B A B C D (pattern of the sequence not followed)

Using option (b)

A B C C D A B C A D A B B C D B A B C D (pattern of the sequence not followed)

Using option (c)

A B C D C A B C C D A B B C D A A B C D (pattern of the sequence not followed)

Using option (d)

A B C D D A B C C D A B B C D A A B C D ({pattern of the sequence followed))

Pattarn is established only in option (d) case. **Hence, option (d) is correct.**

30. Answer: (a)

Following is given in the question

AB and CD are 2-digit numbers. Multiplying AB with CD results in a 3-digit number DEF.

Adding DEF to another 3-digit number GHI results in 975.

E=0 and F=8

A, B, C, D, E, F, G, H, I are distinct digits.

It can be implied from the given information that:

AB x CD = DEF

By substituting E = 0 and F = 8 in DEF we get

D08 + GHI = 975.

As 8+I=15, I=7

Now 1+0+H=7

H=7

D+G=9, ie. (D, G) may be:

(9,0), (8,1), (7,2), (6,3), (5.4)

All above values except (5,4) gets eliminated as A to I are distinct digits.

If D=4 and G =5, 408 + 567 = 975

If D =5 and G =4, 508 + 467 = 975

Now, to verify:

AB x CD = DEF

AB x C5 = 508 (not possible)

AB x C4 = 408 (possible)

So, D=9, G=5

Now analysing the digit till now we are left with digits 1, 2, 3, and 9

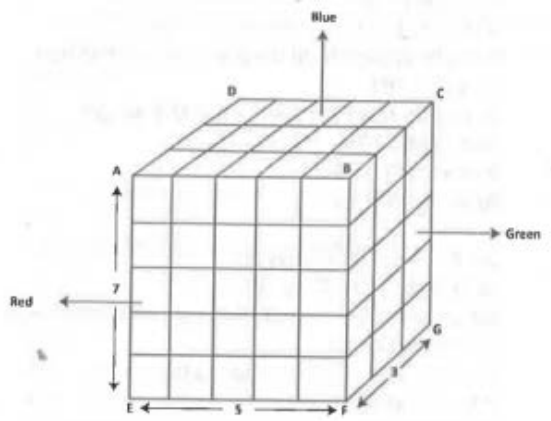
AB x C4 =408

12 x 34 =408

So, A+B+C=1+2+3=6. **Hence, option (a) is correct.**

31. Answer: (a)

As per given in the question: The cuboid has dimensions 7cm x 5cm x 3cm. The opposite faces of the cuboid are painted in different colors: red, green, and blue.



Length dimension: 7cm + 1cm = 7 cubes

Width dimension: 5cm + 1cm = 5 cubes

Height dimension: 3cm + 1cm = 3 cubes

To find the total number of small cubes, we multiply the number of cubes in each dimension:

Total cubes = Length cubes x Width cubes x Height cubes =7 cubes x 5 cubes x 3 cubes = 105 cubes

AEFB and opposite face is red

DABC and opposite face is blue

BCGF and opposite face is green

Statement 1 is true: To determine the number of cubes that are not colored

(5-2) x (3-2) x (7-2)

=3x1x5=15

Statement 2 is false: As shown in the figure, we have only 4 such cubes.

32. Answer: (c) is correct.

The condition given is there are exactly 2 letters between A and E. Either 1st letter, or 5th letter must be A or E because if both A and E are in middle then there cannot be two letters between them.

Hence, only following arrangements are possible:

A_ _ E _

_ A _ _ E

E _ _ A _

_ E _ _ A

Now we need to fill 3 blank spaces in each of these 4 arrangements by 3 letters. The three blank spaces can be filled by three distinct letters in 3 x 2 x 1 = 6 ways. So, we can fill the vacant spaces in each arrangement by 6 ways. So, total possible arrangements = 6 x 4 = 24. **Hence, option (c) is correct.**

33. Answer: (c) is correct.

Number of ways to go from A to C =3

According to the information given in question:

Number of ways to go from A to B = 3 (say a, b and c)

Number of ways to go from B to C = 4. So, Number of ways to go from A to C, via B=3x4= 12,

Because let suppose we go from A to B via road "a" we can go from B to C from any of 4 roads so via road "a" we have 4 ways from A to C. Similarly, we have 4 roads from A to C via road "b" and road "c". Hence, we have total 12 ways from A to C via B.

So total no. of ways from A to C = 12+3 = 15 ways. **Hence, option (c) is correct.**

34. Answer: (c)

The tournament starts with 150 players.

In 1st round 75 matches will be played between 150 players: 75 players are eliminated, and 75 remain. Now in 2" round 37 matches will be played between 74 players and one player will get bye: 37 players are eliminated, and 38 remain.

In 3rd round 19 matches will be played between 38 players: 19 players are eliminated, and 19 remain. In 4th round 9 matches will be played between 18 players and 1 will get bye: 9 players are eliminated, and 10 remain.

In 5th round matches will be played between 10 players: 5 players are eliminated, and 5 remain.

In 6th round 2 matches will be played between 4 players 1 will get a bye: 2 players are eliminated, and 3 remain.

In 7th round 1 match will be played between 2 players 1 will get a bye: 1 player is eliminated, and 2 remain.

In 8th round 1 match will be played between 2 players: 1 player is eliminated, and 1 remains.

So, total number of matches = 75+ 37 + 19+ 9+ 5+ 2 + 1+ 1=149. **Hence, option (c) is correct.**

Question Hack:

We can use a formula "In any knockout tournament with n teams/players total numbers of match played is n-1" In given question n= 150. Hence, total no. of matches = 150-1 = 149.

35. Answer: (d) is correct.

First check the statements one by one.

Statement 1: /3 is not an integer. X may or may not be an integer. For example, If x = 2, then x/3 will not be an integer.

If x= 1/3 then also x/3 is not an integer. For both the values of x (2 and 1/3), x/3 is not an integer but at one place x is integer at other x is not.

Hence, we can conclude x is an integer or not if x/3 is not an integer.

Statement 2: 3x is an integer.

X may or may not be an integer.

For example, if x = 1/3, then 3x will be an integer.

If x = 2 then 3x is an integer. For both the values of x (2 and 1/3), 3x is an integer but at one place x is integer at other x is not.

Hence, we can conclude x is an integer or not if x/3 is not an integer.

On combining the statements, 3x is an integer, but x/3 is not.

x may or may not be an integer. For example, if x = 2, even then x/3 will not be an integer. But 3x = 6 will be an integer.

If x = 1/3, even then 3x will be an integer. But x/3 = 1/9 will not be an integer.

Hence, we cannot conclude whether x is an integer or not. **Hence, option (d) is correct.**

36. Answer: (c) is correct.

The given series is: AABABCABCDABCDE...

The pattern of the series is: A, AB, ABC, ABCD, ABCDE, ...

It is type of an arithmetic series where the first term has 1 letter, second term has 2 letters, and so on. We have to find 100" term of the series.

If we have an arithmetic series, 1, 2, 3 ..., then:

Sum of first n natural numbers where 'n' is is the number of terms in this series,

Now we need to find the 100"' term.

When n = 13; sum of 1 14 natural numbers is

Hence, 91" letter of the sequence is 13" letter of alphabets which is "M".

And 92" letter will be "A" and sequence starts again.

93 letter will be "8", 94" will be "C" and .... 100" will be "I"

Hence, option (c) is correct.

37. Answer: (b)

According to information given in question:

A's position: 16th from left end

V' position: 18th from right end

G' position: 11th from A towards the right and 3rd from V towards the right.

Now, there are two possible cases with respect to position of A:

Case I: A is right to V. it is given that G is 11th right from A. If Alis to the right of V and G is 11 more positions right of A, then G must be at least 12 positions to the right of V. But as per the question G is only three positions to the right from V.

Hence, this case is impossible.

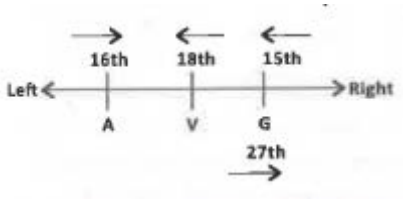
Case II: A is left to V

G is 3rd from V towards the right, so position of G = 18 - 3 = 15th from right end.

G is 11th from A towards the right, so position of G = 16 +11 = 27th from left end.

Hence, total number of boys in the row =27 +15-1=41 (because G is counted twice)

You can see it better in the picture given below:



Hence, option (b) is correct.

38. Answer: (c)

Total no. of letters in arrangement = 20

We can break the arrangement in 4 arrangements of 5 letters each as:

abab_: here next letter will be a because it is sequence of ab.

b_bcb: here next letter will be c because it is sequence of bc.

_dcdc: here next letter will be c because it is sequence of cd.

ded_d: here next letter will be e because it is sequence of de.

The missing letter**s are:** a,c,c,e. Hence, option (c) is correct.

39. Answer: (c)

It is given that 12th January is Sunday.

Let's count no. of days between 12th January and 1st July If it is a normal year: (31-11) +28 +31+30+31+30= 170 days.

If it is leap year: (31-11) + 29+31+30+31+30=171,

Now we will count no. of odd days between 12th January, and 1st July:

If it is a normal year: 170 = 24 x 7 + 2 (2 odd days)

If it is a leap year: 171=24x 7 + 3 (3 odd days)

Hence, day on 1st July if it is normal year: Sunday + 2 = Tuesday

day on 1st July if it is leap year: Sunday + 3 = Wednesday

Now, if it is a normal year: 1st July = 8th July is Tuesday

If 8th July is Tuesday, then 12th July is Saturday and 15t July is Tuesday.

Now, if it is a leap year: 1st July = 8th July is Wednesday

If 8th July is Wednesday, then 12th July is Sunday and 15 July is Wednesday. Hence, option (c) is correct.

40. Answer: (d)

When alphabets are numbered according to their position, then:

GHI =789

DEF =456

ABC =123

According to the question,

GHI=1578 =789 x 2

DEF=912=456x 2

According to question:

ABC = 123 x 2 = 246. Hence, option (d) is correct.

41. Answer: (d)

In the question: 1st letter

A is related to B; B = A+1 by same procedure M will be related to M+1=N

2nd letter:

C is related to E; E = C + 2 by same procedure N will be related to N+2 =P

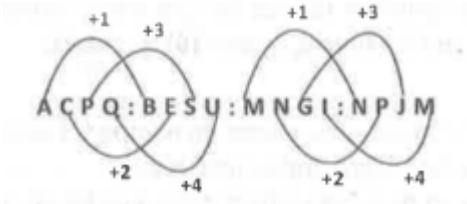
3rd letter:

P is related to S; S = P+3 by same procedure G will be related to G+3 = J

4th letter:

Q is related to U; U = Q+4 by same procedure I will be related to I +4=M

Hence, MNGI is related to NPJM



Hence, option (d) is correct.

42. Answer: The correct answer is option (b)

Mr. 'X' has three children.

The birthday of 1 child falls on 5" Monday of April

April has 30 days

So only two dates are possible on which any day comes 5 times.

The possible dates are 29™ or 30™ April on which 5%

Monday can come

Number of days in May = 31, June = 30, July = 31, August = 31, September = 30, October = 31 and November = 30.

It is given that the birthday of 2" child falls on 5 Thursday of November.

The only possibilities for 5th Thursday in November are 29 or 30

CASE-1: If 29 April is Monday.

Then, Number of days till 29™ November = 1 + 31 +30 +

31+31+30+31+29=214

Divide 244 by 7, we get 4 as s remainder.

So, if 29th April is Monday, then 29th November = Monday + 4 days = Friday. Hence, in this case 5" Thursday is not possible in

November because 4% Thursday will be on 28th November.

CASE-2: If 30 April is Monday.

Then the number of days till 29th November = 31+30+31+31+30+31+29=213 Divide 213 by 7, we get 3 as a remainder.

So, on 29t November, the day is, Monday + 3 days = Thursday.

The birthday of 3rd child falls on 20th December.

The number of days between 29th November and 20

December = 1+20 =21 days

Divide 21 by 7, we get O as remainder. So, on 20th December the day is Thursday.

Hence, option (b) is correct.

43. Answer: The correct option is option (c)

x is greater than or equal to 25

$x > 25$... (1)

y is less than or equal to 40

$y < 40$... (2)

Option (a): x is greater than y. This option is not true.

Take x = 26 and y = 39 .

Here, x is not greater than y.

Option (d): $x + y > 65$. This option is not true.

Take x = 26 and y = 5. Then,

$x + y = 36 + 5 = 31$, which is less than 65.

Now, we will check option (b) and option (c) by finding (y-x).

The maximum value which y can take is 40 and x can take values 25, 26, 27, 28...

In this case, (y-x) can take values like 15, 14, 13...1, 0, -1, -2...

Similarly, if y = 30, then the values of (y-x) are 5, 4, 3, 2, 1, 0, -1, -2...

So, in every case the value of (y-x) is less than or equal to 15. **Hence, option (c) is correct.**

44. Answer: The correct answer is option (b)

Age of Ena 13 years

Ena's mother is 24 years older than Ena

-> Age of Ena's mother= 24 + 13 = 37 years

Ena's mother is 3 years younger than her father.

-> Age of Ena's father= 37+3= 40 years

Ena was born 4 years after her parents' marriage.

-> Ena's father married at the age = 40-13-4=23

So, Ena's father married at 23 years age.

Hence, option (b) is correct.

45. Answer: The correct option is option (a)

The clock shows the correct time = 8:00 a.m. on Monday Number of hours between 8:00 a.m. on Monday to 6:00 p.m. on Wednesday= 24 + 24 + 10 = 58 hr

In every 24 hours, a wall clock moves fast by 10 minutes. Now, number of minutes in 24 hr = 24 x 60 = 1440 min So, when the wall clock moves 1450 min, the actual clock moves 1440 min.

Number of minutes in 58 hours = 58 x 60 = 3480 min

When the wall clock moves 3480 min, then the actual clock moves

$\frac{1440}{1450} \times 3480 = 3456$ min

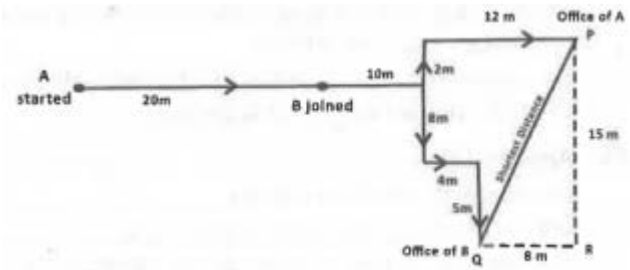
3456 min = 57 hr 36 min

So, when the wall clock moves 58 hr, the actual clock moves by 57 hr 36 min.

The actual time when the wall clock shows 6:00 p.m. = 5:36 p.m. **Hence, option (a) is correct.**

46. Answer: The correct option is option (b)

The information given in the question is represented below in the diagram:



We have to find the shortest distance between the two offices of A and B.

PQ = shortest distance

PR = 15 m

QR = 8 m

Using Pythagoras theorem, in triangle PRQ,

$PQ^2 = PR^2 + QR^2$

$PQ^2 = 15^2 + 8^2$

PQ = 17 m

So, the shortest distance between the two offices = 17 m. Hence option (b) is correct.

47. Answer: The correct answer is option (d)

Number of days in a year = 365 days.

Now, when 365 is divided by 7, one day is left. Number of days in a leap year = 366 days. When 366 is divided by 7, two days are left.

In order to repeat the same calendar year as 2009, the first day of that year should be same as the first day of year 2009 and the number of days should also be same i.e., 365 days or 366 days.

Let the first day of the year 2009 is X.

So, the first day of the year 2010 is (X + 1) day.

Similarly, the first day of the year 2011 is (X + 2) days. Also, the first day of the year 2012 is (X + 3) days. Now comes the leap year 2012 (two days are left). So, the first day of the year 2013 = (X +3)+2 = (X +5) days.

The first day of the year 2014= (X + 6) days.

The first day of the year 2015 = (X +7) days i.e., X.

After every 7 days, the same day appears.

-> The first day of the year 2015 is X which is same as the year 2009.

Also, both the years 2009 and 2015 are not leap years. So, the year 2015 will repeat the same calendar as that of 2009. **Hence, option (d) is correct.**

48. Answer: The correct answer is option (d)

The alphabet is written as: AbCdEfGhIjKIMnOp

QrStUvwWxYz

The first month of the second half of the year is 7th month = JULY.

In this form of alphabet, the required month is written as: jUIY. **Hence, option (d) is correct.**

49. Answer: The correct answer is option (b)

There are many different ways:

All the faces are black. Number of ways = 1

One face is white and other 5 faces are black. Number of ways = 1

Two faces are white, and four faces are black. Number of ways =2

The two white faces can be adjacent or on the opposite sides of the cube.

Three faces are white and three are black. Number of ways =2

All the three white faces are consecutive, and another way is two white faces are on the opposite side of cube.

Four faces are white, and two faces are black. Number of ways =2

The two black faces are adjacent, and they can be on the opposite sides of the cube.

Five faces are white, and one face is black. Number of ways =1

All the faces are white. Number of ways = 1

So, total number of different ways =1+1+2+2+2+1 +1 =10 ways. **Hence, option (b) is correct.**

50. Answer: The correct answer is option (d)

Given that,

$ means 'divided by. @means 'multiplied by' # means 'minus'

Then, the value of 10#5@1$5 = 10-5x15

Using BODMAS rule, we will solve this.

10#5@1$5 = 10-(5 x 15) = 10-(55) = 10-1=9

So, the value of 10#5@1$5 = 9. **Hence, option (d) is correct.**

51. Answer: The correct option is option (c)

Interpretation:

Every student got different marks and D > A, A > B and C>A

C, D >A>B

But, D and A are close competitors. So, C>D>A>B

Also, either C or F scores highest.

Whenever C scores the highest, then E scores the least.

Possibility: C>D>A><B>E

Whenever F scores the highest, B scores the least. Possibility: F>C>D>A>B

If F stands second in the ranking, it means F is not the highest scorer and C scores the highest.

So, when C scores the highest, the possible ranking is C >D>A><B>E

But, given that F stands second in the ranking.

The ranking becomes C>F>D>A>B>E

So, the position of B is fifth. **Hence, option (c) is correct.**

52. Answer: The correct option is option (d)

It is given that B scores the least. It means F scores the highest.

The possible ranking is F>C>D>A>B

But here E is missing.

E can take any position between F and B because E cannot be the highest or least scorer.

CASE-1: F> E >C>D>A>B

CASE-II: F>C>E>D>A>B

CASE-III: F>C>D>E>A>B

CASE-IV: F>C>D>A>E>B

We can clearly see from the above four cases that the position of C is either second or third depending upon the position of E. **Hence, option (d) is correct.**

53. Answer: (b)

If E is ranked third, then from the cases in above question, we can determine the ranking order.

The ranking is: F>C>E>D>A>B (from CASE-II)

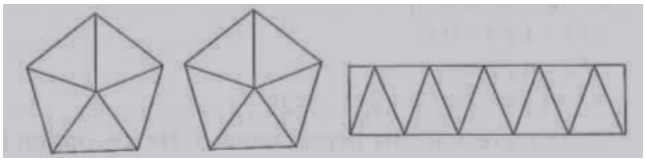
We can clearly see that, E get less marks than C and A is ranked fifth and D is ranked fourth.

It is clear that; C gets more marks than E. **Hence, option (b) is correct.**

Answer: The correct option is option (b)

54. Answer: (b)

The correct option is (b) The upper part, middle part and lower part of the 3-d figure can be represented in 2-d as;



Number of triangles in the topmost part = 5

Number of triangles in the bottom-part = 5

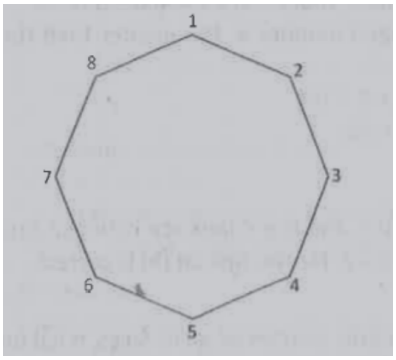
Number of triangles in the middle-part = 10

So, the total number of triangles = 5 + 10 + 5 = 20.

55. Answer: (a)

The correct option is (a).

There are 8 points in an octagon.A diagonal can be drawn from every point to 5 other points in an octagon.



Consider point 1: From there 5 diagonals 1 -3, 1-4, 1-5,1-6 and 1-7.

So, we can make 8 x 5 = 40 diagonals

However, this includes similar pairs i.e., 3-1,4-1,5-1

So, total number of diagonals = 40 / 2 = 20 diagonals.

56. Answer: (b)

The correct option is (b)

Let the line is on the top face of cube.

From figure I and II, line and two dots are common. So, three dots are adjacent to two dots and opposite to single dot.

Similarly, from figure II and III, line and three dots are common. So, four dots face is adjacent to three dots and opposite to two dots.

So, line and cross are opposite faces.

From above explanation, three dots face is opposite to single dot.

57. Answer: (c)

The correct option is (c)

From above explanation, four dots symbol face is opposite to two dots.

58. Answer: (c)

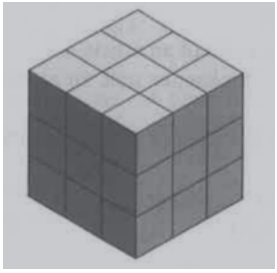
The correct option is (c)

From above explanation, line symbol face is opposite to the cross.

59. Answer: (a)

The correct option is (a)

A solid cube of 3 cm side,painted on all its faces, is cut up into small cubes of 1cm side



On the top layer, 4 (middle) cubes are there whose two

faces are painted.

On the middle layer, 4 (corner) cubes are there whose

two faces are painted.

On the bottom layer, 4 {middle) cubes are there whose

two faces are painted.

So, there are 4+ 4 + 4 = 12 cubes which have exactly two

faces painted.

60. Answer: (b)

The correct option is (b)

Given that, Pen < Pencil, Pencil < Book and Book > Cap

-> Pen < Pencil < Book and Cap < Book

So, Pen < Book is always true.

61. Answer: (c)

The correct option is (c)

Number of equally spaced points lying on the circumference of a circle = 24

The complete circle has 360°

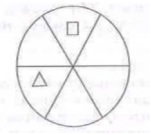
So, each of the 24 points subtend an angle at the center of the circle

We know that 2 consecutive vertices of an equilateral triangle make an angle of 120 at the center.

62. Answer: (b)

The correct option is (b)

We observe that the triangle is moving in anti-clockwise direction and the square is moving in clockwise direction.So, the figure that will appear at VI stage is:



63. Answer: The correct answer is option (d)

A watch loses 2 minutes in every 24 while another watch gains 2 minutes, in 24 hours.

So, the time difference between the two watches in 24 hours = 4 minutes.

The time difference goes on increasing by 4 minutes in every 24 hours. These watches will again show the same time when their time difference is 24 hours i.e.

24 x 60 = 1440 minutes in $\frac{1440}{4}$=360 days

By unitary method, 4 minutes time difference in 1 day 24 hours or 1440 minutes time difference

The two watches show the identical time again on completion of 360 days. **Hence, option (d) is correct.**

64. Answer: (b)

Let each question carry 1 mark.

Total number of questions = 70. Total marks = 70

Passing marks = 60%

Number of problems = 10 arithmetic + 30 algebra + 30 geometry

Problems answer correctly = 70% of arithmetic + 40 of algebra + 60% of geometry

7+12+18=37

Anita's marks = 37

Marks required to get 60% passing marks = 42-37 = 5

So, Anita would have to Answer 5 more questions correctly to earn 60% of passing marks.

Hence, option (b) is correct.

65. Answer: (b)

The three numbers are given as X, Y and Z

X=

=>

Y=

=>

From (i),

Put the value of z in (ii)

Y =

= > Y =

So, y is less than the number x by = 10%.

Hence, option(b) is correct.

66. Answer: (c)

A piece of tin is in the form of a rectangle having length 12 cm and width 8 cm.

Let the side of cube be x

The rectangle is used to construct a closed cube.

-> Surface area of cube = Area of rectangle

$6x^2 = 12 \times 8$

$X^2 = 16$

X = 4cm

So, the side of cube = 4 cm. **Hence, option (c) is correct.**

67. Answer: (d) In a question paper - 5 question - either True/ False.

Each question can be Answered in 2 ways —T or F

Number of ways each set can be Answered in a distinct way =2 x 2 x 2 x 2 x 2 = 32, So, 32 candidates can Answer the 5 questions in distinct sequence. **Hence, option (d) is correct.**

68. Answer: (b)

Let the present sum of ages of 5 family members is X.

3 years ago, the sum of ages of 5 members of a family = 80 years

X—3(5) = 80

X =80+ 15 =95 years

Average age of 5 family members three years ago = 80 / 5 =16 years

Let the age of baby is b years

Average age of family members today (5 members + baby) = 95+b/6

Since the average age of family at present is same as three years ago.

95+b/6

=>95+b=96

=>b=1year

So, the age of baby = 1 year.

Hence, option (b) is correct.

69. Answer: (b)

A person is standing on the 1st step of the ladder from bottom.

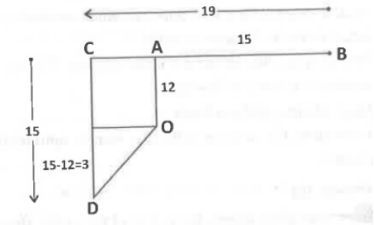
If he climbs 4 more steps, he reaches at the middle of the ladder. It means 5'" step is the middle of the ladder.

So, the total number of steps in the ladder =4+ 1+4=9.

Hence, option (b) is correct.

70. Answer: (a)

A person starts from O and walks 12 km due north reaches at point A and then walks15 km due east, reaches at point B. After that, he walks19 km due west and reaches at C and then 15 km due south and reaches at point D.



We have to find the distance DO:

Using Pythagoras theorem,

$DO^2=3^2+4^2 = 25$

DO = 5

So, he is 5 km away from the starting point.

Hence, option (a) is correct.

71. Answer: (a)

Between 100 and 199: The numbers ending with 2 are: 102, 112, 122,132, 142, 152, 162, 172, 182 and 192 i.e., 10 numbers

Between 200 to 300: 100 numbers start with 2.

We have to find the numbers which either begin or end with 2. So, we cannot double the count as there are some numbers which start with 2 and end with 2 i.e., 202,212...

So, the numbers between 100 and 300 which either begin or end with 2 = 10 + 100 = 110.

Hence, option (a) is correct.

72. Answer: (a)

Four-digit numbers are formed using the digits 1, 2, 3 and 4

Given conditions:

2 and 3 do not to immediately follow each other.

So, the possible numbers: 2_3_, 2_ _3, 3_2_,3_ _2_, 2 _ 3 and_3_2

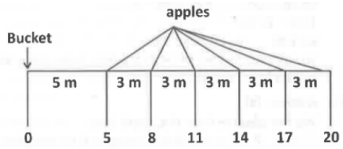
1is not to be immediately followed by 3 and 1 does not come at first place and 4 does not come at last place i.e., 1 _ _ _,_ _13, _ _ _ 4

So, the possible numbers satisfying all the conditions are: 2431, 2143, 3421, 3142, 3412 and 4312.

(6 numbers possible). **Hence, option (a) is correct.**

73. Answer: (d)

6 apples are placed 3 meters apart in a straight line and the bucket is 5 meters away from the first apple.



The competitor can pick one apple at a time and run back to drop in the bucket.

Distance travelled in picking the 1st apple and run back to drop it in the bucket = 5 + 5 = 10

Distance travelled in picking the 2nd apple and run back to drop it in the bucket = 2(5 + 3) = 16

Distance travelled in picking the 3rd apple and run back to drop it in the bucket = 2(5 + 6) = 22

Distance travelled in picking the 4th apple and run back to drop it in the bucket = 2(5 +9) = 28

Distance travelled in picking the 5th apple and run back to drop it in the bucket = 2(5 + 12) = 34

Distance travelled in picking the 6th apple and run back to drop it in the bucket = 2(5 + 15) = 40

So, total distance travelled = 10 + 16 + 22 + 28 + 34 + 40 =150 m.

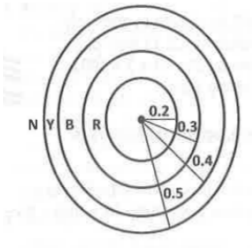
Hence, option (d) is correct.

74. Answer: (c)

Diameter of a round archery target = 1m. So, radius of the band is 0.5m.

It is marked with 4 scoring regions from the centre outward as red, blue, yellow and white.

The radius of red band is 0.20m and the width of all remaining bands is equal. So, the width of each band is 0.1m



Area of the archery target = $\pi r^2 = \pi(0.5)^2 = 0.25\pi$

Area of the red bend = $\pi r^2 = \pi(0.2)^2 0.04\pi$

Probability that the arrow falls in red region of the archery target 0.16.

Hence, option (c) is correct.

75. Answer: (c)

Given that there are two couples in A, B, Cand D.

Statement: "D is B's brother" implies D is male.

A is tallest of all implies A is male.

C is taller than B implies B is shortest.

Hence, A and D are male. B and C are female.

AB and CD are couples.

Option (c) is incorrect as C is male and males are taller than female.

Hence, option (c) is correct.

76. Answer: (b)

Option (a) is incorrect: it is said that daughters were invited to a feast, but they went or not it is not being told.

Option (b) is correct: wife is neither invited to feast and nor she is going

to picnic so she must be at home.

Option (c) is incorrect: as explained above.

Option (d) is incorrect: only wife is surely at home.

Hence, option (b) is correct.

77. Answer: (a)

Radha's statement reflects that meditation makes a person less aggressive. As Naresh has started meditation, his boring performance, which 'require aggression would not have improved. **Hence, option (a) is correct.**

78. Answer: (c)

Option (c): Mangroves have a crucial role in some of the coastal food chains.

This option directly aligns with the information provided, which highlights the role of mangroves in supporting a specific food chain in coastal areas.

79. Answer: (d)

The correct option is option D

Because army personnel's life and civilian life are very different in many aspects like life risk, standard of living,

food habits etc. so we cannot compare these two. Hence, option (d) is correct

80. Answer: (d)

The correct option is option D

Option (a) is not appropriate: No. of buses is not related with no. of accidents.

Option (b) is not appropriate: We cannot conclude this using above statement because accident depend on a lot of things.

Option (c) is not appropriate: We cannot conclude this using above statement because statement doesn't tell about skills of driver. **Hence, option (d) is correct.**

81. Answer: (b)

The correct option is option B

As it is explicitly mentioned in paragraph. "If political leadership fails to emerge, there is likelihood of military taking overpower in developing countries". **Hence, option (b) is correct.**

82. Answer: (d)

Option (d) is correct: Among all the options given, the best justification for "Liberty, therefore, is never real unless the Government can be called to account when it invades rights." would be "In the understanding that liberty and restraints are complementary".

83. Solution: (d)

he given statement says:

"Though quite expensive, television is not a luxury item, as one can learn many things through television."

From this, we can logically infer that **the main conclusion drawn is that television is not considered a luxury item, despite its high cost**, because it serves an educational purpose.

Let's examine the other options:

(a) "All expensive things are regarded as luxury."

This is **not supported**. In fact, the statement gives an example (television) where this is **not** true.

(b) "All essential things for learning are not luxury."

This is **too broad** and not implied by the statement. The statement only talks about television.

(c) "Television is essential for learning."

The statement says **one can learn many things** through television, but it does **not claim it is essential**.

Hence, only **option (d)** correctly captures the **inference** drawn in the statement.

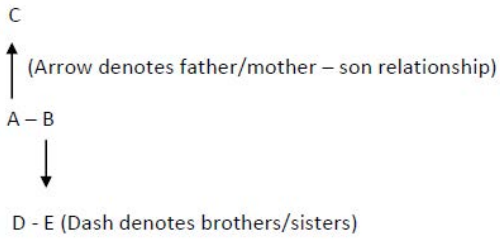
84. Solution: (a)

Simple symbols will get you the answer. R> G>S

Options (b), (c) and (d) are false.

85. Solution: (a)

Draw a family tree diagram and it becomes easy.



86. Solution: (d)

Again simple symbols.

S1: Rama>..... Rani

S2: Ratna>.....Rani

S3: Ratna>...>Rama>...Rani S4: Ratna>Padma>Rama>Rani Ratna scored the highest.

87. Solution: (c)

Take G or Y as the base of the cube and start folding the cube to fit adjacent sides. Use your imagination a little. Or write these letters in that order on a rubber (cuboid/cube) shape to see how the letters will align.

If you take G as the base, these will be opposite faces: GO, YR, BW. Even if you take Y as the base, you will find the same opposite faces.

88. Solution: (a)

Choose a sector that excludes P, but includes all others Q, R and S. It is 6.

89. Solution: (a)

If you observe first two figures, you will know that A, B, H and M are adjacent to K. So, letter opposite to A has to be either H or M.

Now, there are two conditions. One, if M is opposite to A, then H will be opposite to B. But, by third figure, we see it is not possible.

Hence, H has to be opposite to A.

90. Solution: (d)

In first and third set, the black portion of middle image is diagonally opposite to that of first image. This is the pattern. So, second image in second set must be diagonally opposite to the first one. Hence, (d).

91. Solution: (d)

Lets critically analyse each option.

O1: Just because all machines consume energy; and electricity provides energy, does not mean that electricity is the only source of energy. Machines can run on other sources of energy too. So it is wrong.

O2: Electricity provides energy does not mean it is only source providing energy. Hence, O2 is also wrong.

O3: In the light of the above discussion, O3 is also wrong.

O4: From statements 3 and 4, this seems the most reasonable option. Hence, (d)

92. Solution: (d)

Option (a) is incorrect because even if the rich can afford air travel, its not necessary that every rich person will travel by air.

Since (b) is also incorrect, (c) has to be incorrect. Option (d) is the correct answer.

93. Solution: (d)

Simple question. As the total population of Town A and Town B is not given we cannot determine the number of persons with diabetes.

6 Basic Numeracy

Numbers and their relations

1. The digit in the unit place of the number 6129 × 7307 is (CSAT-2026)

(a) 2 (b) 4

(c) 8 (d) 6

2. How many three-digit numbers can be expressed as an integral power of 2? (CSAT-2026)

(a) 1 (b) 2

(c) 3 (d) 4

3. How many times does 5 appear in all two-digit positive integers? (CSAT-2026)

(a) 18 (b) 19

(c) 20 (d) 21

4. If the product of the HCF and LCM of two distinct numbers is the cube of one of the numbers, then which of the following statements is/are correct? (CSAT-2026)

I. The difference of the numbers is an even number.

II. One of the numbers is a perfect square.

Select the answer using the code given below.

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

5. If x and y are two digits and the number 4x5y790 is divisible by 11, then what is the remainder, if x + y is divided by 11? (CSAT-2026)

(a) 1 (b) 3

(c) 5 (d) 7

6. A is a 2-digit number with different digits. B is also a 2-digit number and is obtained by reversing the digits of A. If A − B is a multiple of 27, where A > B, how many such different A's are possible? (CSAT-2026)

(a) 6 (b) 9

(c) 12 (d) 18

Directions for the next 5 (five) items :

Each item in this section contains a question followed by two statements. Answer each item using the following instructions and mark your response on the Answer Sheet accordingly.

(a) Select this option if the question can be answered using one of these statements alone, but cannot be answered using other statement

(b) Select this option if the question can be answered using either statement alone

(c) Select this option if the question can be answered using both the statements together, but cannot be answered using either statement alone

(d) Select this option if the question cannot be answered even using any of the statements

7. X receives three coins of different denominations: 1, 2, 5, 10 and 20. If the total amount received by X is m, does X receive a coin of denomination 5? (CSAT-2026)

Statement I: m is not a prime number.

Statement II: The sum of the digits of m is greater than 5.

8. For two distinct real numbers x and y, which of them is bigger? (CSAT-2026)

Statement I: $x^2 < y < 1$

Statement II: $y < \sqrt{x} < 1$

9. If x and y are integers, then is x even? (CSAT-2026)

Statement I: x^2y^2 is even.

Statement II: $1 + x^2 + y^2$ is odd.

10. X is a collection of certain odd numbers whereas Y is a collection of certain even numbers. T consists of the numbers all of which are either from X or from Y. Is every number of T from Y? (CSAT-2026)

Statement I: The sum of any two numbers belonging to T is even.

Statement II: If both p and q are picked from T, then $(p - 1)q$ is even.

11. If x, y and z are integers, each greater than 1, then is x a prime number? (CSAT-2026)

Statement I: $xy^2 = 116$

Statement II: $xz = 261$

12. X, Y and Z jump forward 4′, 6′ and 5′, respectively. At 8 AM, they all land on mark 199′. How many times will they all land on the same mark (need not be at the same moment) between mark 195′ and 1000′, if all of them cross mark 1000′ by 9 AM? (CSAT-2026)

(a) 11 (b) 12

(c) 13 (d) 14

13. A toy T jumps forward or backward. In each forward jump, it moves 5′ forward whereas in each backward jump, it moves 2′ backward. If in 31 jumps, T moves exactly 15′ forward, then what is the difference of the number of forward and backward jumps? (CSAT-2026)

(a) 6 (b) 7

(c) 8 (d) 9

14. If $10^m \times 1000 \times n = 75^{25} \times 25^{32} \times 32^{75}$, where n is not divisible by 10, then the value of m is (CSAT-2026)

(a) 101 (b) 111

(c) 121 (d) 131

15. If 7*24 = 25 and 12 * 16 = 20, then what is 16 * 63 equal to? (CSAT-2025)

(a) 70 (b) 66

(c) 65 (d) 64

16. The petrol price shot up by 10% as a result of the hike in crude oil prices. The price of petrol before the hike was Rs. 90 per litre. A person travels 2200 km every month and his car gives a mileage of 16 km per litre. By how many km should he reduce his travel if he wants to maintain his expenditure at the previous level? (CSAT-2025)

(a) 180 km (b) 200 km

(c) 220 km (d) 240 km

17. What is the unit digit in the multiplication of 1 x 3 x 5 x 7 x 9 x...x 999? (CSAT-2025)

(a) 1 (b) 3

(c) 5 (d) 9

18. There are n sets of numbers each having only three positive integers with LCM equal to 1001 and HCF equal to 1. What is the value of n? (CSAT-2025)

(a) 6 (b) 7

(c) 8 (d) More than 8

19. Let PQR be a 3-digit number, PPT be a 3-digit number and PS be a 2-digit number, where P, Q, R, S, T are distinct non-zero digits. Further, PQR - PS = PPT. If Q = 3 and T < 6, then what is the number of possible values of (R, S)? (CSAT-2025)

(a) 2 (b) 3

(c) 4 (d) More than 4

20. If n is a natural number, then what is the number of distinct remainders of (1^n+2^n) when divided by 4? (CSAT-2025)

(a) 0 (b) 1

(c) 2 (d) 3

21. Let PQQQ be a 3-digit number. What is the HCF of P and 481? (CSAT-2025)

(a) 1 (b) 13

(c) 37 (d) 481

22. What is the 489th digit in the number 1234567891011121.....? (CSAT-2025)

(a) 0 (b) 3

(c) 6 (d) 9

23. What is X in the sequence 1, 3, 6, 11, 18, X, 42? (CSAT-2025)

(a) 26 (b) 27

(c) 29 (d) 30

24. What is the remainder when (CSAT-2025)

$9^3 + 9^4 + 9^5 + 9^6 + ... + 9^{100}$ is divided by 6?

(a) 0 (b) 1

(c) 2 (d) 3

25. Let p + q = 10, where p, q are integers. (CSAT-2025)

Value-I = Maximum value of p x q when p. q are positive integers.

Value-II = Maximum value of p x q when $p \geq -6$, $q \geq -4$.

Which one of the following is correct?

(a) Value-I < Value-II

(b) Value-II < Value-I

(c) Value-I = Value-II

(d) Cannot be determined due to insufficient data

26. Let x be a real number between 0 and 1. Which of the following statements is/are correct? (CSAT-2025)

I. $x^2 > x^3$

II. $x > \sqrt{x}$

Select the correct answer using the code given below:

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

27. The difference between any two natural numbers is 10. What can be said about the natural numbers which are divisible by 5 and lie between these two numbers? (CSAT-2025)

(a) There is only one such number.

(b) There are only two such numbers.

(c) There can be more than one such number.

(d) No such number exists.

28. Let X be a two-digit number and Y be another two-digit number formed by interchanging the digits of X. If (X+Y) is the greatest two-digit number, then what is the number of possible values of X? (CSAT-2024)

(a) 2 (b) 4

(c) 6 (d) 8

29. Consider the following: (CSAT-2024)

Weight of 6 boys = Weight of 7 girls = Weight of 3 men = Weight of 4 women

If the average weight of the women is 63 kg, then what is the average weight of the boys?

(a) 40 kg (b) 42 kg

(c) 45 kg (d) 63 kg

30. Two persons P and Q enter into a business. P puts Rs. 14,000 more than Q, but P has invested for 8 months and Q has invested for 10 months. If P's share is Rs. 400 more than Q's share out of the total profit of Rs. 2,000, what is the capital contributed by P? (CSAT-2024)

(a) Rs. 30,000 (b) Rs. 26,000

(c) Rs. 24,000 (d) Rs. 20,000

31. 421 and 427, when divided by the same number, leave the same remainder 1. How many numbers can be used as the divisor in order to get the same remainder 1? (CSAT-2024)

(a) 1 (b) 2

(c) 3 (d) 4

32. A can X contains 399 litres of petrol and a can Y contains 532 litres of diesel. They are to be bottled in bottles of equal size so that whole of petrol and diesel would be separately bottled. The bottle capacity in terms of litres is an integer. How many different bottle sizes are possible? (CSAT-2024)

(a) 3 (b) 4

(c) 5 (d) 6

33. Consider the following statements in respect of the sum S = x+y+z, where x, y and z are distinct prime numbers each less than 10: (CSAT-2024)

1. The unit digit of S can be 0.
2. The unit digit of S can be 9.
3. The unit digit of S can be 5.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

34. What is the least possible number of cuts required to cut a cube into 64 identical pieces? (CSAT-2024)

(a) 8 (b) 9

(c) 12 (d) 16

35. In the expression 5*4*3*2*1, * is chosen from, +, -, x each at most two times. What is the smallest non- negative value of the expression? (CSAT-2024)

(a) 3 (b) 2

(c) 1 (d) 0

36. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

Age of each of P and Q is less than 100 years but more than 10 years. If you interchange the digits of the age of P, the number represents the age of Q.

Question:

What is the difference of their ages?

Statement-1:

The age of P is greater than the age of Q

Statement-II:

The sum of their ages is 11/6 times their difference.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the Statements together

37. If in a certain code, 'ABCD' is written as 24 and 'EFGH' is written as 1680, then how is 'IJKL' written in that code? (CSAT-2024)

(a) 11880 (b) 11240

(c) 7920 (d) 5940

38. If in a certain code, 'POT' is written as ATOP and 'TRAP' is written as APART, then how is 'ARENA' written in that code? (CSAT-2024)

(a) AARENA (b) AANREA

(c) AANEAR (d) AANERA

39. What will come in place of * in the sequence 3, 14, 39, 84, *, 258? (CSAT-2024)

(a) 150 (b) 155

(c) 160 (d) 176

40. In some code, letters P, Q, R, S, T represent numbers 4, 5, 10, 12, 15. It is not known which letter represents which number. If Q-S = 2S and T = R+S+3, then what is the value of P+R-T? (CSAT-2024)

(a) 1

(b) 2

(c) 3

(d) Cannot be determined due to insufficient data

41. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

Question:

If the average marks in a class are 60, then what is the number of students in the class?

Statement-1:

The highest marks in the class are 70 and the lowest marks are 50.

Statement-II:

Exclusion of highest and lowest marks from the class does not change the average.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the

Statements together

42. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

There are three distinct prime numbers whose sum is a prime number.

Question

What are those three numbers?

Statement-1: Their sum is less than 23.

Statement-II: One of the numbers is 5.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the Statements together

43. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

Question: Is (x+y) an integer?

Statement-I: (2x+y) is an integer.

Statement-II: (x+2y) is an integer.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the Statements together

44. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

A person buys three articles p, q and r for Rs. 50. The price of the article q is Rs. 16 which is the least.

Question: What is the price of the article p?

Statement-I: The cost of p is not more than that of r.

Statement-II: The cost of r is not more than that of p.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone.

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the Statements together

45. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

P, Q, R and S appeared in a test.

Question:

Has P scored more marks than Q?

Statement-I:

The sum of the marks scored by P and Q is equal to the sum of the marks scored by R and S.

Statement-II:

The sum of the marks scored by P and S is more than the sum of the marks scored by Q and R.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the Statements together

46. Let p and q be positive integers satisfying P<q and p+q = k. What is the smallest value of k that does not determine P and q uniquely? (CSAT-2024)

(a) 3 (b) 4

(c) 5 (d) 6

47. A Question is given followed by two Statements I and II. Consider the Question and the Statements. (CSAT-2024)

Question:

What are the values of m and n, where m and n are natural numbers?

Statement-I:

m +n > mn and m > n.

Statement-II:

The product of m and n is 24.

Which one of the following is correct in respect of the above Question and the Statements?

(a) The Question can be answered by using one of the Statements alone, but cannot be answered using the other Statement alone

(b) The Question can be answered by using either Statement alone

(c) The Question can be answered by using both the Statements together, but cannot be answered using either Statement alone

(d) The Question cannot be answered even by using both the Statements together

48. What is the sum of the first 28 terms in the following sequence?

1, 1, 2, 1, 3, 2, 1, 4, 3, 2, 1, 5, 4,3, 2, …. **(CSAT-2024)**

(a) 83 (b) 84

(c) 85 (d) 86

49. A person buys three articles P, Q and R for Rs. 3,330. If P costs 25% more than R and R costs 20% more than Q, then what is the cost of P? (CSAT-2024)

(a) Rs. 1,000 (b) Rs. 1,200

(c) Rs. 1,250 (d) Rs. 1,350

50. If the sum of the two-digit numbers AB and CD is the three-digit number 1CE, where the letters A, B, C, D, E denote distinct digits, then what is the value of A? (CSAT-2024)

(a) 9

(b) 8

(c) 7

(d) Cannot be determined due to insufficient data

51. Three numbers x, y, z are selected from the set of the first seven natural numbers such that x>2y>3z. How many such distinct triplets (x, y, z) are possible? (CSAT-2024)

(a) One triplet (b) Two triplets

(c) Three triplets (d) Four triplets

52. What is the remainder when 85 x 87 x 89 x 91 x 95 x 96 is divided by 100? (CSAT-2023)

(a) 0 (b) 1

(c) 2 (d) 4

53. A 3-digit number ABC, on multiplication with D gives 37DD where A, B, C and D are different non-zero digits.

What is the value of A+B+C? (CSAT-2023)

(a) 18 (b) 16

(c) 15 (d) Cannot be determined

54. For any choices of values of X, Y and Z, the 6-digit number of the form XYZXYZ s divisible by: (CSAT-2023)

(a) 7 and 11 only (b) 11 and 13 only

(c) 7 and 13 only (d) 7, 11 and 13

55. Let x be a positive integer such that 7x + 96 is divisible by x. How many values of x are possible? (CSAT-2023)

(a) 10 (b) 12

(c) 12 (d) Infinitely many

56. A number N is formed by writing 9 for 99 times. What is the remainder if N is divided by 13? (CSAT-2023)

(a) 11 (b) 9

(c) 7 (d) 1

57. 46. Each digit of a 9-digit number is 1. It is multiplied by itself. What is the sum of the digits of the resulting number? (CSAT-2023)

(a) 64 (b) 80

(c) 81 (d) 100

58. Choose the group which is different from the others: (CSAT-2023)

(a) 17,37,47,97 (b) 31,41,53,67

(c) 71,73,79,83 (d) 83,89,91,97

59. How many natural numbers are there which give a remainder of 31 when 1186 is divided by these natural numbers? (CSAT-2023)

(a) 6 (b) 7

(c) 8 (d) 9

60. What is the value of X in the sequence 20, 10, 10, 15, 30, 75, X? (CSAT-2022)

(a) 105 (b) 150

(b) 120 (d) 225

61. How many 3-digit natural numbers (without repetition of digits) are there such that each digit is odd, and the number is divisible by 5? (CSAT-2022)

(a) 8 (b) 12

(b) 16 (d) 24

62. The increase in the price of a certain item was 25% Then the price was decreased by 20% and then again increased by 10%. What is the resultant increase in the price? (CSAT-2022)

(a) 5% (c) 12.5%

(b) 10% (d) 15%

63. What is the remainder when 91 × 92 × 93 × 94 × 95 × 96 × 97 × 98 × 99 is divided by 1261? (CSAT-2022)

(a) 3 (c) 1

(b) 2 (d) 0

64. Two candidates X and Y contested an election. 80% of voters cast their vote and there were no invalid votes, There was no NOTA (None of the above) option. X got 56% of the votes cast and won by 1440 votes. What is. the total number of voters in the voters list? (CSAT-2022)

(a) 15000 (c) 9600

(b) 12000 (d) 5000

65. What is the smallest number greater than 1000 that when divided by any one of the numbers 6, 9,12, 15,18 leaves a remainder of 3? (CSAT-2022)

(a) 1063 (b) 1073

(c) 1083 (d) 1183

66. Let p be a two-digit number and q be the number consisting of same digits written in reverse order. If p × g = 2430 then what is

the difference between p and q? (CSAT-2022)

(a) 45 (b) 27

(c) 18 (d) 9

67. Consider the following statements in respect of two natural numbers p and q such that p is a prime number and q is a composite number: (CSAT-2022)

1. p × q can be an odd number.
2. q / p can be a prime number.
3. p + q can be a prime number.

Which of the above statements are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

68. What is the value of X in the sequence 2, 12, 36, 80, 150, x? (CSAT-2022)

(a) 248 (b) 252

(c) 258 (d) 262

69. What is the number of numbers of the form 0. XY, where X and Y are distinct non-zero digits? (CSAT-2022)

(a) 72 (b) 81

(c) 90 (d) 100

70. A person X wants to distribute some pens among six children A, B, C, D, E and F. Suppose A gets twice the number of pens received by B, three times that of C, four times that of D, five times that of E and six times that of F. What is the minimum number of pens X should buy so that the number of pens each one gets Is an even number? (CSAT-2022)

(a) 147 (b) 150

(c) 294 (d) 300

71. Three persons A, B and C are standing in a queue not necessarily in the same order. There are 4 persons between A and B, and 7 persons between B and C, s there are 11 persons ahead of C and 13 behind A, what could be the minimum number of persons in the queue? (CSAT-2022)

(a) 22 (c) 32

(b) 28 (d) 38

72. The average weight of A, B, C is 40 kg, the average weight of B, D, E is 42 kg and the weight of F is equal to that of B. What is the average weight of A, B, C, D, E and F? (CSAT-2022)

(a) 40.5 kg

(b) 40.8 kg

(c) 41 kg

(d) Cannot be determined as data is inadequate

73. When 70% of a number x is added to another number Y, the sum becomes 165% of the value of y. When 60% of the number x is added to another number z, then the sum becomes 165% of the value of z. which one of the following is correct? (CSAT-2022)

(a) Z<X<Y (b) X<Y<Z

(c) Y<X<Z (d) Z<Y<X

74. There are two containers X and X contains 100 ml of milk and Y contains 100 ml of water. 20 ml of milk from X is transferred to Y. After mixing well, 20 ml of the mixture in Y is transferred back to X. If m denotes the proportion of milk in X and n denotes the proportion of water in Y then which one of the following is correct? (CSAT-2022)

(a) m=n

(b) m>n

(c) m<n

(d) Cannot be determined due to insufficient data

75. 15 x 14 x13 x....x 3 x 2 x 1= 3^mx n Where m and n are positive integers, then what is the maximum value of m? (CSAT-2022)

(a) 7 (b) 6

(c) 5 (d) 4

76. A pie chart gives the expenditure on five different items A, B, C, D and E in a household. If B, C, D and E correspond to 90°, 50°, 45°, and 75° respectively, then what is the percentage of expenditure on item A? (CSAT-2022)

(a) 112/9 (b) 125/6

(c) 155/9 (d) 250/9

77. The number 3798125P369 is divisible by 7. What is the value of the digit P? (CSAT-2021)

(a) 1 (b) 6

(c) 7 (d) 9

78. A student appeared in 6 papers. The maximum marks are the same for each paper. His marks in these papers are in the proportion of 5:6:7:8:9:10. Overall he scored 60%. In how many number of papers did he score less than 60% of the maximum marks? (CSAT-2021)

(a) 2 (b) 3

(c) 4 (d) 5

79. There are two Classes A and B having 25 and 30 students, respectively. In Class-A the highest score is 21 and lowest score is 17. In Class-B the highest score is 30 and lowest score is 22. Four students are shifted from Class-A to Class-B.

Consider the following statements:

1. The average score of Class-B will definitely decrease
2. The average score of Class-A will definitely increase

Which of the above statements is/are correct? (CSAT-2021)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

80. Half of the villagers of a certain village have their own houses. One-fifth of the villagers cultivate paddy. One third of the villagers

are literate. Four-fifth of the villagers are under 25 years of age. Which one of the following statements is certainly correct? **(CSAT-2021)**

(a) All the villagers who have their own houses are literate.

(b) Some villagers are under 25 years of age are literate.

(c) Only half of the villagers who cultivate paddy are literate.

(d) No villager under 25 years of age has his own house.

81. P scored 40 marks more than Q in an examination. If Q scored 10% less marks than P, then how much did Q score? **(CSAT-2021)**

(a) 360 (b) 380

(c) 400 (d) 420

82. A person P asks one of his three friends X as to how much money he had. X replied, "If Y given me ₹40, then Y will have half of as much as Z, but if Z gives me ₹40, then three of us will have equal amount." What is the total amount of money that X, Y and Z have? **(CSAT-2021)**

(a) ₹420 (b) ₹360

(c) ₹300 (d) ₹270

83. In an objective type test of 90 questions, 5 marks are allotted for every correct Answer and 2 marks are deducted for every wrong Answer. After attempting all the 90 questions, a student got a total of 387 marks.

What is the number of incorrect response? **(CSAT-2021)**

(a) 9 (b) 13

(c) 27 (d) 43

84. Consider the following statements:

1. The sum of 5 consecutive integers can be 100.
2. The product of three consecutive natural numbers can be equal to their sum. (CSAT-2021)

Which of the above statements is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

85. An amount of money was distributed among A, B and ¢ in the ratio p:q:r.

Consider the following statements:

1. A gets the maximum share if p is greater than (q + r)
2. C gets the minimum share if r is less than (p + Q)

Which of the above statements is/are correct? (CSAT-2021)

(a) 1 only (c) Both 1and 2

(b) 2 only (d) Neither 1 nor 2

86. The difference between a 2-digit number and the number obtained by interchanging the positions of the digits is 54.

1. The sum of the two digits of a number can be determined only if the product of the two digits is known.
2. The difference between the two digits of the number can be determined.

Which of the above statements is/are correct? (CSAT-2021)

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

87. If the price of an article is decreased by 20% and then the new price is increased by 25%, then what is the net change in the price? **(CSAT-2021)**

(a) 0%

(b) 5% increase

(c) 5% decrease

(d) Cannot be determined due to insufficient data

88. How many zeroes are there at the end of the following product? 1x5x 10 15 x 20 x 25 x 30 X 35 x 40 x 45 %50 % 55 x 60 (CSAT-2020)

(a) 10 (b) 12

(c) 14 (d) 15

89. Let XYZ be a three-digit number, where (X + Y + Z) is not a multiple of 3. Then (XYZ + YZX + ZXY) is not divisible by **(CSAT-2020)**

(a) 3 (c) 37

(b) 9 (d) (X+Y+2)

90. Let p, g, r and s be natural numbers such that p — 2016 = q+2017 =r-2018 =5 +2019. Which one of the following is the largest natural number? **(CSAT-2020)**

(a) p (b) q

(c) r (d) s

91. How many five-digit prime numbers can be obtained by using all the digits 1, 2, 3, 4 and 5 without repetition of digits? (CSAT-2020)

(a) zero (b) One

(c) Nine (d) Ten

92. How many integers are there between 1 and 100 which have 4 as a digit but are not divisible by 4? **(CSAT-2020)**

(a) 5 (b) 11

(c) 12 (d) 13

93. Let x, y be the volumes; m, n be the masses of two metallic cubes P and Q respectively. Each side of Q is two times that of P and mass of Q is two times that of P. Letu=m/x and v =n /y. Which one of the following is correct? **(CSAT-2020)**

(a) u=4v (b) u=2v

(c) v=u (d) v=4u

94. The average age of a teacher and three students s 20 years. If all the three students are of same age and the difference between the age of the teachers and each student is 20 years, then what is the age of

the teacher? **(CSAT-2020)**

(a) 25 years (b) 30 years

(c) 35 years (d) 45 years

95. A person bought a car and sold it for Rs. 3, 00, 000. If he incurred a loss of 20%, then how much did he spend to buy the car? **(CSAT-2020)**

(a) Rs. 3,60,000 (c) Rs. 3,70,000

(b) Rs. 3,65,000 (d) Rs. 3,75,000

96. In the sum ⊗+1⊗+5⊗+⊗⊗+⊗1=1⊗⊗ for which digit does the symbol ® stand? **(CSAT-2020)**

(a) 2 (b) 3

(c) 4 (d) 5

97. If you have two straight sticks of length 7.5 feet and 3.25 feet, what is the minimum length can your measure? **(CSAT-2020)**

(a) 0.05 foot (b) 0.25 foot

(c) 1 foot (d) 3.25 feet

98. A simple mathematical operation in each number of sequences 14, 18, 20, 24, 30, 32.. results in a sequence with respect to prime numbers. Which one of the following is the next number in the sequence? **(CSAT-2020)**

(a) 34 (b) 36

(c) 38 (d) 40

99. One page is torn from a booklet whose pages are numbered in the usual manner starting from the first page as 1. The sum of the numbers on the remaining pages is 195. The torn page contains which of the following numbers. **(CSAT-2020)**

(a) 5,6 (b) 7,8

(c) 9,10 (d) 11,12

100. Let A3BC and DE2F be four-digit numbers where each letter represents a different digit greater than 3. If the sum of the numbers is 15902, then what is the difference between the values of A and D? **(CSAT-2020)**

(a)1 (b) 2

(c) 3 (d) 4

101. What is the largest number among the following? **(CSAT-2020)**

(a) $(1/2)^{-6}$ (b) $(1/4)^{-3}$

(c) $(1/3)^{-4}$ (d) $(1/6)^{2}$

102. A person x can complete 20% of work in 8 days and another person Y can complete 25% of the same work in 6 days. If they work together, in how many days will 40% of the work by completed? **(CSAT-2020)**

(a) 6 (b) 8

(c) 10 (d) 12

103. How many pairs of natural numbers are there such that the difference of whose squares is 63? **(CSAT-2020)**

(a) 3 (b) 4

(c) 5 (d) 2

104. A digit n >3 is divisible by 3 but not divisible by 6. Which one of the following is divisible by 4? **(CSAT-2020)**

(a) 2n (b) 3n

(c) 2n+4 (d) 3n+1

105. The recurring decimal representation 1.272727 ... is equivalent to **(CSAT-2020)**

(a) 13/11 (b) 14/11

(c) 127/99 (d) 137/99

106. What is the least four-digit number when divided by 3, 4, 5 and 6 leaves a remainder 2 in each case? **(CSAT-2020)**

(a) 1012 (b) 1022

(c) 1122 (d) 1222

107. What is the remainder when 51 x 27 x 35 x 62 x 75 is divided by 100? **(CSAT-2020)**

(a) 50 (b) 25

(c) 5 (d) 1

108. For what value of n, the sum of digits in the number (10n +1) is 2? **(CSAT-2020)**

(a) For n=0 only

(b) For any whole number n

(c) For any positive integer n only

(d) For any real number n

109. If 1 liter of water weighs 1 kg, then how many cubic millimeters of water will weigh 0.1gm.? **(CSAT-2020)**

(a) 1 (b) 10

(c) 100 (d) 1000

110. A vessel full of water weight 40 kg. If it is one-third filled, its weight becomes 20 kg. What is the weight of the empty vessel? **(CSAT-2020)**

(a) 10 kg (b) 15 kg

(c) 20 kg (d) 25 kg

111. A frog tries to come out of a dried well 4.5 m deep with slippery walls. Every time the frog jumps 30 cm, slides down 15 cm. What is the number of jumps required for the frog to come out of the well? **(CSAT-2020)**

(a) 28 (b) 29

(c) 30 (d) 31

112. The number of times the digit 5 will appear while writing the integers from 1 to 1000 is **(CSAT-2019)**

(a) 269 (b) 271

(c) 300 (d) 302

113. In a school every student is assigned a unique identification number. A student is a football player if and only if the identification number is divisible by 4, whereas a student is a cricketer if and only if the identification number is divisible by 6. If every number from 1 to 100 is assigned to a student, then how many of them play cricket as well as football? (CSAT-2019)

(a) 4 (b) 8

(c) 10 (d) 12

114. The ratio of a two-digit natural number to a number formed by reversing its digits is 4:7. The number of such pairs is (CSAT-2019)

(a) 5 (b) 4

(c) 3 (d) 2

115. A printer numbers pages of a book starting with 1 and uses 3089 digits in all. How many pages does the book have? (CSAT-2019)

(a) 1040 (b) 1048

(c) 1049 (d) 1050

116. Number 136 is added to 587 and the sum obtained is 7A3, where A and B are integers. It is given that 7A3 is exactly divisible by 3. The only possible value of B is (CSAT-2019)

(a) 2 (b) 5

(c) 7 (d) 8

117. An 8-digit number 4252746B leaves remainder 0 when divided by 3. How many values of B are possible? (CSAT-2019)

(a) 2 (b) 3

(c) 4 (d) 6

118. Certain 3-digit numbers following characteristics:

1. All the three digits are different.
2. The number is divisible by 7.
3. The number on reversing the digits is also divisible by 7.

How many such 3-digit numbers are there? (CSAT-2017)

(a) 2 (b) 4

(c) 6 (d) 8

119. How many numbers are there between 99 and 1000 such that the digit 8 occupies the units place? (CSAT-2017)

(a) 64 (b) 80

(c) 90 (d) 104

120. The monthly incomes of X and Y are in the ratio of 4: 3 and their monthly expenses are in the ratio of 3: 2. However, each saves 6,000 per month. What is their total monthly income? (CSAT-2017)

(a) ₹28,000 (b) ₹42,000

(c) ₹56,000 (d) ₹84,000

121. There are certain 2-digit numbers. The difference between the number and the one obtained on reversing it is always 27. How many such maximum 2-digit numbers are there? (CSAT-2017)

(a) 3 (b) 4

(c) 5 (d) None of the above

122. What is the total number of digits printed, if a book containing 150 pages is to be numbered from 1 to 150? (CSAT-2017)

(a) 262 (b) 342

(c) 360 (d) 450

123. There is a milk sample with 50% water in it. If 1/3rd of this milk is added to equal amount of pure milk, then water in the new mixture will fall down to (CSAT-2017)

(a) 25% (b) 30%

(c) 35% (d) 40%

124. In aid of charity, every student in a class contributes as many rupees as the number of students in that. class. With the additional contribution of Rs. 2 by one student only, the total collection is Rs. 443. Then how many students are there in the class? (CSAT-2016)

(a) 12 (c) 43

(b) 21 (d) 45

125. An agricultural field is in the form of a rectangle having length X_1 meters and breadth X_2 meters (X_1 and X_2 are variable). If $X_1 + X_2 = 40$ meters, then the area of the agricultural field will not exceed which one of the following values? (CSAT-2016)

(a) 400 sq m (c) 200 sq m

(b) 300 sq m (d) 80 sq m

126. The total emoluments of two persons are the same, but one gets allowances to the extent of 65% of his basic pay and the other gets allowances to the extent of 80% of his basic pay. The ratio of the basic pay of the former to the basic pay of the latter is (CSAT-2016)

(a) 16:13 (c) 7:5

(b) 5:4 (d) 12:11

127. A cube has all its faces painted with different colours, It is cut into smaller cubes of equal sizes such that the side of the small cube is one-fourth the big cube. The number of small cubes with only one of the sides painted is: (CSAT-2016)

(a) 32 (c) 16

(b) 24 (d) 8

128. W can do 25% of a work in 30 days, X can do 1/4 of the work in 10 days, Y can do 40% of the work in 40 days and Z can do 1/3 of the work in 13 days. Who will complete the work first? (CSAT-2016)

(a) W (c) Y

(b) X (d) Z

129. The average monthly income of a person in a certain family of 5 is Rs. 10,000. What will be the average monthly income of a person in the same family if the income of one person increased by Rs. 1, 20,000 per year? (CSAT-2016)

(a) Rs. 12,000 (c) Rs. 20,500

(b) Rs. 16,000 (d) Rs. 34,000

130. A person allows a 10% discount for cash payment from the marked price of a toy and still he makes a 10% gain What is the cost price of the toy which is marked Rs. 770? (CSAT-2016)

(a) Rs. 610 (c) Rs. 630

(b) Rs. 620 (d) Rs. 640

131. If ABC x DEED = ABCABC, where A, B, C, D and E are different digits, what are the values of D and E? (CSAT-2015)

(a) D=2, E=0 (b) D=0, E=1

(c) D=1, E=0 (d) D=1, E=2

132. Price is not the same thing as value. Suppose that on a day the price of everything viz., coal, bread, postage stamps, a day's labour, the rent of houses, etc. were to double. Prices then would certainly rise, but- values of all things except one would not." The writer wants to say that if prices of all things were doubled (CSAT-2014)

(a) the values of all things would remain constant.

(b) the values of the things sold would be doubled.

(c) the values of the things bought would be halved

(d) the value of money only would be halved.

133. The letters L, M, N, O, P, Q, R, S, and T in their order are substituted by nine integers 1 to 9 but not in that order. 4 is assigned to P. The difference between P and T is 5. The difference between N and T is 3. What is the integer assigned to N? (CSAT-2014)

(a) 7 (b) 5

(c) 4 (d) 6

134. How many females voted for the proposal? (CSAT-2011)

(a) 430 (b) 600

(c) 624 (d) 640

135. How many males were undecided? (CSAT-2011)

(a) 31 (b) 227

(c) 426 (d) 581

136. How many females were not in favour the proposal? (CSAT-2011)

(a) 430 (b) 496

(c) 586 (d) 1226

Orders of magnitude

137. $222^{333} + 333^{222}$ is divisible by which of the following numbers? (CSAT-2024)

(a) 2 and 3 but not 37 (b) 3 and 37 but not 2

(c) 2 and 37 but not 3 (d) 2, 3 and 37

138. What is the right most digit preceding the zeros in the value of 30^{30}? (CSAT-2024)

(a) 1 (b) 3

(c) 7 (d) 9

139. How many consecutive zeros are there at the end of the integer obtained in the product $1^2 \times 2^4 \times 3^6 \times 4^8 \times \ldots \times 25^{50}$? (CSAT-2024)

(a) 50 (b) 55

(c) 100 (d) 200

140. $32^5 + 22^7$ is divisible by (CSAT-2024)

(a) 3 (b) 7

(c) 10 (d) 11

141. Three of the five positive integers p, g, 1, s, t are even and two of them are odd (not necessarily in order). Consider the following:

1. p + q + r – s - t is definitely even
2. 2p + q + 2r - 2s + t is definitely odd.

Which of the above statements is/are correct? (CSAT-2023)

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

142. Consider the following in respect of prime number p and composite number c.

1. $\frac{p+c}{p-c}$ can be even
2. 2p+c can be odd
3. pc can be odd.

Which of the statements given above are correct? (CSAT-2023)

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

143. Which number amongst 2%, 321, 438 and 8% is the smallest? (CSAT-2022)

(a) 2^{40} (c) 4^{18}

(b) 3^{21} (d) 8^{12}

144. A has some coins. He gives half of the coins and 2 more to B. B gives half of the coins and 2 more to C. C gives half of the coins and 2 more to D. The number of coins D has now, is the smallest two-digit number. How many coins do A have in the beginning? (CSAT-2022)

(a) 76 (c) 60

(b) 68 (d) 52

145. If 3^{2019} is divided by 10, then what is the remainder? (CSAT-2021)

(a) 1 (b) 3

(c) 7 (d) 9

146. What is X in the sequence 132, 129, 124, 117, 106, 93, X? (CSAT-2019)

(a) 74 (b) 75

(c) 76 (d) 77

147. What is X in the sequence 4, 196, 16, 144, 36, 100, 64, X? (CSAT-2019)

(a) 48 (b) 64

(c) 125 (d) 256

148. In a sequence 1, 5, 7, 3, 5, 7, 4, 3, 5, 7, how many such 5s are there which are not immediately preceded by 3 but are immediately followed by 7? (CSAT-2019)

(a) 1 (b) 2

(c) 3 (d) None

149. The age of Mr. X last year was the square of a number and it would be the cube of a number next year. What is the least number of years he must wait for his age to become the cube of a number again? (CSAT-2017)

(a) 42 (b) 38

(c) 25 (d) 16

150. 30 g of sugar was mixed in 180 ml water in a vessel A, 40 g of sugar was mixed in 280 ml of water in vessel B and 20 g of sugar was mixed in 100 ml of water in vessel C. The solution in vessel B is: (CSAT-2016)

(a) sweeter than that in C (c) as sweet as that in C

(b) sweeter than that in A (d) less sweet than that in C

Miscellaneous

151. P's salary is 20% lower than Q's salary which is 20% lower than R's salary. By how much percent is R's salary more than P's salary? (CSAT-2024)

(a) 48.75% (b) 56.25%

(c) 60.50% (d) 62.25%

152. A number is mistakenly divided by 4 instead of multiplying by 4. What is the percentage change in the result due to this mistake? (CSAT-2024)

(a) 25% (b) 50%

(c) 72.75% (d) 93.75%

153. In an examination, 80% of students passed in English, 70% of students passed in Hindi and 15% failed in both the subjects. What is the percentage of students who failed in only one subject? (CSAT-2024)

(a) 15% (b) 20%

(c) 25% (d) 35%

154. A father said to his son, "n years back I was as old as you are now. My present age is four times your age n years back". If the sum of the present ages of the father and the son is 130 years, what is the difference of their ages? (CSAT-2024)

(a) 30 years (b) 32 years

(c) 34 years (d) 36 years

155. What percent of water must be mixed with honey so as to gain 20% by selling the mixture at the cost price of honey?(CSAT-2024)

(a) 20% (b) 10%

(c) 5% (d) 4%

156. If a + b means a - b, a-b means a × b; a × b means a + b; a+b means a+b, then what is the value of 10+30-100×50+25? (Operations are to be replaced simultaneously) (CSAT-2024)

(a) 15 (b) 0

(c) -15 (d) -25

157. If P means 'greater than (>)'; Q means less than (<); R means 'not greater than (≯); S means 'not less than (≮) and T means 'equal to (=)', then consider the following statements: (CSAT-2024)

1. If 2x (S)3y and 3x(T)4z, then 9y(P)8z.
2. If x (Q)2y and y(R)z, then x(R)z.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

158. What is the unit digit in the expansion of $(57242)^{9x7x5x3x1}$? (CSAT-2023)

(a) 2 (b) 4

(c) 6 (d) 8

159. How many distinct 8-digit numbers can be formed by rearranging the digits of the number 11223344 such that odd digits occupy odd positions and even digits occupy even positions? (CSAT-2023)

(a) 12 (b) 18

(c) 36 (d) 72

160. If p, q, r and s are distinct single digit p positive numbers, then what is the greatest value of (p + q) (r + s)? (CSAT-2023)

(a) 230 (b) 225

(c) 224 (d) 221

161. What is the sum of all digits which appear in all the integers from 10 to 100? (CSAT-2023)

(a) 855 (b) 856

(c) 910 (d) 911

162. Let pp, qq and rr be 2-digit numbers where p < q < r. If pp +q + r = tt0, where tt0 is a 3-digit number ending with zero, consider the following statements:

1. The number of possible values of p is 5.
2. The number of possible values of q is 6.

Which of the above statements is/are correct? (CSAT-2023)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

163. What is the sum of all 4-digit numbers less than 2000 formed by the digits 1, 2, 3 and 4, where none of the digits is repeated? (CSAT-2023)

(a) 7998 (b) 8028

(c) 8878 (d) 9238

164. What is the remainder if 2192 is divided by 6? (CSAT-2023)

(a) 0 (b) 1

(c) 2 (d) 4

165. The digits 1 to 9 are arranged in three rows in such a way that each row contains three digits, and the number formed in the second row is twice the number formed in the first row; and the number formed in the third row is thrice the number formed in the first row. Repetition of digits is not allowed. I only three of the four digits 2, 3, 7 and 9 are allowed to use in the first row, how many such combinations are possible to be arranged in the three rows? (CSAT-2022)

(a) 4 (c) 2

(b) 3 (d) 1

166. A bill for 1,840 is paid in the denominations of 50, 20 and 10 notes. 50 notes in all are used. Consider the following statements: (CSAT-2022)

1. 25 notes of 50 are used and the remaining is in the denominations of 20 and 10.

2. 35 notes of 20 are used and the remaining is in the denominations of 50 and 10.

3. 20 notes of 10 are used and the remaining is in the denominations of 50 and 20.

Which of the above statements are not correct?

(a) 1 and 2 only (c) 1 and 3 only

(b) 2 and 3 only (d) 1,2 and 3

167. From January 1, 2021, the price of petrol (in Rupees per litre) on mt" day of the year is 80 + 0.1m, where m = 1, 2, 3… 100 and thereafter remains constant. On the other hand, the price of diesel (in Rupees per litre) on n' day of 2021 is 69 + 0.15n for any n. On which date in the year 2021 are the prices of these two fuels equal? (CSAT-2021)

(a) 21st May (c) 19th May

(b) 20th May (d) 18th May

168. In a group of 120 persons, 80 are Indians and rest are foreigners. Further, 70 persons in the group can speak English. The number of Indians who can speak English is (CSAT-2021)

(a) 20 (c) 30 or less

(b) 30 (d) 30 or more

169. X said to Y, "At the time of your birth I was twice as old as you are at present." If the present age of X is 42 years, then consider the following statements:

1. 8years ago, the age of X was five times the age of Y.

2. After 14 years, the age of X would be two times the age of Y.

Which of the above statements is/are correct? (CSAT-2021)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

170. Which one of the following will have minimum change in its value if 5 is added to both numerator and the denominator of the fractions 2/3, 3/4, 4/5 and 5/6? (CSAT-2020)

(a) 2/3 (b) 3/4

(c) 4/5 (d) 5/6

171. How many different sums can be formed with the denomination Rs. 50, Rs. 100, Rs. 200, Rs. 500, and Rs. 2,000 taking at least three denominations at a time? (CSAT-2020)

(a) 16 (b) 15

(c) 14 (d) 10

172. P= (40% of A) + (65% of B) and Q = (50% of A) + (50% of B), where A is greater than B.

In this context, which of the following statements is correct? (CSAT-2017)

(a) P is greater than Q.

(b) Q is greater than P.

(c) P is equal to Q.

(d) None of the above can be concluded with certainty.

173. Gopal bought a cell phone and sold it to Ram at 10% profit. Then Ram wanted to sell it-back to Gopal at 10% loss. What will be Gopal's position if he agreed? (CSAT - 2017)

(a) Neither loss nor gain (b) Loss 1%

(c) Gain 1% (d) Gain 0.5%

174. There are some nectar-filled flowers on a tree and some bees are hovering on it. If one bee lands on each flower, one bee will be left out. If two bees land on each flower, one flower will be left out. The number of flowers and bees respectively are: (CSAT-2016)

(a) 2 and 4 (c) 3 and 4

(b) 3 and 2 (d) 4 and 3

175. There is an order of 19000 quantity of a particular product from a customer. The firm produces 1000 quantity of that product per day out of which 5% are unfit for sale. In how. many days will the order be completed? (CSAT-2016)

(a) 18 (c) 20

(b) 19 (d) 22

176. An automobile owner reduced his monthly petrol consumption when the prices went up. The price consumption relationship is as follows:

Price	40	50	60	75
Consumption	60	48	40	32

If the price goes up to Rs. 80 per litre, his expected consumption (in litres) will be: (CSAT-2015)

(a) 30 (b) 28

(c) 26 (d) 24

177. In a test, a candidate attempted only 8 questions ang secured 50% marks in each of the questions. If he. obtained a total of 40% in the test and all questions in the test carried equal marks, how many questions were. there in the test? (CSAT-2015)

(a) 8 (b) 10

(c) 15 (d) 16

178. Candidates in a competitive examination consisted of 60% men and 40% women. 70% men and 75% women cleared the qualifying test and entered the final test where 80% men and 70% women were successful. Which of the following statements is correct? (CSAT-2015)

(a) Success rate is higher for women

(b) Overall success rate is below 50%

(c) More men cleared the examination than women.

(d) Both (a) and (b) above are correct.

179. A cow costs more than 4 goats but less than 5 goats. If a goat costs between Rs. 600 and Rs. 800, which of the following is a most valid

conclusion? **(CSAT-2015)**

(a) A cow costs more than Rs. 2,500

(b) A cow costs less than Rs. 3,600

(c) A cow costs between Rs. 2,600 and Rs. 3,800

(d) A cow costs between Rs. 2,400 and Rs. 4,000

180. Two equal glasses of same type are respectively 1/3 and 1/4 full of milk. They are then filled up with water and the contents are mixed in a pot. What is the ratio of milk and water in the pot? **(CSAT-2015)**

(a) 7:17 (b) 1:3

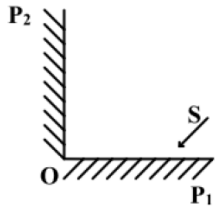
(c) 9:21 (d) 11:23

181. Two glasses of equal volume are respectively half and three-fourths filled with milk. They are then filled to the brim by adding water. Their contents are then poured into another vessel. What will be the ratio of milk to water in this vessel? **(CSAT-2012)**

(a) 1:3 (b) 2:3

(c) 3:2 (d) 5:3

182. Consider the figure given below and answer the items that follows: **(CSAT-2011)**



In the figure shown above, OP 1 and OP 2 are two plane mirrors kept perpendicular to each other. S is the direction of a beam of light falling on the mirror OP 1. The direction of the reflected beam of light from the mirror OP 2 will be

(a) Perpendicular to the direction S.

(b) At 45° to the direction S.

(c) Opposite and parallel to the direction S.

(d) At 60° to the direction S.

183. There are 100 students in a particular class. 60% students play cricket, 30% student play football and 10% student play both the games. What is the number of students who play neither cricket nor football? **(CSAT-2011)**

(a) 25 (b) 20

(c) 18 (d) 15

184. A student on her first 3 tests receives on an average score of N points. If she exceeds her previous average score by 20 points on her fourth test, then what is the average score for the first 4 tests? **(CSAT-2011)**

(a) N + 20 (b) N + 10

(c) N + 4 (d) N + 5

185. In a group of persons, 70% of the persons are male and 30% of the persons are married. If two sevenths of males are married, what fraction of the females is single ? **(CSAT-2011)**

(a) 2/7 (b) 1/3

(c) 3/7 (d) 2/3

186. A natural number N is such that it can be expressed as N = p + q+r, where p, q and r are distinct factors of N. How many numbers below 50 have this property? **(CSAT-2025)**

(a) 6 (b) 7

(c) 8 (d) 9

187. Three prime numbers p, q and r, each less than 20, are such that p-q=q-r. How many distinct possible values can we get for (p+q+r)? **(CSAT-2025)**

(a) 4 (b) 5

(c) 6 (d) More than 6

188. How many possible values of (p+q+r) are there satisfying 1/p + 1/q + 1/r=1, where p, q and r are natural numbers (not necessarily distinct)? **(CSAT-2025)**

(a) None (b) One

(c) Three (d) More than three

189. Team X scored a total of N runs in 20 overs. Team Y tied the score in 10% less overs. Had team Y's average run rate (runs per over) been 50% higher, the scores would have been tied in 12 overs. How many runs were scored by team. X? **(CSAT-2025)**

(a) 72 (b) 144

(c) 216 (d) Cannot be determined

190. The price (p) of a commodity is first increased by k%; then decreased by k%; again increased by k%; and again decreased by k%. If the new price is q, then what is the relation between and q? **(CSAT-2025)**

(a) $p(10^4-k^2)^2=q \times 10^8$ (b) $p(10^4-k^2)^2=q \times 10^4$

(c) $p(10^4-k^2)=q \times 10^4$ (d) $p(10^{84}- k^2)=q \times 10^8$

191. Consider the first 100 natural numbers. How many of them are not divisible by any one of 2, 3, 5, 7 and 9? **(CSAT-2025)**

(a) 20 (b) 21

(c) 22 (d) 23

192. If $4 \leq x \leq 8$ and $2 \leq y \leq 7$, then what is the ratio of maximum value of (x + y) to minimum value of (x - y)? **(CSAT-2025)**

(a) 6 (b) 15/2

(c) -15/2 (d) None of the above

193. What is the maximum value of n such that 7 x 343 x 385 x 1000 x 2401 x 77777 divisible by 35^n? **(CSAT-2025)**

(a) 3 (b) 4

(c) 5 (d) 7

194. If N^2 = 12345678987654321, then how many digits does the number N have? (CSAT-2025)

(a) 8 (b) 9

(c) 10 (d) 11

195. Consider the following statements: (CSAT-2025)

I. There exists a natural number which when increased by 50% can have its number of factors unchanged.

II. There exists a natural number which when increased by 150% can have its number of factors unchanged.

Which of the statements given above is/are correct?

(a) I only (b) II only

(c) Both I and II (d) Neither I nor II

196. In a T20 cricket match, three players X, Y and Z scored a total of 37 runs. The ratio of number of runs scored by X to the number of runs scored by Y is equal to ratio of number of runs scored by Y to number of runs scored by Z. (CSAT-2025)

Value-I = Runs scored by X

Value-II = Runs scored by Y

Value-III = Runs scored by Z

Which one of the following is correct?

(a) Value-I < Value-II < Value-III

(b) Value-III < Value-II < Value-I

(c) Value-I < Value-III < Value-II

(d) Cannot be determined due to insufficient data

197. Consider a set of 11 numbers: (CSAT-2025)

Value-I = Minimum value of the average of the numbers of the set when they are consecutive integers ≥ - 5.

Value-II = Minimum value of the product of the numbers of the set when they are consecutive non-negative integers

Which one of the following is correct?

(a) Value-I < Value-II

(b) Value-II < Value-I

(c) Value-I = Value-II

(d) Cannot be determined due to insufficient data

198. The average of three numbers p, q and r is k. p is as much more than the average as q is less than the average. What is the value of r? (CSAT-2025)

(a) k (b) k-1

(c) k+1 (d) k/2

6 Basic Numeracy_Explanations

Numbers and their relations

1. Answer: (c)

Explanation:

To find the unit digit of the expression $6^{129} \times 7^{307}$, we need to determine the unit digits of 6^{129} and 7^{307} separately and then multiply them.

Step 1: Find the unit digit of 6129

The unit digit of any positive integer power of 6 is always 6. This is because $6 \times 6 = 36$, $36 \times 6 = 216$, and so on. Therefore, the unit digit of 6^{129} is 6.

Step 2: Find the unit digit of 7307

The unit digits of powers of 7 follow a repeating cycle of 4:

- $7^1 = 7$ (unit digit is 7)
- 72 = 49 (unit digit is 9)
- 73 = 343 (unit digit is 3)
- 74 = 2401 (unit digit is 1)
- 75 = 16807 (unit digit is 7, and the cycle repeats)

To find where 7^{307} falls in this cycle, we divide the exponent 307 by the cycle length, which is 4:

307 ÷ 4 = 76 with a remainder of 3

A remainder of 3 means the unit digit corresponds to the 3rd position in the cycle, which is the same as the unit digit of 7^3. Therefore, the unit digit of 7^{307} is 3.

Step 3: Multiply the unit digits

Now, we multiply the unit digits obtained from both parts:

6 × 3 = 18

The unit digit of this product is 8. Thus, the digit in the unit place of the number $6^{129} \times 7^{307}$ is 8.

2. Answer: (c)

Explanation:

To find how many three-digit numbers can be expressed as an integral power of 2, we need to evaluate numbers of the form 2^n (where n is an integer) and check which ones fall in the range of 100 to 999.

Let us list the positive integral powers of 2:

- 21 = 2
- 22 = 4
- 23 = 8
- 24 = 16
- 25 = 32
- 26 = 64
- 27 = 128 (Three-digit number)
- 28 = 256 (Three-digit number)
- 29 = 512 (Three-digit number)
- 210 = 1024 (Four-digit number)

As we can see, the powers of 2 continue to grow, and any power greater than 9 will result in a number with four or more digits. Negative integral powers of 2 result in fractions (e.g., $2^{-1} = 0.5$), which are not three-digit numbers.

Thus, the only three-digit numbers that are integral powers of 2 are 128, 256, and 512. There are exactly 3 such numbers.

3. Answer: (b)

Explanation:

Count the digit 5 separately in tens and units places of two-digit numbers 10–99.

Tens place. The tens digit is 5 in 50, 51, … , 59, that's 10 occurrences.

Units place. The units digit is 5 in 15, 25, 35, 45, 55, 65, 75, 85, 95, that's 9 occurrences.

Total: 10 + 9 = 19.

4. Answer: (c)

Explanation:

Set up using HCF × LCM = product of the two numbers.

Let the numbers be a and b (distinct). We know:

HCF(a, b) × LCM(a, b) = a · b

Given this product equals the cube of one of the numbers, say a^3:

a · b = a3 ⇒ b = a2

So the two numbers are a and a^2 (with $a \geq 2$ for them to be distinct positive integers).

Check the statements.

- **I. Difference is even.** a2 − a = a(a − 1), the product of two consecutive integers, which is always even. ✓ True
- **II. One of the numbers is a perfect square.** b = a2 is a perfect square. ✓ True

Sanity check: a = 2 ⇒ numbers 2, 4; HCF = 2, LCM = 4, product = 8 = 2^3. Difference = 2 (even), and 4 is a perfect square. ✓

5. Answer: (d)

Explanation:

For the 7-digit number 4x5y790, label digit positions from the right: position 1 = 0, 2 = 9, 3 = 7, 4 = y, 5 = 5, 6 = x, 7 = 4.

Apply the divisibility rule for 11.

(Sum of digits at odd positions) − (sum of digits at even positions) must be divisible by 11.

- Odd-position sum: 0 + 7 + 5 + 4 = 16
- Even-position sum: 9 + y + x = x + y + 9

So we need $16 - (x + y + 9) = 7 - (x + y)$ to be a multiple of 11.

Solve.

x, y are digits, so $0 \le x + y \le 18$, giving $7 - (x + y) \in [-11, 7]$. Multiples of 11 in this range: 0 and −11.

- $7 - (x + y) = 0 \Rightarrow x + y = 7$
- $7 - (x + y) = -11 \Rightarrow x + y = 18$

In both cases, $(x + y) \bmod 11 = 7$.

6. Answer: (b)

Explanation:

Let the two-digit number A be $10a + b$, where a and b are digits and $a \neq b$. Since B is obtained by reversing the digits of A, we have:

B = 10b + a

Since B is also a two-digit number, $b \neq 0$. Now,

A − B = (10a + b) − (10b + a) = 9a − 9b = 9(a − b)

It is given that A − B is a multiple of 27. Therefore:

9(a − b) is a multiple of 27

So, a − b must be a positive multiple of 3. Since A > B, we have a > b. The possible values of a − b are:

3, 6, 9

Now count the valid digit pairs with a > b and $b \neq 0$:

For a − b = 3:

(a, b) = (4, 1), (5, 2), (6, 3), (7, 4), (8, 5), (9, 6)

This gives 6 numbers.

For a − b = 6:

(a, b) = (7, 1), (8, 2), (9, 3)

This gives 3 numbers.

For a − b = 9, the only possible pair is (9, 0), but this is not allowed because B would not be a two-digit number. Hence, the total number of possible values of A is:

6 + 3 = 9

Therefore, the correct answer is 9.

7. Answer: (a)

Explanation:

The possible selections of three different coins from 1, 2, 5, 10 and 20 are:

1+2+5 = 8, 1+2+10 = 13, 1+2+20 = 23, 1+5+10 = 16, 1+5+20 = 26

1+10+20 = 31, 2+5+10 = 17, 2+5+20 = 27, 2+10+20 = 32, 5+10+20 = 35

Statement I: m is not a prime number. The possible non-prime values of m are 8, 16, 26, 27, 32 and 35. Here, 8, 16, 26, 27 and 35 involve a coin of denomination 5, but 32 comes from 2 + 10 + 20, which does not involve a coin of denomination 5. So, Statement I alone is not sufficient.

Statement II: The sum of the digits of m is greater than 5. The values of m whose digit sum is greater than 5 are 8, 16, 17, 26, 27 and 35. Each of these sums is obtained from a combination that includes the coin of denomination 5. So, Statement II alone is sufficient to answer the question.

Therefore, the question can be answered using one statement alone, but not using the other statement.

8. Answer: (d)

Explanation:

Statement I alone: $x^2 < y < 1$

This gives $-1 < x < 1$ and $y < 1$, but doesn't pin down which is bigger.

- $x = 0.5, y = 0.3$: satisfies $0.25 < 0.3 < 1 \rightarrow x > y$
- $x = 0.5, y = 0.9$: satisfies $0.25 < 0.9 < 1 \rightarrow y > x$

Insufficient.

Statement II alone: $y < \sqrt{x} < 1$

This gives $0 \le x < 1$ and $y < 1$, again ambiguous.

- $x = 0.25, y = 0.4$: $0.4 < 0.5 < 1$ ✓ $\rightarrow y > x$
- $x = 0.81, y = 0.1$: $0.1 < 0.9 < 1$ ✓ $\rightarrow x > y$

Insufficient.

Both together: $x^2 < y < \sqrt{x}$, with $x \in (0, 1)$.

Note that for $x \in (0, 1)$, $x^2 < x < \sqrt{x}$, so y can lie on either side of x.

- $x = 0.25, y = 0.1$: $x^2 = 0.0625 < 0.1 < 1$ ✓ and $0.1 < \sqrt{x} = 0.5 < 1$ ✓ $\rightarrow x > y$
- $x = 0.25, y = 0.4$: $0.0625 < 0.4 < 1$ ✓ and $0.4 < 0.5 < 1$ ✓ $\rightarrow y > x$

Both statements together still fail to determine which is bigger.

9. Answer: (c)

Explanation:

We need to determine whether x is even.

Statement I: x^2y^2 is even. This means at least one of x^2 or y^2 is even. Hence, at least one of x or y is even. But this does not definitely tell us whether x is even. For example, x could be odd and y could be even. So, Statement I alone is not sufficient.

Statement II: $1 + x^2 + y^2$ is odd. For $1 + x^2 + y^2$ to be odd, $x^2 + y^2$ must be even. This means x^2 and y^2 have the same parity. Therefore, x and y have the same parity: either both are even or both are odd. So, Statement II alone is not sufficient.

Using both statements together:

From Statement I, at least one of x or y is even. From Statement II, x and y have the same parity. Therefore, both x and y must be even. Hence, x is definitely even. So, the question can be answered using both statements together, but not using either statement alone.

10. Answer: (d)

Explanation:

We need to determine whether every number of T is from Y, that is, whether every number of T is even.

Statement I: The sum of any two numbers belonging to T is even. This means that all numbers in T must have the same parity. They may all be even, or they may all be odd. For example:

- If T = {2, 4}, then the sum of any two numbers is even, and every number is from Y.
- If T = {1, 3}, then the sum of any two numbers is also even, but every number is from X, not from Y.

So, Statement I alone is not sufficient.

Statement II: If both p and q are picked from T, then (p − 1)q is even. If all numbers in T are even, then q is even, so (p − 1)q is even. If all numbers in T are odd, then p − 1 is even, so (p − 1)q is also even. Thus, Statement II is satisfied when all numbers are even and also when all numbers are odd. Hence, Statement II alone is not sufficient.

Using both statements together:

Even together, the statements allow both possibilities: T may contain only even numbers (every number from Y), or T may contain only odd numbers (no number from Y). Therefore, the question cannot be answered even using both statements together.

11. Answer: (a)

Explanation:

Step 1: Analyze the given information

We are given that x, y, and z are integers, and each is greater than 1. We need to determine if x is a prime number.

Step 2: Evaluate Statement I

Statement I says: $xy^2 = 116$. Let us find the prime factorization of 116:

$116 = 2 \times 58 = 2 \times 2 \times 29 = 22 \times 29$

Since y is an integer greater than 1, y^2 must be a perfect square greater than 1 that divides 116. Looking at the prime factorization, the only perfect square factor of 116 greater than 1 is $2^2 = 4$. Therefore, $y^2 = 4$, which means y = 2. Substituting $y^2 = 4$ into the equation:

$x \times 4 = 116 \Rightarrow x = 29$

Since 29 is a prime number, we can definitively answer "Yes" to the question. Thus, Statement I alone is sufficient.

Step 3: Evaluate Statement II

Statement II says: xz = 261. Let us find the prime factorization of 261:

$261 = 3 \times 87 = 3 \times 3 \times 29 = 32 \times 29$

Since x and z are integers greater than 1, x can be any factor of 261 other than 1 and 261. The possible values for x are 3, 9, 29, and 87. If x = 3 or x = 29, then x is a prime number. If x = 9 or x = 87, then x is a composite number. Because x can be either prime or composite, we cannot definitively answer the question. Thus, Statement II alone is not sufficient.

Conclusion

The question can be answered using Statement I alone, but cannot be answered using Statement II alone. Therefore, the correct option is A.

12. Answer: (d)

Explanation:

Set up the common landing marks.

At 8 AM all three are on mark 199. After that:

- X lands on 199, 203, 207, … (step 4)
- Y lands on 199, 205, 211, … (step 6)
- Z lands on 199, 204, 209, … (step 5)

A mark m is a common landing iff m − 199 is a multiple of lcm(4, 6, 5) = 60.

List common landings in the range 195 < m < 1000.

m = 199 + 60k

199, 259, 319, 379, 439, 499, 559, 619, 679, 739, 799, 859, 919, 979

The next one, 1039, exceeds 1000.

Count: (979 − 199)/60 + 1 = 13 + 1 = 14

13. Answer: (d)

Explanation:

Let f = forward jumps, b = backward jumps.

Set up.

- Total jumps: f + b = 31
- Net displacement: 5f − 2b = 15

Solve.

Substitute b = 31 − f:

$5f - 2(31 - f) = 15 \Rightarrow 7f - 62 = 15 \Rightarrow f = 11$

So b = 20. The difference is $|f - b| = |11 - 20| = 9$.

14. Answer: (b)

Explanation:

To find the value of m, we need to determine the total power of 10 present in the prime factorization of the right-hand side (RHS) of the equation.

Step 1: Prime factorize the terms on the RHS

- $7525 = (3 \times 25)25 = (3 \times 52)25 = 325 \times 550$
- $2532 = (52)32 = 564$
- $3275 = (25)75 = 2375$

Step 2: Combine the prime factors

Multiplying these together, we get:

$RHS = 325 \times 550 \times 564 \times 2375 = 2375 \times 325 \times 5114$

Step 3: Determine the maximum power of 10 in the RHS

A factor of 10 is formed by multiplying a 2 and a 5. The number of 10s we can form is limited by the smaller of their exponents. The power of 2 is 375 and the power of 5 is 114. The minimum of these is 114. Therefore, we can factor out 10^{114}:

$RHS = (2114 \times 5114) \times 2375{-}114 \times 325 = 10114 \times 2261 \times 325$

Step 4: Compare with the left-hand side (LHS)

The LHS of the equation is given as:

$LHS = 10m \times 1000 \times n = 10m \times 103 \times n = 10m{+}3 \times n$

Equating LHS and RHS:

$10m{+}3 \times n = 10114 \times (2261 \times 325)$

The problem states that n is not divisible by 10, which means n cannot

contain both 2 and 5 as prime factors. The remaining term ($2^{261} \times 3^{25}$) has no factors of 5, so it is indeed not divisible by 10. This means all the powers of 10 have been completely factored out into the 10^{114} term. Thus, we can equate the powers of 10:

$m + 3 = 114 \Rightarrow m = 114 - 3 = 111$

Conclusion: The value of m is 111.

15. Answer: C

Explanation : This is a pattern-based puzzle, not normal multiplication. The pattern used gives $16 \times 63 = 65$ based on a hidden logic or rule, likely involving digits or a coded operation—not standard math.

16. Answer: (b)

Explanation:

We are given:

- **Initial petrol price** = Rs. 90 per litre
- **Hike** = 10% → New price = Rs. 90 + 10% of 90 = Rs. 99 per litre
- **Monthly distance travelled** = 2200 km
- **Car mileage** = 16 km/litre

Step 1: Find original fuel consumption

- Distance = 2200 km
- Mileage = 16 km/litre
- Fuel used = 2200 ÷ 16 = **137.5 litres**

Step 2: Original expenditure

- Price per litre = Rs. 90
- Total cost = 137.5 × 90 = **Rs. 12,375**

Step 3: What distance can he now travel at Rs. 99 per litre with the same budget?

Let's find how many litres he can buy now with the same Rs. 12,375:

- New price = Rs. 99 per litre
- Fuel he can now buy = 12,375 ÷ 99 = **125 litres**

New distance he can travel = 125 × 16 = **2000 km**

Step 4: Reduction in distance

Old distance = 2200 km

New distance = 2000 km

Reduction = 2200 − 200 = 200 km

17. Answer: (c)

Explanation: If an odd number is multiplied by 5 the last digit has to be 5 mathematically. The given expression does not have any even number so the last digit will be odd only.

18. Answer: (a)

Explanation: The number 1001 factors as 7×11×137 \times 11 \times 137×11×13. Each of the three numbers must include at least one of these primes so that their LCM is 1001, and the HCF is 1 (no prime divides all three). Counting all ways to distribute the three primes among three numbers with these conditions gives **6** valid sets.

19. Answer: (b)

Explanation: Using the equation PQR−PS=PPTPQR - PS = PPTPQR−PS=PPT, with given Q=3Q=3Q=3 and T<6T < 6T<6, we find possible digit pairs (R,S)(R, S)(R,S) that satisfy the conditions. After testing valid combinations with distinct digits, there are **3** possible pairs.

20. Answer: C

Explanation : The remainder for 1^n will always be 1 and remainder for 2^n will either be 2 when n is equal to 1 or it will be zero so the only two possibilities are 1 and 3

the error will that many people will make is that they will consider n ki power zero which is not true because any other natural number

21. Answer: C

Explanation : The trick is to know that 111 is divisible by 37 then it's a easy solution

22. Answer: D

Explanation : There are 9 single digit numbers then there are 90 2- digit numbers so when we write from 1 to 99 there are 189 digits which are done

now we are left with 300 digits and 3 digit numbers so the last number which will the 300th digit will be 199 number which will be 199 so the last day will be 9

23. Answer: C

Explanation : Subsequent difference is consecutive prime numbers

24. Answer: A

Explanation: each term in the sum $9^3+9^4+9^5+\cdots+9^{100}$ will have a remainder of 3 when divided by 6.

Number of terms = 100−3+1=98 terms

Sum $\equiv$294 $\equiv$0(mod6)

So remainder = 3 + 3 + 0 (mod6) =0

25. Answer: C

Explanation : In both case, the maximum value is 25 only

26. Answer: A

Explanation: For numbers between zero and one, squaring makes the number larger than cubing it, but the number is less than its square root. So, only the first statement is correct.

27. Answer: C

Explanation:

Let the two natural numbers be nnn and n+10n+10n+10.

Suppose n=3n = 3n=3, so the numbers between 3 and 13 are 4, 5, 6, ..., 12.

Now, which numbers between 3 and 13 are divisible by 5?

- **5** and **10**.

So, in this case, **there are two numbers (5 and 10)** divisible by 5 between 3 and 13.

If n=12n = 12n=12, then the numbers between 12 and 22 are 13, 14, ..., 21.

- The only number divisible by 5 is **15** and **20**.

Again, there are two numbers (15 and 20).

But if n=10n = 10n=10, then between 10 and 20 are 11, 12, ..., 19.

- The only number divisible by 5 is **15**.

Here, there is **only one** number (15).

28. Answer: (d)

Explanation:

Let X be represented as 10a+b, where a and b are the digits of x such that a is the tens digit and b is the units digit. Given that both a and b must satisfy $1 \le a \le 9$ and $0 \le b \le 9$

Let Y be the number formed by interchanging the digits of X. Hence, Y is represented as 10b+a.

Therefore, according to the question

X + Y = 99

10a + b + 10b + a = 99

Or 11a + 11b = 99

Or 11(a + b) = 99

Or a + b = 9

The possible value of X can be anything between 1 to 8.

29. Answer: (b)

Explanation:

Average weight of women = 63 kg.

So, Total weight of 4 women = 63 × 4 = 252 kg.

It is given that the weight of 6 boys = weight of 4 women

So, Total weight of 6 boys = 252 kg. So, Average weight of a boy = 252/6 = 42 kg.

30. Answer: (a)

Explanation:

Let the capital invested by Q as X.

Given that, P puts ₹ 14,000 more than Q. So, P invests X + 14,000

P's investment duration is 8 months

Q's investment duration is 10 months

Total profit is ₹ 2,000

P's share is ₹ 400 more than Q's share

We know that the share of profit is divided based on the product of capital and the time for which it is invested. Let's calculate the shares of P and Q.

First, the product of the capital and the time for each:

P's investment : (X + 14,000) x 8

Qs investment: (X × 10)

Ratio of their investments = (x+14,000) × 8/ (X × 10)

Since the total profit is Rs. 2000 and and P's share is ₹ 400 more than Q's share, let Q's share be y and P's share be y+400

Thus: y+(y+400)=2000

2y + 400 = 2000

2y = 1600

y =800

Qs share is 800 and Ps share is 1200

the ratio of their share is 1200/800 = 3/2

So, (X + 14,000) × 8/X × 10 = 3/2

on solving we get, x= 16,000

So, Qs investment is 16,000. Therefore, P's investment is X + 14000 = 30,000.

31. Answer: (c)

Explanation:

Here, the numbers are 421 and 427

When we subtract 421 and 427 by 1, we get 420 and 426.

Now by taking the prime factors we get:

420 = 2×2×3×5×7×1

426 = 2×3×71×1

The common factors among them are 2, 3, and 6 (multiple of 2 and 3). Hence, we can say that 3 numbers can be used as the divisor.

32. Answer: (b)

Explanation:

Quantity of petrol contained in X = 399

litre Quantity of diesel contained in Y = 532 litres

By taking the factors of 399 and 532 we get:

399 = 1×3×7×19

532 = 1×2×2×7×19

So, the possible bottle sizes are 1 litre, 7 litres, 19 litres and 133 litres. Therefore, the number of possible bottle sizes is 4.

33. Answer: (c)

Explanation:

Given, sum (S)= x + y+ z, here x, y and z are prime numbers and their values are less than 10. This means the numbers could be 2, 3, 5 and 7.

If we take, x+y+z= 2+3+5 = 10

If we take, x+y+z= 2+5+7 = 14

If we take, x+y+z= 2+3+7 = 12

If we take, x+y+z= 5+3+7 = 15

There, the Unit digit of S can be 0. **(Statement 1 is correct)**

There, the Unit digit of S can not be 9. (**Statement 2 is not correct**)

There, the Unit digit of S can be 5. (**Statement 3 is correct**)

34. Answer: (b)

Explanation:

A cube can be divided into smaller cubes by making a series of cuts along its three dimensions (length, width, and height).

To find the minimum number of cuts: We start with a cube and need to divide it into 4 x 4 x 4 = 64 smaller cubes.

We need to make 3 cuts along each dimension to achieve this (since each side of the smaller cubes is ¼ of the original cubes side length.

Make 3 cuts along the length, dividing the cube into 4 slabs.

Make 3 cuts along the width, dividing each slab into 4 slices.

Make 3 cuts along the height, dividing each slice into 4 smaller cubes.

Therefore, the total number of cuts needed is: 3 (cuts along length) + 3 (cuts along width) + 3 (cuts along height) = 9 cuts.

35. Answer: (d)

Explanation:

The given expression is,

$5 \times 4 \times 3 \times 2 \times 1$

here, x is chosen from +, -, x , each at most two times.

so, we can write,

$5 - 4 - 3 + 2 \times 1$

$= 7-7$

= 0, which is the smallest non-negative value of the expression.

36. Answer: (a)

Explanation:

The age of P and Q is less than 100 but more than 10. That is, both P and Q are two-digit numbers.

So, 10 < P, Q < 100

Also, let P = xy, then Q = yx

Using statement-I alone: P > Q There are various possibilities:

81 > 18

72 > 27 …. and so on.

It alone is not sufficient.

Using statement-II alone: P + Q = (11/6) (P - Q)

Or 10x + y + 10y + x = (11/6) (10x + y – 10y - x)

Or 11(x + y) = (11/6) (9x – 9y)

37. Answer: (a)

Explanation:

The code is being calculated by multiplying the position values of the alphabets.

ABCD = 24= 1 x 2 x 3 x 4 = 24

EFGH = 1680= 5 x 6 x 7 x 8 = 1680

IJKL = 9 x 10 x 11 x 12 = 11880

38. Answer: (d)

Explanation:

'POT' is written as 'ATOP'

'TRAP' is written as 'APART',

Here the word is being reversed and prefix 'A' is being added to the front.

Reverse order of POT = TOP

Thus, code of POT will be 'ATOP'.

Reverse order of TRAP = PART Thus, code of TRAP will be 'APART'.

Reverse order of ARENA = ANERA Thus, code of ARENA will be 'AANERA'

39. Answer: (b)

Explanation:

The pattern of the series:

$1^3+1^2+1 = 3$

$2^3+2^2+2 = 14$

$3^3+ 3^2+3 = 39$

$4^3+ 4^2+4 = 84$

$5^3+ 5^2+5 = 155$

$6^3+ 6^2+6 = 258$

40. Answer: (b)

Explanation:

In some code, letters P, Q, R, S, T represent numbers 4, 5, 10, 12, 15.

It is not known which letter represents which number.

Here, Q - S = 2S

Q = 3S …….(i)

The only possible values of S and Q are:

If S = 4, then Q = 12

Or, if S = 5, then Q = 15

Also, T = R + S + 3 …….(ii)

Case 1: If S = 4 and Q = 12

T = R + S + 3 = R + 7

This is only possible if T = 12 and R = 5.

But both Q and T cannot be 12. So, we can eliminate this case.

Case 2: If S = 5 and Q = 15

T = R + S + 3 = R + 8

This is only possible if T = 12 and R = 4. So, we get R = 4, S = 5, T = 12 and Q = 15.

It means P must be 10. So, P + R – T = 10 + 4 – 12 = 2

41. Answer: (d)

Explanation:

Given that the average marks in a class are 60.

Using statement-I alone: The highest marks in the class are 70 and lowest marks are 50. This statement is not sufficient, as the strength of the class can be anything. Using

statement-II alone: Exclusion of highest and lowest marks from the class doesn't change the average. This statement is also not sufficient, as we don't get any information regarding the strength of the class.

This question cannot be answered even by using both the statement together.

42. Answer: (a)

Explanation:

Given that there are three distinct prime numbers whose sum is a prime number.

Using statement-I alone: Their sum is less than 23. Only one triplet of prime numbers satisfies this condition: (3, 5, 11). (3 + 5 + 11 = 19 < 23) Hence, statement I alone is sufficient to answer the question.

Using statement-II alone: One of the numbers is 5. 5 + 7 + 11 = 23 (a prime number) 5 + 11 + 13 = 29 (a prime number) Using this statement alone, we cannot obtain a unique set of prime numbers. Hence, the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

43. Answer: (d)

Explanation:

Statement I: (2x+y) is an integer.

if x=3 and y= 2 then

2x+y = 6+2=8, which is an integer.

and, x + y = 5, which is also an integer.

Now, if we put, x= 1.5 and y=2 then

2x + y = 5

x + y = 3.5 , which is not an integer.

So, using only statement 1, we cannot say that x + y is an integer or not.

Statement 2: (x+2y) is an integer

if x= 3 and y= 2

then, x + 2y= 3+4 = 7 which is an integer.

if x= 2 and y=1.5

x+2y=5

x=y=3.5, which is not an integer.

So, using only statement 2, we cannot say that x + y is an integer or not.

44. Answer: (c)

According to the question, p + r + q= Rs. 50, q= Rs. 16

⇨ p + r + q= Rs. 50

⇨ p+r = 50 – q = 50-16 =34

S1: since cost of is not more than of r.

p <r

p+r=34

It alone is not sufficient

S2: Cost of r is not more than that of p.

r<p

p + r=34

Still not able to find, exact answer,

Combining S1 and S2.

p < r and r< p this can't be exist

So, only existed possibilities p=q.

$p = q = \frac{34}{2} = 17$ ₹

45. Answer: (d)

Explanation:

As per the Statement 1, the sum of the marks scored by P and Q is equal to the sum of the marks scored by R and S.

P + Q = R + S …….(i) So, from the statement –I alone we cannot say that P scored more than Q or not.

As per the statement 2, the sum of the marks scored by P and S is more than the sum of the marks scored by Q and R.

(P + S) > (Q + R) ……..(ii) So, from statement –II alone also, we cannot say that P scored more than Q or not.

Using both the statements, it can not be said that

P scored more that Q or not. Therefore, this question cannot be answered even by using both of the statements together.

46. Answer: (c)

Explanation:

p < q – (1) p+q=k

Now taking options

(a) 3 ⇒ (2, 1) Satisfying condition 1 but no. are unique

(b) 4 ⇒ (3, 1) → Satisfying condition 1 but no. are unique

(c) 5 ⇒ (2, 3) (1, 4) → Satisfying condition 1 and no. are not uniquely.

47. Answer: (c)

Q: What is the value of m and n

S1: m + n>mn and m>n

m → 1, 2, 3

n → 1

1 + 2 > 2

1 + 3 > 3

1 + 4 > 4

∴ (Numerous possibilities) s1 is alone is

not sufficient

S2: product of mn is 24

$$mn \xleftarrow{\begin{matrix}(24,1)\\(12,2)\end{matrix}} (6,4)$$

Three Cases S2 is not sufficient to answer the question

S1 and S2

case mn=(24,1) 24 >1 (m>n)

So, m=24

n = 1

S1 and S2 both will be used for exact answer.

48. Answer: (b)

Explanation:

Option (b) is correct: Series with 28 terms: 1 + 1 + 2, 1 + 3,2,1 + 4,3,2,1 + 5,4,3,2,1 + 6,5,4,3,2,1 + 7,6,5,4,3,2

So, their sum is 84

49. Answer: (d)

Explanation:

Option (d) is correct:

Let Q = 100, then R = 120% of 100 = 120 & P =125% of 120 = 150

So, P:Q:R = 15:10:12

Now, 15 + 10 + 12 = 37 units = Rs. 3330

Or 1 unit = Rs. 90 So, P = 15 units = 15 × 90 = Rs. 1350

50. Answer: (a)

Explanation:

Case 1: There's no carry over from B+D

A + C = 1C

Or A + C = 10 + C

Or A = 10 (not possible)

Case 2: There's carry over 1 from B+D

A + C + 1 = 1C

Or A + C + 1 = 10 + C

Or A = 9

51. Answer: (d)

Explanation:

Option (d) is correct: 4 triplets satisfying the given conditions are: (5,2,1), (6,2,1), (7,2,1) and (7,3,1)

52. Answer: (a)

There are two 5's and one 4 in the expression.

85x87x89x91x95x96

i.e. (85=5x17,95=5x19,96=4x 24)

So, the expression is completely divisible by 5x 5 x 9 or 100.

So remainder = 0. **Hence, option (a) is correct.**

53. Answer: (a)

Let's break down the given number in the question,

37DD= 3000+700+D0+D

If ABC x D= 37DD then,

$$ABC = \frac{3000 + 700 + D0 + D}{D}$$

$$\frac{3700}{D} + 10 + \text{or } \frac{3700}{D} + 11$$

If ABC is a three digit numbers then D may be 1,2,4, and 5 only

Case 1	D = 1	ABC = 3711 → (four digit number)
Case 2	D = 4	ABC = 936 → (A valid option)
Case 3	D = 2	ABC = 1861 → (four digit number)
Case 4	D = 5	ABC = 751 → (B = D = 5, can't be)

So, we have A + B + C = 9 + 3 + 6 =18. **Hence, option (a) is correct.**

54. Answer: (d)

To determine the divisibility of the 6-digit number XYZXYZ, we can analyse its properties and identify the factors that will divide it.

Let's express the number XYZXYZ as a product of its factors:

XYZXYZ = XYZ x 1001

Now, let's consider the factor 1001. We can factorize it as 7 x 11 x 13. This means that any number divisible by 1001 is also divisible by 7, 11, and 13.

Therefore, the 6-digit number XYZXYZ is divisible by 7, 11, and 13 for any choices of values for X, Y, and Z. **Hence, option (d) is correct.**

55. Answer: (c)

7x + 96 is divisible by 'X' only when 96 is divisible by x, Or x is a factor of 96.

By prime factorizing 96, we find that it can be expressed as 25 x 31. This means that 96 is composed of 2 raised to the power of 5 (25) multiplied by 3 raised to the power of 1(31).

The number of factors of a given number can be calculated using the exponents of its prime factorization. To find the number of factors, we add 1 to each exponent (i.e, 5, 1) and multiply the results.

In this case, (5+1) x (1+1) = 6 x 2 = 12. **Hence, option (c) is correct.**

56. Answer: (a)

If a single digit or a set of digits is repeated (p - 1) times, where p is a prime number greater than 1, then the resulting number is divisible by p.

Here, the number N is formed by writing 9 for 99 times. Such that,

N=9999..... 99 times.

Let's consider the rule provided earlier and consider that, P =13

Given in the rule (p - 1), we assume that, (13-1) = 12 times repeat of 9 is also divisible by 13.

Similarly, multiples of 12 times repeat of 9 will also be divisible by 13.

Closest to 99 in the multiples of 12 is 12 x 8 = 96.

It means 9 written 96 times will be completely divisible by 13.

Remaining three 9s can be checked for their divisibility by 13.

999/13 gives divisor 76 and remainder 11.

A number N, that is formed by writing 9 for 99 times, if divided by 13 will give the remainder of 11. **Hence, option (a) is correct.**

57. Answer: (c)

Let's observe the pattern,

1x1=1

11x11=121

111x 111 =12321

1111 x 1111 = 1234321

Similarly, (1111...9 times) x (1111...9 times) = 12345678987654321

1+2+3+4+5+6+7+8+9+8+7+6+5+4+3+2+1 = 81. **Hence, option (c) is correct.**

58. Answer: (d)

'91' = 13 x 7 is the only numbers in option 'd' which is not a prime number. Rest all numbers in every other options are prime numbers. **Hence, option (d) is correct.**

59. Answer: (d)

Let's understand the question in simple way,

We are looking at, 1186 ÷ P = Q (Quotient) and 31 (Remainder)

We have to find possible values of P. it is quite evident that P > 31, and that P should be a factor of 1186 - 31 i.e. 1155

Prime factorization of 1155=1x3x5x7x 11

So, P can be 33, 35, 55, 77, 105, 165, 231, 385 and 1155. (Total 9 such numbers)

3x11=33, 5x7=35, 5x11=55, 7x11=77, 3x5x7=105, 3x5x11=165, 3x7x11=231, 5x7x11=385, 3X5x7x 11=1155. **Hence, option (d) is correct.**

60. Answer: (d) is correct.

The given series is: 20, 10, 10, 15, 30, 75, X, and we need to find the value of X. By observing the given sequence:

1st term of sequence = a_1, = 20

2nd term of sequence = a_2 = 10 hence a_2, = $0.5a_1$

3rd term of sequence = a_3, = 10 hence a_3, = $1.a_2$,

4th term of sequence = a_4 = 15 hence a, = $1.5a_3$,

5th term of sequence = a_5 = 30 hence a = $2a_4$,

6th term of sequence = a_6 = 75 hence a; = $2.5a_5$,

7th term of sequence = a_7 = X, we need to find the value of X.

Here we observe that terms of this sequence are arranged in a pattern that the terms in sequence are some multiple of previous term, i.e

a_{n+1} = m.a_n, (where m is any number)

By observing the given term, we get a sequence of multipliers 0.5, 1, 1.5, 2, 2.5...

In this sequence the next term will be 3. Hence, a_7= m.n_6 and by the observation m = 3, hence a_7 = 3.a_6, a_7= 3.75=225a_7=X=225.

Hence, option (d) is correct.

61. Answer: (b) is correct.

Conditions given in question:

1. Digits are odd: - so digits canbe1, 3, 57 and 9.

2. Number is divisible by 5: - unit digit must be either 0 or 5

But according to 1st condition 0 cannot be the unit digit Hence, unit digit must be 5. Now we need to choose two digits from 4 digits without repetition: $4p_2=\frac{4!}{2!}=\frac{24}{2}$=12 No. of ways to choose 2 digits from 4 digits without repetition =

Hence, option (b) is correct.

62. Answer: (b) is correct.

Let the initial price = 100 Rs. Now, according to question price is increased by 25%.

New price will be 100+ 25% of 100 = 125 (25% of 100 = 25)

Then the price is decreased by 20%.

20% of 125 = $\frac{125*100}{100}$ =25 price is decreased by 25 rs.

New price will be 125 ~ 25 = 100 rs. Again, the price is increased by 10%

Resultant price =100 + 10% of 100 = 110 (10% of 100 = 10) Now, the final price of item is 110 rs (increased by 10 rs) Increase of price in percentage

$\frac{\text{increase in price}}{\text{initial price}}$x100 = $\frac{10}{100}$ x 100 = 10%

Resultant increase in price = 10%. **Hence, option (b) is correct.**

63. Answer: (d)

Here, 1261 = 13×97

Let X = 91×92×93×94×95×96×97×98×99

X = (13×7) ×92×93×94×95×96×97×98×99

We have to divide X by 1261.

Since x contains multiples of 13 and 97, 1261 will completely divide X.

$\frac{x}{1261}=\frac{(13x7)\times92\times93\times94\times95\times96\times97\times98\times99}{13\times97}$

$\frac{x}{1261}$ = 7 ×92×93×94×95×96×97×98×99

The remainder = 0. Hence, **option (d) is correct.**

64. Answer: (a)

Let the total number of voters in the voters list be X.

Number of casted votes = 80% of x = 0.8x

X got 56% of casted vote i.e.S0, Y got (100-56) % i.e. 44% of casted votes

According to question,

X won by 1440 votes, implies X - Y = 1440

0.448x - 0.352x = 1440 0.096x = 1440

Hence, the total number of voters in the voters list = 15000. **Hence, option (a) is correct.**

65. Answer: (c)

LCM of 6, 9, 12, 15 and 18 = 180

Smallest number greater than 1000 which is a multiple of 180 is 180x6 = 1080

So, required number = 1080 + 3 = 1083, which when divided by any one of the numbers 6,9, 12,15, 18 leaves a remainder of 3. **Hence, option (c) is correct.**

66. Answer: (d)

Let p be a 2-digit number is xy i.e. (p=10x+y)

q consists of same digits written in reverse order i.e., yx or q=10y +x.

Given,

P x q = 2430...(a)

The last digit of the product is 0. This implies last digit of p or q is 0, 5.

Case 1: Last digit of p or 9 = 0. This is not possible, otherwise either p or g will be 1-digit.

Case 2: Last digit of p or q = 5. Let y=5

P=10x+5 and q=50+x

Using (a),

(10x +5) (50 + x) = 2430

500 x +250 + $10x^2$+ 5x = 2430

10x2 + 505x + 250 - 2430 = 0

$10x^2$ + 545x - 40x - 2180 = 0

5x(2x + 109) - 20(2x + 109) = 0

(5x-20) (2x+109) =0

5x-20=0,2x+109=0

X=4, - 54.5 (Rejected)

Hence, x=4

p=45 and q=54

The difference between p and q = 54-45 = 9

Hence, option (d) is correct.

Question Hack: Factors of 2160 are 2, 3, 3, 3, 3, 3, and 5.

2430 = (2×3×3×3) × (3×3×5)

2430 = 54×45

2430 = p × q

Where, p = 54 and q = 45 (reverse digits of p).

The difference between p and g = 54-45 = 9.

67. Answer: (d)

Given, p is a prime number i.e., 2,3,5,7,11, 13...

q is a composite number i.e., 4, 6, 8,9, 10...

Statement-1: p × q can be an odd number. This statement is correct.

Take p =3 and q = 9, we have, px g =3 × 9 =27 (odd number)

Statement-2: q/p can be a prime number. This statement is correct. Take p =2 and q = 4, we have, 4/2 = 2 (prime number)

Statement-3: p + q can be a prime number. This statement is also correct.

Take p= and q=3, then, p + q = 2+3=5(prime number) All statements are true.

Hence, Option (d) is correct.

68. Answer: (b)

Given sequence is: 2, 12, 36, 80, 150, X

We need to find the value of X

For this we will make a table of differences

Term	1st differences	2nd differences	3rd differences
2	10(12-2)	14(24-10)	6
12	24	20	6
36	44	26	a-70-26=a-96
80	70	a-70	
150	x-150(let's say 'a')		
x			

Here we observe that 3rd difference of the terms = 6

Hence, a - 96 = 6

Implies a = 102

Now our a =X-150

102 =X-150

Implies X = 150+102 = 252

X = 252. **Hence, Option (b) is correct.**

69. Answer: (a)

We have a number 0. XY, where X and Y are distinct non- zero digits. Possible digits taken by X: 1, 2, 3,4,5,6,7,8, 9 (i.e., 9 possibilities)

The digit which is taken by X cannot be taken by Y. So, Y, has 8 options left.

Therefore, the required numbers of the form 0. XY = 9 × 8 = 72. Hence, option (a) is correct.

70. Answer: (c)

From the given question, we can interpret that A = 28 = 3C = 4D = 5E = 6F

L.C.M. of 2,3, 4, 5, 6 = 60. So, the minimum number of pens A can have is multiple of 60.

Let A have 60 pens. Then,

(even number)

(even number)

(odd number)

Hence, we can't take A = 60

Now, let us take A = 120 pens

(even number)

(even number)

(even number)

(even number)

Therefore, the minimum number of pens X should buy = A+B+C+D+E+F

X=120 + 60 + 40 + 30 + 24 + 20 = 294. **Hence, option (c) is correct.**

71. Answer: (a) is correct.

According to question no. of persons between A and B = 4

No. of persons between B and C=7

No. of persons ahead of C = 11

No. of person behind A= 13

So there are 4 possible cases:

A_4_ B_7__ C (4 between A and B, 7 between B and C)

C_2_A__4_ B (4 between A and B, 7 between B and C)

72. C_7_8B_4 A (4 between A and B, 7 between B and Q)

B_4_A 2_C (4 between A and B, 7 between B and C)

Now, there are 11 persons ahead of C and 13 behind A,

Case (i) is eliminated because there are already more than 11 persons ahead of C.

In case (ii) if there 11 persons ahead of C and 13 behind A then total no. of people will be 28.

In case (iii) 11 persons ahead of C and 13 behind A then total no. of people will be 38.

In case (iv) 11 persons ahead of C and 13 behind A then total no. of people will be 22.

Hence, in case (iv) we have minimum number of people in queue. **Hence, option (a) is correct.**

73. Answer: (a) is correct.

According to the question,

0.7x + y = 1.65y

0.7x = 1.65y-y

0.7 = .65y

=>x<y… (a)

Now, 0.6x+2 = 1.65z

0.6x=1.65z-z

0.6x = 0.65z

x>z or z<x … (b)

From inequalities (a) and (b), we get: Z<X<Y

Hence, option (a) is correct.

74. Answer: (a) is correct.

According to the question, Container X contains 100 ml of milk and container Y contains 100 ml of water.

If 20 ml of milk transferred from container X to Y, then:

Amount of milk left in container X = 100 —20 = 80 ml

Amount of solution in container Y becomes = 100 ml

water + 20 ml milk = 120 ml

Ratio of milk and water in container Y = 20: 100 = 1: 5

Amount of milk in 20 ml solution of container Y = (1/6) x20=3.33ml

Amount of water in 20 ml solution = 20 -3.33 = 16.67 ml If 20 ml of this solution is transferred from container Y to container X, then:

Amount of solution in container X becomes = (80 + 3.33 =83.33 ml) milk + 16.67 mi water

Amount of solution in container Y becomes = (100 — 16.67 = 83.33 ml) water + (20 —3.33 = 16.67ml) milk

According to question, m denotes the proportion of milk in X, and n denotes the proportion of water in Y. So, m = 83.33 ml and n = 83.33ml

Thus, m = n. Hence, option (a) is correct.

75. Answer: (b) is correct.

We have to find the maximum value of min 15 x 14 x 13

x …. x 3 x 2 x 1 = 3^m x n

15 = 3 x 5

12 = 3 x 4

9 = 3 x 3

6 = 3 x 2

3 = 3 x 1

Then, 15 x 14 x 13 x ... x 3 x 2 x 1 =3^mx n

3 x 5 x 14 x 13 x 3 x 4 x 11 x 10 x 3 x 3 x 8 x 7 x 3 x 2 x 5 x

4 x 3 x 2 x 1 = 3^mx n

3^6 x 5 x 14 x 13 x 4 x 11 x 10 x 8 x 7 x 2 x 5 x 4 x 2 x 1 = 3^mx n

On comparing the power of 3 in L.H.S and R.H.S, we get m=6

The maximum value of m = 6

Hence, Option (b) is correct.

76. Answer: (d) is correct.

A pie-chart corresponds to 360°, This implies,

A+B+C+D+E=360°

A+90°+50°+45°+ 75° = 360°

A+260° =360°

A=360°-260°

A =100°

$A = \left(\frac{100}{360} \times 100\right)\% = \frac{250}{9}\%$

So, the percentage of expenditure on item **Hence, option (d) is correct.**

77. Answer: (b)

Given number is 3798125P369. We need to find value of P such that number is divisible by 7.

In this question we will use the rule of divisi for that:

Step 1: break the given number in triplets of digits from right side:

Number can be expressed as: (37) (981) (25P) (369)

Step 2: Add 1st and 3rd triplet and subtract 2nd and 4th triplet.

Which is 369 - 25P + 981 — 37 = 1313 — 25P

Now 25P = 2x100 + 5x10 +P = 200 + 50 +P

1313-25P = 1313 - 250 - P = 1063 - P

When we divide 1063 by 7 we get 6 as remainder Hence, if we subtract 6 from 1063, we get a number which is divisible by 7.

Hence, P = 6. Hence, option (b) is correct.

78. Answer: (b)

Let the maximum marks in each subject = 100

Total marks in 6 subjects = 100 + 100 + 100 + 100 + 100 +100 = 600

Marks scored = 60% of 600 = 360

His marks in these papers are in the proportion of 5 : 6 : 7 : 8 : 9 : 10

Let the marks in each paper be 5x, 6x, 7x, 8x, 9x, 10x

5x + 6x + 7x + 8x + 9x + 10x = 360

45x =360

x=8

Marks in each of the 6 subjects are 5x8, 6x8, 7x8, 9x8, 10x8 = 40, 48, 56, 64, 72, 80. So, in 3 subjects he scored less than 60% marks (40, 48 and 56). Hence, option (b) is correct.

79. Answer: (a)

Students in Class-A = 25

Students in Class-B = 30

Marks of students of Class-A are in the range of 17 to 21.

Marks of students of Class-B are in the range of 22 to 30.

Now, 4 students are shifted from Class-A to Class-B.

We can clearly see that the entire range of marks of students of Class-A is less than marks of students of Class-B. So, the 4 students who were transferred have marks less than marks of students of Class-B.

Average of Class-B will definitely decrease.

So, statement-1 is correct. We do not know which 4 students are transferred from Class A.

If they have marks close to 21, then the average of Class-A will decrease.

If they have marks close to 17, then the average of Class-A will increase.

∴ We cannot determine whether the average of Class-A will increase or decrease.

So, statement-2 is incorrect. Hence, option (a) is correct.

80. Answer: (b)

Half of the villagers have their own house i.e., 50%

One-fifth of the villagers cultivate paddy i.e.,

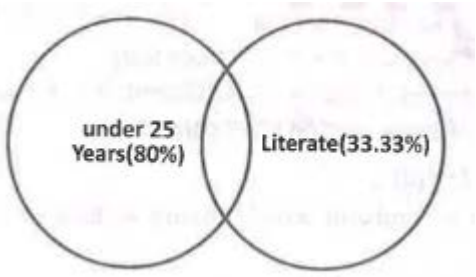
One-third of the villagers are literate i.e.,

Four-fifth of the villagers are under 25 years of age i.e.,

Option-1: All the villagers who have their own houses are literate.

This is definitely incorrect because only 33.33% people are literate and 50% villagers have their own house.

Option-2:



The villagers who are under 25 and literate are 80%+33.33%-100% = 13.33%

Hence, some villagers are under 25 years of age are definitely literate.

Option-3: Only half of the villagers who cultivate paddy are literate.

20% villagers cultivate paddy and 33.33% are literate. This statement cannot be confirmed.

Optipn-4: No villager under 25 years of age has his own house.

This statement is incorrect because 50% villagers have their own house, and 80% villagers are under 25 years of age. So, (50+80-100=30%) at least 30% villagers under 25 year of age have their own house. Hence, option (b) is correct.

81. Answer: (a)

Let the marks of Q be x.

According to question,

P scored 40 marks more than Q in an examination i.e.

marks of P =40+ x ... (1)

Q scored 10% less marks than P

$\Rightarrow x=(40+X)-\frac{10}{100}(40+x)=(40+x)-(4+\frac{x}{10})=36+\frac{9x}{10}$

=> $x-\frac{9x}{10}=36$

=> $\frac{x}{10}=36$

So, Q scored 360 marks. Hence, option (a) is correct.

82. Answer: (b)

Let X, Y and Z have x, y and z amount of money respectively.

According to the question, If Y gives X ₹40, then Y will have half of as much as Z.

->y-40=z/2 ... (1)

If Z gives X ₹40, then all three of them will have equal amount.

->x+40=y=z-40 ... (2)

Put y = z- 40 in equation (1), we get

(z-40)-40=z/2

->z-z/2=80

->z/2=80

->z=160

So, y=z-40=160-40

->y=120

And x+40=y

->x=y-40=120-40= 80

So, the total amount of money that X, Y and Z have = ₹80 + ₹120 + ₹160 = ₹360. Hence, option (b) is correct.

83. Answer: (a)

Let the number of correct questions be x and wrong questions be y.

According to question,

x+y=90 ->x=90-y.. (1)

5x -2y =387..(2)

Put the value of x from equation (1) in equation (2), we get

5(90 - y) - 2y = 38763

450 - 5y -2y = 387

7y=450-387 =63

y=63/7=9

So, the number of incorrect responses = 9. Hence, option (a) is correct.

84. Answer: (c)

Consider statement 1:

Let the 5 consecutive numbers be x =2, x-1, x, x + 1, x +2

We will check if the sum of 5 consecutive numbers cap be 100.

(x-2) + (x+ 1) + x + (x+1) + (x+2) =100

5x =100

x=20

∴ The 5 consecutive numbers are 18, 19, 20, 21 and 22.

So, statement-1 is correct.

Consider Statement-2:

Let the three consecutive numbers be: x = 1, X, x + 1, where x> 1 (Natural numbers are 1, 2 ...)

Now, (x- 1) x (x +1) = (x- 1) + x + (x + 1)

$X^3 - x = 3x$

$X^3 - 4x = 0$

X = -2, 0, 2

The only value which x can take is 3.

∴ The three consecutive numbers are 1, 2 and 3 such that 1+2+3=1x2x3=6

So, Statement-2 is also correct. Hence, option (c) is correct.

85. Answer: (a)

Ratio of money distributed among A, B and C is p:q:r

Now, consider statement 1:

If p>(q+r)

->P>r, p>q

Then, it is sure that p is the largest number.

Hence, A gets the maximum share.

So, statement-1 is correct.

Now, consider statement 2:

If r < (p +q), then it is not necessary that r is the smallest number and C gets the minimum share.

Let p=3, q=1, r=2

Here, 2 < (3+1)

i.e. 2<4

But r > q. This implies, q is the smallest and C may not get the smallest statement.

So, statement-2 is not correct. Hence, option (a) is correct.

86. Answer: (b)

Let the two-digit number be xy i.e., 10x +y.

The number formed by interchanging the two digits be yx i.e., 10y +x

Given, the difference between a 2-digit number and the number obtained by interchanging the positions of the digits is 54.

(10x +y) - (10y +x) = 54

9x - 9y = 54

x-y=6….(1)

The difference between the two digits of the number =6 So, statement-2 is correct.

Statement-1: The sum of the two digits of a number can be determined only if the product of the two digits is known.

The sum of two digits can be determined if x and y is known.

Using equation (1), if x = 9 then y = 3 (xy —yx = 93 -39 =54)

If x =8 then y = 2 (82 — 28 = 54)

If x=7 then y=1(71-17 = 54)

So, Statement-1 is not necessarily true. Hence, option (b) is correct.

87. Answer: (a)

Let the price of the article be x.

Now, the price is decreased by 20%.

Then, new price = x — 20% of x =

New, the new price increased by 25%

Then, the price becomes

∴ The final price of the article = x

But the initial price is also x.

The net change in price = Final price - Initial price = x - x =0. Hence, option (a) is correct.

88. Answer: (a)

Zeros at the end means multiple of 10, i.e. (2 x 5). So, no. of pairs of 2 and 5, and 10s will determine the no. of zeroes at the end of product.

1stly we will factorize the numbers.

We can write the product by factorizing as:

1x5x10x15x 20 x 25 x 30 X 35 x 40 x 45 x 50 x 55 x

60=1x5x10x(3x5) x (2x10) x (5x5) x (3x10) x (5 x 7) x

(10x2x2) x (5x9) x (5x 10) x (5x 11) x (2 x 3 x 10)

Now we need to check number of 10s and pairs of 2 and 5 in above product.

No. of 10s = 6

No. of 2s=4

No. of 5s=8

Pairs of 2s and 5s are = 4 (whichever occur least times)

we have: 6 10s and 4 pairs of 2s and 5s

No. of zeros at the end of product = 6 + 4 = 10. Hence, option (a) is correct.

89. Answer: (b)

XYZ is a three-digit number. So, we can write this as: 100X + 10Y + Z (X is at hundred's place and Y is at ten's place)

We can write YZX as: 100Y + 10Z + X (Y at hundred's place and Zis at ten's place)

We can write ZXY as: 100Z + 10X + Y (Z at hundred's place and X is at ten's place)

Thus, XYZ + YZX + ZXY = (100X + 10Y + Z) + (100Y + 10Z + X) +(100Z + 10X +Y) = 111 (X +Y + Z)

Hence, (XYZ + YZX + ZXY) it is divisible by (X + Y + Z).

Also, 111 is divisible by 3, as well as 37. So, XYZ + YZX + ZXY are not divisible by 9. Hence, option (b) is correct.

90. Answer: (c)

Given information is p — 2016 = q + 2017 =r - 2018 = s +2019

Taking 1st two equation: p — 2016 = q + 2017, we have p—2016=q+2017 or p=q + 2017 + 2016 implies p=q +4033

it means p > q.... conclusion (i)

Now taking q + 2017 = r - 2018, we have

q +2017 = r-2018 or r = g + 2017 + 2018 implies r = q+4035

it means r > q ... Conclusion (ii)

Now taking r - 2018 = s + 2019, we have

r - 2018 =s+2019 or r =s + 2019 + 2018 implies r = s + 4037

it means r > s... conclusion (iii)

Now taking p — 2016 = r - 2018, we have

P - 2016=r - 2018 or r = p - 2016 + 2018 = p + 2 implies

r = p+2

it means r > p.... conclusion (iv)

Now by combining conclusion (i), (ii), (iii) and (iv)

We got 'r' is largest. Hence, option (c) is correct.

Question Hack:

Take s = 0 and put it in equation given in question: p –

2016 =q+2017 =r-2018 =s + 2019

So, p—2016 =5 + 2019 Or p = 2019 + 2016 = 4035 (s = 0), p = 4035

And q+2017 = s +2019 or q = 2019 - 2017 =2 (s =0), q =2

And r - 2018 =2019 Or r = 2019 + 2018 = 4037, r = 4037 Hence, 'r' is the largest number.

91. Answer: (a)

We have to find the numbers which can be obtained by using digits 1, 2, 3, 4 and 5 without repetition of digits.

We need to use 1, 2, 3,4 and 5 in every number.

So, sum of digits of the obtained number in every case will be 1+2+3+4+5 = 15

Now we know that any number will be divisible by 3 is sum of its digit is divisible by 3. Here sum of digits = 15, divisible by 3

Five-digit prime numbers can be obtained by using all the digits 1, 2, 3, 4 and 5 without repetition of digits will be divisible by 3. Hence, we cannot make any prime number from given condition. Hence, option (a) is correct.

92. Answer: (c)

There are 19 integers between 1 and 100 which have 4 as a digit, which are:

4,14, 24,34, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 54, 64, 74, 84 and 94

Out of these 19 integers there are 7 integers which are divisible by 4:

4,24, 40, 44, 48, 64, 84

So, there are 12 integers between 1 and 100 which have 4 as a digit and are divisible by 4. Hence, option (c) is correct.

93. Answer: (a)

Given information is:

Volume of P = x

Volume of Q=y

Mass of P=m

Mass of Q=n

Now it is given that each side of Q is two times that of P

Let us take side of P=a

Then side of Q = 2a

Volume of P = x =a^3

Volume of Q =y = $(2a)^3 = 8a^3$

Implies y = 8x

Also, it is given that mass of Q is two times that of P

Implies n = 2m

Now u=m/x and v=n/y

V=2m/8x implies v = m/4x

V = u/4; OR u = 4v. Hence, option (a) is correct.

94. Answer: (c)

Let the age of each student be x years and the age of the teacher be y years.

Combine age of 3 students = 3x

According to the question average age of teacher and student is 20,

=20 imply 3x +y =80 .... (i)

It is also given that difference between age of teacher and student is 20 years.

Hence, y - x = 20... (ii) it implies y = 20 + x

Put y=20+x in (i)

3x+20+x=80

Implies, 4x = 60, x = 15

Now put x = 15 in (ii)

y=20+15=35

Age of teacher is 35 years. Hence, option (c) is correct.

95. Answer: (d)

Let the cost price be Rs. x

Selling price of car = Rs 3,00,000

Given that loss percent = 20%

Thus, 80% of x = 300000

=300000 imply that 80x = 30000000

Hence, option (d) is correct.

Question Hack:

Let the cost price = 100 rupees

Loss = 20% = 20 rupees

Hence, selling price = 80 rupees

Now we can say that.

When selling price is 80 rupees then cost price = 100 rupees

When selling price is r rupee then cost price =100/80 Now when selling price is 300000 the cost price = (100/80) x 300000 = 3,75,000 rupees.

96. Answer: (b)

Let us take $\otimes$=x

then x + 1x + 5x + xx + x1 = 1xx

Or X+ (10 + x) + (50 + x) + (10x + x) + (10x + 1) = 100 + 10x +x

(Because 15 = 1x10 + 5 or 123 = 1 x 100 + 2.x 10 + 3)

(understand this concept)

Or 24x+61=100 + 11x

Or 13x =39

Or x = 3. Hence, option (b) is correct.

97. Answer: (b)

Given information is:

Length of larger stick (S1) = 7.5 feet

Length of shorter stick (S2) = 3.25 feet

We need to get minimum possible length we can measure using these two sticks, we will use stick S2 and measure the length of stick S1.

Thus, we get: Total length of S1 = 7.5 = 3.25 + 3.25 + remaining length of S1

Or 6.5 + remaining length of S1= 7.5

Or remaining length of S1=7.5 6.5 = 1 foot

Hence, we can measure the multiples of three lengths using these sticks.

These lengths are: 1 foot, 3.25 feet and 7.5 feet.

Now by using 1 foot and 3.25 feet we can measure 0.25 feel.

Reduce 1 foot from 3.25 feet 3 times you will get length 0.25 foot.

Minimum we can measure is 0.25 foot. Hence option (b) is correct.

98. Answer: (c)

Given sequence is: 14, 18, 20, 24, 30, 32...

We can get a pattern in the sequence as:

14=13+1

18=17+1

20=19+1

24=23+1

30=29+1

32=31+1

Now, look at the numbers in bold letters, this is sequence of prime numbers starts from 13:

Sequence is: 13, 17, 19, 23, 29, 31, now the next term of sequence will be next prime number that is: 37.

Hence, next term of our original sequence is 37 + 1 = 38. Hence, option (c) is correct.

99. Answer: (b)

Let the number of pages in the book = n.

Sum of 1st n no. or pages of book = sum of 1st n natural number

Here it is given that sum of pages = 195 after torn of 1 page

We need to find page no. of torn page.

Here we will check the value of n for that value of is near to 195 (value must be greater than 195)

When, n = 20, 210 (nearest to 195, for any other value of n either value is less than 195 or will not be nearest)

Hence, from here we can say that book has 20 pages Now the sum of pages after page torn = 195

Implies that the sum of page numbers of torn page = 210 - 195=15

Now only option (b) gives us sum = 15. Hence, option (b) is correct.

100. Answer: (c)

A3BC + DE2F = 15902 it is given in question where A, B, C, D, E and F are different and greater than 3.

For unit digit

C+F=2: C and F are greater than 3 so C+F cannot be 2 but it can be 12 and carry 1 to tens digit.

So, C+F = 12 imply (C, F) = (4, 8) or (5, 7)

Now for tens digit: B + 2 = 0, now we know that tens digit carry 1 from unit digit Hence,

B+2=9 implies B =7

Now B= 7 so neither C nor F can be 7

Hence, (C, F) = (4, 8)

Now for hundred digits: 3+E= 9, now we know that tens digit carry 1 from unit digit

Hence, 3+E = 8 implies E= 5

Now C=4, B=7, E=5, F=8

Now we have A and D are unknown which can be 6 and 9 (only these values are remaining)

Difference between A and D is 9-6=3. Hence, option (c) is correct.

101. Answer: (c)

Concepts we are going to use in this question:

$$\left(\frac{a}{b}\right)^{-x} = \left(\frac{b}{a}\right)^{x}$$

Option (a): $\left(\frac{1}{2}\right)^{-6} = \left(\frac{2}{1}\right)^{6} = 64$

Option (b): $\left(\frac{1}{4}\right)^{-3} = \left(\frac{4}{1}\right)^{3} = 64$

Option (c): $\left(\frac{1}{3}\right)^{-4} = \left(\frac{3}{1}\right)^{4} = 81$

Option (d): $\left(\frac{1}{6}\right)^{-2} = \left(\frac{6}{1}\right)^{2} = 36$

102. Answer: (a)

According to question:

No. of days taken by x to complete 20% of work is = 8 days

So, no. of days taken by x to complete the work =8 x 5 =40 days

Work done by x in 1 day = 1/40

No. of days taken by y to complete 25% of work is = 6 days

So, no. of days taken by y to complete the work =6 x 4 =24 days

Work done by y in 1 day = 1/24

Work done by x and y together in 1 day = 1/24 + 1/40 = 1/15

Hence, No. of days taken by x and y together to complete the work = 15 days

No. of days to complete the work by x and y together to complete 40% of work = 40% of 15 = (15 x 40)/100 = 6 days. Hence, option (a) is correct.

103. Answer: (a)

Let x and y are the natural numbers such that difference of whose squares is 63.

$x^2-y^2 = 63$ implies $(x - y)(x + y) = 63$ because $x^2-y^2 = (x-y)(x+y)$

Now we know that

63 =21x3, 7x9 and 63x1

Take 63=7x9:

(x-y) (x+y) = 7x9

From here take x -y = 7 and x+y = 9

Which gives x = 8 and y = 1.

Take 63 = 21x3:

(x-y) (x+y) = 3x21

From here take x -y = 3 and x+y = 21

Which gives x = 12 and y =9

Take 63 = 63x1:

(x-y) (x+y) = 1x63

From here take x -y = 1 and x+y = 63

Which gives x =32 and y = 31.

Hence, there are three pairs of natural numbers such that difference of whose squares is 63:

(63, 1), (7, 9) and (21, 3). Hence, option (a) is correct.

104. Answer: (d)

In this type of question, we need to find a number which satisfy the condition given in question.

Condition given: A digit n > 3 is divisible by 3 but not divisible by 6.

n = 9 satisfy the condition given in question.

We now go by option we need to check which option will be divisible by 4.

When n =9, 2n = 18 is not divisible by 4.

When n =9, 3n = 27 is not divisible by 4.

When n =9, 2n+4 = 22 is not divisible by 4.

When n =9, 3n + 1 = 28 is divisible by 4. Hence, option (d) is correct.

105. Answer: (b)

Let x = 1.272727... Multiply both sides by 100 we got

100x = 127.272727…

Now by subtracting 100x - x = 127.272727...- 1.27272 ...

99x =126

X =126 /99 = 14/11. Hence, option (b) is correct.

Question Hack: Check the decimal value of each option.

Option (a): 13/11 = 1.181818... not equal to desired result

Option (b): 14/11 = 1.272727... which is equal to result we need. Hence, option (b) is correct.

106. Answer: (b)

1stty lets find out the L.C.M of 3,4, 5and 6

LCM of 3,4,5 and 6 =60

Now any multiple of 60 give remainder 0 when divide by any of 3,4,5 and 6

Also, (any multiple of 60) + 2 will give remained 2 when divide by any of 3,4, 5 and 6

Now, 17 times of 60 = 17x60 = 1020

1020 gives remainder 0 when divide by any of 3, 4, 5 and 6.

Now 1020 + 2 = 1022 gives remainder 2 when divide by any of 3,4, 5 and 6. Hence, option (b) is correct.

Question Hack: Check the options.

Option (a) and (c) are eliminated because 1012 is divisible by 4 and 1122 is divisible by 3

Now if we check option (d)

1222 when we divide it by 3 it gives remainder 1

1222=407x3 +1 Option (d) is eliminated.

107. Answer: (a)

The expression is: 51 x 27 x 35 x 62 x 75

(51x27x35x62x75)/100=(51x27x35x62x3)/4

= (51x 27 x 35 x 31 x 3) / 2 (by cancelling 1% by 25 to 75 from numerator and 100 from denominator, then by 2)

So, we will find the remainder when 51 x 27 x 35 x 31 x 3 is divided by 2.

The 51 x 27 x 35 x 31 x 3 must be an odd number, because it is a multiplication of all odd numbers.

Hence, when it is divided by 2, we will get 1 as remainder.

But earlier we cancelled out 50.

So, the remainder when 51 x 27 x 35 x 62 x 75 is divided by 100 = 1 x 50 = 50. Hence, option (a) is correct.

Question Hack:

The expression is: 51 x 27 x 35 x 62 x 75.

62 X 75 = 4650; Hence, the unit digit of expression will be 0.

When we divide it by 100 the unit digit of remainder will be the unit digit of expression.

Hence, unit digit of remainder must be 0.

Which we can find in option (a) only that is 50.

108. Answer: (b)

Let $N=10^n+1$

Taken=0

$10^n +1=100+1=1+1=2$

Sum of digit = 2

Hence, for n=0, sum of digits of $10^n + 1$ is 2

Hence, option (c) is eliminated because it only talks about positive integers.

Take n =1

$10^n + 1 =10^1 + 1 = 11$; sum of digits = 1+1= 2.

Hence, for n=0, sum of digits of $10^n + 1$ is 2

Hence, option (a) is eliminated because it only talks about n=0.

Take n=-1

$10^n +1=10^{-1} +1=1/10+1=11/10$

Sum of digits is not equal to 2.

Option (d) is eliminated. Hence, option (b) is correct.

109. Answer: (c)

we know that: | liter = 10^6 cubic milimeters

Given is question, that:

1 liter =1 kg.

Hence, 1 kg. = 10^6 cubic milimeters

1000g = 10^6 cubic milimeters

$1g = 10^6$ cubic milimeters (dividing both sides by 10^3)

$g = 10^6$ cubic milimeters (dividing both sides by 10). Hence, option (c) is correct.

110. Answer: (a)

Let us suppose weight of vessel = x

weight of water in this =y

now given that if the vessel is fully filled then its weight is 40 kg.

So, x+y =40 Or y = 40 - x... (i)

2nd condition is:

It is one third filled then weight is 20 kg.

So, X +y/3 = 20 imply 3x + y = 60... (ii)

Now put the value of y from (i) in (i) we get

3x+40-x=60

2x =20 imply x = 10 kg.

Weight of vessel = 10 kg. Hence, option (a) is correct.

111. Answer: (b)

It is given that total length of the wall = 4.5 m = 450 cm

length covered by frog in one jump = 30cm

frog slides to the length = 15cm net length covered = 30 - 15 = 15 cm

frog has to cover 420 cm and then a jump of 30cm and frog will be out from well.

Number of jumps required by frog to cover 420 cm = 420/15 = 28 jumps

In the 29th jump the frog will reach at the top of the wall and will not slide down.

Hence, total number of jumps required = 29. Hence, option (b) is correct.

112. Answer: The correct option is option (c)

From 1 to 100, the number of times 5 will appear = 20 times

5, 15, 25, 35,45, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 65, 75, 85,95

From 101 to 200, the hundredth place of the number is lie,1__.

So, the number of times 5 will appear on the unit and tens place = 20 times

Similarly, from 201 to 300, 301 to 400, 401 to 499, 600 to 700, 701 to 800, 801 to 900 and 901 to 1000, the number of times 5 will appear = 20 times

Now, from 500 to 599, the hundredth place of the number is 5 i.e., the number is 5 _ _

Number of times 5 will appear on the unit and tens place=20 times and the number of times 5 will appear on the hundredth place = 100 times.

Total number of times 5 will appear from 500 to 599 = 20 +100 = 120.

So, the total number of times 5 will appear from 1 to 1000=20+20+20+20+20+120+20+20+20+20= 300. **Hence, option (c) is correct.**

113. Answer: The correct answer is option (b)

Every number from 1 to 100 is assigned to a student.

A student is football player if he has identification number divisible by 4 i.e., 4, 8, 12, 16 ...

A student is cricketer if he has identification number divisible by 6 i.e., 6, 12, 18, 24. LCM. of 4 and 6 is 12.

-> A student plays cricket as well as football is he has identification number that is divisible by 12.

The identification number of students who plays both cricket and football are: 12, 24, 36, 48, 60, 72, 84 and 96.

So, the total numbers of required students are 8. **Hence, option (b) is correct.**

114. Answer: The correct answer is option (b)

Let the two-digit number be i.e. 10x +y

The number formed by reversing the digits is i.e. 10y +x

It is given that the ratio of a two-digit natural number to

a number formed by reversing its digits is 4:7

$$\frac{10x+y}{10y+x} = \frac{4}{7}$$

7(10x +y) = 4(20y + x)

70x + 7y = 40y + 4x

70x - 4x = 40y- 7y

66x = 33y

2x=y

When x =1,y =2 ->The numbers are 12 and 21.

When x = 2, y =4 -> The numbers are 24 and 42.

When x=3,y -> The numbers are 36 and 63.

4,y = 8-> The numbers are 48 and 84.

When x = ,y =10 -> not possible.

So, 4 such pairs are possible i.e. (12, 21), (24, 42), (36, 63) and (48, 84).

Hence, option (b) is correct.

115. Answer: The correct answer is option (c)

A printer uses 3089 digits in writing the number of pages of a book. Digits required in printing single digit numbers from 1 to 9 = 9

Digits required in printing double digit numbers from 10 to 99 = 902 =180

Digits requires in printing triple digit numbers from 100 to 999 = 900 3 = 2700

Total digits in printing the pages till three-digit numbers = 9+ 180 + 2700 = 2889

Digits left = 3089 — 2889 = 200

After 999, four-digit numbers will start.

The number of 4-digit numbers required = 200/4 = 50

So, total number of pages in a book = 999 + 50 = 1049

Hence, option (c) is correct.

116. Answer: The correct option is option (d)

Given, 7A3 is divisible by 3.

It means 7+ A + 3 = (10+ A) is divisible by 3.

So, A can take values: 2, 5 and 8.

It is given that, when 136 is added to 587, we obtained 7A3.

Unit place: 6+ 7 = 13. One is carried forward to tens place.

Tens Place: 3+ B+ 1(carry) = A

->4+B=A

Hundredth Place: 1+5=7

It is not possible unless, one is carried forward from the tens place.

Possibility-1: when A = 2, then B = 8 and then one is carried forward to next place as 4 + 8 = 12,

So, the possible value of B can be 8

Possibility-2: when A=5, then B=1as 4 + 1 =5 but in this case one is not carried forward to the next place. So, B cannot take 1.

Possibility-3: when A = 8, then B =4 as 4 + 4 = 8 but this

is also not possible because one is not carried forward to the next place.

So, the only possible value of B = 8 such that, 136 + 587 = 723. **Hence, option (d) is correct.**

117. Answer: The correct answer is option (c)

Divisibility test of 3: When the sum of the digits of a number is divisible by 3, then the number of divisible by3.

Given that, an 8-digit number 4252746B leaves remainder 0 when divided by 3.

Now, 4+2+5+2+7+4+6+B=30+ B

If the number 4252746B is divisible by 3, then 30 + B should also be divisible by 3.

There are 4 possible values which B can take i.e. 0, 3, 6 and 9

30, 33, 36 and 39 all are divisible by 3. **Hence, option (c) is correct.**

118. Answer: The correct answer is option (b)

Firstly, we have to list down the3-digit numbers which are divisible by 7.

These are: 105, 112, 119, 126, 133, 140, 147, 154, 161, 168, 175, 182, 189, 196, 203, 210, 217, 224, 231, 238, 245,252, 259.

In this, remove the numbers whose all digits are not different.

These are: 105, 126,140, 147, 154, 168, 175, 182, 189, 196, 203, 210, 217, 231, 238, 245, 259 ...

Now, reverse the digits, we get, 501, 621, 041, 741, 451, 861, 571, 281, 981, 691, 302, 012, 712, 132, 832, 542, 952...

From these digits, only 861 and 952 are divisible by 7. So, only four 3-digit numbers are there i.e., 168, 861, 259 and 952: **Hence, option (b) is correct.**

119. Answer: The correct answer is option (c)

The numbers between 99 to 1000 are 3-digit and we have to find the numbers such that 8 occupy the unit place.

From 100 to 200 — There are 10 numbers in which 8 comes at units place i.e,, 108, 118, 128, 138, 148, 158, 168,178, 188 and 198.

Similarly, from 201 to 300, 301 to 400 ... 901 to 999, there will be 10 numbers in each interval.

So, the number of times 8 will come at the units place between 99 to 1000 = 10 x 9 = 90. **Hence, option (c) is correct.**

120. Answer: The correct answer is option (b)

The monthly incomes of X and Y are in the ratio of 4: 3

Let the income of X and Y be 4x and 3x respectively. The monthly expenses are in the ratio of 3: 2

Let the monthly expense of X and Y is 3y and 2y respectively. Each saves % 6,000 per month

So, 4x — 3y = 6000

3x— 2y = 6000

On solving these two equations, we get, x = 6000 and y = 6000

So, their total monthly income = 4x + 3x = 7x = 7(6,000) = 42,000.

Hence, option (b) is correct.

121. Answer: The correct answer is option (d)

Let the two-digit number be x y i.e., 10x + y and the number formed by reversing the digits is yx i.e., 10y + x.

It is given that the difference between these two numbers is always 27.

(10x +y) - (10y +x) = 27

9x -9y =27

x-y=3

Possible numbers are

When x =9, then

When x = 8, then y

When x=7, then y

6. So, the number is 96

5. So, the number is 85

4. So, the number is 74

When x = 6, then y = 3. So, the number is 63

When x =5, then y = 2. So, the number is 52

When x =4, then y = 1. So, the number is 41

So, 6 numbers are possible. **Hence, option (d) is correct.**

122. Answer: The correct answer is option (b)

There are single digits, double digits and triple digits numbers from 1 to 150. The number of digits from 1to 9= 9 The number of digits from 10 to 99 = 90 x 2 = 180

The number of digits from 100 to 150 = 51x 3 = 153

So, the total number of digits printed from 1to 150 =9 + 180 + 153 = 342.

Hence, option (b) is correct.

123. Answer: The correct answer is option (a)

Let the milk sample = 300 litres

It has 50% water in it. It means 150 litres water and 150 litres milk is present in this sample.

Now, 1/3rd of this milk i.e., 100 litres is added to equal amount of pure milk i.e. 100 litres, then the new mixture =200 litres

So, Water in the mixture = 50 litres

Milk in the mixture = 50 + 100 = 150 litres

Percentage of water in new mixture $= \frac{50}{200} \times 100 = 25\%$

Hence, option (a) is correct.

124. Answer: (b)

Let the total number of students in a class is x

Contribution by each student to charity = Rs. X

One student contributes Rs (x + 2)

-> (x — 1) number of students contribute Rs. X

Total collection = Rs. 443

(x-1) (x) + (1) (x + 2) = 443

X^2-x+x+2=443

X^2 =441

x=21

So, total number of students in the class = 21.

Hence, option (b) is correct.

125. Answer: (a)

The area of the field will be maximum when both the sides of rectangle are equal.

Given, $X_1 + X_2 = 40$

For the area to be maximum, $X_1 + X_2 = 20$

So, maximum area = 20 x 20 = 400 sq m.

Hence, option (a) is correct.

126. Answer: (d)

Let the basic pay of 1st person and 2nd person be x and y respectively.

Emoluments of 1st person = x + 65% of

Emoluments of 2" person = y + 80% of

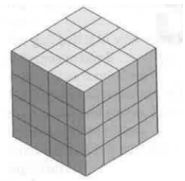
Since the total emoluments of both the person are same.

So, the ratio of basic pay of the former to the basic pay of latter is 12:11.

Hence, option (d) is correct.

127. Answer: (b)

Let the cube is painted with yellow colour on all the sides. It is cut into smaller cubes of equal sizes such that the side of the small cube is one-fourth the big cube. So, there will be 64 small cubes.



There are only 4 cubes (the middle ones) on each face who have only 1 side painted.

So, total number of cubes with only one side painted = 4 x 6 = 24.

Hence, option (b) is correct.

128. Answer: (d)

W can do 25% of a work in 30 days

So, W can do the whole work in 30 x 4 = 120 days

X can do 1/4 of the work in 10 days

So, X can do the whole work in 10 x 4 = 40 days

Y can do 40% of the work in 40 days

So, Y can do 100% of the work in = 100 days

Z can do 1/3 of the work in 13 days

So, Z can do the whole work in 13 x 3 = 39 days.

= > Z will complete the work first. **Hence, option (d) is correct.**

129. Answer: (a)

Average monthly income of a person in a family of 5 = Rs. 10, 000.

So, total monthly income of a family of 5 = 10,000 × 5 = Rs. 50,000

Now, the income of 1 person is increased by Rs. 1, 20,000 per year. It means, his monthly income increases by Rs. 10,000. So, the total monthly income of a family of 5 becomes Rs. 50,000 + Rs. 10, 000 = Rs. 60, 000

The new average monthly income becomes Rs. 60,000/ 5=Rs. 12,000.

Hence, option (a) is correct.

130. Answer: (c)

The cost of a toy is marked = R. 770.

10% discount is given for cash payment. So, the cost becomes =770 - = 693

The person still makes 10% gain

Let the actual cost of a toy = Rs. x

110x = 69300

x=630

So, the actual cost price of the toy is Rs. 630. **Hence, option (c) is correct.**

131. Answer: (c)

ABC x DEED = ABCABC, it is given question.

Now, ABCABC is divisible by 13, 11 and 7.

So, it must be divisible by 13 x 11 x 7 = 1001.

ABC x DEED is divisible by 1001.

DEED = 1001

D = 1 and E = 0. **Hence, option (c) is correct.**

132. Answer: (d)

The correct option is option D

Option (a) and (b) are eliminated: Question statement clearly mentions the value of every things would not rise

or value of every things would not remain constant.

Option (c) is eliminated: The question statement mentions that value of all things except one would rise.

Hence, option (d) is correct.

133. Answer: (d)

The correct option is option D

Given thatP = 4

T-P =5 (The difference between P and T is 5)

Hence, T=9

Now also T- N = 3 (The difference between N and T is 3)

So, N = 6. **Hence, option (d) is correct.**

134. Solution: (b)

Total number of males = 1652. Hence the number of females = 1226.

Given 796 males voted against the proposal out of 1226. Therefore 430 females voted against the proposal. We also know 196 females were undecided.

Therefore, the number of females who voted for the proposal = 1226 – 430 – 196 = 600.

135. Solution: (a)

1425 people voted for the proposal. Hence the number of males who voted for the proposal = 1425

– 600 = 825. Therefore the number of males who were undecided = 1652 – 825 – 796 = 31

136. Solution: a)

Clear from the equations above.

Orders of magnitude

137. Answer: (b)

Explanation:

$222^{333} + 333^{222}$

Even + Odd = Odd

And, the odd number is not divisible by 2.

So, **option (b) is correct.**

138. Answer: (d)

Explanation:

We can break 30^{30} as $3^{30} \times 10^{30}$

10^{30}. would give zeroes.

The non-zero digit would only come from 3^{30} and the unit digit is 9 which is preceded by zeros.

139. Answer: (d)

Explanation:

for zeroes 5 and 2 pair combination should be

for equation $1^2 \times 2^4 \times 3^6 \times 4^8 \times 5^{10} \ldots \times 25^{50}$

Since $\rightarrow 2^4$

$\rightarrow 4^8$ ($24^8 = 2^{16} \ldots$.

So, 2 have more power than 5

So look for 5.

Only $\rightarrow 50^{10}$

$\rightarrow 10^{20} = 5^{26} \times 2^{10}$

$\rightarrow 15^{30} = 5^{30}$

$\rightarrow 20 = 5^{40}$

$\rightarrow 25 = 25^{50} = (5^2)^{50} = 5^{100}$

So, No. of 5's 10+20+30+40+100=200

So, No.of Zeroes 200

140. Answer: (c)

Explanation:

32^5+22^7 is divisible by

$\Rightarrow (32 = 2^5)$

$(2^5)^5 + 2^{27} = 2^{25}+2^{27} = 2^{25}(1 + 22) = 5 \times 2^{25}$

$\Rightarrow 5 \times 2 \times 2^{24}$

$\Rightarrow 10 \times 2^{24}$

Divisible by 10

141. Answer: (a)

Statement 1 is correct: In whatever manner you arrange these numbers, basic nature of having three even and two odd numbers in the expression "p + q + r—s - t" gives the same essence and the result, i.e. an even number.

Statement 2 is incorrect: 2p, 2r, 2s are always even numbers whether you take p, r, s as one odd, two even or two odd, one even or all even ones respectively. But g + t may be even or odd, depending on what you take them as.

142. Answer: (d)

Prime numbers are positive integers greater than 1 that have no factors other than 1 and themselves. such as (2,3,5,7,11,13.....) and; Composite numbers are positive integers greater than 1 that are divisible by more than just 1 and themselves. In other words, a composite number has factors other than 1 and itself, (4,6,8,9,12.....)

Statement 1is correct: Let's take P = 11, c= 9

So, $\frac{P+C}{P-C} = \frac{20}{2} = 10$,

which is an even number.

Statement 2 is correct: Let's take P = 11, c= 9

So, 2p + C = 11 x 2 + 9 = 31, which is an odd number.

Statement 3 is correct: Let's take P = 11, c = 9

So, pc = 11 x 9 = 99, which is an odd number, **Hence, option (d) is correct.**

143. Answer: (b) is correct.

The given numbers are 2^{40}, 3^{21}, 4^{18} and 8^{12} and we have to find the smallest among these. Now observe last two numbers and we find: $4 = 2^2$ and $8 = 2^3$ hence $4^{18} = 2^{36}$ and $8^{12} = 2^{36}$ because $(a^b)^c = a^{bc}$,

We can read 4^{18} and 8^{12} as 2^{36}

Now we can say $2^{40} > 4^{18} = 8^{12}$ (because $4^{18} = 8^{12} = 2^{36}$ and $2^{40} > 2^{36}$)

Hence, 2^{40} cannot be the smallest and eliminated.

Now we need to check smaller no. between 2^{36} and 2^{21}

$2^{36} = (2^{12})^3$ and $3^{21} = (3^7)^3$ Now we just need to check the smaller number between 2^{36} and 3^7.

3^7=2187 and 2^{12} = 4096

Hence $2^{12} > 3^7$ (4096>2187)

3^{21} is smallest. **Hence, option (b) is correct.**

Question Hack: After eliminating 2^{40} we get the Answer directly because 4^{18} and 8^{12} are equal. And we have to find smallest number 4^8 and 8^{12} cannot be the Answer because only one Answer is correct Hence, 4^{18} and 8^{12} are also eliminated.

144. Answer: (d) is correct.

The number of coins D have = 10 (smallest two-digit number)

Now according to question C gives half of the coins and 2 more to D.

No. of coins D have =

Hence, implies : implies C = 16

Now we have information that C had 16 coins. Now according to question B gives half of the coins and 2moreto C. No. of coins C have Hence, implies implies B = 28 New we have information: that B had 28 coins. New according to question A gives half of the coins and 2 more to B. No of coins B have Hence, implies : implies A= 52 Hence, no. of coins

A had initially = 52 coins. **Hence, option (d) is correct.**

145. Answer: (c)

We have to find out the remainder when 3^{2019} is divided by 10.

Now, $3^1 = 3$

$3^2 = 9$

$3^3 = 27$

$3^4 = 81$

$3^5 = 243$

$3^6 = 729$

We can clearly see that, the unit place repeats after every 4 powers.

We can write $2019 = 4 \times 506 + 3$

$3^{2019} = 3^{2016+3} = 3^{2016}3^3 = (3^{4506})(27)$

Since, the unit digit of $3^4 = 1$, the unit digit of $(3^{4506}) = (_1)^{506} = 1$

So, the unit digit of $3^{2019} = (1)(27) = 7$

∴ When 3^{2019} is divided by 10, then remainder is 7. **Hence, option (c) is correct.**

146. Answer: The correct option is option (c)

The given sequence: 132, 129, 124, 117, 106, 93, X We analyse the following pattern:

132 - 129 = 3

129 - 124 = 5

124 - 117 = 7

117 - 106 = 11

106 - 93 = 13

We can infer that 3, 5, 7, 11 and 13 are consecutive prime numbers.

So, the next prime number after 13 would be 17.

93-X = 17

-> X = 93 - 1776. Hence, option (c) is correct.

147. Answer:

We can infer from the given series: 4, 196, 16, 144, 36, 100, 64, X

Odd number place	Even number place
$4 = 2^2$	$196 = 14^2$
$16 = 4^2$	$144 = 12^2$
$36 = 6^2$	$100 = 10^2$
$64 = 8^2$	$x = 8^2 = 64$

It can be clearly observed that, the terms at the odd place are in increasing order of the squares of 2, 4, 6 and 8 and the terms at the even place are in decreasing order of the squares of 14, 12 and 10. Since, X is at even place. So, X = square of (10 —2) = square of 8 = 64. **Hence, option (b) is correct.**

148. Answer: The correct answer is option (a)

The given sequence: 1,5,7,3,5,7,4,3,5,7.

We have to find the number of 5's which are not immediately preceded by 3 but are immediately followed by 7. By looking at the series, we can determine that there is only one 5 which is immediately followed by 7 and not immediately preceded by 3.

That is, 1,5, 7, 3, 5, 7,4,3,5,7. **Hence, option (a) is correct.**

149. Answer: The correct option is option (b)

We have to find two numbers, one of which is a cube of a number and the other is a square of a number and the difference between them must be 2.

Squares: $1^2=1$, $2^2=4$, $3^2=9$, $4^2=16$, $5^2=25$

Cubes: $1^3=1$, $2^3=8$, $3^3=27$, $4^3=61$

We can clearly see that the desired numbers are 25 and 27, where 25 is the square of 5 and 27 is the cube of 3

So, the present age of Mr X is 26 yrs.

Now, the next time his age will be a cube of a number will be when he becomes 64 years old i.e. his age is cube of 4.

Least number of years to wait = 64 — 26 = 38 years.

Hence, option (b) is correct.

150. Answer: (d)

Vessel A-30 g of sugar was mixed in 180 ml water.

-> Sugar in A = 30 g/ 180ml = 0.167 g/ml

Vessel B - 40 g of sugar was mixed in 280 ml of water.

->Sugar in B = 40g / 280ml = 0.143 g/ml

Vessel C— 20 g of sugar was mixed in 100 ml of water.

-> Sugar in C = 20g / 100ml = 0.2 g/ml

Sugar content: B< A< C

So, the solution in vessel B is less sweet than that in A and C.

Hence, option (d) is correct.

Miscellaneous

151. Answer: (b)

Explanation:

Let the salary of R be Rs. 100

Then salary of Q= 80% of Rs. 100 = Rs. 80

And salary of P = 80% of Rs. 80 = Rs. 64 Required percentage = {(100 - 64)/64} × 100 = (36/64) × 100 = 56.25 %

152. Answer: (d)

Explanation:

Let the number be 100.

After multiplying the number by 4, the number will be 4× 100 = 400

But by mistake it was divided by 4. The resultant number will be 100/4 = 25 Required percentage change = {(400 - 25)/400} × 100 = (375/400) × 100 = 93.75%

153. Answer: (b)

Explanation:

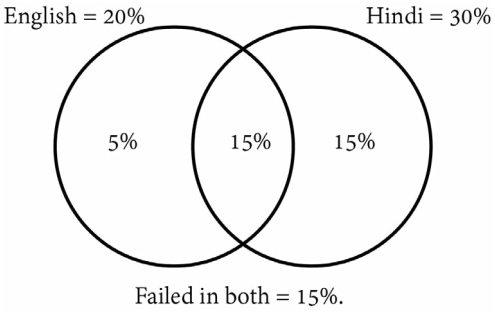
Since question has asked about the failed students.

So student passed in English = 80%.

Failed in English = 20%

Failed in Hindi = 30%

Failed in both = 15%.



Student who failed in both = 5% + 15% = 20

154. Answer: (a)

Explanation:

Let the age of father be X and the age of son be Y.

According to the question,

(X - n) = Y Or X - Y = n …….(i)

And X = 4(Y - n) Or X = 4Y - 4n Or (4Y - X)/4 = n …….(ii)

From equations (i) and (ii),

we get: X - Y = (4Y - X)/4 Or 4X – 4Y = 4Y - X Or 5X = 8Y Or X/Y = 8/5 Or we can say that, X = 8k and Y = 5k

Given that, X + Y = 130 Or 8k +5k = 130 Or 13k = 130 Or k = 10 Hence, age of father (X) = 8k = 80 years And age of son (Y) = 5k = 50 years

Difference in their ages = 80 – 50 = 30 years

155. Answer: (a)

Explanation:

- Let the cost price of honey be ₹1 per liter.
- We want a 20% profit on the honey.
  - That means on the cost price of ₹1, we want to make a profit of 20%:

    Required selling price=Cost Price+20% of Cost Price=1+0.201=₹1.20.
- The mixture (honey + water) will be sold at the cost price of honey, i.e., ₹1 per liter of the mixture.
  - We plan to sell the mixture at **₹1 per liter.**
- Determine the total volume of mixture needed to get a total selling amount of ₹1.20 when sold at ₹1 per liter.
- Volume of mixture required=$\frac{\text{Total amount we want (₹1.20)}}{\text{Price per liter (₹1)}}$ =1.20 liters.
- Since we started with 1 liter of honey, the amount of water to be added is the difference:

  Water to be added=1.20-1=0.20 liters.
- water added as a percentage of the honey originally taken:

  $\frac{0.20}{1} \times 100\% = 20\%$.

156. Answer: (d)

Explanation:

It is given that, a + b means a-b; a-b means a × b; a × b means a + b; a + b means a + b

Therefore, on changing the mathematical signs according to the given instructions, we get:

10 - 30 × 100 50 + 25

10 - 30 × 2 + 25

10 - 60 + 25

35 - 60 = - 25

157. Answer: (d)

Explanation:

From statement 1:

2x(S)3y and 3x(T)4z, then 9y(P)8z

On placing symbols according to the given instructions: 2 × ≥ 3y and 3x = 4z, then 9y > 8z × 3y/2 …….(a)

3x = 4z …….(b)

So, 4z/3 >= 3y/2

Or 8z >= 9y

Statement 1 is not correct

From Statement 2:

x(Q)2y and y(R)z, then x(R}z.

On placing symbols according to the given instructions:

x < 2y and y ≤ z, then x ≤ z.

x < 2y ……(a)

y ≤ z

or 2y ≤ 2z ……..(b)

So, x < 2y ≤ 2z Or x < 2z But from this we cannot deduce that x z Hence, **statement 2 is also not correct.**

158. Answer: (a)

Unit digit for the given expression $(57242)^{9x7x5x3x1}$ or $(57242)^{945}$ totally depends on unit digit of this five digits numbers i.e. 2.

Unit digit for

24n+1=2

24n+2=4

24n+3=8

24n+4=6

24n+5=2

(take n as any whole number)

So, (2)945 is in the form (2)944+1

i.e. 24n+1 so answer here is 2. **Hence, option (a) is correct.**

159. Answer: (c)

In an eight-digit number, there are 4 odd and 4 even positions. We have the number 11223344, which has 4 odd numbers (1, 1, 3, 3) and 4 even numbers (2,2,4,4).

To find the number of distinct arrangements, we calculate the number of ways the odd numbers can be arranged in the odd positions and the even numbers can be arranged in the even positions. The number of ways the 4 odd numbers (1, 1, 3, 3) can be arranged in the 4 odd positions is given by 4!/(2! x 2!)=6.

Here, we divide by 2 twice to account for the repetition of the digit 1 and the digit 3.

Similarly, the number of ways the 4 even numbers (2, 2, 4, 4) can be arranged in the 4 even positions is also 6, calculated as 4!/(2! x 2!).

To find the total number of distinct numbers, we multiply the number of arrangements for odd numbers by the number of arrangements for even numbers: 6 x 6 = 36.

Hence, option (c) is correct.

160. Answer: (b)

As a general rule, if A + B = N, then the product A x B is maximum when A = B = N/2. Similarly, if A+ B+ C= N1,

then the product A x B x C is maximum when A=B=C = N1 /3. For example, A + B+ C =30 then Ax B x Cis maximum when A = B = C = 30/3. So that value is 10 x 10 x 10 = 1000.

(p +q) (r +5) is maximum when p, g, 1,5 i 9, 8, 7, 6 and not necessarily in the same order and (p + q) = (r +s) happens if possible.

If p=9, q=8, r=7, s=6, then (9+8) =17 and (7+6) = 13 Referring to the rule, 17+13=30 and;

A +B = N, then the product A x B is maximum when A = B = N/2 (here, N= 30)

Meaning, (p + q) (r + s) is maximum when (p + q) = (r + s) = 30/2, which is 15; To achieve (p + q) = (r + s), we can assign the values as, p=9, q = 6,r=8,s = 7 with (P +q) =(r + s)

(9+6) (8+7) = (15) (15)

15x15=225

Hence, the greatest value of (p + q) (r + s), if p, g, rand s are distinct single digit positive numbers, is 225. **Hence, option (b) is correct.**

161. Answer: (b)

We have to find sum of all digits in integers from 10 to 100

We have number like

(10 to 19) ..... (i)

(20 to 29) ..... (ii)

(30 to 39) ..... (jii)

(40 to 49) ..... (iv)

(50 to 59) .....(v)

(60 to 69) ..... (vi)

(70 to 79) ..... (vii)

(80 to 89) ..... (viii)

(90 to 99) ..... (ix)

And 100.

So, sum of all digits in unit places:

9K (0+ 1+ 28+ 9) e 9 x 45 = 405

Sum of all digits at ten's places:

10X 1+ 10X 2. 10X 8+ 10X 9

It can be also written as, 10X (1 + 2 + 3.t 8 +9)

i.e. 10 x 45 = 450

When calculate the total by addition, 405 + 450 + 1 (from 100) = 856. **Hence, option (b) is correct.**

162. Answer: (c)

We are given that pp, qq, and rr are two-digit numbers, where p < g <r. This means that each of these numbers is greater than or equal to 10 and less than or equal to 99. We are also given that the sum of pp, qg, and rr is equal to tt0, where tt0 is a three-digit number ending with zero.

Now, let's consider the possible values of tt0.

Since tt0 is the sum of two-digit numbers pp, qg, and rr, the maximum possible value for the sum is when pp =99, qq = 98, and rr = 97.

In this case, the sum would be 99 + 98 + 97 = 294, which is less than 300.

Therefore, the value of tt0 cannot exceed 294, which means the possible

values for tt0 are limited to 110 and 220. These are the only three-digit numbers ending with zero that are less than or equal to 294.

In tt0, first and second digits are same, hence it can either be 110 or 220

Statement 1 and 2 are correct:

Case 1:

t0=110

so, pp +qq + rr = 110

10p + p + 10g + g + 10r + r =110

11(p+q+r) =110

p+q+r=10

Possible values of 'p' as p < q <r, will be 1 or 2.

possible values of 'q' as p < q < r will be 2 or 3 or 4. (3 distinct possible values)

Case 2:

tt0 = 220

so, pp + qq + rr =220

10p + P 10q +q + 10r + r = 220

11 (p +q + r) = 220

p+q+r=20

Possible values of 'p' here are 1 or 2 or 3 or 4 or 5.

Possible values of 'q' here are 6 or 7 or 8 as p<q<r. (3 other distinct possible values)

Total 5 possible distinct values of 'p' and 6 possible distinct values of 'q'. **Hence, option (c) is correct.**

163. Answer: (a)

As directed by the question, thousand place digit will always be 1, where 2, 3, 4 can be arranged in 3! ways giving six possible numbers.

4 digit numbers less than 2000 formed by digits 1,2,3,4 where digits do not repeat are:

```
  1234
  1243
  1432
  1423
  1324
+ 1342
------
  7998
```

Hence, option (a) s correct.

164. Answer: (d)

To find the remainder when 27192 is divided by 6, we can use the modulo function.

Let's consider the following:

21 mod 6 = 2 (2 divided by 6 gives remainder of 2 with quotient 0)

22 mod 6 = 4 (4 divided by 6 gives remainder of 4 with quotient 0)

23 mod 6 = 2 (8 divided by 6 gives remainder of 2 with quotient 1)

24 mod 6 = 4 (2 divided by 6 gives remainder of 4 with quotient 2)

25 mod 6 = 2 (2 divided by 6 gives remainder of 2 with quotient 5)

We observe that, when the 2odd is divided by 6 it gives remainder of 2 and conversely when 2even is divided by 6, it gives remainder of 4.

Conclusion following the logic shown in the pattern we find that, 2192 is an even number.

Hence 2192 divided by 6 will give remainder of 4.

Hence, option (d) is correct.

165. Answer: (c) is correct.

We can only use three of the four digits 2, 3, 7, and 9, in the first row. The first digit in the first row cannot be 7 or 9, as otherwise thrice the number will not be a three digit number. So, the first digit in the first row can either be 2 or 3.

The possible cases for the 1% row are: 237, 273, 239, 293, 279, 297,327,372, 329, 392, 379, or 397. On eliminating (392, 379, 397 and 372 the numbers whose 3 times is not a three-digit number) we are left with the numbers 237, 273,239, 293, 279, 297, 327, and 329.

We will check the above number: -

237 x 2 = 474 (digit repetition, and so eliminated)

273 x 2 = 546;273 x 3 =819 239 x 2 = 478; 239 x 3 = 717 (digit repetition, and so eliminated)

293 x 2 = 586; 293 x 3 = 879 (digit repetition, and so eliminated)

279 x 2 = 558 (digit repetition, and so eliminated)

297 x 2 = 594 (digit repetition, and so eliminated)

327 x 2 = 654; 327 x 3 =981

329x 2 = 658; 329 x 3 = 987 (digit repetition, and so eliminated)

So, only two possible cases. **Hence, option (c) is correct.**

166. Answer: (d) is correct.

Total amount is 1840.

Statement 1 is incorrect: 25 notes of Rs. 50 were used.

So, remaining amount = 1840 — (25 x 50) = 1840 — 1250 = Rs. 590. Even if all the remaining 25 notes are of Rs. 20 denominations, we will only get Rs. 500. So, Statement 1 is incorrect.

Statement 2 is incorrect: 35 notes of Rs. 20 were used.

So, remaining amount = 1840 — (35 x 20) = 1840 - 700 = Rs. 1140. Even if all the remaining 15 notes are of Rs. 50 denominations, we will only get Rs. 750. So, Statement 2 is incorrect.

Statement 3 is incorrect: 20 notes of Rs. 10 were used So, remaining amount = 1840 — (20 x 10) = 1840 - 200 = Rs. 1640.

Even if all the remaining 30 notes are of Rs. 50 denominations, we will only get Rs. 1500. So, Statement 3 is incorrect. **Hence, option (d) is correct.**

167. Answer: (b)

Price of petrol on m*" day of the year = 80 + 0.1m, where m=1,2,3...100 and after that it remains constant.

Price of petrol on 100th day i.e. m = 100 is 80 + 0.1(100) =80+10=90

So, price of petrol is constant after 100" day = 90

Now, price of diesel on n" day of the year = 69 + 0.15n

Price of diesel on 100" day = 69 + 0.15(100) = 69 + 15 = 84

Now, two possibilities arise.

Case - I: When m^{th} day s less than 100 and on m^{th} day price of petrol = price of diesel

80+0.1m = 69 + 0.15n

m = n

80 +0.1m = 69 + 0.15m

80-69 =0.15m - 0.1m

11=0.05m m = 220, which is not possible since we assumed m is less than 100.

Case - 2: When m^{th} day is greater than 100.

Price of petrol on mth day = 90

So, 90 = 69 +0.15m

90-69=0.15m

21=0.15m

m =140

So, on 140th day the price of petrol and diesel will be same.

The date on 140th day in 2021 is 20th May. Hence, option (b) is correct.

168. Answer: (d)

Total number of persons in a group = 120

Number of Indians = 80

Number of foreigners = 120 - 80 = 40

Number of persons who can speak English = 70

We have to find the range for number of Indians who can speak English.

If all the foreigners speak English, then number of Indians who can speak English = 70 - 40 = 30

If no foreigner speak English, then the number of English-speaking persons who are Indians = 70

So, the English-speaking Indians will fall in the range of 30 to 70. Hence, option (d) is correct.

169. Answer: (b)

The present age of X = 42 years.

Let the present age of Y be y years.

According to the question, the age of X at the time of Y's birth is twice the present age of Y.

x-y=2y

42-y=2y

42=3y

Y = 14 years

Now, we will check both the statements.

Statement-1: 8 years ago, the age of X was five times the age of Y.

8 years ago, age of X =42 — 8 = 34 years and age of Y = 14 - 8 = 6 years.

The age of X is not 5 times the age of Y.

So, this statement is not correct.

Statement-2: After 14 years, the age of X would be two times the age of Y.

After 14 years,

Age of X = 42 + 14 = 56 years (i.e.,2Y)

Age of Y = 14 + 14 = 28 years

Clearly, this statement is correct. Hence, option (b) is correct.

170. Answer: (d)

write the decimal values of original fraction:

2/3=0.666667

3/4=0.75

4/5=0.80

5/6=0.833

Now decimal values of new fraction after adding 5 in numerator and denominator:

2/3 will become 7/8 = 0.875 change in value = (0.875 - 0.666) = 0.209

3/4 will become 8/9 = 0.888 change in value = (0.888 — 0.75) = 0.138

4/5 will become 9/10 = 0.90 change in value = (0.9 - 0.8) = 0.1

5/6 will become 10/11 = 0.909 change in value = (0.909 - 0.833) =0.076

We can see that minimum change occur in the value of 5/6. Hence, option (d) is correct.

Question Hack:

Greater are the values of the numerator and denominator, the lesser will be the impact of any addition. For example, 1/1=1and 9/10 = 0.9 By adding 1 in numerator and denominator 1/2=0.5 change in value is 0.5 And 10/ 11 = 0.909 change in value is 0.009.

171. Answer: (a)

Given denominations are: Rs. 50, Rs. 100, Rs. 200, Rs, 500, and Rs. 2,000

We have to find no. of different sums by taking at least three denominations at a time.

Case I: taking all 5 denominations together.

No. of ways = 1

Case II: taking any 4 denominations:

No. of ways for selecting 4 out of 5 items = 5_{c4} = 5 and these sums are:

50 +100+200+500 = 850

50+200+500+2000 = 2750

50+100+500+2000 = 2650

50+100+200+2000 = 2350

100+200+500+2000 = 2800

Case III: taking 3 denominations at a time:

No. of ways for selecting 4 out of 5 items = 5_{c3} = 10

The number of different sums that can be formed =1+ 5 +10 = 16. Hence, option (a) is correct.

172. Answer: The correct answer is option (d)

P = (40% of A} + (65% of b) = Q =(50% of A) + (50% of b) = where A>B

Now, P-Q = P-Q=

We know that, A>B. there are three possibilities:

I. Let A=120and B = 100, then P—Q=15-12 =3 P<Q

II. Let A= 150 and B = 100, then P-Q = 15-15 =0 P=Q

III. Let A=170and B = 100, then P-Q=15-17 =-2P<Q

So, all the three cases are possible. **Hence, option (d) is correct.**

173. Answer: The correct answer is option (c)

Let the cost price of a cell phone = % 100 He sold it to Ram at 10% profit S0, the selling price = Rs. 110. Now, Ram wanted to sell it back to Gopal at 10% loss. Let the selling price = x

$$\text{Loss}=\frac{\text{CP-SP}}{\text{CP}}\times 100$$

$$10=\frac{110-x}{110}\times 100$$

110 = 1100 - 10x

10x = 990

x=99

Ram will sell it back to Gopal at % 99. Now, Gopal is getting the cell phone at % 99 which he was getting at X 100 previously. He registers a gain of 1%, **Hence, option (c) is correct.**

174. Answer: (c)

Let the number of flowers be x and number of bees be y.

If one bee lands on each flower, one bee will be left out.

-> y = x+1...(i)

If two bees land on each flower, one flower will be left out.

-> y =2(x-1) …(ii)

From (i) and (ii), we have

x+1=2x-2

-> x=3 and y=x + 1=4

So, number of flowers = 3 and number of bees = 4.

Hence, option (c) is correct.

175. Answer: (c)

Number of products produced in a day = 1000

Unfit for sale = 5% of 1000= 50

So, effective products produced in a day = 1000 — 50 = 950

Using unitary method,

950 products will be produced in 1 day.

so, 19000/950=22

So, 19000 products will be produced in 22 days. Hence **option (c) corrects.**

176. Answer: (a)

Observe the table given in question:

Monthly cost of petrol is always same that is 40 x 60 = 2400

Hence, when the price is Rs. 80/litre.

Then consumption will be 2400/80 = 30.

Hence, option (a) is correct.

177. Answer: (b)

Let's suppose each question carry 10 marks.

No. of question he attempted = 8.

No. of marks he obtained in each question = 50% of 5 marks.

Total no. of marks he obtained = 5 x 8 = 40.

Now he obtained 40% of total marks.

So, 40 is 40% total marks = 100 (40% of 100 = 40)

Hence, 10 questions carry 100 marks. Each question carries 10 marks.

Hence, option (b) is correct.

178. Answer: (c)

Suppose total no. of persons = 100

No. of men = 60

No. of women = 40.

No. of men cleared the qualifying test = 70% of 60 = 42

No. of women cleared the qualifying test = 75% of 40 = 30

No. of men cleared the final test = 80% of 42 = 33 (almost)

No. of women cleared the final test = 70% of 30 = 21.

Clearly More men cleared the examination than women.

Hence, option (c) is correct.

179. Answer: (d)

Cost of goat = g, Cost of cow =c

Given that, 4g < 5g

Also, 600 < g < 800.

Now 2400< 4g < 3200 and 3000 < 5g < 4000.

From here we can say that 2400<c<4000.

Hence, option (d) is correct.

180. Answer: (a)

Let's take volume of glasses = 12l

Now according to question:

1st glass is 1/3 filled Hence, there is 4l milk in it

And in 2nd glass there is 3l milk.

Water in 1st glass = 8

Water in 2nd glass = 91

Total milk = 7l

Total water = 17l

Ratio of milk to water = 7: 17.

Hence, option (a) is correct.

181. Solution: d)

Lets say both glasses have 100 ml capacity. Glass 1: 50 ml water and 50 ml milk

Glass 2: 25 ml water and 75 ml milk

Total when poured together = 200 ml Milk = 125 ml

Water = 75 ml

Ration = 125:75 = 5:3

182. Solution: (c)

Use the laws of reflection that you learnt in Science. You need not do all the arithmetic and geometry. Just visualize the rays being reflected at equal angles from OP1 and OP2, and you will know how the rays will pass from OP2. Its opposite and parallel to the direction of S.

183. Solution: (b)

If 60% play cricket, and 30% play football then total percentage that plays both = 60 + 30 = 90% But out of these, 10% that play both need to be subtracted as they are counted twice. So, 80% play

both. Those who play neither will be 100-80 = 20%.

184. Solution: (d)

The average score of student in 3 tests is N points. Hence the total score = 3N points.

Given the score in fourth test = N + 20, the average score of student in four tests will be = (3N+N+20)/4 = N+5

185. Solution: (d)

70% males, so 30% females.

If 2/7th of males are married, this means 2/7 (70 %) = 20% males are married. So, 10% females are married (out of 30% total married). So, single females will be 20% out of total 30% females.

The fraction is 2/3.

186. Answer: C

Explanation:

To satisfy the condition N=p+q+r N = p + q + r N=p+q+r, where p,q,rp, q, rp,q,r are distinct factors of NNN, we check each number below 50. Exactly 8 such numbers exist where the sum of any 3 distinct factors equals the number itself.

187. Answer: A

Explanation:

list all valid sets and their sums:

1. (p,q,r) = (7, 5, 3). Sum = 15.
2. (p,q,r) = (11, 7, 3). Sum = 21.
3. (p,q,r) = (19, 11, 3). Sum = 33.
4. (p,q,r) = (17, 11, 5). Sum = 33.
5. (p,q,r) = (19, 13, 7). Sum = 39.

The distinct possible values for (p+q+r) are: 15, 21, 33, 39.

188. Answer: C

Explanation: p,q,r: (3,3,3); (2,3,6); (2,4,4)

189. Answer: (d)

Explanation:

The total overs used by Team Y (10% less than 20) is 18 overs, and the alternate scenario uses 12 overs, but **no consistent value of N satisfies both conditions** without contradiction. Hence, **the data is insufficient to determine N uniquely.**

190. Answer : (a)

Explanation:

Successive equal percentage increase and decrease reduces the value overall. After four such operations, the final price becomes a fixed percentage of the original, leading to the relation in option (a).

191. Answer: C

Explanation: Using the **principle of inclusion and exclusion**, we subtract numbers divisible by 2, 3, 5, 7, and 9 (and their overlaps) from 100. The result is **22 numbers** not divisible by any of them.

192. Answer: D

Explanation : Max x+y=15x + y = 15x+y=15, Min x−y=−3x - y = -3x−y=−3, so ratio = 15 / (−3) = −5.

Since −5 is not in the options, the correct answer is **(D) None of the above**.

193. Answer: B

Explanation: Factorize the product to find powers of 5 and 7 (since 35 = 5 × 7). Count factors of 5 and 7 in the product and find the minimum power among them. The highest power nnn such that 35n35^n35n divides the product is **4**.

194. Answer: (b)

Explanation:

If a number has d digits, then its square root has:

- **d+1\2 digits if d is odd**
- **d\2 digits if d is even**

Since 17 is odd:

Number of digits in N=17+1/2=9

195. Answer is C

Explanation: Statement 1: 12 & 18; Statement 2: 2 & 5

196. Answer: D

Explanation: the values are 9, 12, 16 but they can also be 16, 12,9

197. Answer: C

Explanation: In both case, the min value is 0 only

198. Answer: A

Explanation: Since p is equally above and q equally below the average k, their deviations cancel out. Thus, r must be equal to k to keep the average at k.

7 Arithmetic

1. The class average x in a test increases by 4 when the score of a student is rectified, whose corrected score is 100 instead of 0. Later, the score of another student was found to have been recorded as 81 in place of 56. If there are no other corrections and the final corrected average is y, then y – x is (CSAT-2026)

(a) 2 (b) 3

(c) 5 (d) 6

2. Suppose x, y and z are variables taking positive real numbers as their possible values. It is given that y is directly proportional to x2 and x is inversely proportional to z. For z = 7/25, the values of x and y are 5 and 50, respectively. If y = 98, what is z equal to? (CSAT-2026)

(a) 1/7 (b) 1/5

(c) 5/7 (d) 1

3. In an objective type question paper, 5 marks are awarded for a correct answer and 2 marks are deducted for a wrong answer. A student attempted all the questions and got a score of 69. Had he been awarded 4 marks for a correct answer and 1 mark deducted for a wrong answer, he would have scored 84. How many questions were there in the question paper? (CSAT-2026)

(a) 99 (b) 81

(c) 84 (d) 79

4. A person saves 10% of his salary every month. If his salary increases by 12% and the expenditure increases by 10%, then what will be the change in his saving per month? (CSAT-2026)

(a) 20% increase (b) 30% increase

(c) 03% decrease (d) 02% decrease

5. The ratio of male to female workers in two companies A and B is 13 : 10 and 7 : 5, respectively. If both the companies have the same number of female workers, then what is the ratio of the total number of workers in A to those in B? (CSAT-2026)

(a) 24: 23 (b) 23: 24

(c) 18: 17 (d) 27 : 18

6. An alloy P contains 20% copper and 80% zinc by weight. Another alloy Q contains 60% copper and 40% zinc by weight. A third alloy R is to be prepared from P and Q so that it contains equal amount of copper and zinc. In what ratio, amounts of P and Q be mixed in order to get R? (CSAT-2026)

(a) 1:3 (b) 3: 1

(c) 2:3 (d) 3:2

7. Three partners A, B and C entered into a business. A invested one-third of the capital for one-third duration. B invested one-fourth of the capital for one-fourth duration. C invested the remaining capital for the whole duration. Out of a profit of ₹ 17,000, how much profit will C get? (CSAT-2026)

(a) ₹ 12,000 (b) ₹ 10,000

(c) ₹ 12,500 (d) ₹ 10,750

8. There are two chemicals which do not react with each other. A container contains 10 litres of the chemical A. One litre of this chemical is removed from it and one litre of the chemical B is poured. Then one litre of the mixture is removed from the container and one litre of B is poured. If this process of replacing one litre of the mixture by one litre of B is performed once more, then what is the volume of B that is present in the container approximately (in percentage)? (CSAT-2026)

(a) 25 (b) 27

(c) 29 (d) 31

9. Let p, q, r and s be distinct positive integers. Let p, q be odd and r, s be even. Consider the following statements: (CSAT-2024)

1. $(p-r)^2(qs)$ is even.
2. $(q-s)q^2s$ is even.
3. $(q+r)^2(p+s)$ is odd.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

10. The total cost of 4 oranges, 6 mangoes and 8 apples is equal to twice the total cost of 1 orange, 2 mangoes and 5 apples. Consider the following statements: (CSAT-2024)

1. The total cost of 3 oranges, 5 mangoes and 9 apples is equal to the total cost of 4 oranges, 6 mangoes and 8 apples.
2. The total cost of one orange and one mango is equal to the cost of one apple.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

11. 125 identical cubes are arranged in the form of a cubical block, How many cubes are surrounded by other cubes from each side? (CSAT-2023)

(a) 27 (b) 25

(c) 21 (d) 18

12. A principal P becomes Q in 1 year when compounded half-yearly Y with R% annual rate of interest. If the same principal P becomes Q in 1 year when compounded annually with \$% annual rate of interest, then which one of the following is correct? (CSAT-2023)

(a) $R = S$ (b) $R > S$

(c) $R < S$ (d) $R \leq S$

13. **There are three traffic signals. Each signal changes colour from green to red and then from red to green. The first signal takes 25 seconds, the second signal takes 39 seconds and the third signal takes 60 seconds to change the colour from green to red. The durations for green and red colours are same. At 2:00 p.m, they together turn green. At what time will they change to green next, simultaneously? (CSAT-2023)**

(a) 4:00 p.m. (b) 4:10 p.m

(c) 4:20 p.m. (d) 4:30 p.m.

14. **A biology class at high school predicted that a local population of animals will double in size every 12 years. The population at the beginning of the year 2021 was estimated to be 50 animals. If P represents the population after n years, then which one of the following equations represents the model of the class for the population? (CSAT-2021)**

(a) P=12+50n (c) $P=50(2)12^n$

(b) P=50+12n (d) $P=50(2)^n/12$

15. **In a class, 60% of students are from India and 50% of the students are girls. If 30% of the Indian students are girls, then what percentage of foreign students are boys? (CSAT-2021)**

(a) 45% (b) 40%

(c) 30% (d) 20%

16. **A boy plays with a ball, and he drops it from a height of 1.5m. Every time the ball hits the ground, it bounces back to attain a height 4/5th of the previous height. The ball does not bounce further if the previous height is less than 50 cm. What is the number of times the ball hits the ground before the ball stops bouncing? (CSAT-2021)**

(a) 4 (b) 5

(c) 6 (d) 7

17. **Integers are listed from 700 to 1000. In how many integers is the sum of the digits 10? (CSAT-2021)**

(a) 6 (b) 7

(c) 8 (d) 9

18. **Consider all 3-digit numbers (without repetition of digits) obtained using three non-zero digits which are multiples of 3. Let S be their sum.**

Which of the following is/are correct? (CSAT-2021)

1. S is always divisible by 74.
2. S is always divisible by 9.

Select the correct Answer using the code given below:

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

19. **Consider the following addition problem:**

3P +4P + PP + PP = RQ2; where P, Q and R are different digits.

What is the arithmetic mean of all such possible sums? (CSAT-2021)

(a) 102 (b) 120

(c) 202 (d) 220

20. **Consider the following multiplication problem:**

(PQ) x 3 = RQQ, where P, Q and R are different digits and R≠0.

What is the value of (P+R) ÷ Q? (CSAT-2021)

(a) 1

(b) 2

(c) 5

(d) Cannot be determined due to insufficient data.

21. **There are three points P, Q and R on a straight line such that PQ: QR = 3 : 5. If n is the number of possible values of PQ: QR, then what is n equal to? (CSAT-2021)**

(a) 1 (b) 2

(c) 3 (d) 4

22. **When a certain number is multiplied by 7, the product entirely comprises ones only (1111..). What is the smallest such number? (CSAT-2021)**

(a) 15713 (b) 15723

(c) 15783 (d) 15873

23. **Consider the following data:**

	Average marks in English	Average marks in Hindi
Girls	9	8
Boys	8	7
Overall average marks	8.8	x

What is the value of x in the above table? (CSAT-2020)

(a) 7.8 (b) 7.6

(c) 7.4 (d) 7.2

24. **A bottle contains 20 liters of liquid 4 liters of liquid A is taken out of its and replace by same quantity of liquid B. Again 4 liters of the mixture is taken out and replaced by same quantity of liquid B. What is the ratio of quantity of liquid A to that of liquid B in the final mixture? (CSAT-2020)**

(a) 4:1 (b) 5:1

(c) 16:9 (d) 17:8

25. **The average score of a batsman after his 50th innings was 46.4. After 60th innings, his average score increases by 2.6. What was his average score in the last ten innings? (CSAT-2020)**

(a) 122 (b) 91

(c) 62 (d) 49

26. **As a result of 25% hike in the price of rice per kg, a person is able to purchase 6 kg less rice for Rs. 1,200. What was the original price of rice per kg? (CSAT-2020)**

(a) Rs. 30 (b) Rs. 40

(c) Rs.50 (d) Rs. 60

27. **A shop owner offers the following discount options on an article to a customer:**

1. Successive discounts of 10% and 20%, and then pay a service tax of 10%.
2. Successive discounts of 20% and 10%, and then pay a service tax of 10%.
3. Pay a service tax of 10% first, then successive discounts of 20% and 10%.

Which one of the following i correct? (CSAT-2020)

(a) 1only is the best option for the customer.

(b) 2 only is the best option for the customer.

(c) 3 only is the best option for the customer.

(d) All the options are equally good for the customer.

28. **What is the greatest length x such that are m and m integral multiples of x? (CSAT-2020)**

(a) $1\frac{1}{2}$m (b) $1\frac{1}{3}$m

(c) $1\frac{1}{4}$m (d) $1\frac{3}{4}$m

29. **In adult population of a city, 40% men and 30% women are married. What population is of married adults in city if no man marries more than one woman and no woman marries more than one man; and there are no widows and widowers? (CSAT-2020)**

(a) $33\frac{1}{7}$% (b) 34%

(c) $34\frac{2}{7}$% (d) 35%

30. **A sum of Rs. 2,500 is distributed among X, Y and Z in two ratios $\frac{1}{2}:\frac{3}{4}:\frac{3}{4}$ What is the difference between the '4 maximum share and the minimum share? (CSAT-2020)**

(a) Rs.300 (b) Rs. 350

(c) Rs.400 (d) Rs. 450

31. **In a class, there are three groups A, B and C. If one student from group A and two students from group B are shifted to group C, then what happens to the average weight of the students of the class? (CSAT-2020)**

(a) It increases.

(b) In decreases.

(c) It remains the same.

(d) No conclusion can be drawn due to insufficient data.

32. **A and B are two heavy steel blocks. If B is placed on the top of A, the weight increases by 60%. How much weight will reduce with respect to the total weight of A and B, if B is removed from the top of A? (CSAT-2019)**

(a) 60% (b) 45.5%

(c) 40% (d) 37.5%

33. **When a runner was crossing the 12 km mark, she was informed that she has completed only 80% of the race. How many kilometres was the runner supposed to run in this event? (CSAT-2019)**

(a) 14 (b) 15

(c) 16 (d) 16.5

34. **Raju has ₹9000 with him, and he wants to buy a mobile handset; nut he finds that he has only 75% of the amount required to buy the handset. Therefore, he borrows ₹2000 from a friend. Then (CSAT-2019)**

(a) Raju still does not have enough amount to buy the handset.

(b) Raju has exactly the same amount as required to buy the handset.

(c) Raju has enough amount to buy the handset and he will have 500 with him after buying the handset.

(d) Raju has enough amount to buy the handset and he will have 1000 with him after buying the handset.

35. **In 2002, Meenu's age was one-third of the age of Meera, whereas in 2010, Meenu's age was half the age of Meera. What is Meenu's year of birth? (CSAT-2019)**

(a) 1992 (b) 1994

(c) 1996 (d) 1998

36. **Rakesh and Rajesh together bought 10 balls and 10 rackets. Rakesh spent ₹1300 and Rajesh spent ₹1500. If each racket costs three times a ball does, then what is the price of a racket? (CSAT-2019)**

(a) ₹70 (b) ₹ 80

(c) ₹210 (d) ₹240

37. **In an examination, A has scored 20 marks more than B. if B has scored 5% less marks than A, how much has B scored? (CSAT-2019)**

(a) 360 (b) 380

(c) 400 (d) 420

38. **Seeta and Geeta go for a swim after a gap of every 2 days and every 3 days respectively. If on 1% January both of them went for a swim together, when will they go together next? (CSAT-2019)**

(a) 7th January (b) 8th January

(c) 12th January (d) 13th January

39. **X, Y and Z are three contestants in a race of 1000m. Assume that all run with different uniform speeds. X gives Y a start of 40m and X gives Z a start of 64m. If Y and Z were to compete in a race of 1000m, how many metres start will Y give to Z? (CSAT-2019)**

(a) 20 (b) 25

(c) 30 (d) 35

40. **Rakesh had money to buy 8 mobile handsets of a specific company. But the retailer offered very good discount on that particular handset. Rakesh could buy 10 mobile handsets with the amount he had. What was the discount the retailer offered? (CSAT-2019)**

(a) 15% (b) 20%

(c) 25% (d) 30%

41. **The average marks of 100 students are given to be 40. It was found later that marks of one student were 53 which were misread as 83. The corrected mean marks are (CSAT-2019)**

(a) 39 (b) 39.7

(c) 40 (d) 40.3

42. **If the numerator and denominator of a proper fraction are increased by the same positive quantity which is greater than zero, the resulting fraction is (CSAT-2019)**

(a) Always less than the original fraction

(b) Always greater than the original fraction

(c) Always equal to the original fraction

(d) Such that nothing can be claimed definitely

43. **A family has two children along with their parents. The average of the weights of the children and their mother s 50 kg. The average of the weights of the children and their father is 52 kg. If the weight of the father is 60 kg, then what is the weight of the mother? (CSAT-2019)**

(a) 48 kg (b) 50kg

(c) 52 kg (d) 54 kg

44. **Suppose you have sufficient amount of rupee currency in three denominations: ₹1, ₹10 and ₹50. In how many different ways can you pay a bill of 1072 (CSAT-2019)**

(a) 16 (b) 17

(c) 18 (d) 19

45. **Sunita cuts a sheet of paper into three pieces. Length of first piece is equal to the average of the three single digit odd prime numbers. Length of the second piece is equal to that of the first plus one-third the length of the third. The third piece is as long as the other two pieces together. The length of the original sheet of paper is (CSAT-2019)**

(a) 13 units (b) 15 units

(c) 16 units (d) 30 units

46. **How many triplets (x, y, z) satisfy the equation x + y + z =6, where x, y and z are natural numbers? (CSAT-2019)**

(a) 4 (b) 5

(c) 9 (d) 10

47. **If X is between -3 and -1, and Y is between -1 and 1, then x^2 + y^2 is in between which of the following? (CSAT-2018)**

(a) 9 and 1 (b) -9and 1

(c) 0 and 8 (d) 0 and 9

48. **X and Y are natural numbers other than 1, and Y is greater than X. which of the following represents the largest number?(CSAT-2018)**

(a) XY (b) X/Y

(c) Y/X (d) (X+Y) /XY

49. **A number consist of three digits of which the middle one is zero and their sum is 4. If the number formed by interchanging the first and last digits is greater than the number itself by 198, then the difference between the first and last digits is (CSAT-2018)**

(a) 1 (b) 2

(c) 3 (d) 4

50. **While writing all the numbers from 700 to 1000, how many numbers occur in which the digit at hundred's place is greater than the digit at ten's place is greater than the digit at unit's place? (CSAT-2018)**

(a) 61 (b) 64

(c) 85 (d) 91

51. **A bookseller sold 'a' number of Geography textbooks at the rate of X per book, 'a + 2' number of History textbooks at the rate of X + 2 per book and 'a - 2' number of Mathematics textbooks at the rate of X - 2 per book. What is his total sale in terms of a and X? (CSAT-2018)**

(a) 3aX + 8 (b) 3aX + 4

(c) 3aX + 6 (d) 3aX + 10

52. **A student has to get 40% marks to pass in an examination. Suppose he gets 30 marks and fails by 30 marks, then what are the maximum marks in the examination? (CSAT-2018)**

(a) 100 (b) 120

(c) 150 (d) 300

53. **19 boys turn out for playing hockey. Of these, 11 are wearing hockey shirts and 14 are wearing hockey pants. There are no boys without shirts and/or pants. What is the number of boys wearing full uniform? (CSAT-2018)**

(a) 3 (b) 5

(c) 6 (d) 8

54. **A shopkeeper sells an article at ₹40 and gets X% profit. However, when he sells it at ₹20, he faces same percentage of loss. What is the original cost of the article? (CSAT-2018)**

(a) ₹10 (b) ₹20

(c) ₹30 (d) ₹40

55. **A lift has the capacity of 18 adults or 30 children. How many children can board the lift with 12 adults? (CSAT-2018)**

(a) 6 (b) 10

(c) 12 (d) 15

56. **A person bought a refrigerator worth ₹22,800 with 12.5% interest compounded yearly. At the end of the first year, he paid ₹8650 and at the end of the second year ₹9125. How much will he have to pay**

at the end of the third year to clear the debt? (CSAT-2018)

(a) ₹9,990 (b) ₹10,000

(c) ₹10,500 (d) ₹11,250

57. The average rainfall in a city for the first four days was recorded to be 0.40 inch. The rainfall on the last two days was in the ratio of 4: 3. The average of six days was 0.50 inch. What was the rainfall on the fifth day? (CSAT-2017)

(a) 0.60 inch (b) 0.70 inch

(c) 0.80 inch (d) 0.90 inch

58. A 2-digit number is reversed. The larger of the two numbers is divided by the smaller one. What is the largest possible remainder? (CSAT-2017)

(a) 9 (b) 27

(c) 36 (d) 45

59. Suppose the average weight of 9 persons is 50 kg. The average weight of the first 5 persons is 45 kg, whereas the average weight of the last 5 persons is 55 kg. Then the weight of the 5th person will be (CSAT-2017)

(a) 45 kg (b) 47.5 kg

(c) 50 kg (d) 52.5 kg

60. In a city, 12% of households earn less than 30,000 per year, 6% households earn more than 2,00,000 per year, 22% households earn more than 990 households earn between per year. 1,00,000 per year and 30,000 and ₹ 1,00,000

How many households earn between 1, 00,000 and 2,00,000 per year? (CSAT-2017)

(a) 250 (b) 240

(c) 230 (d) 225

61. There are thirteen 2-digit consecutive odd numbers. If 39 is the mean of the first five such numbers, then what is the mean of all the thirteen numbers? (CSAT-2017)

(a) 47 (b) 49

(c) 51 (d) 45

62. If there is a policy that 1/3rd of population of a community has migrated every year from one place to some other place, what is the leftover population of that community after the sixth year, if there is no further growth in the population during this period? (CSAT-2017)

(a) 16/243rd part of the population.

(b) 32/243rd part of the population.

(c) 32/729th part of the population.

(d) 64/729th part of the population.

63. The sum of income of A and B is more than that of C and D taken together. The sum of income of A and C is the same as that of B and D taken together. Moreover, A earns half as much as the sum of the income of B and D. Whose income is the highest?(CSAT-2017)

(a) A (b) B

(c) C (d) D

64. There are three pillars X, Y and Z of different heights. Three spiders A, B and C start to climb on these pillars simultaneously. In one chance, A climbs on X by 6 cm but slips down 1 cm. B climbs on Y by 7 cm but slips down 3 cm. C climbs on Z by 6.5 cm but slips down 2 cm. If each of them requires 40 chances to reach the top of the pillars, what is the height of the shortest pillar? (CSAT - 2017)

(a) 161 cm (b) 163 cm

(c) 182 cm (d) 210 cm

65. A freight train left Delhi for Mumbai at an average speed of 40 km/hr. Two hours later, an express train left Delhi for Mumbai, following the freight train on a parallel track at an average speed of 60 km/hr. How far from Delhi would the express train meet the freight train? (CSAT-2017)

(a) 480 km (b) 260 km

(c) 240 km (d) 120 km

66. In a class, there are 18 very tall boys. If these constitute three-fourths of the boys and the total number of boys is two - thirds of the total number of students in the class, what is the number of girls in the class? (CSAT-2016)

(a) 6 (c) 18

(b) 12 (d) 21

67. The monthly average salary paid to all the employees of a company was Rs. 5000. The monthly average salary paid to male and female employees was Rs. 5200 and Rs. 4200, respectively. Then the percentage of males employed in the company is (CSAT-2016)

(a) 75% (c) 85%

(b) 80% (d) 90%

68. If R and S are different integers both divisible by 5, then which of the following is not necessarily true? (CSAT-2016)

(a) R - S is divisible by 5. (b) R+ S is divisible by 10.

(c) R x S is divisible by 25. (d) $R^2+ S^2$ is divisible by 5.

69. In a class of 60 students, where the number of girls is twice that of boys, Kamal, a boy, ranked seventeenth from the top. If there are 9 girls ahead of Kamal, the number of boys in rank after him is: (CSAT-2016)

(a) 13 (c) 7

(b) 12 (d) 3

70. Each of A, B, C And D has Rs. 100. A pays Rs. 20 to B, who pays Rs. 10 to C, who gets Rs. 30 from D. In this context, which one of the following statements is not correct? (CSAT-2015)

(a) C is the richest

(b) D is the poorest

(c) C has more than what A and D have together

(d) B is richer than D

71. In a parking area, the total number of wheels of all the cars (four — wheelers) and scooters / motorbikes (two — wheelers) is 100 more than twice the number of parked vehicles. The number of cars parked is (CSAT-2015)

(a) 35 (b) 45

(c) 50 (d) 55

72. The monthly income of Peter and Paul are in the ratio of 4: 3. Their expense are in the ratio of 3:2. If each saves Rs. 6000 at the end of the month, their monthly incomes respectively are (in Rs.) (CSAT-2015)

(a) 24,000 and 18,000 (b) 28,000 and 21,000

(c) 32,000 and 24,000 (d) 34,000 and 26,000

73. A bell rings every 18 minutes. A second bell rings every 24 minutes. A third bell rings every 32 minutes. If all the three bells ring at the same time at 8 o'clock in the morning, at what other time will they all ring together? (CSAT-2014)

(a) 12: 40 hrs (b) 12: 48 hrs

(c) 12: 56 hrs (d) 13: 04 hrs

74. If Sohan, while selling two goats at the same price, makes a profit of 20% on one and incurs a loss of 20% on the other (CSAT-2014)

(a) he makes no profit and no loss.

(b) he makes a profit of 1%.

(c) he suffers a loss of 1%.

(d) he suffers a loss of 2%.

75. Out of a total of 120 musicians in a club, 5% can play all the three instruments, guitar, violin, and flute. It so happens that the number of musicians who can play any and only two of the above instruments is 30. The number of musicians who can play guitar alone is 40. What is the total number of those who can play violin alone or flute alone? (CSAT-2014)

(a) 45 (b) 44

(c) 38 (d) 30

76. In a rare coin collection, there is one gold coin for every three non-gold coins. 10 more gold coins are added to the collection and the ratio of gold coins to non-gold coins would be 1: 2. Based on the information; the total number of corns in the collection now becomes (CSAT-2013)

(a) 90 (b) 80

(c) 60 (d) 50

77. A gardener has 1000 plants: He wants to plant them in such a way that the number of rows and the number of columns remains the same. What is the minimum number of plants that he needs more for this purpose? (CSAT-2013)

(a) 14 (b) 24

(c) 32 (d) 34

78. A sum of RS. 700 has to be used to give seven cash prizes to the students of a school for their overall academic performance. If each prize is Rs. 20 less than its preceding prize, then what is the least value of the prize? (CSAT-2013)

(a) RS. 30 (b) RS. 40

(c) RS. 60 (d) RS. 80

79. Out of 120 applications for a post, 70 are male and 80 have a driver's license. What is the ratio between the minimum to maximum number of males having driver's license? (CSAT-2013)

(a) 1 to 2 (b) 2 to 3

(c) 3 to 7 (d) 5 to 7

80. In a garrison, there was food for 1000 soldiers for one month. After 10 days, 1000 more soldiers joined the garrison. How long would the soldiers be able to carry on with the remaining food? (CSAT-2013)

(a) 25 days (b) 20 days

(c) 15 days (d) 10 days

81. The tank-full petrol in Arun's motor-cycle lasts for 10 days. If he starts using 25% more everyday, how many days will the tank-full petrol last? (CSAT-2013)

(a) 5 (b) 6

(c) 7 (d) 8

82. A person has only Rs. 1 and Rs. 2 coins with her. If the total number of coins that she has is 50 and the amount of money with her is Rs. 75, then the number of Rs. 1 and Rs. 2 coins are, respectively (CSAT-2011)

(a) 15 and 35 (b) 35 and 15

(c) 30 and 20 (d) 25 and 25

83. Three persons start walking together and their steps measure 40 cm, 42 cm and 45 cm respectively. What is the minimum distance each should walk so that each can cover the same distance in complete steps? (CSAT-2011)

(a) 25 m 20 cm (b) 50 m 40 cm

(c) 75 m 60 cm (d) 100 m 80 cm

84. A village having a population of 4000 requires 150 liters of water per head per day. It has a tank measuring 20 m x 15 m x 6 m. The water of this tank will last for (CSAT-2011)

(a) 2 days (b) 3 days

(c) 4 days (d) 5 days

7 Arithmetic_Explanations

1. Answer: (b)

Explanation:

Let the number of students in the class be n. The first correction changes a student's score from 0 to 100, so the total score increases by 100. Since the average increases by 4, we have:

100 / n = 4

Therefore:

n = 25

Now, another student's score was recorded as 81 instead of 56. Correcting this decreases the total score by:

81 − 56 = 25

Since there are 25 students, the average decreases by:

25 / 25 = 1

After the first correction, the average becomes x + 4. After the second correction, the final corrected average is:

y = x + 4 − 1 = x + 3

Hence:

y − x = 3

Therefore, the correct answer is 3.

2. Answer: (b)

Explanation:

Set up the proportionalities.

- $y \propto x2 \Rightarrow y = kx2$
- $x \propto 1/z \Rightarrow x = c/z$, i.e. $xz = c$

Find the constants using z = 7/25, x = 5, y = 50.

- From $y = kx2$: $50 = k \cdot 25 \Rightarrow k = 2$
- From $xz = c$: $c = 5 \cdot (7/25) = 7/5$

So $y = 2x^2$ and $xz = 7/5$.

Use y = 98 to find x.

$2x2 = 98 \Rightarrow x2 = 49 \Rightarrow x = 7$

Find z.

z = c/x = (7/5) / 7 = 1/5

Correct option: 1/5

3. Answer: (b)

Explanation

Let the number of correctly answered questions be x and the number of wrongly answered questions be y. Since the student attempted all questions, the total number of questions in the paper is x + y.

According to the first condition, 5 marks are awarded for a correct answer and 2 marks are deducted for a wrong answer, resulting in a score of 69:

5x − 2y = 69 — (Equation 1)

According to the second condition, if 4 marks were awarded for a correct answer and 1 mark deducted for a wrong answer, the score would be 84:

4x − y = 84 — (Equation 2)

From Equation 2, we can express y in terms of x:

y = 4x − 84

Substitute this value of y into Equation 1:

5x − 2(4x − 84) = 69

5x − 8x + 168 = 69

−3x = 69 − 168

−3x = −99

x = 33

Now, substitute the value of x back into the expression for y:

y = 4(33) − 84

y = 132 − 84

y = 48

The student answered 33 questions correctly and 48 questions wrongly. The total number of questions in the paper is:

x + y = 33 + 48 = 81

Therefore, there were 81 questions in the question paper.

4. Answer: (b)

Explanation:

Let the initial salary of the person be 100.

Given that the person saves 10% of his salary:

- Initial saving = 10% of 100 = 10
- Initial expenditure = Initial salary − Initial saving = 100 − 10 = 90

Now, the salary increases by 12% and the expenditure increases by 10%:

- New salary = 100 + 12% of 100 = 100 + 12 = 112
- New expenditure = 90 + 10% of 90 = 90 + 9 = 99

The new saving will be the difference between the new salary and the new expenditure:

- New saving = New salary − New expenditure = 112 − 99 = 13

Now, we calculate the change in his saving:

- Increase in saving = New saving − Initial saving = 13 − 10 = 3
- Percentage increase in saving = (Increase in saving / Initial saving) × 100%
- Percentage increase = (3/10) × 100% = 30%

Therefore, there is a 30% **increase** in his saving per month.

5. Answer: (b)

Explanation:

Let the number of male and female workers in company A be 13x and 10x, respectively. Let the number of male and female workers in company B be 7y and 5y, respectively.

Given that both companies have the same number of female workers, we can equate the two:

$10x = 5y \Rightarrow y = 2x$

Now, let's find the total number of workers in each company.

- Total workers in company A = 13x + 10x = 23x
- Total workers in company B = 7y + 5y = 12y

Substitute y = 2x into the total for company B: Total workers in company B = 12(2x) = 24x. The ratio of the total number of workers in company A to those in company B is:

23x / 24x = 23 : 24

Alternative Method:

Ratio of male to female workers in A = 13 : 10. Ratio of male to female workers in B = 7 : 5. To make the number of female workers equal in both ratios, we can multiply the ratio for company B by 2:

- New ratio for company B = $(7 \times 2) : (5 \times 2) = 14 : 10$

Now, the number of female workers is 10 units in both companies.

- Total workers in company A = 13 + 10 = 23 units
- Total workers in company B = 14 + 10 = 24 units

Ratio of total workers in A to B = 23 : 24.

6. Answer: (a)

Explanation:

Let the quantity of alloy P to be mixed be x and the quantity of alloy Q to be mixed be y.

In alloy P, the percentage of copper is 20%. In alloy Q, the percentage of copper is 60%. In the final alloy R, the amounts of copper and zinc are equal, which means the percentage of copper in R must be 50%.

Method 1: Using Algebra

The total amount of copper in the mixture is the sum of the copper from P and Q: Total copper = 0.20x + 0.60y. The total weight of the new alloy R is (x + y). Since copper makes up 50% of alloy R, we can set up the following equation:

$0.20x + 0.60y = 0.50(x + y)$

Expanding and solving for the ratio of x to y:

$0.20x + 0.60y = 0.50x + 0.50y$

$0.60y - 0.50y = 0.50x - 0.20x$

$0.10y = 0.30x$

$x / y = 0.10 / 0.30 = 1/3$

Method 2: Using the Rule of Alligation

We can apply the rule of alligation directly to the concentration of copper in the alloys:

- Concentration of copper in P = 20%
- Concentration of copper in Q = 60%
- Desired mean concentration in R = 50%

The ratio of P to Q is calculated by taking the cross-differences: Ratio = $(60 - 50) : (50 - 20) = 10 : 30 = 1 : 3$.

Thus, the amounts of P and Q must be mixed in the ratio 1 : 3.

7. Answer: (a)

Explanation:

To find the profit share of each partner, we need to calculate the ratio of the product of their invested capital and the duration of their investment. Let the total capital be K and the total duration be T.

1. **Investment and Duration of A:**
   - Capital invested by A = (1/3)K
   - Duration of A's investment = (1/3)T
   - Equivalent investment of A = $(1/3)K \times (1/3)T = (1/9)KT$
2. **Investment and Duration of B:**
   - Capital invested by B = (1/4)K
   - Duration of B's investment = (1/4)T
   - Equivalent investment of B = $(1/4)K \times (1/4)T = (1/16)KT$
3. **Investment and Duration of C:**
   - Capital invested by C = Remaining capital = $K - ((1/3)K + (1/4)K) = K - (7/12)K = (5/12)K$
   - Duration of C's investment = Whole duration = T
   - Equivalent investment of C = $(5/12)K \times T = (5/12)KT$
4. **Ratio of Profits:**

   The profit-sharing ratio of A, B, and C is the ratio of their equivalent investments:

   A : B : C = 1/9 : 1/16 : 5/12

   To simplify this ratio, we multiply by the Least Common Multiple (LCM) of the denominators 9, 16, and 12, which is 144.
   - A's share in ratio = $(1/9) \times 144 = 16$
   - B's share in ratio = $(1/16) \times 144 = 9$
   - C's share in ratio = $(5/12) \times 144 = 60$

   The simplified profit ratio is 16 : 9 : 60.
5. **Calculating C's Profit:**
   - Total parts in the ratio = 16 + 9 + 60 = 85 parts
   - Total profit = ₹ 17,000
   - Value of 1 part = 17,000 / 85 = 200

   C's profit share = 60 parts × 200 = ₹ 12,000

   Therefore, C will get a profit of ₹ 12,000.

8. Answer: (b)

Explanation:

To find the final volume of chemical B, we can first calculate the remaining volume of chemical A after the replacement process is completed.

Step 1: Understand the given parameters

- Initial volume of chemical A (V) = 10 litres.
- Total volume of the mixture in the container remains constant at 10 litres.
- Volume replaced in each step (x) = 1 litre.
- Number of times the process is performed (n) = 3 (The first replacement, then a second replacement, and then performed "once more").

Step 2: Apply the replacement formula

The formula for the amount of the original liquid remaining after n replacements is given by:

Final Amount of A = Initial Amount × (1 − x/V)n

Substitute the values into the formula:

A3 = 10 × (1 − 1/10)3 = 10 × (9/10)3

A3 = 10 × 729/1000 = 7.29 litres

Step 3: Calculate the volume and percentage of chemical B

Since the total volume of the mixture is always 10 litres, the volume of chemical B present in the container after 3 operations is:

Volume of B = Total Volume − Final Amount of A = 10 − 7.29 = 2.71 litres

Now, convert this volume into a percentage of the total mixture:

Percentage of B = (2.71 / 10) × 100 = 27.1%

Approximately, the percentage of chemical B is 27%. Therefore, the correct option is B.

9. Answer: (c)

Explanation:

Here, p and q are odd numbers. Similarly, r and s are even numbers.

Some key facts to consider:

- The square of an odd number is odd.
- The square of an even number is even.
- The product of an odd number with an even number is even.
- The sum or difference of an odd and an even number is odd.
- The sum or difference of two odd numbers or two even numbers is even.

Statement 1: $(p-r)^2(qs)$ is even.

- p is odd and r is even, so p-r is odd.
- Squaring an odd number gives an odd number.
- q is odd and is even, so is even.
- Multiplying an odd number by an even number yields an even result.

Thus, **Statement 1 is true.**

Statement 2: $(q-s)q^2$ is even.

- q is odd and is even, so q - s is odd.
- Squaring an odd number (q^2) gives an odd number.
- Multiplying two odd numbers (odd × odd) gives an odd result.

Thus, **Statement 2 is false.**

Statement 3: $(q+s)^2(p+s)$ is odd.

- q is odd and s is even, so p+s is odd. Its square remains odd.
- is odd and is even, so is odd.
- The product of two odd numbers (odd × odd) is odd.

Thus, **Statement 3 is true.**

10. Answer: (c)

Explanation:

Let the costs of oranges, mangoes and apples be Q, M and A.

4Q + 6M + 8A = 2(Q + 2M + 5A)

Or 4Q + 6M + 8A = 2Q + 4M + 10A Or

2Q + 2M = 2A Or

Q + M = A

Statement 1: 3Q + 5M + 9A = 4Q + 6M + 8A Or A = Q + M It is true.

Statement 2: Q + M = A It is true. **Hence, both statements 1 & 2 are true.**

11. Answer: (a)

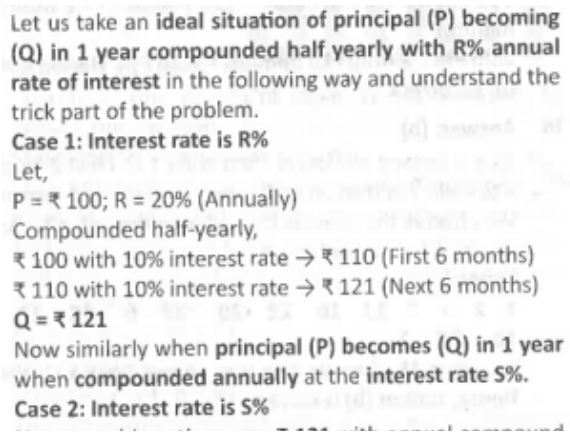
There are 125 cube, which means 53, i.e. 5x5x5. Hence n=5.

In a standard 3-dimensional cube, the inner layer can be obtained by reducing the length, width, and height of the larger cube by 1 unit on each side. This reduction accounts for removing the outermost layer on all sides.

The correct calculation for the number of internal cubes in a 3-dimensional cube is (n-2)3, where "n" represents the length of the larger cube.

In the case of 125 identical cubes arranged in a cubical block, the length of the larger cube is 5. Thus, the number of internal cubes = (5-2)3 = (5-2) x (5-2) x (5-2) =33 =27. **Hence, option (a) is correct.**

12. Answer: (c)


Let us take an **ideal situation of principal (P) becoming (Q) in 1 year compounded half yearly with R% annual rate of interest** in the following way and understand the trick part of the problem.
Case 1: Interest rate is R%
Let,
P = ₹ 100; R = 20% (Annually)
Compounded half-yearly,
₹ 100 with 10% interest rate → ₹ 110 (First 6 months)
₹ 110 with 10% interest rate → ₹ 121 (Next 6 months)
Q = ₹ 121
Now similarly when **principal (P) becomes (Q) in 1 year** when **compounded annually** at the **interest rate S%.**
Case 2: Interest rate is S%

Now, to achieve the **same ₹ 121** with annual compound interest rate, compounded annually, the interest rate would be:
Compounded annually,
₹ 121 with **21% interest rate → ₹ 121** (in 1 year)
where, **P = 100; Q = 121**, then **S%** becomes **21%.**
An interest rate of 20% compounded half-yearly will yield same result as interest rate of 21% compounded annually. Hence, R < S. Hence, option (c) is correct.

13. Answer: (b)

The cycle times for the three signals are as follows:

Signal 1: 25 seconds

Signal 2: 39 seconds

Signal 3: 60 seconds

To find the next time when all three signals will turn green simultaneously, we need to find the least common multiple (LCM) of these cycle times.

Prime factorization of each cycle time is:

25 = 52

39 = 3 x 13

60 = 22 x 3 x 5

To calculate the LCM, we take the highest power of each prime factor:

LCM=22x3x52x13

4x3x25 x 13 = 3900 seconds

To convert the time to a standard format, we divide 3900 seconds by 60 to get the number of minutes:

3900 seconds ÷ 60 = 65 minutes or 1 hour and 5 minutes

This is the time signals will take to turn from green to red simultaneously.

It will take another 65 minutes or 1 hour 5 minutes. to turn from red to green simultaneously. The traffic signals will simultaneously change to green again at:

65 minutes + 65 minutes = 130 minutes or 2 hours 10 minutes.

2:00 PM + 2 hours 10 minutes = 4:10 PM. **Hence, option (b) is correct.**

14. Answer: (d)

The population at the beginning of the year 2021 = 50

P represents population and n represents years.

Let be some constant.

The population of animals will double in size every 12 years.

The equation will be exponential in the form: $P = 50(2)^{an}$

When n =0, $P = 50(2)^0 = 50(1) = 50$

When n = 12 years, P = 2P = 2(50) = 100

$100 = 50(2)12\alpha$

$2 = (2)12\alpha$

On comparing the powers of 2, we get $1=12\alpha=\alpha=1/12$

So, the equation becomes: $P = 50(2)^{n/12}$. Hence, option (d) is correct.

15. Answer: (d)

Let the total number of students in the class = 100

60% of students are from India = 60

-> Number of foreign students = 100 - 60 = 40

50% of these students are girls i.e., 50 students are girls.

If 30% Indian students are girls i.e., 30% of girls are Indian.

Number of foreign girls are 50 - 18 = 32

∴ Number of foreign boys = 40-32 = 8

% of foreign students who are boys – 40 X100 - 20%.

Hence, option (d) is correct.

16. Answer: (b)

Given height from where the ball is dropped = 1.5 m = 1.5 x 100 cm = 150 cm

When ball hits the ground for the 1st time, its height = (4/5) x 150 = 120 cm

When ball hits the ground for the 2nd time, its height = (4/5) x 120 =96 cm

When ball hits the ground for the 3rd time, its height = (4/5) x 96 = 76.8 cm

When ball hits the ground for the 4th time, its height = (4/5) x 76.8 = 61.44 cm

When ball hits the ground for the 5th time, its height = (4/5) x 61.44 = 49.15 cm (< 50 cm)

After that the ball will not bounce back. So, after the ball hits the ground for the 5th time, it doesn't bounce back Hence, option (b) is correct.

17. Answer: (d)

We have to find out all the integers from 700 to 1000 whose sum of the digits is 10.

The integers are: 703, 712, 721, 730, 802, 811, 820, 901 and 910.

So, there are 9 integers from 700 to 1000 whose sum of digits is 10. Hence, option (d) is correct.

18. Answer: (c)

Three non-zero digits which are multiples of 3 are 3, 6, and 9.

Without repetition of digits, we can make 6 three-digit numbers (3! = 3x2x1=6)

The 3-digit numbers are 369, 396, 639, 693, 936 and 963.

S is the sum of these 3-digit numbers.

So, S =369 + 396 + 639 + 693 + 963 = 3996

Also, 3996 =2x2x3x3x3x37

3996=6x9x74

We can clearly see that 3996 is divisible by both 74 and 9. Hence, option

(c) is correct.

19. Answer: (c)

We have,

3P=30+P

4P=40+P

PP=10P +P=11P

Now, 3P +4P + PP + PP = RQ2

30+P+40+P+11P +11P =RQ2

70 + 24P = RQ2

We can clearly see that the last digit of resultant is 2. So,

P can be 3 or 8.

If P =3, then 70 + 24(3) = 70 + 72 = 142

If P =8, then 70 + 24(8) = 70 + 192 = 262

Possible sums are 142 and 262.

$=\frac{142+262}{2}=\frac{404}{2}=202$. So, arithmetic mean of all possible sums. Hence, option (c) is correct.

20. Answer: (b)

We know, PQ=10P+Q

RQQ =100R +10Q + Q= 100R + 11Q

Given, (PQ) x 3 =RQQ

(10P +Q) x3=100R + 11Q

30P+3Q=100R +11Q

30P = 100R + 8Q....(1)

The last digit of 30P and 100R is 0. So, the last digit of 8Q must be 0.

Hence, Q =5 and 8Q = 8x5=40

Put the value of 8Q in equation (1)

30P = 100R + 40

3P = 10R + 4

P, Q and R are different digits.

Let R=1, then P=14/3

Let R=2, then P=24/3=8

So, P=8, Q=5 and R=2

Also, when R =5, then P = 18 (2-digit). So, this is rejected.

(P+R)÷Q = (8+2)÷5 = 10÷5 = 2 Hence, option (b) is correct.

21. Answer: (b)

There are two ways to arrange three points P, Q and R on a straight line such that PQ: QR=3:5

P---3---Q---5---R (internally)

Q---3---P---2---R (externally)

Q can divide the line PR internally and externally. So, n =2. Hence, option (b) is correct.

22. Answer: (d)

In this question, we will proceed by checking all the options.

We will multiply all numbers given in the options by 7 and see which of the product entirely comprises 1's only.

15713 x 7 = 109991

15723 x 7 = 110061

15783 x 7 = 110481

15873 x 7=111111 (all 1's)

Hence, option (d} is correct.

23. Answer: (a)

Let no. of boys = b

No. of girls = g

Marks obtained by all girls in English = 9g

Marks obtained by all boys in English = 8g

Marks obtained by all girls in Hindi = 8g

Marks obtained by all boys in Hindi = 9g

Overall average marks in English

Or 9g + 8b = 88 + 8.8b

Or 0.2 = 0.8 or g = 4B

Now, x = average marks in Hindi

Hence, x = 7.8. Hence, option (a) is correct.

24. Answer: (c)

Initial quantity in bottle is: 20 liter of liquid A and 0 liter of liquid B.

1st Mixing:

Now we take 4 liters of liquid A from bottle and replaced with liquid B.

Quantities after 1st mix: 16 liters of liquid A and 4 liters of liquid B.

2nd Mixing:

Again 4 liters of the mixture is taken out and replaced by same quantity of liquid B.

Now 4 liters of mixture contain:

20 liters of mixture contain 16 liters of liquid A and 4 liters of liquid B

1 liter of mixture contain: 16/20 liters of liquid A and 4/ 20 liters of liquid B

4 liters of mixture contain: 16/5 liters of liquid A and 4/5 liter of liquid B.

Quantities after 2nd mix:

Liquid A: 16 — 16/5 = 64/5 liters

Liquid B: 8 — 4/5 = 36/5 liters

Now ratio of liquids in mixture:

Liquid A: Liquid B = 64/5: 36/5 = 16:9. Hence, option (c) is correct.

25. Answer: (c)

Given that, average score after 1st 50 innings = 46.4

So, total after 50 innings = 50 x 46.4 = 2320

Average score after 60 innings = 46.4 + 2.6 = 49

So, total runs after 60 innings = 60 x 49 = 2940

Runs made in last ten innings = 2940 - 2320 = 620

Average score in last 10 innings = 620/10 = 62. Hence, option (c) is correct.

26. Answer: (b)

Let the original price = x/kg

Quantity purchased for Rs. 1200 = (1200/x) kg

Hike in price = 25%

New price = x + 25% of x = x + 25x/100 = 5x/4

Quantity purchased for Rs. 1200 on new price = 1200 x 4/5x = 960/x

Now according to question: (1200/x) - (960/x) = 6

240/x = 6, implies x = 240 / 6 = 40

Original price = 40/kg. Hence, option (b) is correct.

27. Answer: (d)

Let's take the base price = 100

Option 1:

After 1st discount = 100 — 10% of 100 = Rs. 90

After 2nd discount = 90 — 20% of 90 = Rs. 72

After Service tax = 72 + 10% of 72 = Rs. 79.2

Option 2:

After 1st discount = 100 — 20% of 100 = Rs. 80

After 2nd discount = 80 — 10% of 80 = Rs. 72

After Service tax = 72 + 10% of 72 =Rs. 79.2

Option 3:

After Service tax = 100 + 10% of 100 = Rs. 110

After 1st discount = 110 - 20% of 110 = Rs. 88

After 2nd discount = 88 — 10% of 88 = Rs. 79.2

In every option customer has to pay 79.2.

All options are equal for the customer. Hence, option (d) is correct.

28. Answer: (d)

To find the greatest length of which these lengths are integer multiple we need to find

H.CF. of

To find H.C.F. of we need to find

H.C.F. of a and c (numerator), and

L.C.M of b and d (denominator)

In this question a = 7, b=2, c=35 and d=4

HCF of 7 and 35= 7

LCM of 2 and 4 = 4

Hence, the H.C.F 7/2 of 35/4=7/4=13/4m Hence, option (d) is correct.

29. Answer: (c)

Let the number of women in the city = 100

Then no. of married women = 30% of 100 = 30

Now, no. of married man = no. of married women Hence, no. of married men = 30

And it is given that there are 40% married men.

If no. of men in the city = x

Then 40% of x = 30

40x/100 = 30 imply x = 75

Hence, no. of men = 75 Total population = 100 + 75 = 175

Married population = 30 + 30 = 60

Now we need to check what percent of 175 is 60.

Let 60 is y percent of 175

Then 175y/100= 60

y = 6000/175 = $34\frac{2}{7}$% Hence, option (c) s correct.

30. Answer: (c)

Given that share of X, Y and Z is in ration.

Multiply the ratio expression by 12 we get =6:9:10

Now we got that share of X, Y and Z is in ratio 6:9:10.

Let's take share of X = 6x

Then share of Y = 9x and

Share of Z=10x

Now by question 6x+9x+10x = 2500 imply x = 100

Hence, share of X = 6x = 600

share of Y = 9x = 900

share of Z = 10x = 1000

So, difference between maximum and minimum = 1000 - 600 = 400. Hence, option (c) is correct.

31. Answer: (c)

$$SS = \frac{\text{total weight of the class}}{\text{no. of students in the class}}$$

Average weight of the students of the class

As per the information given in the question only shifting of students take place.

Hence, no student left the class, no new student joined.

So, the total weight and number of students remain the same.

The average weight remains the same. Hence, option (c) is correct.

32. Answer: (d)

Let the weight of A and B is x and y, respectively.

If B s placed on the top of A, the weight increases by 60% i.e., the weight becomes

Now, B is removed from the top of A.

Decrease in weight = $\frac{\text{total weight -weight of A}}{\text{total weight}} \times 100\%$

$=\frac{\frac{8}{5}x-x}{\frac{8}{5}x} \times 100\%$

$=\frac{\frac{3}{5}x-x}{\frac{8}{5}x} \times 100\%$

$=\frac{3}{8} \times 100\% = \mathbf{37.5\%}$

Hence, option (d) is correct.

Question Hack:

Let the weight of A = 100 kg

If B is placed on the top of A, the weight increases by

60% The combined weight of A + B = 160 kg

Now, B is removed from the top A.

Then the decrease in weight

33. Answer: The correct answer is option (b)

Let the distance which the runner is supposed to run = km

It is given that, when the runner crossed the 12km mark she has completed 80% of the race.

80% of race = 12 km

$=>\frac{80}{100}x=12$

x=15 km

So, 15 kilometres were supposed to run in this event.

Hence, option (b) is correct.

34. Answer: The correct option is option (a)

Total money with Raju = 29000

Let the cost of mobile handset = x

Raju has only 75% of the amount required to buy the handset.

$=> \frac{8}{5}x=9000$

x=12000

∴ The cost of mobile handset = ₹12000

Now, Raju borrows ₹2000 from his friend.

Total money with Raju = ₹9000 + ₹2000 =₹11000

But the cost of mobile handset is ₹12000

He needs ₹12000 - ₹11000 = ₹1000 more to buy the mobile handset.

So, Raju still does not have enough amount to buy the handset. **Hence, option (a) is correct.**

35. Answer: The correct option is option (b)

In 2002, let the age of Meera and Meenu be x and y, respectively.

Given,

$y = \frac{x}{3}$ .. (1)

In 2010 i.e., 8 years later, Meenu's age was half the age of Meera

$(y + 8) = \frac{x+8}{2}$

Put the value of y from equation (1)

$\frac{x}{3}+8 = \frac{x+8}{2}$

$\frac{2x}{3}+16=x-8$

$x-\frac{2x}{3}=16-8$

$\frac{x}{3}=8$

x=24

From (1), $y=\frac{x}{3}=\frac{24}{3}=8$

The age of Meenu in 2002 = 8 years

Meenu's year of birth = 2002 — 8 = 1994, **Hence, option (b) is correct.**

36. Answer: The correct answer is option (c)

Let the price of a ball = x

The price of racket is three times the price of ball i.e., 3x

Now, the price of 10 balls = 10x

The price of 10 rackets = 30x

Total money spent by Rakesh and Rajesh in buying 10

balls and 10 rackets = 1300 + 31500 = 32800

=>10x + 30x = 2800

=>40x = 2800

=>x=70

So, the price of ball = 270

The price of racket = 3x = ₹210. **Hence, option (c) is correct.**

37. Answer: The correct answer is option (b)

Let the marks scored by B be x.

A has scored 20 marks more than B

Marks scored by A = x + 20

Given that, Marks of B = 5% less than A

=>x=(x+20) (x +20)

=>100x = 100x + 2000 — 5x - 100

=>5x=1900

=>x=380

So, marks scored by B = 380.

Hence, option (b) is correct.

38. Answer: The correct answer is option (d)

On January 1%, both Seeta and Geeta went for swimming together.

Seeta and Geeta go for a swim after a gap of every 2 days and every 3 days respectively.

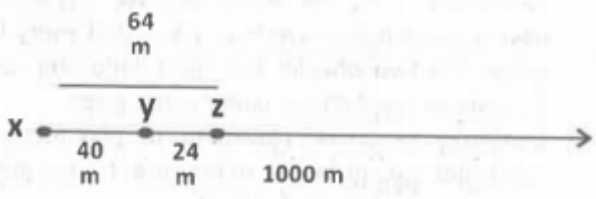
Seeta went for swimming on dates: 1st, 4th, 7th, 10th, 13th and 16th... (gap of 2 days)

Similarly, Geeta went for swimming on dates: 1st, 5th, 9th, 13th and 17th... (gap of 3 days)

So, they both went for swimming together on 13th January. **Hence, option (d) is correct.**

39. Answer: The correct answer is option (b)

In a 1000m race, gives a start of 40m, and gives a start of 64 m.



This implies, Y covers a distance of 1000 - 40 = 960m

For 960m, Y gives Z a start of 60 - 40 = 24m

So, for 1000m race, Y gives Z a start = 25m

Hence, option (b) is correct.

40. Answer: The correct option is option (b)

Let Rakesh had x amount money.

He can buy 8 mobile handsets with that amount of money.

Cost of 1 mobile handset = x/8

Now, he got a discount and he could buy 10 mobile handsets with the same amount of money.

So, the cost of each mobile handset becomes = x/10

$$\text{Discount} = \frac{\text{actual price-sold price}}{\text{actual price}} \times 100$$

$$\text{Discount} = \frac{\left(\frac{x}{8}\right)-\left(\frac{x}{8}\right)}{\frac{x}{8}} \times 100$$

Discount = 20%

The discount offered by the retailer is 20%. Hence,

option (b) is correct.

41. Answer: The correct answer is option (b)

The average marks of 100 students = 40+x_2

$x_1+x_2\ldots+x_{100}=400$

It is found out later that marks of one student 53 which were misread as 83

$x_1+x_2\ldots+x_{99}+83=4000$

$x_1+x_2\ldots+x_{99}=3917$

Hence option (b) is correct

42. Answer: The correct option is option (b)

Proper fraction: when the numerator is less than the denominator.

Let is a proper fraction

=> a < 1

Let us add a positive quantity x on both the sides

=> a + x < b + x

Now, because (b-a) >0

$$\frac{a+x}{b+x} > \frac{a}{b}$$

The resulting fraction is always greater than the original fraction. **Hence, option (b) is correct.**

43. Answer: The correct option is option (d)

Given, weight of father = 60 kg

Let the weight of both the children be x.

The average of the weights of the children and their father is 52 kg.

$$\frac{x+60}{3}=52$$

x+60 =156

x= 156 - 60 = 96 kg

The weight of both the children = 96 kg

It is also given that the average of the weights of the children and their mother is 50 kg

Let the weight of mother be y kg

$$\frac{96+y}{3}=50$$

96 + y=50

y = 150 - 96 = 54 kg

So, the weight of mother = 54 kg. **Hence, option (d) is correct.**

44. Answer: The correct answer is option (c)

The various possibilities are given below:

CASE-I: Two ₹50 notes. There is only 1 possibility i.e.

2(₹50)+7(₹1)=₹107

CASE-II: One ₹50 note. There are 6 possibilities. These

are:

₹50+57(₹1)=₹107

₹50+ ₹10+ ₹47(₹1) = ₹107

₹50+2(₹10)+37(₹1) = ₹107

₹50+₹3(₹10) + ₹27(₹1) = ₹107

₹50+₹4(₹10)+₹17(₹1) = ₹107

₹50+₹5(₹10)+₹7(₹1) =₹107

CASE-III: No ₹50 note. There are 11 possibilities. These are:

₹10(₹10)+7(₹1)=₹107

₹9(₹10)+17(₹1)=₹107

₹8(₹10)+27(₹1) = ₹107

₹7(₹10) + 37(₹1) = ₹107

₹6(₹10) + 47(₹1) = ₹107

₹5(₹10)+57(₹1) = ₹107

₹4(₹10)+67(₹1) = ₹107

₹3(₹10)+77(₹1) = ₹107

₹2(₹10)+87(₹1) = ₹107

₹1(₹10)+97(₹1) = ₹107

₹0(₹10)+107(₹1) = ₹107

So, total number of possibilities = 1 +6 +11=18. Hence, option (c) is correct.

45. Answer: The correct option is option (d)

Sunita cuts a sheet of paper into three pieces.

Length of 1º' piece = Average of 3 single digit odd prime numbers i.e., average of 3, 5 and 7.

So, length of 1 piece = $\frac{3+5+7}{3}=\frac{15}{3}=5$ units

Let the length of second piece be x units.

Length of 3rd piece = length of 1 piece + length of 2" piece =5+x

Also, length of 2" piece = length of 1% piece + one-third of length of 3rd piece

x = 5 + (5=x)

3x = 20 + x

2x = 20

x = 10

The length of 2 piece = 10 units

The length of 3rd piece =5 + x = 5 + 10 = 15units

So, the length of original sheet of paper =5+ 10 + 15 = 30 units. **Hence, option (d) is correct.**

46. Answer: The correct answer is option (d)

Natural numbers are the numbers that starts from 1 i.e., 1,2,3..

We have x + y + z = 6

When x = y = z = 2 then x + y + z = 2 + 2 + 2 = 6

(2,2, 2) is one of the triplets.

When x = 1, y = 1 and z = 4, then x + y + z = 1 + 1 + 4 = 6

Three combinations are possible.

These are: (1,1, 4), (1,4,1) and (4,1, 1)

When x = 1, y = 2 and z = 3, then x + y + z = 1 + 2 + 3 = 6

Here, six combinations are possible.

These are (1, 2, 3), (1, 3,2),(2,1,3),(2,3,1), (3,1, 2) and (3,2,1)

So, total number of triplets, (x, y, z) =1 + 3 + 6 = 10.

Hence, option (d) is correct.

47. Answer: (d)

The correct option is (d).

It is given that,

-3 < X < -1

1 < X^2 < 9

-1 < Y < 1

0 < y^2 < 1

Maximum value of Y^2 is 1

1-1<x^2- Y^2 < 9-1

0 < x^2 - y^2<8

Minimum value of y^2 is 0

1<x^2 - y^2 < 9

So, x^2 - y^2 lies between 0 and 9

48. Answer: (a)

The correct option is (a)

X and Y are natural numbers other than 1, and Y is greater than X.

Y>X>1

Let X=2 and Y=4

Now,

XY = 8 (maximum)

X/Y=1/2=0.5

Y/X=4/2=2

(X+Y)/XY=6/8=3/4=0.75

So, XY represents the largest number.

49. Answer: (b)

The correct option is (b)

Let the 3-digit number be xyz.

The middle digit is 0 i.e.,y=0 and x+y+z=4

→z = 4 - x

So, the number is x 0 (4 - x) i.e., 100x + (4 - x) = 99x + 4

The number formed by interchanging the first and last digit is (4 -x) 0 x

Interchanged number = 100(4 - x) + x = 400 — 99x

Also, the interchanged number is 198 greater than the original number.

s correct

->400-99x

=99x + 4 + 198

->400 - 202 = 99x + 99x

->198 = 198x

->x=1

So, the number is 103 and the difference between first and last digit is 3 - 1 = 2.

50. Answer: (c)

The correct option is (c)

We have to find the numbers from 700 to 1000 such that

the digit at hundred's place is greater than the digit at

ten's place is greater than the digit at unit's place.

From 700 to 800:

700 – 709	710 – 719	720 – 729	730– 739	740 - 749	750 - 759	760- 770	771- 800
0	1	2	3	4	5	6	0
	710	720, 721	730, 731, 732	740, 741, 742, 743	750, 751, 752, 753, 754	760, 761, 762, 763, 764, 765	

Total required numbers from 700-800=1+2+3+4 +5+6=21

Similarly, from 801 - 900:

801- 809	810 – 819	820 – 829	830 – 839	840 – 849	850 – 859	860 – 869	870 – 879	880 – 900
0	1	2	3	4	5	6	7	0

Total required numbers from 801 -900=1+2 +3 +4 +5+6+7=28

Similarly, from 901 — 1000:

901- 909	910 – 919	920 – 929	930 – 939	940 – 949	950 – 959	960 – 969
0	1	2	3	4	5	6

Total required numbers from 901 -1000=1+2+3+4 +5+6+7+8=36

So, total required numbers from 700 to 1000 = 21 + 28 +36 = 85.

51. Answer: (b)

The correct option is (b)

A shopkeeper sold Geography, history, and Mathematics textbooks.He sells 'a' number of Geography textbooks at the rate ₹x per book.

He sells 'a + 2" number of History textbooks at the rate ₹(x +2) per book.

He sells 'a - 2" number of Mathematics textbooks at the rate of ₹(x - 2) per book.

Total sale = Sale of Geography + History + Mathematics textbooks

=ax+(a+2)(x+2)+(a-2)(x-2)

=ax+(ax+2x+2a+4) + (ax-ax-2x - 4)

=3ax+8

52. Answer: (c)

The correct option is (c)

Let the maximum marks in the examination by x. passing marks

Marks obtained by student = 30 and he failed by 30 marks.

=>x=150

So, the maximum mark in the examination is 150.

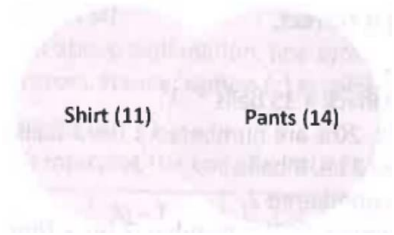
53. Answer: (c)

The correct option is (c)

Total boys = 19

Number of boys wearing hockey pants = 14

Number of boys wearing hockey shirt = 11



To solve this, we will use the principle of set theory. Let:

S= 11 (boys wearing shirts)

P= 14 (boys wearing pants)

N= 19 (total boys)

F= boys wearing both shirts and pants

We know:

S+P−F=N

S + P - F = N

S+P−F=N

Substitute the values:

F=6 (Answer)

54. Answer: (c)

The correct option is (c)

Let the cost price of an article is a

When the shopkeeper sells an article at % 40, he got X% profit.

When the shopkeeper sells an article at Rs. 20, he got X% loss.

Divide (i) by (ii), we get

40 - a=a - 20

2a = 60

A = 30

So, the original cost of article is X 30.

55. Answer: (b)

The correct option is (b)

A lift has the capacity of 18 adults or 30 children.

=>18A=30C

This means 3 adults and 5 children have the same capacity.

When 12 adults board the lift then 18 - 12 = 6 adult's capacity is still remaining in the lift.

3 adults have the same capacity as 5 children.

-> 6 adults have the same capacity as 5 x 2 = 10 children.

S0, 10 children can board the lift with 12 adults.

56. Answer: (d)

The correct option is (d)

Cost of refrigerator = 22800.

Interest rate = 12.5%

Interest of 1st year = 12.5% Of 22800 = 22800 x 12.5/100 = 2850.

Total debt at the end of 1%t year = 22800 + 2850 = 25650.

Amount paid after 1 year = 8650.

Principal amount for the 2nd year = 25650 - 8650 = 17000

Interest of 2nd year = 12.5% of 17000 = 2125.

Total debt = 17000 + 2125 = 19125.

Amount he paid = 9125.

Principal amount at the starting of 3rd year = 19125 - 9125 = 10000.

Interest for 3rd year = 12.5% of 10000 = 1250.

Amount to be paid after end of 3rd year = 10000 + 1250 = 11250.

57. Answer: The correct answer is option (c)

The average rainfall for first four days = 0.40 inch

$= \frac{D_1 + D_2 + D_3 + D_4}{4} = 0.40$

$D_1 + D_2 + D_3 + D_4 = 1.6$ ...(1)

Also.

$D_5 : D_6 = 4:3$ ...(2)

So, according to question,

$\frac{D_1 + D_2 + D_3 + D_4 + D_5 + D_6}{6} = 0.50$

$D_1 + D_2 + D_3 + D_4 + D_5 + D_6 = 3$

From (1)

$1.6 + D_5 + D_6 = 3$

$D_5 + D_6 = 1.4$

$D_6 = 1.4 - D_5$

Put the of D_6 in (2)

$3D_5 = 5.6 - 4D_5$

$7D_5 = 5.6$

$D_5 = 0.80$

->So, the rainfall on the fifth day 0.80 inch. **Hence, option (c) is correct.**

58. Answer: The correct answer is option (d)

We have to maximise the remainder. For that to happen, one number should be as large as possible. Let the two numbers are of the form 9x and x9. When x = 1, the numbers are 91 and 19, the remainder

=15 When x = 2, the numbers are 92 and 29, the remainder

=3 When x = 3, the numbers are 93 and 39, the remainder

=15 When x = 4, the numbers are 94 and 49, the remainder

= 45(max) When x 5.5, the numbers are 95 and 59, the remainder =36 After that the remainder is decreasing. So, the largest possible remainder is 45. **Hence, option (d) is correct.**

59. Answer: The correct option is option (c)

Let the weight of 9 persons be W1, W2, W3 ... W9,

The average weight of 9 persons = 50kg

=W1+W2+W3+...\W9 =50 x 9 = 450kg... (i)

The average weight of first 5 persons is 45kg

=W1+W2+ W3+ W4 +WS5+=45x5 = 225kg... (ii)

The average weight of last 5 persons is 55kg

=WS5 + W6 + W7 + W8 + W9 = 55 x Skg... (iii)

Adding (ii) and (iii), we get

WI1+W2+W3+W4A+WS5+W5+W6+W7+W8+W9=225+275

WI+W2+W3+W4+2W5+W6+W7+W8+W9=500

Using (i),

450 + W5 = 500

WS5 = 50kg

So, the weight of the 5th person = 50 kg. Hence, option (c)is correct.

60. Answer: The correct answer is option (b)

Let the total number of households =

Households who earn less than ₹30,000 per year = 12%

-> Households who earn more than ₹30,000 per year = 100-12 = 88%

Households who earn more than ₹1, 00,000 per year = 22%

->Households who earn between ₹30,000 to ₹1, 00,000 per year = 88 — 22 = 66%

It is given that there are 990 households who earn between Rs. 30,000 and Rs. 100000 per year.

=>x=1500

So, total number of households = 1500

Households who earn more than ₹2, 00,000 per year = 6%

So, households who earn between ₹1, 00,000 to ₹2,00,000 per year = 22 — 6 = 16% of households

$= \frac{16}{100} \times 1500 = 240.$

Hence, option (b) is correct.

61. Answer: The correct answer is option (a)

Let the 5 consecutives off numbers be: x -4, x -2, x+2 and x+4

Their means =39

=>5x=195

=>x=39

So, the first five consecutive odd numbers of the series are: 35, 37, 39, 41 and 43.

The thirteen consecutive odd numbers of the series are:

35, 37, 39, 41, 43, 45, 47,49, 51, 53, 55, 57 and 59.

So, the mean of all the thirteen numbers = 47. **Hence, option (a) is correct.**

62. Answer: The correct answer is option (d)

It is given that one-third of the population migrated every year.

Let the initial population be x.

After one year the population world be After 2nd year the population world be

After 3rd year the population world be Similarly, After 6rd year the population world be

So, the leftover population of that community after the sixth year is 64/729th part of the population. **Hence, option (d) is correct.**

63. Answer: The correct answer is option (b)

Using the information given in the question, we infer

that:

A+B>C+D...(i) A+C=B+D... (ii)

=> B+ D = 2A

Put the value of B + D in (ii)

A+ C = 2A

=> A=C

Using (i)

A+B>C+D

C+B>C+D

=>B>D

From (iii), we can conclude that the value of A lines

between B and D.

So, the order is B>A=C>D

The income of B is highest. **Hence, option (b) is correct.**

64. Answer: (b)

- **Spider A (Pillar X):** Climbs 6 cm and slips 1 cm in one chance. Net climb per chance 5cm. In the last chance, it only climbs 6 cm since it doesn't slip.
- **Spider B (Pillar Y):** Climbs 7 cm and slips 3 cm in one chance. Net climb per chance = 7−3=4 cm. In the last chance, it only climbs 7 cm since it doesn't slip.
- **Spider C (Pillar Z):** Climbs 6.5 cm and slips 2 cm in one chance. Net climb per chance = 6.5−2=4.5 cm. In the last chance, it only climbs 6.5 cm since it doesn't slip.
- Each spider takes **40 chances to reach the top** of their respective pillars.

In the first 39 chances, each spider climbs the net height per chance:

- **Spider A:** 39×5=195 cm
- **Spider B:** 39×4=156 cm
- **Spider C:** 39×4.5=175.5 cm

In the 40th chance, each spider climbs their full climbing distance (without slipping):

- **Spider A:** 195+6=201 cm195 + 6 = 201
- **Spider B:** 156+7=163 cm156 + 7 = 163
- **Spider C:** 175.5+6.5=182 cm175.5 + 6.5 = 182

The heights of the pillars are:

- X=201 cm
- Y=163 cm
- Z=182 cm

Thus, the shortest pillar is **163 cm**.

65. Answer: The correct answer is option (c)

Speed of freight train = 40 km/hr, Speed of express train = 60 km/hr.

Let the distance travelled by express train and freight train when they meet is x km.

Let the time taken by freight train to travel x km = t hours Express train started after 2 hours. Time taken by express train to meet the freight train = (t-2) hours

We know, distance = speed x time

So, 40 x t = 60 x (t - 2)

40t = 60t - 120

20t = 120

t = 6 hours

So, the distance travelled = 6 x 40 = 240 km. **Hence, option (c) is correct.**

66. Answer: (b)

Let the total number of students in the class = x

Total number of boys = two-third of the number of students =

So, total number of girls =

Number of very tall boys = 18

These are three-fourth of the boys =

x=36

Total number of students = 36

Number of girls = 36/3 = 12.

Hence, option (b) is correct.

67. Answer: (b)

Let the number of male and female employees of a company are x and y, respectively.

Let the total salary of all the male and all the female employees be M and F respectively.

The monthly average salary paid to all the employees = Rs. 5000

=>M+F=5000 (x +y) ... (i)

The monthly average salary paid to male and female employees was Rs. 5200 and Rs. 4200 respectively.

=>=5200 = > M = 5200x

And = 4200 = > F = 4200y

So, equation (i) becomes,

5200x + 4200y = 5000x + 5000y

=>200x = 800y

=>x=4y

So, the percentage of male employess **Hence, option (b) correct.**

68. Answer: (b)

It is given that R and S are different integers and both are divisible by 5.

Let R = 5x and S = 5y, where x, y can take 1,2, 3 ..

Now, R—S=5x—5y =5 (x—v) Clearly, it is divisible by 5.

R × S = 5x × 5y = 25xy. Clearly, it is divisible by 25.

$R^2 + S^2 = (5x)^2 + 5y^2 = 25 (x^2 + y^2)$. Clearly, it is divisible by 5.

R+S = 5x + 5y = 5 (x + y).

This may not be divisible by 10,

when (x +y) is odd.

Hence, **option (b) is correct.**

69. Answer: (b)

Total number of students = 60

Let the number of girls and boys be x and y respectively.

Given, x = 2y

So, number of girls and boys in a class are 40 and 20 respectively

Kamal ranked 17" from top and there are 9 girls ahead of him.

->There are 16 -9 = 7 boys ahead of him.

Total number of boys = 20

So, number of boys after Kamal = 20 - 7 - 1 (himself) = 12.

Hence, option (b) is correct.

70. Answer: (c) in starting A=B=C=D = 100

A pays Rs. 20 to B then A =80 and 8 = 120

B pays 10 to C then B = 110 and C = 110.

C gets 30 from D then C= 140 and D =70

Option (c) is incorrect as A+ D = 150 and C = 130

Hence, A+C> D. **Hence, option (c) is correct answer.**

71. Answer: (c)

Let's suppose no. of cars = x

No. of two wheelers = y

Total parked vehicles = x + y

Total no. of wheels = 4x + 2y

Now according to question: total number of wheels of all the cars (four — wheelers) and scooters / motorbikes (two — wheelers) is 100 more than twice the number of parked vehicles.

Hence, 4x + 2y = 2(x +y) + 100

2x = 100

Hence, option (c) is correct.

72. Answer: (a)

Let's take monthly income of peter and Paul is 4x and 3x, respectively.

And their monthly expenses are 3y and 2y, respectively.

Then according to information:

4x - 3y = 6000 and 3x— 2y = 6000 by solving these two equations

We get x = 6000 and y = 6000.

73. Answer: (b)

The correct option is option B

1st bell rings in every 18 minutes

2nd bell rings in every 24 minutes

3rd bell rings in every 32 minutes To find the time when they ring together, we need to find

L.C.M of 18, 24 and 32

L.C.M of 18, 24 and 32 = 288

Hence, after 288 minutes they will ring together. 288 minutes = 4 hours and 48 minutes

It is given that they rang at 8 o'clock in the morning.

So next time they will ring together at 8 hours 0 minutes + 4 hours 48 minutes = 12: 48 hrs. **Hence, option (b) is correct.**

74. Answer: C

The correct option is option C

Let cost of 1% goat be Rs. 100.

Sohan made 10% profit in 1 goat so it's selling price = 110 = selling price

of 2" goat. In 2" goat Sohan suffers 10% loss.

Then the cost price of 2" goat = 110/0.9 = 122.22

Hence, total cost price = 122.2 + 100 = 222.2

Total selling price = 110×2 = 220

Total loss = 2.2 Rs. He suffered a loss of almost 1%.

Hence, option (c) is correct

75. Answer: b

The correct option is option B

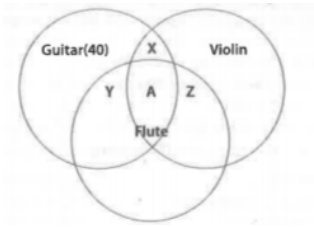
Total no. of musicians = 120

No. of musicians who can play all 3 instruments = 5% of 120=6

Number of musicians who can play guitar alone = 40

The number of musicians who can play any and only two of the above instruments = 30

We can use a Venn's diagram to solve this:



In the Above Venn's diagram, A= 6

X+Y+Z (who play only two) = 30

GH+V+F+(X+Y4Z+A) = 120 (G, V, F are the people who can play only guitar, violin or flute)

By putting all data, we have 40 +V+F+30+6=120

V+F = 44. **Hence, option (b) is correct.**

76. Solution: (a)

Find the Initial ratio = Gold (G):Non-Gold(N) = 1:3

(i) After adding 10 gold coins, the ratio becomes (10+G):N = 1:2

(ii) On solving (i) & (ii), you get, G=20 and N=60.

Therefore, now the total number of coins in the collection = (10+G)+N= 10+20+60=90

77. Solution: (b)

If the number of rows and number of columns are to be equal, then the total number of trees should be a perfect square. As 1000 is not a perfect square, find a perfect square nearest to 1000.

It's 1024, which is square of 32. So he needs to add 24 more trees to get 1024.

78. Solution: (b)

Let the least value price be X. Then the next value is X+20, and the next value is X+40 and on and on up to 7 values.

It's nothing but an arithmetic Progression with sum 700. You can do it either manually or using the formula of AP.

Technique 1: (Manually)

X + X+20 + X+40 +… X+120=700 => 7X+(20+40+…+120)=700

=> 7X+20(1+2+3+…+6)=700 => 7X+20(21)=700 => 7X=280 =>X=40

Technique 2: (Using AP Formula)

Number of prizes = 7 (N).

So sum till nth term will be: n/2 (2X+ (n-1) x 20) = 700. Put the value of N as 7. 7/2 (2X + 120) = 700 => X = 40

79. Solution: (c)

Number of females = 120 -70 = 50

So, a case where minimum number of males have driving license would mean all females have driving license. All 50 females have driver's license. So, only 30 males with license (minimum number).

Now, where maximum number of males will have license, all 70 should have license (only 10 females would have license then).

Ratio: 30/70 = 3 to 7

80. Solution: (d)

After 10 days, the remaining food would be sufficient for the 1000 soldiers for 20 more days

So, If 1000 more soldiers are added, it shall be sufficient for only 10 days (as the no. of soldiers is doubled, the days are halved)

81. Solution: (d)

Total consumption is say N. So daily consumption is N/10. Now if daily increases by 25%, then totally its 1.25N. So, when you divide 10 by 1.25N, it will be N/8. It will go for 8 days.

82. Solution: (d)

Let the Rs. 1 and Rs. 2 coins be x and y in number respectively. Given x + y = 50 and x + 2y = 75

Solving the above two equations, you get x = 25 and y = 25.

83. Solution: (a)

It's a simple LCM problem. Take the LCM of 40, 42 and 45 to know at what point will they find a common multiple. It is 2520 cm i.e. 25 m 20 cm.

84. Solution:

Total requirement = 4000 X 150 = 6,00,000 litres

Volume (capacity) of tank = L x B x H (for cuboids) = 20 X 15 X 6 = 1800 m3 1m = 1000 litres

So, capacity = 18,00, 000

So water will last for 18,00,000/6,00,000 = 3 days.

8 Geometry & Mensuration

1. A cut on a solid object divides the object into two parts where the new surfaces thus produced are plane. On the other hand, one single cut can be used to cut more than one object at a time. In an experiment, the total number of pieces produced by applying n cuts is denoted by xn. The experiment is performed on a solid cube where pieces remain unmoved after each cut. In this experiment, if after the third cut, the pieces are identical, then which of the following is not a possible value for x4? (CSAT-2026)

(a) 16 (b) 12

(c) 8 (d) 5

2. There are three types of rectangular tiles: 3′ × 3′, 3′ × 7′ and 3′ × 11′. An area of rectangular shape of dimensions 3′ × 100′ is to be covered using these tiles without breaking them. If x and y are the maximum and minimum numbers of tiles of various sizes, respectively, that can be used to cover the area exactly, then x – y is (CSAT-2026)

(a) 20 (b) 12

(c) 10 (d) 7

3. A rectangular floor measures 4 m in length and 2.2m in breadth. Tiles of size 140 cm by 60 cm have to be laid such that the tiles do not overlap. A tile can be placed in any orientation so long as its edges are parallel to the edges of the floor. What is the maximum number of tiles that can be accommodated on the floor? (CSAT-2023)

(a) 6 (b) 7

(c) 8 (d) 9

4. There are eight equidistant points on a circle. How many right-angled triangles can be drawn using these points as vertices and taking the diameter as one side of the triangle? (CSAT-2022)

(a) 24 (c) 12

(b) 16 (d) 8

5. Consider the following statements in respect of a rectangular sheet of length 20 cm and breadth 8 cm: (CSAT-2022)

1. It is possible to cut the sheet exactly into 4 square sheets.
2. lt is possible to cut the sheet into 10 triangular sheets of equal area.

Which of the above statements is are correct?

(a) 1 only (c) Both 1 and 2

(b) 2 only (d) Neither 1 nor 2

6. A cubical vessel of side 1 m is filled completely with water. How many millilitres of water is contained in it (neglect thickness of the vessel)? (CSAT-2021)

(a) 1000 (b) 10000

(c) 100000 (d) 1000000

7. A pie diagram shows the percentage distribution of proteins, water, and other dry elements in the human body. Given that proteins correspond to 16% and water corresponds to 70%. If both proteins and the other dry elements correspond to p%, then what is the central angle of the sector representing p on the pie diagram? (CSAT-2021)

(a) 54° (b) 96°

(c) 108° (d) 120°

8. Consider the following statements:

1. The minimum number of points of intersection of a square and a circle is 2.
2. The maximum number of points of intersection of a square and a circle is 8.

Which of the above statements is/are correct? (CSAT-2020)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

9. A solid cube is painted yellow, blue and blacks such that opposite faces are of same colour. The cube is then cut into 36 cubes of two different sizes such that 32 cubes are small and the other four cubes are big. None of the faces of the bigger cube is painted blue. How many cubes have only one face painted? (CSAT-2019)

(a) 4 (b) 6

(c) 8 (d) 10

10. The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of four parallel lines, is (CSAT-2019)

(a) 18 (b) 24

(c) 32 (d) 36

11. P, Q and R are three towns. The distance between P and Q is 60 km, whereas the distance between P and R is 80 km. Q is in the West of P and R is in the South of P. What is the distance between Q and R? (CSAT-2019)

(a) 140 km (b) 130 km

(c) 110 km (d) 100 km

12. Twelve equal squares are placed to fit in a rectangle of diagonal 5 cm. There are three rows containing four each. No gaps are left between adjacent squares. What is the area of each square? (CSAT-2018)

(a) 5/7 sq cm (b) 7/5 sq cm

(c) 1 sq cm (d) 25/12 sq cm

13. The outer surface of a 4 cm x 4 cm x 4 cm cube is painted completely in red. It is sliced parallel to the faces to yield sixty-four 1 cm x 1 cm x 1 cm small cubes. How many small cubes do not have painted faces? (CSAT-2017)

(a) 8 (b) 16

(c) 24 (d) 36

14. Two walls and a ceiling of a room meet at right angles at a point P. A fly is in the air 1 m from one wall, 8 m from the other wall and 9 m from the point P. How many meters is the fly from the ceiling? (CSAT-2017)

(a) 4 (b) 6

(c) 12 (d) 15

15. A person X was driving in a place where all roads ran either north-south or east-west, forming a grid. Roads are at a distance of 1 km from each other in a parallel He started at the intersection of two roads, drove 3 km north, 3 km west and 4 km south. Which further route could bring him back to his starting point, if the same route is not repeated? (CSAT-2016)

(a) 3 km east, then 2 km south

(b) 3 km east, then 1 km north

(c) 1 km north, then 2 km west

(d) 3 km south, then 1 km north

16. AB is a vertical trunk of a huge tree with A being the point where the base of trunk touches the ground Due to a cyclone the trunk has been broken at C which is at a height of 12 meters; broken part is partially attached to the vertical portion of the trunk at C. If the end of the broken part B touches the ground at D which is at 5 distances of 5 meters from A, then the original height of the trunk is (CSAT-2016)

(a) 20 m (c) 30 m

(b) 25 m (d) 35 m

17. In a plane, line X is perpendicular to line Y and parallel to line Z, line U is perpendicular to both lines V and W, line X is perpendicular to line V. Which one of the following statements is correct? (CSAT-2015)

(a) Z, U and W are parallel

(b) X, Vand Y are parallel

(c) Z, V and U are all perpendicular to W

(d) Y, Vand W are parallel

18. A gardener increased the area of his rectangular garden by increasing its length by 40% and decreasing its width by 20%. The area of the new garden (CSAT-2014)

(a) has increased by 20%.

(b) has increased by 12%.

(c) has increased by 8%.

(d) is exactly the same as the old area.

8 Geometry & Mensuration_Explanations

1. Answer: (a)

Explanation:

The condition says that after the third cut, the pieces are identical. It does not necessarily mean that there are 8 pieces. For example, three parallel equally spaced cuts can produce 4 identical slabs, and other arrangements can produce 6 or 8 identical pieces.

A fourth cut can only increase the number of pieces by the number of existing pieces it actually cuts. Hence:

x4 = x3 + (number of existing pieces cut by the 4th plane)

Now check the options:

- $x_4 = 5$ is possible: after three parallel cuts, there are 4 identical slabs; the fourth cut may cut only one slab, giving 4 + 1 = 5.
- $x_4 = 8$ is possible: after three parallel cuts, there are 4 identical slabs; the fourth cut may cut all 4 slabs, giving 4 + 4 = 8.
- $x_4 = 12$ is possible: after three cuts producing 6 identical pieces, the fourth cut may cut all 6 pieces, giving 6 + 6 = 12.
- $x_4 = 16$ would require $x_3 = 8$ and the fourth cut to cut all 8 pieces. But after three cuts producing 8 identical pieces, the cube is divided into a 2 × 2 × 2 arrangement, and a single plane cannot pass through the interiors of all 8 pieces.

Therefore, 16 is not a possible value of x_4.

2. Answer: (a)

Explanation:

The problem requires us to cover a rectangular area of 3′ × 100′ using tiles of sizes 3′ × 3′, 3′ × 7′, and 3′ × 11′.

Since the width of the area is 3′ and all the available tiles also have a width of 3′, the tiles must be placed end-to-end along the 100′ length. They cannot be rotated because a width of 7′ or 11′ would not fit inside the 3′ wide area.

Thus, the problem simplifies to finding the number of tiles of lengths 3′, 7′, and 11′ that add up exactly to 100′. Let a, b, and c be the number of 3′, 7′, and 11′ tiles used, respectively. We need non-negative integers a, b, c such that:

3a + 7b + 11c = 100

We need to find the maximum (x) and minimum (y) total number of tiles (a + b + c).

Step 1: Find the maximum number of tiles (x)

To maximize the total number of tiles, we should use as many of the smallest tiles (3′) as possible.

- If we use 33 tiles of 3′, the length covered is 33 × 3 = 99′, leaving 1′, which cannot be filled by 7′ or 11′ tiles.
- If we use 32 tiles of 3′, the length covered is 32 × 3 = 96′, leaving 4′, which cannot be filled.
- If we use 31 tiles of 3′, the length covered is 31 × 3 = 93′, leaving 7′. This remaining length can be exactly filled by one 7′ tile.

So, a valid combination is a = 31, b = 1, c = 0. The maximum number of tiles is x = 31 + 1 + 0 = 32.

Step 2: Find the minimum number of tiles (y)

To minimize the total number of tiles, we should use as many of the largest tiles (11′) as possible.

- If we use 9 tiles of 11′, the length covered is 9 × 11 = 99′, leaving 1′, which cannot be filled.
- If we use 8 tiles of 11′, the length covered is 8 × 11 = 88′, leaving 12′. This remaining length can be exactly filled by four 3′ tiles (4 × 3 = 12′).

So, a valid combination is a = 4, b = 0, c = 8. The minimum number of tiles is y = 4 + 0 + 8 = 12.

(Note: We can mathematically verify that no combination yields fewer than 12 tiles. If a + b + c = 11, the maximum possible length is 11 × 11 = 121. To get exactly 100, we check the equation 11(a + b + c) − (3a + 7b + 11c) = 8a + 4b. Substituting the sums gives 121 − 100 = 21, which means 8a + 4b = 21. Since 8a + 4b is always even and 21 is odd, 11 tiles is impossible.)

Step 3: Calculate x − y

x − y = 32 − 12 = 20

3. Answer: (d)

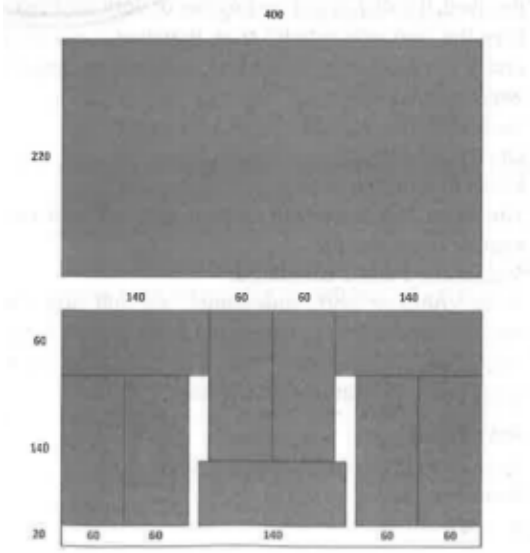
Let's arrange convert the measurement from meters to centimeters.

Length of the floor =4 m =400 cm

Breadth of the floor = 2.2 m =220 cm

Size of each tile = 140 cm by 60 cm

Tiles can be arranged in following ways:



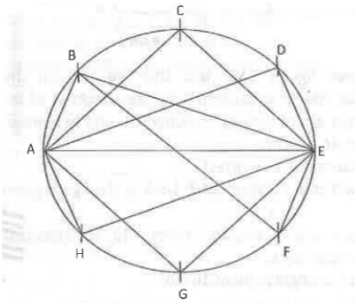
Hence, option (d) is correct.

4. Answer: (a)

Firstly, we have to draw 8 equidistant points on a circle — A, B, C, D, E, F, G, and H.

We know that the angle subtended by a diameter/ semicircle on any point of a circle is 90°.

Consider AE as the diameter and one side of the right- angled triangle, we can draw 6 right-angled triangles. (ABE, ACE, ADE, AHE, AGE, AFE)



Similarly, we consider BF, CG and DH as diameters and with each diameter 6 right-angled triangles can be drawn.

With BF: (BCF, BDF, BEF, BGF, BHF, BAF)

With CG: (CDG, CEG, CFG, CHG, CAG, CBG)

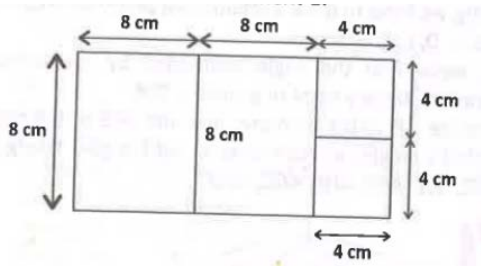
With DH: (DEH, DFH, DGH, DAH, DBH, DCH)

So, the total number of right-angled triangles that can be drawn = 6 + 6 + 6 + 6 = 24.

Hence, option (a) is correct.

5. Answer: (c)

The dimensions of rectangle is 20cm × 8cm Firstly, we will check Statement-1:



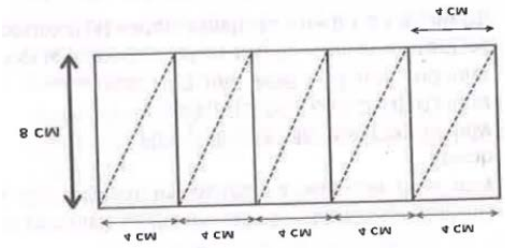
From above figure, it is clear that we can cut the rectangular sheet exactly into 4 square sheets in which two squares are of dimension 8cm×8cm and two are of dimension 4cm×4cm.

Hence, Statement-1 is correct.

Now we will check Statement-2: Look at the figure given below,

We can cut the rectangular sheet into 10 triangular sheets of equal area.

Are of each triangular sheet 16 cm?



Statement-2 is also correct. **Hence, Option (c) is correct.**

6. Answer: (d)

Side of cubical vessel = 1 m

Volume of cube = side x side x side =1x1x1=1 m^3

We know, 1 m^3 = 1000 litre and 1 litre = 1000 ml

So, 1 m^3 = 1000 x 1000 ml = 1000000 millilitres. Hence, option (d) is correct.

7. Answer: (c)

A human body consists of proteins, water and other dry elements.

% of proteins + % of water + % of dry elements 100%

Percentage of proteins in human body = 16%

Percentage of water = 70%

So, Percentage of other dry elements = 100% - 16% - 70% =14%

Given, the percentage of proteins and other dry elements =p%

->P=16+14=30%

In a pie diagram, total angle = 360°

This means, 100% corresponds to 360° or 30% corresponds to

The central angle of the sector representing p on the pie diagram = 108°, Hence, option (c) is correct.

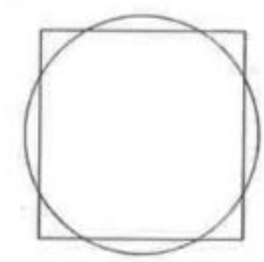
8. Answer: (b)

Statement 1 is incorrect: The minimum number of points of intersection of a square and a circle is 1.

Because there can be a case where circle just touch the square then point of intersection is 1.

Statement 2 is correct: The maximum number of points of intersection of a square and a circle is 8. A circle can pass through a straight line maximum 2 times.

There are 4 straight lines so a circle can pass through maximum 8 times. As shown in figure:



Statement 2 is correct. Hence, option (b) is correct.

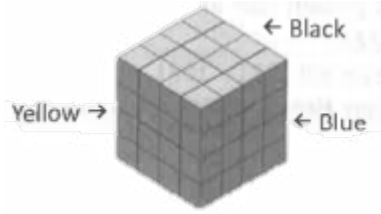
9. Answer: The correct option is option (c)

We have a solid cube painted yellow, blue, and black such that opposite faces are of same colour.

Now, the cube is cut into 36 pieces with 32 small and 4 big cubes.

No face of the 4 big cubes painted with blue

The given information is shown below:



Let the length of cube is 4 units. So, there are 32 cubes with 1 unit length and 4 cubes with 2-unit length with no face painted blue.

We have to find the number of cubes with only one face painted

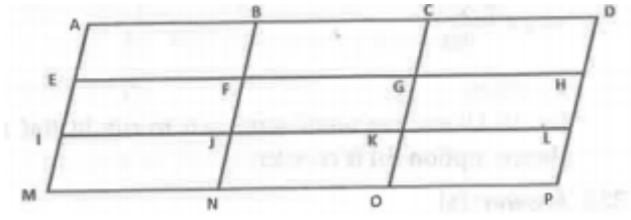
The middle 4 cubes of unit length on the blue face have only one face pained. Similarly, on the opposite side also

So, the total number of such cubes = 4 + 4 = 8. Hence,

option (c) is correct.

10. Answer: The correct option is option (d)

We have to find the number of parallelograms formed from a set of four parallel lines intersecting another set of four parallel lines Let's count the parallelograms from the diagram given below:



The parallelograms of 1 x 1 are: ABFE, EFJI, INM, BCGF, FGKJ, JKNO, CDHG, GHLK and KLPO. Total — 9

The parallelograms of 1 x 2 are: ACGE, BDHF, EGKI, FHLJ, IKOM and JLPN. Total - 6 The parallelograms of 1 x 3 are: ADHE, EHLL and ILPM

Total =3 The parallelograms of 2 x 1 are: ABJI, EFNM, BCKJ, FGON, CDLK and GHPO. Total - 6

The parallelograms of 2 x 2 are: ACKI, EGOM, BDLJ and FHPN.

Total -4

The parallelograms of 2 x 3 are: ADLI and EHPM

Total — 2

The parallelograms of 3 x 1 are: ABNM, BCON and CDPO.

Total -3

The parallelogram's of 3 x 2 are: ACOM and BDPN

Total -2

The parallelogram of 3 x 3is ADPM i.e. 1

So, total number of parallelograms =9+6 +3+6 + 4+ 2 +3+2+1=36. **Hence, option (d) is correct.**

11. Answer: The correct option is option (d)

Given, Qs in the West of P and R is in the South of P

Distance between P and Q = 60 km and distance between P and R = 80 km

As we can observe from the above figure, QPR is a right- angled triangle.

Using Pythagoras Theorem,

$PQ^2 + PR^2 = QR^2$

$60^2 + 80^2 = QR^2$

The distance between QR = 100 km. **Hence, option (d) is correct.**

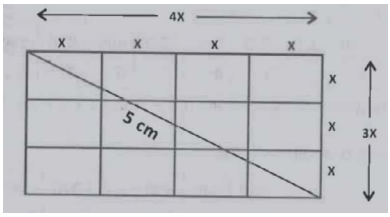
12. Answer: (c)

The correct option is (c)

Let the side of each square be x.

12 squares are placed to fit in a rectangle of diagonal 5

cm in three rows containing 4 each.



The length and breadth of rectangle are 4x and 3x respectively.

Using Pythagoras Theorem,

$(\text{length})^2 + (\text{breadth})^2 = (\text{diagonal})^2$

$(4x)^2+(3x)^2=5$

$16x^2 + 9x^2 = 25$

$25x^2=25$

$x^2=1$

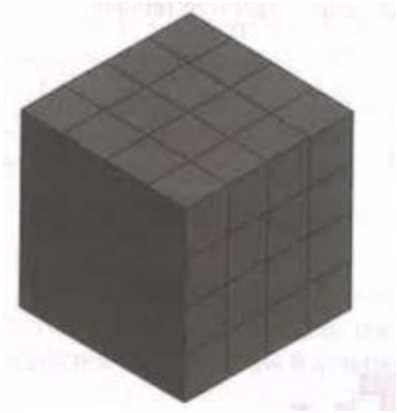
x=1 cm

So,the area of each square= 1cm x 1cm = 1 sq cm.

13. Answer: (a)

The dimensions of cube are 4 cm x 4 cm x 4 cm which is painted red.

Now it is sliced into 64 1 cm x 1 cm x 1 cm small cubes.

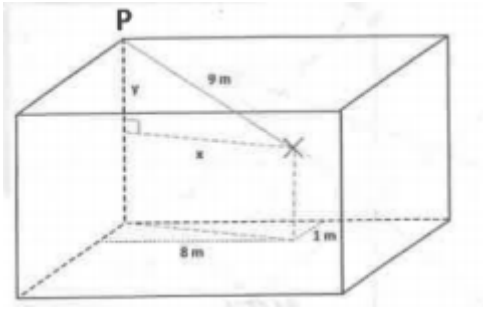


All the outer faces are painted red. Only the middle cubes of 2nd and 3rd row and 2nd and 3rd columns, has no face painted red. These are 4 + 4 = 8 cubes. **Hence, option (a) is correct.**

14. Answer: The correct answer is option (a)

Two walls and a ceiling of a room meet at right angles at point P.

Let the fly is at point X as shown in the figure, which is 1m from one wall and 8m from another wall. Also, the distance of fly from point P is 9m



Let x is the distance of fly from the intersection of two

walls and y be the distance of fly from ceiling.

Using Pythagoras theorem,

$x^2 = 82 + 12 = 65$

Again, using Pythagoras theorem,

$x^2 + y^2 = 92$

$65 + y^2 = 81$

$y^2 = 16y = 4$

So, the distance of fly from the ceiling, y = 4m. Hence, option (a) is correct

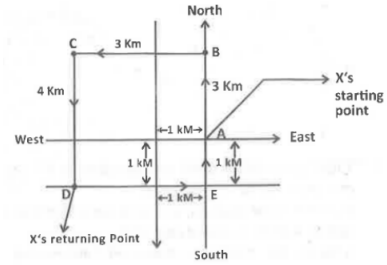
15. Answer: (b)

Let the person X starts from a point A.

He drove 3 km north and reaches at point B.

Then he drove 3 km west and reaches at point C.

Further, he drove 4 km south and reaches at point D.



Now, he has to come back at his starting point A and cannot take the same route.

He drives 3 km east and reaches at point E and from there he drives 1 km north to reach at his starting position.

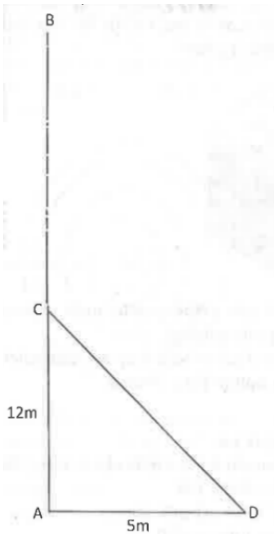
So, he should drive 3 km east, then 1 km north.

Hence, option (b) is correct.

16. Answer: (b)

It is given that AB is the vertical trunk of huge tree.

AC=12m, AD =5m



Using Pythagoras theorem,

$AC^2 + AD^2 = DC^2$

$12^2 + 5^2 = DC^2$

$DC^2 = 144 + 25 = 169$

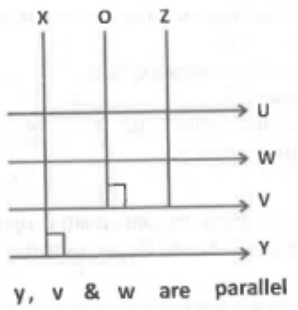
$DC = 13$

So, the original height of the trunk = AC + CD = 12m + 13m = 25m.

Hence, option (b) is correct.

17. Answer: (d)

We can easily solve this question by making a picture of given information:



Hence, it is very clear from the photo that y, v and w are parallel.

Hence, option (d) is correct.

18. Answer: (b)

The correct option is option B

Let's suppose initial length and breadth of garden = 10m Area of garden initially = 10×10 = 100m sq

Now length is increased by 40%

New length = 10 + 40% of 10 = 14m Breadth is decreased by 20%

New breadth = 10 - 20% of 10 = 8m

New area = 14×8 = 112m sq

Change in area = 112 - 100 = 12m sq. (increased)

Change in percentage = 12/100 x 100 = 12%. **Hence, option (b) is correct.**

9 Permutation, Combination & Probability

1. There are four types of weights, namely 1 kg, 2 kg, 5 kg and 10 kg. What is the maximum number of different ways one can measure 20 kg, if at least eight but not more than eleven weights of 1 kg are to be used while measuring? (CSAT-2026)

   (a) 7 (b) 8

   (c) 9 (d) 10

2. How many words can one form by shuffling the letters of the word QUEUE, if Q is always followed by U? The words thus formed need not necessarily have any meaning. (CSAT-2026)

   (a) 6 (b) 8

   (c) 10 (d) 12

3. The top of a table is rectangular and its dimensions are $6' \times 10'$. Two rectangular portions of the table top are painted in blue colour; both these portions have dimensions $2.5' \times 8'$ and each of them has exactly two sides common with two edges of the table top. If the table is fixed to the ground and the remaining portion of the table top is painted in white, how many different patterns are possible when observed from above? (CSAT-2026)

   (a) 2 (b) 4

   (c) 6 (d) 8

4. Let both p and k be prime numbers such that (p2+k) is also a prime number less than 30. What is the number of possible values of k? (CSAT-2025)

   (a) 4 (b) 5

   (c) 6 (d) 7

5. Raj has ten pairs of red, nine pairs of white and eight pairs of black shoes in a box. If he randomly picks shoes one by one (without replacement) from the box to get a red pair of shoes to wear, what is the maximum number of attempts he has to make? (CSAT-2023)

   (a) 27 (b) 36

   (c) 44 (d) 45

6. In how many ways can a batsman score exactly 25 runs by scoring single runs, fours and sixes only, irrespective of the sequence of scoring shots? (CSAT-2023)

   (a) 18 (b) 19

   (c) 20 (d) 21

7. ABCD is a square. One point on each of AB and CD; and two distinct points on each of BC and DA are chosen. How many distinct triangles can be drawn using any three points as vertices out of these six points? (CSAT-2023)

   (a) 16 (b) 18

   (c) 20 (d) 24

8. A box contains 14 black balls, 20 blue balls, 26 green balls, 28 yellow balls, 38 red balls and 54 white balls. Consider the following statements:

   1. The smallest number n such that any n balls drawn from the box randomly must contain one of full group of at least one colour is 175.
   2. The smallest number m such that any m balls drawn from the box randomly must contain at least one hall of each colour is 167.

   Which of the above statements is/are correct? (CSAT-2023)

   (a) 1 only (b) 2 only

   (c) Both 1 and 2 (d) Neither 1 nor 2

9. What is the number of selections of 10 consecutive things out of 12 things in a circle taken in the clockwise direction? (CSAT-2023)

   (a) 3 (b) 11

   (c) 12 (d) 66

10. In an examination, the maximum marks for each of the four papers namely P, Q, R and S are 100. Marks scored by the students are in integers. A student can score 99% in n different ways. What is the value of n? (CSAT-2023)

    (a) 16 (b) 17

    (c) 23 (d) 35

11. A flag has to be designed with 4 horizontal strips using some or all of the colours red, green and yellow. What is the number of different ways in which this can be done so that no two adjacent stripes have the same colour? (CSAT-2023)

    (a) 12 (b) 18

    (c) 24 (d) 36

12. There are five persons, P, Q, R, S and T each one of whom has to be assigned one task. Neither P nor Q can be assigned Task-1. Task-2 must be assigned to either R or S. In how many ways can the assignment be done? (CSAT-2023)

    (a) 6 (b) 12

    (c) 18 (d) 24

13. 36. Let A, B and C represent distinct non-zero digits. Suppose x is the sum of all possible 3-digit numbers formed by A, B and C without repetition. (CSAT-2022)

    Consider the following statements:

    1. The 4-digit least value of x is 1332.
    2. The 3-digit greatest value of x is 888.

    Which of the above statements is/are correct?

    (a) 1 only (c) Both 1 and 2

    (b) 2 only (d) Neither 1 nor 2

14. There is a numeric lock which has a 3-digit PIN. The PIN contains digits 1 to 7. There is no repetition of digits. The digits in the PIN from left to right are in decreasing order. Any two digits in the PIN differ by at least 2. How many maximum attempts does one need to find out the PIN with certainty? (CSAT-2022)

(a) 6 (c) 10

(b) 8 (d) 12

15. One non-zero-digit, one vowel and one consonant from English alphabet (in capital) are to be used in forming passwords, such that each password has to start with a vowel and end with a consonant. How many such passwords can be generated? (CSAT-2022)

(a) 105 (b) 525

(c) 945 (d) 1050

16. There are 9 cups placed on a table arranged in equal number of rows and columns out of which 6 cups contain coffee and 3 cups contain tea. In how many ways can they be arranged so that each row should contain at least one cup of coffee? (CSAT-2022)

(a) 18 (b) 27

(c) 54 (d) 81

17. The sum of three consecutive integers is equal to their product. How many such possibilities are three? (CSAT-2022)

(a) Only one (b) Only two

(c) Only three (d) No such possibility is there

18. On a chess board, in how many different ways can 6 consecutive squares be chosen on the diagonals along a straight path? (CSAT-2021)

(a) 4 (b) 6

(c) 8 (d) 12

19. Using 2, 2, 3, 3, 3 as digits, how many distinct numbers greater than 30000 can be formed? (CSAT-2021)

(a) 3 (b) 6

(c) 9 (d) 12

20. There are 6 persons arranged in a row. Another person has to shake hands with 3 of them so that he should not shake hands with two consecutive persons. In how many distinct possible combinations can the handshakes take place? (CSAT-2021)

(a) 3 (b) 4

(c) 5 (d) 6

21. How many different 5-letter words (with or without meaning) can be constructed using all the letters of the word 'DELHI' so that each word has to start with D and end with I? (CSAT-2020)

(a) 24 (b) 18

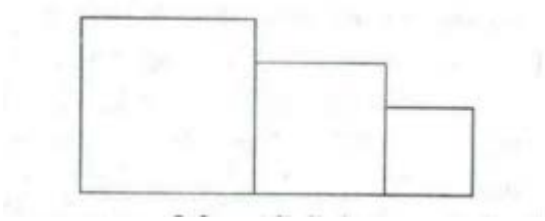
(c) 12 (d) 6

22. A bag contains 15 red balls and 20 black balls. Each ball is numbered either 1 or 2 or 3. 20% of the red balls are numbered 1 and 40% of them are numbered 3. Similarly, among the black balls, 45% are numbered 2 and 30% are numbered 3. A boy picks a ball at random. He wins if the ball is red and numbered 3 or if it is black and numbered 1 or 2. What are the chances of his winning? (CSAT-2018)

(a) 1/2 (b) 4/7

(c) 5/9 (d) 12/13

23. For sports meet, a winner's stand comprising three wooden blocks is in the following form:



There are six different colours available to choose from and each of the three wooden blocks is to be painted such that no two of them has the same colour. In how many different ways can the winners' stand be painted? (CSAT-2018)

(a) 120 (b) 81

(c) 66 (d) 36

24. A bag contains 20 balls. 8 balls are green, 7 are white and 5 are red. What is the minimum number of balls that must be picked up from the bag blindfolded (without replacing any of it) to be assured of picking at least one ball of each colour? (CSAT-2017)

(a) 17 (b) 16

(c) 13 (d) 11

25. A selection is to be made for one post of Principal and two posts of Vice-Principal. Amongst the six candidates called for the interview, only two are eligible for the post of Principal while they all are eligible for the post of Vice-Principal. The number of possible combinations of selectees is (CSAT-2015)

(a) 4 (b) 12

(c) 18 (d) None of the above

26. A student has to opt for 2 subjects out of 5 subjects for a course, namely, Commerce, Economics, Statistics, Mathematics I and Mathematics II. Mathematics II can be offered only if Mathematics I is also opted. The number of different combinations of two subjects which can be opted is (CSAT-2015)

(a) 5 (b) 6

(c) 7 (d) 8

27. A person ordered 5 pairs of black socks and some pairs of brown socks. The price of a black pair was thrice that of a brown pair. While preparing the bill, the bill clerk interchanged the number of black and brown pairs by mistake which increased the bill by 100%. What was the number of pairs of brown socks in the original order? (CSAT-2015)

(a) 10 (b) 15

(c) 20 (d) 25

28. If all the three males' are selected, then how many combinations of four member teams are possible? (CSAT-2013)

(a) 1 (b) 2

(c) 3 (d) 4

29. There are some balls of red, green and yellow colour lying on a table. There are as many red balls as there are yellow balls. There are twice as many yellow balls as there are green ones. The number of red balls (CSAT-2013)

(a) is equal to the sum of yellow and green balls.

(b) is double the number of green balls.

(c) is equal to yellow balls minus green balls.

(d) cannot be ascertained.

9 Permutation, Combination & Probability_ Explanations

1. Answer: (b)

Explanation:

Let a, b, c, d be the number of 1, 2, 5, 10 kg weights. We need

$a + 2b + 5c + 10d = 20$, $a \in \{8, 9, 10, 11\}$, $b, c, d \geq 0$.

Case a = 8: $2b + 5c + 10d = 12$

- d = 0: (b, c) = (6, 0), (1, 2) → 2 ways
- d = 1: (b, c) = (1, 0) → 1 way

Subtotal: 3

Case a = 9: $2b + 5c + 10d = 11$ (odd, so c must be odd)

- d = 0: (b, c) = (3, 1) → 1 way

Subtotal: 1

Case a = 10: $2b + 5c + 10d = 10$

- d = 0: (b, c) = (5, 0), (0, 2) → 2 ways
- d = 1: (b, c) = (0, 0) → 1 way

Subtotal: 3

Case a = 11: $2b + 5c + 10d = 9$ (c odd)

- d = 0: (b, c) = (2, 1) → 1 way

Subtotal: 1

Total: 3 + 1 + 3 + 1 = 8

2. Answer: (d)

Explanation:

We need arrangements of the letters of QUEUE (one Q, two U's, two E's) such that Q is immediately followed by U.

Glue Q to a U.

Treat "QU" as a single block. The remaining letters are one U and two E's. So we are arranging four objects: {QU, U, E, E}.

Count arrangements with the repeated E.

4! / 2! = 24 / 2 = 12

(The two E's are identical; the U inside the QU-block is distinguishable from the lone U by the constraint that one is preceded by Q and the other isn't.)

3. Answer: (b)

Explanation:

Let the table top have dimensions $10' \times 6'$. Each blue rectangular portion has dimensions $8' \times 2.5'$.

Since each blue rectangle has exactly two sides common with two edges of the table top, each blue rectangle must be placed at a corner of the table. Also, its $8'$ side must lie along the $10'$ side of the table, and its $2.5'$ side must lie along the $6'$ side of the table.

Thus, there are four possible corner positions for a blue rectangle: bottom-left, bottom-right, top-left and top-right. However, the two blue rectangular portions cannot overlap. Now check the possible pairs of corner placements:

- Bottom-left and bottom-right overlap, so this is not possible.
- Top-left and top-right overlap, so this is not possible.
- Bottom-left and top-left do not overlap, so this is possible.
- Bottom-right and top-right do not overlap, so this is possible.
- Bottom-left and top-right do not overlap, so this is possible.
- Bottom-right and top-left do not overlap, so this is possible.

Hence, the number of possible patterns is 4. Therefore, the correct answer is 4.

4. Answer: (b)

Explanation:

Possible values can be -

$2^2+3 = 7$

$2^2+7 = 11$

$2^2+13 = 17$

$2^2+19 = 23$

$3^2+2 = 11$

5. Answer: (d)

In the given scenario, to maximize the number of attempts raj has to make, simply take the worst case scenario.

Suppose he randomly picks all white and black shoes i.e. 9 x 2 +8 x 2 = 34 shoes, he is left with only ten pairs of red shoes i.e. twenty total number of red shoes - ten left one and ten right one.

Take the worst case again. Suppose he picks all ten left foot red shoes; he still needs to pick one from the other ten right foot red shoes to make sure he has one pair of perfect red shoes to ware. So the required number of attempts =34 +10+1=45

Hence, option (d) is correct.

6. Answer: (b)

To solve this problem, let's introduce three variables: x represents the number of singles taken, y represents the number of fours scored, and z represents the number of sixes hit.

We get following equation $x + 4y + 6z = 25$, representing the total runs scored.

Analysing different cases based on the value of z:

Case 1: z = 0 (no sixes hit)

In this case, we have the equation $x + 4y = 25$. By checking different combinations of x and y, we find the following valid combinations: (1, 6), (5, 5), (9, 4), (13, 3), (17, 2), (21, 1), and (25, 0). Thus, there are 7 possible combinations for this case.

Case 2: z=1 (one six hit)

The equation x + 4y = 19 holds in this case. By evaluating different values of x and y, we find the following valid combinations: (3, 4), (7, 3), (11, 2), (15, 1), and (19, 0). Hence, there are 5 possible combinations for this case.

Case 3: z = 2 (two sixes hit)

By substituting z = 2 into the equation x + 4y = 13, we find the valid combinations (1, 3), (5, 2), (9, 1), and (13, 0). Therefore, there are 4 possible combinations for this case.

Case 4: z = 3 (three sixes hit) For z=3, the equation x + 4y = 7 holds. Evaluating different values of x and y, we find the valid combinations (3, 1) and (7, 0). Hence, there are 2 possible combinations for this case.

Case 5: z = 4 (four sixes hit) When z = 4, we have x + 4y = 1. Evaluating the equation, we find the only valid combination as (1, 0). Thus, there is 1 possible combination for this case.

Adding up the possibilities from all cases,

we find that there are a total of 7+ 5 +4+2+1=19 possible ways to score exactly 25 runs using single runs, fours, and sixes. Hence, option (b) is correct.

7. Answer: (c)

To determine the number of distinct triangles that can be formed, we will apply the principles of combinatorics.

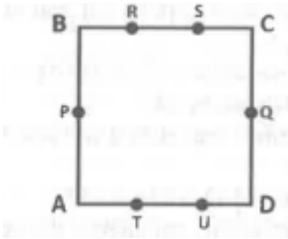
Let's label the selected points as follows:

The point on AB as P.

The point on CD as Q.

The two points on BC as R and S

The two points on DA as T and U.



To form a triangle, we need to choose 3 points out of the 6 given points: P, Q, R, S, T, and U.

The total number of ways to select 3 points out of 6 can be determined using the combination

Formula:

C(n, r)=n! / (r! * (n-r)!)

Here, 'n' represents the total number of items (6 points), and 'r' represents the number of items being selected (3 points).

Applying the combination formula, we have:

$$C(6,3) = \frac{6!}{(3!(6-3)\cdot 1)}$$

Evaluating the factorials:

6! = 6*5*4*3*2*1=720

3! = 3*2*1=6

(6-3)! = 3! = 6

Now, substituting these values into the formula:

C(6,3) = 720/(6*6)

720/36=20

The Total number of distinct triangles that can be formed using any three points as vertices out of the six given points (P, Q, R, S, T, and U) is 20. **Hence, option (c) is correct.**

8. Answer: (c)

Statement 1 is correct: 14 black balls, 20 blue balls 26 green balls, 28 yellow balls, 38 red balls and 54 whitg balls are there.

To get are full group of at least one colour here take the worst case scenario by selecting one less ball from each, group, i.e. 13 black ball, 19 blue balls, 25 green ball, 27 yellow balls, 37 red balls and 53 white balls i.e. 174 balls. Here even after drawing 174 balls we don't have 5 fill group here.

If we add one more ball, any one group will be complete. 174 + 1 = 175 (Smallest number of balls that can be drawn to complete at least one full group with certainty) Statement 2 is correct: To find the minimum possible number of balls that must be drawn from the box to ensure that at least one ball of each color is selected, let's consider the worst-case scenario.

In this scenario, we aim to maximize the number of balls drawn without selecting any black ball. Since the number of black balls is the lowest among all the colors, we can draw the maximum possible number of balls from the other colors.

By selecting 20 blue balls, 26 green balls, 28 yellow balls, 38 red balls, and 54 white balls, we obtain a total of 20 + 26 + 28 + 38 + 54 = 166 balls.

After drawing these 166 balls, only the black balls remain in the box. Therefore, the next ball we select will undoubtedly be a black ball, ensuring that we have at least one ball of each color.

In this worst-case scenario, the minimum number of balls required to be drawn is 166 + 1 = 167.

9. Answer: (c)

The number of required selection here in the question will be 12 as 10 consecutive things have to be taken out of 12 in a circle which paves our ways like this:

X1 X2.....X9, X10
X2 X3.....X10, X11
X3 X4.....X11, X12
X4 X5.....X12, X1
X5 X6.....X1, X2
X6 X7.....X2, X3
X7 X8.....X3, X4
X8 X9.....X4, X5
X9 X10.....X5, X6
X10 X11.....X6, X7
X11 X12.....X7, X8
X12 X1.....X8, X9

There are 12 possible selections. **Hence, option (c) is correct.**

10. Answer: (d)

Given:

Maximum marks per paper = 100

Total papers = 4

Percentage scored by the student = 99%

To find: The number of ways (n) the student can score, 99%.

To score 99%, the student has to score 99% in all four,

subjects.

Hence, 99 +99 + 99 + 99 = 396

Determine the ways the student can lose 4 marks (396 - 400):

(4, 0, 0, 0): Scored 100 in three papers and 96 in one paper.

Possible ways = 4! /3! =4.

(3, 1, 0, 0): Scored 100 in two papers, 99 in one paper,

and 97 in one paper.

Possible ways =4! /2! =12,

(2, 1, 1, 0): Scored 100 in one paper, 99 in two papers,

and 98 in one paper.

Possible ways =4! /2! =12,

(2, 2, 0, 0): Scored 100 in two papers and 98 in the

remaining two papers.

Possible ways = 4! / (2! 2!) = 6.

(1, 1, 1, 1): Scored 99 in all four papers. Possible ways = 1,

Calculate the total number of ways: n=4+12+12+6 +1 = 35.

Hence, option (d) is correct.

11. Answer: (c)

For first stripe we have three options for second stripe

we are left with only two options now.

For third stripe we can't use the one used in second

one, so we are left with two options again.

For fourth stripe we again will have two options.

Total possible ways = 3 x 2 x 2 x 2 = 24.

Hence, option (c) is correct.

12. Answer: (d)

To assign the tasks, we have the following conditions:

o Neither P nor Q can be assigned Task-1.

o Task-2 must be assigned to either R or S.

Tasks	1	2	3	4	5
Case 1	'S' or 'T' Two possible ways	'R'	Three possible ways	Two possible ways	One possible way
Case 2	'S' or 'T' two possible ways	'S'	Three possible ways	Two possible ways	One possible way

2x1x3x2x1 = 12 ways ←
2x1x3x2x1 = 12 ways ←
Total 12 + 12 = 24 ways. Hence, option (d) is correct.

13. Answer: (a)

Let the three-digit number is represented by ABC where A, B, and C are non-zero digits.

We have to find x which is the sum of all possible 3-digit numbers formed by A, B, and C without repetition.

Total 6 three-digit numbers can be formed using distinct digits of A, Band C (3! = 3x2=6) i.e. ABC, ACB, BAC, BCA, CAB and CBA.

x = ABC + ACB + BAC + BCA + CAB + CBA

Let's do a bit of hit and trial, to find the value of x closest to 1000.

Consider the minimum possible digitsi.e. 1, 2and 3. Then, x =123 + 132 + 213 + 231 + 312 + 321 = 1332 (4-digit).

This is the least possible value of x. Hence, statement-1 is correct. Statement 2 is incorrect: The least possible value of x is 1332 which is a 4-digit number. So, we cannot find any three digits value of x. Hence, statement 2 is not correct. **Hence, option (a) is correct.**

14. Answer: (c)

The PIN contains 3 digits (without repetition) out of 1, 2,3,4,5,6 and 7 and the digits are in decreasing order from left to right with the difference between any two digits in the PIN differ by at least 2.

Let us consider the various possible cases:

Case 1: Let the greatest digit is 7 as the leftmost digit. Possible combinations: 753, 752, 751, 742, 741, 731 (i.e 6 combinations)

Case 2: Let 6 as the leftmost digit. Possible combinations: 642,641,631 (i.e. 3 combinations)

Case 3: Let 5 as the leftmost digit Possible combinations: 531 (i.e. 1 combination)

Case 4: Let 4 as the leftmost digit No possible combination because the difference between any two digits should be at least 2 and the digits

should be in decreasing order.

So, the total number of possible combinations of the PIN =6+3+1=10. **Hence, option (c) is correct.**

15. Answer: (c)

The password has one non-zero-digit, one vowel and one consonant from English alphabet.

One non-zero digit = 1, 2, 3, 4, 5,6, 7, 8, 9 (i.e., 9 possibilities)

Vowel in English alphabet = A, E, 1, 0, U (i.e., 5 possibilities)

Consonants in English alphabet = 21

The password is of the form Vowel non-zero-digit consonant

Number of such passwords = 5 × 9 × 21 = 945

The number of such passwords which can be generated is 945. **Hence, option (c) is correct.**

16. Answer: (d)

There are 9 cups out of which 6 cups contains coffee and 3 cups contains tea.

These cups are arranged in equal number of rows and column i.e., 3 rows and 3 columns.

Now, let's consider all the possible cases (each row has at least one cup of coffee)

CASE-1: Each row has 2 cups of coffee and 1 cup of tea.

C	C	T
C	C	T
C	C	T

No. of ways in which One teacup can be placed in a row =3 ways

So, Possible arrangements wherein all rows have 1 cup of tea = 3×3×3 = 27 ways

Case 2: One row has 3 cups of coffee.

In this case, one of row will have 2 cups of coffee and the other will have 1 cup of coffee.

If the first row has 3 coffee cups, then the possible arrangements = 3×3 = 9 ways

The rows having 3, 2 and 1 coffee cups can be arranged in 3×2×1 = 6 ways

So, total no. of arrangements wherein one row has 3 coffee cups =9 × 6 ways = 54 ways

Therefore, total arrangements = 27 + 54 = 81 ways.

Hence, option (d) is correct.

17. Answer: (c)

Let the three consecutive numbers by (x=1), x, (x +1)

Given that the sum of three consecutive integers is equal to their product.

(x=1) +x+(x+1) = (x-1) x (x+1)

3x=x3-x

X3-4x=0

X(x-2) (x+2) = 0

x=0,2-2

There are only three such possibilities. **Hence, option (c) is correct.**

Question Hack:

Let the numbers be -1, 0 and 1, then -1+ 0 + 1=-1x 0 x1-0

Let the numbers be 1, 2and 3, then 1+2+3=1×2×3 =6

Let the numbers be -3, -2 and -1, then (3) + (-2) +(-1) = (-3)×(-2)×(-1)=-6

3 possibilities are possible.

18. Answer: (b)



Here is the 8x8 chess board. There are two diagonals with 8 squares each.

On one diagonal, 6 consecutive squares can be chosen in 3 ways i.e. A8B7C6DSE4F3, B7C6D5E4F3G2, and C6DSE4F3G2H1

So, total number of ways of choosing 6 consecutive squares on both the diagonals along a straight path =3 + 3 = 6. Hence, option (b) is correct.

19. Answer: (b)

Using 2, 2, 3, 3, and 3 as digits we have to form distinct numbers greater than 30000.

So, every number should start with 3 in order to be. greater than 30000 i.e. 3 _ ___ _

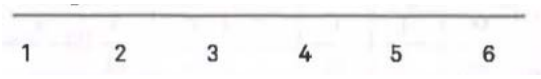
Now, these 4 blanks have to be filled by two 2's and two 3's.

Number of ways to fill these 4 places = 4!/2!2 = 6 ways

These numbers are 32233, 32323, 32332, 33223, 33232 and 33322. Hence, option (b) is correct.

20. Answer: (b)

Let the six persons be numbered as 1, 2, 3, 4, 5 and 6 are, arranged in a row.



So, the other person shakes hand with 3 of them but should not shake hands with two consecutive persons. The other person shakes hand with the person numbered.

1, 3, 5

1, 3, 6

2, 4, 6

1, 4, 6

So, there are 4 distinct possible combinations. Hence, option (b) is correct.

21. Answer: (d)

it is given that position of D and I are fixed Words will be like :D_ _ _|

re we have 3 spaces to fill with three letters (E, L and H)

The no. of ways we can fill three spaces with three letters are: 3! = 3x2x1 = 6. Hence, option (d) is correct.

22. Answer: (b)

The correct option is (b)

Bag = 15 Red + 20 Black = 35 balls

Out of 15 red balls, 20% are numbered 1 i.e. 3 balls and 40% are numbered 3 i.e. 6 balls.

So, 6 red balls are numbered 2.

15 Red balls — Number 1 (3) + Number 2 (6) + Number 3(6)

Out of 20 black balls, 45% are numbered 2 i.e. 9 balls and

30% are numbered 3 i.e. 6 balls. So, 5 black balls are numbered 1.

20 Black balls — Number 1 (5) + Number 2 (9) + Number3(6)

A ball is picked at random by a boy. He wins if the ball is red and numbered 3 or if it is black and numbered 1 or 2.

Probability that the ball is red and numbered

Probability that the ball is black and numbered

Probability that the ball is black and numbered

23. Answer: (a)

The correct option is (a)

There are 6 different colours and 3 are to be chosen for each of the three wooden blocks.

Number of ways to choose 3 colours out 6 colours

Number of ways in which the winners' stand can be painted = 20x3! = 120.

24. Answer: The correct answer is option (b)

Total balls = 20 = 8G + 7W + 5R

If we pick 8 balls, there is a possibility that all of them are green. No white and red balls are picked

If we pick 15 balls, there is a possibility that 8 are green and 7 are white. No red ball is picked.

If we pick 16 balls, at least one of them is red ball if the other 15 are green and white balls.

So, at least 16 balls must be picked so that there is at least one ball of each colour. Hence, option (b} is correct.

25. Answer: (b)

No. of candidates who are eligible for principal post =2

No. of ways to select 1 person out of 2 = 2 ways.

Now 1 person is selected, remaining persons = 5

All are eligible for post of vice principal.

No. of ways to select 2 out of 5 persons = $5C_2$=10.

Total no. of ways to selection = 10 + 2 = 12.

Hence, option (b) is correct.

26. Answer: (c)

It is given that Mathematics II can be offered only if Mathematics I is also opted.

Case I: math II is not opted:

No. of ways to select 2 subjects out of 4 = 4C2= 6

Case II: math II is opted:

Then there is only 1 way.

Hence, total no. of ways = 6 + 1 = 7.

Hence, option (c) is correct.

27. Answer: (d)

Let the Price of Brown socks = z

And the number of Brown socks =n

According to question, the price of black pair was thrice that of a brown pair Price of Black pair = 3z

Now, wrong bill is 100% more means it is twice than the correct bill.

Hence, nx3z+5x2=2(nxz+5x3z)

H 3n+5=2n+30=>n = 25. **Hence, option (d) is correct.**

28. Solution: b)

If male A, B, C are selected, then one more person is needed to make up a four member team.

Since B is selected, W cannot be selected and as C is selected, Z cannot be selected. So the either X or Y has to be chosen.

(A,B,C,X) or (A,B,C,Y) . So these two only can be formed.

29. Solution: b)

Simple arithmetic. R=Y= 2G

10 Time and Distance & Time and Work

1. X and Y are two runners who run for the same duration of time on the same circular track. They started running at the same time in the same direction with uniform speeds. When X completed 7 rounds, Y did exactly 5. After completing 5 rounds, Y changed his direction and started running in the opposite direction with speed which is double of his earlier speed. On the other hand, X continued to run with the same speed. They stopped running when X completed exactly 21 rounds. How many times did X and Y meet after they had started and before they finally stopped? (CSAT-2026)

   (a) 35 (b) 34
   (c) 31 (d) 29

2. An explosion takes place at a certain distance from an army camp. As soon as the sensor in the camp receives the sound of the explosion, a drone starts flying towards the spot of explosion. The drone clicks a picture from the spot and the camp receives it at the same time. Immediately another drone starts flying to the spot and it also sends a picture as soon as it reaches the spot. The two pictures were received at 5:02 PM and 5:05 PM, respectively. If the speed of the drones is 30 m/s, at what time did the explosion take place? Assume that the speed of sound is 300 m/s. (CSAT-2026)

   (a) 4:59:00 PM (b) 4:59:02 PM
   (c) 4:58:42 PM (d) 4:56:32 PM

3. X travels 6 km on a bicycle with average speeds of 5 km per hour, 10 km per hour and 4 km per hour during the first 1 km, the next 2 km and the remaining 3 km, respectively. Y travels the same distances with average speeds of 4 km per hour, 10 km per hour and 5 km per hour, respectively. How many minutes early will Y complete the journey if both X and Y start at the same time? (CSAT-2026)

   (a) 3 (b) 4
   (c) 5 (d) 6

4. A person facing the East travels 4 km straight and then turns right and travels 3 km, then further turns left and travels 2 km and finally turns left and travels 3 km. The minimum distance between the final point and the initial point, and the direction in which the person is facing at the final point are, respectively (CSAT-2026)

   (a) 12 km, East (b) 6 km, East
   (c) 8 km, North (d) 6 km, North

5. A train has to complete a journey of 800 km. If it meets a minor accident, its speed becomes half of the existing speed. If there is a mechanical defect, the speed becomes one-fourth of the existing speed. On its way, the train meets with a minor accident after 200 km; and 400 km thereafter, it develops a mechanical defect. Had the train developed the mechanical defect after 200 km and met the minor accident 400 km thereafter, it would have taken 4 more hours to reach its destination. What was the original speed of the train in km per hour? (CSAT-2026)

   (a) 200 (b) 190
   (c) 150 (d) 100

6. The speed of a train T is 100 km per hour and the speed of a person P is 4 km per hour. T crosses P in 15 seconds, if P travels along the direction of motion of T. If P travels along the opposite direction of T, then in how much time does T cross P, in seconds, approximately? (CSAT-2026)

   (a) 13.51 (b) 13.65
   (c) 13.85 (d) 14.05

7. P and Q walk along a circular track. They start at 5:00 am. from the same point in opposite directions. P walks at an average speed of 5 rounds per hour and Q walks at an average speed of 3 rounds per hour. How many times will they cross each other between 5:20 a.m. and 7:00 a.m. ? (CSAT-2025)

   (a) 12 (b) 13
   (c) 14 (d) 15

8. A tram overtakes 2 persons X and Y walking at an average speed of 3 km/hr and 4 km/hr in the same direction and completely passes them in 8 seconds and 9 seconds respectively. What is the length of the tram? (CSAT-2025)

   (a) 15 m (b) 18 m
   (c) 20 m (d) 24 m

9. A set (X) of 20 pipes can fill 70% of a tank in 14 minutes. Another set (Y) of 10 pipes fills 3/8th of the tank in 6 minutes. A third set (2) of 16 pipes can empty half of the tank in 20 minutes. If half of the pipes of set X are closed and only half of the pipes of set Y are open, and all pipes of the set (Z) are open, then how long will it take to fill 50% of the tank? (CSAT-2025)

   (a) 8 minutes (b) 10 minutes
   (c) 12 minutes (d) 16 minutes

10. X can complete one-third of a certain work in 6 days, Y can complete one-third of the same work in 8 days and Z can complete three-fourth of the same work in 12 days. All of them work together for n days and then X and Z quit and Y alone finishes the remaining work in 8 and 2/3 days. What is n equal to ? (CSAT-2025)

    (a) 3 (b) 4
    (c) 5 (d) 6

11. There are 7 places A, B, C, D, E, F and G in a city connected by various roads AB, AC, CD, DE, BF, EG and FG. A is 6 km south of B. A is 10 km west of C. D is 5 km east of E. C is 6 km north of D. F is 9 km west of B. F is 12 km north of G. A person travels from D to F through these roads. What is the distance covered by the person? (CSAT-2025)

    (a) 20 km (b) 25 km

(c) 31 km (d) 37 km

12. A certain number of men can complete a piece of work in 6k days, where k is a natural number. By what percent should the number of men be increased so that the work can be completed in 5k days? (CSAT-2024)

(a) 10% (b) (50/3)%

(c) 20% (d) 25%

13. X, Y and Z can complete a piece of work individually in 6 hours, 8 hours and 8 hours respectively. However, only one person at a time can work in each hour and nobody can work for two consecutive hours. All are engaged to finish the work. What is the minimum amount of time that they will take to finish the work? (CSAT-2024)

(a) 6 hours 15 minutes (b) 6 hours 30 minutes

(c) 6 hours 45 minutes (d) 7 hours

14. A, B, C working independently can do a piece of work in 8, 16 and 12 days respectively. A alone works on Monday, B alone works on Tuesday, C alone works on Wednesday; A alone, again works on Thursday and so on. Consider the following statements:

1. The work will be finished on Thursday.
2. The work will be finished in 10 days. Which of the above statements is/are correct? (CSAT-2023)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

15. On one side of a 1.01km long road, 101 plants are planted at equal distance from each other. What is the total distance between 5 consecutive plants? (CSAT-2022)

(a) 40 m (b) 40.4 m

(c) 50 m (d) 50.5 m

16. A man started from home at 14:30 hours and drove to village, arriving there when the village clock indicated 15:15 hours. After staying for 25 minutes, he drove back by a different route of length 1.25 times the first route at a rate twice as fast reaching home at 16:00 hours As compared to the clock at home, the village clock is (CSAT-2022)

(a) 10 minutes slow (c) 10 minutes fast

(b) 5 minutes slow (d) 5 minutes fast

17. 24 men and 12 women can do a piece of work in 30 days. In how many days can 12 men and 24 women do the same piece of work? (CSAT-2022)

(a) 30 days

(b) More than 30 days

(c) Less than 30 days or more than 30 days

(d) Data is inadequate to draw any conclusion

18. X and Y run a 3 km race along a circular course of length 300m. Their speeds are in the ratio 3:2. If they start together in the same direction, how many times would the first one pass the other (the start-off is not counted as passing)? (CSAT-2022)

(a) 2 (b) 3

(c) 4 (d) 5

19. Consider the Question Statements given below in respect of and two three cities P, Q and R in a State: (CSAT-2022)

Question: How far is city P from city Q?

Statement-1: City Q is 18 km from city R.

Statement-2: City P is 43 km from city R.

Which one of the following is correct in respect of the Question and the Statements?

(a) Statement-1 alone is sufficient to Answer the Question.

(b) Statement-2 alone is sufficient to Answer the Question.

(c) Both Statement-1 and Statement-2 are sufficient to Answer the Question.

(d) Both Statement-1 and Statement-2 are not sufficient to Answer the Question.

20. A person X from a place A and another person Y from a place B set out at the same time to walk towards eachi other. The places are separated by a distance of 15km. X walks with a uniform speed of 1.5km/hr and Y walks with a uniform speed of 1km/hr in the first hour, with a uniform speed of 1.25km/hr in the second hour and with a uniform speed of 1.5km/hr in the third hour and so on. Which of the following is correct? (CSAT-2021)

1. They take 5 hours to meet.
2. They meet midway between A and B.

Select the correct Answer using the codes given below:

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

21. A man completes 7/8 of a job in 21 days. How many more days will it take him to finish the job if quantum of work is further increased by 50%? (CSAT-2021)

(a) 24 (b) 21

(c) 18 (d) 15

22. A man walks down the backside of his house straight 25 meters, then turns to the right and walks 50 meters again; then he turns towards left and again walks 25 meters. If his house faces to the East, what is his direction from the starting point? (CSAT-2020)

(a) South-East (c) North-East

(b) South-West (d) North-West

23. A car travels from a place X to place Y at an average speed of V km/hr, from Y to X at an average speed of 2v km/hr., again from X to Y at an average speed of 3v km/hr. and again from Y to X at an average speed of 4v km/hr. Then the average speed of the car for the entire journey. (CSAT-2020)

(a) is less than v km/hr.

(b) lies between v and 2v km/hr.

(c) lies between 2v and 3v km/hr.

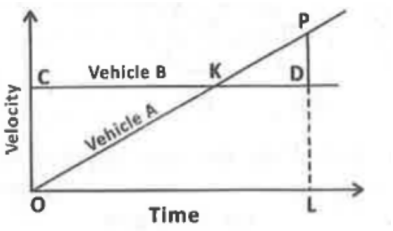
(d) lies between 3v and 4v km/hr.

24. A man takes half time in rowing a certain distance downstream than upstream. What is the ratio of the speed in still water to the speed of current? (CSAT-2020)

(a) 1:2 (b) 2:1

(c) 1:3 (d) 3:1

25. The figure drawn below gives the velocity graphs of two vehicles A and B. The straight line OKP represents the velocity of vehicle A at any instant, whereas PD= 1/2LD: the horizontal straight-line CKD represents the velocity of vehicle B at any instant. In the figure, D is the point where perpendicular from P meets the horizontal line CKD such that



What is the ratio between the distance covered by vehicles A and B in the time interval OL? (CSAT-2018)

(a) 1:2 (b) 2:3

(c) 3:4 (d) 1:1

26. A train 200 meters long is moving at the rate of 40 kmph. In how many seconds will it cross a man standing near the railway line? (CSAT-2018)

(a) 12 (b) 15

(c) 16 (d) 18

27. Two persons, A and B, are running on a circular track. At the start, B is ahead of A and their positions make an angle of 30° at the centre of the circle. When A reaches the point diametrically opposite to his starting point, he meets B. What is the ratio of speeds of A and B, if they are running with uniform speeds? (CSAT-2018)

(a) 6:5 (b) 4:3

(c) 6:1 (d) 4:2

28. P works thrice as fast as Q, whereas P and Q together can work four times as fast as R. If P, Q and R together work on a job, in what ratio should they share the earnings? (CSAT-2017)

(a) 3:1:1 (b) 3:2:4

(c) 4:3:4 (d) 3:1:4

29. A class starts at 11:00 am and lasts till 2:27pm. Four 'periods of equal duration are held during this interval, After every period, a rest of 5 minutes is given to the students. The exact duration of each period is: (CSAT-2016)

(a) 48 minutes (c) 51 minutes

(b) 50 minutes (d) 53 minutes

30. Four friends A, B, C and D need to cross a bridge. A maximum of two persons can cross it at a time. It is night and they just have one lamp. Persons that cross the bridge must carry the lamp to find the way. A pair must walk together at the speed of slower person. After crossing the bridge, the person having faster speed in the pair will return with the lamp each time to accompany another person in the group. Finally, the lamp has to be returned at the original place and the person who returns the lamp has to cross the bridge without lamp. The time taken by them is as follows: A: 1 minute, B: 2 minutes, C: 7 minutes and D: 10 minutes. What is the total minimum time required by all the friends to cross the bridge? (CSAT-2016)

(a) 23 minutes (c) 21 minutes

(b) 22 minutes (d) 20 minutes

31. A daily train is to be introduced between station A and B starting from each end at 6 AM and the journey is to be completed in 42 hrs. What is the number of trains needed in order to maintain the shuttle service? (CSAT-2016)

(a) 2 (c) 4

(b) 3 (d) 7

32. Ram and Shyam work on a job together for four days and complete 60% of it. Ram takes leave then and Shyam works for eight more days to complete the job. How long would Ram take to complete the entire job alone? (CSAT-2016)

(a) 6 days (b) 8 days

(c) 10 days (d) 11 days

33. A cylindrical overhead tank of radius 2m and height 7 m is to be filled from an underground tank of size 5.5m x 4m x 6m. How much portion of the underground tank is still filled with water after filling the overhead tank completely? (CSAT-2016)

(a) 1/3 (c) 1/4

(b) 1/2 (d) 1/6

34. A and B walk around a circular park. They start at 8 a.m. from the same point in the opposite directions. A and B walk at a speed of 2 rounds per hour and 3 rounds per hour respectively. How many times shall they cross each other after 8.00 am and before 9.30 a.m.? (CSAT-2016)

(a) 7 (c) 5

(b) 6 (d) 8

35. In a 500 meters race, B starts 45 meters ahead of A, but A wins the race while B is still 35 meters behind. What is the ratio of the speeds of A and B assuming that both start at the same time? (CSAT-2015)

(a) 25:21 (b) 25: 20

(c) 5:3 (d) 5:7

36. **Two pipes A and B can independently fill a tank completely in 20 and 30 minutes, respectively. If both the pipes are opened simultaneously, how much time will they take to fill the tank completely? (CSAT-2015)**

(a) 10 minutes (c) 15 minutes

(b) 12 minutes (d) 25 minutes

37. **In a 500 meters race, B starts 45 meters ahead of A, but A wins the race while B is still 35 meters behind. What is the ratio of the speeds of A and B assuming that both start at the same time? (CSAT-2015)**

(a) 25:21 (b) 25:20

(c) 5:3 (d) 5:7

38. **Two cities A and B are 360 km apart. A car goes from A to B with a speed of 40 km/hr. and returns to A with a speed of 60 km/hr. What is the average speed of the car? (CSAT-2015)**

(a) 45 km/hr. (b) 48 km/hr.

(c) 50 km/hr. (d) 55 km/hr.

39. **A worker reaches his factory 3 minutes late if his speed from his house to the factory is 5 km/hr. If he walks at a speed of 6 km/hr then he reaches the factory 7 minutes early. The distance of the factory from his house is (CSAT-2014)**

(a) 3 km (b) 4 km

(c) 5 km (d) 6 km

40. **Two cars start towards each other, from two places A and B which are at a distance of 160 km. They start at the same time 08:10 AM. If the speeds of the cars are 50 km and 30 km per hour respectively, they will meet each other at (CSAT-2014)**

(a) 10:10 AM (b) 10:30 AM

(c) 11:10 AM (d) 11:20 AM

41. **A thief running at 8 km/hr is chased by a policeman whose speed is 10 km/hr. If the thief is 100 m ahead of the policeman, then the time required for the policeman to catch the thief will be (CSAT-2013)**

(a) 2 min (b) 3 min

(c) 4 min (d) 6 min

42. **A train travels at a certain average speed for a distance of 63 km and then travels a distance of 72 km at an average speed of 6 km/hr more than its original speed. If it takes 3 hours to complete the total journey, what is the original speed of the train in km/hr? (CSAT-2013)**

(a) 24 (b) 33

(c) 42 (d) 66

43. **Four cars are hired at the rate of Rs. 6 per km plus the cost of diesel at Rs. 40 a litre. In this context, consider the details given in the following table: (CSAT-2013)**

Car	Mileage (km/l)	Hours	Total Payment (Rs.)
A	8	20	2120
B	10	25	1950
C	9	24	2064
D	11	22	1812

Which car maintained the maximum average speed?

(a) Car A (b) Car B

(c) Car C (d) Car D

44. **A person can walk a certain distance and drive back in six hours. He can also walk both ways in 10 hours. How much time will he take to drive both ways? (CSAT-2013)**

(a) Two hours (b) Two and a half hours

(c) Five and a half hours (d) Four hours

45. **Mr. Kumar drives to work at an average speed of 48km/hr. The time taken to cover the first 60% of the distance is 10 minutes more than the time taken to cover the remaining distance. How far is his office? (CSAT-2012)**

(a) 30km (b) 40km

(c) 45km (d) 48km

46. **A contract on construction job specifies a penalty for delay in completion of the work beyond a certain date is as follows: Rs. 200 for the first day, Rs. 250 for the second day, Rs. 300 for the third day etc., the penalty for each succeeding day being 50 more than that of the preceding day. How much penalty should the contractor pay if he delays the work by 10 days ? (CSAT-2011)**

(a) Rs. 4950 (b) Rs. 4250

(c) Rs. 3600 (d) Rs. 650

47. **If a bus travels 160 km in 4 hours and a train travels 320 km in 5 hours at uniform speeds, then what is the ratio of the distances travelled by them in one hour? (CSAT-2011)**

(a) 8 : 5 (b) 5 : 8

(c) 4 : 5 (d) 1 : 2

10 Time and Distance & Time and Work_ Explanations

1. Answer: (a)

Explanation:

Let's break the problem into two phases: before and after Y changes direction.

Phase 1: Running in the same direction

- We are given that when X completes 7 rounds, Y completes exactly 5 rounds.
- Let this duration be T.
- The speeds of X and Y are VX = 7 rounds per T and VY = 5 rounds per T.
- Since they are running in the same direction, their relative speed is VX − VY = 7 − 5 = 2 rounds per T.
- The number of times they meet is equal to the relative distance covered in rounds. In time T, they cover a relative distance of 2 rounds, meaning they meet exactly 2 times.
- These meetings occur at t = 0.5T and t = 1.0T. The meeting at t = 1.0T happens exactly when X completes 7 rounds and Y completes 5 rounds (both are at the starting point).

Phase 2: Running in opposite directions

- After t = 1.0T, Y changes direction and doubles his speed.
- Y's new speed is V′Y = 2 × 5 = 10 rounds per T.
- X continues in the same direction with the same speed, VX = 7 rounds per T.
- Because they are now running in opposite directions, their new relative speed is VX + V′Y = 7 + 10 = 17 rounds per T.
- X stops when he completes exactly 21 rounds. Since he already completed 7 rounds in Phase 1, he must run 21 − 7 = 14 more rounds in Phase 2.
- The time taken for Phase 2 is the distance X runs divided by his speed: 14 rounds ÷ (7 rounds per T) = 2T.
- In this duration of 2T, the number of times they meet is their relative speed multiplied by the time: 17 × 2 = 34 times.
- These 34 meetings happen strictly after t = 1.0T. The very last meeting (the 34th in this phase) occurs exactly at the end of the 2T duration (at t = 3.0T), which is when X completes his 21st round and they both finally stop.

Total Number of Meetings:

- The question asks for the number of meetings after they had started and before they finally stopped.
- We exclude the start (t = 0) and the final stop (t = 3.0T).
- Meetings in Phase 1: 2 (at t = 0.5T and t = 1.0T).
- Meetings in Phase 2: 34 − 1 = 33 (excluding the final stop at t = 3.0T).
- Total valid meetings = 2 + 33 = 35.

Therefore, X and Y met 35 times.

2. Answer: (c)

Explanation:

Let T be the time of the explosion and d the distance from the camp to the explosion site.

Time interval between the two pictures.

The second drone leaves at 5:02 PM (when the first picture arrives) and reaches the spot in d/30 seconds, with its picture received the moment it lands. So:

d / 30 = (5:05 − 5:02) = 180 s ⇒ d = 5400 m

Working backward from the first picture (5:02 PM).

After the explosion at T:

- Sound takes d/300 = 18 s to reach the camp.
- The first drone then takes d/30 = 180 s to reach the spot (picture received instantly).

Total: 18 + 180 = 198 s = 3 min 18 s.

T = 5:02:00 − 0:03:18 = 4:58:42 PM

3. Answer: (d)

Explanation:

For X, the time taken is:

TX = 1/5 + 2/10 + 3/4

Since 2/10 = 1/5, we get:

TX = 1/5 + 1/5 + 3/4 = 2/5 + 3/4

Taking LCM as 20:

TX = 8/20 + 15/20 = 23/20

So, X takes 23/20 hours. For Y, the time taken is:

TY = 1/4 + 2/10 + 3/5 = 1/4 + 1/5 + 3/5

Taking LCM as 20:

TY = 5/20 + 4/20 + 12/20 = 21/20

So, Y takes 21/20 hours. The time saved by Y is:

TX − TY = 23/20 − 21/20 = 2/20 = 1/10

Thus, the time saved is 1/10 hour. Converting hours into minutes:

(1/10) × 60 = 6

Therefore, Y completes the journey 6 minutes earlier than X.

4. Answer: (d)

Explanation:

Let the starting point be the origin (0, 0).

1. The person faces East and travels 4 km straight. Current position: (4, 0). Facing: East.

2. The person turns right. Since they were facing East, a right turn means they are now facing South. They travel 3 km. Current position: (4, −3). Facing: South.
3. The person turns left. Since they were facing South, a left turn means they are now facing East. They travel 2 km. Current position: (4 + 2, −3) = (6, −3). Facing: East.
4. The person turns left again. Since they were facing East, a left turn means they are now facing North. They travel 3 km. Current position: (6, −3 + 3) = (6, 0). Facing: North.

The final point is (6, 0) and the initial point is (0, 0). The minimum distance between the final and initial point is the straight-line distance:

Distance = √((6 − 0)2 + (0 − 0)2) = √36 = 6 km

At the final point, the person is facing North. Therefore, the minimum distance is 6 km and the direction the person is facing is North. This matches Option D.

5. Answer: (a)

Explanation:

Let the original speed of the train be v km/h.

The phrase **"existing speed"** means the speed of the train at that specific moment. Therefore, if the train suffers a second incident, the new speed will be calculated based on its current reduced speed (the effects compound).

The total journey is 800 km. We can divide the journey into three parts: the first 200 km, the middle 400 km, and the final 200 km.

Case 1: Minor accident first, then mechanical defect

- 0 to 200 km: The train travels at its original speed v. Time taken = 200/v.
- 200 to 600 km: After the minor accident, the speed becomes half of the existing speed, i.e., v/2. Time taken = 400/(v/2) = 800/v.
- 600 to 800 km: After the mechanical defect, the speed becomes one-fourth of the existing speed (v/2), i.e., (v/2) × 1/4 = v/8. Time taken = 200/(v/8) = 1600/v.

Total Time (T1) = 200/v + 800/v + 1600/v = 2600/v

Case 2: Mechanical defect first, then minor accident

- 0 to 200 km: The train travels at its original speed v. Time taken = 200/v.
- 200 to 600 km: After the mechanical defect, the speed becomes one-fourth of the existing speed, i.e., v/4. Time taken = 400/(v/4) = 1600/v.
- 600 to 800 km: After the minor accident, the speed becomes half of the existing speed (v/4), i.e., (v/4) × 1/2 = v/8. Time taken = 200/(v/8) = 1600/v.

Total Time (T2) = 200/v + 1600/v + 1600/v = 3400/v

Notice that the time taken for the first 200 km and the last 200 km is identical in both cases. The difference in time comes entirely from the middle 400 km.

Calculating the Original Speed:

The problem states that Case 2 takes 4 more hours than Case 1.

T2 − T1 = 4

3400/v − 2600/v = 4

800/v = 4

v = 800/4 = 200 km/h

The original speed of the train was 200 km/h.

6. Answer: (c)

Explanation:

Find the length of the train using the same-direction crossing.

Relative speed when P moves in the same direction as T:

vsame = 100 − 4 = 96 km/h = 96000/3600 = 80/3 m/s

Length of train L = v_{same} × 15 = (80/3) × 15 = 400 m.

Time to cross when moving in opposite directions.

Relative speed:

vopp = 100 + 4 = 104 km/h = 104000/3600 = 260/9 m/s

Time to cross:

t = L / vopp = 400 / (260/9) = (400 × 9) / 260 = 3600/260 ≈ 13.846 s

So t ≈ 13.85 s.

7. Answer: B

Explanation:

P and Q together cover 8 rounds per hour in opposite directions.

In 1 hour 40 minutes, they cross about 13 times, so the answer is 13.

8. Answer: C

Explanation: Using relative speed and time to pass each person, the tram's length LLL is the same in both cases. Calculating with both speeds gives L=20L = 20L=20 meters.

9. Answer: D

Explanation : The capacity of pipe X is T/400 capacity of pipe Y is T/160 and capacity of Z is T/640

if you arrange the total number of pipes available the total net capacity comes to T/32 so it will take 32 minutes to fill the whole tank so to fill the 50% of tank it will take 16 minutes

10. Answer: B

Explanation: the equation is: 23n/144+26/72=1

11. Answer: C

Explanation: Two paths: DCABF and DEGF both gives 31 kms as distance

12. Answer: (c)

Explanation:

From the question statement,

Earlier the time taken was 6k days, now 5k days.

so time has become 6 to 5, so the number of people should go from 5 to 6.

there will be increase in 1/5 x 100 = 20%

The number of men should be increased by 20% to complete the piece of work in 5k days.

13. Answer: (c)

Explanation:

X, Y and Z can complete a piece of work individually in 6 hours, 8 hours and 8 hours respectively.

To finish the work in minimum possible time, we need to ensure that X is utilized the most, as he's the most efficient worker. So, we will use X alternatively every second hour. We will also start with him.

The sequence of their work will be X, (Y or Z), X, (Y or Z), X, …..and so on.

X can complete the work in 6 hours. So, in one hour he can complete 16.67% of the work.

Similarly, Y (or Z) can complete the work in 8 hours. So, in one hour he can complete 12.5% of the work.

Now, X + Y + X + Y + X + Y = 3X + 3Y = 50 + 37.5 = 87.5%

12.5% of the work is left and it's X's turn to do it. X can do it in (12.5/16.67) hours, i.e. 0.75 hours, or 45 minutes. So, the entire task can be done by them in the minimum possible time of 6 hours 45 minutes.

14. Answer: (a)

Statement 1 is correct and Statement 2 is incorrect: Let LCM (8, 16, 12) = 48 units is the total amount of work.

A does 48/8 = 6 units/day

B does 48/16 = 3 units/day

C does 48/12 = 4 units/day

Three-day work = 6 + 3 + 4 = 13 units

(Mon + Tue + Wed) = 13 units

(Thu + Fri + Sat) = 13 units

(Sun + Mon + Tue) = 13 units

Amount of work left =48-13x3=9

Wed + Thursday = 6 + 3 = 9 units respectively.

So, work will get finished on Thursday last mentioned.

15. Answer: (b) is correct.

Length of the road = 1.01 km = 1010 m 101 plants are planted at equal distance from each other. So, No. of gaps between 101 plants = 100 gaps length of

each gap = $\frac{\text{total distance}}{\text{totap no. of gap}} = \frac{1010}{100} = 10.1$ m

Now, we need to find total distance between 5 consecutive plants. There must be 4 gaps between 5 consecutive plants.

So, required distance = 4 x 10.1 = 40.4 m. **Hence, option (b) is correct.**

16. Answer: (d)

Initially, Let the distance between point A (Home) and B (village) is x metre and the speed be U and time taken to reach at point B is t_1.

We know that, distance = speed X time .... (a)

$X_e = ut_1$ ...(b)

then, he stayed there for 25 minutes. New the time village clock is 15:15 + 25 minutes = 15:40

the person drove back with a different route Which g 1.25 times the first route i.e., 1.25x and with twice Speed

i.e., 2u Using (a), $1.25x = 2ut_2$,

Using (b),

$1.25(ut_1) = 2ut_2$,

$1.25/2t_1 = t_2$

$5/8\ t_1 = t_2$...(c)

The person started from A (home) at 14:30 and reached back at home at 16:00 and in between he stayed at his village for 25 minutes.

Total travel time = (16:00 — 14:30) - 25 minutes = (90 - 25) minutes = 65 minutes.

$t_1 + t_2 = 65$

$t_1 + 5/8t_1 = 65$

$13/8\ t_1 = 65$

$t_1 = 40$ minutes

$t_2 = 25$ minutes

But the time shown in the clock was 15:15. This means the clock is 5 minutes fast (15:15 - 15:10 = 5 minutes). **Hence, option (d) is correct.**

17. Answer: (d)

From the statement, 24 men and 12 women can do a piece of work in 30 days we cannot determine that in how many days 12 men and 24 women can do the same piece of work.

Here, we do not know about the work done by a single man and single women alone. So, we cannot compare the efficiencies of man and women. The data is inadequate to draw any conclusion. **Hence, option (d) is correct.**

18. Answer: (b) is correct.

It is given that speed of X and Y is in ratio 3:2.

Let their speed of X be 3 m/sec and speed of Y is 2 m/sec.

So, their relative speed =3 -2 = 1 m/sec

X will cross Y when he covers an extra 300 m (1 complete round).

Y= $\frac{\text{distance}}{\text{relative speeds}} = \frac{300}{1}$ =300 seconds =5 minutes
it means that

So, the time taken by X to cross

It means that X will Y every 300 seconds, or 5 minutes,

ce= $\frac{\text{total distance}}{\text{speeds of X}} = \frac{3000}{3}$ =1000 seconds
Now, the time taken by X to complete the entire race During these 1000 seconds X will cross Y 3 times- after 300 seconds, 600 seconds, and 900 seconds. **Hence, option (b) is correct.**

19. Answer: (d) is correct.

Explanation:

Clearly, statement 1 does not tell anything about distance between cities P and Q.

Clearly, statement 2 also does not tell anything about distance between cities P and Q.

Even if we combine both the statements, we cannot clearly conclude the distance between P and Q.

Because we can have two possibilities if we combine both the statements

Possibility 1: P _____ 25 _____ Q _____ 18 _____ R

Possibility 2: P _____ 43 _____ R _____ 18 _____ Q

In possibility 1 distance between P and Q is 25km and in possibility 2 that distance is 61km.

Hence, option (d) is correct.

20. Answer: (c)

We know that distance = speed x time

Distance between places A and B = 15 km

Speed of X = 1.5 km/hr

So, distance covered by X in 5 hours = 1.5x5 = 7.5 km

Speed of Y in 1st hour = 1 km/hr

Distance covered by Y in 1st hour = 1 km

Speed of Y in 2nd hour = 1.25 km/hr

Distance covered by Y in 2nd hour = 1.25 km

In the same way, distance covered by Y in 3rd hour = 1.5 km

Distance covered by Y in 4th hour = 1.75 km

Distance covered by Y in 5th hour = 2 km '

∴ Distance covered by Y in 5 hours =1+1.25 + 1.5+ 1.75 +2=7.5km

-> A and B meet midway in 5 hours. Hence, option (c) is correct.

21. Answer: (d)

Let the complete job = x units

Given that, A man completes 7/8 of a job in 21 days i.e.

Then, the complete job x is done in

If the quantum of work is increased by 50%, then the complete work

= x + 50% of x =

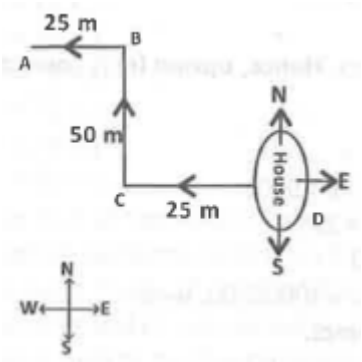
Time taken to complete unit of work = 36 days

7/8th of the job is completed in 21 days

∴ The number of extra days taken to finish the job = 36 - 21 = 15 days. Hence, option (d) is correct.

22. Answer: (d)

Check the picture given below:



As shown in above picture house is at point "0" facing east.

A man walks down from backside of house which means he is facing west and walk 25 meters and reached at point "C".

Now he take right, means towards north and walks 50 m to reach at point "B". Then he takes left, means towards west and walks 25 meters to reach point "A".

We need to find direction of "A" from point "O". As we can check in sub picture the direction is "NORTH — WEST". Hence, option (d) is correct.

23. Answer: (b)

Let the distance between X and Y be 12 km and v = 1 km/ hr. (I took the distance = 12 because 12 is L.C.M of 1,2,3 and 4 so it will be easy to divide and will avoid fraction)

Time taken in 1st journey from X to Y = Distance/Speed = 12/v =12/1=12hrs

Time taken in 2nd journey from Y to X = 12/2v = 12/2 = 6 hrs. (v=1)

Time taken in 3rd Journey from X to Y= 12/3v = 12/3 = 4hrs.

Time taken in 4th Journey from X to Y = 12/4v = 12/4 = 3hrs.

Now, total distance travelled = 12 x 4 = 48 km

Average speed

(this value lies between 1 and 2)

So, the average speed of the car for the entire journey lies between v and 2v. Hence, option (b) is correct.

24. Answer: (d)

Let the speed of man in still water = x

The speed of current = y.

Let the total distance = d.

So, speed in downstream = x +y, and speed in upstream =x-y

Time taken by man in downstream.

Time taken by man in upstream.

Now, according to the question, time taken by man in upstream = 2 x time taken implies 2x + 2y = x -y implies x=3y

x:y = 3:1. Hence, option (d) the correct.

Question Hack:

If time taken in downstream is half of time taken in upstream, then the speed in downstream must be double of speed in upstream, as the distance is same.

Hence, x + y = 2(x -y)

Or x=3y thus x:y = 3:1.

25. Answer: (c)

The correct option is (c).

it is given that PD = LD

Let PD =x. so, LD = 2x

LP=LD+DP =x+2x = 3x

Distance travelled by vehicle A and B is the area under the velocity-time graph.

Distance travelled by vehicle A = Area under velocity A curve

= area of triangle OLP = PD

Distance travelled by vehicle B = Area under velocity B

curve = OL x LD = 2x(OL)

$$OL = \frac{\frac{3}{2}(OL)(x)}{2x(OL)} = \frac{3}{4} = 3:4$$

So, the ratio of distance covered by vehicles A and B in the time interval

26. Answer: (d)

The correct option is (d).

Length of train = 200 m

Velocity of train = 40 kmph =

We know that,

Distance = speed X time

Time = 18 sec

So, the train will cross a man standing near the railway line in 18 sec.

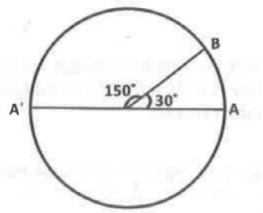
27. Answer: (a)

The correct option is (a)

Let the speed of A and B be x and y respectively.

Let the radius of circular track is x and after time 't' both A And B meet at A'

At the start, B is ahead of A and their positions make an angle of 30" at the centre of the circle.



Distance travelled by A to reach the diametrically opposite end

Distance travelled by B to reach the diametrically opposite end

We know that distance = speed X time

Divide (i) by (ii), we get.

28. Answer: The correct answer is option (a)

Let P, Q and R work for p, q and r units a day.

Given,

p =3q... (i) p+q=4r.. (i)

Put (i) in (ii),

3p+q=4r

q=r So, the ratio of their work is p:q:r = 3p:qiq = 3:1:1

P, Q and R will earn with respect to their work. So, their

earnings ratio is also 3: 1: 1. **Hence, option (a) is correct.**

29. Answer: (a)

A class starts at 11:00 am and lasts till 2:27pm.

Total number of minutes = 60 + 60 + 60 +27 = 207 minutes.

Total periods = 4 and rest between periods = 5 minutes.

Total rest time = 5 x 3 = 15 minutes.

~ Minutes left for 4 periods = 207 - 15 = 192 minutes.

Using unitary method,

4 periods in 192 minutes

1 period in 192/4 = 48 minutes

So, the exact duration of each period = 48 minutes. **Hence, option (a) is correct.**

30. Answer: (a)

The time taken by each person in crossing the bridge is given as –

A - 1 minute, B - 2 minutes, C - 7 minutes and D - 10 minutes

Minimum time would be achieved only when the fastest person would be assigned to return the lamp. The fastest person in this case is A.

Time taken by A and B = 2 minutes + 1 minute (return time of A) = 3 minutes

Time taken by A and C = 7 minutes + 1 minute (return time of A) = 8 minutes

Time taken by A and D = 10 minutes + 1 minute (return time of A) = 11 minutes

Time taken by A to cross the bridge alone in the end = 1 minute

So, the total time taken by all the friends to cross the bridge = 3 + 8 + 11 + 1 = 23 minutes.

Hence, option (a) is correct.

31. Answer: (c)

Train A and train B leaves the station from each end at 6AM

Journey time = 42 hours = 1 day + 18 hours

So, the train A and B can start their return journey of the third day at 6 AM.

The number of trains required in order to maintain the shuttle service = 2 (1% day) + 2(2" day) = 4. **Hence, option (c) is correct.**

32. Answer: (c)

Let the whole work = x

Work done by both in 4 day = 60% of x or 60x /100

Remaining work = 40x / 100

Shyam did 40x / 100 work in 8 days

Work of Shyam in 1 day = x/20

Work done by Shyam in 4 days = x/5

Work done by Ram in 4 days = 2x/5

Work done by Ram in 1 day = 2x/20 =x/10

No. of days Ram will take to complete the work = 10 days.

Hence, option (c) is correct.

33. Answer: (a)

The radius and height of a cylindrical overhead tank is 2m and 7m respectively.

Volume of cylindrical overhead tank = Dimensions of underground tank is 5.5m x 4m x 6m

Volume of underground tank = $5.5 \times 4 \times 6 = 132\ m^3$

Now, the cylindrical tank in filled from an underground tank.

So, water left in underground tank = $132 - 88 = 44m^3$

Ratio of water left to full tank. **Hence, option (a) is correct.**

34. Answer: (a)

A and B starts at 8 a.m. from the same point in opposite direction. Total minutes between 8 a.m. to 9.30 a.m. = 90 minutes

Speed of A = 2 rounds per hour

Speed of B = 3 rounds per hour

Relative speed = Speed of A + Speed of B = 5 rounds / hour

Distance covered by A and B in 90 minutes = 5/60 * 90 =7.5 rounds

So, they cross each other 7 times between 8.00 a.m. to 9.30 a.m.

Hence, option (a) is correct.

35. Answer: (a)

Distance covered by A in given time (t) = 500m

Distance covered by B in given time (t) = 500 - 45 - 35 = 420.

Ratio of their speed = (500/t): (420/t)

Or 500: 420 =25: 21. **Hence, option (a) is correct.**

36. Answer: (b)

Pipe A can fill the tank in = 20 minutes. Pipe B can fill the tank in = 30 minutes

In 1 minute, pipe A and pipe B can fill 1/20 and 1/30 of tank respectively.

If they opened simultaneously for 1 minute they can fill: 1/20 + 1/30 = 1/12 of tank. If they opened together, they can fill the tank in 12 minutes.

Hence, option (b) is correct.

37. Answer: (a)

Distance covered by A in given time (t) = 500m

Distance covered by B in given time (t) = 500 — 45 - 35 =420.

Ratio of their speed = (500/t): (420/t)

Or 500: 420 =25: 21. **Hence, option (a) is correct.**

38. Answer: (b)

Distance travelled by car from A to B = 360km

Speed of car = 40km/hr.

Time taken = 360/40 = 9 hrs.

Distance travelled by car from B to A = 360km.

Speed of car = 60km/hr.

Time taken = 360/6 = 6 hrs.

Total distance travelled by car = 720kms.

Total time taken by car = 9 + 6 = 15 hrs.

Average speed = 720/15 = 48km/hr.

Hence, option (b) is correct.

39. Answer: (c)

The correct option is option C

Let's us take distance between factory and house = x km

Ideal time he takes to reach factory = y hours

Note: In this question change minutes in hours and meters in kilometers South

So, if the speed is Skm/hr. then he is 3 minutes late or 3/60 hours late. We can make a diagram to solve this question using

So, X/5 =y +3/60... i) given information

And if the speed is 6km/hr. then he reach 7 minutes early or 7/60 hours early

So, x/6 =y — 7/60... (ii)

Now solve (i) and (ii)

We get x/5 - x/6 = (y + 3/60) - (y — 7/60) x/30=1/6

Or x = 5. **Hence, option (c) is correct.**

40. Answer: (a)

The correct option is option A

Distance between A and B = 160km

Speed of A = 50km/hr.

Speed of B = 30km/hr.

A and B are moving in opposite direction. So, their ! relative speed = 50 + 30 = 80km/hr.

So, time taken by cars to cover 160km = 160/80 = 2 hrs.

They start at the same time 08: 10 AM and they will meet after 2 hrs. at 10: 10 A.M. **Hence, option (a) is correct.**

41. Solution:

Use relative speed concept. Relative speed of policeman w.r.t the thief is 2 Km/hr (10-8). His Speed in m/min will be (2 X1000)/60 = 100/3 m/min

He has to cover 100 m. So time needed is 3 mins.

42. Solution: c)

Let the original speed be 'S'.

Total time taken = 3 hr => 63/S + 72/S+6 = 3

You can solve the question easily if you just do an option check. Otherwise the long way is to solve the equation completely. Save time.

As we have 63 in the numerator, we can assume that S may be a multiple of 7. The only option available is 4. So, let us try with that.

63/42 + 72/48 = 3/2 + 3/2 = 3

You have the answer.

43. Solution: a)

You need to find the total distance travelled by and divide it by total time. To find the distance, first form a correct equation that sums up the relation of all variables of each Car – A , B, C , D.

Let the distance be N. So diesel consumed for car A will be N/8 litres. Cost will be (N/8) X 40 = 5N

Moreover, 6 Rs. needs to paid per km. So additional cost will be 6N Total cost = 11N = Rs. 2120 => N = 192.7 Km

Average speed of Car A = 192.7/20 = Around 9.6 km/hr

Now, instead of repeating this process for every car, form a general equation: Total cost = [(Distance/mileage) x Fuel Cost] + Distance x per km. Cost]

Find distance from here and then average speed. You will get the answer as A.

44. Solution: (a)

2-way walk = 10 hrs, therefore 1-way walk = 5 hrs

1-way walk + 1-way Drive = 6 hrs; therefore 1-way drive = 6-5= 1 hr Hence, 2-way drive = 2 hrs

45. Solution: b)

Let the distance be N.

First 60% distance is covered in 10 more minutes more time than that of the rest 40% distance

So, 0.6 x N distance is covered in 10 more minutes more than that of the rest 0.4 x N distance

Which means 0.2 x N is covered in 10 minutes i.e., 1/6 hr Hence, Average speed = distance/time = 0.2 x N /(1/6) = 48

=> x = 48/6 x 0.2 =40 km

46. Solution: (b)

The penalty for delay on first day is 200, for second day is 250, for third day is 300 and so on.

The penalties form an arithmetic progression (AP) with a common difference of 50 and first element

(a) as 200.

The sum of penalties for 10 days is thus same as the sum of numbers in the A.P. with n as 10. Sum of AP (formula) = n/2 [2a + (n-1) d] = 10/2 [2 X 200 + (10-1) X 50] = 4250

47. Solution: (b)

Take the ratio of both the speeds. First bus = 40 kmph, so 40 Kms in one hour. Second bus = 64 kmph, so 64 kms in one hour.

Ratio = 40:64 = 5:8

11 Data Interpretation

Charts

1. Consider the following data:

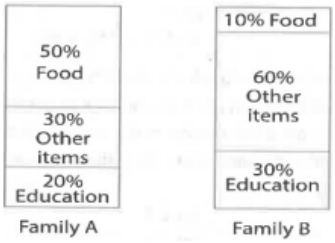
Year	Birth Rate	Death Rate
1911-1921	48.1	35.5
1921-1931	46.4	36.3
1931-1941	45.2	31.2
1941-1951	39.9	27.4
1951-1961	41.7	22.8
1961-1971	41.1	18.9
1971-1981	37.1	14.8

For which period was the natural growth rate maximum? (CSAT-2020)

(a) 1911-1921 (b) 1941-1951

(c) 1961-1971 (d) 1971-1981

2. The proportion of expenditure on various item by two families A and B are represented in the following Bar Charts: (CSAT-2015)



Total expenditure Rs.20,000 Total expenditure Rs. 10,0000

From these charts, we can conclude that ~ (CSAT-2015)

(a) Family A spent more money on food than family B.

(b) Family B spent more money on food than family A.

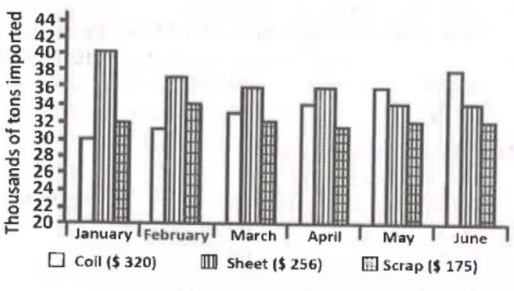
(c) Family A and Family B spent the same amount on food.

(d) The expenditure on food by family A and family B cannot be compared.

Graphs

Directions for the following 3 (three) items:

The following three items are based on the graph given below which shows imports of three different types of steel over a period of six months of a year. Study the graph and answer the three items that follows.



3. **By how much (measured in thousands of tons) did the import of sheet steel exceed the import of coil steel in the first three months of the year? (CSAT-2018)**

   (a) 11 (b) 15

   (c) 19 (d) 23

4. **What was the approximate total value (in $) of sheet steel imported over the six months period? (CSAT-2018)**

   (a) 45,555 (b) 50,555

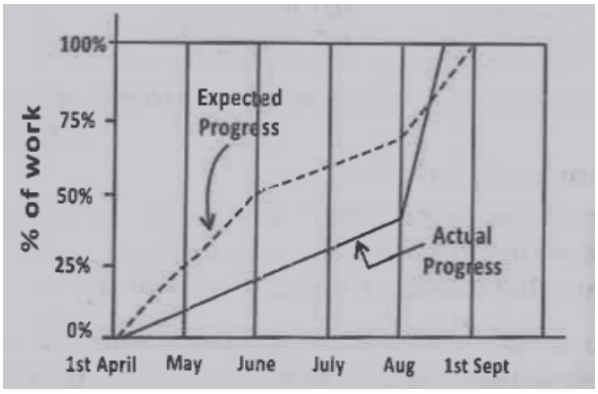
   (c) 55,550 (d) 65,750

5. **What was the approximate ratio of sheet steel and scrap steel imports in the first three months of the year? (CSAT-2018)**

   (a) 1:1 (b) 1.2:1

   (c) 1.4:1 (d) 1.6: 1

6. **Consider the given graph:**



Which one of the following statements is not **correct with reference to the graph given above? (CSAT-2018)**

(a) On 1 June, the actual progress of work was less than expected.

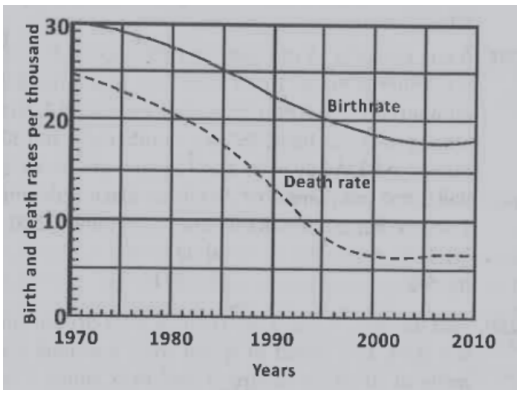
(b) The actual rate of progress of work was the greatest during the month of August.

(c) The work was actually completed before the expected time.

(d) During the period from 1 April to 1 September, at no time was the actual progress more than the expected progress.

For the next two questions:

Consider the following graph in which the birth rate and death rate of a country are given and answer the two items that follows.



7. **Looking at the graph, it can be inferred that from 1990 and 2010 (CSAT-2018)**

   (a) Population growth rate has increased.

   (b) Population growth rate has decreased.

   (c) Growth rate of population has remained stable.

   (d) Population growth rate shows no trend.

8. **With reference to the above graph consider the following statements considering 1970 as base year:**

   1. Population has stabilized after 35 years.
   2. Population growth rate has stabilized after 35 years.
   3. Death rate has fallen by 10% in the first 10 years.
   4. Birth rate has stabilized after 35 years.

   Which of the above are the most logical and rational statements that can be made from the above graph? (CSAT-2018)

   (a) 1 and 2 only (b) 1, 2 and 3 only

   (c) 3 and 4 only (d) 2 and 4 only

9. **Average hourly earnings per year (E) of the workers in a firm are represented in figures A and B as follows:**

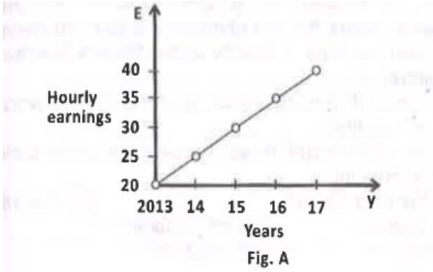


Fig. A

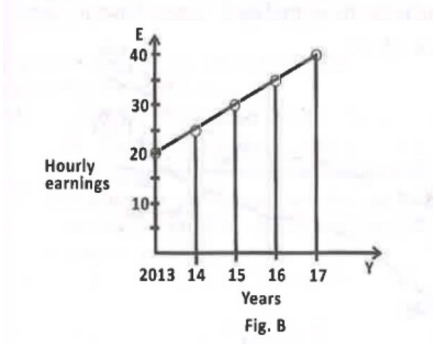


Fig. B

From the figures, it is observed that the **(CSAT-2018)**

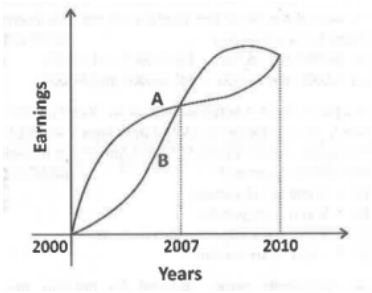
(a) values of E are different.

(b) ranges (i.e., the maximum difference between the maximum and minimum) of E are different.

(c) slopes of the graphs are same

(d) rate of increase of E are different

10. The graph below depicts the earnings of A and B over the period 2000 to 2010: **(CSAT-2015)**



From the graph, which one of the following can be concluded? (CSAT-2015)

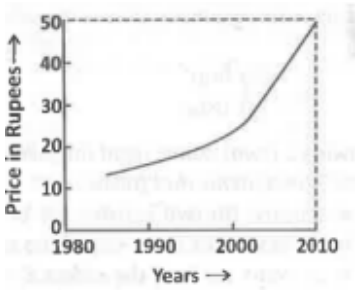
(a) On the average A earned more than B during this period.

(b) On the average B earned more than A during this period.

(c) The earnings of A and B were equal during this period.

(d) The earnings of A were less as compared to B during this period.

11. Year - wise variation of the price of a certain commodity is shown in the following graph:



The price of the commodity in the year 1990 (CSAT-2015)

(a) Must have been Rs. 10

(b) Must have been Rs. 12

(c) Must have been anywhere between Rs. 10/- and 20/-

(d) Is higher than that in the year 1991

12. In which year is the average profit of A and B same? (CSAT-2014)

(a) 1995 (b) 1996

(c) 1997 (d) 1998

13. What is the difference between the average profit of B and A in the year 1998? (CSAT-2014)

(a) Rs. 100 (b) Rs. 1,000

(c) Rs. 600 (d) Rs. 300

14. How much more average profit did A make in the year 2000 than in the year 1999? (CSAT-2014)

(a) Rs. 200 (b) Rs. 1,000

(c) Rs. 1,500 (d) Rs. 2,000

15. What is the trend of the average profit of B from the 1997 to the year 2000? (CSAT-2014)

(a) Non-increasing

(c) Steady Year, (CSAT»ZOM)

(b) Non -decreasing

(d) Fluctuating

16. How many Physics professors belong to the age group 35 - 44? (CSAT-2013)

(a) 18 (b) 16

(c) 14 (d) 12

17. Which one of the following disciplines has the highest ratio of males to females? (CSAT-2013)

(a) Physics (b) Mathematics

(c) Chemistry (d) Economics

18. What percentage of all Psychology professors are females? (CSAT-2013)

(a) 40% (b) 50%

(c) 60% (d) 70%

19. If the number of female Physics professors in the age group 25 - 34 equals 25% of all the Physics professors in that age group, then what is the number of male Physics professors in the age group 25 - 34? (CSAT-2013)

(a) 9 (b) 6

(c) 3 (d) 2

20. If the Psychology professors in the University constitute 2% of all the professors in the University, then what is the number of professors in the University? (CSAT-2013)

(a) 400 (b) 500

(c) 600 (d) 700

Tables

21. In a recruitment process, the selection of candidates is based on their performance in three components. The weightages of the components 1, 2 and 3 are 0.2, 0.3 and 0.5, respectively. Use the data given below and find the cutoff score if exactly three candidates are to be selected: (CSAT-2026)

Candidate	Score in component 1	Score in component 2	Score in component 3
1	5	4	6
2	4	6	5
3	3	2	8
4	9	4	3
5	8	8	2

(a) 5.1 (b) 5.2

(c) 5.3 (d) 5.4

22. Consider the following table:

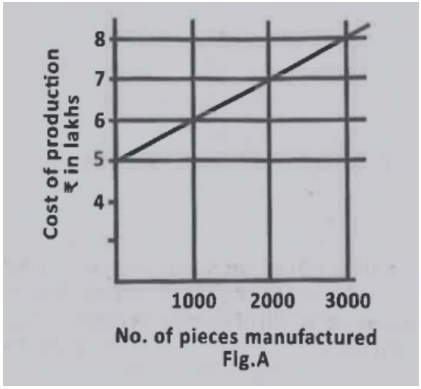
Player	Runs scored in the First Innings	Balls faced in the First Innings	Runs scored in the Second Innings	Balls faced in the Second Innings
A	61	99	14	76
B	05	12	50	85
C	15	75	20	50
D	13	55	12	50

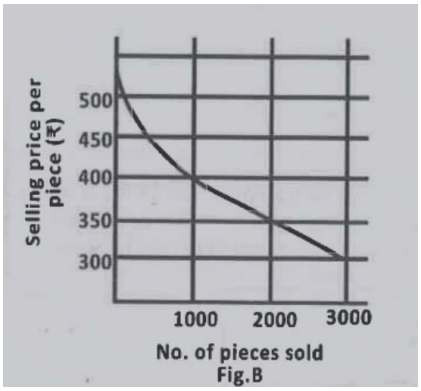
Who is the fastest run scorer in the Test Match? (CSAT-2021)

(a) A (b) B

(c) C (d) D

23. Consider the figures A and B:



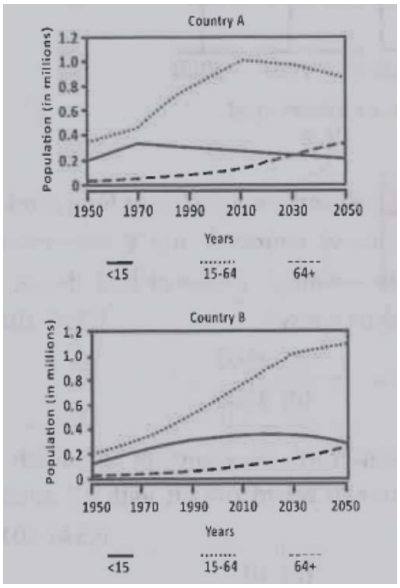


The manufacturing cost and projected sales for a product are shown in the above figures A and B respectively. What is the minimum number of pieces that should be manufactured to avoid a loss? **(CSAT-2018)**

(a) 2000 (b) 2500

(c) 3000 (d) 3500

24. Consider the following graphs. The curves in the graphs indicate different age groups in the population of two countries A and B over a period of few decades:



With reference to the above graphs, which of the following are the most logical and rational inferences that can be made?

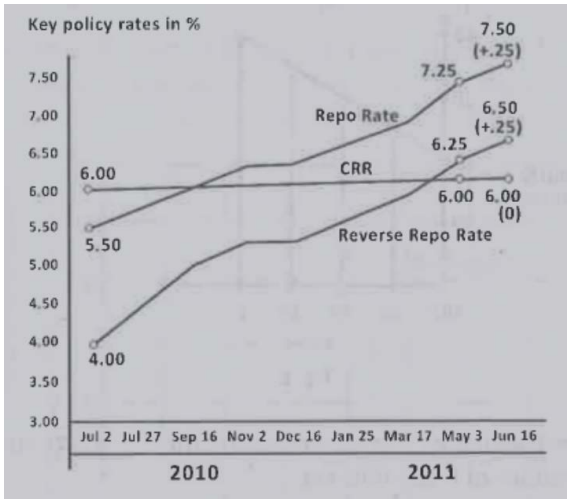
1. Over the last two and a half decades, the dependency ratio for country B has decreased.
2. By the end of the next two and a half decades, the dependency ratio of country A will be much less than that of country B
3. In the next two decades, the workforce relative to its total population will increase in country B as compared to country A.

Select the correct answer using the code given below. (CSAT-2018)

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2, and 3

25. The graph given below indicates the changes in key policy rates made by the central bank several times in a year:



Which one of the following can be the most likely reason for the central bank for such an action? (CSAT-2018)

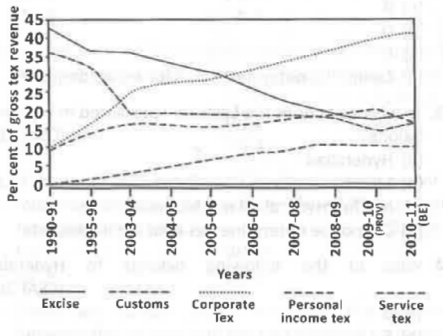
(a) Encouraging foreign investment

(b) Increasing the liquidity

(c) Encouraging both public and private savings

(d) Anti-inflationary stance

The following graph indicates the composition of our tax revenue for a period of two decades:



26. With reference to the above graph, which of the following is/are the most logical and rational inference(s) that can be made?

1. During the given period, the revenue from Direct Taxes as a percentage of gross tax revenue has increased while that of Indirect Taxes decreased.
2. The trend in the revenue from excise duty demonstrates that the growth of manufacturing sector has been negative during the given period

Select the correct answer using the code given below. (CSAT-2018)

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

For the next 2 items: The following table gives the GDP growth rate and Tele-density data of different states of a country in a particular year. Study the table and answer the two items that follow.

States	Per capita income ($)	GDP growth rate (%)	Tele – density
State 1	704	9.52	70.27
State 2	419	5.31	35.88
State 3	254	10.83	50.07
State 4	545	9.78	5.94
State 5	891	10.8	76.12
State 6	1077	11.69	77.5
State 7	900	8.88	104.86
State 8	395	5.92	6
State 9	720	7.76	82.25
State 10	893	9.55	96.7
State 11	363	4.7	57.7
State 12	966	7.85	63.8
State 13	495	9.37	52.3
State 14	864	5.46	97.9
State 15	497	7.48	62.3
State 16	777	7.03	93.8
State 17	335	5.8	49.9
State 18	599	7.49	47.84

27. With reference to the above table, which of the following is/are the most logical and rational inference(s) that can be made?

1. Higher per capita income is generally associated with higher Tele-density.
2. Higher GDP growth rate always ensures higher per capita income.
3. Higher GDP growth rate does not necessarily ensure higher Tele-density.

Select the correct answer using the code given below. (CSAT-2018)

(a) 1 only (b) 2 and 3

(c) 1 and 3 (d) 3 only

28. If for a sample data Mean < Median < Mode, then the distribution is (CSAT-2017)

(a) symmetric

(b) skewed to the right

(c) neither symmetric nor skewed

(d) skewed to the left

29. The following table shows the marks obtained by two student in different subject:

Subjects	Student A	Maximum marks	Student B	Maximum marks
English	60	100	80	150
Psychology	70	100	70	100
English	30	100	60	100
Psychology	50	50	15	25

The difference in the mean aggregate percentage marks of the students is (CSAT-2014)

(a) 2.5%
(b) 13.75%
(c) 1.25%
(d) Zero

30. The following table gives population and total income of a city for four years:

Year	1992	1993	1994	1995
Population In lakhs	20	21	22	23
Income In crores(Rs.)	1010	1111	1225	1345

Which one of the following statements corrects in respect of the above data? (CSAT-2014)

(a) Population increased by 5% or more every year.
(b) Income increased by 10% or more every year.
(c) Per capita income was always above 5,000.
(d) Per capita income was highest in 1994.

Data sufficiency

31. There are four letters and four envelopes and exactly one letter is to be put in exactly one envelope with the correct address. If the letters are randomly inserted into the envelopes, then consider the following statements:

1. It is possible that exactly one letter goes into an incorrect envelope.
2. There are only six ways in which only two letters can go into the correct envelopes.

Which of the statements given above is/are correct? (CSAT-2023)

(a) 1 only (c) Both 1and 2

(b) 2 only (d) Neither 1 nor 2

32. Consider the following including the Question and the Statements:

There are S members A, B, C, D, E in a family.

Question: What is the relation of E to B?

Statement-1: A and B are a married couple.

Statement-2: D is the father of C

Statement-3: E is D's son

Statement-4: A and C are sisters.

Which one of the following is correct in respect of the above Question and Statements? (CSAT-2023)

(a) Statement-1, Statement-2 and Statement 3 are sufficient to answer the Question.

(b) Statement-1, Statement-3 and Statement-4 are sufficient to answer the Question.

(c) All four statement together are sufficient to answer the Question.

(d) All four statements are not sufficient to answer the Question.

33. Two statements S1 and S2 are given below followed by a Question:

S1: There are not mare than two figures on any page of a 51-page book.

S2: There is at least one figure on every page.

Question: Are there more than 100 figures in that book? Which one of the following is correct in respect of the above Statements and the Question? (CSAT-2020)

(a) Both S1 and S2 are sufficient to Answer the Question, but neither S1 nor S2 alone are sufficient to Answer the Question.

(b) S1 alone is sufficient to Answer the Question.

(c) S1 and S2 together are not sufficient to Answer the Question.

(d) S2 alone is sufficient to Answer the Question.

34. Consider two statements S1 and S2 followed by aquestion:

S1: p and q both are prime numbers.

S2: p + q is an odd integer.

Question: Is pq an odd integer?

Which one of the following is correct? (CSAT-2019)

(a) S1 alone is sufficient to Answer the question

(b) S2 alone is sufficient to Answer the question

(c) Both S1 and S2 taken together are not sufficient to Answer the question

(d) Both S1 and S2 are necessary to Answer the question

Directions for the following 2(two) items: Read the following statements S1and S2 and Answer the two items that follow:

s1: Twice the weight of Sohan is less than the weight of Mohan or that of Rohan.

s2: Twice the weight of Rohan is greater than the weight of Mohan or that of Sohan.

35. Which one of the following statements is correct? (CSAT-2019)

(a) Weight of Mohan is greatest

(b) Weight of Sohan is greatest

(c) Weight of Rohan is greatest

(d) Whose weight is greatest cannot be determined .

36. A five-storeyed building with floors from I to V is painted using four different colours and only one colour is used to paint a floor. Consider the following statements:

1. The middle three floors are painted in different colours.
2. The second (/1) and the fourth (IV) floors are painted in different colours.
3. The first (1) and the fifth (V) floors are painted red.

To ensure that any two consecutive floors have different colours (CSAT-2019)

(a) Only statement 2 is sufficient.

(b) Only statement 3 is sufficient.

(c) Statement 1 is not sufficient, but statement 1 along with statement 2 is sufficient.

(d) Statement 3 is not sufficient, but statement 3 along with statement 2 is sufficient.

37. If x-y = 8, then which of the following must be true?

1. Bothxandy must be positive for any value of x and y.
2. If x is positive, y must be negative for any value of x and y.
3. If x is negative, y must be positive for any value of x and y.

Select the correct Answer using the code given below. (CSAT-2018)

(a) 1only (b) 2 only

(c) Both 1and 2 (d) Neither 1 nor 2 nor 3

Miscellaneous

38. Five friends P, Q, X, Y and Z purchased some notebooks. The relevant information is given below: (CSAT-2022)

1. Z purchased 8 notebooks more than X did.
2. P and Q together purchased 21 notebooks.
3. Q purchased 5 notebooks less than P did.
4. X and Y together purchased 28 notebooks.
5. P purchased 5 notebooks more than X did.

If each notebook is priced 40, then what is the total cost of all the notebooks?

(a) 2,600 (c) 2,400

(b) 2,400 (d) 2,320

39. Seven books P, Q, R, S, T, U and V are placed side by side. R, Q and T have blue covers and other books have red covers. Only S and U are new books, and the rest are old. P, Rand S are law reports; the rest are Gazetteers. Books of old Gazetteers with blue covers are (CSAT-2021)

(a) Q and R (b) Q and U

(c) Q and T (d) T and U

Directions for the following 2 (two) items:

Read following information and answer the two items that follow.

The plan of an office block for six officers A, B, C, D, E and F is as follows: Both B and C occupy officers to the right of the corridor (as one enters the office block) and A occupies on the left of the corridor. E and F occupy officers in opposite sides. the corridor, but their offices do not face each. The offices of C And D face each other. E does not have a corner office. F's office is further down the corridor than A's but on the same side.

40. If E sits in his offices and faces the corridor, whose office is to his left? (CSAT-2018)

(a) A (b) B

(c) C (d) D

41. Who is / are F's immediate neighbour / neighbours? (CSAT-2018)

(a) A only (b) A and D

(c) C only (d) B and C

Directions for the following 6 (six) items:

Read the information given below and answer the six items that follow.

A, B, C and D are students. They are studying in four different cities, viz., P, Q, R and S (not necessarily in that order). They are studying in Science college, Arts college, Commerce and Engineering college (not necessarily in that order) which are situated in four different states viz., Gujarat, Rajasthan, Assam and Kerala (not necessarily in that order). Further, it is given that:

1. D is studying in Assam
2. Arts college is located in city S which is in Rajasthan
3. A is studying in Commerce college
4. B is studying in city Q
5. Science college is located in Kerala

42. A is studying in (CSAT-2018)

(a) Rajasthan (b) Gujarat

(c) City Q (d) Kerala

43. Science college is located in (CSAT-2018)

(a) City Q (b) City S

(c) City R (d) City P

44. C is studying in (CSAT-2018)

(a) Science college (b) Rajasthan

(c) Gujarat (d) City Q

45. Which one of the following statements is correct? (CSAT-2018)

(a) D is not studying in the city S

(b) A is studying in Science college

(c) A is studying in Kerala

(d) Engineering college is located in Gujarat

46. Which one of the following statements is correct regarding Engineering college? (CSAT-2018)

(a) C is studying there. (b) B is studying there.

(c) It is located in Gujarat (d) D is studying there

47. Which one of the following statements is correct? (CSAT-2018)

(a) Engineering college is located in Assam

(b) City Q is situated in Assam

(c) C is studying in Kerala

(d) B is studying in Gujarat

48. Lakshmi, her brother, her daughter, and her son are badminton players. A game of doubles is about to begin:

(i) Lakshmi's brother is directly across the net from her daughter.

(ii) Her son is diagonally across the net from the worst player's sibling.

(iii) The best player and the worst player are on the same side of the net.

Who is the best player? (CSAT-2018)

(a) Her brother (b) Her daughter

(c) Her son (d) Lakshmi

49. A question paper must have a question on one of the eight poets: A, B, C, D, E, F, G or H. The first four belong to the medieval period while the rest are considered modern poets. Generally, modern poets figure in the question paper in alternate years. Generally, those who like H like G also; and those who like F like E also. The paper-setter does not like to ask about F as he has written a book on F, but he likes F. Last year, the paper contained a question on A. On the basis of the information given, this year's paper is most

likely to contain a question on (CSAT-2014)

(a) C (b) E

(c) F (d) H

50. In a group of six women there are four dancers, four vocal musicians, one actress, and three violinists. Girija and Vanaja are among the violinists while Jalaja and Shailaja do not know how to play the violin. Shailaja and Tanuja are among the dancers. Jalaja, Vanaja, Shailaja, and Tanuja are all vocal musicians and two of them are also violinists. If Pooja is an actress, who among the following is certainly a dancer and a violinist? (CSAT-2014)

(a) Jalaja (b) Pooja

(c) Shailaja (d) Tanuja

11 Data Interpretation_Explanations

Charts

1. Answer: (d)

Natural growth rate = Birth rate — Death rate

By the given table:

Natural growth rate in the period 1911 -1921 = 48.1 — 355=12.6

Natural growth rate in the period 1941 -1951 = 39.9 — 27.4=12.5

Natural growth rate in the period 1961 -1971 = 41.1 — 18.9=22.2

Natural growth rate in the period 1971 -1981 = 37.1 — 14.8=22.3

For the period 1971 — 1981, natural growth rate was the maximum. Hence, option (d) is correct.

2. Answer: (c)

Let's make a table to understand this:

Family	Education	Food	Other items
A	20% of 20,000 = 40,00	50% of 20,000 = 10,000	30% of 20,000 = 60,00
B	30% of 1 lakh = 30,000	10% of 1 lakh = 10,000	60% of 1 lakh = 60,000

From the above table we can conclude that family spent on food equal to family B.

Hence, option (c) is respectively. correct.

Graphs

3. Answer: (c)

The correct option is (c)

Using graph, during the first three months, Sheet steel imports = 40 + 37 + 36 = 113 thousand tons =113 TT

Coil steel imports = 30 + 31 + 33 = 94 thousand tons =94 TT

So, the sheet steel imports exceed the coil steel imports by 113 - 94 = 19 thousand tons.

4. Answer: (c)

The correct option is (c)

Total imported sheet steel over the six-month period = 40+37 +36 +36 + 34+ 34 =217 thousand tons = 217 TT

Average cost per ton of sheet steel = $ 256. So, the cost of total imported sheet steel = 256 x 217 = $ 55,552 thousands.

5. Answer: (b)

The correct option is (b)

Total imports of sheet steel in the first three months = 40+37 +36 =113 thousand tons

Total imports of scrap steel in the first three months = 32+34 + 32 =98 thousand tons

So, their ratio (approximately)

6. Answer: (d)

The correct option is (d)

Using the given graph, we can observe that

Option (a): On 1% June, the actual progress of work was less than expected. This statement is correct..

Option (b): The actual rate of progress of work was the greatest during the month of August. This statement is correct.

Option (c): The work was actually completed before the expected time. This statement is also correct.

Option (d): During the period from 1* April to 1 September, at no time was the actual progress more than the expected progress. This statement is not correct as during a brief window in August, the actual progress was more than the expected progress.

7. Answer: (d)

The correct option is (d)

Using graph, we can observe that during 1990 to 2010,population growth rate shows no trend.

Both birth rate and death rates are decreased at first and then stabilised.

8. Answer: (d)

The correct option is (d)

We can clearly observe form the graph that population growth rate has stabilized after 35 years as both the birth rate and death rate is stabilized.

Population growth = Birth rate — Death rate

So, the most logical and rational statements are statement-2 and statement-4.

9. Answer: (c)

The correct option is (c)

From graph,

Slope of first graph

Slope of second graph

So, the slopes of the graph are same. Hence, option (c) is correct

10. Answer: (a)

We can observe from the graph that:

Earning of A is more than earning of B for the period of (2000 - 2007) for 7 years and earning of B is more than earning of A for period of (2007 — 2010) 3 years only. By the above statements we can conclude that on the average A earned more than B during given period.

Hence, option (a) is correct.

11. Answer: (c)

Nothing much to explain you can see clearly from graph that price of commodity in 1990 is somewhere between Rs. 10/- and 20/-.

Hence, option (c) is correct.

12. Answer: (b)

The correct option is option B

Average profit of A and B will be same when these two curves meet or cut each other.

Now from graph it is evident that both the curves meet when the year is 1996. **Hence, option (b) is correct**

13. Answer: (c)

The correct option is option C

By looking at the graph we can see that in year 1998 or when x axis is on 1998 curve A is under curve B.

Hence, in year 1998 average profit of B is more than average profit of A.

Hence, Average profit of B-Average profit of A>0

There is only one option which is option (c) which gives us positive result. **Hence, option (c) is correct.**

14. Answer: (d)

The correct option is option D

Observe the graph carefully and look at curve A which gives us the average profit of A:

Average profit of A in year 1999 = 4000

Average profit of A in year 2000 = 6000

Difference = 2000. **Hence, option (d) is correct.**

15. Answer: (b)

The correct option is option B

Average profit of B in 1997 = 4000

Average profit of B in 1998 4000 (same as previous year)

Average profit of B in 1999 5000 (increasing)

Average profit of B in 2000= more than 5000 (increasing) Now see the trend 1st two year it is constant Hence, not steady.

After 2 years it is increasing so it is never decreasing Hence, it is neither fluctuating nor non increasing. Hence, trend is non-decreasing. **Hence, option (b) is correct.**

16. Solution: (b)

From figure 2, you can see that 40% of Physics professors belong to the age group 35-44. From figure 1, we can find that there are 32 male and 8 female Physics professors.

Totally there are 40 Physics professors - out of which 40% belong to 35-44 age group. 40% of 40 will be 16 – answer.

17. Solution: (a)

Simple mathematics. Just take the concerned ratios.

18. Solution: (c)

There are 4 male and 6 female Psychology professors - totally 10 Psychology professors. There are 6 female professors out of all Psychology professors. So, (6/10) X 100 = 60%

19. Solution: (a)

Look at figure2 - 30% of Physics professors belong to the age group 25-34. Then from figure1, we know there are 32 male and 8 female Physics professors.

So, totally 40 Physics professors, out of that 30% belong to 25-34 age group.

So there are 12 (30% of 40) Physics professors in the age group 25-34 .....S1

Further its given that the number of female Physics professors in the age group 25 - 34 equals 25% of all the Physics professors in that age group, so, 25% of 12 i.e. 3 are female professors in that age group S2

From S1 and S2: 12-3=9 male professors belong to 25-34 age group.

20. Solution: (b) Simple arithmetic.

Tables

21. Answer: (a)

Explanation:

To find the cutoff score, we first need to calculate the total weighted score for each candidate. The formula for the total score is:

Total Score = (Score in component 1 × 0.2) + (Score in component 2 × 0.3) + (Score in component 3 × 0.5)

Let's compute the total score for each candidate step-by-step:

- **Candidate 1:** 5 × 0.2 + 4 × 0.3 + 6 × 0.5 = 1.0 + 1.2 + 3.0 = 5.2
- **Candidate 2:** 4 × 0.2 + 6 × 0.3 + 5 × 0.5 = 0.8 + 1.8 + 2.5 = 5.1
- **Candidate 3:** 3 × 0.2 + 2 × 0.3 + 8 × 0.5 = 0.6 + 0.6 + 4.0 = 5.2
- **Candidate 4:** 9 × 0.2 + 4 × 0.3 + 3 × 0.5 = 1.8 + 1.2 + 1.5 = 4.5
- **Candidate 5:** 8 × 0.2 + 8 × 0.3 + 2 × 0.5 = 1.6 + 2.4 + 1.0 = 5.0

Now, let's arrange the total scores of all candidates in descending order to identify the top performers: Candidate 1: 5.2, Candidate 3: 5.2, Candidate 2: 5.1, Candidate 5: 5.0, Candidate 4: 4.5.

The recruitment process requires exactly three candidates to be selected. This means we must select the top three scorers. The top three scores are 5.2, 5.2, and 5.1.

To ensure exactly three candidates make the list, the cutoff score (the minimum score required to be selected) must be set at the score of the third-highest candidate. Therefore, the cutoff score is 5.1.

22. Answer: (b)

Player	Total run scored in both the innings (I+II)	Total balls faced in both innings (I+II)	Total runs / Total balls (Run rate)
A	61 + 14 = 75	99 + 76 = 175	75/175 = 0.428
B	5 + 50 = 55	12 + 85 = 97	55/97 = 0.567
C	15 + 20 = 35	75 + 50 = 125	35/125 = 0.28
D	13 + 12 = 25	55 + 50 = 105	25/105 = 0.238

∴ Fastest run scorer in Test Match (highest run rate = 0.567) is player. Hence, option (b) is correct.

23. Answer: (a)

The correct option is (a)

In order to avoid any loss, selling price per piece must be greater than or equal to cost price per piece.The minimum loss is when the selling price is equal to the cost price.

Using graphs, the cost price of 2000 pieces = ₹7 lakhs.

The selling price of 2000 pieces = 2000 x 350 = ₹7 lakhs

So, the minimum number of pieces that should be manufactured to avoid a loss is 2000.

24. Answer: (c)

The correct option is (c)

Dependency ratio is defined as the ratio of the non working population to the working population.(15 to 64 is working population)

Statement 1 is correct: The dependency ratio of country B has decreased in the last two decades as working the population has increased during this period.

Statement 2 is incorrect: The dependency ratio of country A is more than the dependency ratio of CountryB because the working population of country B has increased more than the working population of country A.

Statement 3 is correct: by the graph we can see that the working population of country B to the overall population has increased more than the working population of country A to the overall population.

25. Answer: (d)

The correct option is (d)

By the graph we can observe that central bank is increasing the rates to suck excessive liquidity from economy.

Hence, central bank is trying to curb inflation.

26. Answer: (a)

The correct option is (a)

Statement 1 is correct: direct tax includes income tax and corporate tax while other 3 constitutes indirect taxes,now observe the graph and each curve we can see that During the given period, the revenue from Direct Taxes as percentage of gross tax revenue has increased while

that of Indirect Taxes decreased.

Statement 2 is incorrect: observe the curve of excise we can clearly see that in last segment of curve it is increasing,this statement is not correct

27. Answer: (c)

The correct option is (c)

Statement 1 s correct: observe the state15 and state 16,per capita income of state 15 is more than that of state 16 but tele —density of state 15 is less but this happens in very few cases but by the table you can say that generally Higher per capita income is generally associated with

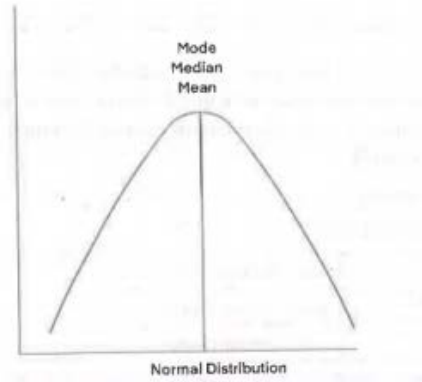
higher Tele-density.

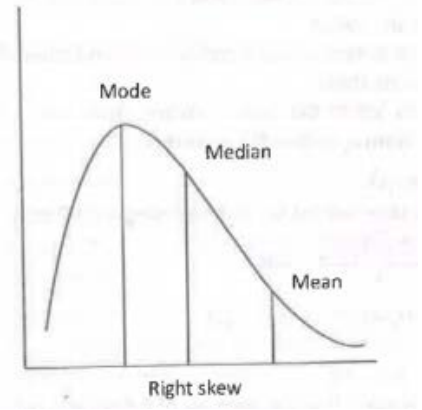
Statement 2 is incorrect: observe the state l5 and state 16, state 16 has higher GDP than state15 but per capita income of state 15 is more than that of state 16.

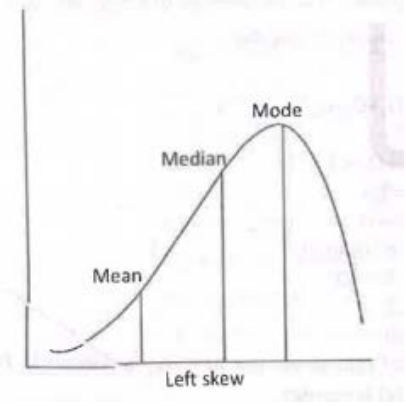
Statement 3 is correct: check for state 2 and state 8 GDP of state 2 is less than GDP of state 8 but tele - density of state 2 is higher than state 8.

28. Answer: The correct option is option (d)

When Mean < Median < Mode, then the distribution is skewed to the left.







29. Answer: (d)

The correct option is option D

To find mean aggregate marks we 1st count the total marks and maximum marks of both students.

Total marks obtained by A= 60+ 70+30 +50 = 210 Maximum marks for A= 100+ 100+ 100+50 = 350. Means aggregate percentage of marks A get= Total marks obtained by B=80+ 70+60 + 15 = 225 Maximum marks for B = 150+ 100+100+25=375 Mean aggregate percentage of marks B get

Hence, both students got same mean aggregate percentage marks.

Hence, option (d) is correct.

30. Answer: (c)

The correct option is option C

Let's check option (a):

Population in 1992 = 20 lakhs

Population is 1993 = 21 lakhs, change in population = increased by 1 lakh

Population in 1994 = 22 lakhs, change in population = increased by 1 lakh

Now increase in percentage in 1993 =

$$\frac{\text{Change in Population}}{\text{Population of 1993}}\times 100=\frac{1}{20}\times 100=5\%$$

Now increase in percentage in 1994 =

$$\frac{\text{Change in Population}}{\text{Population of 1994}}\times 100=\frac{1}{21}\times 100=4.9\%$$

Now in 1995 percentage increase in less than 5% Hence, option (a) is incorrect.

Let's check option (b):

By the pattern given above just change population by income and check you will find that in year 1995 the income is increased by less than 10 %.

As income in 1994 = 1225

Income in 1995 = 1345

Change in income = 1345 - 1225 = 120

Percentage change = $\frac{\text{Change in Population}}{\text{Population of 1994}}\times 100=\frac{120}{1225}\times 100=<10\%$

Hence, option (b) is not correct.

Let's check option (c):

Per capita income = total income/ population

Per capita income in 1992 = 101000/20 = 5050 (1crore = 100 lakhs, as population is given in lakhs)

Per capita income in 1993 = 111100/21 = 5290.4

Per capita income in 1994 = 122500/22 = 5568

Per capita income in 1995 = 134500/23 = 5380

Hence, in ever year per capita income is more than 5000.

Hence, option (c) is correct.

Data sufficiency

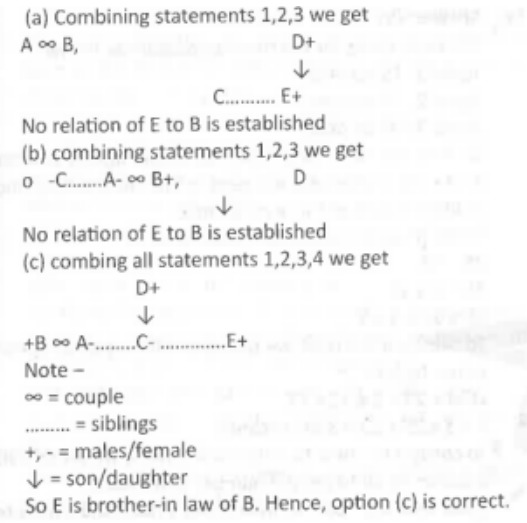
31. Answer: (b)

Statement 1 is incorrect: At minimum two letters will get misplaced as shown below:

A	B	C	D
B^1	A^1	C^1	D^1

Statement 2 is correct: If two letters go in correct envelopes only then above figure gives the correct demonstration. So, the idea is to simply choose two things out of four i.e. AB, AC, AD, BC, BD and CD or 4C2 ways i.e. 6 ways are there for it.

32. Answer: (c)



33. Answer: (c)

Statement 1: There are not more than two figures on any page of a 51-page book.

From this statement we cannot Answer the question whether there are more than 100 figures in the book because:

Situation 1: every page has two figures then no. of figures = 102

Situation 2: zero figure on all pages then no. of figures in book = 0.

Hence, by statement 1 we cannot conclude about the number of figures in book.

Statement 2: There is at least one figure on every page.

Situation1: every page has two figures then no. of figures =102

Situation 2: 1 figure on all pages then no. of figures in book=1.

Hence, we got two different Answers in both situations. Hence, by statement 2 we cannot conclude about the number of figures in book.

By combining both the statements:

Combined statement is: there are at least one figure and maximum two figures on a page.

Situation1: every page has two figures then no. of figures =102

Situation 2: 1 figure on all pages then no. of figures in book =51.

We got two different Answers in both situations. Hence, by combining both statements we cannot conclude about the number of figures in book. Hence, option (c) is correct.

34. Answer: The correct option is option (b)

Using S1: Both p and q are prime numbers.

So, p and q can take values 2(even), 3, 5, 7, 11, 13, 17...

We have to determine if pq is an odd integer.

Take p = 2 and q = 3, then pq = 6(even)

Take p = 3 and q = 5, then pq =15(odd)

So, Statement 1 is not sufficient to Answer whether pq is even or odd.

Now, consider S2: p + q is an odd integer.

This is possible when one of p or q is even and the other is odd.

Since, even + even even and odd + odd = even

Also, multiplication of (odd) (even) = even (always). So, pq is not an odd integer.

S2 alone is sufficient to Answer the question. **Hence, option (b) is correct.**

35. Answer: The correct answer is option (d)

Interpretation:

Let the weight of Rohan, Sohan and Mohan be x, y and z respectively.

According to S1: 2y <x or 2y <z

2y < x or z

y <x or z

According to S2: 2x > y or 2x > z

y < 2x or z < 2x

From S1: We know that y is less than x and z but we cannot determine whether x is greater than z or z is greater than x.

It means; the greatest weight among the three cannot be determined from S1.

From S2: We get, y < 2x or z < 2x.

We are given that z is less than 2x.

->z can be greater than x or can be less than x also while being less than 2x.

Let z = 5 and x = 4. Here z is greater than x but less than 2x

Let z 5 and x = 6. Here z is less than x and 2x.

So, we cannot determine whose weight is greatest. **Hence, option (d) is correct.**

36. A five-storeyed building with floors from I to V is painted using four different colours and only one colour is used to paint a floor. Consider the following statements:

1. The middle three floors are painted in different colours.
2. The second (/1) and the fourth (IV) floors are painted in different colours.

3. The first (1) and the fifth (V) floors are painted red.

To ensure that any two consecutive floors have different colours (CSAT 2019)

(a) Only statement 2 is sufficient.

(b) Only statement 3 is sufficient.

(c) Statement 1 is not sufficient, but statement 1 along with statement 2 is sufficient.

(d) Statement 3 is not sufficient, but statement 3 along with statement 2 is sufficient.

37. Answer: (d)

The correct option is (d)

Statement 1 is incorrect: take x = 1 then 1-y = 8 Hence, y = -9

Statement 2 is incorrect: take x = 10 then 10 - y

Hence, y = 2 (positive).

Statement 3 is incorrect: take x =-1 then-1-y=8 Hence,

9=-9 (negative).

Miscellaneous

38. Answer: (a) is correct.

According to the information provided in the given question,

Z=X+8...(a)

P+Q=21…(b)

Q=P-5…(c)

X+Y=28…(d)

P=X+5…(e)

We need to find the value of P+ Q+ X +Y + Z (the total number of notebooks)

Using (b) and (c), we get:

P+Q=21

P +(P-5)=21(from (c))

2P=21+5=26

P=13

From(c), Q=P-5=13-5=8

From (e), P=X+5

13=X+5

X=13-5=8

From (a), Z=X+8=8+8=16

From (d), X +Y=28

Y=28-X=28-8=20

Therefore, The total number of notebooks = P + Q + X + Y + 7 = 13 + 8 + 8 + 20 + 16 = 65

Price of 1 notebook = 40. So, Price of 65 notebooks = 65x40 = 2,600

Her{ce, the total cost of all notebooks is 2,600. **Hence, option (a) is correct.**

39. Answer:(c)

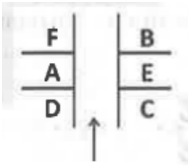
Using the information in the question, we will Make a table.

	P	Q	R	S	T	U	V
COVER	Red	Blue	Blue	Red	Blue	Red	Red
OLD/NEW	Old	Old	Old	New	Old	New	Old
LAW/ Gazetteers	Gazetteers	Gazetteers	Law	Law	Gazetteers	Gazetteers	Gazetteers

So, books of Old, Gazetteers and blue covers are Q and T. Hence, **option (c) is correct.**

40. Answer:(c)

The correct option is (c)



Using the given interpretation, we observe that

F, A and D are on the same side and B, E and C are on the opposite side.

The opposite officers are F and B, A and E, D and C.

Using the above explanation, if E sits in his offices and

So, CCIDD is the code for DELHI. Hence, option (a) is

faces the corridor, C is on his left.

41. Answer: (a)

The correct option is (a)

Using the above explanation, the immediate neighbour of F is A.

42. Answer: (b)

The correct option is (b)

Using the given information:

A		Commerce	Gujarat
B	City Q	Science	Kerala
C	City S	Arts	Rajasthan
D		Engineering	Assam

From table, A is studying in Gujarat

43. Answer: (a)

The correct option is (a)

From table, Science college is located in city Q.

44. Answer: (b)

The correct option is (b)

From table, C is studying in Rajasthan.

45. Answer: (a)

The correct option is (a)

From table, A Is studying in Commerce college in Gujarat and Engineering college is in Assam and C is studying in city S.

46. Answer: (d)

The correct option is (d)

From table, D is studying in Engineering College

47. Answer: (a)

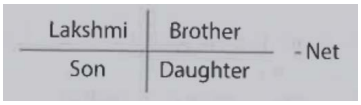
The correct option is (a)

Fromstable, engineering college is in Assam and city Q is in Kerala and C is studying in Rajasthan and B is studying in Kerala.

48. Answer: (a)

The correct option is (a)

We can visualize the given statements using a picture:



By using this picture and above statements we can conclude that her brother is best player.

49. Answer: b

The correct option is option B

From the question:

A, B, C, D - Medieval poet

E, F, G, H=Modern poet

Also it is given in question that:

Modern poets come alternate years.

Last year A came (medieval poet) which means this year, a modern poet will come.

Now paper setter does not like to ask about F. so, F is out of scene. Itis also given that paper setter likes F thus he also like F because F and E are liked together. Hence, E is the most probable poet for the year. **Hence, option (b) is correct**

50. Answer: (d)

The correct option is option D

Let's make a table by using information given in question:

NAME	PROFESSION
Girija	Violin
Vanaja	Violin, Vocal
Jalaja	Vocal, No violin
Shailaja	No violin, Dancer, Vocal
Tanuja	Dancer, Vocal, Violin
Pooja	Actress

Hence, by the table it is clear that Tanuja is certainly a dancer and a violinist. **Hence, option (d) is correct.**